# TFPT Verification Catalog

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Source registry edition

This reference volume lists the registered verification modules and their stated checks. It is generated from the script registry. A row or a passing finite check is not a proof of the complete physical theory. Claim status and scope remain governed by `verification/status_ledger.csv` and original sources.

The reading guide, research contracts and dated publication history are separate documents. See https://www.fixpoint-theory.com/documentation for the central source and evidence registry.

| Script | What it machine-checks |
|---|---|
| `v1_e8_glue.py` | glue theorem $E_8{=}(D_5{\oplus}A_3){+}\mu_4$: disc $=\mathbb{Z}_4$, $q(D_5){+}q(A_3){=}\frac{5}{4}{+}\frac{3}{4}{=}2$, $240{=}16{\cdot}5{\cdot}3$, $248{=}240{+}8$ |
| `v2_carrier_pascal.py` | $g_{\rm car}{=}5$ unique Pascal solution; $16{=}1{+}5{+}10$, $N_{\rm fam}{=}3$, rank $E_8{=}8$, $\Omega_{\rm adm}{=}48$, $b_1{=}\frac{41}{10}$ |
| `v3_em_alpha.py` | EM closure: unique root of $F_{U(1)}(\alpha){=}0$, $\alpha^{-1}{=}137.0359992168$, existence/uniqueness, inverse test, ablation |
| `v4_flavor_matrix.py` | residue matrix $R$: $\det{=}8$, minors $\{2,3,5\}$, $\chi_R{=}t^3{-}9t^2{+}10t{-}8$, SNF $\mathrm{diag}(1,1,8)$, $\|R\|_F^2{=}78$, $\sum L{=}40$ |
| `v5_e8_cascade.py` | cascade $D_n{=}60{-}2n$: endpoints, exponent rungs $\to 240$, IR tail 46080, variance 6552 |
| `v6_bootstrap.py` | reverse glue $\mu^2{-}5\mu{+}4{=}0$; $g_{\rm car}{=}5$ three ways; $8{=}\,\mathrm{rank}\,E_8{=}h(D_5){=}\varphi(30)$ |
| `v7_gravity_cosmo.py` | scalaron exponent 7, $M^2_{\rm scal}/\bar{M}^2_{\rm Pl}{=}c_3^7$, $A_s, n_s, r$, $\Omega_b$, $\sin^2\theta_{12}$ |
| `v8_horizon.py` | horizon code: $\frac{1}{2\pi}{=}4c_3$, $1920{=}\lvert W(D_5)\rvert$, $S_{dS}$, Page, $\beta_{\rm rad}{=}0.2424^\circ$ |
| `v9_neutrino_texture.py` | solar angle: explicit $\mu\tau$ Majorana texture $\to \sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\rm ret}}{2}$, $\theta_{23}{=}45^\circ$, $\theta_{13}{=}0$ |
| `v10_projection_involution.py` | $Q,\Sigma$ algebra: $K{=}R{+}Q\Sigma$, $L{=}R{+}Q(I{+}\Sigma)$; $\chi_Q,\chi_K$; det ladder $3{\cdot}4{\cdot}8{\cdot}20{=}1920{=}\lvert W(D_5)\rvert$; $a^\top K a{=}41$ |
| `v11_unique_KQ.py` | $K,Q$ are the *unique* nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration) |
| `v12_mass_generation_polynomials.py` | sector/generation polynomials of $K$ and the anchor-block det ladder $(9,10,16,40)$ |
| `v13_open_gates.py` | gate closures: $M{=}41{=}10b_1{=}\sum L{+}N_\Phi$; $Q_+ = A_3$ exponents, $Q_-^2{=}N_{\rm fam}$ |
| `v14_carrier_uniqueness.py` | $g_{\rm car}{=}5$ unique ($N_{\rm fam}\in\mathbb{Z}^+$, rank $E_8{=}8$); split $(3,2)$ from $b{+}s{=}5, b^2{+}s^2{=}13$; $\mathrm{Tr}\,Y{=}\mathrm{Tr}\,Y^3{=}0$ |
| `v15_bootstrap_classification.py` | $D_5{\oplus}A_3$ unique familyful cyclic-glue of $E_8$ (16-spinor + 3 families), glue $\mathbb{Z}_4$ |
| `v16_solar_dual_anchor.py` | $a^\top R^{-1}{=}a^\top L^{-1}{=}(-\frac{1}{2},-\frac{1}{2},1)$; $\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\rm ret}}{2}$; full PMNS |
| `v17_hexagonal_resolvent.py` | finite resolvent $D_y^{-1}{=}(\sum y^{5-m}\delta^m U_6^m)/(y^6{-}\delta^6)$ (quark $c$ backbone) |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v18_quark_yukawa.py` | quark source ratios $\frac{m_u}{m_d}$=0.470085, $\frac{m_c}{m_s}$=13.61, $\frac{m_t}{m_b}$=40.80 exact; lepton $c$=($\frac{16}{7}$, $\frac{4}{3}$, $\frac{7}{6}$); full hierarchy |
| `v19_monodromy_moduli.py` | exact pole $z_\star=\frac{794-7\sqrt{9961}}{2187}$ ($3^7$); $D_4$ $SU(3)$ monodromy constructible but $D_4$-fixed locus positive-dim ⇒ exact quark $c$ reduce to ($U$) |
| `v20_lepton_c_derivation.py` | lepton $c$'s derived: $\delta=\frac{1}{2}$ resolvent $c=\lvert\mu_4\rvert^w/(\frac{5}{4}-\cos\frac{r\pi}{3})$ ⇒ $\frac{16}{7}$, $\frac{4}{3}$; product $\frac{2^{g_{\rm car}}}{N_{\rm fam}^2}=\frac{32}{9}$ ⇒ $\frac{7}{6}$ |
| `v21_solar_product_quark.py` | solar coeff = $q(A_3)=\frac{3}{4}$ (glue-norm, $q(D_5)+q(A_3)=2$), $\sin^2\theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\rm ret}}{2}$; product $\frac{2^{g_{\rm car}}}{N_{\rm fam}^2}$; quark law fails ($\frac{1}{7}$ vs 0.724) |
| `v22_open_gates_audit.py` | residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual) |
| `v23_anchor_generator.py` | anchor $a$=(1, 1, 2): $e_k$=(4, 5, 2), $c_3=\frac{1}{2e_1\pi}$; $p_n$=2+$2^n$ ⇒240, 248, binary ladder; inputs → $\{a,\pi\}$ |
| `v24_quark_ratio_closure.py` | quark ratios $\frac{55}{117}$, $\frac{34}{47}$, $\frac{3}{26}$ from TFPT blocks; mass ratios match < 0.03%; rational $c$ gauge, $\Lambda_q$ in $O(1)$ |
| `v25_frontier_conjectures.py` | conjectures: Koide $Q_{\rm src}+\frac{\varphi_0^{\rm ret}}{24}\approx\frac{2}{3}$; axion $f_a=M_{\rm scal}/128\to m_a\approx23.8\,\mu$eV |
| `v26_flavor_frontier_unification.py` | the "11" is not uniquely forced (⇒ [C]); flavor frontier unifies to one ($U$) gate; cusp config on the parabolic stability wall |
| `v27_wall_representative.py` | explicit balanced wall rep $W_{\rm wall}$ (cols = perm(0, 1, 2), rows $w$=(2, 1, 1), pardeg 0); selector det $R$=8, Spec($Q_+$)=$\{1,2,3\}$ |
| `v28_gravity_fR.py` | $R$+$R^2$ closed: $f(R)$, $M=c_3^{7/2}\bar{M}_{\rm Pl}$ scalaron, $N_\star$=57→$n_s, r, A_s$; full metric measure open; $248c_3^2<6\log\frac{3}{2}$ |
| `v29_research_contract_certs.py` | research-contract certificates: $C_U^{(1)}$ wall enumeration (1296→144→5 orbits; symmetry alone does not pick $W_{\rm wall}$) and $G5$ gap-dominance $2\lVert V\rVert<\Delta$ |
| `v30_d4_character_variety.py` | $C_U^{(2)}$: full $D_4$-fixed $SU(3)$ character variety is positive-dimensional ⇒ selector needs the $R(\rho)$ dictionary (H2; exact since v118–v122, residue = the R4′ selector establishment) |
| `v31_R_dictionary.py` | $R(\rho)$ characterized: case A alive (irreducible ⇒ mixing); $R$ integer but $\rho$-invariants continuous ⇒ $R(\rho)$ is the transcendental non-abelian-Hodge map (residual = Hitchin solve) |
| `v32_rh_splitting.py` | RH route: $\sum_k U^k A_0 U^{-k}$=4 diag($A_0$) exact ⇒ splitting $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2\Leftrightarrow$diag($A_0$)=($\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{4}$) (algebraic); $\prod M_k=I$ + $R$-bridge stay open ($c_u/c_d$ not forced) |
| `v33_explicit_flat_bundle.py` | explicit valid ($U_{\rm wall}$) flat bundle (RH solve): cusp + $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$ + $\lVert M_\infty-I\rVert\sim10^{-9}$ ($\prod M_k=I$) + irreducible (case A) |
| `v34_h2_bridge_attempt.py` | H2 bridge: explicit per-puncture $M_k$ (cusp, $\prod M_k=I$); honest negative — natural extraction ≠ lepton amplitudes, the $\Gamma^{\rm min}$ dictionary is the remaining input ($c_u/c_d$ not obtained) |
| `v35_quark_qcd_boundary.py` | the open gates are the discrete↔continuous boundary: quark cross-ratio cleanness tracks generation/QCD sensitivity (compiler hands off to continuous physics) |
| `v36_spectral_action_g2.py` | G2: spectral action → $R$+$R^2$ ($a_2\sim-R/3$, $a_4\lvert_{R^2}\sim R^2/72$); $M^2/\bar{M}_{\rm Pl}^2=6(4\pi)^2/f_0$, $c_3^7$ fixes $f_0$; gap-decoupling $\Delta_{\rm eff}$=1.648>0 ⇒ admissible IR closed under RP, ambient (G6) = strict-TOE completion target |
| `v37_plucker_anchor.py` | anchor-plane Plücker apparatus: $\lVert{\rm Pl}(K)\rVert_1$=11, $\lVert{\rm Pl}_R(K)\rVert_1$=26, $\sum{\rm Pl}_R(L)$=60; pencil det($K$+$xQ$)=$3x^3$+$7x^2$+6$x$+4; $K_e-K_u=a$, $L_e-L_u$=Exp($A_3$); lepton ring $c_ec_\tau=\lvert\mathbb{Z}_2\rvert c_\mu$; Koide $u\to\frac{53}{54}u$ ([C]) |
| `v38_uwall_killswitch.py` | ($U_{\rm wall}$) U2 kill-switch hardened: $D_4$-cusp rep irreducible at all 400 sampled points (commutant=1), mixing ≥0.35, reducible locus measure-zero ⇒ case A generic, ($U_{\rm wall}$) survives; amplitude dictionary still research-level |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v39_uwall_selectors.py` | ($U_{\text{wall}}$) selector side closed on the explicit point: splitting type = anchor $a{=}(1,1,2)$, pardeg $=0$; $\det R{=}8{=}n{\cdot}a$, $\mathrm{Spec}(Q_+){=}\{1,2,3\}{=}3\alpha{+}1$ read off the bundle; only the amplitudes $c_u/c_d$ ($\Gamma^{\min}$ Hitchin) remain |
| `v40_harmonic_metric.py` | ($U_{\text{wall}}$) harmonic metric = FINITE linear algebra (polystable⇒unitary⇒$\Phi{=}0$): unique invariant form $H{>}0$, unitarised $\widetilde{M_k}$ cusp-class with $\lvert\widetilde{M_k}\rvert{\in}\{0,\frac{1}{2},\frac{1}{\sqrt{2}}\}$; the feared Hitchin PDE collapses to unitarisation + combinatorial $R$ |
| `v41_leg_assignment.py` | ($U_{\text{wall}}$) final leg test: amplitudes = $\delta{=}\frac{1}{2}$ resolvent (closed); $\Lambda_\mu{>}1$ rules out holonomy moduli; harmonic holonomy $\lvert\mathrm{diag}\rvert{=}(0,\frac{1}{2},\frac{1}{2})$ supplies $\delta{=}\frac{1}{2}$ (suggestive); honest negative on literal $c_u/c_d$ reproduction |
| `v42_exterior_leg.py` | ($U_{\text{wall}}$) Exterior Leg Lemma: $u,d$ share $(r,w){=}(1,1)$ so any scalar leg gives $c_u/c_d{\equiv}1$; the $u/d$ split is the $\Lambda^2F$ area $\mathrm{Pl}_{\mathbf{1},a}(K)$, $\lVert\cdot\rVert_1{=}11$ generated ⇒ $\frac{5{\cdot}11}{9{\cdot}13}{=}\frac{55}{117}$ **[E]**; phys. identification **[C]** |
| `v43_exterior_bridge.py` | ($U_{\text{wall}}$) $\Lambda^2F$ bridge: $\Lambda^2(\widetilde{M}){=}\overline{\widetilde{M}}$ (exterior leg $=\overline{\mathbf{3}}$) confirmed; continuous action ≠ integer $\mathrm{Pl}(K)$ ⇒ bridge is the discrete H2 invariant ($\det R{=}8$, $\mathrm{Spec}(Q_+)$ confirmed), not continuous |
| `v44_carrier_exterior.py` | ($U_{\text{wall}}$) Lie-level grounding: $16{=}\Lambda^{\text{even}}(5){=}1{+}10{+}5$; $u^c{\in}\Lambda^2(5)$, $d^c{\in}\Lambda^4(5)$ ⇒ exterior degrees $(2,4)$, sum$=6{=}\lvert R^+(A_3)\rvert$, diff$=2{=}\lvert\mathbb{Z}_2\rvert$; the exterior leg is the carrier's own $\Lambda^2$ grading (no special math) |
| `v45_family_exterior.py` | ($U_{\text{wall}}$) the "11" is Pascal: $\lvert\mathrm{Pl}(K)\rvert{=}(1,4,6){=}\{\Lambda^0,\Lambda^1,\Lambda^2\}(\mu_4)$; $\lVert\mathrm{Pl}(K)\rVert_1{=}11{=}\sum_{k\le2}\binom{4}{k}{=}16{-}g_{\text{car}}$ (cumulative ext of $\mu_4$ up to $\deg u^c{=}2$); $15{=}\dim\mathfrak{su}(4){=}10{+}\bar{5}{=}N_{\text{fam}}g_{\text{car}}$ |
| `v46_grand_mass_volume.py` | absolute mass-scaling is **[E]**, not a product rule: sector dets $(\varphi_0^{\text{ret}})^6,(\varphi_0^{\text{ret}})^9,(\varphi_0^{\text{ret}})^{10}$ (exponents $\lvert R^+A_3\rvert,N_{\text{fam}}^2,\mathcal{A}_\Lambda$) ⇒ $\det M_{\text{SM}}{\sim}(\varphi_0^{\text{ret}})^{25}{=}(\varphi_0^{\text{ret}})^{g_{\text{car}}^2}$; $Q$-rows $(4,5,6)$ sum $15{=}\dim A_3$; $25{+}15{=}40{=}\sum L$ |
| `v47_selection_theorem.py` | Boundary Carrier Selection (Thm A): $\mu_4{\to}A_3$, $N_{\text{fam}}{=}3$; 16-spinor$\to D_5$; $q(D_5){+}q(A_3){=}2$, glue $4{=}\lvert\mu_4\rvert$; only $D_5{\oplus}A_3$ meets all four conditions ($D_8$, $E_7{+}A_1$, $E_6{+}A_2$, $A_4{+}A_4$, $A_8$ excluded) **[E]**/**[C]** |
| `v48_em_ward.py` | EM boundary Ward (Thm C): $F_{U(1)}{=}$ Maxwell $\alpha^3$ − Calderón $2c_3^3\alpha^2$ − transport $8b_1c_3^6\log\frac{1}{\varphi_{\text{seam}}}$; $8b_1{=}\frac{4}{5}M{=}\frac{164}{5}$, $-\frac{5}{4}{=}q(D_5)$ **[E]**; Ward origin **[C]** |
| `v49_readout_rigidity.py` | Readout Rigidity (Thm U2): $c_u/c_d{=}\frac{55}{117}$ is constant on the discrete selector stratum $\mathcal{S}_{8,Q}$ ⇒ no point-uniqueness needed; $(U_{\text{wall}}){=}U_{\text{unitary}}{+}U_{\text{H2}}{+}U_{\Lambda^2}{+}U_{\text{point}}$, only $U_{\text{point}}$ open |
| `v50_q_geometry.py` | Q geometry (Thm Q): $Q_+{=}$ diag$(3,2,1)$-block $\chi{=}(t{-}1)(t{-}2)(t{-}3)$, $Q_-$ $\chi{=}t(t^2{-}3)$; $Q$ unique under rows$(4,5,6)$/cols$(9,5,1)$/spectra; geom. origin **[C]** |
| `v51_boundary_half_step.py` | glue norms $=(g_{\text{car}},N_{\text{fam}})/\lvert\mu_4\rvert$: $q(D_5){=}\frac{5}{4}$, $q(A_3){=}\frac{3}{4}$; four ops forced → {sum$=$ 2 root norm, diff$=\frac{1}{2}{=}\lvert\mathbb{Z}_2\rvert/\lvert\mu_4\rvert{=}\delta$, prod$=\frac{15}{16}$, ratio$=\frac{5}{3}$}. So $\delta{=}\frac{1}{2}$ is the carrier−family count over the glue index, not chosen **[E]** |
| `v52_pencil_endpoints.py` | $K{+}xQ$ pencil endpoints: $P(-1){=}2{=}\lvert\mathbb{Z}_2\rvert$, $P(0){=}4{=}\lvert\mu_4\rvert$, $P(1){=}20{=}\det L$, $P(2){=}68{=}2p_5(a)$; $\det(K{-}Q){=}2$, $\mathrm{tr}{=}N_{\text{fam}}$ (audit ext. of v37) **[E]** |
| `v53_compiler_core.py` | Big picture: the whole integer skeleton from $(g_{\text{car}},N_{\text{fam}}){=}(5,3)$ — $\mathrm{rank}\,E_8{=}8$, $\lvert\mathbb{Z}_2\rvert{=}2$, Pythagorean $\Delta_Y{=}g_{\text{car}}^2{=}25{=}N_{\text{fam}}^2{+}\lvert\mathbb{Z}_2\rvert{\cdot}\mathrm{rank}\,E_8{=}9{+}16{=}\dim S^+$ (diff. of squares); anchor char-poly $(t{-}1)^2(t{-}2)$ unique **[E]** |
| `v54_seam_horizon_keystones.py` | The "8" in $c_3{=}\frac{1}{8\pi}$ overdetermined and gravitationally aligned $(2\lvert\mu_4\rvert{=}\mathrm{rank}\,E_8{=}h(D_5){=}\varphi(30){=}8\pi$ Hawking/Einstein); one boundary transport operator (spec $\{1,(2/3)^6,(1/3)^6\}$, $\lambda_2{=}(2/3)^6$) governs *both* SM flavor gap and horizon Page recovery **[E]** |
| `v55_coxeter_cycle.py` | Computed $E_8$ Coxeter element: order exactly $30{=}\lvert\mathbb{Z}_2\rvert N_{\text{fam}}g_{\text{car}}$, eigenphases = exponents = totatives of 30, so $\mathrm{rank}\,E_8{=}8{=}\varphi(30){=}$ live phases of the order-30 cycle; one $\alpha^{-1}$ sets EM, $S_{dS}$ and $\Lambda$ ($S_{dS}\rho_\Lambda{=}32\pi^4$) **[E]** |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v56_unique_attractor.py` | Gapped boundary transport (gap $6\log\frac{3}{2}>0$) ⇒ *unique* attracting fixed point, geometric rate $(2/3)^6$ (two starts → same direction); Coxeter in $\lvert\mu_4\rvert=4$ planes; entropy reset = rank-1 projector [E] |
| `v57_horizon_crosslinks.py` | Horizon cross-links: $c_3=\frac{1}{8\pi}$ is the Einstein/Jacobson coefficient ($1/4=1/\lvert\mu_4\rvert$ entropy density); Hod QNM $\omega_R/T_H=\ln 3=\ln N_{\rm fam}$; Hawking power denom $1920=\lvert W(D_5)\rvert$; one $\alpha^{-1}$ for EM/$S_{dS}$/Λ/Hubble [E] (+ [C] identification) |
| `v58_seam_horizon_chain.py` | Seam-horizon chain (precise: seam = abstract normaliser, locally realised as a horizon): one-sided $S^2$ Gauss–Bonnet $c_3=1/(2\cdot4\pi)=\frac{1}{8\pi}$; KMS $4c_3\kappa$; $S_{BH}=M^2/2c_3$; RT in seam units. Seam–Horizon Theorem (area-law from the seam kernel) [O] open |
| `v59_area_law_evidence.py` | Area-law necessary condition met: a reflection-positive Gaussian (Calderón-type) kernel yields an area law (gapped block-EE saturates; gapless $\sim\frac{c}{3}\log L$), by the *same* gap that fixes the attractor; $8=\lvert\mathbb{Z}_2\rvert\lvert\mu_4\rvert$. The $c_3$ coefficient stays [O] open |
| `v60_lambda_metrology_branch.py` | Λ branch selection: $(8\pi)^2\delta_{\rm top}=\frac{3}{4\pi^2}$, mis-scale $2c_3/\delta_{\rm top}=\frac{64\pi^3}{3}=661.47$ ⇒ $G_N$ pinned as output; 123 orders = 119.028+3.920; ACT DR6 birefringence $0.215°\pm0.074°$ vs $0.2424°$ ($0.37\sigma$) [E] |
| `v61_cft_bridge.py` | Boundary-CFT mirror: level-1 WZW central charges $c(E_8)=8$, $c(D_5)=5$, $c(A_3)=3$ = (rank $E_8$, $g_{\rm car}$, $N_{\rm fam}$); conformal embedding $c_{\rm coset}=0$; $SU(3)\leftrightarrow SU(4)$ reconciliation $N_{\rm fam}=\lvert\mu_4\rvert-1$, 4→3+1 (one geometry $\mathbb{P}^1\setminus\mu_4$, two sides) [E] |
| `v62_data_scorecard.py` | TFPT vs 2024/25 data: $\alpha^{-1}$, $\sin^2\theta_{12}$, $\beta_{\rm rad}$ match; $\sin^2\theta_{13}$ ($2.0\sigma$) and Starobinsky $n_s$ vs DESI ($\sim2\sigma$) mild tensions; $\theta_{23}$ open; $r\approx0.004$ future falsifier [E] |
| `v63_seam_engineering_index.py` | Exact Seam-Engineering Index $\Xi=2\lVert V\rVert/\Delta=\frac{31}{24\pi^2\log(3/2)}\approx0.323$ ($2\lVert V\rVert=\frac{31}{4\pi^2}$, $248=\dim E_8$, $31=1+h^\vee$), $\Delta_{\rm eff}\approx1.648$; four-class possibility taxonomy [E]/[O] |
| `v64_causal_boundary_nogo.py` | Causal Boundary Engineering *conditional no-go*: gapped transfer has no periodic orbit (discrete no-CTC); RP+OS+gap ⇒ ANEC ⇒ Topological Censorship ⇒ no traversable shortcut; only ER=EPR bridge survives [E] (core)/[C] (chain) |
| `v65_falsification_layer.py` | Prediction layer with explicit failure thresholds: 3 core ($v_{\rm GW}=c$, no 4th gen, seam=8), 4 observational ($\beta$, $r$, $n_s$, $\theta_{12}$), no-go + unitarity; $N_\star$ demoted to [C] input; none violated [E] |
| `v66_e8_casimir_degrees.py` | Organizing simplification: the compiler atoms *are* the $E_8$ Casimir degrees $\{2,8,12,14,18,20,24,30\}$ ($2=\lvert\mathbb{Z}_2\rvert$, 8= rank, $14=\dim G_2$, $20=\det L$, $24=\lvert W(A_3)\rvert$, $30=h(E_8)$); $\sum=128=2^7$ (dS prefactor), $\sum\exp=120=\lvert R^+(E_8)\rvert$ [E] |
| `v67_area_law_coefficient.py` | Central-theorem *structural closure*: Fursaev–Solodukhin $S=4\pi kA$, $k=c_3/2$ ⇒ $S=2\pi c_3A=\frac{1}{4}A$ ($2\pi c_3=\frac{1}{4}$); $c_3=\frac{1}{8\pi}$ is the *unique* value with replica-coeff = BH $\frac{1}{4}$. Reduces to the one coeff $k=c_3/2$ (then *forced*, v73) [E] |
| `v68_seeley_dewitt_residual.py` | Residual *resolved as an anchor*: $a_2=-\frac{d}{192\pi^2}R$ ⇒ the *absolute* $\frac{1}{16\pi G}$ is UV-sensitive ($\propto d\Lambda^2 f_2$), so the *scale* $1/G$ is the one dimensionful anchor (the dimensionless $k=c_3/2$ is forced, v73); cutoff-indep. $R^2$/scalaron derived [E] |
| `v69_d4_q_geometry.py` | $D_4$-equivariant Q-geometry (gate [C]/[E]): on $H_1(\mathbb{P}^1\setminus\mu_4)=B_1\oplus E$, $Q_+=3w+1=\mathrm{diag}(1,2,3)$ (cusp weights on $\mu_4=\mathbb{Z}_4$ eigenspaces), $Q_-=E$-coupling $\sqrt{N_{\rm fam}}$ ⇒ $t(t^2-3)$; Σ-split $=D_4=\mathbb{Z}_4\rtimes\mathbb{Z}_2$ [E] |
| `v70_q_integer_lift.py` | Q integer-lift sharpened: $\mathbb{Z}_4$ puncture action $R$ unimodular (eigenvalues $\{-1,i,-i\}$); $\det Q=3=N_{\rm fam}$, SNF $\mathrm{diag}(1,1,3)$ ⇒ residual = one lattice invariant $\det Q=N_{\rm fam}$, not closable by $D_4$ alone [E] |
| `v71_simple_r_bridge.py` | *Simple* R-bridge (gate #3): selector stratum $\mathcal{S}_{8,Q}$ now fully derived ($\det R=8=\mathrm{rank}\,E_8$ lattice + $\mathrm{Spec}(Q_+)=\{1,2,3\}$ from v69) ⇒ quark *ratios* are pure integer Plücker readouts ($\frac{55}{117}$, no monodromy, v49 rigidity); residual $U_{\rm point}=$ absolute norm. = an anchor [E]/[O] |
| `v72_q_det_from_cusp.py` | $\det Q=N_{\rm fam}$ derived from the cusp class: $\det Q=\lvert\mathrm{coker}\,Q\rvert=$ cusp-class order = denominator of the weights $\{0,\frac{1}{3},\frac{2}{3}\}$ (same data as $\mathrm{Spec}\,Q_+$); $\mathrm{coker}\,Q=\mathbb{Z}/N_{\rm fam}=$ triality deck group ⇒ integer-lift residual closed [E] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v73_k_c3_half.py` | Central coefficient $k{=}c_3/2$ *forced*: (1/2) variational × $1/(\lvert\mathbb{Z}_2\rvert 2\pi\chi(S^2))$ Gauss–Bonnet topology ($\chi(S^2){=}2$), both cutoff-indep ⇒ Fursaev–Solodukhin $S{=}A/4$; only the absolute 4D Newton scale stays the anchor (v68) [E] |
| `v74_compiler_micro_lemmas.py` | Spine quotient ladder ($\frac{2}{3},\frac{3}{4},\frac{4}{5},\frac{5}{4},\frac{4}{3}$ = adjacent quotients of $(2,3,4,5)$); pencil differences $(2,16,48){=}(\lvert\mathbb{Z}_2\rvert,\dim S^+,\Omega_{\rm adm})$ sheet→1gen→3gens; anchor QF $a^\top Ka{-}\mathbf{1}^\top K\mathbf{1}{=}41{-}25{=}16{=}\dim S^+$ (EM budget = mass vol + one gen); solar dual anchor $L{=}R{+}6\mathbf{1}e_1^\top$, $a^\top R^{-1}\mathbf{1}{=}0$ (Sherman–Morrison rank-one invariance) [E] |
| `v75_upoint_to_vgeo.py` | **Gate 1 complete:** $U_{\rm point}{\to}v_{\rm geo}$. Ratios closed (v49/v71) + lepton $c$ exact (v20) + sector products = $\varphi_0^{\rm ret}$-powers (v46); (ratios, product)⇔(individuals) bijection ⇒ absolute amplitudes fixed up to *one* scale $v_{\rm geo}$ = the same anchor as $1/G$ (v68); two $[A]\to$ one [E]/[O] |
| `v76_gmetric_reduction.py` | **Gate 2:** Tier A (IR) closed under RP+gap (Decoupling Thm: $\Delta_{\rm eff}{=}6\log\frac{3}{2}{-}\frac{31}{4\pi^2}{=}1.648{>}0$, $\lVert V\rVert{\le}248c_3^2{=}\frac{31}{8\pi^2}$); Tier B (strict TOE) holographically reduced from bulk to a finite seam-*boundary* measure (conditional: RP+tightness ⇒ closes) [E]/[C] |
| `v77_e8_conformal_net.py` | **G6 route:** the seam boundary measure = the $(E_8)_1$ *lattice net* (rigorous, holomorphic $c{=}8$); level-1 $c(E_8){=}\frac{248}{31}{=}8$, $c(D_5){=}5$, $c(A_3){=}3$, embedding $5{+}3{=}8$ (coset $c{=}0$); imports $G6$ into RCFT/conformal-net rigor [E]/[C] |
| `v78_vgeo_floor.py` | $v_{\rm geo}$ **floor:** no pure number is dimensionful ⇒ $v_{\rm geo}$ irreducible by logic; all scales are ratios × one $\bar{M}_{\rm Pl}$; $S_{dS}\rho_\Lambda{=}32\pi^4$ pins it from one measurement; theory at the minimum (one scale + $\pi$) [E]/[O] |
| `v79_review_identities.py` | Four validated review identities + one firewall: hypercharge Lucas–Binet $D_n{=}(3^n{-}(-2)^n)/5{=}1,1,7,13,55,133$ (roots = carrier charges); Inverse Anchor Thm ($\mathbf{1}^\top M^{-1}\mathbf{1}{=}1$/atom, $a^\top M^{-1}a{=}1$ for $R,K,L$); MacWilliams $(1,10,5)$; $6Y^2{-}Y{-}1{=}(2Y{-}1)(3Y{+}1)$. Rejects $m_p/m_e$, $\eta_B$ near-formulas (firewall) [E] |
| `v80_operator_pencil_geometry.py` | Operator-pencil geometry: Anchor Singularity Thm $\det B_{K+xQ}{=}(N_{\rm fam}x{+}\lvert\mathbb{Z}_2\rvert)(N_{\rm fam}x{+}g_{\rm car}){=}(3x{+}2)(3x{+}5)$ ($\frac{2}{3}$ Koide/gap = anchor-plane singularity); buildup chain $(3,10,16,20)$; type checker $\det B_{Q,K,R,L}{=}(9,10,16,40)$; audit: differential spectrum, $F_4{\times}G_2$ shadow $248{=}52{+}14{+}26{\cdot}7$ [E]/[C] |
| `v81_singular_pencil_matrices.py` | Double cover $y^2{=}\det B_{K+xQ}{=}(3x{+}2)(3x{+}5)$: Koide $x{=}{-}\frac{2}{3}$ and carrier $x{=}{-}\frac{5}{3}$ are the two branch points (deck $2{=}\lvert\mathbb{Z}_2\rvert$; disc$=81{=}N_{\rm fam}^4$; separation$=1$ transport period). Clearing matrices $C_{2/3}{=}3K{-}2Q$ ($D_5{\oplus}A_3$ glue, $B{=}(5,7)^\top(9,11)$, $\sum{=}240$), $C_{5/3}{=}3K{-}5Q$ (neutral, $\chi{=}(\lambda{+}2)(\lambda^2{+}\lambda{+}6)$); scalaron 7= mixed anchor disc/$N_{\rm fam}$ [E] |
| `v82_koide_attractor_splitting.py` | Koide attractor *forced*: the branch-divisor-preserving Möbius map fixing $q{=}2,5$ is unique, multiplier $(2/3)^6{=}\lambda_2$ of the established transfer operator ⇒ $F'(2){<}1$ attractor, $F'(5){>}1$ repeller; three postulates → one [C]. Splitting trichotomy disc 81/49/40: the clean split is non-generic (anchor-first) [E]/[C] |
| `v83_e8net_holomorphic_uniqueness.py` | Holomorphy pins $(E_8)_1$: #primaries = det Cartan ($D_8{:}4$ vs $E_8{:}1$), and a holomorphic $c{=}8$ chiral CFT is the lattice theory of the *unique* even unimodular rank-8 lattice (Minkowski–Siegel mass $1/\lvert W(E_8)\rvert$); carrier Pascal truncation $K{=}(g{-}1)/2$ = half-spinor split [E] |
| `v84_frozen_registry.py` | Blind-prediction registry frozen 2026-06-09 (25 digits, machine-enforced lock formula↔value): exactly *one* $\theta_{12}$ prediction of record; $r$, $n_s$ as $N_\star$ *bands*; conditional entries carry their hypothesis by name [E] |
| `v85_master_cover.py` | Master-cover theorem: $\det B$ is GL(2)-covariant on $\mathrm{span}\{K,Q\}$ ⇒ exactly *one* double cover up to Möbius (disc = $N_{\rm fam}^4\det(G)^2$); the disc-81 family is its orbit ($-\frac{8}{3}{=}{-}{\rm rank}E_8/N_{\rm fam}$ = carrier − one period); $\mu_4$ is *not* a 4:1 cover (ladder of double covers); branch trace fixes only the scalaron *scale* [E] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v86_nstar_reheating.py` | $N_\star$ band → point [C]: $\Gamma{=}4M^3/48\pi\bar{M}^2{=}128\,\mathrm{GeV}$, $T_{\rm reh}{=}9.6{\times}10^9$ GeV, $N_\star(0.05/\mathrm{Mpc}){=}51.4 \Rightarrow n_s{=}0.9611$, $r{=}0.0045$ (inside the frozen band); honest tensions: $n_s$ $+0.9\sigma$, $A_s$ dichotomy (slow point $-11.4\sigma \Rightarrow$ near-instant reheating or exponent fails) [C] |
| `v87_bulk_uniqueness_reduction.py` | Red-team Target A residual (ii) conditionally closed: holomorphic ⇒ Rep = Vect ⇒ 2D bulk *unique* (LR/KLM/BKLR); machine contrast $SO(16)_1$ admits *six* modular invariants (incl. both $E_8$ pairings) ⇒ Target A = one residual [E]/[C]/[O] |
| `v88_cp_phase_audit.py` | Target-D CP residual quantified: frozen $\delta{=}\pi/3{+}3\lambda^2$ survives at $+0.98\sigma$ vs $\gamma_{\rm PDG}$; the 0.07°-near alternative $\pi/3{+}2\lambda^2$ recorded as look-elsewhere trap, *not* adopted; decision at $\sigma_\gamma{\le}0.96°$ [E] |
| `v89_carrier_index_lemma.py` | Carrier Index Lemma: KLM $\mu_A{=}[B{:}A]^2\mu_B \Rightarrow$ Jones index $[(E_8)_1{:}(D_5)_1{\times}(A_3)_1]{=}4{=}\lvert\mu_4\rvert$ (glue order *is* the index); $\mu$-additivity $16/4^2{=}1 \Rightarrow$ holomorphy *follows*; Gate A (H)⇔(H′) index-4 extension [E] |
| `v90_conical_defect_chain.py` | Fursaev–Solodukhin factor $4\pi$ *derived* (smoothed-cone Gauss–Bonnet, exactly smoothing-independent); replica $S{=}4\pi kA$, $k{=}c_3/2 \Rightarrow S{=}A/4 \Leftrightarrow c_3{=}\frac{1}{8\pi}$ (sympy-solved); residual isolated to "seam determinant ⇒ EH form" [E]/[O] |
| `v91_spine_tetrahedron.py` | Spine tetrahedron $\{2,3,4,5\}$ = anchor + zeroth moment; edges/faces/volume = the integer grammar, volume $120{=}\lvert R^+(E_8)\rvert$; $240{=}\lvert\mu_4\rvert\lvert E(K_4)\rvert\lvert E(K_5)\rvert$ with $K_6$ negative control; dual cuts typed tautological; sub-grammar incompleteness honest [E] |
| `v92_glue_uniqueness.py` | Glue uniqueness: of all 15 subgroups of the discriminant form exactly two Lagrangian glues (the two *chiralities*, swapped by either single spinor conjugation, fixed by the simultaneous one) and exactly one halfway extension = $SO(16)_1$; tower carrier→$SO(16)_1$→$(E_8)_1$, nothing else [E] |
| `v93_koide_relaxation_toy.py` | Koide relaxation toy (P2 narrowed): basin lemma (every physical configuration in the attractor basin); exact rate $(2/3)^6$; source at $\rho{=}-\varphi_0^{\rm ret}/24$ to 0.8% ([C]); honest negatives: flow time $t{=}2.84 \notin \mathbb{Z} \Rightarrow$ the missing object is a *continuous* generator [E]/[C] |
| `v94_sheet_diamond.py` | Sheet diamond: all four flavor operators on *one* surface $M(s,t){=}R{+}Q\,\mathrm{diag}(s,t,t)$ (pencil = the cut $(1{+}x, x{-}1)$; diagonal-cut disc 153 negative control); winding line $\det B{=}2\det$ for all $s$; cofactor seam normal $(5,-9,6)$; Plücker canonicalisation: 11 *and* 26 both live in $K$; spine-quotient firewall *rejected* [E] |
| `v95_centered_diamond.py` | Centered cross: $Q{=}U{+}V$, $R/L{=}C{\mp}U$, $K/F{=}C{\mp}V$ (five objects → three); $\mathrm{Spec}\,V{=}\{0,1,2\}{=}N_{\rm fam}{\cdot}$cusp; $\det C{=}14$, $\sum C{=}31{=}2^{g_{\rm car}}{-}1$; $\mathrm{Pl}_R(C){=}7(2,3,1) \parallel \mathrm{Pl}_R(L){=}10(2,3,1)$; $G_2$ reading audit-typed [E] |
| `v96_branch_kernel_selection.py` | Branch-kernel selection (P1 unblocked): rank-1 anchor blocks at the branch points have *integer* kernels (carrier kernel = **1**); collapse lemma ⇒ $(-1,1,0)$: up = −down, *lepton pairing* = 0 (leptons on the ramification); sector zero ladder (up $-\frac{3}{2}$ doubly, down 0 & $-\frac{9}{5}$); anchor-placement controls [E] |
| `v97_sheet_conjugation_bridge.py` | Sheet-conjugation bridge (P1 → one [C]): the anchor forces the cusp conjugation uniquely ⇒ integer $T_A$, $\det{=}{-}1$; $a{=}e_2{+}e_3$ (conjugation-symmetric); one deck action on $\mathbb{R}^3/\mathrm{span}\{\mathbf{1}, a\}$; $\langle T_A, \Sigma\rangle{=}D_4$; sheet-index lemma (intertwiner dets even, min $2{=}\lvert\mathbb{Z}_2\rvert$); remaining [C]: $Q_+$ grading = $A_3$ discriminant grading [E]/[C] |
| `v98_discriminant_dictionary.py` | Dictionary *derived* (P1 closed modulo GATE.QGEO): $G{=}T_A\Sigma$ acts integrally on the cusp basis ($Ge_1{=}{-}e_1$, $Ge_3{=}e_2$, $Ge_2{=}{-}e_3 = B_1{\oplus}E$); cusp-0 line = self-conjugate $\mathbb{Z}_4$-class-2 line; $T_AGT_A^{-1}{=}G^{-1}$ ($k{\mapsto}{-}k$); $q_{A_3}(1){=}q_{A_3}(3){=}\frac{3}{8}$, $q_{A_3}(2){=}\frac{1}{2}$; reflection classes = glue-swap vs glue-fix parities ⇒ P1 carries no separate [C]; residual = the one existing $Q$-geometry gate [E] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v99_koide_flow_time.py` | Koide flow time: canonical generator $\dot{q}=(\Delta/N_{\rm fam})(q-2)(q-5)=(\Delta/N_{\rm fam})\det B(q)$, time-1 map $=F$ [E]; non-integrality of $t=2.84$ is *data-fragile* ($Q$ crosses $\frac{2}{3}$ at $+0.43\sigma(m_\tau)$, $t$ passes every integer $\geq 3$ within $+0.5\sigma$) [E]; conditional steps: $n=2$ excluded ($-2.9\sigma$), $n=3=N_{\rm fam}\Rightarrow m_\tau=1776.9427\,\text{MeV}$ ($+0.14\sigma$), decision at $\sigma(m_\tau)\sim 0.01\,\text{MeV}$ [C]; $\ln(46080)$ scale reading destroyed by controls (firewall) |
| `v100_numerology_null_mc.py` | Numerology null test (look-elsewhere corrected): exact census of the full declared formula grammar over the 13 scored frozen-registry observables (conservative data windows) gives joint formula-fishing probability $\prod p_i = 10^{-25.8}$, robust to budget/window variation; $2\times 2\cdot 10^5$ Monte-Carlo pseudo-theories reach at most 5/13 hits vs. TFPT 13/13; power controls (seed perturbation, data shuffle) collapse the scorecard; all 94 500 $F_{U(1)}$ variants solved, exactly *one* hits CODATA $\Rightarrow$ total $\leq 10^{-30.7}$, *conditional on the declared grammar* [E] |
| `v101_horizon_anchor.py` | The maximal black hole is the anchor (SdS in seam units): Nariai cubic $t^3-3t+2=(t-1)^2(t+2)$, roots $(1,1,-2)$ = traceless anchor; $S_{\rm Nariai}/S_{dS}=\frac{2}{3}=\lvert\mathbb{Z}_2\rvert/N_{\rm fam}$ (per horizon $\frac{1}{3}$); interpolation $(x^2+1)/\Phi_3(x)$; three-sheet total $\lvert\mathbb{Z}_2\rvert S_{dS}$ conserved; $Q_{\rm geom}\in[\frac{3}{8},\frac{1}{2}]$ = the $SU(4)_1$ weights; mass-line cover $(1-3m)(1+3m)$, deck = horizon swap; $\tau_{BH}=128\,g_{\rm car}M^3/c_3$; six atom landings, zero free parameters [E]/[C] (bulk reading) |
| `v102_seam_orientation.py` | One orientation: the anchor is the *stationary repeller* in both sectors — flavor relaxation = gradient flow of a cubic potential with critical points = the branch points, curvatures $V''(2)=+\Delta$, $V''(5)=-\Delta$ (= the gap), inflection at scalaron/2, Lyapunov rate $\Delta$ constant; SdS entropy has Nariai as its *unique* stationary point, curvature $\frac{2}{9}=\lvert\mathbb{Z}_2\rvert/N_{\rm fam}^2$; evaporation = entropy ascent away from the anchor [E]; "one variational principle of the seam" stays a reading [C] |
| `v103_trisection_normal_form.py` | Trisection normal form (the canonical coordinate): SdS cubic $\Leftrightarrow \cos 3\theta=-3m$ (angle trisection; $\mathbb{Z}_3$ deck = triality); $m=\cos\psi/N_{\rm fam}$ and $S_{\rm tot}/S_{dS}=\frac{4}{3}-\frac{2}{3}\cos\frac{2\psi}{3}$ — *one* cosine of glue atoms; canonical curvature $(\frac{2}{3})^3=(\lvert\mathbb{Z}_2\rvert/N_{\rm fam})^{N_{\rm fam}}$ (Koide constant to the family power); invariant slope $d\sigma/dm\rvert_N=-\frac{8}{9}=-\operatorname{rank}E_8/N_{\rm fam}^2$; bridge: flavor rate $(2/3)^{2N_{\rm fam}}$ vs gravity curvature $(2/3)^{N_{\rm fam}}$, missing = one *clock* [E]/[C] |
| `v104_nariai_clock.py` | The *classical* Nariai clock is the anchor (third appearance): static pin $\phi=\rho$ solves the $dS_2$ static-patch equation with $m^2=-2\Lambda=-\lvert\mathbb{Z}_2\rvert\Lambda$ exactly (Ginsparg–Perry tower: one negative mode); in Hubble units $\chi_{\rm clock}=(\lambda-1)(\lambda+2)$ = the anchor quadratic, eigenvalues $\{1,-2\}$; Nariai cubic $=(t-1)\cdot\chi_{\rm clock}$; entropy-deviation rate $2H=\lvert\mathbb{Z}_2\rvert H$; honest (2/3)-test *negative* for the classical clock — the quantum clock is the one remaining [C] of the seam variational principle [E]/[C] |
| `v105_residual_inventory.py` | The residual inventory: *one constant* $2/3=\lvert\mathbb{Z}_2\rvert/N_{\rm fam}$ exact in seven places across both sectors), *one anchor* (three dynamical appearances: configuration, Nariai roots, clock spectrum), *relocation theorem* (cosmological quantum clock suppressed by the $\Lambda$ hierarchy, 121.5 orders $\approx 2\alpha^{-1}/\ln 10\Rightarrow$ the clock must be the seam transfer operator — one [C] identification), and the *complete machine-pinned gap list*: exactly five structural objects + one scale + $\pi$; a sixth gap would fail the script [E]/[C] |
| `v106_review_validation.py` | External-review validation: seed normal form $\varphi_0^{\rm ret}=\frac{\lvert\mu_4\rvert}{N_{\rm fam}}c_3+\Omega_{\rm adm}c_3^{\lvert\mu_4\rvert}$ exact [E]; hypercharge moment $\operatorname{Tr}X^2=120=5!$ computed from the charge table; factorial spine $5!=\lvert R^+(E_8)\rvert=\sum$exponents, $240=2\cdot 120$, $1920=2^4\cdot 5!$; degree-2 inventory (Pascal $K=2$ unique at $g=5$, glue norms sum 2, $c_3=\frac{1}{2}\times\frac{1}{4\pi}$, $\binom{5}{2}=10$); named [C] hypothesis *Quadratic Boundary Locality* ($\Rightarrow K=2$ forced; would upgrade the Pascal selection to [E]); audit $240=16\times 15$ non-unique [E]/[C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v107_quantum_clock_target.py` | Quantum-clock target made quantitative: the classical Ginsparg–Perry clock decays on $\{0,-1,-2\} = -\operatorname{Spec}(V) = -N_{\text{fam}}\times\text{cusp}$ (the transfer operator's own grading) [E]; exact target identity $(1/3)^6 = \left((2/3)^6\right)^{\log_{3/2} 3}$ — required bend $\log_{9/4} 3 = 1.355$ per weight step; coupling landscape $\kappa = c/(24\pi)\cdot\Lambda/\bar{M}_{\text{Pl}}^2$: $10^{-122}$ cosmological vs $1/(3\pi)$ at seam curvature (the only viable, borderline-nonperturbative regime) [E]; identification stays [C] |
| `v108_pascal_ladder.py` | Pascal Ladder Theorem: $2^{g-1}=\sum_{k\le K}\binom{g}{k} \Leftrightarrow g=2K+1$ (odd-$g$ midpoint identity + even-$g$ straddle) $\Rightarrow$ carrier selection $g=5$ *exactly equivalent* to degree bound $K=2$ — the Target-B residual = the one QBL hypothesis; neighbour worlds: $K=1$ one-family, $K=3$ nine-family, $K=4$ inconsistent; only $K=2$ also satisfies $g+N_{\text{fam}}=8$ (v14 overdetermination) [E]/[C] |
| `v109_sheet_pairing.py` | Sheet-Pairing Lemma (QBL anchored): character-exact multisets from $(\pm\frac{1}{2})^5$ — no scalar within a sheet (zero-weight mult of $S^+\otimes S^+ = 0$; odd $g$ flips parity); $S^+\otimes S^- = \Lambda^0\oplus\Lambda^2\oplus\Lambda^4$ exactly ($256=1+45+210$, zero-modes $(1,5,10)$ = the code); sheet-diagonal = odd forms $(2\Lambda^1+2\Lambda^3+\Lambda^5)$; tower tops at $\Lambda^{2K}$, $K=2 \Rightarrow$ the seam scalar kernel is necessarily a *sheet pairing* [E]/[C] |
| `v110_calderon_sheet.py` | Calderón-sheet selection theorem: a Calderón involution $\varepsilon$ on $S^+\oplus S^-$ certifies a scalar two-point datum *iff* it is sheet-odd; sheet-odd $\Rightarrow$ exactly $2=\lvert\mathbb{Z}_2\rvert$ invariant kernels = the two Lagrangian glues; ladder genericity ($g=3,5,7$): sheet-oddness forces $K=(g-1)/2$, *not* $g=5$ (anti-overclaim); QBL residue split into *interface* ("$\varepsilon$ sheet-odd", natural from the one-sided collar) + *analytic core* (slot-degrees $\le 2$) [E]/[C] |
| `v111_quadratic_transport.py` | Quadratic-Transport Theorem: in the integer Jordan–Wigner model linears are sheet-odd, the 45 quadratics ($=\mathfrak{so}(10)$, the $\Lambda^2$ term of $1+45+210$) sheet-even, generate the code from the vacuum *and* span the full $\operatorname{End}(S^+)=256$ at word length $2 \Rightarrow$ *transport degree exactly 2* (degree $\le 1$ generates nothing) — the transport half of the QBL core is a theorem; $g=3$ control: degree selected, not rank [E]/[C] |
| `v112_selfcounting_channel.py` | Self-Counting Channel: $w \mapsto (w,-w)$ is a bijection code states $\leftrightarrow$ neutral certified kernels, so #kernels $= 2^{g-1}$ *exactly*; graded by pair-degree the same set has sizes $\binom{g}{m} \Rightarrow$ the Pascal closure $2^{g-1}=\sum_{m\le K}\binom{g}{m}$ is *two countings of one set* (an identity, not a requirement); Hodge fold $\binom{g}{2j}=\binom{g}{\min(2j,g-2j)}$; QBL residue = one input ("a single scalar 2-point kernel"), selection carried by rank-8+integrality [E]/[C] |
| `v113_quasifree_kernel.py` | One kernel is the whole net: carrier $(D_5)_1\times(A_3)_1 = SO(10)_1\times SO(6)_1 = 16$ free Majoranas, $c=5+3=8$ at every tower level (only the glue grows, index $2\times2=4=\lvert\mu_4\rvert$); Wick/Pfaffian exact (all $210+210$ higher correlators = Pfaffians of the one kernel); kernel = Calderón involution, rank $P=5=g_{\text{car}}$, seam level $8=\operatorname{rank} E_8$ ($c$ = rank of the kernel); vacuum unique $\Rightarrow$ QBL input = R2 premise: gate closes $\Rightarrow$ carrier choice closes [E]/[C] |
| `v114_torsion_delta.py` | Torsion normal form + $\delta=\frac{1}{2}$ theorem: flatness of the $(U_{\text{wall}})$ $\mathbb{Z}_4$-family $\Leftrightarrow$ $(MU)^4=\mathbf{1}$ ($\mu_4$ = order of the twisted cusp generator); on the involutive branch ($T^2=\mathbf{1}$) the cusp trace forces $\operatorname{diag} M=(0,\frac{i}{2},-\frac{i}{2})$ with spec $\{1,\omega,\omega^2\}$ automatic $\Rightarrow$ $\delta=\frac{1}{2}$ exact and constant (v20's hand value = torsion identity); census: $\{1,i,-i\}$ branch varies, v33 point in its $d_1=0$ slice; residue = bundle-side branch selection [E]/[C] |
| `v115_anchor_residue.py` | The anchor pins the residue: $\mu_4$-average = diagonal projection $\Rightarrow$ anchor splitting $\Leftrightarrow$ $\operatorname{diag} A_0=a/\lvert\mu_4\rvert$; off-weights uniquely $(8,0,5)/144$ $(=(\operatorname{rank} E_8, 0, g_{\text{car}})/(\lvert\mu_4\rvert N_{\text{fam}})^2$, total $13=\Delta_Q$); *exact* residue $A_0^\star$ with $\chi=\lambda(\lambda-\frac{1}{3})(\lambda-\frac{2}{3})$ is the RH solution ($\lVert M_\infty-\mathbf{1}\rVert\sim10^{-10}$); anchor forces $d_1=0 \Rightarrow$ harmonic diagonal anchor+torsion-forced [E] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v116_resonance_uniqueness.py` | Resonance theorem (v115 [N]→[E], linear — no Gröbner): twisted $\mu_4$ averages collapse to $B_1{=}4a_{12}E_{12}{+}4\overline{a_{13}}E_{31}$; anchor exponents diag$(2,1,1)$ give resonance gap 1 with obstruction $4\overline{a_{13}} \Rightarrow M_\infty{=}\mathbf{1} \Leftrightarrow a_{13}{=}0$; + $(8,0,5)/144$ + Diagonaleichung ⇒ flat anchor locus = *one* gauge orbit = exact $A_0^\star$ — $U_{\text{wall}}$ bundle datum algebraically unique; sharpness control $\|M_\infty{-}\mathbf{1}\| \sim 8.5|a_{13}|$ [E] |
| `v117_monodromy_weyl_a3.py` | The monodromy is $W(A_3)$: the unitarised holonomy of $A_0^\star$ is *exact* (entries in $\frac{1}{2}\mathbb{Z}[i]$), diag $\widetilde{M}_0{=}(0,-\frac{i}{2},\frac{i}{2}) \Rightarrow d_1{=}0$, $\delta{=}\frac{1}{2}$ are matrix entries; $\langle U,\widetilde{M}_0\rangle$ enumerated exactly: order 24, statistics $(1,9,8,6)$, characters $(3,-1,0,1) = S_4{=}W(A_3)$ (standard⊗sign) — the flavor wall realises the $A_3$ glue factor as its Weyl group; ODE match $3.5{\times}10^{-10}$ [E] |
| `v118_hexagon_family_dictionary.py` | First exact H2 piece: $\text{spec}(\widetilde{M}_0) \cup \text{spec}(-\widetilde{M}_0) = \mu_6$ (hexagon = family spectrum + sheet twist, $\mathbb{Z}_6{=}\mathbb{Z}_2{\times}\mathbb{Z}_3$); $\det(\mathbf{1}{-}t\widetilde{M}_0){=}1{-}t^3 \Rightarrow \det_\mp{=}(\frac{7}{8},\frac{9}{8})$; lepton coefficients = resolvent determinants ($c_e{=}|\mu_4|/(2\det_-)$, $c_\mu{=}N_{\text{fam}}/(2\det_+)$, $c_\tau{=}|\mu_4|\det_-/N_{\text{fam}}$, product $= |\mu_4|/\det_+ {=}\frac{32}{9}$); $\langle U,\widetilde{M}_0,-\mathbf{1}\rangle$ has order $48{=}\Omega_{\text{adm}}$; the address table stays open [E]/[C] |
| `v119_review_validation_2.py` | Second review validation: anchor ratio triad $(e_3,e_1,e_2)/p_0{=}(\frac{2}{3},\frac{4}{3},\frac{5}{3})$ = (Koide branch, seed gain, carrier branch; flow critical points $(2,5){=}(e_3,e_2)$); the 121 audit lemma $(\mathbf{1}{+}a)^T R(\mathbf{1}{+}a){=}121{=}11^2{=}\|\,\text{Pl}(K)\|_1^2$ (orderings $\{105,121,135\}$, only canonical = 121; audit, not load-bearing); micro-identities $\Omega_{\text{adm}}{=}2p_1p_2$, $10b_1{=}1{+}p_1p_3$, $\Delta_Y{=}e_2^2{=}p_0^2{+}2\,\text{rank}$; flavor surface = v94/v95 (presentational) [E] |
| `v120_address_table.py` | Address table (frozen v18/v20 integers): lepton words $(8,5,3){=}(\text{rank}\,E_8, g_{\text{car}}, N_{\text{fam}})$; addresses = divmod by the hexagon ($p_2{=}6$): $e{\to}(2,1)$, $\mu{\to}(5,0)$, $\tau{\to}(3,0)$, $\sum r{=}p_3$; sheet parity: $\tau$ sits at the real twisted eigenvalue $-1$ (= the product-rule site); quark sums: up $= p_3$, down $= p_1{+}p_3$, all quarks $= 24{=}|W(A_3)|$, all nine $= 40{=}p_1p_3{=}a^TRa$; top at the vacuum site $(0,0)$; the assignment derivation stays open [E]/[C] |
| `v121_address_pinning.py` | Address pinning: $L{=}R{+}2U$ (v95) ⇒ the word table = the residue operator $R$ + the generation-1 winding, $R$ column 1 = the anchor $a$; margins = atoms (rows $(e_1,\text{rank},p_3)$, columns $(e_1,\Delta_Q,g_{\text{car}})$); *census theorem*: among all hexagon matrices with atom margins (17 of them) exactly *one* has $\det{=}8$ — and it is $R$ (also unique with SNF $(1,1,8)$; all 17 determinants distinct) ⇒ the address table carries *zero* residual information; the H2 residue = the six atom margins [E]/[C] |
| `v122_margin_theorem.py` | Margin theorem: the *established* selectors — the $D_4$ annihilator $n{=}(5,-9,6)$ with $n^TR{=}(8,0,0)$ (v94), $\det R{=}8$, $\det K{=}4$ (the v71 quartet) — pin $R$ *uniquely* among hexagon matrices ($64 \to 12 \to 1$) ⇒ the atom margins, the anchor column and all sum rules are *consequences* (their input status is revoked); bonus: $\det L{=}20$ holds automatically on the locus (the quartet is redundant); H2 introduces *no* new residual class [E]/[C] |
| `v123_inventory_update.py` | Residual inventory updated (post v110–v122): thirteen new ledger rows machine-pinned; R2 carries *three* loads (metric + carrier + QBL: one theorem, three doors); the table re-pins to *five* classes — R1 clock, R2 index-4 net, R3 EH, R4′ selector establishment (replacing the *discharged* H2 dictionary), R5 $Q$ realisation — + $v_{\text{geo}}$, $\pi$; a sixth class fails the script [E] |
| `v124_resummed_clock.py` | Resummed clock (the R1 bend in closed form): transfer eigenvalues = exactly $(1{-}n/N_{\text{fam}})^{p_2} \Rightarrow \text{rate}(n){=}{-}p_2\ln(1{-}n/N_{\text{fam}})$ reproduces $\{0,\Delta,6\ln 3\}$, the bend $\log_{3/2}3$ in one line; the linearisation carries slope $|\mathbb{Z}_2|{=}2$ relative to the classical GP clock $\{0,-1,-2\}$; the pole at $n{=}N_{\text{fam}}$ forces the three-level spectrum; the R1 target is now closed-form [E]/[C] |
| `v125_glue_qsystem.py` | Glue Q-system: $\mathbb{C}[\mathbb{Z}_4]$ satisfies all Longo Q-system axioms exactly (associativity, unit, Frobenius, specialness $mm^*{=}|\mathbb{Z}_4|\mathbf{1}$) ⇒ Jones index $4{=}|\mu_4|$ (the v89 value); locality = isotropy ($q(k(1,1)){=}k^2 \in \mathbb{Z}$, v92); halfway sub-Q-system $\mathbb{C}[\mathbb{Z}_2]$ = the $SO(16)_1$ step, $4{=}2{\times}2$; R2 sharpens from "find" to "identify" [E]/[C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v126_clock_wall_bridge.py` | Clock and wall share one spectrum: the resummation weights $n/N_{\text{fam}}=\{0,\frac{1}{3},\frac{2}{3}\}$ = exactly spec $A_0^\star$ (the parabolic MS weights); $\lambda_n=(1-\alpha_n)^{p_2}$; checkpoint 1 passed *classically* ($p_2/N_{\text{fam}}=2=\lvert\mathbb{Z}_2\rvert$ = the v104 entropy rate $2H$); $\det(\mathbf{1}-A_0^\star)=\frac{2}{9}$ = the entropy curvature (v102); the quantum content of R1 is isolated in the geometric tail [E]/[C] |
| `v127_ring_resummation.py` | The clock is a log determinant: rate $=-p_2\,\mathrm{Tr}\ln(\mathbf{1}-\alpha P)$ — six identical ring (RPA) towers, *one per hexagon site*, coupling = the parabolic weight; the $k=1$ ring = the classical term, the $k=2$ ring = the first quantum correction; $\kappa_{\text{seam}}=\frac{1}{3\pi}=\alpha_1/\pi$ (audit); the wall = the RPA instability at $\alpha\to1$; v107's "O(1) resummation" is thereby *typed* (ring/log-det), its derivation open [E]/[C] |
| `v128_graded_hull.py` | Graded hull + zero-mode multiplicity: the 240 $E_8$ roots explicit in $D_5\oplus A_3$+glue coordinates, coset decomposition $240=52+64+60+64$; the grading is exactly additive on *all* 6720 root-pair sums ⇒ $E_8$ is a $\mathbb{Z}_4$-*graded* Lie algebra over the carrier (grading = glue = Q-system); ring multiplicity $6=2(2l+1)\lvert_{l=1}$ = the sheet-doubled GP zero modes [E]/[C] |
| `v129_entropy_power_law.py` | The clock is an entropy power law: $\lambda_n=(S_n/S_{dS})^{p_2}$ with the three Nariai entropy levels $S/S_{dS}\in\{1,\frac{2}{3},\frac{1}{3}\}$ (v101); rate $=p_2\ln(S_{dS}/S)$; triple identity $\alpha=n/N_{\text{fam}}=$ MS weight = entropy-deficit share ⇒ the RPA coupling is nowhere free; orientation consistent (attractor = max entropy); R1 thermodynamic: $\Gamma\propto(S/S_{dS})^{p_2}$ (a Gibbons–Hawking power law, $h=N_{\text{fam}}$ audit) [E]/[C] |
| `v130_born_square.py` | Born square: amplitude $\sim(S/S_{dS})^{n_{\text{zero}}/2}=(S/S_{dS})^3$ (one $S^{1/2}$ per zero mode), probability $=\lvert A\rvert^2=(S/S_{dS})^6$ ⇒ $h=n_{\text{zero}}/2=3=N_{\text{fam}}$, $p_2=2h$ (the Born rule) — the v129 weight audit is *derived*; the 3 = the dimension of the $l=1$ space = the number of proper conformal Killing vectors of $S^2$; the ×2 = the horizon pair (the v101 deck); the R1 residue = the standard zero-mode measure [E]/[C] |
| `v131_measure_is_area.py` | The measure is the area: $\lVert Y_{1m}\rVert^2=r^2=A/(4\pi)$ *exactly* (all three $l=1$ modes integrated) ⇒ the per-mode Jacobian = the mode norm $\sim S^{1/2}$ ⇒ amplitude $S^3$, Born $S^6$ — every exponent factor with a derivation origin; the $4\pi$ = the same Gauss–Bonnet primitive as in $c_3$, cancelling in all ratios; the R1 residue = *one* det-ratio cancellation $\det'_n/\det'_{dS}=1+O(\text{deficit})$ [E]/[C] *(cancellation since derived within the SdS family, v144)* |
| `v132_detratio_anomaly.py` | Det-ratio anomaly = the Koide constant: $\zeta(0)\lvert_{\det'}$ of the GP sphere operator $-\Delta-2$ (the 3 zero modes removed) $=a_1-3=\frac{7}{3}-3=-\frac{2}{3}=-\lvert\mathbb{Z}_2\rvert/N_{\text{fam}}$ exactly ($a_1=\frac{1}{N_{\text{fam}}}+\lvert\mathbb{Z}_2\rvert$; numerical continuation $\sim10^{-7}$) — the eighth 2/3 appearance, the first as a *spectral* anomaly; consequence $\det'(cL)=c^{\zeta(0)}\det'(L)$; the R1 residue = *one* $\zeta(0)$ budget statement (the 4d sum) [E]/[C] |
| `v133_zeta_budget.py` | The $\zeta$ budget, both routes exact (Euclidean Nariai = $S^2\times S^2$): the reduced route gives $-\frac{2}{3}$ per sector, total $-\frac{4}{3}=-e_1/p_0$ (= minus the seed gain — two of the three triad values); the naive 4d route gives $a_2=\frac{161}{45}$, $\zeta(0)=-\frac{109}{45}$ (*no* atom; continuation $\sim10^{-8}$); zero modes 3+3 = the horizon pair ⇒ the budget structurally favours the *reduced* seam reading; remainder = graviton/ghost shares (Volkov–Wipf class) [E]/[C] |
| `v134_dual_anchor.py` | Dual anchor lemma: $d:=a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2},-\frac{1}{2},1)$, $d\cdot\mathbf{1}=0$, $d\cdot a=1$, $(1,1,-2)=-2d$ — the inverse flavor response *is* the Nariai root (v101); the invariance is exact via Sherman–Morrison (⇔ orthogonality to $R^{-1}\mathbf{1}=\frac{1}{4}(1,1,-1)$; controls: $\mathbf{1},e_1,n$ do *not* lie on the plane); $(d,n)$ = the dual normal pair of the flavor boundary; a third purely *algebraic* leg of the flavor↔horizon bridge [E]/[C] |
| `v135_det_surface.py` | Determinant surface: $\det M(s,t)=3s(t+1)(t+2)+t^2+5t+8$ — two $s$-independent iso-volume walls $t=-2\Rightarrow2=\lvert\mathbb{Z}_2\rvert$, $t=-1\Rightarrow4=\lvert\mu_4\rvert$ ($\det K=4$ is a *layer*); winding line $6s+8$ ⇒ $(8,14,20)$, $\det F=32=2^{g_{\text{car}}}$; $\det B(s,0)=2\det M(s,0)$ exactly, $B$-wall only at $t=-2$; row-sum axes $C=(7,11,13)$, $U=(3,3,3)$, $V=(1,2,3)=\mathrm{Spec}(Q_+)$; micro-identities $e_1^2+e_2^2=41=10b_1$, $p_1^2+p_2^2=52=$ the carrier coset root count (v128) [E]/[C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v136_dual_normal_selector.py` | Dual-normal selector: $(d,n)$ pins $R$ *columnwise* — $d{\cdot}c_j{=}a_j$, $n{\cdot}c_j{=}8\delta_{1j}$, kernel = primitive $(6,8,7)$, census: each column unique (1 of 216) ⇒ R4′ shrinks from $6^9$ to three column problems; the $A_0^\star$ zero mode carries $\sigma{=}(2,-9,5){=}(\lvert\mathbb{Z}_2\rvert,-N_{\rm fam}^2,g_{\rm car})$ ($\lVert v\rVert^2{=}\frac{16}{9}$, $16{=}\dim S^+$; adj $=\frac{2}{9}P_v$ = the SdS curvature); $n{\cdot}\sigma{=}121{=}11^2$; $W(A_3)$ orbit control: the naive lift $\sigma{\to}n$ fails — a genuine mechanism, not a group element [E]/[C] |
| `v137_qplus_cohomology.py` | The $Q_+$ grading is cohomology: $H^1(\mathbb{P}\setminus\mu_4){=}\chi_1{\oplus}\chi_2{\oplus}\chi_3$ ($\omega_k{=}z^{k-1}dz/(z^4{-}1)$, character $i^k$, residues $\zeta^k/4$); degrees $(1,2,3){=}\mathrm{Spec}(Q_+){=}$ the $A_3$ exponents ($h{=}4{=}\lvert\mu_4\rvert$); cusp weights $\{0,\frac{1}{3},\frac{2}{3}\}{=}\mathrm{spec}(A_0^\star)$ ⇒ MS weights = cusp classes = characters = $Q_+$: *one* spectrum, four readouts; the QGEO residue = the naturality of the canonical map [E]/[C] |
| `v138_vw_firewall.py` | Volkov–Wipf firewall (R1 typed closed): external 4d value $\alpha_{\rm PI}{=}{-}\frac{98}{45}{=}\frac{22}{45}{-}\frac{8}{3}$ (arXiv:2506.02142 Tab. 4.1 = VW hep-th/0003081); **edge match**: $\alpha_{\rm edge}{=}{-}\frac{8}{3}{=}2{\times}(-\frac{4}{3})$ — two identical edge copies (the horizon pair), per copy exactly the reduced seam budget (v133, the seed gain); bulk $\frac{22}{45}$ = the graviton gas (not the clock); difference $-\frac{38}{45}$ no atom ⇒ typing: the seam clock = an edge object, the full 4d determinant = the gravity firewall [E]/[C] |
| `v139_selector_triangle.py` | Selector triangle: $d{=}\frac{3}{2}a{-}2\cdot\mathbf{1}$ exactly (pure anchor data, no $R$) ⇒ the first selector is *derived*; the frame $(\mathbf{1},a,\sigma)$ has $\det{=}11{=}\lVert\mathrm{Pl}(K)\rVert_1$, and $n$ is the *unique* covector with atom pairings $(2,8,121){=}(\lvert\mathbb{Z}_2\rvert,\mathrm{rank}\,E_8,11^2)$; on the constrained line the pairing is $11(11{+}t)$, a square only at the true $n$; review lift exact (audit): $n{=}\mathrm{reverse}(\sigma){+}\lvert\mu_4\rvert e_3$; R4′ residue = three pairings [E]/[C] *(since v142: two — the third pairing is integrality-forced)* |
| `v140_canonical_map.py` | The canonical map exists and is rigid: the plain deck $U$ (characters $(0,1,3)$) cannot match $H^1$; the *sheet-twisted* $-U$ ($(2,3,1)$) and the cusp exponential $i^{Q_+}$ ($(1,2,3)$) both carry the $H^1$ character set $\{1,2,3\}$ (the v118 sign twist); multiplicity-freeness ⇒ Schur-rigid equivariant isomorphism (unique up to scalars); $i^{Q_+}$ pulls the Euler degree back to $Q_+$, $-U$ to $Q_+\circ\rho$ — the QGEO residue collapses to *one* $\mathbb{Z}_3$ choice [E]/[C] *(the choice since derived, v141)* |
| `v141_deck_selection.py` | Deck Selection Theorem (R5): the established integer deck $G{=}T_A\Sigma$ (v97/v98) pairs the $Q_+{=}1$ line with the self-conjugate character 2 ⇒ of the three $\mathbb{Z}_3$ rotations only the sheet-twisted $(2,3,1)$ = chars$(-U)$ survives; $i^{Q_+}$ excluded (same spectrum, wrong grading pairing); residual freedom = the established sheet $\mathbb{Z}_2$ (v92); Euler degree = $Q_+\circ\rho$ [E]/[C] |
| `v142_frame_integrality.py` | Frame Integrality (R4′): pairing triples of *integer* covectors against $(\mathbf{1},a,\sigma)$ form an index-11 sublattice ($x{-}3y{+}z\equiv0$ mod 11, $11{=}\lVert\mathrm{Pl}(K)\rVert_1$); with $(x,y){=}(\lvert\mathbb{Z}_2\rvert,\mathrm{rank}\,E_8)$ the third pairing is forced $\equiv0$ mod 11, primitivity forces even $t$; Cramer reductions for the two line pairings; R4′: three pairings → two [E]/[O] *(since v145: the values are atom identities — residue = one lift map)* |
| `v143_graded_frobenius.py` | Graded Frobenius (R2 at the finite Lie level): coset duality $-C_k{=}C_{-k}$ exact on all 240 roots ⇒ Killing pairing nondegenerate per $g_k\times g_{-k}$; glue average = carrier projector (index $\lvert\mathbb{Z}_4\rvert{=}4$); each sector one $W(D_5){\times}W(A_3)$ orbit = simple current; fusion $=\mathbb{C}[\mathbb{Z}_4]$ (v125); residual = the one conformal-net statement [E]/[C] |
| `v144_detratio_family.py` | Det-ratio cancellation within the SdS family (R1, the v131 step): $e_2$-rigidity gives $r_b r_c{=}1{-}\Delta^2/3$ exactly ⇒ $\det'$-ratio $=(1{-}\Delta^2/3)^{4/3}$ (via $\zeta(0){=}{-}\frac{2}{3}$, v132) — no first-order term in the split; deficit coefficient $\frac{32}{27}{=}\lvert\mu_4\rvert(\frac{2}{3})^3$ (audit); finite-weight absorption stays open with the $6\to\frac{14}{3}$ obstruction stated sharply [E]/[C] *(the bend since shown $\det'$-clean, v147)* |
| `v145_pairing_atoms.py` | R4′ values are *atom identities*: with the lift $n{=}w_0(\sigma){+}\lvert\mu_4\rvert e_3$ ($w_0{=}$ the $A_3$ exponent duality), $n\cdot\mathbf{1}{=}\lvert\mu_4\rvert{-}\lvert\mathbb{Z}_2\rvert{=}\lvert\mathbb{Z}_2\rvert$, $n\cdot a{=}\lvert\mu_4\rvert a_3{=}8$ via the new closure $\lvert\mu_4\rvert{+}g_{\rm car}{=}N_{\rm fam}^2$ ($4{+}5{=}9$, $\sigma\perp w_0(a)$), $n\cdot\sigma{=}\lVert\sigma\rVert^2{-}N_{\rm fam}^2{+}\lvert\mu_4\rvert g_{\rm car}{=}121$; residue = derive the *one* lift map [E]/[C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v146_moebius_d4.py` | Möbius $D_4$ realisation (R5 at its floor): the full dihedral $\langle z{\mapsto}iz,\ z{\mapsto}1/z\rangle$ on $H^1(\mathbb{P}^1\backslash\mu_4)$ matches the integer model $\langle G, T_A, \Sigma\rangle$ exactly — $\iota^*$ swaps $\chi_1{\leftrightarrow}\chi_3$, fixes $\chi_2$ with $+1$, $\det{=}{-}1$ ($=T_A$ in the cusp basis); $\delta\iota{=}\Sigma$ class; parities match (v98) — the `GATE.QGEO` premise reduces to the module identification [E]/[C] |
| `v147_clock_gaussian.py` | The clock is a Gaussian zero-mode integral (R1): variance ratio $(1{-}\alpha)$ (norm = area, v131) $\Rightarrow$ Born$^2$ $\Gamma$-ratio $(1{-}\alpha)^{p_2}$, rate $= -p_2\ln(1{-}\alpha)$ = the v127 ring sum term by term; the bend $\log_{3/2}3$ lives between the two Nariai levels of *one* geometry $\Rightarrow$ *det′-clean*; the $6{\to}\frac{14}{3}$ obstruction is localised to the $\alpha{=}0$ reference [E]/[C] |
| `v148_fock_census.py` | NS/R sector census (R2 scoped, corrected): the untwisted (NS) module of the 16 Majoranas carries only the *even* classes $\{0,2\}{\times}\{0,2\}$ — the odd glue sectors $k(1,1)$ are *Ramond* modules; the R zero-mode module ($2^8{=}256$) splits $128{+}128$ into the odd sectors of the *two* Lagrangian glues (v92): choosing the glue *is* choosing the R-projection ($128{=}$ one $SO(16)$ chirality), $248{=}120_{\rm NS}{+}128_{\rm R}$; R2 is irreducibly a twisted-sector net statement [E]/[C] |
| `v149_cusp_normal.py` | Both dual normals are *anchor data* (R4′): in the cusp frame $n$ pairs to $(6,3,5) = (p_2,p_0,e_2)(a)$ — one anchor invariant per $Q_+$ line ($d=\frac{3}{2}a-2\cdot\mathbf{1}$ in the generation frame, v139); equivalent to the $(2,8,121)$ frame data and the $w_0$-lift (v145); audit: $\sigma\to(5,1,2)$, sum $14=\dim G_2$; residue $=$ *one* assignment [E]/[C] |
| `v150_replica_eh_model.py` | The EH form from the replica variation, at the gapped-model level (first R3 movement): conical deficit exact and $t$-independent (image sum $(N^2{-}1)/(12N) = C(2\pi/N)$); the gap makes the $\zeta$-variation *finite and cutoff-independent*, $\Delta\log\det' = 2C(\gamma)\ln m$; linearisation $= (\ln m/12\pi)\int\sqrt{g}R$ — the EH form; target equation $k{=}c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit); Calderón transfer + normalisation stay open — `SEAM.THEOREM.01` narrows, does *not* close [E]/[O] *(Calderón transfer since answered — the DtN kernel is conically clean, v151)* |
| `v151_bfk_split.py` | BFK split (the Calderón transfer of v150 answered): the Kac corner constant is boundary-condition *independent* ($C_N{=}C_D$, derived from reflection doubling) $\Rightarrow$ the conical deficit of the Calderón/DtN jump determinant *vanishes identically*, $C_{\rm DtN}{=}C_{\rm cone}(\gamma){-}2C_D(\gamma/2){=}0$; via Burghelea–Friedlander–Kappeler the seam-reduced action inherits the v150 EH form verbatim — the EH content sits in the *local* half determinants, the nonlocal kernel is conically clean; residue = the $q(A_3)$ normalisation + the kernel premise [E]/[O] |
| `v152_norm_is_anchor.py` | R3's normalisation *is* the dimensionful anchor: the EH coefficient is $k = \ln(m/\mu)/(12\pi)$ — $\ln m$ alone is scale-ambiguous, only the ratio $m/\mu$ is physical; $k{=}c_3/2 \Leftrightarrow m/\mu{=}e^{3/4}$ is the induced-gravity anchor $1/G$ (v68), *not* a separate gap; the irreducibles stay {one scale, $\pi$}; audit: the log-ratio $=\frac{3}{4}=q(A_3)$ is the target value, NOT a derivation [E]/[O] |
| `v153_no_unit_theorem.py` | No-Unit Theorem (v78/v152 formalised): dimensionless data are scale-invariant under $L\mapsto\lambda L$, a mass / $1/G$ is not, so a dimensionless compiler *cannot* select an absolute scale; the three residuals collapse — $U_{\rm point}\sim v_{\rm geo}$, $1/G\sim v_{\rm geo}^2$, $m/\mu = e^{3/4}$ (a ratio); $v_{\rm geo}$ is an irreducible metrology primitive, not an open gap; irreducibles stay {one unit, $\pi$} [E]/[O] |
| `v154_simple_current_theorem.py` | Simple-Current Extension Theorem (the central $G_{\rm net}$ theorem, assembled): $A{=}(D_5)_1{\otimes}(A_3)_1$ extended by the isotropic glue $L{=}\langle(1,1)\rangle$ has index $|L|{=}4$, $c{=}5{+}3{=}8$, $\mu(B){=}\mu(A)/|L|^2{=}16/16{=}1 \Rightarrow$ holomorphic $\Rightarrow B{=}(E_8)_1$ (unique even-unimodular rank-8; $SO(16)_1$ excluded); finite realisation in hand (v125/v128/v143/v148); residue = the seam-side identification [E]/[C] |
| `v155_quasifree_boundary.py` | Quasi-free in $\Rightarrow$ quasi-free out (Python-only): the boundary marginal of a Gaussian bulk is the compression $P\Gamma P$ (a contraction, hence a valid covariance), so the $G_{\rm net}$ analytic premise (v110(b)/v113) *follows* from 'the seam bulk is a reflection-positive free field' — the *same* premise as v59 (area law) and v150/v151 (EH); the seam-side identification reduces to ONE shared physical premise [E]/[O] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v156_seam_net_construction.py` | Constructive route (Route 3) to its exact endpoint: the DtN/Calderón operator of the 2d Laplacian is $\omega_k{=}\lvert k\rvert$ (free chiral dispersion); 16 free Majoranas give $c{=}8{=}5{+}3$; the $E_8$ currents are $248{=}120_{\mathrm{NS}}{+}128_{\mathrm{R}}$; and the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1{+}248q{+}4124q^2{+}\dots) = j^{1/3}$ (verified to order 4) makes $B \cong (E_8)_1$ a *checked* identity, not an assertion; residual = the free-bulk premise (v155) + net existence [E]/[C] |
| `v157_rigid_fixed_point.py` | Freeness as a *rigid fixed point* (simplicity-first reframing of premise A): (i) the DtN/Calderón symbol is universally $\lvert k\rvert$ (Lee–Uhlmann, bulk-detail-independent — the free dispersion is not a premise); (ii) a holomorphic $c{=}8$ theory has *no* marginal $(1,1)$ field (all $\bar{h}{=}0$) ⇒ rigid, isolated, no room for an interaction; (iii) the same $\lvert\mathbb{Z}_2\rvert$ that halves $8\pi$ in $c_3$ IS the Ramond projection giving $\mu{=}1$; (iv) holomorphic $c{=}8$ unique = $(E_8)_1$ = free. Premise (A) shrinks from 'prove Gaussianity' to 'the gapped flow reaches the holomorphic fixed point' [E]/[C] |
| `v158_fixed_point_stable.py` | The free chiral $c{=}8$ fixed point is *stable* (exact dimension count, the finite core of (A)): Majorana $h{=}\frac{1}{2}$, bilinear current $h{=}1$, quartic $h{=}2$; the relevant window $(0,1)$ for bosonic operators is *empty*, the $h{=}1$ currents are chiral (not $(1,1)$-marginal, v157), the lowest interaction (quartic) is $h{=}2{>}1$ irrelevant ⇒ isolated stable fixed point; (A) reduces to a basin-of-attraction statement [E]/[C] |
| `v159_pyrate_gauge_crosscheck.py` | PyR@TE cross-check of the $F_{\mathrm{transfer}}$ gauge inputs: the carrier/SM content gives $(b_1,b_2,b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ by the textbook one-loop formula; $b_1{=}\frac{41}{10}$ holds *three* independent ways — carrier algebra $10b_1{=}g_{\mathrm{car}}2^{g_{\mathrm{car}}-2}{+}1{=}41$, the $U(1)$ hypercharge index $\sum \frac{3}{5}Y^2$, and the external RGE generator PyR@TE 3 (`arXiv:2007.12700`) which writes $\beta_{g_1}{=}(\frac{41}{10})g_1^3$ verbatim; the $M{=}41$ split $10b_1{=}40_{(\mathrm{ferm})}{+}1_{(\mathrm{Higgs})}$ reproduces v13 Gate 1; evidence in `verification/pyrate`. The transfer functor's gauge inputs are not free knobs [E]/[C] |
| `v160_seam_gaussianity_from_pf.py` | Seam Gaussianity as a *conditional* fixed-point theorem (the proposed closure of premise A, honestly typed): (1) quasi-free ⇒ all higher connected cumulants $\kappa_{2n}{=}0$ (signed set-partition/Pfaffian inversion, exact on a pure Majorana covariance); (2) Wick functoriality ⇒ a kernel-preserving transport fixes the whole quasi-free state (v113); (3) the Perron–Frobenius dichotomy (a *fixed* $\kappa_{2n}$ is a SECOND fixed ray, breaking uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only gap$>$0 — the value $(2/3)^6$ only sets the rate. **The load** [O]: v56's gapped PF uniqueness lives on the 3-dim scalar transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$, while the seam correlation cone has $\binom{16}{4}{=}1820$, $\binom{16}{6}{=}8008$ cumulant directions ⇒ (A) reduces to the single internal statement 'the admissible TFPT transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum' [E]/[O] |
| `v161_one_particle_reduction.py` | The cone reduces to one particle (discharging v160's scope premise via Bogoliubov second quantisation): a quasi-free transport is $T{=}\Gamma(t)$, the second quantisation of a one-particle contraction $t$; $\Gamma(t)$ acts on the degree-$m$ correlator sector as the exterior power $\Lambda^m(t)$, so (1) $\mathrm{spec}(\Gamma(t)\lvert_{\deg m}) =$ products of $m$ one-particle eigenvalues [E]; (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly [E]; (3) the eigenvalue-1 sector = wedges of the fixed subspace (disconnected Wick powers), no independent fixed higher cumulant, uniqueness closed by v113 (one polarization ⇔ one state) [E]; (4) quasi-freeness *forced* not assumed — the $120{=}\dim \mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant [C]. **Residual** [O]: v160's three full-cone premises collapse to ONE 16-dim one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 $K_\Sigma$, gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open [E]/[O] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v162_seam_transport_identification.py` | The final identification, forced as far as it goes — the v161 one-particle statement carries *no* new open content: (1) the gap $(2/3)^6$ is *forced* — cusp weights = spec$(A_0^\star)$ = $\{0, \frac{1}{3}, \frac{2}{3}\}$ (parabolic residue, v115), transfer eigenvalue $(1-\alpha)^{p_2}$, $p_2$=6=$\lvert R_+(A_3)\rvert$, all carrier data [E]; (2) $\mu_4$-equivariance forces the generator diagonal in the cusp basis — $\sum_k U^k X U^{-k} = \lvert\mu_4\rvert\,\mathrm{diag}(X)$, commutant of $U$ = the diagonal algebra — so the gapped $\mu_4$-equivariant generator has spectrum = the cusp weights, gap $(2/3)^6$ [E]; (3) the cusp grading *is* the $A_3$ grading on the 16 Majoranas (v137) [E]; (4) the fixed space = the rank-8 Calderón polarization (v113/v156) [E]. **Closure** [C]/[O]: with v160+v161, (A) closes given only $\mu_4$-equivariance + admissibility, whose two non-derived inputs are the *already-open* A2 (net existence) and R1 (seam clock, `HOR.CLOCK`) — so (A) factors exactly into A2 + R1, adds *no* new open item, and is closed conditionally on them (a unification, not an unconditional proof) [E]/[O] |
| `v163_rg_stability_flavor.py` | $F_{\text{transfer}}$ scale-tagging: the lepton-mixing seeds ($\sin^2\theta_{12}$=$\frac{1}{3}-\frac{\varphi_0}{2}$=0.3067, $\sin^2\theta_{13}$=$\varphi_0 e^{-5/6}$=0.0231) are *RG-stable* — PMNS running is $y_\tau^2$-suppressed ($\sim 1.8\times10^{-5} \ll$ precision), so seam value = measured value; the quark mass-ratio readouts ($m_t/m_b\sim$41) run through $y_t^2 \sim O(10\text{–}15\%)$ and carry a scale tag. The only mixing-rotating Yukawa term $\frac{3}{2}YY^\dagger Y$ carries $y_{\max}^2$ (PyR@TE SM RGEs) [E]/[C] |
| `v164_qcd_scale.py` | the QCD confinement scale is parameter-free from the carrier $b_3$=−7: running $\alpha_s(M_Z)$ down (two-loop, $n_f$ thresholds) gives $\alpha_s(m_b)\sim$0.22, $\alpha_s(m_c)\sim$0.38 and confinement ($\alpha_s\sim$1) at ∼0.5 GeV — so the QCD-scale *numerator* of $m_p/m_e$ is independently sourced (dimensional transmutation), while the ratio 1836 stays the honest 'Open / not forced' frontier non-claim ($m_p$=$O(1)\Lambda$ is lattice) [E]/[O] |
| `v165_premise_a_closure.py` | Premise (A), the honest maximal closure — the two residual gates pinned to their floor, so (A) carries *zero* independent open content: (1) A2 (net existence) closes by *assembly* of established mathematics [C]— 16 free Majoranas give a free-fermion net (Buchholz–Mack–Todorov / Böckenhauer–Evans), the holomorphic $c$=8 lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the $\mu_4$ extension is a Longo Q-system of index 4 (v125), $\mu(B)$=16/16=1, 248=120+128, character $E_4/\eta^8=j^{1/3}$ (v156); the only TFPT input (seam Calderón kernel = free-fermion two-point function) is done via DtN = $\lvert k\rvert$ (v156/v157); (2) R1 (seam clock) reduces to `GATE.QGEO` [C]— $\mu_4$-equivariance forces diagonal-in-cusp, the flat $\mu_4$-equivariant generator is *unique* = $A_0^\star$ (v115/v116) so the gap $(2/3)^6$ is forced, residual = 'seam boundary = the $\mu_4$-punctured sphere $\mathbb{P}^1\backslash\mu_4$' = `GATE.QGEO` (cohomology established, v137); (3) the irreducible core stays $\{\pi, v_{\text{geo}}\}$, a *theorem* by the No-Unit Theorem (v153). **Final accounting**: (A) = A2 assembly [C] + R1=`GATE.QGEO` + the $\{\pi, v_{\text{geo}}\}$ core ⇒ ZERO independent open gates; a unification + re-typing, not an unconditional proof [E]/[O] |
| `v166_higgs_free_seam.py` | the Higgs quartic from the *free seam* (premise A / v158): a Gaussian seam UV forces $\lambda(M_{\text{seam}})$=0 *and* $\beta_\lambda(M_{\text{seam}})$=0 (Shaposhnikov–Wetterich double criticality, here *motivated* not assumed). Two-loop PyR@TE-sourced SM running with the measured $(m_H, m_t)$ gives $\lambda(\bar{M}_{\text{Pl}})\sim$0.002, $\beta_\lambda(\bar{M}_{\text{Pl}})\sim$0 — the famous SM near-criticality, now *explained* by the free seam (2-loop sharpens $\lambda(\bar{M}_{\text{Pl}})$ from ∼0.022 to ∼0.002). The double condition predicts $m_H\sim$129–134 GeV; at the scalaron scale it gives ∼107 GeV (too low) ⇒ the BC lives at $\bar{M}_{\text{Pl}}$ (seam=Planck) [E]/[C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v167_inventory_post_closure.py` | The terminal residual inventory (post premise-(A) closure; line v105 → v123 → v167): after the (A)-chain (v160–v165), premise (A) is *no longer* an independent structural class (`GATE.METRIC.16-19`). The residual collapses from v123's FIVE classes to: **Tier 0** — the irreducible core $\{\pi, v_{\text{geo}}\}$, a *theorem* (No-Unit, v153), nothing to close; **Tier 1** — ONE hinge (seam = flavour = horizon): $A_2$ net existence (assembly of established net constructions) + `GATE.QGEO` (seam boundary = $\mathbb{P} \setminus \mu_4$, cohomology $b_1{=}N_{\text{fam}}{=}3$); **Tier 2** — folded ($\to v_{\text{geo}}$, $\to G_{\text{net}}$); **Tier 3** — $F_{\text{transfer}}$, one functor with four typed conditional interfaces ($\eta_B$ carries the most open room). Completeness: a sixth independent structural class fails the script — a unification + re-typing, not an unconditional proof [E]/[C] |
| `v168_qgeo_rigidity.py` | The QGEO *Rigidity Theorem* — the finite half of the Tier-1 hinge `GATE.QGEO`, hardened. Exact: $\mu_4{=}\{1, i, -1, -i\}$ has cross-ratio 2 and Möbius stabiliser $\langle z{\to}iz, z{\to}1/z\rangle$ of order $8{=}\lvert D_4\rvert$ (faithful) ⇒ four marks with faithful $D_4$ are the $\mu_4$ square up to Möbius; $H^1(\mathbb{P} \setminus \mu_4)$ has rank $4{-}1{=}3{=}N_{\text{fam}}$; the forms $\omega_k{=}z^{k-1}dz/(z^4{-}1)$ $(k{=}1,2,3)$ are $\mu_4$-eigenforms with $z{\to}iz$ eigenvalue $i^k$ = the three nontrivial characters $\chi_1, \chi_2, \chi_3$, weights $(1,2,3)$ = $A_3$ exponents = $\text{Spec}(Q_+)$. `QGEO.RIGID.01` [E] (math half closed); `QGEO.REALIZE.01` [C] (seam-collar = $\mathbb{P} \setminus \mu_4$ stays the realisation premise) [E]/[C] |
| `v169_etaB_boltzmann_interface.py` | The $\eta_B$ leptogenesis transfer *operationalised* (Tier-3 $F_{\text{transfer}}$, not a new class): type-I thermal leptogenesis $\eta_B \sim 0.96{\times}10^{-2}\,\varepsilon_1\kappa_f$ (sphaleron 28/79) fed by TFPT's NO neutrino spectrum + the predicted $\delta_{CP}{=}240°$. Davidson–Ibarra $\varepsilon_1{=}\frac{3}{16\pi}M_1 m_3/v^2 \sim 8.5{\times}10^{-7}$ for $M_1{=}10^{10}$ GeV; efficiency $\kappa_f(\tilde{m}_1) \sim 0.03$–$0.19$. **Kill test**: over natural $M_1 \in [3{\times}10^9, 3{\times}10^{10}]$ GeV and washout, $\eta_B$ spans $[8.4{\times}10^{-11}, 4.7{\times}10^{-9}]$ which *brackets* the observed $6.1{\times}10^{-10}$ (canonical point gives $6.0{\times}10^{-10}$, not tuned) ⇒ the route is viable; $M_1$/washout are scenario inputs, so $\eta_B$ stays [C]. If excluded, the route falls, not the theory [C] |
| `v170_diamond_slice_compression.py` | E8 Slice Compression Lemma: the seven $E_8$ audit slices are *one* projection of two alphabets — anchor power sums $P{=}(3,4,6,10)$ ($p_n{=}2{+}2^n$) and Sheet-Diamond determinants $D{=}(3,4,8,14,20,32)$. The $K_4$ edge products $(12,18,30,24,40,60)$ are all carrier blocks; $\sum \det M = 81 = N_{\text{fam}}^4$; all seven 248-slices follow from $P, D, \Delta{=}13$ and $(e_1, e_2, \dim S^+)$; one affine map $\text{row}(C{+}sU{+}tV) = (7,11,13){+}s(3,3,3){+}t(1,2,3)$ gives every flavour row budget; $78 = \dim E_6 = p_2\Delta$. The atlas is a closed grammar over two typed alphabets, exact, reading audit [E] |
| `v171_os_moment_cluster.py` | Atomic OS Moment Lemma + Sugawara Gap Safety: $\text{spec}(T){=}\{1, \frac{64}{729}, \frac{1}{729}\}$ makes $C(n){=}A{+}Br^n{+}Cs^n$ a positive moment sequence, so every OS Hankel matrix factorises as a Vandermonde Gram ⇒ reflection positivity (clean proof, not numeric Gram); cluster $\frac{729}{665}$, $\xi{=}0.41105$, Källen–Lehmann masses $6\log\frac{3}{2}, 6\log 3$; gap safety $\Delta_{\text{eff}}{=}6\log\frac{3}{2}{-}\frac{31}{4\pi^2}{>}0$ with $31{=}1{+}h^\vee(E_8)$, $c((E_8)_1){=}248/31{=}8$ [E]/[C] |
| `v172_trace_anomaly_seed.py` | Trace-anomaly origin of the seed prefactor $\frac{4}{3}$: over the seam $S^2$ ($\chi{=}2$) the $(E_8)_1$ ($c{=}8$) Weyl anomaly is $c\chi/6{=}\frac{8}{3}$, halved by the one-sided $\lvert\mathbb{Z}_2\rvert$ to $\frac{4}{3} = \dim S^+ / \dim \mathfrak{g}_{SM} = 16/12$, the seed base $(\frac{4}{3})c_3{=}\frac{1}{6\pi}$. Plus QCD $b_3{=}11{-}\frac{2}{3}N_f{=}7$ = scalaron exponent $\Omega_{\text{adm}}{-}10b_1{=}48{-}41$ (confinement↔inflation), and the Volkov–Wipf factorisation $-98/45{=}-2\cdot 7^2/\dim(D_5)$ [E] |
| `v173_pfaffian_cp_car.py` | Strong CP as Pfaffian reality: a self-dual CAR (16-Majorana) model demonstration of $\theta_{\text{eff}}{=}0$ — the sheet-odd Calderón involution $\{D, \Sigma\}{=}0$ pairs the spectrum $(\pm\lambda)$, $\gamma_5$-Hermiticity $D^\dagger{=}\gamma_5 D \gamma_5$ makes the Pfaffian real ($\arg \in \{0, \pi\}$), and reflection positivity ($Z{>}0$) selects the positive branch ⇒ $\theta_{\text{eff}}{=}0$ [E] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v174_seam_fock_readings.py` | Seam Fock-space readings: (1) CAR cone compression — $\dim \Lambda^{\text{even}}(\mathbb{R}^{16})=2^{15}=32768$ reduces to a 16-dim one-particle problem under quasi-free Bogoliubov ($2^{15}/16=2^{11}$), the explicit count behind v161; (2) Witten index = the Plücker norm $11=\sum_{k\le 2}\binom{4}{k}$, with $11=\dim S^{+}-g_{\text{car}}$ (full $2^4=16$) and $c_u/c_d=55/117$ reading the QBL-protected boundary state space; (3) Affleck–Ludwig $g$-theorem $c_{UV}=5+3=8=c_{IR}((E_8)_1)$ so $\Delta c=0$ forces an exact isometry (boundary-entropy form of v157/v158) [E] |
| `v175_net_existence_full_cone.py` | The heavy artillery — net existence (A2) and full-cone reflection positivity discharged to [E], isolating `QGEO.REALIZE.01` as the single open premise (nothing fabricated): (1) **full-cone RP for all** $m$ — the CAR second-quantisation functor $\Gamma(t)=\bigoplus_m \Lambda^m(t)$ of the one-particle seam contraction $t$ acts on the complete Fock space $\Lambda^{\bullet}(\mathbb{R}^{16})$ ($\dim 2^{16}=65536$); the exterior-power spectrum theorem makes every eigenvalue a subset product of $\text{spec}(t)$, verified on all $2^{16}$ products to lie in $[0,1]$ (contraction), with eigenvalue-1 multiplicity $2^8=256$ (wedges of the rank-8 $K_\Sigma$ polarisation) and sub-leading = the one-particle gap $(2/3)^6$, so full-cone RP reduces *for all* $m$ to the one-particle contraction (Shale–Stinespring CAR functor), discharging the v160 scope premise by a theorem; (2) **A2 net existence** assembled + verified — $(E_8)_1$ character $E_4/\eta^8=1+248q+4124q^2+34752q^3+213126q^4+1057504q^5$ (level-1=248), embedding $248=120+128$, $E_8$ Cartan even unimodular ($\det 1$); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system); (3) both collapse to the same seam-kernel identification = `QGEO.REALIZE.01`, finite half v168, realisation half left [O][E][O] |
| `v176_seam_collar_realisation.py` | The Seam Collar Realisation Theorem assembled as ONE central statement + reduction certificate (the single structural proof obligation, formulated cleanly — *not* a fabricated proof): given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful $D_4$ marks and admissible $c=8$ data, *then* its conformal boundary is Möbius-equivalent to $\mathbb{P}\setminus\mu_4$ with $H^1$ grading $(1,2,3)$, the Calderón contraction is the rank-8 $K_\Sigma$ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, FIVE already [E]: geometry/uniformisation (v168: cross-ratio 2, faithful Möbius $D_4$ order 8, $b_1=N_{\text{fam}}=3$), Calderón DtN symbol $\lvert k\rvert$ (v156), $H^1$ character = generation grading $(1,2,3)=A_3$ exponents$=\text{Spec}(Q_+)$ (v137/v168), $\mu_4$-equivariant contraction diagonal with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1\otimes(A_3)_1\rtimes\mu_4=(E_8)_1$ + full-cone RP all $m$ (v154/v175); the whole structural residual reduces to the ONE open premise `QGEO.REALIZE.01` (the physical seam collar's boundary *is* this $\mathbb{P}\setminus\mu_4$), left honestly [O][E][O] |
| `v177_seam_marking_kernel.py` | The QGEO proof tree — the v176 single realisation premise split into its honest constituents: REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET, with FOUR nodes closed *constructively* [E] and the open residual sharpened into exactly TWO named obligations. [E] UNIFORM (Lemma 2, constructive): an order-4 Möbius map conjugates to $z\mapsto iz$, a non-fixed orbit scales to $\mu_4$, the reflection $z\mapsto 1/z$ gives $\sigma\rho\sigma=\rho^{-1}$ ($D_4$, order 8) — a genus-0 four-marked faithful-$D_4$ boundary *is* $(\mathbb{P},\mu_4)$; [E] COHOM: $\rho^*\omega_k=i^k\omega_k$ so the $H^1$ grading is $(1,2,3)=A_3$ exponents$=\text{Spec}(Q_+)$; [E] MODULE: the $H^1$ →generation iso is *unique* (multiplicity-free + residue normalisation; reflection $\omega_1\leftrightarrow\omega_3$, $\omega_2$ fixed); [E] NET (v154/v175): index-4 $\mu_4$ extension of $(D_5)_1\otimes(A_3)_1=(E_8)_1$. The TWO genuine open obligations [O]: `QGEO.MARKS.01` (raw seam → genus-0 four-marked $D_4$ boundary) and `QGEO.KERNEL.01` (raw seam Calderón kernel = the $\mu_4$-equivariant free gapped contraction as an *operator*); the whole structural residual is now exactly these two doors [E][O] |

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| Script | What it machine-checks *(continued)* |
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| `v178_marks_kernel_attempt.py` | An honest *attempt* at the two open obligations `QGEO.MARKS.01` + `QGEO.KERNEL.01` — *not* a closure (genuine constructive-geometry/AQFT statements, not finite computations), but a deeper reduction: each finite core is closed [E] and each obligation reduced to ONE sharper irreducible premise [O]. *Marks core* [E]: a genus-0 double with an order-4 clock $\rho$ ($\sim z \mapsto iz$, fixed points $\{0,\infty\}$) and reflection $\tau$ forces the marks to be a generic $\rho$-orbit $\{1,i,-1,-i\}$=$\mu_4$ (not the fixed points), distinct iff $b_1$=4−1=3=$N_{\mathrm{fam}}$, $\tau\rho\tau$=$\rho^{-1}$ ⇒ faithful $D_4$ order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* [E]: on the 3-dim multiplicity-free $H^1$ the $\mu_4$ characters $(1,2,3)$ are distinct, so by Schur a $\mu_4$-equivariant operator is forced diagonal and *completely determined by its eigenvalue per character* — two such with the forced transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ (v162) and the unique residue-normalised basis (v177) are *equal as operators*, not merely isospectral (the spectrum-vs-operator worry vanishes on the finite block); so KERNEL reduces to the single full-operator-reduction premise (principal symbol $\lvert k\rvert$, Lee–Uhlmann/v156; no drift, v158). Neither obligation closed; both sharpened [E][O] |
| `v179_conformal_realisation.py` | The two residual premises (MARKS-residual + KERNEL-residual) collapse to ONE geometric premise `QGEO.CONF.01` (honest unification; the conformal realisation stays [O]). MARKS's residual is not three independent assumptions: (1) genus-0 *is* P1 — $c_3$=1/(\|$\mathbb{Z}_2$\| $2\pi\,\chi(S^2)$)=1/(8$\pi$) with $\chi$=2 ⇒ $g$=0, the same $\chi$=2 that gives the '8' (v58/v73); (2) the order-4 clock *is* the carrier — Coxeter element of $W(A_3)$=$S_4$, order $h(A_3)$=4=$\lvert\mu_4\rvert$ (v117); (3) marks = one orbit is *forced* — parabolic marks ≠ the two elliptic fixed points $\{0,\infty\}$, a free order-4 orbit has exactly 4=$e_1(a)$ points. So MARKS reduces to 'the carrier clock acts conformally on the genus-0 double'. KERNEL then follows from that geometry: DtN symbol $\lvert k\rvert$ (Lee–Uhlmann, v156) ⇒ rank-8 $c$=8 fixed polarisation; no drift (v158); the cusped-surface residual data is $H^1$=$\mathbb{C}^3$=$N_{\mathrm{fam}}$, the 3-dim gapped transfer block (Schur, v162/v178). Both ⇒ ONE premise `QGEO.CONF.01`: the raw RP seam double is conformally $(\mathbb{P},\mu_4)$ with the carrier clock as the order-4 Möbius map — the single structural residual of the whole theory, left honestly [O][E][O] |
| `v180_clock_is_mobius.py` | Cracking `QGEO.CONF.01` one level further — the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder*, non-circular premise `QGEO.ISO.01`: 'the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric'. (1) Uniformisation (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, v179) *is* $\mathbb{P}$ — the target is produced, not assumed; (2) Kerékjártó (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of $S^2$ is conjugate to a rotation, so the order-4 clock ($h(A_3)$=4) is topologically $z \mapsto iz$; (3) isometry⇒conformal⇒Möbius: an orientation-preserving isometry of a Riemannian surface is holomorphic, Aut($\mathbb{P}$)=PSL$(2,\mathbb{C})$; (4) order-4 Möbius = $z \mapsto iz$ (verified: projective order 4, fixed $\{0,\infty\}$, multiplier $i$, free orbit $\mu_4$). So `QGEO.CONF.01` is *implied* by the milder isometry premise; the new residual `QGEO.ISO.01` (the seam transport is a metric-compatible holonomy preserving the RP inner product) does not name $\mathbb{P}/\mu_4$ and is milder, its surface realisation from the raw seam still [O][E][O] |

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| Script | What it machine-checks *(continued)* |
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| `v181_clock_is_conformal_symmetry.py` | The FINAL honest reduction + the explicit identification of BEDROCK. The isometry premise `QGEO.ISO.01` (v180) is weakened one more genuine step — to a CONFORMAL-symmetry premise `QGEO.SYM.01` — via equivariant uniformisation (Nielsen realisation, finite cyclic on genus-0): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. The clock is the Coxeter element of $W(A_3)$=$S_4$ (v117), order $h(A_3)$=4=$\lvert\mu_4\rvert$, the $\mu_4$ deck, and a deck transformation preserves the pulled-back conformal structure by definition. So below `QGEO.SYM.01` the residual is *definitional*, not a missing theorem: 'the seam's conformal structure *is* the $\mu_4$-deck structure of the carrier clock' (the carrier $\mu_4$ glue is the conformal monodromy of the seam) — a physical/definitional identification tying P2 to the seam's conformal geometry, not reducible further without relabeling. The chain REALIZE(v175)→theorem(v176)→MARKS+KERNEL(v177)→finite cores(v178)→`CONF.01`(v179)→`ISO.01`(v180)→`SYM.01`(v181) has reached bedrock; we stop here honestly [E][O] |
| `v182_reviewer_residual_map.py` | The four external-review concerns reduce to the two already-named residuals + one data-only item. $A$ (mapping $2D \to 4D$) [E]/[C]: the Plücker readout $11 = \sum_{k\le 2}\binom{4}{k} = \dim S^+ - g_{\rm car} = 16-5$ (QBL count, v174) and the holographic reduction give $A$ = `QGEO.SYM.01` + $F_{\rm transfer}$. $B$ (UV gravity) [E]: 'gravity = geometric boundary readout' *is* `QGEO.SYM.01`. $C$ (grammar tuning) [E]/[C]: frozen (v84) + minimal + over-determined (the anchor $a$=$(1,1,2)$ in $\ge 6$ independent not-selected-for structures) is look-elsewhere-resistant, but the residual falls only to out-of-sample data (JUNO/CMB-S4). $D$ (Koide Möbius flow) [E]/[C]: the Möbius nature is forced by Aut($\mathbb{P}$)=PSL$(2,\mathbb{C})$ on the deck, $t$∼2.84 only locates the point, and the missing RG generator *is* $F_{\rm transfer}$, so $D$ = `QGEO.SYM.01` + $F_{\rm transfer}$. NET: three reduce onto {`QGEO.SYM.01`, $F_{\rm transfer}$}, one waits for data — a unification of the concern structure, not a closure (premises stay [O]/[C]) [E][C] |
| `v183_koide_f_corner_transfer.py` | The Koide source→pole factor $\frac{53}{54}$ has an *operator* origin (external-review proposal, validated): $u_{\rm pole}/u_{\rm source} = a^T(R{+}Q)\mathbf{1}/(2\,\mathbf{1}^T R a) = 53/(2{\cdot}27) = \frac{53}{54}$. (1) $\mathbf{1}^T R a = 27$ is the $E_6{\times}A_2$ cubic family block (248 = dim $E_6$ 78 + det $R$ 8 + 2·27·$N_{\rm fam}$); (2) $F$=$R$+$Q$ is the *missing* Sheet-Diamond corner $M(1,1)$ with det $B_F = 52 = \dim F_4$ (v80) and anchor response $a^T(R{+}Q)\mathbf{1} = 53 = 52{+}1$; (3) $\frac{53}{54} = 1-\frac{1}{2N_{\rm fam}^3}$ exactly (54=2·27); (4) negative control: only $F$ gives 53 ($R/Q/K/L \to 32/21/35/56$). Upgrades `FR.KOIDE.03` from a bare guess to an operator identity; [E] the factor is exact, [C] the leptonic source→pole reading the $F$ corner stays an $F_{\rm transfer}$ conjecture [E][C] |
| `v184_etaB_anchored_boltzmann.py` | Honest test of the external-review $\eta_B$ proposal ($M_1$=$M_R\phi_0^4$, $\tilde m_1$=$m_3/A_\Lambda$) — firewall verdict: *sharper scenario*, not a closure. The washout anchors plausibly ($\tilde m_1$=$m_3/A_\Lambda$=$m_3/10 \approx 5$ meV, $A_\Lambda$=10=$\lvert E(K_5)\rvert$, in the strong-washout window) [C]; but $M_1$ does *not* — $M_R$ is the seesaw scale $\sim v^2/m_3 \approx 6{\times}10^{14}$ GeV and the nearest clean TFPT powers of $\bar M_{\rm Pl}$ both miss it ($c_3$-power off ∼75%, $\phi_0$-power off ∼40%), so $M_1$=$M_R\phi_0^4$ merely relocates the free input $M_1 \to M_R$ (the reviewer's $6.0{\times}10^{-10}$ uses $M_R$=$1.3{\times}10^{15}$; the seesaw value gives $3.4{\times}10^{-10}$, the tell that $M_1$ is unpinned). $\eta_B$ stays in the observed band so the route is viable, but the proposal cuts the free inputs from two to one, not to zero — $\eta_B$ stays [C], a sharper scenario [C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v185_axion_relic_solver.py` | Axion relic, honest misalignment estimate: the determinant-line axion $f_a{=}M_{\rm scal}/128 \approx 2.39{\times}10^{11}$ GeV ($m_a \approx 23.8\,\mu$eV) can provide all dark matter, but *only* near the $\theta_i \approx 170°$ hilltop where the relic is exponentially sensitive — so the axion DM branch stays a [C] scenario. The harmonic misalignment under-produces at this $f_a$ (prefactor $\sim 0.026/\theta^2$, $\theta{\sim}O(1)$ gives $\sim$21% of $\Omega_{\rm DM}$, need $\theta^2_{\rm eff}{\sim}4.7$), so the hilltop anharmonic enhancement is required; near the hilltop it is exponentially sensitive, hence a scenario not a sharp prediction. $F_{\rm relic}(f_a,\theta_i,N_{\rm DW}){=}^{?}\Omega_{\rm DM}$ is a kill test for the *branch*, not the theory (no free singlet WIMP: $128{=}(16,4){+}(16,4)$) [C][O] |
| `v187_ftransfer_laws.py` | The $F_{\rm transfer}$ discipline guard (CI-style firewall): the four continuous-transfer frontier observables — Koide, $\eta_B$, the axion relic, $m_p/m_e{+}\Lambda_{\rm QCD}$ — must *never* be typed as primitive [E] compiler predictions; each carries a [C]/[O] marker. The guard reads `status_ledger.csv` and enforces it (exact algebraic sub-parts — Koide 53/54 v183, $b_3{=}{-}7$ v159 — may be [E], the physical prediction never is). Records the four interfaces $F_{\rm pole}/F_{\rm Boltzmann}/F_{\rm relic}/F_{\rm QCD}$ and that the raster rule + frozen registry + null model bind, so no future edit can quietly promote a pretty frontier number to a closed compiler output [E] |
| `v188_frontier_wording_guard.py` | The frontier-wording guard (a prose sentinel, no new physics): v187 protects the ledger *typing*, this guard protects the *prose*. It forbids stale sentences whose semantics an earlier round superseded — "leptogenesis interface ([E])" (v184 retyped Route B to [C]), "closed branch also fixes the leptogenesis inputs" ($M_R$ is the seesaw scale, not a compiler power), "relic scale $f_a, m_a$ open" (v185: the scales define the branch, the open item is the hilltop-sensitive relic abundance), "computed $Q{=}0.664\ldots$ near, not exact" (v183 gave 53/54 an exact operator origin), "Suite now 187 modules" (the count is 187, highest ID `v188`; v186 skipped). Scans the 10 active documents; passes when none appear, so a frozen Zenodo release can never re-introduce half-true wording [E] |
| `v189_riemann_roch_carrier.py` | The Riemann–Roch carrier reading: $(g_{\rm car}, N_{\rm fam}) = (5,3)$ are the two canonical invariants of *one* object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$. [E] arithmetic: $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg{+}1 = 5$ (function side) and $\operatorname{rank} H_1(\mathbb{P}^1 \backslash \mu_4) = \deg{-}1 = 3$ (cycle side, already `QGEO.COHOM.01`/v177); $g_{\rm car} = 5 = \operatorname{rank}(D_5)$ (the carrier is the $\mathfrak{so}(10)$ half-spinor, $\operatorname{rank} D_5{+}\operatorname{rank} A_3 = 5{+}3 = 8$, v1), $N_{\rm fam} = 3$. [C]: reading the carrier as the meromorphic seam-mode space $H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ (less number, more geometry) is matched by $\operatorname{rank}(D_5) = 5$, but the physical identification is conjectural and does *not* replace the over-determined $g_{\rm car} = 5$ (Pascal/bootstrap/Lean) — a fourth, geometric reading [E][C] |
| `v190_nariai_entropy_bound.py` | The one-line Nariai entropy bound: the Schwarzschild–de Sitter entropy ratio $S_{\rm tot}/S_{dS} = (x^2{+}1)/\Phi_3(x)$ ($x{=}r_b/r_c$; $\Phi_3{=}x^2{+}x{+}1$ the $N_{\rm fam}{=}3$ cyclotomic, already in `tfpt_horizon_readouts`) satisfies $S_{\rm tot} \geq \frac{2}{3} S_{dS}$ with equality exactly at the Nariai merge $x{=}1$, via the perfect square $3(x^2{+}1) - 2\Phi_3(x) = (x-1)^2 \geq 0$ — a variation-free proof of the entropy floor (global minimum on $x{>}0$ is $2/3 = |Z_2|/N_{\rm fam}$) [E] |
| `v191_universal_branch_line.py` | The universal branch line, honestly typed [C] — an exact affine *relabeling*, not a theorem. The map $q = \frac{7}{2} + \frac{N_{\rm fam}^2}{2} m$ carries the SdS branch points $m = \pm 1/N_{\rm fam}$ onto the flavor branch points $\{2,5\} = \{|Z_2|, g_{\rm car}\}$ ($m{=}0{\to}7/2$). The arithmetic is exact, but the map exists for *any* two pairs of points (two intervals are always affinely equivalent), so the slope $N_{\rm fam}^2/2$ and midpoint $7/2$ are *determined* by the data, not forced predictions; a decoy pair $\{3,4\}$ admits an equally exact map (negative control). The genuine shared content — the $2/3 = |Z_2|/N_{\rm fam}$ ramification of the mass↔transport double cover — is already established. Recorded as a [C] alignment, never promoted to [E] [C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v192_qgeo_energy_clock.py` | The energy-conserving-clock reformulation, honestly typed: `QGEO.ENERGY.01` ($\rho^*\Lambda_\Sigma\rho=\Lambda_\Sigma$ — the carrier clock preserves the Calderón/DtN energy form) *relocates* the bedrock `QGEO.SYM.01` to a more checkable operator identity, but does *not* eliminate it. The downstream chain (conformality → Möbius → $z\mapsto iz$ → $\mu_4$ → `QGEO.SYM.01`) already exists (v180/v181); the new upstream link (energy invariance ⇒ conformality in 2D) is checkable on the DtN operator $\Lambda=\lvert k\rvert$. But '$\rho$ preserves $\Lambda$' is plausibly equivalent to '$\rho$ is the conformal deck', so it is a sharper restatement, not a theorem, unless $\rho$ is independently defined and the invariance independently proven (open). Bedrock stays [O] |
| `v193_qgeo_energy_commutator.py` | `QGEO.ENERGY.02`: the energy-*commutator* contract (the non-circular sharpening of v192): $[\rho,\Lambda_\Sigma]=0$ on the rank-8 Calderón polarisation, $\rho$ the carrier clock, $\Lambda_\Sigma$ the DtN energy form of the *raw* RP seam. Three sub-claims: (1) [E] $\rho$ is carrier-defined (Coxeter of $W(A_3)=S_4$, order $4=\lvert\mu_4\rvert$, v117), not imported from the seam; (2) [O] the non-circularity hinge — $\Lambda_\Sigma$ must be the *raw* RP-seam DtN, not the $\mu_4$ normal form; (3) [E] $[\rho,\Lambda_\Sigma]=0$ forces the $(1,2,3)$ grading on $H^1$ (`QGEO.COHOM.01`). Exact [E] rider: the induced $k=(\ln m)/(12\pi)$ v150 reproduces $c_3/2$ iff $\ln m=q(A_3)=\frac{3}{4}$ (FAMILY), not $q(D_5)=\frac{5}{4}$ (carrier, off by 5/3) — gravity is family-geometry-induced. Proof target [O]; if proven, `QGEO.SYM.01` becomes an energy-rigidity theorem [O] |
| `v194_raw_seam_dtn_rp.py` | `QGEO.DTN.01` — subclaim 2 of v193 ("the raw RP-seam DtN is RP-definable") tackled: a *reduction*, not a closure. (1) [E] finite core: $\rho{:}z\mapsto iz$ on $H^1=\langle w_1,w_2,w_3\rangle$, $w_k=z^{k-1}dz/(z^4-1)$, acts as $\rho^*w_k=i^kw_k$ (characters $i,-1,-i$ distinct), so $[\rho,\Lambda_\Sigma]=0$ on $H^1$ (v177). (2) [E] hinge met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so $\Lambda_\Sigma$ is RP-definable without assuming $\mathbb{P}^1\backslash\mu_4$. (3) [O] residual relocates: full-$L^2$ commutation ⇔ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (must be intrinsic, else re-circulates). `QGEO.SYM.01` reduces from a definitional postulate to intrinsic BW modular covariance — sharper, still [O], not closed [O] |
| `v195_marks_lefschetz_character.py` | `QGEO.MARKS.02` — a Lefschetz/character derivation of the four seam marks (review path 2): instead of positing $D_4$ marks, they are *forced* to be a free $\mu_4$ orbit. [E]: $\rho{:}z\mapsto iz$ (order 4) fixes only $\{0,\infty\}$; $H^1=\chi_1+\chi_2+\chi_3$ (the $A_3$ exponents $(1,2,3)$, v177) has no trivial character ⇒ no puncture at a fixed point; $\operatorname{rank}H_1=n-1=3\Rightarrow n=4\Rightarrow$ exactly one free orbit ⇒ marks $=\mu_4$ (cross-ratio 2); $\operatorname{Tr}(\rho\lvert H^1)=i+i^2+i^3=-1$, $L(\rho)=1+1=2=\#\text{fixed}$. [O] residual relocates from "the seam is $\mathbb{P}\backslash\mu_4$ (geometric)" to "$H^1$ carries the $A_3$-exponent rep" — less circular [E][O] |
| `v196_seam_energy_functional.py` | `QGEO.VARI.01` — the $E_{\text{fail}}$ variational functional (review path 1), a numerically-testable sharpening of the v194/`ENERGY.02` target: $E_{\text{fail}}=\lVert[\rho,\Lambda_\Sigma]\rVert^2_{HS}+\lVert\rho^4-1\rVert^2+\lVert\Theta\rho\Theta-\rho^{-1}\rVert^2$, with $E_{\text{fail}}=0\Leftrightarrow$ the `QGEO.SYM.01` conditions. [E] on the finite $H^1$ block ($\rho=\operatorname{diag}(i,-1,-i)$, $\Lambda$ diagonal, $\Theta$=conj) all three vanish ⇒ $E_{\text{fail}}=0$; [E] a $\mu_4$-breaking $\Lambda$ or non-order-4 $\rho$ gives $E_{\text{fail}}>0$ ($\mu_4$ a true minimiser); [E] the zero locus is the seam-deck conditions; [O] the full-operator minimisation is the v194 Bisognano–Wichmann residual — a discretisable handle on the open bedrock, a target not a closure [E][O] |
| `v197_rr_carrier_clifford_d5.py` | `ARCH.RRCAR.02` — hardening the Riemann–Roch carrier from $g_{\text{car}}$ to $D_5$ itself (review point 7): the 5-dim carrier mode space $C_{\text{car}}:=H^0(\mathbb{P}^1,\mathcal{O}(\mu_4))$ generates the $\mathfrak{so}(10)=D_5$ half-spinor by its even Clifford algebra. [E] $\dim\Lambda^{\text{even}}(\mathbb{C}^5)=\binom{5}{0}+\binom{5}{2}+\binom{5}{4}=1+10+5=16=2^{g_{\text{car}}-1}=\dim S^+$; the $\mathfrak{so}(10)=D_5$ half-spinor $2^4=16=S^+=\Lambda^{\text{even}}(C_{\text{car}})$; closes the chain $\mu_4\to g_{\text{car}}\to D_5$ from one four-point divisor (cycle side $\operatorname{rank}H_1=3=N_{\text{fam}}(A_3)$; function side $h^0=5=g_{\text{car}}$; Clifford spinor $16=D_5$ half-spinor = carrier). [C] "carrier = Clifford spinor of the seam mode space" stays physical (as v189), does not change the over-determined $g_{\text{car}}=5$ [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v198_modular_commutator_reduction.py` | `QGEO.MODULAR.01` — cracking the bedrock to a state-invariance. The v194 residual ($[\rho,\Lambda_\Sigma]$=0 on full $L^2$, framed as Bisognano–Wichmann with a circularity worry) is (a) made *exact* at the principal-symbol level and (b) reduced via Tomita–Takesaki to a clean state-invariance. [E] the DtN principal symbol $\lvert k\rvert=\sqrt{-d^2}$ is diag($\lvert n\rvert$) and the clock $\rho{:}z\mapsto iz$ is diag($i^n$) in the Fourier basis — both diagonal ⇒ $[\rho,\lvert k\rvert]$=0 exactly on *all* of $L^2$; [E] Tomita–Takesaki: $[\rho,\Lambda_\Sigma]$=0 follows from $\omega\circ\rho=\omega$ (the modular $\Delta$ is intrinsic to $(M,\Omega)$; a state-preserving symmetry commutes with it), needing *no* conformal covariance — removes the BW circularity; [E] the finite $H^1$ block is already $\mu_4$-invariant (v177). [O] residual: the full raw seam state is $\mu_4$-invariant ($\omega\circ\rho=\omega$) ⇐ the Gaussian bulk measure is deck-invariant = operational `QGEO.SYM.01`. Sharpest non-circular form; not a closure (a foundational symmetry cannot be derived from nothing) [E][O] |
| `v199_seam_state_invariance.py` | `QGEO.STATE.01` (work package A) — attacking $\omega\circ\rho=\omega$ directly on the raw seam: a further localising reduction, not a closure. [E] setup non-circular ($\rho$ = Coxeter of $W(A_3)=S_4$, $\rho^4=1$, an automorphism of the raw CAR algebra; both carrier-side, no seam geometry); [E] ground-state reduction $\omega\circ\rho=\omega \Leftrightarrow [\rho,H_1]=0$ ($C_\Sigma=\frac{1}{2}(1+\operatorname{sgn}H_1)$); [E] character criterion: $[\rho,H_1]=0 \Leftrightarrow H_1$ block-diagonal in the four $\mu_4$-character classes $\{n=r \bmod 4\}$; principal symbol $\lvert k\rvert=\operatorname{diag}(\lvert n\rvert)$ passes automatically. [O] residual: the bounded sub-principal symbol has *no* off-character matrix elements — non-circular, sharply typed, holds on $H^1$ (v177); not a closure [E][O] |
| `v200_seam_variational_scan.py` | `QGEO.VARI.02` (work package B) — the discretised variational test: v196 had $E_{\text{fail}}$=0 at the $\mu_4$ config on the fixed block; here the clock angle is free and $E_{\text{fail}}$ is minimised over it. [C] scanning $E_{\text{fail}}(\theta)=\lVert\rho(\theta)^4-1\rVert^2$+faithfulness over $\theta\in(0,\pi)$ gives the unique minimum at $\theta=\pi/2$ ($E_{\text{fail}}$=0: order-4 *and* three distinct characters $i,-1,-i$ = the $(1,2,3)$ grading); decoys $(0,\pi,\pi/3,2\pi/3)$ all lose. So $\arg\min E_{\text{fail}}$ = the $\mu_4$ normal form. [E] consistent with v196/v199/v198. [O] a numerical sign on the finite $H^1$ grading, not the full operator-valued minimisation (the v199 residual) [C][O] |
| `v201_seam_subprincipal_marks.py` | `QGEO.SUBPRIN.01` — the v199 bedrock residual reduced one genuine step further: block-diagonality of the sub-principal symbol is *not* an independent postulate. [E] the seam DtN is $\Lambda=\lvert k\rvert+M_f$, $M_f$ = multiplication by a curvature function $f(\theta)$ (standard Steklov sub-principal), so $\langle n\rvert M_f\rvert n'\rangle=f_{n-n'}$. [E] $M_f$ is $\mu_4$-character-block-diagonal ⇔ $f$ has Fourier support only on modes $\equiv 0 \pmod 4$ ⇔ $f$ is $\mathbb{Z}_4$-invariant ($\lvert k\rvert$ already commutes, v198). [E] a $\mu_4$-mark sum $f(\theta)=\sum_{j=0}^{3} g(\theta-2\pi j/4)$ has $f_m=4g_m[m\equiv 0 \bmod 4]$ for *any* profile $g$ — automatically $\mathbb{Z}_4$-invariant, so (v195 marks=$\mu_4$ orbit) *forces* the sub-principal symbol block-diagonal; negative controls: 3 marks ($\mathbb{Z}_3$) and 4 generic marks break it. [O] new residual: $\omega\circ\rho=\omega$ reduces to 'the DtN sub-principal symbol is mark-local (seam flat away from the $\mu_4$ marks = conformal deck v177)', the narrowest, structurally-definitional form of `QGEO.SYM.01` [E][O] |

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| Script | What it machine-checks *(continued)* |
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| `v202_rare_kaon.py` | The rare kaon sector $K \to \pi\nu\bar{\nu}$ from the TFPT CKM point ([C] downstream flavor readout, *not* a compiler power). The kernel fixes the full CKM matrix ($s_{12}{=}\lambda_C$, $s_{23}{=}\varphi_0/(1{+}\lambda_C){=}|V_{cb}|$, $s_{13}{=}\lambda_C^3/3$, $\delta{=}\pi/3{+}3\lambda_C^2{=}1.19823$ rad; v18/v88) with no flavor fit; feeding the exact point into the standard Brod–Gorbahn–Stamou short-distance formulas gives $\mathrm{BR}(K^+ \to \pi^+\nu\bar{\nu}){=}9.45 \times 10^{-11}$ and $\mathrm{BR}(K_L \to \pi^0\nu\bar{\nu}){=}3.33 \times 10^{-11}$. [E] CKM point + exact apex $(\bar{\rho},\bar{\eta}){=}(0.1374, 0.3509)$, $\lambda_t{=}V_{ts}^*V_{td}$; [C] the branching ratios (*not* parameter-free — $X_t, P_c, \kappa_\pm$ are external EW/QCD input); [E] $\mathrm{BR}(K^+)$ $\mathrm{BR}(K^+){=}9.45 \times 10^{-11}$ lands ON the NA62 2016-2024 combination $(9.6^{+1.9}_{-1.8}) \times 10^{-11}$ (La Thuile 2026), $+0.08\sigma$ (the 2016-2022 obs $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$ was $\sim 1.2\sigma$ above) — a strong consistency hit, but an $F_{\mathrm{transfer}}$ bridge, not a unique TFPT-vs-SM discriminator (SM $(8.6 \pm 0.4) \times 10^{-11}$ also inside the NA62 error); [E] Grossman–Nir ratio $0.352 \ll 4.3$; [X] dated kill test: a stable NA62 $\mathrm{BR}(K^+)$ outside $[7, 12] \times 10^{-11}$, or a KOTO-II $\mathrm{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched) [C][X] |
| `v203_eht_achromatic.py` | The EHT achromatic polarization-rotation intercept $\beta_{\mathrm{BH}}$ around a horizon-scale black hole — the surviving core of the old UFE/BH notes (*not* the RN-type metric, superseded by Nariai/seam=horizon v101–v104/v190; only the polarization signature). [E] the coupling is a compiler number: $\beta_{\mathrm{BH}}(r){=}Q_e^{\mathrm{eff}}Q_m^{\mathrm{eff}}/(256\pi^4 r^2){=}16c_3^4(Q_eQ_m/r^2)$, $16c_3^4{=}1/(256\pi^4)$ exactly. [E] $\alpha$-kernel cross-link: $16c_3^4{=}\delta_{\mathrm{top}}/3$, $\delta_{\mathrm{top}}{=}48c_3^4{=}3/(256\pi^4)$ — the *same* top-form coefficient that fixes the $\alpha$-kernel precision-zone correction (one row from $\beta_{\mathrm{rad}}{=}\varphi_0/4\pi$). [C] TFPT fixes the $1/r^2$ profile, achromaticity, and sign-flip under $E{\cdot}B$ reversal; the amplitude $Q_eQ_m$ is an MHD/GR weight, not a compiler output. [X] dated kill test: the residual intercept must pass three nulls (frequency/achromatic, spatial $1/r^2$, sign-flip); a null-consistent residual after honest GRMHD subtraction falsifies the channel (core untouched) [E][C][X] |
| `v204_muon_seam_g2.py` | The muon anomalous magnetic moment $a_\mu{=}(g_\mu{-}2)/2$ as a seam vertex readout ([C] downstream, *not* a compiler power; archive integration of the old muon-$g{-}2$ note). The carrier fixes the second-order defect $\delta_2{=}B\gamma\,\delta_{\mathrm{top}}^2{=}\frac{5}{4}\delta_{\mathrm{top}}^2$ ($\delta_{\mathrm{top}}{=}\Omega_{\mathrm{adm}}c_3^4{=}48c_3^4{=}3/256\pi^4$, v2/v23; $B\gamma{=}\frac{3}{2}{\cdot}\frac{5}{6}{=}\frac{5}{4}$, v51); the seam-loop projection $a_\mu^{\mathrm{seam}}{=}\delta_2/(2\pi){=}45/(524288\pi^9) \approx 2.879 \times 10^{-9}$. [E] the value is an exact compiler number ($\delta_2{=}4!\,\mathrm{Tr}_{S^+}(X^2)c_3^8$, Tr$=$120$=$5!); [C] the identification of $\delta_2/(2\pi)$ as a magnetic vertex ($1/2\pi{=}4c_3$) is a physical bridge; [E] $0.81\sigma$ vs dispersive $\Delta a_\mu{=}(2.49{\pm}0.48) \times 10^{-9}$; [C] HONEST: lattice/CMD-3 HVP shrinks it ($\sim 1.5 \times 10^{-9}$), the fixed value then $\sim 1.5\sigma$ high; [X] dated kill test: a converged $\Delta a_\mu$ outside $2.879 \times 10^{-9}{\pm}0.5 \times 10^{-9}$ excludes the mechanism [E][C][X] |
| `v205_xi_threequarter.py` | $\xi{=}c_3/\varphi_{\mathrm{tree}}{=}3/4$: an *independent* route to the gravitational 3/4 (archive integration of the old Eliminating-$K$ note). The torsion-compression ansatz $\kappa^2{=}\xi\varphi_0/c_3^2$ with the Einstein limit $\kappa^2{=}8\pi G$ fixes $\xi{=}8\pi c_3^2/\varphi_0{=}c_3/\varphi_0$ (since $8\pi c_3^2{=}c_3$), and at tree level $\varphi_{\mathrm{tree}}{=}1/(6\pi)$ this is exactly 3/4. [E] tree identity $\xi{=}c_3/\varphi_{\mathrm{tree}}{=}3/4$; [E] Einstein-limit reduction $8\pi c_3^2{=}c_3 \Rightarrow G$ is an output; [E] over-determination $3/4{=}q(A_3){=}N_{\mathrm{fam}}/|\mu_4|{=}\ln(m/\mu)$ (v152 $k{=}c_3/2$) — the torsion-compression $\xi$ and the gapped EH replica are two independent appearances of the same gravitational 3/4; [E] with the retained seed $\xi{=}c_3/\varphi_0{=}0.748303$ ($-0.226\%$); [C] the $\kappa^2{=}\xi\varphi_0/c_3^2$ ansatz is the bridge, the modern derivation is v152/v68, $1/G$ stays the one anchor (v60/v153) [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v206_h0_lambda_branch.py` | The cosmological-constant Letter branch: the seam number $\delta_\Sigma{=}m_1 e^{-2\pi\alpha^{-1}/3}$ with the lowest block $m_1{=}\Omega_{\rm adm}{=}48$, the Planck-convention split, and the quantitative $H_0$=66.5–67.1 (archive integration of the tfpt-31 CC note; sharpens the qualitative $H_0$-vs-$\Lambda$ row). [E] $\delta_\Sigma^{\rm lead}{=}48\,e^{-2\pi\cdot 137.036/3}\approx 1.085\times10^{-123}$; [E] convention split $M_{\rm Pl}^4/\bar{M}_{\rm Pl}^4{=}(8\pi)^2{=}631.65$ (unreduced vs reduced — must be displayed); [E] $\rho_\Lambda{=}M_{\rm Pl}^4\delta_\Sigma\approx 2.4\times10^{-47}\,{\rm GeV}^4$ (observed vacuum order); [E] 123-orders cross-link $\lvert\log_{10}\delta_\Sigma\rvert\approx 122.96$, consistent with the v60 split 122.948; [C] under flatness closure $H_0^{\rm lead}$=66.51, $H_0^{\rm disc}$=67.10 — 1.6–0.5$\sigma$ below Planck, $\sim 6\sigma$ below SH0ES: the Hubble tension is *not* relieved by raising the vacuum scale; $H_0$ uses imported $\Omega_b,\omega_{\rm DM}$, a [C] readout [E][C] |
| `v207_asymptotic_safety.py` | Asymptotic Safety as a *second*, independent, [O] route to the metric sector (archive integration of the old Five-Problems note). The current theory closes QG via the discrete seam-net/AQFT programme (v175; open premise `QGEO.REALIZE.01`); this records a complementary continuum-FRG argument that a non-Gaussian UV fixed point exists from the same axioms — a plausibility cross-check, *not* load-bearing, does *not* close $G_{\rm metric}$. [E] the axion coupling is fixed and small $y^2{=}16c_3^2{=}1/(4\pi^2){=}0.025330$ exactly ($g_{a\gamma\gamma}{=}-4c_3$ from the cubic); [E] existence by IVT: $\partial_t g/g{=}2{+}\eta_R$ is $+2{>}0$ as $g{\to}0$ and $<0$ at large $g$ (torsion $K^2$ damping) $\Rightarrow$ a root $g_*{>}0$; [E] $g_*{=}2/a_1{>}0$ finite, Reuter-type $O(1)$, no new free input; [O] HONEST: a continuum-FRG ansatz the discrete-compiler theory did *not* adopt, primary stays the seam-net (v175) [E][O] |
| `v208_bh_thermodynamics.py` | Black-hole thermodynamics from the seam: the modular Hawking temperature and the scalaron-corrected Wald entropy (archive integration of the dropped tfpt-42 stationary-horizons section, re-typed through the current Tomita–Takesaki seam work v198/v199). [E] modular $T_H$: $\Delta^{it}{=}e^{-2\pi t K_H}$ (Bisognano–Wichmann) $\Rightarrow$ KMS at $\beta{=}2\pi$ and $T_H{=}\kappa/(2\pi)$; the $2\pi$ is the seam unit $1/(4c_3)$ ($1/2\pi{=}4c_3$, v58), reproducing $T_H{=}c_3/M$ (v57); seam↔horizon identification [C]; [E] Wald entropy: with the induced $f(R){=}R{+}R^2/(6M_s^2)$ ($M_s{=}c_3^{7/2}\bar{M}_{\rm Pl}$, v28), $S_W{=}(f_R A)/(4G){=}(A/4G)(1{+}R_h/3M_s^2)$ — leading $A/4G$ is the $c_3$ area law ($1/4{=}1/\lvert\mu_4\rvert$), the correction an exact scalaron consequence; [E] SdS $T_H{=}(1/4\pi r_h)(1{-}\Lambda r_h^2)$, $1/4\pi{=}2c_3$; [E] area law $S_{BH}{=}M^2/(2c_3){=}4\pi M^2{=}A/4G$ [E][C] |
| `v209_bh_regular_core.py` | The black hole as a seam fixed point (topological defect), *not* a singularity: the honest [O] synthesis of the old Five-Problems BH picture, with the superseded torsion/RN-vortex metric dropped (RN-type metric superseded by Nariai/seam=horizon v101–v104/v190; cf. v203). Built entirely from established suite atoms. [E] no singularity = seam fixed point: the gapped attractor multiplier $\lambda_2{=}(2/3)^6{=}64/729$ with gap $\Delta{=}6\log(3/2){>}0$ (v56) — the core is a fixed point, $\rho{\to}\infty$ replaced by $\varphi{\to}\varphi_*$ [C]; [E] maximal BH = anchor: Nariai cubic $t^3{-}3t{+}2{=}(t{-}1)^2(t{+}2)$, roots $(1,1,-2)$ = the traceless projection of $a{=}(1,1,2)$ (v101/v190); [E] information stays in the boundary field: Page recovery decays at $\lambda_2{=}(2/3)^6$ per step (v54) [C]; [E] holographic remnant $S_{BH}{=}A/4G$, $1/4{=}1/\lvert\mu_4\rvert$ [C]; [O] HONEST: a qualitative structural synthesis (reuses v54/v56/v57/v101/v190), *not* load-bearing, *not* a metric [E][O] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v210_mark_local_dtn.py` | A numerical *sharpening* of v201/`QGEO.SUBPRIN.01` on *realistic* Steklov curvature profiles, carried to the quasi-free *state* level (the actual $\omega\circ\rho{=}\omega$, not just $[\rho,\Lambda]{=}0$); *not* a closure of `QGEO.SYM.01`. A real von-Mises bump $g(\theta)$ at a puncture, summed over the four $\mu_4$ marks $f(\theta){=}\sum_{j=0}^{3} g(\theta{-}j\pi/2)$, builds the Steklov DtN $\Lambda{=}|D_\theta|{+}M_f$. [E] every off-(mod 4) Fourier coefficient of the real $f$ vanishes to $< 10^{-16}$ (exact 4-fold cancellation), although $g$ has all modes. [C] $\|[\rho,\Lambda]\| < 10^{-15}$. [C] *state invariance* (new vs v201): the quasi-free covariance $C{=}\frac{1}{2}(1{+}\operatorname{sgn} H_1)$ satisfies $\rho C\rho^\dagger{=}C$ to $< 10^{-13}$ — the state, not just the operator, is clock-invariant. [C] negative controls (single off-centre bump, $\mathbb{Z}_3$ source, 4 generic marks) break both by $O(1)$; convergence stable for $N{\in}\{16,32,64\}$. [O] certifies 'mark-local DtN $\Rightarrow$ $\omega\circ\rho{=}\omega$' for concrete profiles *and* at the state level; the premise that the physical seam *is* mark-local stays [O] — the narrowest definitional form of `QGEO.SYM.01`, not its closure [C][O] |
| `v211_axion_spine_angle.py` | Axion DM, the spine-angle branch $\theta_i = \pi N_{\rm fam}/g_{\rm car} = 3\pi/5$ (108°): an alternative, more robust misalignment angle than the determinant-line hilltop $\theta_i \approx 170°$ (v185, which over-produces $\Omega_a h^2 \approx 0.66$). [E] rational motive $\theta_i = 3\pi/5$ exactly (spine quotient 3/5, no fit); [E] the v185 finite-T solver ($\theta{=}1 \to 0.03$) reaches $\Omega_{\rm DM}$ at $\theta \approx 106°$, so 108° is on target (harmonic $0.03\theta^2{=}0.107$); [C] *robust*: 108° is 62° below the 170° hilltop (mild-anharmonic, not exponentially sensitive); [O] an alternative ansatz to $\theta_i{=}\pi(1{-}\varphi_{\rm seam}){\approx}170°$ (mutually exclusive, the full solver decides, `DM.AXION.SPINE.01`); [X] kill test: $\Omega_a h^2$ outside $\sim[0.08, 0.16]$ demotes the branch [E][C][O][X] |
| `v212_leptogenesis_decuple.py` | Leptogenesis, the scalaron-decuple branch: both Boltzmann inputs share the decuple $A_\Lambda{=}10{=}|E(K_5)|$ — $M_1{=}M_{\rm scal}\varphi_0^2/A_\Lambda \approx 8.65{\times}10^9$ GeV and $\tilde m_1{=}m_3/A_\Lambda \approx 5$ meV. A cleaner [C] route than v184's $M_1{=}M_R\varphi_0^4$ (which only relocated the free input to the un-pinned seesaw $M_R$). [E] shared decuple $A_\Lambda{=}10$; [E] $M_1{=}8.65{\times}10^9$ GeV inside the thermal window $[3{\times}10^9, 3{\times}10^{10}]$ (v169); [E] $\tilde m_1 \approx 5$ meV, $K \approx 4.6$ strong washout; [E] Davidson–Ibarra $\varepsilon_1 \approx 8.5{\times}10^{-7}$, $\eta_B \approx 1.2{\times}10^{-9}$ brackets observed $6.1{\times}10^{-10}$; [C] HONEST: $M_1$ uses only $\{M_{\rm scal}, \varphi_0, A_\Lambda\}$ but the coupling mechanism is posited not derived; [X] kill test falls the route, not the theory (`FR.ETAB.01` independent) [E][C][X] |
| `v213_ftransfer_functor.py` | The `F_transfer` *functor contract* (`CONTRACT.F.01`): the four frontier transfers are ONE typed functor $F_{\rm transfer}{=}F_{\rm observable}\circ F_{\rm threshold}\circ F_{\rm RG}$ with four structural axioms, each instance a discrete compiler kernel [E] composed with an external continuous solver [C]; consolidates v183 ($F_{\rm pole}$), v212 ($F_{\rm Boltzmann}$), v211 ($F_{\rm relic}$), v164/`FR.MPME` ($F_{\rm QCD}$), the structural complement of the v187 typing guard. [E] axiom 1 $\mu_4$-deck equivariance ($\lambda_2{=}(2/3)^6{=}64/729$ the deck transfer eigenvalue); [E] axiom 2 Plücker preservation ($53{=}a^T(R{+}Q)\mathbf{1}$, $\mathrm{Pl}(F){=}(1,22,30)$, $\|\mathrm{Pl}(K)\|_1{=}11$); [E] axiom 3 positivity/stochasticity ($\operatorname{spec} T{=}\{1,(2/3)^6,(1/3)^6\}$ positive, $\kappa_f{\in}(0,1)$); [E] axiom 4 external modules explicit ($b_3{=}{-}7$ a carrier output, $|b_3|{=}\Omega_{\rm adm}{-}10b_1{=}7$); [C] verdict: a typed functor, not a bag of open topics; an architectural consolidation guarded by v187, not a closure [E][C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v214_seam_pillowcase.py` | The *pillowcase* reduction of `QGEO.SYM.01` (`QGEO.PILLOW.01`): the two separately-tracked open QGEO residuals — v180/`QGEO.ISO.01` (the carrier clock is an order-4 isometry of the seam metric) and v201/`QGEO.SUBPRIN.01` (the DtN sub-principal symbol is mark-local, the seam flat away from the $\mu_4$ marks) — are ONE statement, 'the seam carries the uniformising flat orbifold (pillowcase) metric', and the order-4 clock is *derived* from cross-ratio 2 (v168), not assumed. **[E]** $\chi_{\text{orb}}(S^2(2,2,2,2))=2-4(1-\frac{1}{2})=0 \Rightarrow$ flat (Troyanov cone-metric uniformisation, unique up to scale; Gauss–Bonnet four deficits $\pi$ sum to $4\pi=2\pi\chi(S^2) \Rightarrow \int K_{\text{smooth}}=0$, flat away from the marks). **[E]** NEW LINK: cross-ratio($\mu_4$)=2 (v168) $\Rightarrow j(\lambda)=256(\lambda^2-\lambda+1)^3/(\lambda^2(\lambda-1)^2)$ with $j(2)=1728$ (all six harmonic cross-ratios $\{2,-1,\frac{1}{2}\}$ give 1728) $\Rightarrow$ the square modulus, hence the order-4 clock, is a *consequence* of cross-ratio 2. **[E]** $j=1728 \Leftrightarrow \tau=i$ (square torus) $\Leftrightarrow$ CM by $\mathbb{Z}[i] \Leftrightarrow$ Aut$=\mathbb{Z}/4$ ($j=0$ gives $\mathbb{Z}/6$; generic $\mathbb{Z}/2$). **[E]** explicit CM $(x,y) \mapsto (-x,iy)$ on $y^2=x^3-x$ has order $4=z \mapsto iz$. **[E]** negative control $\lambda=3 \Rightarrow j=21952/9 \notin \{0,1728\} \Rightarrow$ Aut$=\mathbb{Z}/2$, no clock. **[E]** UNIFICATION: 'seam = uniformising flat pillowcase metric' implies both v201 mark-locality and v180 `QGEO.ISO.01`. **[O]** does *not* close `QGEO.SYM.01` — the canonical constant-curvature-metric premise stays open, milder/more canonical; net existence + full-cone RP remain v175's **[E]** |
| `v215_seam_deck_killtest.py` | The bedrock `QGEO.SYM.01` ($\omega\circ\rho=\omega$) recast as a *falsifiable* kill-test (`QGEO.KILL.01`), not a proof-promise — a foundational symmetry cannot be derived from nothing (v181/v153), and Bisognano–Wichmann would presuppose the covariance it should produce. Non-circular reduction (v198/v199/v201): $\omega\circ\rho=\omega \Leftrightarrow [\rho,C]=0$; the principal symbol $\lvert k\rvert$ commutes exactly, so the residual is the sub-principal $M_f$ with $[\rho,M_f]=0 \Leftrightarrow f$ $\mathbb{Z}_4$-invariant; measure form = the raw Gaussian bulk form $A$ is $\mu_4$-deck-invariant. Four levers (carrier-side **[E]**, seam-side **[O]**): **K1** exactly 4 marks (order-4 clock free orbit $\mu_4$ + $b_1=\#\text{marks}-1=3=N_{\text{fam}} \Rightarrow \#\text{marks}=4=\lvert\mu_4\rvert$); **K2** square config (cross-ratio $2 \Rightarrow j=1728$, Aut$=\mathbb{Z}/4$; order 4 selects $\mathbb{Z}/4$ not the $j=0$ $\mathbb{Z}/6$; generic $\Rightarrow \mathbb{Z}/2$); **K3** mod-4 sub-principal support ($\mu_4$-mark sum $\Rightarrow f_m=0$ for $m \not\equiv 0 \bmod 4 \Rightarrow [\rho,M_f]=0$); **K4** transfer gap $(2/3)^6=64/729$ (cusp weights $\{0,\frac{1}{3},\frac{2}{3}\}$). Freeze-bound (`freeze_file.csv seam_deck_symmetry`). **[O]** the decisive kill is *deferred* (`QGEO.REALIZE.01`): the raw $\mathbb{P}\setminus\mu_4$ collar DtN must yield 4 square marks *without* inputting them; carrier-side K1–K4 are green regression guards. A kill-test, not a closure |
| `v216_marks_gauss_bonnet.py` | The four seam marks *emerge* from Gauss–Bonnet (`QGEO.MARKS.03`), executing the K1 lever of the v215 kill-test. **[E]** mark count $= 2\chi$: $\mathbb{Z}_2$ branch points (deficit $\pi$) on a flat sphere ($\chi=2$) give $n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 = \lvert\mu_4\rvert = e_1(a) = N_{\text{fam}}+1$ (the same $\chi=2$ as the 8 in $c_3=1/(8\pi)$). **[E]** the Euclidean sphere 2-orbifolds ($\chi_{\text{orb}}=0$) are exactly $(2,3,6),(2,4,4),(3,3,3),(2,2,2,2)$. **[E]** all-order-2 (the $\lvert\mathbb{Z}_2\rvert$ sheet branch) $\Rightarrow$ uniquely $(2,2,2,2)$; **[E]** $N_{\text{fam}}=3$ (rank $H^1=\#\text{marks}-1$) $\Rightarrow$ uniquely $(2,2,2,2)$, picking the 4-mark square over the 3-mark hexagonal $(3,3,3)$; **[E]** a free order-4 deck $\Rightarrow$ uniquely $(2,2,2,2)$ — three independent selectors converge. **[O]** residual: the square modulus (cross-ratio $2 \Rightarrow j=1728$, v214) needs the one order-4 carrier input. The mark count + equality are now derived; does *not* close `QGEO.REALIZE.01` |

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| Script | What it machine-checks *(continued)* |
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| `v217_marks_emergence_scan.py` | The *numerical* emergence scan (`QGEO.EMERGE.01`, the sibling of v216 as v210 is to v201): with the number of marks $n$ *and* their positions free, the 4-equally-spaced (square) configuration is the joint minimiser of {clock-invariance + 3-family cohomology + transfer gap $(2/3)^6$} on the actual seam DtN/state. [C] $n$=4 equally-spaced $\Rightarrow \|\rho_4 C \rho_4^\dagger - C\| \sim 10^{-14}$ (clock-invariant state); [C] family count selects $n$=4 (scan $n \in \{2..8\}$, only $n-1=3=N_{\mathrm{fam}}$; $n$=3 → 2 families = hexagonal, excluded; $n$=5 → 4); [C] gap: $\mu_4$-equivariant ⇒ diagonal in the cusp basis (deck-average off-diagonal $\sim 10^{-16}$), cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ ⇒ transfer $(1-w)^6=\{1, (2/3)^6, (1/3)^6\}$, gap $(2/3)^6$=64/729; [C] position scan: the square minimises the defect (square $\sim 10^{-14}$ vs off-square $\sim 0.4$); [C] negative controls $n$=3, 5, 6. [O] confirms the emergence on the real DtN (number scanned, positions free); selector = the 3-family/gap structure (carrier input); executes K1 numerically, does *not* close `QGEO.REALIZE.01`. Python-only |
| `v218_diamond_axis_geometry.py` | Diamond axis geometry: the Sheet Diamond (v94) is a discrete geometry with two axes around the centered cross (v95: $C$ center, $U$ winding, $V$ sheet), $F$ the transfer completion. Three exact [E] lemmas + honest audit/heuristic typing, no new numbers. [E] AXIS CURVATURE: $\det(C+xU)=14+6x$ linear (slope $6=\|R^+(A_3)\|$), $\det(C+yV)=14+14y+4y^2$ quadratic, 2nd difference 8=rank $E_8$; $\det B(C+xU)=2\det(C+xU)$, anchor-block sheet curvature $6=\|R^+(A_3)\|$. [E] PLÜCKER LADDER: $\mathrm{Pl}(K)=(-1,6,4) \to \mathrm{Pl}(C)=(0,14,14) \to \mathrm{Pl}(F)=(1,22,30)$, steps $(1,8,10)=(N_\Phi, \mathrm{rank}\, E_8, A_\Lambda)$ and $(1,8,16)=(N_\Phi, \mathrm{rank}\, E_8, \dim S^+)$ (identity [E], transfer reading [C]). [E] SPECTRAL RAMIFICATION: $\mathrm{Disc}(\chi)=q(r)^2\mathrm{Disc}(q)$ for $Q, K, C, F$; squares $(1,3,4,6)$, kernels $(13, 48, 65, 105)$ with $48=\Omega_{\mathrm{adm}}$, $105=N_{\mathrm{fam}} g_{\mathrm{car}} \cdot 7$. [C] heuristic: $F$ the transfer-completion corner, a search principle, not a hard filter; the $G_2/F_4$ pair-sum labels stay audit-only (cf. v94/v95) [E][C] |
| `v219_icosahedral_mckay.py` | McKay bedrock: *why* the atoms are $\{2,3,5\}$. $E_8$ is the exceptional *top* of the McKay tower of finite $SU(2)$ subgroups ($2T \to \hat{E}_6$, $2O \to \hat{E}_7$, $2I \to \hat{E}_8$). The McKay graph is built *from the group*: the 120 icosians close to the binary icosahedral $2I$ (element orders $\{1,2,3,4,5,6,10\}$, containing the 2, 3, 5 axis orders), 9 conjugacy classes, Dixon irrep degrees $\{1,2,2,3,3,4,4,5,6\}$ (sum $30=h(E_8)$, sum of squares $120=\|R^+(E_8)\|$); the natural-rep McKay adjacency *is* the affine $E_8$ graph with Kac marks = the degrees. A backward certificate of the closed $E_8$ [E], the reading that it explains $(2,3,5)$ [C], the raw-seam origin [O] |
| `v220_cp_hexagonal_modulus.py` | CP lives in the *hexagonal* phase fiber, not the square seam deck (a geometric sharpening of red-team Target D). [E] the seam deck is the square modulus $j(i)$=1728, Aut=$\mathbb{Z}/4$ (the $\mu_4$ clock); its exceptional partner is the hexagonal modulus $j(\rho)$=0, $\rho=e^{i\pi/3}$, Aut=$\mathbb{Z}/6$, CM by $\mathbb{Z}[\omega]$, $\arg\rho=\pi/3$. [E] the frozen leading CKM reading $\delta=\pi/3+3\lambda_C^2=68.654°$ (v88) has leading term $\pi/3=\arg\rho$; the $\|\mathbb{Z}_2\|$ alternative $\pi/3+2\lambda_C^2$ is the frozen look-elsewhere trap. The deck stays $\mathbb{Z}/4$ (v215); CP is the $\mathbb{Z}/6$ fiber, the bridge [C][E][C] |
| `v221_seam_qecc.py` | The seam as a *finite recoverability* code (recovery rate $(2/3)^6$=64/729). [E] code dimension $\dim S^+=16=2^{g_{\mathrm{car}}-1}$; the gapped transport $T$ on the cusp-weight 3-space (deviations $(1,-1,0)$ and the Nariai anchor $(1,1,-2)$) is symmetric, doubly stochastic (CPTP), spectrum $\{1, (2/3)^6, (1/3)^6\}$; on the trace-zero subspace $\|T\delta\| \le (2/3)^6\|\delta\|$ hence $\|T^n\delta\| \le (2/3)^{6n}\|\delta\|$ (a Knill–Laflamme-type bound); a free ratio breaks it. [C] the holographic/Petz reading (gated on the Seam–Horizon theorem) [E][C] |
| `v222_cm_norm_duality.py` | CM-norm duality: the two exceptional moduli give 41 and 7. [E] SQUARE (Gaussian $\mathbb{Z}[i]$, $j$=1728): $N(g_{\mathrm{car}}+i\|\mu_4\|)=5^2+4^2=41=10b_1$ — the EM index *is* the Gaussian norm of the carrier-glue vector; $N(3+2i)=13=\Delta_Q$. [E] HEX (Eisenstein $\mathbb{Z}[\omega]$, $j$=0): $N(3+2\omega)=3^2-3\cdot2+2^2=7$= scalaron. The $(3,2)$=(colour,weak) split generates $(5,6,7,13)$ by sum / product / Eisenstein / Gauss norm; the rings are rigid (Eisenstein of $(5,4)=21\neq41$). The architecture bridge square↔EM, hex↔scalaron is [C][E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v223_coxeter_totative_clock.py` | The $\mu_4$ clock is the order-4 character of the $E_8$ Coxeter cycle. [E] $(\mathbb{Z}/30)^\times=\{1,7,11,13,17,19,23,29\}=$ the $E_8$ exponents = live Coxeter phases; $\varphi(30)=8=\operatorname{rank} E_8$, $30=2\cdot3\cdot5$. [E] $(\mathbb{Z}/30)^\times\cong\mathbb{Z}/4\times\mathbb{Z}/2$ with order-4 generator 7 ($7^2=19$, $7^4=1 \bmod 30$): a literal $\mu_4$ clock $\langle7\rangle=\{1,7,13,19\}$ on the exponents; the conjugate pairs $m+(30-m)=30$ give $\lvert\mu_4\rvert=4$ invariant planes the clock permutes. Only $E_8$ has $\varphi(h)=\operatorname{rank}$ *and* $h=2\cdot3\cdot5$; the seam-clock identification is [C][E][C] |
| `v224_diamond_ftransfer_path.py` | $F_{\text{transfer}}$ as *one* path on the Sheet Diamond, not four external interfaces. [E] family $M(s,t)=R+Q\,\mathrm{diag}(s,t,t)$: the corners $K=M(1,-1), C=M(1,0), F=M(1,1)$ lie on the sheet axis ($s=1$); the sheet axis is curved $\det M(1,t)=4t^2+14t+14$ (2nd diff $8=\operatorname{rank} E_8$), the winding axis $\det M(s,0)=6s+8$ flat (slope $6=\lvert R^+(A_3)\rvert$); Plücker steps $(1,8,10),(1,8,16)$. [C] $F_{\text{transfer}}$ is the $K\to C\to F$ sheet geodesic, Koide source→pole its 1-D Möbius projection (v37/v183) [E][C] |
| `v225_dual_normal_frame.py` | The dual normal frame $(d,n)$: one boundary compass for flavor and horizon, with CP as orientation. [E] $d=a^\top R^{-1}=a^\top L^{-1}=(-\frac12,-\frac12,1)=-\frac12(1,1,-2)$ (the Nariai/horizon normal; $d\cdot\mathbf{1}=0$, $d\cdot a=1$); $n=(5,-9,6)$ with $n^\top R=(\det R,0,0)=(8,0,0)$, $n^\top L=(20,0,0)$ (the first-generation/torsion selector). [E] oriented volume $\det(\mathbf{1},d,n)=21=N_{\text{fam}}\cdot7$. [C] CP as orientation: $\mathrm{Im}\det(\mathbf{1},d,\rho n)=21\sin(\pi/3)$ for $\rho=e^{i\pi/3}$ — CP sits where the magnitude bijection (Target D) fails [E][C] |
| `v227_degree_exponent_channel_split.py` | $248=120+128$ as a magnitude/phase channel typing (replaces the over-strong $S^-$-dark-matter reading). [E] $E_8$ exponents sum $120=\lvert R^+(E_8)\rvert$ (*magnitude* channel); degrees $\{2,8,12,14,18,20,24,30\}$ sum $128=2^{\text{rank}-1}=\operatorname{rank}\cdot\dim S^+$ (*phase/glue* channel); $248=120+128=\mathrm{adj}(D_8)+\mathrm{spinor}(D_8)$; $\sum\deg-\sum\exp=\operatorname{rank}=8$. [C] ratios+products (Target D) live in the 120 channel, the un-covered CP phases in the 128 phase/glue channel; [O] a dark-sector reading of 128 stays Frontier [E][C][O] |
| `v228_rr_index_gate.py` | P2 as the mode space of a degree-4 seam divisor (a Riemann–Roch *index gate*; bedrock language for P2, not a proof from nothing). [E] $\deg D=4=\lvert\mu_4\rvert$ on $\mathbb{P}^1\Rightarrow h^0(\mathcal{O}(D))=5=g_{\text{car}}$ (Riemann–Roch, genus 0); $\operatorname{rank} H_1(\mathbb{P}^1\backslash4)=3=N_{\text{fam}}$; $h^0+\operatorname{rank} H_1=8=\operatorname{rank} E_8$, $h^0-\operatorname{rank} H_1=2=\lvert\mathbb{Z}_2\rvert$; $\Lambda^{\text{even}}(\mathbb{C}^5)=1+10+5=16=\dim S^+$. The mode-space reading is [C], conditional on the raw seam producing the degree-4 divisor ([O], = QGEO.SYM.01) [E][C][O] |
| `v229_lepton_frobenius_algebra.py` | The charged leptons as an étale Frobenius algebra over the $\mu_6$ resolvent. [E] exact coefficients $(c_e,c_\mu,c_\tau)=(\frac{16}{7},\frac43,\frac76)$, ring closure $c_ec_\tau=\frac83=\lvert\mathbb{Z}_2\rvert c_\mu$, product $\frac{32}{9}=2^{g_{\text{car}}}/N_{\text{fam}}^2$. [E] $A=\mathbb{Q}[t]/(m)$, $m=\prod(t-c)$: the trace-form Gram is nondegenerate ($\det=\mathrm{disc}(m)\neq0$, $A$ étale) — a commutative Frobenius algebra; the $C_6$ shift has characteristic polynomial $t^6-1$ (spectrum $\mu_6=\mu_3$ family $\times\,\mu_2$ sheet). [C] the PMNS extension via $\mathrm{Aut}(A)$ and the hexagonal CM point [E][C] |
| `v230_center_budget_norms.py` | The Sheet-Diamond center budget $(7,11,13)$ is the three *local norms* of the theory. [E] $C=R+Q\,\mathrm{diag}(1,0,0)$, row sums $(7,11,13)$: $7=N_{\mathbb{Z}[\omega]}(3+2\omega)$ (Eisenstein/hex = scalaron), $13=N_{\mathbb{Z}[i]}(3+2i)$ (Gauss/square = $\Delta_Q$), $11=\binom40+\binom41+\binom42=1+4+6$ (QBL boundary Fock count on the 4 $\mu_4$ corners = $\dim S^+-g_{\text{car}}=16-5$). The two CM rings (v222) and the seam boundary count meet in one operator center; the reading is [C][E][C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v231_cp_mu6_phases.py` | Both CP phases are $\mu_6$ powers of *one* hexagonal CM unit, split by the sheet — a structural reduction of red-team Target D (sharper than v220/v225). With $\rho{=}e^{i\pi/3}$ (the $j{=}0$ Eisenstein CM unit): [E] $\rho$ is a primitive 6th root ($\rho^6{=}1$, $\rho^3{=}{-}1$ the $\mathbb{Z}_2$ sheet half-turn); [E] $\delta_{\mathrm{CKM}}$ leading $=\arg(\rho^1){=}\pi/3$ (quark), $\delta_{\mathrm{PMNS}}=\arg(\rho^4){=}4\pi/3$ (lepton, the *phase lattice*); [E] $\rho^4{=}{-}\rho \Rightarrow \delta_{\mathrm{PMNS}}{=}\delta_{\mathrm{CKM,lead}}{+}\pi{=}$(CM unit)$\times$($\mathbb{Z}_2$ sheet) — the two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; [E] the $C_6/\mu_6$ monodromy has charpoly $t^6{-}1$ (spectrum $\mu_6$ carries $\rho$); [E] orientation $\mathrm{Im}\det(\mathbf{1},d,\rho^1 n){=}{+}21\sin(\pi/3)$, $\mathrm{Im}\det(\mathbf{1},d,\rho^4 n){=}{-}21\sin(\pi/3)$ (sheet-flipped Jarlskog, v225); [E] NEG $\mu_3$ has no $-1$ (no sheet; $\mu_6{=}\mu_3$ family $\times\,\mu_2$ sheet). REDUCTION: 2 CP phase inputs $\to$ 1 hexagonal unit + the sheet; the physical CP derivation stays [C], the deck stays $\mathbb{Z}/4$ (v215) [E][C] |
| `v232_e8_kleinian_seam.py` | The seam as the $E_8$ *Kleinian singularity* — a canonical algebraic-geometric model for the open seam-realisation premise `QGEO.REALIZE.01` (the du Val side of the McKay correspondence, v219). [E] the McKay graph of $2I$ is affine $E_8$; dropping the unique trivial-rep node (the single Kac mark 1) leaves the *finite* $E_8$ Dynkin on 8 nodes = the dual intersection graph of the eight exceptional $\mathbb{P}^1$'s of the minimal resolution of $\mathbb{C}^2/2I$ (trivalent node, arms $1,2,4$). [E] eight exceptional $\mathbb{P}^1$'s $=\mathrm{rank}\,E_8 = g_{\mathrm{car}}{+}N_{\mathrm{fam}} = \#$ nontrivial $2I$ irreps — a fourth reading of the 8 in $c_3{=}1/(8\pi)$. [E] the negated intersection form is the $E_8$ Cartan (even, $\det{=}1$, positive definite; each $\mathbb{P}^1$ a $(-2)$-curve). [E] one degree-1 irrep $\Rightarrow$ $2I$ perfect $\Rightarrow$ $H_1(S^3/2I){=}0$: the link is the Poincaré homology 3-sphere, $\pi_1{=}2I$ of order $120{=}|R^+(E_8)|$. [C] seam reading: the raw RP seam ($\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for `QGEO.REALIZE.01`, not a P2 proof (presupposes $E_8$); bedrock [O][E][C][O] |
| `v233_cp_triality_phase.py` | The CP phase is the *universal* family/triality phase, only sheet-split — one step past v231 in $\mu_6{=}\mu_3$ (family/triality) $\times\,\mu_2$ (sheet). [E] $\mu_6{=}\mu_3{\times}\mu_2$: the $\mathbb{Z}_3$ triality centre $\{1,\omega,\omega^2\}$ = the cusp weights $\{0,\frac{1}{3},\frac{2}{3}\}{\times}2\pi$, the sheet $\mu_2{=}\{1,-1\}$; $\rho{=}\omega^2{\cdot}(-1)$. [E] SAME FAMILY CLASS: $\rho^4{=}\rho^1{\cdot}\rho^3$ with $\rho^3{\in}\mu_2$, so the quark ($\rho^1$) and lepton ($\rho^4$) CP phases have the *same* image in $\mu_6/\mu_2{\cong}\mu_3$ and *opposite* sheet — the CP phase *is* the universal triality phase $\omega^2$ (the 2/3 cusp-weight $\mathbb{Z}_3$ centre), not a free power choice. [E] $\omega$ is the $\mu_3$ factor of the order-30 $(2,3,5)$ Milnor/Coxeter monodromy that builds $E_8$ (v232). Reduces Target D from 'one hexagonal unit + sheet' (v231) to 'the universal triality phase + sheet'; the physical CP derivation stays [C][E][C] |
| `v234_seam_holomorphy_selection.py` | The Seam-Holomorphy selection certificate: the entire structural residual is *one* condition with *three* equivalent faces, all forcing $E_8$. The condition: *the seam carries no nontrivial abelian sector.* [E] (i) AQFT — the boundary net is holomorphic ($\mu$-index 1); (ii) geometry — the seam link is a homology 3-sphere $\Leftrightarrow$ the binary group is perfect $\Leftrightarrow$ $2I$; (iii) rep theory — exactly one 1-dim irrep. [E] $\#$(mark-1 affine nodes) $=|\Gamma^{\mathrm{ab}}|=|H_1(S^3/\Gamma)|=\#$(1-dim irreps): $A_n{\to}n{+}1$, $D_n{\to}4$, $E_6{\to}3$, $E_7{\to}2$, $E_8{\to}1$ — only $E_8$ gives 1 ($2I$ the unique perfect $SU(2)$ subgroup, Poincaré the unique homology-sphere space form). [E] holomorphic $c{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}} \Rightarrow$ the unique even unimodular rank-8 lattice $=E_8$. [E] the three faces (v83/v154, v232, v219) are the SAME $E_8$-selector, so Target A, $G_{\mathrm{net}}$, the carrier P2 and the Kleinian seam are *one* gate. [O] the closing theorem (stated): RP+gap+chirality $\Rightarrow$ quasi-free bulk (v160) $\Rightarrow$ invertible bulk $\Rightarrow$ holomorphic boundary $\Rightarrow$ $E_8$; the single residual analytic step is 'free bulk $\Rightarrow$ holomorphic boundary' [E][O] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v235_seam_chern_simons.py` | The closing step in abelian Chern–Simons language: holomorphic ⇔ $\det K{=}1$ — a genuine reduction (not a closure) of `GATE.HOLO.01` into standard anyon-condensation physics. [E] a free gapped bosonic 2+1d bulk = an even integer $K$-matrix with #anyons = $\lvert\det K\rvert$, edge $c$ = signature($K$), holomorphic ⇔ $\lvert\det K\rvert{=}1$. The TFPT extension tower (v92) *is* the condensation tower read by determinant: carrier $D_5{\oplus}A_3$ ($\det 16$, 16 anyons) → $SO(16)_1{=}D_8$ ($\det 4$) → $(E_8)_1$ ($\det 1$, holomorphic), all rank 8, $c$ = sig$=8=g_{\rm car}{+}N_{\rm fam}$. [E] holomorphic ⇔ $\det 1$ ⇔ $E_8$ = the Kitaev $E_8$ quantum-Hall state. [E] condensation: a Lagrangian glue of order $\sqrt{16}{=}\lvert\mu_4\rvert{=}4$ takes $\det 16{\to}1$ (a partial order-2 isotropic only → $4{=}D_8$). [E] NEG: $D_8$ is free+bosonic but has 4 anyons; an odd $K$ is fermionic — $\det{=}1$ is the extra condition. [O] residual: 'the free RP seam condenses the order-$\lvert\mu_4\rvert$ Lagrangian glue' = the sheet/QGEO selection, now with holomorphy⇔ $\det 1$ a theorem [E][O] |
| `v236_brieskorn_capstone.py` | The $(2,3,5)$ Brieskorn singularity $x^2{+}y^3{+}z^5$ is the *one* generator of the discrete skeleton — the capstone of the icosahedral round (v55/v219/v223/v232/v233); its exponents *are* the atoms $(\lvert\mathbb{Z}_2\rvert, N_{\rm fam}, g_{\rm car}){=}(2,3,5)$. [E] Milnor number $\mu{=}(2{-}1)(3{-}1)(5{-}1){=}1{\cdot}2{\cdot}4{=}8{=}\operatorname{rank}E_8$ = the 8 in $c_3{=}1/(8\pi)$ (a fifth independent origin). [E] the Milnor monodromy eigenvalues are the eight primitive 30th roots $e^{2\pi i m/30}$, $m$ the $E_8$ exponents (charpoly $\Phi_{30}$, $\deg\varphi(30){=}8{=}\mu$) — so it *is* the order-30 $E_8$ Coxeter element (v55). [E] both clocks: the monodromy group $\langle h\rangle{=}\mathbb{Z}/30{=}\mathbb{Z}/2{\times}\mathbb{Z}/3{\times}\mathbb{Z}/5$ carries the $\mu_3$ triality (CP, v233) as $h^{10}$; the Galois group $(\mathbb{Z}/30)^\times{=}\mathbb{Z}/2{\times}\mathbb{Z}/4$ carries the $\mu_4$ clock (×7, v223). [E] Milnor lattice = $E_8$, link = Poincaré sphere. [C] one singularity generates atoms + 8 + $E_8$ + order-30 + both clocks + seam; a unification, not a closure [E][C] |
| `v237_seam_sre_closure.py` | The closing step as a *physical* condition: no topological ground-state degeneracy ⇔ $\det K{=}1$ ⇔ the seam bulk is short-range-entangled (SRE/invertible) — sharpening `GATE.HOLO.02`'s residual from definitional to falsifiable. [E] the $K$-matrix ground-state degeneracy on a genus-$g$ surface is $\lvert\det K\rvert^g$ (carrier → 16, $D_8$ → 4, $E_8$ → 1 on the torus), so 'no degeneracy on any closed surface' ⇔ $\det K{=}1$. [E] among rank-8 even positive-definite $K$ ($c{=}8{=}g_{\rm car}{+}N_{\rm fam}$) only $E_8$ has $\det 1$ — the unique nontrivial bosonic SRE 2+1d phase (Kitaev $E_8$), edge holomorphic; $D_8$/carrier are LRE. [E] a unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$, the torus does), so the residual is 'the seam unique-vacuum extends to the torus' = SRE. [E] NEG: an LRE bulk ($D_8$) has 4-fold torus degeneracy + non-holomorphic boundary. [O] sharpened residual: 'the free RP gapped seam is short-range-entangled (no topological order)' = $\det K{=}1$ = the Kitaev $E_8$ phase — a yes/no statement, sharper than v235/`QGEO.SYM.01`; not a closure [E][O] |
| `v238_modular_lindblad_dynamics.py` | `DYN.SEMIGROUP.01` — the static→dynamic flip: the seam's modular flow (reversible) and its recovery channel as a Markov/Lindblad generator (irreversible), the dissipative gap = the OS mass gap. Companion to v196/v198 (modular) and v221/v64 (recovery), which state the structure *statically*; this reads the same objects *dynamically*. [E] the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) and the $\mu_4$ clock is conserved ⇔ $[\rho,\Lambda_\Sigma]=0$ (the v198 state-invariance), so the static commutator is the conservation law of modular time. [E] $L=\log T$ is a doubly-stochastic generator ($e^L=T$, $e^{L\tau}\to$ the democratic projector = the arrow of time, cf. v64). [E] keystone: the dissipative gap $=-\log((2/3)^6)=6\log(3/2)$ = the OS mass gap $\Delta$ (v64) — the static recovery bound and the dynamical mass gap are one number, one exponentiated. [E] GKSL split (imaginary + real$\le 0$ spectrum). [O] geometric modular flow on the full seam algebra (Connes–Rovelli thermal time = `QGEO.SYM.01`, v198) — a target, not a closure. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v239_kms_thermal_time.py` | `DYN.KMS.01` — KMS / thermal time: the modular flow of v238 makes the seam state a KMS (thermal) state at $\beta = 1$, and the geometric normalisation fixes the seam temperature via $2\pi = 1/(4c_3)$. [E] KMS at $\beta = 1$: $\omega(A\,\sigma_i(B)) = \omega(BA)$ ($\omega(X) = \mathrm{Tr}(\rho X)$, $\rho = e^{-\Lambda_\Sigma}/Z$), so modular time (v238) is a thermal time (Connes–Rovelli); NEG $\beta \neq 1$ fails (Tomita–Takesaki pins $\beta = 1$). [E] the seam unit $\beta_{\mathrm{phys}} = 2\pi = 1/(4c_3)$ (v58) makes the KMS temperature the horizon temperature $T_H = c_3/M$ (v57/v208). [E] one object with v238. [O] the geometric-boost identification = `QGEO.SYM.01` — a target, not a closure. |
| `v240_gns_os_reconstruction.py` | `DYN.GNS.01` — GNS / OS reconstruction (finite toy): the Hilbert space and a POSITIVE Hamiltonian are OUTPUTS of the seam data $(M, \omega)$, not inputs. [E] GNS: $\langle A, B\rangle = \mathrm{Tr}(\rho A^\dagger B)$ Hermitian positive-definite ($\rho > 0$), $\dim n^2$, identity cyclic+separating. [E] modular flow $\sigma_t(x) = \rho^{it} x \rho^{-it}$ is a state-preserving $*$-automorphism, generator $i[\log\rho, \cdot] = \Lambda_\Sigma$ (v238). [E] OS Hamiltonian $H_{\mathrm{OS}} = -\log T$ (the v221 transfer) is POSITIVE — eigenvalues $\{0, 6\log(3/2), 6\log 3\} \geq 0$, gap $= 6\log(3/2)$ = the v64 mass gap. [E] $H_{\mathrm{OS}} = -L$: minus the v238 dissipative generator (recovery semigroup = Euclidean time). [O] the geometric/operator-level step = `QGEO.SYM.01`/BW + the standard OS theorem under RP/gap — a target, not a closure. |
| `v241_dhr_sectors.py` | `DHR.SECTORS.01` — particles = DHR superselection sectors: the carrier discriminant/anyon content and the v235/v237 condensation tower as an abelian MTC. Carrier form (Lean-verified, `FORM.GLUE.01`): on $\mathbb{Z}_4 \times \mathbb{Z}_4$, $q(x,y) = (5x^2{+}3y^2)/8 \bmod 1$, $\theta(a) = e^{2\pi i q(a)}$. [E] 16 sectors = $\mathbb{Z}_4 \times \mathbb{Z}_4$ ($= \det K_{\mathrm{carrier}} = \dim S^+$), fusion = abelian group law. [E] condensable $\Leftrightarrow \theta = 1 \Leftrightarrow$ isotropic $\Leftrightarrow 5x^2{+}3y^2 \equiv 0 \bmod 8$ — the six elements $\{(0,0),(1,1),(1,3),(2,2),(3,1),(3,3)\}$. [E] tower = v235/v237: Lagrangian $\langle(1,1)\rangle$ (order 4, self-dual) $\to$ 1 sector $= (E_8)_1$; order-2 $\langle(2,2)\rangle \to 4 = D_8$; carrier $16 \to D_8\,4 \to E_8\,1$. [E] NEG: a non-isotropic anyon $(1,0)$ is confined. [C] sector↔SM-multiplet map is the interpretation; the MTC is exact. |
| `v242_spin_statistics_modular.py` | `DHR.MODULAR.01` — spin-statistics and the modular $(S,T)$ data of the carrier MTC: statistics + the fusion ring recovered from modular data + the anyon central charge $c = 8$. Same MTC as v241 ($\mathbb{Z}_4{\times}\mathbb{Z}_4$, $q = (5x^2{+}3y^2)/8$, $\theta = e^{2\pi i q}$). [E] spin-statistics: $T = \mathrm{diag}(\theta_a) \to 6$ bosons ($\theta = 1$), 2 fermions ($\theta = -1$) $= \{(0,2),(2,0)\}$, 8 anyons. [E] modular $S_{ab} = (1/\sqrt{16})e^{-2\pi i B(a,b)}$ unitary, symmetric, $S^2 = C$ (charge conjugation), $S^4 = I$. [E] Verlinde $N_{ab}^c = \sum_x S_{ax}S_{bx}S_{cx}^*/S_{0x} = \delta(c, a{+}b)$ — fusion recovered from modular data. [E] Gauss–Milgram $(1/\sqrt{16})\sum\theta_a = e^{2\pi i c/8}$ with $c = 8 = g_{\mathrm{car}} + N_{\mathrm{fam}}$ = the $(E_8)_1$ central charge (v156/v175). [C] the boson/fermion split is the kinematic skeleton; the sector↔multiplet map is the interpretation. |
| `v243_haag_ruelle_braiding.py` | `DHR.SMATRIX.01` — scattering: the Haag–Ruelle skeleton + the braiding S-matrix. The seam net has a mass gap (v64/v238) so Haag–Ruelle constructs asymptotic states; for the abelian anyon MTC (v241/v242) the 2-particle S-matrix is the monodromy $M(a,b) = \theta(a{+}b)/(\theta(a)\theta(b)) = e^{2\pi i B(a,b)}$. [E] mass gap $\Delta = 6\log(3/2) > 0 \to$ HR in/out states. [E] 2-particle S = monodromy: $\lvert M\rvert = 1$, diagonal in conserved labels $\to$ no particle production, factorised (integrable) scattering. [E] Yang–Baxter: $M$ is a bicharacter $\to$ multi-particle factorisation. [E] crossing+symmetry: $M(a,b) = M(b,a)$, $M(a,-b) = M(a,b)^*$. [E] holomorphic point: on the condensed $(E_8)_1$ (single sector) $M = 1$, trivial braiding (det 1/SRE). [O] the 4d S-matrix / cross-sections are the open holographic uplift (Phase 3). |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v244_spectral_action_contract.py` | `CONTRACT.QFT4D.01` — the 4d Lagrangian via Connes' spectral action: the honest contract for the Phase-3 gap (2d seam net → 4d physics). A single Dirac operator whose $\mathrm{Tr}\, f(D/\Lambda)$ yields gravity AND matter. [E] carrier spinor = one generation: $SO(10)$ half-spinor = $\dim S^+ = 2^{g_{\rm car}-1} = 16$ $(15+\nu_R)$, $N_{\rm fam}=3$. [E] SM gauge group inside the carrier: $SU(3)\times SU(2)\times U(1)$ (dim 12) $\subset SU(5)$ (24) $\subset SO(10)=D_5$ (45); colour $SU(3)\subset A_3=SU(4)$ (Pati–Salam); $\mathrm{rank}(D_5)+\mathrm{rank}(A_3)=8=\mathrm{rank}\,E_8$. [E] one Dirac operator → gravity ($a_2\sim R$, $a_4\sim R^2$, v36) + matter ($a_4\sim F^2$ Yang–Mills $+|D\phi|^2$ Higgs, Gilkey). [X] hooks: $g_3=g_2=\sqrt{5/3}\,g_1$ + a Higgs-quartic relation. [O] `CONTRACT.QFT4D.01`: obligations (seam→Dirac $D$; finite geometry = carrier; spectral action reproduces SM couplings) → the 4d SM+gravity Lagrangian is an OUTPUT — the Phase-3 target, not closed. |
| `v245_spectral_matter_content.py` | `QFT4D.MATTER.01` — the computable core of `CONTRACT.QFT4D.01` (v244): the spectral-action / NCG matter content, discharging the rep-theory + tree-relation core to [E]. The carrier half-spinor is the $SO(10)$ **16**. [E] **16** = one generation: $SO(10)\,\mathbf{16}=SU(5)(\mathbf{10}+\bar{\mathbf{5}}+\mathbf{1})$; SM dims $6+3+1+2+3+1=16=\dim S^+$, $N_{\rm fam}=3$ (15 Weyl fermions $+\nu_R$). [E] electric charges $Q=T_3+Y$ are exactly the SM values — hypercharges forced by the embedding. [E] anomaly-free: $\sum Y=0$, $\sum Y^3=0$, $[SU(2)]^2U(1)=0$, $[SU(3)]^2U(1)=0$ (exact). [E] unification: $\sin^2\theta_W=(\sum T_3^2)/(\sum Q^2)=3/8$, i.e. $g_3=g_2=\sqrt{5/3}\,g_1$. [E] Higgs = the $(1,2)_{1/2}$ inner fluctuation (one doublet). [O] residual: the seam→Dirac realisation + measured-coupling reproduction stay the open contract. |
| `v246_unification_data_crosscheck.py` | `QFT4D.RGTEST.01` — the honest data cross-check for the spectral-action prediction (v244/v245): does $g_1=g_2=g_3$, $\sin^2\theta_W=3/8$ at $\Lambda$ survive the *measured* SM couplings? Not in the plain SM. Inputs (PDG, GUT-normalised, $M_Z$): $\alpha_i^{-1}=(59.01, 29.59, 8.47)$, SM 1-loop $b=(41/10,-19/6,-7)$ (v159). [E] the prediction: $\alpha_1=\alpha_2=\alpha_3$, $\sin^2\theta_W=3/8$ at $\Lambda$. [X] tension: pairwise crossings spread $\sim 4$ decades ($\sim 10^{13}$–$10^{17}$ GeV), residual spread at $2\times10^{16}$ $\Delta\alpha^{-1}\sim 8.8$ — $g_1=g_2=g_3$ not realised by the plain SM. [X] $\sin^2\theta_W$ 3/8 vs measured 0.23122. [X] 2-loop also misses (min spread $\sim 3.2$ at $1.7\times10^{14}$ GeV). [O] no rescue: SM gauge content + no new states (the $E_8$ spine is arithmetic, not a threshold tower) ⇒ no admissible thresholds; closing it needs new matter (vs 'no new state') or a non-GUT/boundary reading — tension like NCG-SM, likely falsified unless extended. A kill-test, [X]/[O], never [E]-as-success. |
| `v247_e8_branching_no126.py` | `PS.E8BRANCH.01` — the $E_8$ audit hull forbids the **126** and supplies exactly $\{1,10,16,45\}$ for $SO(10)$. Carrier $D_5\oplus A_3=SO(10)\times SU(4)\subset E_8$ (via $SO(16)$); $\mathbf{248}=(45,1)+(1,15)+(10,6)+(16,4)+(\overline{16},\bar4)=248$. [E] ranks $5+3=8$; [E] dims $=248$ ($SO(16)$: 120 adj +128 spinor); [E] $SO(10)$ content $\{1,10,16,45\}$, the **126** ABSENT ⇒ $126_H$ algebraically forbidden, $10+16+45$ $E_8$-supplied not fitted; [E] caveat only one 45/one 15 → proton-marginal; [O] the field-content principle + NCG complement (v250). |
| `v248_ps_algebra.py` | `PS.ALGEBRA.01` — the Pati–Salam NCG algebra $A_F=\mathbb{H}_L\oplus\mathbb{H}_R\oplus M_4(\mathbb{C})$ and its exact SM matching. [E] unimodular group $SU(2)_L\times SU(2)_R\times SU(4)_c$ (rank 5, dim 21); [E] $\mathbf{16}=(4,2,1)+(\bar4,1,2)=8+8$; [E] $Y=T_{3R}+(B-L)/2$ gives the exact SM hypercharges and $Q=T_{3L}+Y$ the exact charges; [E] matching $\alpha_1^{-1}=\frac{3}{5}\alpha_{2R}^{-1}+\frac{2}{5}\alpha_4^{-1}$, $\sin^2\theta_W=3/8$. [O] $A_F$ uniquely from seam+sheet+$\mu_4$; [O] caveat $SU(4)_c\subset SO(10)=D_5$ is not the carrier $A_3=SU(4)$ (family). |
| `v249_ps_unification.py` | `PS.RGTEST.01` — carrier-native Pati–Salam/$SO(10)$ two-step unification kill-test + negative controls (suite check of `experiments/gauge-unification`). [E] matching + a valid 1-loop unification solution; [X] NEG SM-only fails (spread $\sim 8.8$ @ $2\times10^{16}$); [C] $M_{PS}/M_s=1.0$–$1.7$ (1-loop; 2-loop $\sim 1.0$–$1.2$) — gauge unification meets the gravitational scalaron scale; [X] NEG 126 breaks the match ($b_{2R}>0$; also $E_8$-forbidden v247); [X] NEG proton decay selects content (minimal $M_{GUT}\sim 2\times10^{15}$ excluded; $+(15,1,1)$ safer). [O] the gauging fork, exact $\kappa$, the one-45 marginality. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v250_dirac_triple_contract.py` | `CONTRACT.QFT4D.DIRAC.01` — the spectral-triple contract that would make 'carrier gauged' a theorem. Target $(A, H, D, J, \gamma)$ almost-commutative, $A_F$ = PS (v248), $H_F = 3{\times}16$, $D_F$ from $(R, L, \varphi_0,$ Majorana). [E] $H_F = 3{\times}16 = 48$; [E] the Yukawas are already fixed (v18) so the open test is NCG consistency; [O] the 7 NCG axioms (first-order condition = the historic $SO(10)$ obstruction); [O] $\sigma$ = scalaron hypothesis (rescues first-order + $\nu_R$ Majorana; $M_{PS}{=}\kappa M_s$) — caveat $\sigma \leftrightarrow \nu_R\nu_R$ not automatic; [O] inner fluctuations → 10+16+45 not 126. Reduces 4d QFT to one object + finite NCG checks. |
| `v251_first_order_sigma.py` | `PS.DIRAC.01` — the first-order condition is violated *exactly* by the Majorana/$\sigma$ term (Chamseddine–Connes–van Suijlekom 'beyond first order' → Pati–Salam + $\sigma$), exhibited on a faithful lepton-block model; advances v250 from contract to mechanism. Honest correction: $\sigma$ does NOT 'rescue' first order — one DROPS it and the inner fluctuations generate PS + $\sigma$. [E] order-zero; [E] real-even ($J^2{=}+1$, $\gamma^2{=}1$, $D\gamma{=}-\gamma D$, KO-6 signs); [E] first order HOLDS for the Yukawa block; [E] first order VIOLATED by $M \neq 0$, supported exactly on the $\nu_R\nu_R^c$ blocks ($\sigma$ the unique violator). [C] $\sigma$ = scalaron (v249). [O] the full 96-dim SM/PS triple + spectral action (v250). |
| `v252_full_finite_triple.py` | `PS.DIRAC.02` — the full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ built explicitly with the NCG axioms verified; advances v250 from a 7-obligation contract to a concrete object discharging 5 to [E]. $H_F$ = particle⊕antiparticle doubling of three $SO(10)$ **16**-plets (dim 96). [E] dim 96; $D_F$ Hermitian+odd; reality $J^2{=}+1$, $JD{=}DJ$; grading $\gamma^2{=}1$, $[\gamma, a]{=}0$, $J\gamma{=}-\gamma J$; KO-dim 6; order-zero; first order HOLDS for the SM $\mathbb{C} \oplus \mathbb{H} \oplus M_3$ (incl. Majorana) and VIOLATED for Pati–Salam exactly by the Majorana on $\{\nu_R, e_R\}$ (CCvS beyond-first-order → PS+$\sigma$); spectrum = fermion masses + seesaw $m_{\text{light}}{=}m_D^2/M_R$. [C] Poincaré duality (intersection form non-degenerate). [O] orientability + no-junk + spectral-action expansion. |
| `v253_kappa_scalaron_ps.py` | `PS.KAPPA.01` — pinning $\kappa$ in $M_{PS} = \kappa\, M_s$, discharging residual 6(b) of v249. $M_s = c_3^{7/2}\bar{M} \approx 3.06 \times 10^{13}$ GeV (exact). [E] scalaron scale exact; [E] $\kappa$ pinned by RG (scan PDG errors × the two $E_8$-allowed contents) to a tight $O(1)$ interval $\sim 1.3$ (1-loop $[1.0, 1.7]$, 2-loop $[1.0, 1.2]$) — pinned, not free. [C] heat-kernel origin $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\text{grav}}}$, a ratio of heat-kernel traces over the 96-dim $H_F$ (v252) — $\Lambda$ cancels, $\kappa$ scale-free and $O(1)$. [X] the **126**-content drives $\kappa$ out of the $O(1)$ window (content-selective). [O] the exact decimal needs the cutoff moments $f_2/f_0$ fixed. |
| `v254_inner_fluctuations.py` | `PS.NCG.FLUCT.01` — the inner fluctuations $\Omega^1_D(A_F)$ of the finite Dirac operator computed explicitly (reuses the v252 triple); closes v250 #5 (Higgs content, not **126**) + the no-junk obligation + the $\sigma$ quantum numbers. [E] $\Omega^1_{D_F}$ purely chirality-odd → finite fluctuations are scalars; [E] no junk / no **126**: SM-algebra 1-forms sit exactly in the Higgs-doublet pattern (the single $(1,2)_{1/2}$); [E] the $\sigma$ ($\nu_R$ Majorana) direction is absent for the SM algebra but present for Pati–Salam — $\sigma$ is the CCvS beyond-first-order scalar; [E] $\sigma$ is an SM singlet $(1,1,0)$ with $B{-}L = 2$; [C] $\sigma$ = scalaron consistency. |
| `v255_spectral_action_expansion.py` | `PS.SPECACT.01` — the spectral-action heat-kernel (Seeley–DeWitt) expansion $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$ over the finite triple; closes v252 #11 (spectral-action expansion) and makes $\kappa(f_2/f_0)$ explicit. [E] expansion structure ($a_0$ cosmological, $a_2$ Einstein–Hilbert+Higgs-mass, $a_4$ Yang–Mills+Weyl$^2$ + $R^2$+Higgs-quartic, $N{=}96$); [E] trace inputs $b/a^2 = 1/3$ (top-dominated, Chamseddine–Connes); [E] $\sin^2\theta_W = 3/8$ (cross-check v245); [C] Higgs quartic $\lambda(\Lambda) \sim g^2 b/a^2 \to \sim$125 GeV with $\sigma$; [E] scalaron = $R^2$ spin-0 mode (gravitational $a_4 \sim f_0 N$); [C] $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}$ explicit; [O] exact decimal needs the cutoff moments fixed. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v256_orientability_poincare.py` | `PS.NCG.ORIENT.01` — orientability and Poincaré duality treated honestly (no faked proof). KO-dim 6 forces the intersection form $\mathrm{cap}(p,q)=\mathrm{Tr}(\gamma\,\pi(p)\,J\pi(q)J^{-1})$ to be antisymmetric. [E] skew form forced by $J\gamma{=}{-}\gamma J$ + order-zero + reality (verified); [E] on the SM K-theory $\mathbb{Z}^3$ the skew form has rank 2, corank 1, null $\sim(1,1,-1)$ — the structural obstruction; [E] orientability local condition $[\gamma_F,\pi(a)]=0$, full orientability is a degree-4 cycle on $M\times F$; [C] the canonical hypercharge-faithful SM/PS triple satisfies both (van Suijlekom; CCM) — literature; [O] a self-contained machine proof needs the complete canonical bimodule. |
| `v257_quasiorient_modpd.py` | `PS.NCG.ORIENT.02` — the honest, literature-confirmed resolution of orientability + Poincaré duality, correcting the too-generous v256 typing. Published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192): the CCM SM finite triple (KO-6, $3\,\nu_R$) genuinely FAILS strict orientability and strict Poincaré duality, satisfying the weakened versions. [E] strict orientability fails ($\gamma\notin\mathrm{span}\{\lambda(a)\rho(b)\}$); [E] quasi-orientability holds (left/right $A$-irrep boxes disjoint by chirality: $L=H$ doublet, $R=\mathbb{C}$ singlet $\Rightarrow L^{\mathrm{LR}}(\mathrm{even},\mathrm{odd})=\{0\}$); [E] strict Poincaré fails (KO-6 skew form, corank 1, the equal-$L/R$-neutrino case); [C] modified Poincaré duality (CCM) / Stephan's PD-restoring triple. A known model-class feature, not a TFPT defect — no faked closure. |
| `v258_dirac_covariance_induction.py` | `PS.DIRAC.03` — $D_F$ is the modular/covariance *induction* of the boundary seam state, not an independent posit; closes the 'posited Yukawa vs derived operator' gap of v250/v252 ( Q2: Dirac as Kasparov product + covariance formula). Quasi-free/modular dictionary: a Gaussian state is fixed by its covariance $C$ (contraction, spec $\subset(0,1)$) with one-particle modular Hamiltonian $H=\log((1{-}C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C=(1{+}e^{H})^{-1}$ ($\beta{=}1$, v239). [E] inversion identity $\log((1{-}C)C^{-1})=H$ exactly (random Hermitian + a chirality-graded Yukawa block); [E] $C_F=(1{+}e^{D/\mu})^{-1}$ a positive contraction; [E] chirality-odd ($D_F^{\mathrm{ind}}\gamma=-\gamma D_F^{\mathrm{ind}}$); [E] unitarily equivalent to v252 (inducing $C_F$ from the 9 Yukawas returns the same spectrum); [E] Majorana = off-diagonal covariance (seesaw $m_{\mathrm{light}}{=}m_D^2/M_R$); [C] $C_F=P_FC_\Sigma P_F$, so $D_F$ is the KK shadow $[D_\Sigma]\widehat{\otimes}_{(E_8)_1}[\mathcal{K}_{\mathrm{car}}]$ and the v18 Yukawas are the readout of $C_\Sigma$. |
| `v259_modular_cutoff_kappa.py` | `PS.SPECACT.02` — the spectral-action cutoff *is* the seam KMS weight, fixing $f_2/f_0{=}1$ exactly and reducing the last open scheme freedom (v253 #5, v255 #7) from a free function to a finite-triple trace ratio ( Q4: the cutoff is modular, not chosen). By Tomita–Takesaki/BW the seam state is KMS at $\beta{=}1$ (v239) and $2\pi{=}1/(4c_3)$ (v58) fixes $\beta$, so the heat-kernel cutoff $f(u){=}e^{-u}$ is *derived*. [E] KMS cutoff moments $f_0{=}f(0){=}1$, $f_2{=}\int f{=}1$, $f_4{=}\int uf{=}1$, so $f_2/f_0{=}f_4/f_2{=}1$; [E] seam-fixed $\beta{=}1$ (no new parameter beyond $c_3$); [E] non-vacuous neg control: a Gaussian cutoff $e^{-u^2}$ gives $f_2/f_0{=}\sqrt{\pi}/2\neq1$; [E] $\kappa{=}\sqrt{(f_2/f_0)c_{PS}/c_{\mathrm{grav}}}{=}\sqrt{c_{PS}/c_{\mathrm{grav}}}$, no cutoff function left; [C] the RG $\kappa{\sim}1.3$ pins $c_{PS}/c_{\mathrm{grav}}{\sim}1.7$ ($O(1)$); the [O] free scheme becomes a [C] finite computation over the built triple; [C] the same logic fixes $\lambda(\Lambda)$. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v260_k3_kummer_unification.py` | `ARCH.K3.01` — one Kummer/K3 surface carries the *seam*, the *carrier-16* and the $E_8$ lattice as facets of one smooth object, the geometric unification of the v1 'glue $D_5\oplus A_3+\mu_4\Rightarrow E_8$' step (the K3/Kummer proposal). The seam square torus $\tau{=}i$ ($j{=}1728$, v214) under the elliptic involution $z\mapsto -z$ squares to the Kummer K3. [E] the $E_8$ Cartan is even, $\det{=}1$, positive-definite (the unique even unimodular pos-def rank-8 lattice); [E] the K3 lattice $L_{K3}{=}U^3\oplus E_8(-1)^2$ has rank $22{=}b_2$, $\det{=}-1$ (unimodular), is even, signature $(3,19)$, with $E_8(-1)^2$ an orthogonal summand; [E] seam $= T^2/(z\mapsto -z)$: $4{=}|(\mathbb{Z}/2)^2|$ fixed points, pillowcase $S^2(2,2,2,2)$, orbifold Euler char 0 (the v214/v216 four marks); [E] carrier-16 = Kummer nodes $|A[2]|{=}2^4{=}16 \to 16$ exceptional $\mathbb{P}^1 = \dim S^+ = \Lambda^{\text{even}}(\mathbb{C}^5) = 16$ (v197), $16{=}4{\times}4$; [E] honesty: the 16 node classes span the rank-16 Kummer lattice, *not* literally $E_8(-1)^2$ (distinct sublattices of $H^2(K3)$); [C] two faces of $E_8$ in K3 (local du Val $\mathbb{C}^2/2I$, v232/v236; global $H^2$ summand); [C] unification (lattice [E], carrier↔node identification [C]). |
| `v261_modular_spectral_closure.py` | `QFT.MSC.01` — the *Modular Spectral Closure*: the assembly/reduction certificate turning the whole QFT layer (v238–v260) into *one* relative object $\mathsf{TFPT}_{\mathrm{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\mathrm{rel}})$ reduced to *one* open premise (like v176/v234, an assembly certificate, not new physics). [E] one index = marks = fixed points = glue order: Jones $[B{:}A]{=}4$ (v89) $= |\mu_4|{=}4$ (v92) = seam marks $2\chi{=}4$ (v216) = pillowcase/Kummer fixed points $|(\mathbb{Z}/2)^2|{=}4$ (v260); [E] one carrier-16: $\dim S^+{=}\Lambda^{\text{even}}(\mathbb{C}^5)$= Kummer nodes $2^4{=}16$, $H_F{=}2N_{\text{fam}}\cdot 16{=}96$; [E] one gap $\Delta{=}-\log((2/3)^6){=}6\log(3/2){>}0$ = the OS mass gap (v240) = the recovery rate (v221) = the Lindblad gap (v238); [E] the three closures re-derived (Kasparov $\log((1{-}C)C^{-1}){=}H$, v258; KMS cutoff $f_2/f_0{=}1$, v259; K3 lattice signature $(3,19)/\det -1/\text{even}$, v260); [E] relative-object manifest (10/10 component scripts present); [O] the single residual: the whole QFT layer reduces to `QGEO.SYM.01` (seam realisation), the ambient QG measure `QG.AMB.01` kept separate; [C] the master statement (Modular Spectral Universality) holds modulo `QGEO.SYM.01` — the honest sense in which the QFT layer is complete. |
| `v262_fqcd_mp_me.py` | `FR.MPME.02` — the $F_{\mathrm{QCD}}$ transfer pipeline for $m_p/m_e$, implemented as a real solver (the one $F_{\mathrm{transfer}}$ pipeline that was contract-only). *Not* a compiler power — a [C] standard-physics transfer reproduction. [E] $b_3{=}-7{=}-(11{-}2n_f/3)$ at $n_f{=}6$ (the only TFPT number in the run); [E] a two-loop $\alpha_s(M_Z){=}0.1179$ run with thresholds gives $\alpha_s(2\,\mathrm{GeV}){\sim}0.30$ (real ODE solve); [C] $\Lambda^{(3)}_{\overline{\mathrm{MS}}}{\sim}0.33\,\mathrm{GeV}$ × the lattice bridge $C_p{=}m_p/\Lambda$ (external $O(1)$) over the closed $m_e$ gives $m_p/m_e{\sim}1824$ vs the measured 1836.15 within the $\sim 7\%$ external budget; [X] firewall: $m_p/m_e$ is not an admissible compiler integer ($SU(9)$ absent from $D_5\oplus A_3$). Advances `FR.MPME.01` ([O]) to `FR.MPME.02` ([C]). |
| `v263_mnu_from_df.py` | `FLAV.PMNS.02` — the light neutrino matrix $M_\nu$ as the type-I seesaw of $D_F$, advancing the solar-angle texture (v9) from a posited matrix to a seesaw *object* built from $D_F$'s blocks. [E] $M_\nu = -m_D M_R^{-1} m_D^\top$ symmetric + seesaw-suppressed (the light masses an output of $D_F$); [E] a $\mu\tau$-symmetric $(m_D, M_R)$ gives a $\mu\tau$-symmetric $M_\nu$ (the v9 texture is in the seesaw image); [C] diagonalising gives $\theta_{23}{=}45^\circ$ and $\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\mathrm{ret}}}{2}{=}0.30675$ under the seam misalignment; [O] $\theta_{13}{=}0$ in the leading texture (physical $\sin^2\theta_{13}{\sim}\varphi_0^{\mathrm{ret}}e^{-5/6}$ a separate readout), the CP magnitude $\delta_{\mathrm{PMNS}}{=}240^\circ$ is the $\mu_6$/triality phase (v231/v233), and $M_R$ is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v264_raw_seam_marklocal.py` | `QGEO.ENERGY.03` — the raw-seam ⇒ mark-locality *reduction* (not a closure of `QGEO.SYM.01`). Composing v216+v201+v198 on the canonical pillowcase: [E] Gauss–Bonnet 4 cone points (deficit $\pi$ each, sum $4\pi{=}2\pi\chi(S^2)$), curvature *only* at the 4 marks; [E] the curvature sourced at the $\mu_4$ orbit $\{0,\pi/2,\pi,3\pi/2\}$ has Fourier support *only* on $4\mathbb{Z}$ (FFT), so mark-locality is automatic for the pillowcase; [E] $M_f$ then commutes with the clock $\rho{=}\mathrm{diag}(i^n)$; [E] with $\lvert k\rvert$ already commuting (v198), $[\rho,\Lambda]{=}0 \Rightarrow \omega\circ\rho{=}\omega$ on this metric; [O] the residual *is* the metric — the chain assumes the raw seam carries the uniformising flat pillowcase metric (=`QGEO.SYM.01`), the negative control (curvature off the $\mu_4$ orbit) breaks it. A reduction/sharpening, not a closure. |
| `v265_qft4d_fork_freeze.py` | `QFT4D.FORK.01` — freeze the 4D-QFT fork as a *branch policy* (not an open ambiguity); writes `qft4d_fork_freeze.json` + a prose guard, no new physics. [C] A: boundary-only is the canonical 4D reading (no 4D-GUT claim by default); [X] SM-only 4D-GUT unification is killed (1-loop spread ∼8.8 at $2{\times}10^{16}$, v246); [C]/[O] B: carrier-native Pati–Salam is a falsifiable UV branch ($E_8$ content $\{1,10,16,45\}$ no **126** v247, $\sin^2\theta_W{=}3/8$ v248, $\kappa$ $O(1)$ band v249/v253, threshold + proton kill test); [X] proton safety rule (no fake $\tau_p$ window); [E] wording guard (bare 4D over-claims absent, scoped policy phrases present). A frozen decision tree, not an open ambiguity. |
| `v266_ps_threshold_proton.py` | `PS.PROTON.02` — the explicit Pati–Salam threshold model + proton-decay window that DECIDES branch B of the v265 fork; writes `ps_threshold_proton.json`. Two-step PS solve + $d{=}6$ $\tau_p(p{\to}e^+\pi^0)$: the carrier-native PS UV branch survives proton decay *only* with the $E_8$-allowed $(15,1,1)$ [the single 45]. [E] explicit thresholds: minimal 16H → $M_{GUT}{\sim}2.4{\times}10^{15}$, $+(15,1,1)$ → $M_{GUT}{\sim}5.9{\times}10^{15}$; [X] minimal KILLED ($\tau_p{\sim}4.4{\times}10^{33}$ yr < Super-K $2.4{\times}10^{34}$, even high-band); [C] $+(15,1,1)$ survives now ($\tau_p{\sim}1.6{\times}10^{35}$ yr); [X] *Hyper-K decisive* ($\tau_p$ at the $\sim 1.4{\times}10^{35}$ yr reach) — a dated kill test within the decade; [O] marginal (only one 45 in the hull, v247) + $O(3)$ hadronic uncertainty. A (boundary-only) stays canonical. |
| `v267_qgeo_rigidity_minimal_axiom.py` | `QGEO.SYM.02` — the rigidity / minimal-axiom form of the bedrock `QGEO.SYM.01`: the ONE bare statement ( the raw RP seam admits the carrier $\mu_4$ clock as a conformal symmetry permuting its 4 marks, $=\omega\circ\rho=\omega$ ) *forces* everything downstream; it does *not* close the axiom. [E] an order-4 Möbius orbit $\{a,ia,-a,-ia\}$ has cross-ratio 2 (independent of $a$); [E] cross-ratio $2 \Leftrightarrow j{=}1728$ (only $\lambda\in\{-1,\frac{1}{2},2\}$) $\Leftrightarrow$ the order-4 CM automorphism $(x,y)\mapsto(-x,iy)$ on $y^2{=}x^3{-}x$; [E] downstream forced: square ⇒ the unique flat pillowcase (Troyanov, v214) ⇒ curvature at the 4 marks ⇒ $\mathbb{Z}_4$-Fourier DtN (v201/v264) ⇒ $[\rho,\Lambda]{=}0 \Rightarrow \omega\circ\rho{=}\omega$ (v198); [E] neg controls (generic $j\neq 1728$, only $\mathbb{Z}_2$; hexagonal $j{=}0$ has $\mathbb{Z}_6$, order $6\neq 4$); [C] arithmetic rigidity ($\{1,i,-1,-i\}$ over $\mathbb{Q}$, $j{=}1728$ CM by $\mathbb{Z}[i]$, Belyi/dessins); [O] the bare order-4 symmetry stays the one foundational postulate (like $c$=const). Closes the reduction (axiom⇒everything), not the axiom. |
| `v268_theta13_carrier_trace.py` | `FLAV.TH13.01` — the reactor angle $\theta_{13}$ *exponent* is the carrier hypercharge trace. The value $\sin^2\theta_{13}=\varphi_0^{\mathrm{ret}}e^{-5/6}=0.0231$ was already checked (v16); here the exponent origin is closed exactly: [E] $\gamma=\mathrm{tr}_E Y^2=3(1/3)^2+2(1/2)^2=5/6$ over the 5-slot carrier hypercharge $Y=\mathrm{diag}(-\frac{1}{3},-\frac{1}{3},-\frac{1}{3},\frac{1}{2},\frac{1}{2})$ (roots of $6Y^2-Y-1=0$), complement 1/6 the neutrino ratio; [C] value 0.02311 vs PDG ∼0.0222; [E] own channel — the $\mu\tau$-breaking $\varepsilon=3\varphi_0^{\mathrm{ret}}/4$ induces only $\sim 1.6\times10^{-3}\ll 0.023$, so $\theta_{13}$ is a separate carrier-trace channel, not the seesaw $\mu\tau$ breaking (corrects the tempting '$\theta_{13}$ from $M_\nu$' route); [O] CP magnitude + absolute $\nu$ scale remain open. |

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| Script | What it machine-checks *(continued)* |
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| `v269_spert_paqft_skeleton.py` | `QFT4D.SPERT.01` — the perturbative 4D S-matrix ($S_{\text{pert}}$) as an Epstein–Glaser / pAQFT skeleton: the honest middle layer of the three-layer S-matrix ($S_{\text{top}}$ 2d DHR braiding v243 / $S_{\text{pert}}$ 4d EG of the spectral action / $S_{\text{phys}}$ LSZ on the OS Wightman functions v240), *not* a nonperturbative closure. [E] three distinct layers (the 2d braiding $\|M\|=1$ is statistics, not 4d cross-sections); [E] every spectral-action $a_4$ operator has mass dimension $\leq 4$ in 4D (power-counting renormalizable); [E] a finite counterterm basis (7 marginal +3 relevant, no infinite tower); [C] Epstein–Glaser existence — a renormalizable interaction has an order-by-order $S(g)=T\exp(i\int g\, L_{\text{int}})$ by causal factorisation + finite local extension (no path integral, no UV divergences); [C] adiabatic limit/gap ($\Delta=6\log(3/2)>0$) $\to S_{\text{phys}}$ via LSZ; [O] perturbative only — the ambient measure `QG.AMB.01` stays open. A contract/skeleton, explicitly perturbative. |
| `v270_pmns_jarlskog_assembly.py` | `FLAV.PMNS.03` — the complete complex PMNS matrix + the leptonic Jarlskog $J_{\text{PMNS}}$ (the CP *strength*), assembled from the four TFPT channels ($\sin^2\theta_{12}=\frac{1}{3}-\frac{\phi_0}{2}$, $\sin^2\theta_{23}=\frac{1}{2}$, $\sin^2\theta_{13}=\phi_0 e^{-5/6}$, $\delta=4\pi/3$) into one unitary object. [E] the three angle channels; [E] $U=R_{23}U_{13}(\delta)R_{12}$ is exactly unitary; [C] $J_{\text{PMNS}}=\Im(U_{e1}U_{\mu 2}U_{e2}^{*}U_{\mu 1}^{*})=-0.02965$ (formula and full matrix agree), $J_{\max}=0.03424$ — a *derived* CP strength, the leptonic analogue of the CKM $J$ (v88); [C] data: $J_{\max}$ vs NuFIT 0.0332 ($\sim 3\%$), $J$ vs best fit $-0.026$, the negative sign reproduced; [O] $\delta$ is the $\mu_6$/triality phase (assembled input, *not* from $M_\nu/D_F$) and the absolute $\nu$-mass scale ($\Sigma m_\nu$, $M_R$) stays open. Assembles the PMNS matrix + CP strength; does not derive $\delta$ from $D_F$ nor fix the scale. |
| `v271_eg_oneloop_quartic.py` | `QFT4D.SPERT.02` — a concrete Epstein–Glaser order-2 (one-loop) quartic renormalization, taking the v269 $S_{\text{pert}}$ skeleton from *exists* to an actual EG number (no path integral). [E] $T_1=:L_{\text{int}}$: needs no extension, the first extension is $T_2$; [E] the massless propagator has scaling degree sd=2 in $d=4$, the one-loop bubble (two propagators) has sd=4=$d$ so the UV singular order $\omega=\text{sd}-d=0$; [E] $\omega=0\Rightarrow$ exactly one logarithmic local counterterm ($C(\omega+d,d)=1$, renormalizable, no infinite tower); [E] the loop factor $\Omega_3/(2(2\pi)^4)=1/(16\pi^2)$ exactly — the same factor that normalises the spectral-action $a_4$ geometric quartic; [C] the $\phi^4$ one-loop $\beta_\lambda=3\lambda^2/(16\pi^2)$ (symmetry factor 3 = the $s,t,u$ channels), EG initial condition $\lambda_\Phi(\text{UV})=1/(16\pi^2)$; [O] order-2/one-loop only — the all-order EG construction and the nonperturbative measure `QG.AMB.01` stay open. Turns the skeleton into a concrete one-loop number. |
| `v272_nu_mass_scale.py` | `FLAV.NUSCALE.01` — the absolute neutrino-mass scale, typed honestly (it does *not* fabricate a $\Sigma m_\nu$ number, which would contradict the v184 firewall). [E] the absolute scale reduces to *one* seesaw ratio $m_3=(y_\nu v_{\text{EW}})^2/M_R$ — one irreducible UV input like $v_{\text{geo}}$ (v153), while the ratios ($m_2/m_3$, ordering, PMNS angles v270) are pinned; [C] the observed $m_3=0.050$ eV with the TFPT PS→GUT window $M_R\in[4.2\times10^{13},2.4\times10^{15}]$ GeV (v249) needs a third-generation Dirac Yukawa $y_\nu\in[0.26,2.0]$ — $O(1)$, no tuning; [O] $M_R(y=1)=6.1\times10^{14}$ has $\log_{c_3}(M_R/M_{\text{Pl}})=2.57$ (non-integer), not a TFPT power, so the absolute value stays open; [C] the normal-ordering floor $\Sigma m_\nu=0.0586$ eV is consistent with cosmology ($<0.072$–$0.12$ eV); [X] a cosmological $\Sigma<0.0586$ eV excludes the normal-ordered $m_1\sim 0$ spectrum. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio), not the absolute number; sharpened by v481 (carrier normalisation $y_\nu=y_t$: one-parameter, window-bracketed candidate). |

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| Script | What it machine-checks *(continued)* |
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| `v273_eg_gauge_running.py` | `QFT4D.SPERT.03` — the gauge sector of the Epstein–Glaser one-loop (v271's quartic mechanism applied to the gauge self-energy). [E] the one-loop gauge 2-point has scaling degree sd=4=$d$ so $\omega$=0 → one counterterm per coupling (same marginal mechanism + loop factor $1/(16\pi^2)$ as v271); [E] the one-loop $b_i$ from the explicit carrier/SM content: $b_3=-7$, $b_2=-19/6$, $b_1$=41/10 (the same 41 as the carrier algebra, v159/`CAR.SM.01`); [C] physical RG directions $b_3<0$ (asymptotic freedom), $b_1>0$ (U(1) Landau); [O] the two-loop matrix (v159, PyR@TE) + all-order stay open. Shows the gauge running is an EG order-2 result. Exact $b_i$ mirrored in Wolfram. |
| `v274_scale_overdetermination.py` | `SCALE.OVERDET.01` — the single mass anchor is *over-determined*, and the absence of a derivable absolute scale is a *theorem*. [E] one anchor (No-Unit, v153): every dimensionful output = pure number $\times M_{\rm Pl}^d$ ($\rho_\Lambda/M_{\rm Pl}^4$, $M_{\rm scal}/M_{\rm Pl}=c_3^{7/2}$, the spectrum); [C] two independent routes to $M_{\rm Pl}$ agree: gravity $(8\pi G)^{-1/2}$=2.4353×$10^{18}$ GeV and cosmology (invert $\rho_\Lambda/M_{\rm Pl}^4=(3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4}$=2.24 meV) =2.438×$10^{18}$ GeV, to 0.11%; [E] the anchor is the gravitational unit ($G$), the one measured dimensionful input of all physics; [E] theorem not gap: the seam is a conformal $c$=8 CFT (v157/v158), scale-invariant by construction → no intrinsic scale; [O]/[E] scale-complete up to one over-determined anchor = maximal predictivity. Sharpens v153/v78. |
| `v275_qgamb_roadmap.py` | `QGAMB.ROADMAP.01` — the obligation roadmap for `QG.AMB.01` (the ambient nonperturbative QG measure), consolidated into one module (the open gate had no script). [E] Tier A gap decoupling: $\Delta_{\rm eff}=6\log(3/2)-2{\cdot}248\,c_3^2$=1.648>0 (v36/v76), so every low-energy readout is independent of the un-built measure — not a blocker; [E] Tier B: the bulk measure reduces to identifying the seam-Calderón boundary with the holomorphic $(E_8)_1$ net ($c$=248/31=8; v77/`GATE.METRIC.03`), reduced not closed (`REDTEAM.A.01`); [C] the perturbative layer is in hand (v269/v271/v273); [O] the nonperturbative ambient measure is *not* constructed — the one genuine structural frontier; [X] closes IFF the seam net = $(E_8)_1$ *and* the bulk reconstruction is proved. A roadmap, not a closure. |
| `v276_qgeo_flat_closes_commutator.py` | `QGEO.SYM.03` — the flat $\tau=i$ pillowcase premise *closes* the modular-commutator chain to all orders, collapsing the bedrock `QGEO.SYM.01` to *one* geometric postulate (it does *not* prove the premise). For a quasi-free seam state `QGEO.SYM.01` = ($\omega\circ\rho=\omega$) ⇔ $[\rho,C]=0$ ⇔ $[\rho,H]=0$. [E] isometry⇒commutator: if $H=f(\Delta_{\rm flat})$ and the order-4 deck $\rho$ ($z\to iz$) is a flat-$\tau=i$ isometry then $[\rho,\Delta]=0\Rightarrow[\rho,H]=0$ *exactly* (all orders), verified on the discretised square torus ($\rho^4=I$, $[\rho,\Delta]=0$, $[\rho,\sqrt{-\Delta}]=0$; generic non-isometry fails); [E] this upgrades v198 (principal symbol) + v201 (subprincipal) to the full $H$ (flatness removes the curvature terms); [C] $\tau=i$ is forced by order-4 ($j$=1728; hexagonal $j$=0 is order-6, excluded; v214/v267); [O] `QGEO.SYM.01` collapses to the single statement 'the raw seam is the flat $\tau=i$ pillowcase quasi-free state' — needs a human constructive-QFT proof or stays an honest axiom. Sharpest non-circular form. |
| `v277_seam_calderon_e8_match.py` | `QGAMB.TIERB.01` — the seam-Calderón → $(E_8)_1$ matching certificate, the concrete Tier-B step of `QG.AMB.01` (v275): it computes the full $(E_8)_1$ matching target and pins the residual to *one* bit (it does *not* close it). [E] $c$=8 (Sugawara, 248/31=$g_{\rm car}+N_{\rm fam}$), but $c$=8 alone does not pin the net; [E] the discriminator is holomorphy — $\det\mathrm{Cartan}(E_8)$=1 (1 primary, holomorphic) vs the $c$=8 counterexample $\det\mathrm{Cartan}(D_8{=}SO(16))$=4 (4 primaries, non-holomorphic), so the single discriminator is $\lvert\det K\rvert$=1; [E] the $(E_8)_1$ character $E_4/\eta^8=j^{1/3}=q^{-1/3}(1+248q+\dots)$ has a single primary + exactly 248=$\dim E_8$ weight-1 currents (240 roots); [E] equivalence chain holomorphic ⇔ $\det$ Cartan 1 ⇔ unique even unimodular rank-8 (v83) ⇔ homology-sphere link ⇔ $\lvert\det K\rvert$=1 (v235); [C] seam-side match via v276 (flat $\tau=i$) + v260 (Kummer/K3); [O] residual = the single holomorphy bit — reduced to one bit, *not* closed. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v278_lsz_bridge_unitarity.py` | `QFT4D.SPERT.04` — the $S_{\text{pert}} \to S_{\text{phys}}$ LSZ bridge + one-loop unitarity: connects the EG perturbative S-matrix (v269/v271/v273) to the physical asymptotic S-matrix ($S_{\text{phys}}$ = LSZ on the OS Wightman functions, v240). [E] LSZ bridge: $S_{\text{phys}}$ = amputate+on-shell-residue of the connected time-ordered products, and the EG $T$-products of $S_{\text{pert}}$ *are* those correlators (OS Wick-rotation, v240) $\Rightarrow S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$; [C] the gap $\Delta{=}6\log(3/2){>}0$ gives Haag–Ruelle asymptotic states so LSZ is valid; [E] *one-loop unitarity* (optical theorem): the bubble discontinuity $\operatorname{Im} B(s)/\pi = x_+{-}x_- = \sqrt{1-4m^2/s}$ *exactly* = the two-body phase space, so $2\operatorname{Im} M = \sum\!\int d\Pi |M|^2$ (cutting rule) holds ⇒ matter+gauge unitary; [E] threshold: $\operatorname{Im} B{=}0$ below $s{=}4m^2$, turning on at the physical two-particle threshold; [O] gravity caveat — the Stelle ghost makes the gravity perturbative S-matrix non-unitary →`QG.AMB.01`; [C] finite field-strength $Z$ from the EG scheme. |
| `v279_qgeo_obligation_lemma.py` | `QGEO.OBLIG.01` — the `QGEO.SYM.01` bedrock written as *one* precise constructive-QFT lemma with a proof-tree completeness check (the formal obligation write-up; does *not* close it). Lemma: given the premise 'the raw seam carries the flat $\tau{=}i$ pillowcase metric $g_{\text{flat}}$ (Troyanov-unique, cone angles $\pi$), so $\Lambda_\Sigma = \sqrt{-\Delta_{\text{flat}}}$', then $\omega_\Sigma \circ \rho = \omega_\Sigma \to$ mark-locality $\to E_8$. [E] the reduction DAG from 'free RP seam' has 8 closed nodes and *exactly one* open leaf ('the raw seam state is the flat state'); [E] the geometric side is fixed (flat DtN diagonal, $\rho$ commutes to all orders, Lean `FORM.QGEO.04`), so the obligation is state-identification not geometric ambiguity; [C] two proof routes (RP-state uniqueness on the flat orbifold, or Troyanov + OS); [E] the all-orders step is a Lean theorem (`SeamDeckClosure.flat_all_orders_clock`, standard axioms, no `sorry`); [O] one constructive-QFT statement closes `QGEO.SYM.01`. |
| `v280_pillowcase_steklov.py` | `QGEO.STEKLOV.01` — a direct numerical experiment on the flat $\tau{=}i$ pillowcase orbifold (the seam geometry): the self-investigable test of the `QGEO.SYM.01` obligation (does *not* prove the premise). [E] pillowcase = the Z2-even sector of the flat torus, 4 cone points (2-torsion, deficit $\pi$ each, sum $4\pi{=}2\pi\cdot 2$); [E] the order-4 deck $\rho$ ($z{\to}iz$) descends ($[\rho, Z_2]{=}[\rho, P_{\text{even}}]{=}0$) and permutes the 4 cone points as 2 fixed + 1 swapped pair; [E] the geometric chain is exact: $[\rho,\Delta]{=}0 \Rightarrow [\rho,H]{=}0 \Rightarrow [\rho,C]{=}0 \Rightarrow \omega\circ\rho{=}\omega$ (the v276/v279 all-orders closure on the real orbifold); [C] the Steklov DtN is positive self-adjoint, fixed by the flat metric (route-(i) evidence); [O] confirms the conclusion given flat geometry, the premise stays open. |
| `v281_holomorphy_modular_data.py` | `QGAMB.MODULAR.01` — the modular-data / anyon-condensation layer of the `QG.AMB.01` Tier-B holomorphy bit. [E] #anyons = $|\det \text{Gram}|$: $E_8{\to}1$ (holomorphic), $D_8{=}SO(16){\to}4$, carrier $D_5{\oplus}A_3{\to}16$; [E] the anyon-condensation tower $D_5{\oplus}A_3$ (16) → $D_8$ (4) → $E_8$ (1), each step condensing a Lagrangian $Z_2$ boson, the Kitaev $E_8$ endpoint (v92/v235); [E] Gauss–Milgram recovers $c{=}8$; [E] $E_8$ is the unique holomorphic $c{=}8$ (det 1) — $SO(16)$ has the same $c$ but 4 anyons, so holomorphy is THE discriminator (v277); [O] the open bit: does the seam-Calderón net condense to $\det K{=}1$? Reduced to one bit. |
| `v282_e8_tau_i_unification.py` | `QGAMB.UNIFY.01` — the $E_8$-at-$\tau{=}i$ unification: the candidate *simplification* reducing the two open obligations to *one* object (the raw seam is the $(E_8)_1$ holomorphic CFT at $\tau{=}i$). $\chi_{E_8}(\tau){=}\Theta_{E_8}/\eta^8{=}j^{1/3}$. [E] $\tau{=}i$ is the order-4 CM point ($j{=}1728$, the `QGEO.SYM.01` flat geometry); [E] $\chi_{E_8}{=}j^{1/3}$ ($\chi^3{=}q\,j$ as $q$-series), so $\chi_{E_8}(i){=}1728^{1/3}{=}12$ (the $(E_8)_1$ character at the SAME $\tau{=}i$, the `QG.AMB.01` holomorphy); [E] one object, two faces; [C] obligation count 2→1: prove 'the raw seam is $(E_8)_1$ at $\tau{=}i$' and *both* follow; [O] the unified premise stays open but is now one object. |

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| Script | What it machine-checks *(continued)* |
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| `v283_live_killtest_scorecard.py` | `PRED.LIVE.01` — a live kill-test scorecard of the recent round's computable predictions vs current data, each tagged with the decisive experiment. [C] PMNS angles $\sin^2\theta_{12}$=0.30675 (+0.31$\sigma$), $\sin^2\theta_{13}$=0.02311 (+1.45$\sigma$), $\theta_{23}$ maximal; [C] leptonic CP $J_{\rm PMNS}$=−0.0297, $\delta$=240° vs ∼ 232(30); [X] $\Sigma m_\nu$ NO floor 0.0586 eV (DESI/CMB-S4 decisive); [X] proton decay $\tau_p$=1.55×$10^{35}$ yr for +(15, 1, 1) (Hyper-K decisive); [E] closed readouts $\alpha^{-1}$, $\Omega_b$, $\beta$; all rows consistent < 2$\sigma$, two sharp near-term kill tests. A consolidation, not new physics. |
| `v284_route_i_rp_uniqueness.py` | `QGEO.ROUTEI.01` — Route (i) (RP-state uniqueness) decomposed into a 6-lemma chain with every dischargeable lemma discharged and the one irreducible lemma isolated (does *not* prove the open premise). [E] 5/6 lemmas are existing [E]/[O] results — L1 RP→DtN (v194), L2 the four marks force the pillowcase class (v195/v216/v214), L3 the flat orbifold Steklov operator = $\sqrt{-\Delta_{\rm flat}}$ (verified: flat-strip DtN($k$)/$k$ → 1, principal symbol $\lvert\xi\rvert$ with no curvature term), L4 covariance $C = (1+e^H)^{-1}$ from the DtN (v258), L5 $\mu_4$-invariance (v276/Lean/v280); [C] Troyanov reduction: 4 cone points of angle $\pi \Rightarrow \chi_{\rm orb} = 0 \Rightarrow$ the unique constant-curvature representative is flat ($\tau$=$i$), so $L_{\rm open}$ reduces to 'the raw seam is the constant-curvature geometric state'; [O] the one open lemma — the raw RP seam state is the constant-curvature/geometric state (the state-identification step). |
| `v285_route_ii_seam_condensation.py` | `QGAMB.ROUTEII.01` — Route (ii) (seam-net condensation) decomposed into a 4-lemma chain, with the key result that its open lemma *coincides* with Route (i)'s (the v282 unification at the lemma level). [E] 3/4 lemmas discharged — M1 no abelian sector ⇔ $\lvert\det K\rvert$=1 ⇔ holomorphic (v234/v235), M2 holomorphic $c$=8 ⇔ $(E_8)_1$ with the $SO(16)$ det-4 counterexample (v277/v281), M3 the condensation tower 16→4→1 = the unique $\mu_4$-Lagrangian glue (v92/v281/v235); [E] the discriminator det 1 vs det 4; [E] Routes (i) and (ii) share *one* open lemma — both say 'the raw seam is the $(E_8)_1$-at-$\tau$=$i$ object' (v282), so proving either closes both; [O] the unified open lemma: construct a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau$=$i$. $M_{\rm open}$ = `QGEO.SYM.01`. |
| `v286_seam_equivalence_contract.py` | `SEAM.EQUIV.01` — the Seam Equivalence contract: concentrates the whole open structure on *one* named arrow (does *not* prove it). Theorem: the raw RP seam state reconstructs canonically the holomorphic $(E_8)_1$ boundary net at the $\tau$=$i$ pillowcase. [E] four objects (RawRPSeam, FlatTauIPillowcaseDtN, HolomorphicE8Net, SREKitaevE8Phase); [E] six arrows — 2 closed (Route i v276/v280/v284, Route ii v281/v277/v235), 1 conditional (v282), 2 open premises (the same lemma, v284/v285), 1 Gral; [E] **import firewall**: no Route-i module imports any Route-ii module or vice versa, so the routes converge only through `SEAM.EQUIV.01` (kills the '$E_8$ into the geometry and back' circularity); [E] exact-label taxonomy (Formal/Numerical/Conditional Exact, Open Selection); [O] the one open arrow 'free Gaussian RP bulk ⇒ holomorphic single-sector boundary'. |
| `v287_free_rp_bulk_to_holomorphic_boundary.py` | `SEAM.EQUIV.A01` — Route A (AQFT) theorem-dependency checker for the open Gral arrow 'free Gaussian RP bulk ⇒ holomorphic single-sector boundary net': types the 5-lemma chain and isolates the *one* missing standard theorem. [E] L1 RawRPSeam→OS→CAR/quasi-free boundary (v155/v160/v240/v175); [E] L2 gap+chirality ⇒ gapped chiral bulk (v64/v156); [O] L3 invertible/SRE Gaussian bulk ⇒ single-sector boundary ($\mu$-index 1) — the one missing standard theorem (the AQFT form of $\det K$=1); [E] L4 $\mu$=1+$c$=8 ⇒ holomorphic (v234); [E] L5 holomorphic $c$=8 ⇒ $(E_8)_1$ (v83/v277/v281). Verdict: 4/5 discharged, Route A reduces `SEAM.EQUIV.01` to one cited theorem. |

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| Script | What it machine-checks *(continued)* |
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| `v288_full_l2_subprincipal_z4.py` | `SEAM.EQUIV.B01` — Route B (DtN): the full-$L^2$ lift of the sub-principal $\mathbb{Z}_4$ block-diagonality, proving (numerically, on the full Toeplitz operator) the 'probably provable' step and lifting v201/v284 from the finite block to $L^2$. [E] character orthogonality: a mark-sourced curvature $f = \sum_{j<4} g(\theta - j\pi/2)$ has Fourier support only on $m \equiv 0 \pmod 4$; [E] full-$L^2$ block-diagonality: $\Lambda = |n| + M_f$ commutes with $\rho = \mathrm{diag}(i^n)$, $\|[\rho, \Lambda]\| \sim 2\times10^{-15}$ on $-50..50$; [E] negative control: a generic curvature gives $\|[\rho, \Lambda]\| = 4.44 > 0$; [E] Lean consistency (`geom_sum_fourth_root` + `markLocal_blockDiagonal`); [O] the sharper residual — why is the raw seam sub-principal term mark-local? |
| `v289_marklocal_raw.py` | `MARKLOCAL.RAW.01` — decompose the one remaining Route-B residual ('why is the raw seam sub-principal term mark-local?') into a 5-lemma chain and isolate the single open analytic lemma. [C] L1 conic parametrix: conic Steklov DtN $= \|D_\theta\| + M_f +$ cone sources (b-calculus/Melrose, imported); [O] L2 **Flat-Away** — the one open analytic lemma: RP+gap+4 marks ⇒ curvature vanishes away from the marks (the flat-pillowcase realisation); [E] L3 orbit-source: the 4 marks form one $\mu_4$ orbit (v195/v216); [E] L4 Fourier-support: $\mu_4$-orbit curvature ⇒ support only on $4\mathbb{Z}$ (v264/v288, with a non-orbit negative control); [E] L5 full-operator: ⇒ $[\rho, \Lambda]$=0 on full $L^2$ (v288). 4/5 discharged; Route B reduces to exactly L2. |
| `v290_z4_smooth_curvature_adversary.py` | `SEAM.ADVERSARY.01` — the Route-B red-team that $\mathbb{Z}_4$ block-diagonality is *not* mark-locality. The strongest adversary, a smooth $\mathbb{Z}_4$-symmetric off-mark curvature $f = \varepsilon\cos(4\theta)$: [E] it is $\mathbb{Z}_4$-symmetric (Fourier support $\{\pm4\} \subset 4\mathbb{Z}$) but *not* mark-local ($f(\pi/4) = -\varepsilon \neq 0$); [E] it *passes* the v288 commutator $\|[\rho, \Lambda]\| = 0$ (same as mark-local) — so $[\rho, \Lambda]$=0 is necessary but *not* sufficient; [E] but it shifts the DtN spectrum off the flat ladder ($\max|\Delta\lambda| = 0.155$) and the heat trace ($\Delta = 0.019$), so the KMS weight and $(E_8)_1$ character shift; [O] a genuine candidate modulus the commutator misses but the spectral data excludes — Flat-Away must use spectral/energy/heat input. |
| `v291_flataway_contract.py` | `FLATAWAY.RP.01` — the Flat-Away lemma as its own named mini-theorem with three proof routes opened by v290. Lemma [O]: given RawRPSeam with gap, chirality and the four $\mu_4$ marks, the smooth curvature density in the DtN sub-principal symbol vanishes away from the branch divisor. [E] *spectral-rigidity*: scanning $f = \varepsilon\cos(4\theta)$, the spectrum deviation from the flat ladder is 0 only at $\varepsilon$=0 — flat is the unique spectral match; [E] *heat-kernel*: the heat-trace deviation is 0 only at $\varepsilon$=0; [C] *RP-energy*: RP+gap is a Dirichlet-energy minimisation in the Troyanov conformal class ⇒ constant-curvature minimiser (Troyanov; cf. v284). [O] three independent routes to one lemma; closing any one closes Route B. |
| `v292_flataway_heat_reduction.py` | `FLATAWAY.HEAT.01` — the Heat-Kernel route made precise: reduces Flat-Away to *one* named spectral invariant. [E] the heat-trace deviation $\Delta\mathrm{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a positive-definite quadratic form in the off-mark curvature ($c_k(t) > 0$ for every $\mathbb{Z}_4$ mode $4k$; positive on all combinations), so $\Delta\mathrm{Tr} = 0 \Leftrightarrow f = 0$; leading invariant $\|f\|_{L^2}$ (Parseval). [O] reduction: Flat-Away ⇐ 'the seam fixes the $a_2$ heat coefficient to its flat value' ⇒ $\int f^2 = 0 \Rightarrow f = 0$ off the marks — one external invariant. |
| `v293_flataway_spectral_hessian.py` | `FLATAWAY.SPEC.01` — the Spectral-Rigidity route upgraded from one direction (v291) to the *full* smooth-$\mathbb{Z}_4$ space. [E] splitting mechanism: the degenerate flat Steklov level $\|n\|$ is split by mode $4k$ at $\|n\|$=$2k$ by $\pm\varepsilon/2$; [E] $S = \sum(\Delta\lambda)^2 \sim 0.50\varepsilon^2$; [E] the Hessian over modes $\{4, 8, 12\}$ has eigenvalues $\{0.497, 0.500, 0.503\} > 0$, so the flat seam is a *strict* isolated minimum over the whole $\mathbb{Z}_4$ space; [E] a non-$\mathbb{Z}_4$ mode breaks the commutator outright. [O] pinning the seam's Steklov spectrum excludes every smooth $\mathbb{Z}_4$ deformation ⇒ Flat-Away. |

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| Script | What it machine-checks *(continued)* |
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| `v294_flataway_rp_energy.py` | `FLATAWAY.ENERGY.01` — the RP-energy / Troyanov route: the flat pillowcase is the unique constant-curvature minimiser at the fixed mark divisor. [E] prescribed divisor + Gauss-Bonnet: $S^2$ with 4 cone angles $\pi$, $\sum(2\pi-\theta)=4\pi=2\pi\chi$; [C] Troyanov/Euclidean-cone uniqueness (Troyanov 1991) excludes smooth off-mark curvature; [E] strictly convex curvature energy $E[f]=\int f^2$, Hessian $2\pi I$ (PD) $\Rightarrow$ unique minimiser; [E] no zero-cost deformation. [O] premise: 'RP+gap realises the energy-minimiser', converging with v292/v293 on one selection principle; closing any one closes Route B. |
| `v295_flataway_a2_exact.py` | `FLATAWAY.A2.01` — the *exact* analytic $a_2$ coefficient, upgrading the v292 positive-definiteness from numerical to a proof. [E] first variation vanishes: $\partial_\varepsilon \mathrm{Tr}\, e^{-t\Lambda(\varepsilon)}|_0 = -t\hat{f}(0)\sum e^{-t\lvert n\rvert}=0$ (zero-mean off-mark curvature, Gauss-Bonnet); [E] convexity $\Rightarrow$ global minimum: $\phi(x)=e^{-tx}$ strictly convex $\Rightarrow \Lambda\mapsto \mathrm{Tr}\, e^{-t\Lambda}$ convex (Klein/Peierls), so $\varepsilon$=0 is the global minimum and $\Delta\mathrm{Tr}\geq 0$ for all $\varepsilon$, $=0$ iff $f=0$; [E] exact $a_2$ kernel via the divided difference of $e^{-tx}$ (Duhamel), matched to numerics to rel.err $<10^{-6}$; [E] degenerate-split closed form $2e^{-2kt}(\cosh(tg/2)-1)\geq 0$; [O] reduction now analytic: Flat-Away $\Leftarrow$ 'the seam fixes $a_2$'. Lean-mirrored (`FORM.FLATAWAY.01`). |
| `v296_flataway_a2_closed.py` | `FLATAWAY.A2.CLOSED.01` — the exact $a_2$ coefficient in *closed form*. [E] the operator tails telescope geometrically ($C_k+D_k e^{-4kt}=4kt(1-e^{-4kt})$), giving $W_k^{\mathrm{tail}}=t(1-e^{-4kt})/(8k(1-e^{-t}))$; [E] a finite $(4k-1)$-term middle (incl. the degenerate diagonal $\Phi(2k,2k)=(t^2/2)e^{-2kt}$); [E] closed form $W_k(t)=\frac{t(1-e^{-4kt})}{8k(1-e^{-t})}+\frac{1}{2}\sum_{m=1}^{4k-1}\Phi(m,4k-m)$ ($W_1(t)=t(2te^t+e^{3t}+3e^{2t}+e^t-1)e^{-3t}/8$), reproducing v295 to machine precision; [E] manifestly positive; [E] small-$t$: $W_k=t/2+O(t^3)$ (mode-independent leading $1/2$, the $t^2$ term cancels). |
| `v297_route_a_literature_stack.py` | `SEAM.EQUIV.A.LIT.01` — Route A's 'one standard import' as a precise citable theorem stack. [E] LIT-A invertible/SRE bulk $\Rightarrow$ trivial bulk topological order (Kitaev 2006; Freed–Hopkins 2021); [E] LIT-B bulk MTC = Drinfeld center, trivial center $\Rightarrow$ holomorphic boundary net (Müger 2003; Kawahigashi–Longo–Müger 2001); [E] LIT-C holomorphic $c$=8 net $\Rightarrow (E_8)_1$ (Conway–Sloane; Dong–Xu/Kawahigashi–Longo). [E] the three compose into 'invertible bulk $\Rightarrow (E_8)_1$', all legs established; [O] the one open hypothesis is the input of LIT-A, 'the raw quasi-free seam bulk is invertible (SRE)' = Route B's Flat-Away. [E] non-circular with Route B (the v286 firewall holds). |
| `v298_e8_network_attractor.py` | *E8 as a network attractor* — a computational-substrate (Wolfram-program) reading of the $(2,3,5)$/McKay hull, on the v219 bedrock; an honest audit, not a derivation. [E] the E8 Kac marks $(1,2,3,4,5,6,4,2,3)$ are the *unique* attractor of the minimal lazy graph update $m\leftarrow(A+2I)/4\cdot m$ on the affine-$E_8/(2,3,5)$ McKay graph (converges from any positive start; $A\,m=2m$, top eigenvalue 2). [E] the spherical McKay tower terminates at $E_8$ ($2T\to E_6$, $2O\to E_7$, $2I\to E_8$, $h=30$); [E] the cascade is nested-subgraph coarse-graining ($240\to 126\to 72\to 40\to 20$). [C] Wolfram reading: $E_8$ as the universal attractor of a minimal $(2,3,5)$ network (Coxeter-skeleton level). [O] a minimal hypergraph *rewrite* whose attractor is the full TFPT structure stays open; $\varphi_0=(4/3)c_3=1/(6\pi)$ is the analytic seed, not an edge fraction. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v299_hypergraph_0d_autonomous.py` | The hypergraph/Wolfram bridge (hardened): the 3-arm $(2,3,\cdot)$ branch type is the irreducible seed on which a local witness-based spectral growth *reaches* $E_8$ as the last finite point ($E_6 \to E_7 \to E_8$, then stops at the affine boundary $\hat{E}_8$) — *not* 'P2 derived'. [E] icosahedron = 3-uniform hypergraph ($\lvert\mathrm{Aut}\rvert = 120 = \lvert I_h\rvert$; the McKay group is the $SU(2)$ binary lift $2I$, a different order-120 group); [E] the $1 \to 4$ subdivision rewrite is equivariant (120 symmetries at every scale); [E] the $2I$ fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine $E_8$ McKay graph; [E] Smith: $\rho \le 2$ selects (affine) ADE; [E] Collatz–Wielandt: $\rho \le 2 \Leftrightarrow$ a positive witness $m$ with $Am \le 2m$ (local); [E] the autonomous rule (diffusion witness + local gate $Am \le 2m$) runs $E_6 \to E_7 \to E_8 \to \hat{E}_8$ under the $(2,3,\cdot)$ seed; [E] $A_n$ ($\rho = 2\cos\frac{\pi}{n+1}$) and $D_n$ ($\rho = 2\cos\frac{\pi}{2n-2}$) controls are $< 2$ for all $n$, never reaching $E_8$; [O] the seed = P2. A second network reading of the $(2,3,5)/E_8$ core (v219/v298). |
| `v300_flataway_rigid.py` | `FLATAWAY.RIGID.01` — the attack on Flat-Away. [E] the flat fingerprint: $\mathrm{spec}(\lvert D_\theta\rvert)$ is the integer ladder, every level $n \ge 1$ of multiplicity exactly 2 (a discrete fingerprint, not a norm); [E] the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (sympy $2\times 2$ $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$), validated vs the full Toeplitz operator; [E] any $g_k \neq 0$ lifts the resonant degeneracy (mult $2 \to 1{+}1$) and changes the spectral multiset, so preserving the flat ladder *with* its mult-2 degeneracies forces every $g_k = 0 \Rightarrow f = 0$ — a discrete obstruction, strictly stronger than the v291/v293 'deviation grows' scan; [E] with the v295 convexity, flat is isolated analytically *and* discretely; [C] the pin derived: the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1 + 248q + \dots)$ has integer $q$-coefficients (248 at level 1; $E_8$ even unimodular $\Rightarrow$ integer weights), and by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam, so $(E_8)_1$ rationality $\Rightarrow f = 0$ (conditional on Route A); [O] Flat-Away carries *zero* new open content beyond Route A — the two `SEAM.EQUIV.01` routes converge on one rationality fact. |
| `v301_route_a_invertible.py` | `SEAM.EQUIV.A.INV.01` — discharge Route A's last open hypothesis ('the raw quasi-free seam bulk is invertible/SRE') via the free-fermion classification; since v300 collapsed Route B into Route A, this is the last unconditional gap of `SEAM.EQUIV.01`. [E] carrier = 16 Majoranas, $c = 8$ ($g_{\mathrm{car}}{+}N_{\mathrm{fam}} = 5{+}3 = 8 = 16/2$, v148); [E] $SO(16)_1 \supset (E_8)_1$: $\dim \mathfrak{so}(16) = 120$ currents $+\,128$ spinor $= 248 = \dim E_8$ (v154); [C] free-fermion invertibility (Kitaev): a gapped Gaussian/quasi-free 2+1D fermion phase (class D) is $\mathbb{Z}$-classified by the chiral Majorana number and is *always* invertible (no intrinsic anyons; topological order needs interactions), so a gapped quasi-free bulk is invertible automatically; [E] the invariant $\lvert\det K\rvert = \#\mathrm{anyons}$ is 1 for $E_8$ (invertible) vs 4 for the gauged $D_8{=}SO(16)$ (anyons), which free fermions cannot produce; [O] LIT-A's 'invertible bulk' now follows from 'gapped 16-Majorana quasi-free bulk' (v148/v160+the gap), so the residual changes from a topological-order statement to the spectral input 'is the quasi-free bulk gapped?'; [O] with v300 the whole open content of `SEAM.EQUIV.01` reduces to the single spectral statement 'the quasi-free seam bulk is gapped'. |

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| Script | What it machine-checks *(continued)* |
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| `v302_seam_gap.py` | `SEAM.EQUIV.GAP.01` — the last lever: after v300/v301 the whole open content of `SEAM.EQUIV.01` was 'the quasi-free seam bulk is gapped'; that gap is *not* a free input but the established Recovery gap of the seam transfer operator. [E] $\mathrm{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ (v160/v162), a simple leading Perron eigenvalue 1; [E] the gap is derived: $\Delta = -\ln((2/3)^6) = 6\ln(3/2) = 2N_{\mathrm{fam}}\ln(3/2) \approx 2.4328 > 0$ (multiplicative gap $1-(2/3)^6 = 665/729$; $6 = 2N_{\mathrm{fam}}$, $(2/3)^6$ the $\mu_4$-deck transfer eigenvalue forced by the carrier); [E] the positivity margin $\Delta - 31/(4\pi^2) = 1.6476 > 0$ (the v76 decoupling margin); [C] by Osterwalder–Schrader / quasi-free clustering a transfer gap $\Delta > 0$ is a mass gap of the reconstructed Hamiltonian; [O] so the last TFPT-internal input is the derived Recovery gap $6\ln(3/2) > 0$, and `SEAM.EQUIV.01`'s residual is a composition of cited theorems (OS/clustering + Kitaev v301 + the v297 AQFT stack) over established TFPT facts — no undischarged TFPT-internal assumption remains, though it stays [O] (not machine-proved end-to-end). |
| `v303_ftransfer_dynamics.py` | `FR.DYNAMICS.01` — the discrete→dynamic lens on `F_transfer`: the main-branch mechanism (a gapped, positivity-preserving relaxation to a *unique* attractor; the local-averaging update → the $E_8$ marks, gap $6\ln(3/2)$) is the shared *shape* of all four instances. [E] F_pole (Koide) *is* the seam dynamics — the source→pole/branch relaxation is the Möbius map with fixed-point multiplier $F' = (2/3)^6 = \lambda_2$, the seam-clock subleading eigenvalue (v82/v54); $\lvert F'\rvert < 1 \Rightarrow$ a unique attractor at the seam rate; [C] F_Boltzmann ($\eta_B$) is a gapped positive contraction (washout $\kappa_f \in (0,1)$, H-theorem; compiler fixes $A_\Lambda = 10$, rate thermal); [C] F_relic (axion) freezes at an adiabatic fixed point (seed $\theta_i = 3\pi/5$, rate cosmological); [E] F_QCD ($m_p/m_e$) is the 1-loop RG flow with the carrier $b_3 = -7$ (asymptotic freedom $\Rightarrow$ the Gaussian UV attractor, $\Lambda_{\mathrm{QCD}}$ generated). [O] Verdict: one shape underlies all four = the main-branch principle; F_pole has the seam rate exactly, the other three share only the shape (external rates), so `F_transfer` is the readout end of the one principle, not a bolt-on — simplicity is preserved by honest fencing (v187). |
| `v304_idg_nonlocal_ghost.py` | `QGAMB.IDG.01` — the KMS spectral-action cutoff suggests a *nonlocal* (infinite-derivative) graviton form factor; under entire analyticity the Stelle ghost is a *truncation artefact* — a [C] narrowing of the red-team `rt_F` perturbative-gravity attack, *not* a closure of `QG.AMB.01` (no new parameter: the cutoff is the v259 seam KMS weight). [E] the local four-derivative propagator $1/(p^2(p^2+M^2)) = (1/M^2)[1/p^2 - 1/(p^2+M^2)]$ has a healthy massless graviton (residue $+1/M^2$ at $p^2{=}0$) and a massive spin-2 mode with *negative* residue $-1/M^2$ at $p^2{=}-M^2$ (the Stelle ghost, reproducing v278); [E] for an IDG operator $p^2 a(p^2)$ with $a(z) = e^{z/M^2}$ entire and nowhere zero, the dressed propagator $1/(p^2 a)$ keeps its *only* pole at $p^2{=}0$ — no ghost (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006); [E] negative control: a polynomial truncation $a(z) = 1 + z/M^2$ (the $R{+}R^2$ order) has a real zero at $z{=}-M^2$ = the ghost pole, so 'entire vs polynomial' is the discriminator; [C] the cutoff is the $\beta{=}1$ seam KMS weight $f(u) = e^{-u}$ (v259), not a choice — truncating $\mathrm{Tr}\, e^{-D^2/\Lambda^2}$ at $a_4$ is what makes the local $R{+}R^2$ and the ghost; [C] **conditional narrowing**: *if* the resummed form factor is entire (Biswas–Mazumdar–Siegel class) *then* the perturbative graviton is ghost-free and `rt_F`'s attack is a truncation artefact; [O] **open**: (i) the trace regulator $f$ is not the kinetic form factor $a(\Box)$ — entire analyticity of the resummation is unproven; (ii) a ghost-free *perturbative* graviton is not the nonperturbative ambient measure — `QGAMB.IDG.01` stays [C]/[O], not a `QG.AMB.01` closure. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v305_witness_independence.py` | Witness-independence / generator-economy audit (structural anti-numerology firewall, companion to v100): every headline integer is a documented image of the single anchor $a=(1,1,2)$ ($e=(4,5,2)$, $p_n=2+2^n$; v23/v228), $\pi$ the only transcendental primitive (`ARCH.CORE.01`) — 13 atoms / 2 free primitives = 6.5× compression made explicit (the recurring $2/3=\lvert\mathbb{Z}_2\rvert/N_{\rm fam}$ and $(2/3)^6$ are *one* ratio); $E_8/g_{\rm car}$/anchor are overdetermined ($\geq 3$ structurally independent witnesses), the 5 pure-seed readouts single-witness (correlated, scored jointly by v100); $\alpha^{-1}$'s '137' is foreign to the anchor family = an independent witness **[E]** |
| `v306_seed_crossval.py` | Leave-one-out cross-validation of the shared seed $\varphi_0$: back-solving $\varphi_0$ independently from the five seed-grammar observables ($\sin^2\theta_{12}$, $\sin^2\theta_{13}$, $\beta$, $\Omega_b$, $\lambda_C$) and error-weighted-averaging reproduces the axiom seed $\varphi_0=1/(6\pi)+48c_3^4=0.0531720$ to 0.04% (one real seed, not a per-observable fit); leave-one-out predicts every held-out observable within $3\sigma$ out of sample (most-tensioned = $\sin^2\theta_{13}$, $\sim 2\sigma$); a data↔formula shuffle explodes the $\chi^2$ by $>100\times$ **[E]** |
| `v307_data_watchdog.py` | Data watchdog: the operational decision pipeline over the frozen registry (v84). Classifies all 20 entries (16 predictions + 2 assigned textures + 2 bands) PASS/WATCH/TENSION/KILL by live $n_\sigma$ vs repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18); no decidable prediction in KILL (theory alive); watch items $\alpha^{-1}$ ($1.9\sigma$), $s_{23}$ ($1.8\sigma$) and the most-tensioned core $\sin^2\theta_{13}$ ($2.0\sigma$); the prediction-of-record $\sin^2\theta_{12}=0.306747$ compatible (JUNO, the fastest falsifier); $r\in[0.0033, 0.0048]$ below the BK18 limit and the kill 0.01; $w=-1$ flagged the dark-energy watchdog (DESI) **[E]** |
| `v308_seam_equiv_chain.py` | TOE attack 1 — the `SEAM.EQUIV.01` keystone as a composition certificate: the v297 stack as one well-typed chain gap$>0$ → invertible → holomorphic $c=8$ → $(E_8)_1$ with exact discriminators $\lvert\det \mathrm{Cartan}(E_8)\rvert=1$ vs $\lvert\det \mathrm{Cartan}(D_8)\rvert=4$, carrier tower $4\cdot4=16\to4\to1$, $c=g_{\rm car}+N_{\rm fam}=8$; the input gap is the derived recovery gap $6\ln(3/2)>0$. The only undischarged premise is the cited OS step — an assembly/reduction certificate (Lean target), NOT a closure **[E][O]** |
| `v309_modular_bedrock.py` | TOE attack 2 — the bedrock $\omega\circ\rho=\omega$ via an explicit modular Hamiltonian, non-circular: marks = 4 from Gauss–Bonnet (v216) + the order-4 clock from cross-ratio 2 (v214) ⇒ $\mu_4$ derived, not assumed; the quasi-free modular Hamiltonian $K=\log((1-C)/C)$ round-trips to the seam operator and $[\rho,K]=0\Leftrightarrow\omega\circ\rho=\omega$ (a $\mathbb{Z}_2$ profile breaks it). Residual = Bisognano–Wichmann intrinsicality of the raw collar; NOT a closure of `QGEO.SYM.01` **[E][O]** |
| `v310_carrier_sm_anomaly.py` | TOE attack 3 — the carrier half-spinor is one anomaly-free SM generation with the SM RG: $16=\Lambda^{\rm even}(\mathbb{C}^5)=1+10+5\to SU(5)\subset SO(10)$ as $10+\bar5+1$ = one generation; gauge-anomaly-free ($\sum Y=0$, $\sum Y^3=0$, $[SU(2)]^2U(1)=[SU(3)]^2U(1)=0$, exact) and reproduces $(b_1,b_2,b_3)=(\frac{41}{10},-\frac{19}{6},-7)$; a fourth family shifts $b_1$, so $N_{\rm fam}=3$ pins the RG. Closes the spectrum/anomaly/$\beta$ sub-question; the full EG $S$-matrix + Yukawa stay open **[E][C]** |
| `v311_gap_decoupled_measure.py` | TOE attack 4 — the physical QG measure as a gap-decoupled, exponentially-clustering object: the transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ gives geometric clustering at rate $(2/3)^6$, correlation length $1/(6\ln(3/2))$, finite susceptibility 729/665; the v76 margin $6\ln(3/2)-31/(4\pi^2)\approx1.648>0$ decouples the physical sector from the un-built ambient. Spectral-action $a_2=\mathrm{EH}/a_4=R^2/72$ (v36) + KMS $\beta=1$ (v259); `QG.AMB.01` stays **[O]** (gap-decoupled, inert) **[E][O]** |

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| Script | What it machine-checks *(continued)* |
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| `v312_hypergraph_substrate.py` | TOE attack 5 — the hypergraph substrate sharpened: the affine-$E_8$ Kac marks are the Perron eigenvector ($A\,m{=}2m$) and the subleading eigenvalue is exactly $2\cos(\pi/5)$= the golden ratio (the 5-fold/icosahedral signature), but the recovery rate $(2/3)^6$ is *not* any eigenvalue and $\varphi_0$ is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight $2/3{=}|\mathbb{Z}_2|/N_{\text{fam}}$ (gap $6\ln(3/2)$) and the analytic seed $\varphi_0$. The open question is precisely delimited, not closed [E][O] |
| `v313_golden_atoms_spectral.py` | The golden ratio is the $g_{\text{car}}{=}5$ signature: the $(2,3,5)$ atoms *are* the affine-$E_8$ network spectral angles (v312 follow-up, exact). [E] the charpoly factors $x(x^2{-}4)(x^2{-}1)(x^2{-}x{-}1)(x^2{+}x{-}1)$, the golden minimal polynomial $x^2{-}x{-}1$ divides it ($\phi{=}2\cos(\pi/5)$ an exact eigenvalue); the non-trivial eigenvalues are $2\cos(\pi/k)$ for $k\in\{2,3,5\}{=}\{|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}\}$ (giving $0,1,\phi,1/\phi$), so the golden ratio is the $g_{\text{car}}{=}5$ (5-fold) signature — a new spectral witness for $g_{\text{car}}$ (raises the v305 multiplicity). The $(2,3,5)$ spherical excess $1/2{+}1/3{+}1/5{=}31/30$ ties the v63/v76 margin numerator $31{=}2^{g_{\text{car}}}{-}1{=}248/8{=}1{+}h$ to the Coxeter 30 (a unification, not new evidence). [E] honest null: $\phi$ is structural, not a readout. [C] direction: $\phi$ is the $E_8$-quasicrystal (Elser–Sloane) ratio, a candidate substrate [E][C] |
| `v314_rate_translation.py` | Do the discrete-kernel and dynamic rates translate? An exact number-field split. The affine-$E_8$ adjacency spectrum splits into the rational part $\{0,\pm1,\pm2\}$ in $\mathbb{Q}$ and the golden part $\{\pm\phi,\pm1/\phi\}$ in $\mathbb{Q}(\sqrt{5})$; the dynamic transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ is entirely rational. [E] the family rates DO translate (the rational angles $2\cos(\pi/2){=}0{=}|\mathbb{Z}_2|$, $2\cos(\pi/3){=}1{=}N_{\text{fam}}$ build the recovery rate $(2/3)^6{=}(|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}}$ in $\mathbb{Q}$), but the carrier 5-fold (golden, $\mathbb{Q}(\sqrt{5})$) has *no* rational dynamic image — a field obstruction, so it is static-only. The shared object is the order-30 Coxeter element ($30{=}g_{\text{car}}(2N_{\text{fam}}){=}5{\cdot}6$); the $\phi$ quasicrystal carries the carrier, not the cusp-weight family dynamics [E][C] |
| `v315_coxeter_coupling.py` | Couple or factorize? The order-30 Coxeter sectors couple as a cyclotomic field (v314 follow-up). [E] GROUP FACTORIZES: $\gcd(5,6){=}1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5\times\mathbb{Z}/6$ (direct product, group-decoupled); [E] EXPONENTS FACTOR: E8 exponents = totatives(30) = totatives(5)× totatives(6), $\phi(30){=}4{\cdot}2{=}8{=}$ rank $E_8$; [E] FIELD COUPLES: $[\mathbb{Q}(\zeta_{30}){:}\mathbb{Q}]{=}8{=}$ rank $E_8$, $\mathbb{Q}(\zeta_{30})$ is the compositum of the carrier $\mathbb{Q}(\zeta_5)$ ($\ni\sqrt{5}$) and the family $\mathbb{Q}(\zeta_3)$; [E] GALOIS $=\mu_4\times\mathbb{Z}_2$: $(\mathbb{Z}/30)^\times=(\mathbb{Z}/5)^\times\times(\mathbb{Z}/3)^\times=\mathbb{Z}/4\times\mathbb{Z}/2$ (order 8, max order 4), $\mu_4{=}(\mathbb{Z}/5)^\times$ the carrier Galois (the seam clock = the carrier golden-field Galois), $\mathbb{Z}_2{=}(\mathbb{Z}/3)^\times$ the family Galois; [E] $\sqrt{5}{=}\phi{+}1/\phi$ = the Gauss sum in $\mathbb{Q}(\zeta_5)$. [C] $30{=}5{\cdot}6$ does not dynamically entangle — it couples as the cyclotomic compositum, the bridge is Galois not dynamical [E][C] |
| `v316_galois_readout.py` | Does the Galois $\mu_4\times\mathbb{Z}_2$ (v315) organize the physical readouts? Yes — the CP phases. [E] the hexagonal CP unit $\rho{=}\zeta_6{=}\zeta_{30}^5$ is the order-6$=2N_{\text{fam}}$ FAMILY factor $c^5$ (the carrier is $\zeta_5{=}\zeta_{30}^6$); [C] $\delta_{\text{CKM,lead}}{=}\pi/3{=}\arg\zeta_6$, $\delta_{\text{PMNS}}{=}4\pi/3{=}\arg\zeta_6^4$ (power $4{=}|\mu_4|$), $\zeta_6^4{=}-\zeta_6$ the sheet flip ($\mu_6{=}\mu_3\times\mu_2$); [E] the Galois $\mathbb{Z}_2{=}\text{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q}){=}\{\zeta_6\mapsto\zeta_6^5{=}\bar{\zeta}_6\}$ IS CP conjugation ($\delta\mapsto-\delta$); [E] the carrier factor $\zeta_5$ (golden $\sqrt{5}$) carries no phase (static); [E] the magnitudes ($\sin^2\theta_{12}{=}1/3{-}\varphi_0/2$, transcendental $\varphi_0$) are not cyclotomic. So the Galois bridge reaches phenomenology through the family factor (phases); the carrier is static; the magnitudes are the analytic seed [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v317_galois_family.py` | Are the 3 generations a $\mu_3$/Galois orbit? Yes ($\mu_3$), refined 1+2 by Galois (v316 follow-up). [E] the 3 cusp weights $\{0, 1/3, 2/3\}$ (Spec$(Q_+)$=$\{1,2,3\}$=$A_3$ exponents) map under $w \mapsto e^{2\pi i w}$ to the cube roots $\{1, \zeta_3, \zeta_3^2\}$; the cyclic family symmetry $w \mapsto w{+}1/3$ is a transitive $\mu_3$ orbit (the family IS the cube-root orbit). [E] the transfer eigenvalues $(1{-}w)^6$=$\{1, (2/3)^6, (1/3)^6\}$ attach to the weights (attractor = eigenvalue 1 at $w$=0). [E] the Galois $\mathbb{Z}_2$ ($\zeta_3 \mapsto \zeta_3^2$, = CP conjugation v316) fixes $w$=0 (rational) and swaps $w$=1/3 ↔ 2/3 ⇒ orbit 1+2; so the Galois-FIXED generation is the recovery ATTRACTOR (the law), the Galois PAIR the decaying hierarchy. [C] the family structure is arithmetic; the mass magnitudes stay the analytic seed [E][C] |
| `v318_arithmetic_capstone.py` | The capstone: the SM structural sector is $\mathbb{Q}(\zeta_{30})$+ Galois $\mu_4 \times \mathbb{Z}_2$, and the complete input is $\{a, \pi, v_{\rm geo}\}$ (synthesis of the v313–v317 arc). [E] the structural sector (3 generations, 2 CP phases, orbit/hierarchy) lives in $\mathbb{Q}(\zeta_{30})$: $[\mathbb{Q}(\zeta_{30}){:}\mathbb{Q}]$=$\varphi(30)$=8= rank $E_8$, 30=$\lvert\mathbb{Z}_2\rvert N_{\rm fam} g_{\rm car}$=$h(E_8)$, Galois $(\mathbb{Z}/30)^\times$=$\mu_4 \times \mathbb{Z}_2$; [E] rigidity: forced by the anchor atoms $\{2,3,5\}$=$e(a)$; [E] magnitude reduction: $\varphi_0$=$(4/3)c_3{+}48c_3^4$ is a pure function of $\pi$ (no free parameters), so the magnitudes are $\{a, \pi\}$-determined, not free; [E] free inventory: 0 dimensionless free parameters, $\pi$ the unique transcendental, $v_{\rm geo}$ the one dimensionful unit. [O] residual: the raw-seam realisation stays open (`QGEO.SYM.01/SEAM.EQUIV.01`) [E][O] |
| `v319_translation_clock.py` | The translation clock: the discrete↔dynamic bridge *is* the order-30 Coxeter element, a clock with two coprime hands $5 \times 6$, read law-inclusive (0..5) or live-only (1..5). [E] the only object holding both the static carrier ($\mathbb{Q}(\sqrt{5})$) and the dynamic family (rational) data is the order-30 Coxeter element (v314); $30 = h(E_8) = g_{\rm car}(2N_{\rm fam}) = 5 \cdot 6$, $\gcd(5,6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (v315). [E] DYNAMIC HAND ($\mathbb{Z}/6 = 2N_{\rm fam}$, ticks 0..5): recovery exponent exactly $6 = 2N_{\rm fam}$, rate $(2/3)^6$; order-6 generator $c^5$ ($\zeta_6 = \zeta_{30}^5$). [E] LAW VS LIVE: the resummed clock rate$(n) = -p_2 \ln(1 - n/N_{\rm fam})$ (v124) has rate$(0) = 0$ (the attractor/law) and live modes from $n$=1; the $E_8$ live phases (totatives of 30) start at 1, so $0, 1, 2, 3, 4, 5$ is law-inclusive, $1, 2, 3, 4, 5$ live-only. [E] STATIC HAND ($\mathbb{Z}/5 = g_{\rm car}$, ticks 1..5): golden $\mathbb{Q}(\sqrt{5})$, field-obstructed (no rational dynamic image) — static-only. [E] one full turn = $\mathrm{lcm}(5,6) = 30 = h(E_8)$ (rank $E_8 = 8 = \varphi(30)$, $\lvert\mu_4\rvert$=4 planes, v55); the "it is one clock" reading is [C]. |
| `v320_galois_cp_relation.py` | A NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\rm PMNS} = \delta^{\rm lead}_{\rm CKM} + \pi$. [E] both phases are powers of one hexagonal unit $\rho$=$\zeta_6$ (v316): $\delta^{\rm lead}_{\rm CKM}$= arg $\rho$=$\pi/3$ (60°), $\delta_{\rm PMNS}$= arg $\rho^4$=$4\pi/3$ (240°), $\rho^4 = -\rho$, so $\delta_{\rm PMNS} - \delta^{\rm lead}_{\rm CKM}$=$\pi$ exactly (the $\mathbb{Z}_2$ sheet flip) — the quark and lepton leading CP phases are NOT independent. [C] prediction of record $\delta^{\rm lead}_{\rm PMNS}$=240° (upgrades the assigned texture to a Galois-forced relation to the measured quark phase); [C] currently within the broad NuFIT 6.0 range; [X] kill test: $\delta_{\rm PMNS}$ robustly away from 240° ($> 3\sigma$; DUNE/Hyper-K/JUNO) falsifies the Galois-CP organization [E][C][X] |
| `v321_killtest_board.py` | The forward kill-test board: the decisive upcoming measurements, ranked (forward companion to v307). [E] each entry is locked to a `freeze_file.csv` kill row: #1 JUNO $\sin^2\theta_{12}$=0.306747 (fastest), #2 CMB-S4/LiteBIRD $r$ (kill $r$>0.01), #3 DESI/Euclid $w$=−1, #4 DUNE/Hyper-K $\delta_{\rm PMNS}$=240° (the new v320 Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering+$m_{\beta\beta}$, #6 PSI n2EDM, #7 JUNO $\sin^2\theta_{13}$ ($\sim 2\sigma$ now), #8 LiteBIRD $\beta$. A forward decision ledger complementing v307; no new physics [E] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v322_cp_lock_sharpened.py` | The v320 CP-lock prediction sharpened into a quantitative kill test. [E] the leading CP phase sits on the six hexagonal nodes $\arg \rho^k = k \cdot 60°$; $\delta^{\text{lead}}_{\text{CKM}}=60°$ (node 1), $\delta^{\text{lead}}_{\text{PMNS}}=240°$ (node 4). [E] the node selector is the deck order: $\delta^{\text{lead}}_{\text{PMNS}}=\lvert\mu_4\rvert\,\delta^{\text{lead}}_{\text{CKM}}=4(\pi/3)=4\pi/3=\delta^{\text{lead}}_{\text{CKM}}+\pi$ ($\rho^4=-\rho$). [C] sub-leading budget: $\delta^{\text{full}}_{\text{CKM}}=\pi/3+3\lambda^2$ ($\sim 68.7°$); the lepton sub-leading is a separate open holonomy term bounded by $\sim 3\lambda^2 \approx 8.7° \Rightarrow$ band $240\pm \sim 9°$. [C] current tension: vs NuFIT 6.0 NO ($\delta_{CP}=212^{+26}_{-41}{}°$) the leading 240° sits at $+1.08\sigma$, band edge $\sim 231°$ at $+0.74\sigma$. [X] kill: the nearest wrong node is 60° away $\gg$ budget; DUNE/Hyper-K ($\sim$ 5–15°) discriminate, $> 3\sigma$ outside kills the lock. [E] an anarchic $\delta_{CP}$ hits the band with $p \sim 5\%$ and node 4 1-in-6 [E][C][X] |
| `v323_bw_geometric_modular.py` | The keystone pushed one step: Bisognano–Wichmann makes the seam modular flow *geometric*, so the $\mu_4$ deck invariance is not an independent axiom. [E] the clock $\rho=\text{diag}(i^n)=\exp(i(\pi/2)L)$ is a geometric rotation ($L=\text{diag}(n)$, $\rho^4=I$, $\mu_4 \subset U(1)_{\text{rot}}$). [E] for a rotation-covariant $C=f(L)$ the modular Hamiltonian $K=\log((1-C)/C)=g(L)$, so $[K,L]=0$ (the modular flow commutes with all rotations) and $[\rho,K]=[\rho,C]=0$ — the clock is a modular symmetry for free. [E] discriminator vs v309: a period-4 curvature preserves the clock but its modular flow is *not* geometric ($[K,L]\neq 0$); BW is strictly stronger. [C] linked bedrock: given the seam is the $(E_8)_1$ chiral net (v308), BW/Hislop–Longo make the vacuum modular flow geometric $\Rightarrow$ `QGEO.SYM.01` ($\omega \circ \rho=\omega$) is downstream of `SEAM.EQUIV.01` + a rotation-invariant vacuum. [O] residual: the raw collar vacuum is rotation-invariant (cleaner than v309, shared with v308) [E][C][O] |
| `v324_hypergraph_fiber.py` | The minimal hypergraph substrate that works: a $(2,3,5)$-network *fibered* by a 3-fold cusp channel reproduces both the $E_8$ skeleton and the recovery gap (the constructive v312 follow-up). [E] network factor: $T_{\text{net}}=(A+2I)/4$ fixes the Kac marks (eigenvalue $1=E_8$ skeleton), subleading $(\varphi+2)/4$ (golden); $(2/3)^6$ is *not* in its spectrum (the v312 negative). [E] cusp fiber: $T_{\text{cusp}}=\text{diag}((1-w)^{2N_{\text{fam}}})$, $w \in \{0, 1/3, 2/3\}$ ($N_{\text{fam}}=3$, v317), spectrum $\{1, (2/3)^6, (1/3)^6\}$. [E] the fibered substrate $T=T_{\text{net}} \otimes T_{\text{cusp}}$ has top eigenvalue 1 (eigenvector marks $\otimes$ ($w=0$), skeleton $\times$ democratic cusp) *and* $(2/3)^6$ as a genuine eigenvalue (network attractor $\times$ cusp subleading). [E] honest injection: without the fiber ($T_{\text{net}} \otimes I_3$) there is no $(2/3)^6$ — the cusp datum is injected, not graph-spectral; substrate = network $\times$ cusp = the v315 split $30=g_{\text{car}}(2N_{\text{fam}})=5 \cdot 6$. [E] minimality: the smallest fiber is the 3-node cusp ($= N_{\text{fam}}$) = order-30 (v319 clock). [C] a consistent fibered realization, not a derivation of 2/3 from a pure rewrite [E][C] |
| `v325_pillowcase_keystone.py` | The seam keystone as ONE citable theorem: the raw seam state *is* the flat $\tau=i$ pillowcase state, with **exactly one** open lemma. IF (raw RP-collar seam state = flat $\tau=i$ pillowcase / rotation-invariant state) THEN 4 square marks + $\mu_4$ deck + $\omega \circ \rho=\omega$ + holomorphic $c=8 \Rightarrow (E_8)_1$, with the carrier recovery gap as input. [E] pieces all exact: $n=2\chi(S^2)=4$ (v216); cross-ratio $2 \Rightarrow j=1728$ (v214/v267); flat orbifold $S^2(2,2,2,2)$, $\chi_{\text{orb}}=0$ (Troyanov); DtN sub-principal at $m \equiv 0 \bmod 4$ (v201/v280); $\det \text{Cartan}(E_8)=1$ vs $D_8=4$ (v83/v143/v308); gap $6\ln\frac{3}{2} > 0$ (v76/v302). [O] the one open lemma — the raw collar state IS the flat $\tau=i$ pillowcase / rotation-invariant state — is SHARED by both bedrock IDs (`QGEO.SYM.01`, `SEAM.EQUIV.01`; v323), machine-pinned in Lean (`FORM.SEAMEQUIV.01` + `FORM.QGEO.BW.01`); an assembly/reduction certificate (like v176/v261), not a closure [E][O] |

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| Script | What it machine-checks *(continued)* |
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| `v326_ftransfer_suite.py` | The $F_{\text{transfer}}$ solver suite: the four transfer readouts as ONE runnable harness, each a gapped relaxation to a unique attractor (productises v303). [E] $F_{\text{pole}}$ (Koide): Moebius contraction to $Q^*=2/3$ with multiplier $(2/3)^6$ = the seam-clock subleading eigenvalue, unique attractor at the SEAM rate. [C] $F_{\text{Boltzmann}}$ ($\eta_B$): washout contraction to the balance $6.1\times10^{-11}$, monotone (H-theorem), external thermal rate. [C] $F_{\text{relic}}$ (axion): the comoving number $N=E/\omega$ is the adiabatic invariant, the relic freezes (seed $\theta_i=3\pi/5$), cosmological rate. [E] $F_{\text{QCD}}$ ($m_p/m_e$): 1-loop RG with carrier $b_0=7$ flows to the Gaussian UV fixed point, $\Lambda_{\text{QCD}}$ generated. [C] $F_{\text{RareKaon}}$: $\text{BR}(K^+)=9.45\times10^{-11}$ on NA62 2016-2024 (v202). [O] only $F_{\text{pole}}$ has the seam rate $(2/3)^6$; the other four share the SHAPE with external rates honestly fenced (v187/v303) — the frontier is measurable without claiming the external rates [E][C][O] |
| `v327_hypergraph_rewrite.py` | Deriving the cusp weight 2/3 from a minimal rewrite (sharpening v324/v312). A minimal local rule — a family channel with $N_{\text{fam}}=3$ slots, one absorbing attractor ($w=0$) and $\lvert\mathbb{Z}_2\rvert=2$ surviving — supplies the cusp fiber. [E] $M$ has spectrum $\{1,2/3,0\}$; $2/3=(N_{\text{fam}}-1)/N_{\text{fam}}=\lvert\mathbb{Z}_2\rvert/N_{\text{fam}}$ EMERGES from the rule arity, not injected. [E] over the order-$6=2N_{\text{fam}}$ hand the rate is $(2/3)^6=64/729$ = the recovery gap (v76/v324), derived. [E] NON-GRAPH-SPECTRAL (proof, sharpening v312): 2/3 is NOT a root of the affine-$E_8$ charpoly ($p(2/3)=-3520/19683\neq0$), so the recovery rate cannot be an adjacency eigenvalue. [E] the absorbing state (eigenvalue 1) is the democratic $w=0$ law (v317). [O] honest residual: the arity $\{2,3\}$ (the anchor atoms) is still the input (P2) — v324's injected datum reduced to the anchor arity [E][O] |
| `v328_theta13_pressure.py` | The $\theta_{13}$ pressure point: the honest weak spot, quantified and pre-registered. [C] $\sin^2\theta_{13}=\varphi_0^{\text{ret}}e^{-5/6}=0.023108$ vs NuFIT 6.0 NO ($0.02195\pm0.00058$) is $+2.0\sigma$ — the most-tensioned CORE prediction (v62/v307). [E] $\theta_{13}$ is the carrier hypercharge trace (exponent $\gamma=\text{tr}_E Y^2=5/6$ EXACT, v268), NOT the seesaw $\mu\tau$ breaking that sets $\theta_{12}$ — distinct channels. [C] no cheap relief: the back-solved seed $\varphi_0^{\text{ret}}(\theta_{13})=0.0505$ is ∼5% below the axiom, PMNS $\theta_{13}$ RG running is $<1\%$ (NO), and 5/6 is exact — the prediction stays frozen (`REG.FREEZE.01`). [X] pre-registered kill: a stable $\sin^2\theta_{13}$ away from 0.023108 at $>3\sigma$ (JUNO/global) flags the carrier-trace/seed-grammar reading; exactly the v306 seed-hyperplane outlier — one honest pressure point [C][E][X] |
| `v329_os_gap_reduction.py` | Closing the OS step: the bulk gap *is* the transfer gap (second quantization), and assembling the single sufficient premise of the whole bedrock. [E] OS GAP = TRANSFER GAP: the many-body OS Hamiltonian $d\Gamma(-\log T)$ has spectrum = sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy $=6\ln\frac{3}{2}$, independent of system size — exactly "OS bulk gap = transfer gap" (the v308 open input), discharged at the many-body level. [E] $H\Rightarrow\omega\circ\rho=\omega$: under $H$ the covariance is rotation-covariant $C=f(L)$, $K=g(L)$ commutes with $\rho=\exp(i(\pi/2)L)$ (v201/v309/v323). [E] $H\Rightarrow$ SRE $\Rightarrow(E_8)_1$: $\det\text{Cartan}(E_8)=1$ vs $D_8=4$ (v237/v308). [E] both bedrock IDs follow from the ONE premise $H$ (Lean-pinned `FORM.SEAMEQUIV.01` + `FORM.QGEO.BW.01`). [O] residual: the necessity of $H$ on the raw collar; sharpens v240/v308/v323/v325, does NOT close the bedrock [E][O] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v330_qgamb_admissible_measure.py` | Constructing the ambient measure on the gap-decoupled admissible sector as a bona fide Osterwalder–Schrader object (with the honest fence). [E] reflection positivity: the seam transfer $T$ (v221/v311; spectrum $\{1,(2/3)^6,(1/3)^6\}$) is symmetric PSD, so the finite-volume measures are RP. [E] exponential clustering: $C(n)/C(n-1)\to\lambda_2=(2/3)^6$ ($\xi=1/\Delta$, $\Delta=6\ln\frac{3}{2}$). [E] tightness ⇒ projective limit: $\chi=\sum_n C(n)=729/665$ finite, so $\mathrm{Var}(S_\Lambda)/\Lambda\to\chi$ ⇒ uniform tightness (Prokhorov) ⇒ the projective limit $\mu=\lim\mu_\Lambda$ EXISTS on the admissible sector. [E] OS reconstruction: $H_{\mathrm{OS}}=-\log T\geq 0$, unique vacuum, gap $6\ln\frac{3}{2}$ (v240). [E] NEG: an ungapped transfer sends $\chi\to\infty$, tightness fails. [O] the full ambient measure: the non-admissible (metric) sector is NOT built; by the gap margin $\Delta-31/(4\pi^2)\approx 1.648>0$ (v76/v275) the physical readouts do not need it, but `QG.AMB.01` stays open. A genuine PARTIAL construction, NOT a closure [E][O] |
| `v331_necessity_of_H.py` | The *necessity* of $H$ reduced: rotation-invariance of the seam DtN is EQUIVALENT to the four marks at the square ($\tau=i$), so the one open lemma is exactly the order-4 CM selection (the necessity companion to v329's sufficiency). [E] square ⇒ $[\rho,\Lambda]=0$: marks at $\{0,\pi/2,\pi,3\pi/2\}$ make the curvature $\mathbb{Z}_4$-symmetric (Fourier support 0 mod 4), $M_f$ couples only modes differing by $4\mathbb{Z}$ (the v201/v210 sufficiency as a DtN $\lvert k\rvert+M_f$ operator). [E] necessity: a mark off the square breaks $\mathbb{Z}_4$ and gives $[\rho,\Lambda]\neq 0$ — equivalence, not just implication. [E] order-4 CM: cross-ratio$(1,i,-1,-i)=2$ ⇒ $j=1728$, the unique 4-config with an order-4 automorphism (v214/v267); $H\Leftrightarrow$ square $\Leftrightarrow\mu_4$. [E] NEG: a $\mathbb{Z}_2$ mark set commutes with $\rho^2$ but not $\rho$. [O] residual: the dynamical *selection* of the square (the 4 marks forced by v216) is the irreducible postulate; does NOT close the bedrock [E][O] |
| `v332_qgamb_metric_sector.py` | The metric sector of `QG.AMB.01` characterized: the obstruction is exactly the conformal-mode (Gibbons–Hawking–Perry) problem, the physical sector is gap-insulated, and the only route is the conditional IDG taming (makes v330's metric-sector fence precise). [E] conformal-mode wrong sign: $c_{\mathrm{conf}}(D)=-(D-1)(D-2)/4$ is negative for $D\geq 3$ ($c_{\mathrm{conf}}(4)=-3/2$), TT positive — the conformal-factor problem. [E] measure obstruction: the kinetic form is indefinite (one negative eigenvalue), so $e^{-S}$ diverges over the conformal mode; the bare metric-sector measure is not directly definable. [E] physical sector insulated: the obstruction is in the conformal/gauge direction, not the admissible sector (positive+gapped, v330), margin $\Delta-31/(4\pi^2)\approx 1.648>0$ (v76). [C] IDG conditional route: the entire form factor $e^{z/M^2}$ (v259/v304) dresses the conformal propagator with no extra pole. [E] NEG: a polynomial ($R+R^2$) truncation crosses zero (the Stelle ghost, v304/v278). [O] metric sector open: the full `QG.AMB.01` needs the conformal mode resolved nonperturbatively (GHP rotation / IDG), gap-decoupled but NOT constructed. A precise characterization, NOT a closure [E][C][O] |
| `v333_cm_selection.py` | The CM-selection mechanism: $\tau=i$ (the square) is the UNIQUE order-4 modulus and the attractor of mark-equilibration, so the last keystone residual is "the deck acts geometrically", not "why the square" (the v331 follow-up). [E] order-4 modular element: $S=\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$ in $\mathrm{SL}(2,\mathbb{Z})$ has $S^2=-I$, $S^4=I$ and fixes $\tau=i$ — the unique order-4 elliptic point of $\mathrm{PSL}(2,\mathbb{Z})$ (the other, $\rho$, is order 3). [E] deck order → CM point: order-4 → cross-ratio 2 → $j=1728$ → $\tau=i$ (square); order-3 → $j=0$ → $\tau=\rho$ (hexagon); $\lvert\mu_4\rvert=4$ selects $\tau=i$. [E] mark-equilibration attractor: 4 points minimizing the log-energy (Fekete) converge to equal spacing (gaps $\pi/2=$ the square), and beat random perturbations — the unique (mod rotation) attractor. [E] NEG: an order-3 deck selects $\rho$, not the square. [O] residual: only "the deck acts *geometrically* (= `QGEO.SYM.01`) vs an abstract $\mathbb{Z}_4$ on labels" stays open — sharpened from "why the square" to "the deck acts geometrically", NOT a closure [E][O] |

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| Script | What it machine-checks *(continued)* |
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| `v334_conformal_resolution.py` | The conformal-mode resolution: the Gibbons–Hawking–Perry contour rotation (IR sign) + the infinite-derivative (IDG) dressing (UV pole) give a CONDITIONALLY convergent metric-sector measure; the residual is the nonperturbative validity of the contour (the v332 follow-up). [E] the obstruction: $c_{\mathrm{conf}}(4) = -3/2 < 0$, the unrotated conformal Gaussian diverges. [E] GHP rotation: $\phi \to i\phi$ flips the sign, $\int e^{-\lvert c\rvert\phi^2} = \sqrt{\pi/\lvert c\rvert}$ FINITE (GHP 1978). [E] IDG dressing: the entire form factor $e^{p^2/M^2} > 0$ (no ghost pole, v304), the dressed per-mode action positive post-GHP and UV-suppressed. [E] NEG: a polynomial truncation crosses zero (the Stelle ghost, v304/v278). [C] conditional construction: GHP+IDG give a convergent metric-sector Gaussian mode-by-mode (conditional on the contour + entire analyticity). [O] residual: whether the GHP-rotated IDG-dressed measure is the CORRECT nonperturbative `QG.AMB` measure (the contour's nonperturbative validity); gap-decoupled (v76), blocks no test, NOT a closure [E][C][O] |
| `v335_seam_equiv_unify.py` | The keystone unification: there is ONE open theorem (`SEAM.EQUIV.01`), `QGEO.SYM.01` is its COROLLARY (a conformal-net axiom), and `QG.AMB.01` is a decoupled general QG problem, NOT a TFPT gate. [E] `QGEO.SYM.01` = corollary: a chiral conformal net has a Moebius-covariant vacuum (the unique invariant positive-energy vector, a NET AXIOM); rotations $U(1) <$ Moebius $\Rightarrow$ rotation-invariant vacuum; the $\mu_4$ clock = rotation by $\pi/2$ fixes it $\Rightarrow \omega \circ \rho = \omega$. So `SEAM.EQUIV.01`⇒`QGEO.SYM.01` with NO extra premise (the v323 "rotation-invariant vacuum" is a net axiom); the two bedrock items collapse to ONE. [E] the one keystone theorem: `SEAM.EQUIV.01` = a construction theorem ($c$=8, $\det \mathrm{Cartan}(E_8)$=1, gap $6 \ln \frac{3}{2} > 0$ derived) with EXACTLY one open continuum lemma. [E] `QG.AMB.01` decoupled: margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0 \Rightarrow$ no TFPT prediction depends on it; the general Euclidean-QG conformal-factor problem (GHP 1978), inherited not TFPT-specific. [E] collapsed status: structural items $2 \to 1$; live residual $= v_{\mathrm{geo}} + G_{\mathrm{net}}$(=`SEAM.EQUIV.01`) $+ F_{\mathrm{transfer}}$. [O] the one residual = `SEAM.EQUIV.01`'s continuum OS reconstruction (Lean-pinned). A synthesis/inventory; does NOT close `SEAM.EQUIV.01` [E][O] |
| `v336_continuum_limit.py` | The ONE open lemma of `SEAM.EQUIV.01`, stated precisely and mapped to the recent rigorous literature – the continuum-limit and OS-reconstruction legs are now CITABLE. [C] continuum leg: Osborne–Stottmeister (*CFT from Lattice Fermions*, CMP **398** (2023) 219, doi:10.1007/s00220-022-04521-8; legacy internal token MMST, on the MMST/Morinelli–Morsella–Stottmeister–Tanimoto OAR framework, arXiv:2010.11121) give the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur → Virasoro, correct $c$); WZW range rank $\le c \le D$. $(E_8)_1$: $c$=8, rank 8, $D$=16 Majoranas ($SO(16)_1 \to (E_8)_1$), $8 \le 8 \le 16$ in range; the collar is a gapped quasi-free (CAR) state. [C] OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms (linear growth) for unitary VOAs incl. lattice VOAs; $(E_8)_1$ is the even self-dual rank-8 lattice VOA. [C] construction + covariance (Adamo–Giorgetti–Tanimoto): arXiv:2506.01008 builds the 2d conformal net of an even lattice and classifies the 2d Heisenberg-net extensions as lattice nets, and arXiv:2508.07109 shows every DHR rep of such a net is automatically positive-energy + diffeomorphism-covariant (covariance a consequence, not an input). [E] target pinned: $c = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8$, $\det \mathrm{Cartan}(E_8)$=1, the $D$=16 Majorana $SO(16)_1$ IS the carrier-16 and the $\mu_4$ extension IS the seam glue. [O] open residual: $c$=8 alone does NOT pin the net ($SO(16)_1$ also $c$=8, det 4), so the one open lemma = the holomorphy discriminator $\det K$=1. A literature-mapping/reduction; does NOT close `SEAM.EQUIV.01` [C][E][O] |

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| Script | What it machine-checks *(continued)* |
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| `v337_decoupling_theorem.py` | THE DECOUPLING THEOREM (consolidated, citable): every TFPT dimensionless readout factors through the gapped admissible sector and is INDEPENDENT of the ambient QG measure – the theorem that makes the `QG.AMB.01` reclassification (v335) honest. [E] gap margin: $\Delta - 2\dim(E_8)c_3^2 = 6\ln\frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$. [E] factorization: every readout (mass ratios, mixings, $\alpha^{-1}$, $R{+}R^2$) is a function of the admissible gapped transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ + the discrete kernel alone. [E] the theorem: margin $> 0 \Rightarrow$ exponential clustering ($\lambda_2{=}(2/3)^6$), finite susceptibility $\chi{=}729/665$, so the ambient contribution is gap-suppressed $\Rightarrow$ no readout depends on the ambient measure $\Rightarrow$ `QG.AMB.01` decoupled. [E] neg control: gap closed $\Rightarrow$ $\chi$ diverges $\Rightarrow$ fails (non-vacuous). [E] corollary: zero frozen kill tests depend on the ambient measure. Consolidates v76/v311/v330/v332; does NOT construct the ambient measure [E] |
| `v338_theta13_budget.py` | `THETA13.BUDGET.01`: the $\theta_{13}$ sub-leading budget vs the current global fit – the $+2\sigma$ pressure point (v328) quantified against BOTH NuFIT 6.0 NO variants and JUNO; neither defuses nor excludes, fences correctly. [C] SK-dependent tension: $\sin^2\theta_{13} = (\varphi_0^{\rm ret})\, e^{-5/6} = 0.023108$ is $+2.0\sigma$ vs the no-SK central (0.02195) and $+1.7\sigma$ vs the with-SK NO best fit (0.02215) – not a clean $2\sigma$. [E] within $3\sigma$: $0.023108 <$ the with-SK $3\sigma$ upper 0.02388 (not falsified). [C] budget: moving to the central needs $-4.1\%$; RG $< 1\%$ and the exponent 5/6 is exact $\Rightarrow$ no freeze-respecting relief (v328). [X] JUNO decides: sub-1% precision turns $1.7$–$2.0\sigma$ into $\sim 6\sigma$ – a near-term pre-registered kill/confirm. One honest pressure point (v306/v328), fenced [C][E][X] |
| `v339_firstprinciples_boundary.py` | The honest first-principles boundary, mapped precisely – answers (a) can $M_{\rm Planck}/v_{\rm geo}$ be computed? and (b) can $F_{\rm transfer}$ be made first-principles? Both NO, with the exact reason per residual. [E] $v_{\rm geo}/M_{\rm Planck}$ FORBIDDEN BY THEOREM (No-Unit, v153): a mass is dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs a scale; over-determined (gravity = dark energy to 0.11%, v274); every dimensionless RATIO is derived, only the unit is not. [E] $m_p/m_e$ structure in place (v262): the carrier $b_3{=}{-}7$ drives the QCD run + the closed electron source; two inputs stay external: $\alpha_s(M_Z)$ and the lattice $C_p{=}m_p/\Lambda^{(3)}{\sim}2.83$. [X] $\alpha_s(M_Z)$ external BECAUSE OF A TENSION: SM content does NOT unify (1-loop spread $\sim$8.8 at $2{\times}10^{16}$ GeV, v246). [C] $C_p$ parameter-free but lattice-only (v35). [C] $\eta_B$/axion = cosmological initial conditions; Koide $Q{=}2/3$ structural [E]. [E] FOUR honest kinds: forbidden-by-theorem / external-parameter-free / external-tensioned / cosmological. A boundary audit, NOT a solution [E][C][X] |
| `v340_As_reconcile.py` | $A_s$ is predicted (not missing), is DIMENSIONLESS (so it does NOT define an absolute mass), and the apparent $-11\sigma$ tension is a reheating-channel question – reconciled with the archived old derivation (TFPT v4.5/v4.6). [E] $A_s = N_\star^2(M_{\rm scal}/\bar{M}_{\rm Pl})^2/(24\pi^2) = N_\star^2 c_3^7/(24\pi^2)$ (v86); $M_{\rm scal}/\bar{M}_{\rm Pl} = c_3^{7/2} = 1.2565{\times}10^{-5}$ is a PURE RATIO, so $A_s$ carries NO absolute scale – it pins $N_\star$, NOT $\bar{M}_{\rm Pl} \Rightarrow A_s$ does NOT define mass (No-Unit v153 stands). [C] the $-11\sigma$ is the slow-vs-fast reheating dichotomy ($N_\star{=}51.44 \to A_s{=}1.764{\times}10^{-9}$; $A_s$-preferred $N_\star{=}56.2$). [C] ARCHIVE: old v4.5/v4.6 used the algebraic $N_\star{=}3/\varphi_0^{\rm ret} - 1{=}55.42$ and reported $A_s{=}2.124{\times}10^{-9}$; at that $N_\star$ the current normalisation gives $2.047{\times}10^{-9}$ ($-1.9\sigma$). [E] nothing lost: the current theory DERIVES $N_\star$ (record band $[50, 60]$, v84). [X] decisive test: fast preheating or slow reheating confirmed [E][C][X] |

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| Script | What it machine-checks *(continued)* |
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| `v341_alpha_quillen.py` | The $\alpha$ fixed point reformulated as the STATIONARITY of a $U(1)$ determinant line, every ingredient identified as an index / heat-kernel coefficient / discriminant form – NOT a free knob (the "derive the FORM" step). $F_{U(1)} = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6 M \ln(1/\varphi_{\text{seam}}) = 0$ in Quillen form: $[\alpha^3 - 2c_3^3\alpha^2]$ (det-line variation) $= [8b_1c_3^6]$ (anomaly counterterm) $\times[\ln(1/\varphi_{\text{seam}})]$ (seam response). [E] $M$=41 a $U(1)$ INDEX (EM Ward, v48): $\frac{4}{5}M = 8b_1$, $b_1$=41/10 the SM 1-loop $U(1)_Y$ beta (v159/v246). [E] $c_3$=$1/(8\pi)$ the GAUSS–BONNET boundary coefficient (8= rank $E_8$, v216); $c_3^{3,6}$ heat-kernel powers. [E] $-5/4 = -q(D_5)$ a DISCRIMINANT form ($q(D_5)$+$q(A_3)$=2, v1/v154); $\varphi_{\text{base}}$=$1/(6\pi)$=$\frac{4}{3}c_3$ carries $q(A_3)$: $c_3/\varphi_{\text{base}}$=3/4. [E] $\delta_{\text{top}}$=$\Omega_{\text{adm}}c_3^4$. [E] the Quillen split holds at the $\alpha^{-1}$=137.0359992 root. [E] NO NEW GATE: $\alpha^{-1}$ stays [E]; v341 SHARPENS the pre-existing [O] origin `EM.WARD.01` (no new $\alpha$ problem). [O] the one still-open part IS `EM.WARD.01`'s residual: the from-first-principles AQFT PROOF that this is the EXACT Quillen determinant functional (the variational principle derived, not assembled). Sharpens v3/v48/`EM.WARD.01` from "fixed point" to "stationarity of a named determinant line" [E][O] |
| `v342_em_ward_heatkernel.py` | The heat-kernel sharpening of `EM.WARD.01`: DERIVES the ORIGIN and STRUCTURE of the $F_{U(1)}$ determinant-line terms from textbook Seeley–DeWitt / Gilkey coefficients, advancing the EM-Ward origin from [C] to [E] for the STRUCTURE; NO new gate. [E] $c_3$=$1/(8\pi)$ the boundary GAUSS–BONNET coefficient (seam = flat pillowcase $\chi$=2, $\oint K$=$4\pi$, v216). [E] the $\alpha^2$ term is the GILKEY gauge-curvature coefficient: the $U(1)$ Laplacian $a_4$ has $30\Omega^2/360 = \frac{1}{12}\Omega^2$ (Gilkey Thm 4.8.16), $\Omega$=$i\alpha F \Rightarrow -(\alpha^2/12)F^2$, so the Calderon $-2c_3^3\alpha^2$ is the gauge-curvature piece, not an ansatz. [E] the $c_3$-ladder $\{0,3,6\}$ is arithmetic (step 3 = 3 channels; 6=$\lvert R^+(A_3)\rvert$ $C_6$ hexagon). [E] multiplicities forced (2=$\lvert\mathbb{Z}_2\rvert$, $8b_1$=(rank $E_8$)$b_1$). [E] $\pi$-power test: $2c_3^3$=$1/(256\pi^3)$ carries $\pi^3 \Rightarrow$ THREE boundary insertions. [C] the exact coefficient needs the seam $F$-normalisation. [O] the cubic $\alpha^3$ Maxwell moment + full identification = `EM.WARD.01` residual (tied to `SEAM.EQUIV.01`). $\alpha^{-1}$ stays [E] (v3) [E][C][O] |
| `v343_four_routes_analysis.py` | The honest investigation of the four black-hole-birth solution routes (A finite causal diamond, B Carlip–Cardy near-horizon CFT, C modular/thermal flow, D self-reproduction attractor) – direct answer to "which closes ALL problems and is 100% provable": NONE individually, but they CONVERGE on the same two irreducible residuals as v335 (`SEAM.EQUIV.01` + the decoupled QG contour). [E] ROUTE A reframes `QG.AMB` to a FINITE thermal trace (de Sitter finite, gapped $\chi$=729/665) but $c_{\text{conf}}(4) = -3/2 < 0$ still needs the GHP contour (v334, [O]); does NOT touch `SEAM.EQUIV.01`. [C] ROUTE B: Carlip+Cardy consistent with $c$=8 but $L_0$ free (ambiguity) and gives $c$ only, not holomorphy $\det K$=1 – inherits `SEAM.EQUIV.01`. [E] ROUTE C: modular flow KMS $\beta$=1 (v239), geometric identification = `SEAM.EQUIV.01` (v335) – a reduction. [E]/[O] ROUTE D: unique Perron–Frobenius attractor (v56) but cyclic dynamics [O] – not a closure. [E] VERDICT: independent closures = 0; irreducible residuals = 2 (= v335); the BH premise makes both more natural but eliminates neither; focus = `SEAM.EQUIV.01` [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v344_detk_synthesis.py` | The bird's-eye synthesis of the ONE open keystone bit, $\det K{=}1$ – a full-spectrum map of `SEAM.EQUIV.01`'s only open residual. [E] $\det K{=}1$ is ONE statement with SIX equivalent faces (holomorphy / homology-sphere link $H_1{=}0$ / one 1-dim irrep / det Cartan $= 1$ / full $\mu_4$ condensation / $\tau{=}i$ CM), unified by $\lvert\det \text{Cartan}\rvert = \lvert H_1(\text{link})\rvert = \#(\text{1-dim irreps})$ across ADE ($A_n \to n{+}1$, $D_n \to 4$, $E_6 \to 3$, $E_7 \to 2$, $E_8 \to 1$) – ONLY $E_8$ gives 1 ($2I$ the unique PERFECT ADE group, v232/v219). [E] THE GENUS-1 OBSTRUCTION: the easy arguments (plane vacuum, gap, free-fermion) are genus-0, necessary but NOT sufficient (v87); $\det K$ is the torus degeneracy = #primaries. [E] THREE genus-1 access routes: R1 continuum-modular (v336), R2 anyon-condensation (v281), R3 hypergraph-homotopy. [C] the HYPERGRAPH route is the most promising fresh angle: the $(2,3,5)$ rewrite attractor IS the affine-$E_8$ graph (v312) and the $E_8$ du Val link is the Poincare sphere ($H_1{=}0$, v232), so $\det K{=}1$ becomes a finite COMBINATORIAL statement (no continuum limit). [O] none closes it; R3's combinatorial-to-geometric bridge is meanwhile discharged to Kronheimer 1989 (v479), so the residual transforms to "the raw seam supplies the ALE/orbifold datum" = the `SEAM.EQUIV.01` content. A synthesis/roadmap; closes nothing [E][C][O] |
| `v345_hypergraph_homotopy.py` | EXECUTE the combinatorial core of route R3 (the hypergraph-homotopy route to $\det K{=}1$, v344): computes explicitly that the $(2,3,5)$-rewrite attractor graph's plumbing link is the POINCARE homology sphere ($H_1{=}0$), only $E_8$ among ADE; R3's combinatorial half is now [E]-complete, the geometric bridge (= `SEAM.EQUIV.01`) stays [O]. Plumbing topology (Brieskorn): link $H_1 = \text{coker}(\text{Cartan})$. [E] SNF($E_8$ Cartan) = identity $\Rightarrow \text{coker}(E_8){=}0 \Rightarrow$ the $E_8$ plumbing boundary is an INTEGRAL HOMOLOGY SPHERE (the genus-1 meaning of $\det E_8{=}1$). [E] only $E_8$: $\text{coker}(A_n){=}\mathbb{Z}/(n{+}1)$, $D_n \to \mathbb{Z}/4$, $E_6 \to \mathbb{Z}/3$, $E_7 \to \mathbb{Z}/2$ – all nontrivial, only $E_8{=}0$. [E] $\pi_1{=}2I{=}SL(2,5)$ PERFECT (commutator subgroup order $120{=}\lvert G\rvert$, computed) $\Rightarrow H_1{=}0 \Rightarrow$ THE Poincare sphere; $\lvert 2I\rvert{=}120{=}\lvert\mu_4\rvert\, h(E_8)$. [O] does NOT close $\det K{=}1$: the graph→geometry half is meanwhile Kronheimer-cited (v479); what stays open is the raw seam supplying the ALE/orbifold datum = `SEAM.EQUIV.01` [E][O] |
| `v346_seam_geometric_bridge.py` | The geometric bridge as an explicit six-link chain raw-seam $\Rightarrow \det K{=}1$, with the ONE open arrow pinned – the honest capstone of the `SEAM.EQUIV.01` investigation. L1 [E]/[C] raw seam $\to \mu_4$ + four marks (v216); L2 [O] $\mu_4$ + $(2,3,5)$ atoms $\to 2I$ orbifold $\mathbb{C}^2/\Gamma$ (THE OPEN ARROW = `SEAM.EQUIV.01`); L3 [E] $2I \to$ affine $E_8$ McKay (v219); L4 [E] $2I \to E_8$ du Val (v232); L5 [E] $\mathbb{C}^2/2I \to$ Poincare link $H_1{=}0$ (v345); L6 [E] $H_1{=}0 \to \det K{=}1$. [E] 5 of 6 links are theorems, only L2 open. [E] the group is FORCED by the seam atoms: axis signatures $(2,3,3)/(2,3,4)/(2,3,5) \to 2T/2O/2I$, $(2,3,5)$ unique to $2I$, and $(\lvert\mathbb{Z}_2\rvert, N_{\text{fam}}, g_{\text{car}}){=}(2,3,5)$ ARE those axes. [E] downstream of L2 forced. [O] L2 (the seam IS an ADE orbifold point) = `SEAM.EQUIV.01`, NOT closed – the keystone reduced to one geometric-realisation arrow, group pinned by $(2,3,5)$ [E][C][O] |
| `v347_seam_closure_modes.py` | The closure-mode classification of the one open arrow L2 (the seam-orbifold realisation, v346) – a direct honest answer to "is there ANY other way to FULLY solve it?". [E] THE PRECISE LOCUS: the seam gives the pillowcase base $S^2(2,2,2,2)$, $\chi_{\text{orb}}{=}0$ (EUCLIDEAN, v216), but the $2I/E_8$ structure is the Seifert base $S^2(2,3,5)$, $\chi_{\text{orb}}{=}1/30$ (SPHERICAL); L2 must bridge a FLAT 4-marked base to a SPHERICAL 3-marked base – why L2 needs the carrier $(2,3,5)$ input. [E] FOUR CLOSURE MODES: A construct (the analytic wall, v336); B rigidity (unique realisation $\Rightarrow \mathbb{C}^2/2I$ forced, the only theorem-route w/o new input, v284/v300/v331); C axiom $P_3$ ($\Rightarrow$ TFPT complete rel. 3 axioms); D empirical. [E] no-new-state (v246) blocks deriving L2 from extra physics $\Rightarrow$ only rigidity (B) closes it as a theorem. [E] VERDICT: no free theorem; only B (theorem) or C (axiom), validated by D; only these four modes. [O] NOT closed [E][O] |

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| Script | What it machine-checks *(continued)* |
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| `v348_seam_rigidity_route.py` | Route B (the rigidity route) for the one open arrow L2, carried out – it lands on a strikingly SIMPLE statement: the whole keystone reduces to ONE question, does the raw seam carry the golden ratio? [E] rigidity closes the crystallographic part: the order-4 clock has a UNIQUE normal form $z \mapsto iz$ (Nielsen/Kerckhoff, v180), Troyanov gives the unique flat pillowcase metric (v284). [E] downstream is a LATTICE identity: $E_8$ = the ring of icosians (Conway–Sloane), rank 8, so $2I \to E_8$ is a lattice identity (stronger than McKay v219). [E] the single bridge is $\varphi$: crystallographic orders $\{1,2,3,4,6\}$, the 5-fold non-crystallographic; $\mu_4$ crystallographic, $2I$ (5-fold) not, $\varphi = 2\cos(\pi/5)$ extends one to the other (quasicrystal). [O] the keystone reduces to ONE question: "does the raw seam carry $\varphi$?" – if yes, $\mu_4{+}\varphi \to 2I \to E_8$ closes L2. [O] honest subtlety: TFPT's $\varphi$ is currently $E_8$-side (v312/v313); the raw-seam data (cusp $\{0, 1/3, 2/3\}$, recovery $(2/3)^6$, seed $\varphi_0$) are not manifestly golden, so showing the RAW seam carries $\varphi$ non-circularly is the residual (= L2). NOT closed [E][O] |
| `v349_raw_seam_golden_test.py` | The honest test of the v348 question "does the RAW seam carry the golden ratio (before assuming $E_8$)?". Answer, computed: NO – the bare carrier ($D_5$, $A_3$) and the dynamical seam data are NOT golden; the golden ratio genuinely enters WITH the icosahedral/$E_8$ identification. A NEGATIVE result that prevents a false closure. [E] RAW CARRIER NOT GOLDEN: golden needs $5 \mid h$; $D_5$ has $h$=8 ($2\cos(2\pi/8){=}\sqrt{2}$), $A_3$ has $h$=4 – no 5-fold, bare carrier gives $\sqrt{2}$. [E] DYNAMICAL DATA NOT GOLDEN: cusp $\{0, 1/3, 2/3\}$ rational, recovery $(2/3)^6$ rational, seed $\varphi_0$ transcendental. [E] GOLDEN IS THE ICOSAHEDRAL/$E_8$ SIDE: $5 \mid h$ only for $A_4{=}SU(5)$ ($h$=5), $H_3$ ($h$=10), $E_8$ ($h$=30). [E] $\varphi$ alone insufficient: $2I \neq C_5 \times \mu_4$ ($20 \neq 120$). [O] RESOLUTION: L2 reduces to "is $g_{\mathrm{car}}$=5 a PENTAGON (5-fold ⇒ golden ⇒ icosahedral) or a COUNT (rank $D_5 \Rightarrow \sqrt{2}$)?" – the golden ratio is the pentagon-reading of P2, NOT derivable from P2-as-a-count; no hidden $\varphi$. Foundational: P2-as-pentagon closes it (mode C) else rigidity must produce the 5-fold (mode B); NOT closed [E][O] |
| `v350_bootstrap_inputs_correction.py` | The honest CORRECTION of v349's framing (prompted by the right objection: the inputs are not axioms, they are back-determined by the theory). [E] THE INPUTS ARE BOOTSTRAP-FORCED, NOT FREE AXIOMS (v6): $g_{\mathrm{car}}$=5 over-determined three ways (rank-fill, Coxeter-match $g_{\mathrm{car}}$= max prime of $h(E_8)$=30, Pascal). [E] $g_{\mathrm{car}}$=5 IS THE ICOSAHEDRAL 5-FOLD, not the $D_5$ rank: $g_{\mathrm{car}}$= max prime of $h(E_8)$=30=2·3·5; the atoms $(2,3,5)$ ARE the primes of the icosahedral Coxeter number 30. [E] THE GOLDEN RATIO IS EMERGENT FROM THE GLUE, not external: $D_5$ ($h$=8)/$A_3$ ($h$=4) non-golden alone, but the $\mu_4$ extension $D_5{\oplus}A_3 \to E_8$ (v154) gives $h(E_8)$=30 (golden, $5 \mid 30$) – the seam's own $\mu_4$ produces the 5-fold; v349 checked the un-glued pieces, the wrong object. [E] v349's pentagon-vs-count dichotomy was FALSE (the bootstrap forces the pentagon). [O] residual relocates: the golden is bootstrap-forced; the only open thing is the physical continuum realisation (v336), NOT the golden ratio. Caveat: self-consistency within the framework, not from-nothing. Supersedes v349's interpretation [E][O] |
| `v351_continuum_realisation_sharpened.py` | The continuum realisation sharpened – the "$c$=8 ambiguity" flagged open in v277/v344 ($E_8$ vs $SO(16)$ both $c$=8) is RESOLVED, non-circularly, by the seam's ORDER-4 $\mu_4$ clock. [E] $E_8$ (det 1, holomorphic) vs $D_8{=}SO(16)$ (det 4). [E] the order-4 clock (the four Gauss–Bonnet marks, $\lvert\mu_4\rvert{=}4{=}h(A_3)$, v216): index-4 extension $D_5{\oplus}A_3{=}E_8$ (v154), index-2 = $SO(16)$; so order-4 selects $E_8$ (16→4→1) not $SO(16)$ (16→4) – the order (4 not 2) is the discriminator. [E] bootstrap agrees: $h(E_8)$=30 (max prime 5) vs $h(D_8)$=14 (max prime 7). [E] which-net ambiguity RESOLVED ($\det K{=}1$ is then a consequence). [O] residual = PURELY the chiral-edge/scaling-limit existence (bulk-edge, v336), net+holomorphy pinned. Does NOT close `SEAM.EQUIV.01` [E][O] |

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| Script | What it machine-checks *(continued)* |
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| `v352_framework_irreducible.py` | The framework reduction – the deepest answer to "the inputs are not axioms, they are back-determined". [E] P2 ($g_{\rm car}$=5) FORCED three ways (v6): rank-fill, Coxeter-match ($g_{\rm car}$= max prime of $h(E_8)$=30), Pascal. [E] P1 ($c_3$=1/(8π)) FORCED: $c_3$ is the boundary Gauss–Bonnet coefficient (v216/v342), the 8 over-determined four ways (8= rank $E_8$=$h(D_5)$=$\phi(30)$= det $R$, v54/v6); only $\pi$ is non-forced (the boundary transcendental). [E] the $E_8$ hull is forced (unique even unimodular rank-8, v83/v154). [E] so the ONLY irreducibles are the FRAMEWORK (a discrete reflection-positive boundary with a finite Clifford carrier) + $\pi$ – the precise content of "the inputs are not axioms". [O] the framework is the META-AXIOM (not math-reducible, the foundational stance, tested by its consequences not proved). TFPT's true input is ONE framework + $\pi$ [E][O] |
| `v353_selfloop_capstone.py` | The bird's-eye rethink (refining v352): TFPT is not a LINEAR theory (axioms → theorems) but a CLOSED SELF-CONSISTENT LOOP whose outputs fix its own inputs, with ZERO free adjustable dimensionless parameters. [E] IT IS A LOOP: outputs fix inputs ($g_{\rm car}$=5= max prime of $h(E_8)$=30; the 8 in $c_3$= rank $E_8$) and inputs fix outputs; the arrows close into a circle (v6) with a unique fixed point (v56). [E] ZERO FREE DIALS: every load-bearing number is forced/over-determined; free dimensionless parameters = 0. [E] $\pi$ is a FIXED geometric constant (the boundary 2-sphere, $\oint K$=4π ⇒ $c_3$=1/(8π)), not a dial. [C] refining v352: $\pi$ is not a free input and the framework is a SELECTION PRINCIPLE (the unique self-consistent loop), not a free meta-axiom. [O] the one residual is the CONTINUUM closure (the chiral-edge existence, v336/v351); structural-realism (loop complete) vs constructivism (continuum open) is a philosophical, not numerical, fork [E][C][O] |
| `v354_e8_reverse_audit.py` | The REVERSE numerology audit + what the golden ratio means. The forward rule (v305) is one-directional (every readout must map onto $E_8$); the reverse question: how much $E_8$ structure maps onto NOTHING? [E] $E_8$ has 8 Casimir invariant degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$; THREE feed a readout (2 metric, 8 rank→ $c_3$, 30 Coxeter→ $g_{\rm car}$) and FIVE are UNMAPPED $\{12, 14, 18, 20, 24\}$ – 3/8 load-bearing, 5/8 not. [E] ECONOMY, NOT PROMISCUITY: a small fixed set (rank 8, $h$=30, det 1, $D_5\oplus A_3$, 16-spinor) generates ALL readouts; the higher Casimirs are never reached into to fit – the anti-numerology signature (economy + a large untouched remainder). [E] the golden ratio is UNMAPPED (the affine-$E_8$ attractor eigenvalue v312, the icosian lattice v348; no frozen readout is $\varphi$, the sole [C] exception $\cos(\theta_i{=}3\pi/5){=}-1/(2\varphi)$ on the axion spine branch, refined in v429) ⇒ NOT numerology (which needs a number fitted to an observation); $\varphi$ is the algebraic signature of the 5-fold $h$=30=2·3·5, the carrier announcing it is icosahedral. [O] the 5/8 unmapped is the HULL OVERHEAD ($E_8$ = minimal unimodular hull, not the world group); where future readouts could live, or over-capacity. The reverse complement of v305 [E][O] |
| `v355_e8_unmapped_bandwidth.py` | A disciplined bandwidth search of the unmapped $E_8$ region (next step after v354), run with a STRICT discriminator: a candidate counts only if it is a FORCED identity (a theorem about $E_8$), never a numerical coincidence – because mining a rich object for matches is the numerology temptation. [E] FORCED SET-LEVEL (the real collective content): $\sum$deg = 128 = dim $S^+$ the spinor half of 248=120+128; $\sum$exp = 120 = #pos roots; $\prod$deg = $\lvert W(E_8)\rvert$; exponents = the $\varphi(30)$=8 totatives of 30 (v223); max deg = 30 = $h(E_8)$. [C] SUGGESTIVE not forced: 12=$h(E_6)$, 18=$h(E_7)$, 30=$h(E_8)$ (the McKay ladder v219), but 'deg($E_8$) contains subalgebra Coxeter numbers' is not a theorem – flagged. [O] DECLINED: the per-degree atom matches (14= dim $G_2$, 20= det $L$, 24=$\lvert W(A_3)\rvert$, 12=$\lvert R(A_3)\rvert$, 18=$p_4(a)$; v66 flagged 12, 18 fittable) each admit multiple readings, and the $\lvert W\rvert$ prime 7='scalaron exponent' is one of many – declined. [E] RECONCILES v66 (degrees=atoms) and v354 (only 2, 8, 30 primary): NO new forced physical hit; the disciplined decline IS the anti-numerology result [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v356_continuum_mmst_applicability.py` | Direction A: the continuum chain of `SEAM.EQUIV.01` assembled theorem-by-theorem, residual reduced to ONE lattice-realization input tied to the seam's one-sidedness. Chain: gapped (v302) quasi-free 16-Majorana collar (v155/v160) → invertible (Kitaev, v301) → the NONTRIVIAL invertible phase ($c_-$=8) → a non-gappable anomalous chiral edge (anomaly inflow: edge EXISTENCE follows, reducing the v336/v351 'does the edge exist?' residual) → chiral-CFT scaling limit (MMST v336, 8≤8≤16) → order-4 $\mu_4$ clock + AQFT stack pin $(E_8)_1$ (v351/v297). The only non-theorem link is the seam one-sidedness = the $c_3$ origin ($c_-\neq 0$ = one-sided = the $\lvert\mathbb{Z}_2\rvert \oint K$=8 of $c_3$=1/(8π), v58). [C]/[E]/[O]; ADVANCES `SEAM.EQUIV.01` to one realization input, does not close. (Direction B was found already closed by v40.) [C][E][O] |
| `v358_grav_entropy_equilibrium.py` | The entanglement-equilibrium derivation (Jacobson 2015 / Faulkner et al., $\delta S = \delta\langle K\rangle$) with TFPT's atoms ⇒ the LINEARISED Einstein equation falls out PARAMETER-FREE. [E] $G_{ab} = (1/c_3)T_{ab}$, $1/c_3 = 8\pi$ fixed (Unruh $1/(2\pi)$=$4c_3$, EH $1/(16\pi)$=$c_3/2$, Einstein $8\pi$=$1/c_3$, Bekenstein $1/4$=$1/\lvert\mu_4\rvert$). [E] THERMO=GEO coincidence (new): the first law's $2\pi/\eta$ = the geometric $\lvert\mathbb{Z}_2\rvert 2\pi\chi$ iff $\lvert\mu_4\rvert$=$\lvert\mathbb{Z}_2\rvert\chi$=4 – the thermodynamic and geometric origins of $c_3$ are ONE. [C] J3 solved: the CHM ball modular Hamiltonian (boost via BW v323) gives the matter flux $\delta\langle K_B\rangle = \frac{8\pi^2 R^4}{15}\delta\langle T_{00}\rangle$. [E] entropy density atom-fixed: $1/4 = 1/\lvert\mu_4\rvert$ + central charge $c = g_{\rm car} + N_{\rm fam} = 8$ (form v59). [O] residual: the equation-of-state status + the global ambient measure (C7) + the $v_{\rm geo}$ unit (v359 closes the FULL covariant equation); no free dimensionless gravity dial remains [E][C][O] |
| `v359_grav_nonlinear_einstein.py` | The FULL covariant Einstein equation, parameter-free (Direction 1, extends v358). The Jacobson-2015 fixed-VOLUME entanglement equilibrium brings in the EINSTEIN TENSOR $G_{ab}$ (not just $R_{ab}$) – the unique divergence-free metric tensor (Lovelock) – so equilibrium yields the full covariant $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with conservation built in. [C][E] fixed-volume → Einstein tensor (trace = $-R$ in $d$=4). [E] Lovelock ⇒ $\nabla^a T_{ab}$=0 is an output. [E] coeff 1: $8\pi = 1/c_3$ fixed (v358). [E] coeff 2: Λ from $\alpha$ – $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$, prefactor $(8\pi)^2 \cdot 48 c_3^4 = 3/(4\pi^2)$ (v60); BOTH coefficients TFPT-fixed. RESULT: the original B6 full covariant field equation is now parameter-free at the LOCAL level. [O] residual: the equation-of-state status + the global ambient measure (C7) [E][C][O] |
| `v360_grav_gap_corrections.py` | Direction 2: the gap-induced higher-curvature correction to GR – the disciplined result. Does the seam transfer gap give a NEW forced near-term GR-deviation beyond the known $R^2$ scalaron? With the v354/v355 discriminator: NO. [E] the leading higher-curvature term IS the $R^2$ scalaron ($a_4 = R^2/72$ v36; $M_{\rm scal}^2/\bar{M}_{\rm Pl}^2 = c_3^7$, exp 7=$g$+$N$−1) – already in TFPT, already falsifiable via $n_s = 1 - 2/N_\star$, $r = 12/N_\star^2$ (CMB-S4). [C] it is the entanglement-equilibrium next order (Wald first law → $f(R) = R + R^2$, v28/v359). [E] the gap $\Delta = 6\ln(3/2)$ is dimensionless, so a distinct gap term sits at $\xi\, v_{\rm geo}$ ∼ seam/Planck scale, not near-term observable. [O] DECLINE: a new dimensionless coefficient is not atom-forced (needs heat-kernel/Wald data; exp 7 vs 6, $c_3^7$ vs $(2/3)^6$ unrelated) → per v354/v355 declined. No new near-term forced prediction; the answer is the scalaron [E][C][O] |
| `v361_grav_backreaction.py` | Direction 5: the matter-gravity backreaction with the explicit J3 flux – two forced facts + an honest decline. [E] the carrier's gravitational anomaly is FORCED: $c_-$=8 (16 Majoranas = $\dim S^+ = 2^{g_{\rm car}-1}$; chiral central charge 16/2=8=$g_{\rm car}$+$N_{\rm fam}$, the same $c_-$ as v358). [E] the backreaction is FINITE → Λ from $\alpha$: the matter vacuum energy through $G_{ab} = c_3^{-1} T_{ab}$ is gap-suppressed (Decoupling Thm v337) to $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$ (v60), 123 orders = 119.03+3.92 – NOT the $M_{\rm Pl}^4$ catastrophe; the loop (v359+v60+carrier vacuum) closes. [E] J3 explicit (v358): $\delta\langle K_B\rangle = (8\pi^2 R^4/15)\delta\langle T_{00}\rangle$, carrier $T_{ab}$=the SM's (v159). [C] SM trace anomaly reproduced (content=SM). [O] DECLINE: no new atom-forced read-out of individual SM quantum numbers beyond $c_-$=8 and Λ → per v354/v355 declined. Finite & consistent backreaction, no new dial [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v364_vgeo_sharpen.py` | Direction 6: $v_{\text{geo}}$ is the SINGLE dimensionful input of the WHOLE theory – confirmed across matter AND gravity – and irreducibly primitive. The No-Unit Theorem (v153) showed the dimensionless compiler cannot output a scale ($v_{\text{geo}}$ a forced unit). SHARPENING after the parameter-free gravity (v358/v359/v361): the gravity sector adds NO new scale either – $1/c_3{=}8\pi$ and the $\Lambda$ prefactor $(8\pi)^2{\cdot}48c_3^4 = 3/(4\pi^2)$ are dimensionless, so $1/G \sim v_{\text{geo}}^2$ and $\Lambda \sim (\text{dimensionless})v_{\text{geo}}^4$ reduce to $v_{\text{geo}}$ times atoms. [E] FINAL TALLY: every dimensionful quantity = (dimensionless ratio)×(power of $v_{\text{geo}}$); free content $= \{v_{\text{geo}}, \pi\}$, ZERO dimensionless dials, vs the SM's ∼26. [O] $v_{\text{geo}}$ irreducibly primitive (No-Unit; $1/G$ UV-sensitive Sakharov v68). Clarity, not a new derivation; the last dimensionful anchor named [E][O] |
| `v365_qg_oneloop_saddle.py` | Direction 3: the QG measure (gate C7/`QG.AMB.01`) as one-loop GAUSSIAN fluctuations around the PARAMETER-FREE saddle (v358/v359). [E] the fluctuation stiffness is $1/c_3 = 8\pi$ (no free dial), so the one-loop problem is well-posed (it was NOT before v358). [E] the fluctuation operator is gapped $M \geq \Delta_{\text{eff}} = 6\log\frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) ⇒ mode-by-mode Gaussian convergence. [E] finite model log-det $\mathrm{Tr}'\log M = 6\log\frac{9}{2}$ on the OS spectrum $\{0, 6\log\frac{3}{2}, 6\log 3\}$. [C] the conformal mode converges on the GHP contour ($\int e^{-|c|\phi^2} = \sqrt{2\pi/3}$, v334). [E] the admissible projective limit exists ($\chi = 1/(1-(2/3)^6) = 729/665$, v330). [E] 0 new dimensionless dials (v364). [O] residual: the full non-perturbative projective limit (G6/`QG.AMB.01`) on the ambient sector. Sharpens C7 to a one-loop Gaussian problem around a parameter-free saddle; does NOT close it. [E][C][O] |
| `v366_mmst_seam_collar.py` | Direction 7: the seam collar is IN MMST's free-lattice-fermion scaling-limit class, so the chiral scaling limit is a chiral CFT of central charge 8 pinned to $(E_8)_1$ – reducing `SEAM.EQUIV.01` to the single S3 lattice-realisation input. Osborne–Stottmeister (arXiv:2107.13834, CMP **398** (2023) 219; legacy internal token MMST, on the MMST/Morinelli–Morsella–Stottmeister–Tanimoto OAR framework): free lattice fermions (CAR quasi-free) → chiral-CFT massless scaling limit with correct $c$, range $\text{rank} \leq c \leq D$. [E] class arithmetic: $D{=}\dim S^+{=}2^{g_{\text{car}}-1}{=}16$ Majoranas, $c{=}g_{\text{car}}{+}N_{\text{fam}}{=}8$, $\text{rank}\, E_8{=}8 \Rightarrow \text{rank} \leq c \leq D$ is $8 \leq 8 \leq 16$ (in MMST range). [E] gapped ($\Delta{=}6\log\frac{3}{2}{>}0$, derived v302). [C] quasi-free CAR class (v155/v160). [E] chirality $c_-{=}D/2{=}8 \neq 0$ (non-gappable anomalous edge, v356 S4). [C] the MMST scaling-limit theorem applies (in-class & gapped & chiral & in-range) ⇒ chiral CFT $c{=}8$ (v336). [E] target pinned to $(E_8)_1$ by $c{=}8$ + the order-4 $\mu_4$ clock ($\det K{=}1$ vs $SO(16)$ $\det K{=}4$, v351). [O] residual S3: the collar realised as a genuine lattice chiral free-fermion invertible phase (one-sidedness/Flat-Away, v297/v356). Verifies MMST in-class; ADVANCES `SEAM.EQUIV.01`, does NOT close it. [E][C][O] |
| `v367_seam_s3_lattice.py` | Track 1: the EXPLICIT lattice realisation of the S3 input – a concrete gapped chiral-Majorana $(p+ip)$ lattice model whose edge is the 16-Majorana $SO(16)_1$ carrier, $\mu_4$-condensed to $(E_8)_1$; turns the abstract S3 residual (v356/v366) into a RUNNABLE model with a numerically computed topological invariant. [E] bulk gapped (band gap $2 > 0$ at $M{=}1$). [E] non-trivial Chern number (Fukui–Hatsugai–Suzuki: $\lvert C\rvert{=}1$ at $M{=}1$, $C{=}0$ at $M{=}3$ trivial control). [E] $c_-{=}8$: each chiral Majorana edge carries $c_-{=}C/2{=}1/2$, $N_{\text{Maj}}{=}\dim S^+{=}2^{g_{\text{car}}-1}{=}16$ copies ⇒ $c_-{=}16/2{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$. [E] edge K-matrix $SO(16)_1$ (det Cartan $D_8{=}4$) → $(E_8)_1$ (det Cartan $E_8{=}1$) by the order-4 $\mu_4$ clock (v351/v369). [C] the $\mu_4$ condensation $SO(16)_1 \to (E_8)_1$ (v154/v351). [O] residual: the continuum scaling-limit existence (MMST v336). Reduces S3 to a runnable lattice model; does NOT close `SEAM.EQUIV.01`. [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v368_seam_s3_inflow.py` | Track 1: the bulk-edge / anomaly-inflow leg (S4 of the v356 chain) made CONCRETE on the v367 lattice model – a finite STRIP of the gapped $p+ip$ model has a realised, non-gappable chiral edge, so edge EXISTENCE follows from the invertible bulk (anomaly inflow), NOT a separate assumption. [E] topo in-gap edge states ($M$=1 strip $\min_{k_x}\lvert E\rvert \sim 0$) vs trivial clean gap ($M$=3, neg control); [E] the near-zero state is edge-localised (weight within 4 sites of a boundary, edge≫bulk); [E] bulk-edge correspondence (edge branch count = bulk Chern $\lvert C\rvert$=1) ⇒ 16-Majorana edge, $c_-$=8. [C] anomaly inflow: a $c_- \neq 0$ invertible bulk cannot have a gappable edge ⇒ discharges the v336/v351 'does the edge exist?' residual on the lattice. [O] residual: the continuum scaling limit (MMST v336). Establishes edge EXISTENCE on the lattice; does NOT close `SEAM.EQUIV.01`. [E][C][O] |
| `v369_qgamb_redundancy.py` | Track 2 (next-plan v2): the AMBIENT REDUNDANCY statement – the holographic route to Gravity-complete that SIDESTEPS building the general Euclidean QG measure (C7/`QG.AMB.01`). Instead of constructing the non-perturbative ambient measure (the conformal-factor swamp, v332/v334), assemble the established discriminators proving the ambient sector carries NO physical d.o.f. relative to the holomorphic boundary algebra, so `QG.AMB.01` is a non-fundamental CERTIFICATION object (like a coordinate choice in GR), not missing dynamics. [E] HOLOMORPHIC ⇒ DHR = Vec: #DHR primaries = $\lvert\det \text{Cartan}\rvert = 1$ for $E_8$ (ONE primary = vacuum) vs 4 for $SO(16)/D_8$ – no nontrivial superselection charges ⇒ no hidden boundary sector a bulk could carry invisibly. [E] NO TORUS GSD: ground-state degeneracy on $T^2$ = #anyons = $\lvert\det K\rvert = 1$ for $E_8$ ⇒ unique ground state ⇒ no topologically protected invisible bulk d.o.f. (the genus-1 obstruction of v344). [E] FINITE PETZ RECOVERY: the deviation subspace contracts at $(2/3)^6$=64/729 < 1 (v221) ⇒ $\text{Rec}_\Sigma$ bounded. [E] GAP-DECOUPLING MARGIN: $\Delta_{\text{eff}}=6\log\frac{3}{2}-\frac{31}{4\pi^2}\approx 1.648>0$ (v76/v337) ⇒ the ambient sector is energetically separated. [C] RECONSTRUCTION MAP: Bisognano–Wichmann gives the modular boost (v258/v309/v323) ⇒ $\text{Rec}_\Sigma$ : (bulk) → $\mathcal{A}_\Sigma^{\mu_4}$. [C] THE REDUNDANCY STATEMENT: under `SEAM.EQUIV.01` + BW, $\mathcal{O}^{\text{bulk}}_{\text{phys}}/\text{Diff} \cong \mathcal{A}_\Sigma^{\mu_4}$ ⇒ `QG.AMB.01` carries no extra physical d.o.f. [O] residual: the two consumed premises – `SEAM.EQUIV.01` (the S3 lattice input, v366) + the BW intrinsicality of the raw collar (v329); provides the ALTERNATIVE (holographic) route, does NOT construct the measure. Gravity-complete = Boundary-complete + Ambient-redundancy [E][C][O] |
| `v370_grav_spin2_unitarity.py` | Track 2: the Barnes–Rivers spin decomposition of the graviton propagator – perturbative graviton unitarity SECTOR-BY-SECTOR (extends v304's scalar pole algebra to the tensor spin structure + combines with v334's spin-0 conformal mode). [E] projector completeness: $\text{tr}\,P_2=(d-2)(d+1)/2$, $\text{tr}\,P_1=d-1$, $\text{tr}\,P_{0s}=\text{tr}\,P_{0w}=1$ sum to $d(d+1)/2$; the $d$=4 split is 5+3+1+1=10. [E] EH structure: spin-2 coefficient +1, spin-0 scalar $-1/(d-2)=-1/2$. [E] the Stelle ghost is a SPIN-2 state (a local $1+p^2/M^2$ dressing puts the negative-residue pole at $p^2=-M^2$). [E] the entire KMS form factor $e^{p^2/M^2}$ (v304/v259) makes spin-2 ghost-free (only the $p^2$=0 pole). [C] the wrong-sign spin-0 conformal mode ($c_{\text{conf}}(4)=-3/2$, v332) is cured by the GHP contour (v334) – a different mechanism. [C] ⇒ perturbative graviton unitarity sector-by-sector, conditional on v304's entire-analyticity assumption. [O] perturbative-only + the open analyticity assumption; NOT the non-perturbative measure (`QG.AMB.01`). [O] decline: $\text{tr}\,P_2$=5 is dimension counting, NOT $g_{\text{car}}$ (v354/v355). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v371_ftransfer_pole.py` | Track 3 ($F_{\text{transfer}}$): the Koide source→pole transfer as a typed solver – promotes the `experiments/ftransfer` solve into the ledger with a kill test (does NOT re-prove 53/54 v183 or $Q^*$=2/3 v82/v101). [E] $Q_{\text{pole}}$=Koide(PDG poles)= 2/3 to $3\times10^{-6}$, $Q_{\text{src}}(\varphi_0^{\text{ret}}\text{-ladder})$= 0.66446 (0.33% below). [E] QED runs the wrong way: $dQ/d\epsilon > 0$ pushes $Q$ ABOVE 2/3; the $\epsilon$ mapping $Q_{\text{pole}} \to Q_{\text{src}}$ is NEGATIVE while QED gives $\epsilon$=+(3/2)($\alpha/\pi$) > 0 – wrong sign for any $\alpha \in [1/137, 1/128]$. [C] verdict: $F_{\text{pole}}$ is NOT standard QED running; the transfer is the non-QED 53/54 operator / $(2/3)^6$ Moebius generator (v183/v82). Stays [C], plain running excluded as the kill. [E][C] |
| `v372_ftransfer_boltzmann.py` | Track 3 ($F_{\text{transfer}}$): $\eta_B$ from the INTEGRATED Buchmueller–Di Bari–Pluemacher Boltzmann network at the TFPT-frozen heavy scale – promotes the `experiments/ftransfer` solve, replacing the v326 toy washout. Frozen $M_1$=$M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$ ($\sim 8.65\times10^9$ GeV, v212), $\tilde{m}_1$=$m_3/A_\Lambda$=5 meV, $\delta_{CP}$=$4\pi/3$. [E] integrated efficiency $\kappa_f \sim 0.09$ (strong washout $K \sim 4.6$) agrees with the BDP fit to $\sim 25\%$. [C] $\eta_B$(ODE)$\sim 6.5\times10^{-10}$ at the frozen $M_1$ lands within a factor $\sim 1.1$ of the observed $6.1\times10^{-10}$, NO $M_R$ dial. [X] kill: $\eta_B$ outside [obs/3, 3 obs] for all seesaw-consistent $M_R$ would drop the leptogenesis branch. Stays [C] (a transfer target). [E][C][X] |
| `v373_ftransfer_relic.py` | Track 3 ($F_{\text{transfer}}$): the axion relic from the FINITE-T misalignment ODE – promotes the `experiments/ftransfer` solve and DECIDES the two TFPT angle branches, replacing the v326 adiabatic toy. $f_a$=$M_{\text{scal}}/128 \sim 2.4\times10^{11}$ GeV, $m_a \sim 23.8\,\mu$eV, lattice $\chi(T)$ $n$=8.16. [E] candidate scales fixed by the determinant-line atoms (128=2 dim $S^+|\mu_4|$). [C] the frozen SPINE angle $3\pi/5$=108°=$\pi N_{\text{fam}}/g_{\text{car}}$ gives $\Omega_a h^2 \sim 0.12$ in [0.08, 0.16] (lands on $\Omega_{\text{DM}}$ untuned) – the robust dominant-DM branch. [X] the HILLTOP 170° over-produces ($\sim 5.4\times$) – disfavoured. [X] kill: a finite-T solve robustly outside [0.08, 0.16] for the spine across $\chi(T)/g_*$ drops the dominant axion-DM branch. Stays [C]/[X]. [E][C][X] |
| `v374_ftransfer_qcd.py` | Track 3 ($F_{\text{transfer}}$): the $m_p/m_e$ transfer as a typed uncertainty-budget + kill test – the falsifiability companion to v262 (the precise 2-loop point), replacing the v326 toy (which hardcodes 1836.15). [E] $b_3$= − (11 − $2n_f/3$)= − 7 ($n_f$=6 carrier). [C] two declared externals (v339): $\alpha_s(M_Z)$=0.1179±0.0009 and $C_p$=2.83±0.15, each $\sim 5.3\%$. [C] $m_p/m_e$ central $\sim$ 1824 (v262) with a propagated ±7.5% band [$\sim$ 1690, $\sim$ 1960] CONTAINING the observed 1836.15 – no free dial. [X] kill: a future lattice $C_p$±0.03 + tighter $\alpha_s$ shrink the band to $\sim \pm2\%$; a band then excluding 1836.15 drops the QCD-matching route ($b_3$= − 7 + the $SU(9)$-integer firewall v262). Distinct from v262's point; stays [C]. [E][C][X] |
| `v375_observatory_registry.py` | Track 4: the prediction OBSERVATORY – a status-typed CI over the frozen prediction registry (`freeze_file.csv`) that makes the falsifiability surface machine-checkable (no new physics). [E] falsifiability complete: every registry row carries a non-empty kill criterion (no orphan claims). [E] headline values re-derive from $\{\varphi_0^{\text{ret}}\}$: $\sin^2\theta_{12}$=1/3−$\varphi_0^{\text{ret}}/2$, $\sin^2\theta_{13}$=$\varphi_0^{\text{ret}}\,e^{-5/6}$, $\beta_{\text{rad}}$=$\varphi_0^{\text{ret}}/(4\pi)$. [C] live scorecard: $\theta_{12}$ vs JUNO first run (0.3092±0.0087)= 0.28$\sigma$ (compatible); $\theta_{13}$ vs NuFIT 6.0 NO (0.02195±0.00058)= 2.0$\sigma$ (the flagged most-tensioned core prediction); $\beta_{\text{rad}}$ vs ACT DR6 (0.215±0.074)= 0.37$\sigma$ (hint); $r$=$12/N_\star^2 \sim 0.004$ < BK18 (compatible). [E] registry integrity: the single machine-readable falsifiability surface. Tooling for the closed/open pieces. [E][C] |
| `v376_seam_s3_centralcharge.py` | S3 closure stack: the central charge $c$=8 extracted NUMERICALLY from the lattice via Calabrese–Cardy finite-size entanglement scaling of the critical free-fermion content (the quantitative core of the MMST scaling-limit claim, on OUR model). [E] one critical complex mode gives $c$=1.0000 (Peschel correlation-matrix EE); [E] the collar is 16 Majoranas = 8 complex modes (v367), $c$ additive ⇒ $c$=8=$g_{\text{car}}$+$N_{\text{fam}}$; [E] chiral $c_-$=8 (Chern $\lvert C\rvert$=1 per Majorana v367, non-gappable edge v368). [O] the specific net $(E_8)_1$ is v377/v378; the continuum existence is the cited MMST theorem (v336). Numerical $c$=8 from the lattice; does NOT close `SEAM.EQUIV.01`. [E][O] |

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| Script | What it machine-checks *(continued)* |
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| `v377_seam_s3_e8character.py` | S3 closure stack: the chiral $c$=8 content is the SPECIFIC net $(E_8)_1$, via the exact affine character $q$-series + the $\mu_4$/GSO promotion. [E] $\chi{=}E_4/\eta^8{=}q^{-1/3}(1{+}248q{+}4124q^2{+}\dots)$: leading $-c/24{=}-1/3 \Rightarrow c{=}8$; level-1 $=248{=}\dim E_8$ (one primary). [E] the $\mu_4$/GSO promotion $SO(16)_1 \to (E_8)_1$: $248{=}240{+}8$, $240{=}\dim S^+\, g_{\rm car}\, N_{\rm fam}{=}16{\cdot}5{\cdot}3$; $248{=}120{+}128{=}\dim SO(16)$+spinor. [E] discriminator: $(E_8)_1$ has 1 primary/248 currents vs $SO(16)_1$ 4 primaries/120. [O] continuum existence (v336); the $\mu_4$ projection = the seam one-sidedness (S3). Pins the target net; does NOT close `SEAM.EQUIV.01`. [E][O] |
| `v378_seam_s3_modular.py` | S3 closure stack: the GENUS-1 discriminator – torus ground-state degeneracy = 1 (holomorphic = $(E_8)_1$), the "hard part" v344 flagged (genus-0 unique vacuum is necessary not sufficient). [E] KLM condensation: $\mu(\text{carrier}){=}\det C(D_5)\det C(A_3){=}4{\cdot}4{=}16$; the order-4 $\mu_4$ glue ($\lvert L\rvert{=}4$) $\Rightarrow \mu(E_8){=}16/16{=}1$ (torus GSD = 1); index-2 $\Rightarrow \mu(SO16){=}16/4{=}4$. [E] $\lvert\det C(E_8)\rvert{=}1$ vs $\lvert\det C(D_8)\rvert{=}4$. [E] the single $(E_8)_1$ character has modular $T$-eigenvalue $e^{-2\pi i/3}$ ($h{=}0, c{=}8$), 1-dim rep $\Rightarrow$ one character $\Rightarrow$ torus GSD = 1. [O] continuum existence (v336). Pins the net at genus-1; does NOT close `SEAM.EQUIV.01`. [E][O] |
| `v379_seam_s3_rp.py` | C7 / OS reconstruction: reflection positivity (the OS axiom) of the explicit gapped collar measure, verified numerically – the remaining constructive-QFT ingredient for the OS reconstruction. [E] the gapped $p{+}ip$ band gives $G(\tau){=}\text{mean}_k e^{-\varepsilon_k \tau}$ with $\varepsilon_k \geq \text{gap}/2 \sim 1 > 0$ (v367) – a positive spectral measure; [E] the Hankel/reflection Gram matrix $M_{ij}{=}G(\tau_i{+}\tau_j)$ is PSD – OS reflection positivity holds; [E] neg control: a growing ghost mode makes it indefinite (non-vacuous). [C] RP + gap (v76/v337) + GNS (v240) + $c$=8 (v376) + $(E_8)_1$ (v377/v378) give the OS inputs; C7/`QG.AMB.01` is then DISCHARGED by the redundancy theorem v369, not built. [O] the full constructive measure stays the cited step. [E][C][O] |
| `v380_grav_kms_hessian.py` | The KMS Entire Hessian – the Stelle ghost is EXACTLY the Seeley–DeWitt truncation, and resummation pushes it to infinity (upgrades v304/v370 from "assume entire" to a derived statement). [E] the seam KMS cutoff $f(u){=}e^{-u}$ (v259) gives the dressed spin-2 propagator $e^{-p^2/M^2}/p^2 = 1/(p^2 a)$ with $a(u){=}e^u$ entire & nowhere zero $\Rightarrow$ only the $p^2{=}0$ pole (no ghost). [E] each finite truncation carries the ghost: the $R{+}R^2$ order $T_1{=}1{+}u$ has the Stelle zero $u{=}-1$, the Weyl$^2$ order $T_2{=}1{+}u{+}u^2/2$ the pair $-1{\pm}i$; the entire $a$ has none. [E] the ghost DECOUPLES: $\lvert$nearest zero$\rvert$ of $T_n$ grows monotonically with $n$ $(1, 1.41, \dots, 3.07$ for $n{=}1..8) \to \infty$, so "entire" = "do not truncate". [C] $\Rightarrow$ perturbative graviton unitarity. [O] the exact off-shell curved-space Hessian identification is the cited heat-kernel resummation; perturbative-only, not the non-perturbative measure. [E][C][O] |
| `v381_qft4d_eg_allorder.py` | The all-order Epstein–Glaser / BRST contract for the perturbative 4D S-matrix $S_{\rm pert}$ – the "all-order closing statement" for the 4D perturbative QFT leg. Types the standard causal-perturbation-theory closure for the TFPT finite interaction class; it does *not* build a path integral. [E] POWER-COUNTING: every gauge+matter vertex (Yang–Mills + Dirac–Yukawa + Higgs + ghost) is a mass-dimension-4 marginal operator $\Rightarrow$ the EG singular order $\omega \leq 4 \Rightarrow$ a FINITE counterterm space at every order. [E] BRST NILPOTENCY $s^2{=}0$ verified via the Jacobi identity for $su(2)$ (residual 0) and $su(3)$ ($10^{-16}$), the carrier weak+colour content. [E] the seam gap $\Delta{=}6\ln(3/2){>}0$ (v302) gives the IR-safe adiabatic limit $g \to 1$. [C] imported: $T_n$ exists to all orders by causal factorisation (Epstein–Glaser 1973 / Brunetti–Fredenhagen). [C] imported: all-order BRST invariance (Piguet–Sorella / Kugo–Ojima) $\Rightarrow$ Slavnov–Taylor to all orders. [E] scope fence: matter+gauge ONLY – the $R^2$/Weyl$^2$ gravity sub-sector is v304/v370/v380, not a local EG vertex; $S_{\rm pert}$ is not the ambient measure `QG.AMB.01` (v369). NET [C] (imported EG+BRST on the TFPT dim-4 class + gap), prerequisites [E]; extends v269/v271/v273/v278 from 1-loop to all-order [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v382_alpha_quillen_exact.py` | Names the Quillen determinant-line VARIATION $\delta_\tau\big(\log\det_\zeta \Delta_{U(1)} + 8b_1c_3^6\log\varphi_{\text{seam}}\big)=0$ as a tracked target (review Point 10), elevating `EM.WARD.01`'s residual to a citable [O]. [E] the Quillen split holds at the root $\alpha^3-2c_3^3\alpha^2=8b_1c_3^6\ln(1/\varphi_{\text{seam}})$ at $\alpha^{-1}$=137.0359992 ($F_{U(1)}$=0 IS the stationarity of the $U(1)$ determinant line, v341). [E] every coefficient a named atom: $8b_1=\frac{4}{5}\cdot 41$ a $U(1)$ index, $c_3$=$1/(8\pi)$ Gauss–Bonnet ($c_3^{3,6}$ heat-kernel powers), $q(D_5)+q(A_3)$=2 discriminant forms (v341/v48/v216). [E] the value $\alpha^{-1}$=137.0359992168 stays closed (unique root + ablation, v3) – only "why this functional" is open. [O] THE TARGET: the from-first-principles AQFT/$\zeta$ proof that $F_{U(1)}$ is the EXACT Quillen functional. [C] a FACE of `SEAM.EQUIV.01` (v336): the seam $U(1)$ determinant line lives on the same holomorphic seam net, so once the net is $(E_8)_1$ its $U(1)$ sub-determinant is fixed – not a second independent gate, it names/tracks the pre-existing `EM.WARD.01` residual (v341/v342). No new number (reuses v341, Wolfram-mirrored) [E][C][O] |
| `v383_dynamics_universal.py` | DYNAMICS.UNIVERSAL.01: the UNIVERSAL spectral-gap / Perron–Frobenius principle – every TFPT sector is the SAME object (a gapped operator with a UNIQUE leading attractor = the physics, a spectral gap = the reason there is no free parameter), extending v303 from $F_{\text{transfer}}$ to the whole theory. [E] FLAVOR (computed): the cusp transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ has a simple Perron leading 1 and gap $6\ln(3/2)>0$ (v56/v82). [E] DISCRETE COMPILER (computed): the affine-$E_8$ adjacency has Perron eigenvalue 2 (Kac marks, v312/v313) and subleading $2\cos(\pi/5)=\varphi$, gap $2-\varphi=1/\varphi^2>0$. [E] DECOUPLING MARGIN: $\Delta_{\text{eff}}=6\ln(3/2)-31/(4\pi^2)\approx 1.648>0$ (v76/v337). [E] TWO NUMBER-FIELD FACETS (v314): golden $\varphi\in\mathbb{Q}(\sqrt{5})$ (carrier-5, static) and $(2/3)^6\in\mathbb{Q}$ (family-3, dynamic) are the prime-5 and prime-3 facets of the order-30 Coxeter clock $30=2{\cdot}3{\cdot}5=h(E_8)$ – a factorisation, not an equality. [C] the other sectors are the same structure: gravity (v358/v359), QG saddle (v365), the holomorphic $c$=8 RG fixed point (v157/v158), the all-order S-matrix (v381), recovery (v221), $F_{\text{transfer}}$ (v303), $\tau=i$ (v333). [C] META-THEOREM: gapped ⇒ unique attractor ⇒ parameter-free, theory-wide – one spectral-gap statement wearing many hats. A synthesis, no new number [E][C] |
| `v384_residual_certification.py` | RESIDUAL.CERTIFICATION.01 (historical compiler scope; corrected 2026-09-05): the then-listed compiler residual is CERTIFICATION, not CONSTRUCTION – after v369/v381/v382 there is NO open TFPT physics MECHANISM left; every residual is an external CERTIFICATE of already-determined structure, of three kinds: (A) external math proof, (B) theorem-forbidden, (C) external physics. A ledger-CI audit. [E] LEDGER CI: every residual claim id is present (no orphan). [E] NO-UNIT (B): $c_3$=$1/(8\pi)$ and $g_{\text{car}}$=5 are mass-dimension 0, so by No-Unit (v153/v364) no dimensionless compiler outputs an absolute scale ⇒ $v_{\text{geo}}$ is forbidden by theorem, not a gap. [E] GAP-DECOUPLED (A): $\Delta_{\text{eff}}\approx 1.648>0$ (v76/v337) ⇒ every readout is independent of the ambient measure (`QG.AMB.01` a [C] redundancy, v369). [E] CLASSIFICATION COMPLETE: `SEAM.EQUIV.01` (continuum existence = cited MMST/Adamo, v336)→A, `ALPHA.QUILLEN.EXACT.01` (v382)→A, `QG.AMB.01` (v369)→A, $v_{\text{geo}}$ →B, $F_{\text{transfer}}$ (v371–v374, firewall v187)→C; ZERO open internal mechanisms within that scoped compiler list. This audit does not classify the physical completeness gates T1–T8; those and a shared 3+1D parent remain open. [C] CONVERGENCE applies only to the historical compiler residual. An audit, no new number [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v385_uvbranch_killtest.py` | UVBRANCH.KILLTEST.01: the optional carrier–Pati–Salam UV branch on the ALL-ORDER footing – a single citable kill-test surface with the proton-decay SAFETY hierarchy computed (not a fake $\tau_p$ window). A consolidation (does *not* re-run the RGE v246 nor re-derive the PS scale v249/v253). [E] carrier content = SM ($b = (41/10, -19/6, -7)$, v159/v246, no new states) and $S_{\rm pert}$ closes to all orders (v381) ⇒ the SM-non-unification is all-order-robust. [X] plain-SM 4D-GUT killed ($\sin^2\theta_W = 3/8$ vs 0.231; v246). [C] optional carrier-PS branch: $M_{PS} = \kappa M_{\rm scal}$, $M_{\rm scal} = c_3^{7/2}\bar{M}_{\rm Pl} \approx 3.06{\times}10^{13}$ GeV, $\kappa \in [1.0, 1.2]$ (v253); content $\{1, 10, 16, 45\}$, no 126 (v247/v248); not the default (v265). [E] proton-decay SAFETY hierarchy: minimal PS gauge bosons are $SU(4)$ leptoquarks ($B{-}L$-conserving $q \leftrightarrow l$, rare LFV $K_L \to \mu e$), *not* the $p \to e^+\pi^0$ dim-6 operator (needs an $SO(10)/SU(5)$ diquark, absent); the binding bound $M_{PS} \gtrsim 10^6$ GeV is cleared by $M_{PS}/M_{\rm bound} \sim 10^7 \gg 1$. [X] frozen kill tests (v265): $\kappa$ leaving $O(1)$ / a forbidden rep / a new gauge-charged state / a rare-LFV signal below the bound; no fake $\tau_p$. A consolidation, no new number [E][C][X] |
| `v386_grav_amplitude.py` | GRAV.AMPLITUDE.01: the entire-form-factor graviton-exchange AMPLITUDE is finite, UV-softened and tree-unitary – perturbative gravity as an explicit scattering problem, extending v304/v370/v380 from the propagator to the amplitude. The dressed spin-2 propagator $P = e^{-p^2/M^2}/p^2 = 1/(p^2 a)$, $a(u) = e^u$, $M = M_{\rm scal}$. [E] one pole, ghost-free: $a$ entire & nowhere zero (v380) ⇒ only pole at $p^2 = 0$, residue $e^0 = 1 > 0$. [E] UV-softened: ratio $e^{-u}$ at $u = 1, 5, 10, 20 = 0.37, 6.7{\times}10^{-3}, 4.5{\times}10^{-5}, 2.1{\times}10^{-9}$ (faster than any power). [E] low-energy = GR + a calculable $-p^2/M^2$ correction ($M = M_{\rm scal} = c_3^{7/2}\bar{M}_{\rm Pl}$, v253). [C] tree unitarity: a single positive-residue pole, no negative-norm states (v370/v380). [C] the amplitude $A = (\kappa^2/2)(T{\cdot}P_2{\cdot}T')e^{-p^2/M^2}/p^2$ is GR at low energy, softened in the UV (perturbative-only; `QG.AMB.01` a [C] redundancy v369). No new number [E][C] |
| `v387_corrections_gap.py` | CORRECTIONS.GAP.01: the practical harvest of v383 – the same spectral gap that makes each sector parameter-free ALSO sets the SIZE of its first correction (correction$_n \sim (\lambda_2/\lambda_1)^n$). [E] FLAVOR: $\lambda_2 = (2/3)^6 = 64/729$ ⇒ first Koide correction $\sim 0.088$ (v56/v82). [E] RECOVERY: the same $(2/3)^6$ Petz rate (v221). [E] QG: capped by $\chi = 1/(1-(2/3)^6) = 729/665$ (v337). [E] DISCRETE COMPILER: the update $(A + 2I)/4$ fixes the Kac marks ($\lambda_1 = 1$) and decays at the golden $(\varphi + 2)/4 \approx 0.905$ (carrier-5 facet, v303/v312). [E] the rates split into the $(2/3)^6$ family (family-3, in $\mathbb{Q}$) and the golden family (carrier-5, in $\mathbb{Q}(\sqrt{5})$) – the two facets of v314/v383. [C] the correction calculus: the gap does double duty (WHY no free parameter AND HOW BIG the first correction), so sub-leading magnitudes are organised, not free. No new number [E][C] |
| `v388_corrections_budget.py` | CORRECTIONS.BUDGET.01: the gap correction (v387) is a *typed* budget, NOT a uniform band on every prediction – the $(\lambda_2/\lambda_1)^n$ size is a property of the distance from a gapped operator's leading eigenvector, so it is only defined where $\lambda_2$ exists. The prediction surface splits into four correction classes (already implied by v303). [E] SEAM-GAPPED: Koide ($F_{\rm pole}$ v303), recovery (v221), the QG bound (capped by $\chi = 729/665$, v337) carry $(2/3)^6$; the discrete compiler decays at the golden $(\varphi + 2)/4$ (v312). [E] KOIDE FIRST CORRECTION: $F_{\rm pole}$ is the ONE seam-gapped prediction, rate $(2/3)^6 \approx 0.0878$ (v371), source quotient 0.33% below 2/3 – the rate is the suppression SCALE, not the residual. [E] EXACT-IDENTITY class ($\det R = 8$, $N_\Phi = 1$, $\theta_{\rm eff} = 0$, $\sin^2\theta_{13}$): no $\lambda_2$ ⇒ correction 0 STRUCTURALLY (a band would contradict [E]). [E] EXTERNAL-RATE class ($\eta_B$/axion/$m_p, m_e$): the gapped shape with external rates (only $F_{\rm pole}$ has the seam rate, v303), firewalled (v187). [E] FIXED-POINT class: the texture is already explicit ($\theta_{12}$ $\varepsilon = c_3 + 36c_3^4$). [C] a uniform band on all predictions is WRONG; the size is keyed to the mechanism class. A typing of v387, no new number [E][C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v389_grav_loop_finiteness.py` | GRAV.LOOP.FINITE.01: the entire-form-factor graviton is UV-finite at the *loop* level by power counting – the honest next step past v386 (tree), extending v304/v370/v380/v386 from the propagator to the superficial degree of divergence. Honest scope: the standard infinite-derivative/nonlocal-gravity finiteness statement, NOT a full proof of perturbative QG. [E] super-polynomial falloff $p^k e^{-p^2/M^2} \to 0$ for every $k$. [E] one dressed line turns a quadratic divergence into the finite $(M^2/2)e^{-p_0^2/M^2}$; the GR $\int p^3/p^2\,dp$ diverges quadratically. [E] the superficial degree $\to -\infty$ at every loop order (dressed cutoff-ratio $\to 1$ vs GR $\to 4$). [E] the regulator is the SEAM scale $M = M_{\rm scal} = c_3^{7/2}\bar{M}_{\rm Pl}$ (v253/v259), not a dial. [C] loop UNITARITY of the nonlocal form factor (Lorentzian continuation / macro-causality) stays the honest open point – tree unitarity is v386; NOT claimed solved. No new number [E][C][O] |
| `v390_prime2_facet.py` | COXETER.PRIME2.01: the prime-2 facet of the order-30 Coxeter clock, completing v383. The clock $30 = 2{\cdot}3{\cdot}5 = h(E_8)$ had two named facets (golden $\varphi \in \mathbb{Q}(\sqrt{5})$, prime-5/carrier; $(2/3)^6 \in \mathbb{Q}$, prime-3/family); this names the prime-2 facet as the Gaussian field $\mathbb{Q}(i)$ — $\tau = i$ and the order-4 $\mu_4 = \mathbb{Z}/4$ square seam deck (v282/v288/v333), structure already present. [E] $h(E_8) = 30$ factors into exactly three primes. [E] one quadratic facet field per prime, each ramified at its prime: $\operatorname{disc}\mathbb{Q}(i) = -4$, $\operatorname{disc}\mathbb{Q}(\sqrt{-3}) = -3$, $\operatorname{disc}\mathbb{Q}(\sqrt{5}) = 5$; ramified primes $\{2,3,5\}$, product $= 30$. [E] prime-2 Gaussian facet: min. poly $x^2+1$, $\tau = i$, $j(i) = 1728 = 12^3$, $\lvert\mu_4\rvert = 4 = 2^2$ (the static seam-orientation facet). [E] prime-5 golden (v383): $2\cos(\pi/5) = \varphi$ (pentagon). [E] prime-3 family (v383/v220): $(2/3)^6$ plus the Eisenstein $\zeta_3$ (hexagonal CP $\pi/3$). [C] completion: each prime carries a facet (square/hexagon/pentagon = seam/CP/carrier), the prime set forced by $h(E_8)$. A synthesis, no new number [E][C] |
| `v391_alpha_quillen_progress.py` | ALPHA.QUILLEN.PROGRESS.01: an *honest attempt* at the external target ALPHA.QUILLEN.EXACT.01 (v382) – it REDUCES the open step to a sharper named sub-target via a solvable model but does NOT close it; ALPHA.QUILLEN.EXACT.01 stays [O] (external math, type A, v384). [E] solvable model $\det_\zeta(-\partial_\tau^2+m^2)$ on a circle $= (2\sinh(\beta m/2))^2$, $m$-variation $\beta\coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \dots$ a closed heat-kernel/Laurent series ($m^3$ coeff $-1/360$ verified). [E] a $\zeta$-det variation is structured heat-kernel data, so v382's $8b_1c_3^6$ counterterm (a Gauss–Bonnet heat-kernel power) is the right *kind* of object. [E] the v382 split re-verified at $\alpha^{-1} = 137.0359992$. [E] reduction: the open step reduces to the named sub-target "seam Seeley–DeWitt coefficient $= 8b_1c_3^6$". [O] NOT CLOSED: the actual seam $\zeta$-det proof stays open, SEAM.EQUIV.01 continuum existence (v336) the other external [O]; $\alpha^{-1}$ stays [E]. An attempt that reduces, does not close [E][O] |
| `v392_seam_s3_scalinglimit.py` | SEAM.S3.SCALINGLIMIT.01: an *honest attack* on the one open lemma of SEAM.EQUIV.01 (the continuum scaling-limit existence, v336) – it does NOT close it (the abstract existence is the cited MMST theorem); it STRENGTHENS the case with the existence-relevant observable v376 did not test: that the finite-size data *converge*. [E] the Calabrese–Cardy EE slope $c_1(L)$ at $L = 66\dots514$ is a monotone Cauchy-like sequence approaching 1 ($\lvert c_1(L)-1\rvert$ strictly decreasing). [E] a $1/L$ Richardson fit gives $c_1(\infty) \approx 1.000$, so the 16-Majorana collar $c = 8c_1 \approx 8.00 = g_{\rm car}+N_{\rm fam}$ is reached *in* the limit. [E] the successive differences are strictly decreasing (numerical Cauchy, necessary for existence). [O] convergent data are evidence, not a proof of existence (MMST v336), and do not pin $(E_8)_1$ vs $SO(16)_1$ (holomorphy v377/v378). Strengthens the residual, does NOT close it [E][O] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v393_corrections_numeric.py` | CORRECTIONS.NUMERIC.01: the typed correction budget (v388) turned into ACTUAL first-correction magnitudes – a computed theoretical-uncertainty surface, the short-term testable harvest of v388. [E] FIXED-POINT band $(\theta_{12}/\theta_{13}/\lambda_C/\beta_{\rm rad})$ = the explicit $\varphi_0$ puncture term $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$ (exact; = $\theta_{12}$'s $\varepsilon$ first-correction). [E] SEAM-GAPPED band (Koide/recovery) $= (2/3)^6 \approx 8.78\%$; the QG bound capped by $\chi - 1 = 64/665 \approx 9.62\%$ (v337). [E] EXACT-IDENTITY band = 0 structurally ($\det R = 8$, $N_\Phi = 1$, $\theta_{\rm eff} = 0$). [E] EXTERNAL-RATE band = external physics quoted ($m_p/m_e$ ±7.5% v374; $n_s/r$ $N_\star$ band; $\eta_B$ 1.07× v212), fenced v187. [C] each prediction gets "central value ± computed first correction". A harvest of v388, no new number [E][C] |
| `v394_coxeter_atoms.py` | COXETER.ATOMS.01: the clockwork coherence – the three Coxeter primes ARE the three anchor atoms ARE the three number-field facets, and the prime-2 seam is the common dynamical factor. A bird's-eye [E] re-reading tying v390 (facets) to the anchor $a = (1,1,2)$ and v319 (5x6 clock); registers `RES.COXETER.SYMMETRY.01` as a tracked [O]. [E] $a = (1,1,2)$: $e_3$=2=$\lvert Z_2\rvert$, $p_0$=3=$N_{\rm fam}$, $e_2$=5=$g_{\rm car}$, product 30=$h(E_8)$. [E] atoms = facet fields (v390): 2→$\mathbb{Q}(i)$, 3→$\mathbb{Q}(\sqrt{-3})$, 5→$\mathbb{Q}(\sqrt{5})$, each ramified at its prime. [E] $\chi$=729/665=$3^6/(3^6-2^6)$, $(2/3)^6$=$2^6/3^6$, exponent 6=$2N_{\rm fam}$. [E] both dynamic rates carry prime-2: 2/3=$\lvert Z_2\rvert/N_{\rm fam}$, $(\varphi+2)/4$=$(\varphi+2)/\lvert\mu_4\rvert$, $\lvert\mu_4\rvert$=4=$2^2$. [E] 30=5×6=$g_{\rm car}(2N_{\rm fam})$. [C] the seam is the universal dynamic engine. [O] `RES.COXETER.SYMMETRY.01` registered, not closed [E][C][O] |
| `v395_hypergraph_coupled.py` | HYP.COUPLED.01: the coupled hypergraph rewrite – one local rule unifies v299/v327/v324 (carrier × family). [E] LOCAL COUPLED MICRO-RULE: state on 9 affine-$E_8$ nodes × 3 family slots; one micro-step = network lazy diffusion on each cusp column + v327 $M$ on each node's family vector; steps commute, $T_{\rm micro} = T_{\rm net} \otimes M$. [E] JOINT ATTRACTOR: top eigenvalue 1 with eigenvector marks⊗$e_0$; iteration from any positive start converges. [E] ONE HAND = RECOVERY: one network step + $2N_{\rm fam}$ family micro-steps gives $(2/3)^6$ on marks⊗$v_{\rm surv}$. [E] v324 EMERGES: $T_{\rm full} = T_{\rm net} \otimes T_{\rm cusp}$ shares the same invariants; $T_{\rm cusp}$ is the 6-hand readout in the Galois/cusp basis (v317). [E] GROWTH + FIBER: v299 $E_6$→$E_7$→$E_8$→$\hat{E}_8$ unchanged with $M$-fiber attached. [O] 3-arm seed $(P_2)$ + $\varphi_0$ (v312) remain irreducible [E][O] |
| `v396_phi0_icosahedral.py` | HYP.PHI0.01: the $\varphi_0$ leading term from icosahedral combinatorics (sharpening v312/v395). [E] ICO COUNTS ANCHOR-FORCED: $V$=12, $E$=30=$h(E_8)$, $F$=20=$g_{\rm car}(g_{\rm car}-1)$, $\lvert\,\mathrm{Aut}\,\rvert$=120=$4h$=$\lvert 2I\rvert$. [E] LEADING TERM EXACT: $\varphi_0^{\rm tree}$=1/(6π)=(4/3)$c_3$=$F/(4h\pi)$=$F/(\lvert\,\mathrm{Aut}\,\rvert\pi)$=$(F/(g_{\rm car}N_{\rm fam}))c_3$. [E] $F/h$=$\lvert Z_2\rvert/N_{\rm fam}$; leading term = $(F/h)/(4\pi)$=$c_3(g_{\rm car}-1)/N_{\rm fam}$. [E] GAUSS–BONNET: icosahedron deficit sum $4\pi$; $c_3$=$1/(\lvert Z_2\rvert 4\pi)$ (v216/v342). [E] HONEST NEGATIVE: puncture $48c_3^4$ not a rational hypergraph fraction. [C] tree-level seed = (hyperedges)/$(g_{\rm car}N_{\rm fam})$ × $c_3$, not an independent injection; puncture [O] |
| `v397_external_clock_probe.py` | FR.CLOCK.PROBE.01: a *falsifiable* probe of how external the "external physics" really is – do the frontier scheme-numbers land on the $\{2,3,5\}$ clock (the anchor atoms, v394) or not? Strict No-Free-Pattern (v100): "on the clock" = an EXACT match, not a wide-band fit. [E] $\eta_B$ decuple $A_\Lambda$=10=$\lvert Z_2\rvert g_{\rm car}$ ON CLOCK (v212). [E] Koide rate $(2/3)^6$=$(\lvert Z_2\rvert/N_{\rm fam})^6$ ON CLOCK (v303). [E] axion spine angle $3\pi/5$=$\pi N_{\rm fam}/g_{\rm car}$ ON CLOCK (v211). [E] proton $C_p \sim 2.83\pm0.15$ NOT discriminating ($2^{3/2}$, 17/6, 14/5 all fit the band) ⇒ EXTERNAL (firewall v187/v374). [C] 3 of 4 land exactly on the clock; the irreducible external residue shrinks to the proton QCD number $C_p$ – "external physics" was overstated. [E] anti-numerology: refuses to assign $C_p$ a clock value [E][C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v398_seam_state_rigidity.py` | **Seam State Rigidity Theorem** (Paper A, the $G_{\text{net}}$ closure contract): the scattered state-invariance reductions (v177/v198/v199/v308/v309/v335) as ONE named target. [E] RP-definability hinge (OS canonical transfer $T{=}e^{-H}$ from RP + reflection alone, v54, on a quasi-free state v155, no $\mu_4$ normal form); [E] band-limited core $[\rho,\lvert k\rvert]{=}0$ and character-block-diagonal sub-principal symbol swept $N{=}4..64$ (beyond the 3-dim $H^1$); [E] holomorphy $\Rightarrow (E_8)_1$ ($c{=}8$, $\lvert\det \text{Cartan}\rvert{=}1$ vs 4, index $4{=}\lvert\mu_4\rvert$, gap $6\log\frac{3}{2}{>}0$). [O] the one residual: $[\rho,\Lambda_\Sigma]{=}0$ on the full $L^2$ (intrinsic Bisognano–Wichmann). A contract+evidence module, NOT a closure of `SEAM.EQUIV.01` (continuum = cited MMST v336) |
| `v399_gravity_complete.py` | **Gravity-complete** = **Boundary-complete** + **Ambient-redundancy** (Paper B accounting): combines the parameter-free local equation (v358/v359) and the ambient redundancy (v369). [E] $G_{ab}{+}\Lambda g_{ab}{=}(1/c_3)T_{ab}$, $1/c_3{=}8\pi$ (four $c_3$-roles), $\Lambda$ prefactor $3/(4\pi^2)$; [E] redundancy discriminators ($\lvert\det \text{Cartan}\rvert{=}1$ vs 4, torus GSD$=1$, Petz $(2/3)^6$, margin $1.648{>}0$); [E] $\mathcal{O}_{\text{phys}} \subset \mathcal{A}_\Sigma$ over five readout classes (Newton $8\pi$, $\Lambda$, scalaron $c_3^7$, horizon $1/\lvert\mu_4\rvert$, recovery $(2/3)^6$) – all boundary/seam/gap. [C] the verdict; [O] `SEAM.EQUIV.01` + intrinsic BW (no new gate) |
| `v400_grav_nonlocal_action.py` | **Nonlocal spectral-gravity action** with the local $R{+}R^2$ as its IR projection (Paper C, consolidating v304/v370/v380/v386). [E] the entire form factor $a{=}e^u$ gives the dressed propagator $1/(p^2 a)$ ONE pole (residue 1, ghost-free); [E] every truncation carries a Stelle zero whose nearest-zero modulus grows monotonically (v380); [E] the IR order $a{=}1{+}u{+}u^2/2{+}\dots$ reproduces the spectral-action $R{+}R^2$ (v28/v36) with the atom-fixed scale $M^2_{\text{scal}}/\bar{M}^2_{\text{Pl}}{=}c_3^7$ ($7{=}g_{\text{car}}{+}N_{\text{fam}}{-}1$, v253). [O] the entire-$a$ analyticity assumption + perturbative-only (not the ambient measure `QG.AMB.01`, a [C] redundancy v369) |
| `v401_metrology_closure.py` | **Metrology closure** of the final input set $\{a,\pi,v_{\text{geo}}\}$ (Paper D, consolidating v153/v274/v364/v384). [E] $a{=}(1,1,2) \to (e_1,e_2,e_3){=}(4,5,2){=}(\lvert\mu_4\rvert, g_{\text{car}}, \lvert Z_2\rvert)$, $c_3{=}1/(2e_1\pi){=}1/(8\pi)$, 0 dimensionless dials + 1 unit; [E] dimensional bookkeeping (No-Unit v153) mass/$1G$/$\Lambda$/$U_{\text{point}}$/$(m/\mu)$= number $\times v_{\text{geo}}^{1,2,4,1,0}$; [E] gravity adds no unit (v364); [E] $G$-vs-$\Lambda$ over-determination 0.11% (v274). [O] $v_{\text{geo}}$ theorem-forbidden ($\text{TOE}_{\text{strict}}{=}\text{TOE}_{\text{dimensionless}}{+}[v_{\text{geo}}]$, v384 type B) |
| `v402_ftransfer_suite.py` | $F_{\text{transfer}}$ **functor suite** (Paper E): the four frontier sectors as ONE functor $F_{\text{transfer}}{=}F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ (consolidating `FR.POLE/BOLTZMANN/RELIC/QCD.SOLVE`). [E] the composition law (each observable factors source→threshold→RG→observable); [E] exact atom sources ($Q_{\text{src}}$ v93; $A_\Lambda{=}2g_{\text{car}}{=}10$, $\delta_{CP}{=}4\pi/3$ v9/v212; $f_a{=}M_{\text{scal}}/128$ v185; $b_3{=}{-}7$ v164/v262); [E] a committed kill test per sector. [C] the firewall: physical predictions stay [C]/[O], only the algebraic source is [E] (never re-pressed into the compiler, v187) |
| `v403_facet_compositum.py` | ARITH.HULL.01: the three number-field facets (v390) COMPOSE to $K=\mathbb{Q}(i,\sqrt{-3},\sqrt{5})$, a degree-8$=\text{rank}\,E_8$ abelian field – the arithmetic hull dual to the $E_8$ lattice hull. [E] the squarefree parts $-1,-3,5$ are independent in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, so $[K:\mathbb{Q}]=2^3=8=\text{rank}\,E_8=g_{\text{car}}{+}N_{\text{fam}}$, Galois group $(\mathbb{Z}/2)^3$ order 8; $2^{\omega(h(E_8))}=2^3=8=\text{rank}\,E_8$ is $E_8$-special (the $E_7$ control $h{=}18$, $2^2{=}4{\neq}7$ fails). [E] exactly 7 quadratic subfields $\{-1,-3,5,3,-5,-15,15\}$, split 4 imaginary $=\lvert\mu_4\rvert$ (CM/dynamic) + 3 real $=N_{\text{fam}}$ (RM/static). [E] ramified only at $\{2,3,5\}$ = the atoms = the primes of $h(E_8){=}30$. [C] $K$ is the abelian $(\mathbb{Z}/2)^3$ arithmetic hull dual to $E_8$; the 7 quadratic subfields a candidate for the 7 $E_8$ audit slices [E][C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v404_e8_char_cm_duality.py` | E8.CMDUAL.01: the $(E_8)_1$ vacuum character $\chi_{E_8}{=}E_4/\eta^8{=}j^{1/3}$ at the two CM points – 12 at the seam ($\tau{=}i$), 0 at the family ($\tau{=}\omega$), extending v282. [E] $\chi_{E_8}^3{=}j$ (q-series). [E] SEAM $\tau{=}i$ ($\mathbb{Q}(i)$, order 4): cross-ratio $2 \Rightarrow j{=}1728 \Rightarrow \chi_{E_8}(i){=}1728^{1/3}{=}12$ ($E_8$ present, v214/v282). [E] FAMILY $\tau{=}\omega$ ($\mathbb{Q}(\sqrt{-3})$, order 6): the equianharmonic $\lambda$ ($\lambda^2{-}\lambda{+}1{=}0$) $\Rightarrow j{=}0 \Rightarrow \chi_{E_8}(\omega){=}0$ ($E_4$ vanishes, v220/v267). [E] $\tau{=}i, \tau{=}\omega$ are the ONLY elliptic points of $\mathrm{PSL}(2,\mathbb{Z})$, so the values are forced. [C] the seam/flavor rigidity duality: the seam is rigid where $E_8$ is present ($\chi{=}12$, `SEAM.EQUIV.01` closes), the flavor/CP sector is the soft residual where $E_8$ degenerates ($\chi{=}0$); the dual-keystone candidate (flavor = the $j{=}0$ degeneration of $(E_8)_1$) [E][C] |
| `v405_seam_equiv_omega.py` | **SEAM.EQUIV.02** – the dual keystone at $\tau{=}\omega$: the family/flavor sector as the order-3 Eisenstein/$A_2$ face of $E_8$, dual to the seam = $(E_8)_1$ at $\tau{=}i$ (v404). A named contract (genre of v286/v398), NOT a closure. [E] $\chi_{E_8}(\omega){=}0$ ($E_8$ degenerates at the family CM point) vs $\chi_{E_8}(i){=}12$. [E] $A_2$ is the $\tau{=}\omega$ Eisenstein lattice (Cartan $\det{=}3{=}N_{\mathrm{fam}}$, $h{=}3{=}N_{\mathrm{fam}}$, $\mu_6{=}\mu_3{\times}\mu_2$). [E] $R$ lives in $E_6{\times}A_2$: $248{=}\|R\|_F^2{+}\det R{+}2(\mathbf{1}^\top Ra)N_{\mathrm{fam}}{=}78{+}8{+}2{\cdot}27{\cdot}3$, $\|R\|_F^2{=}78{=}\dim E_6$, $\det R{=}8{=}\dim A_2{=}\mathrm{rank}\,E_8$. [E] $\delta_{\mathrm{PMNS}}{=}4\pi/3$ is the $\tau{=}\omega$ hexagonal CM phase ($\arg\zeta_6{=}\pi/3$, v220/v233). [E] the cusp weights $\{0, 1/3, 2/3\}$ are the order-3 deck $= N_{\mathrm{fam}} = \det Q$ (v72), $\mathrm{Spec}(Q_+){=}(1,2,3)$ (v69/v137). [C] the family/flavor sector is the order-3 Eisenstein/$A_2$ deck at $\tau{=}\omega$; the two keystones sit at the only two elliptic points of $\mathrm{PSL}(2,\mathbb{Z})$. [O] residual = the canonical $\mathbb{Z}_3$ deck map (v140) + the $v_{\mathrm{geo}}$-class scale (No-Unit v153) [E][C][O] |
| `v406_pmns_phase_origin.py` | FLAV.PMNS.CLOSE.01: the PMNS dynamics closed modulo the $\tau{=}\omega$ keystone (v405) – the CP-phase ORIGIN is the family CM point and the absolute scale is one $v_{\mathrm{geo}}$-class unit, so PMNS has zero free flavor dials beyond $v_{\mathrm{geo}}$. [E] the three angles ($\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\mathrm{ret}}}{2}$, $\theta_{23}{=}45°$, $\sin^2\theta_{13}{=}\varphi_0^{\mathrm{ret}}e^{-5/6}$; v9/v268) assemble to one unitary $U_{\mathrm{PMNS}}$ (v270). [C] CP-PHASE ORIGIN (advance over v270): $\delta_{\mathrm{PMNS}}{=}4\pi/3$ is the $\tau{=}\omega$ hexagonal CM phase (v405/v220/v233), upgrading v270's [O] assembled input to a keystone consequence. [C] $J_{\mathrm{PMNS}}{=}{-}0.0297$ derived. [E] the absolute scale = one seesaw ratio $m_3{=}(y_\nu v_{\mathrm{EW}})^2/M_R$ = a $v_{\mathrm{geo}}$-class UV input (v272/v153), NOT a flavor gap. [C] net: PMNS = angles + $\delta$ [keystone] + $J$ [derived] + scale [$v_{\mathrm{geo}}$], zero free flavor dials. [O] residual = the seesaw operator realisation ($M_R$ not a clean $\bar{M}_{\mathrm{Pl}}$ power, v272/v184; one-parameter carrier-normalised candidate v481) + the normal-ordering $\Sigma m_\nu$ floor kill test [E][C][O] |
| `v407_dn_pairings_omega.py` | FLAV.SELECTOR.CLOSE.01: the R4′ residue folds into the $\tau{=}\omega$ keystone (v405) – the three $(d,n)$ selector pairings ARE the $\tau{=}\omega$ family-slice atoms. [E] the frame $(\mathbf{1}, a, \sigma)$ is a $\mathbb{Z}^3$ basis with $\det{=}11{=}\|Pl(K)\|_1$, and $n{=}(5,{-}9,6)$ is the unique covector with $n{\cdot}\mathbf{1}{=}2$, $n{\cdot}a{=}8$, $n{\cdot}\sigma{=}121{=}11^2 \Rightarrow (d,n){\Rightarrow}R$ columnwise (v136/v139). [E] $n{\cdot}a{=}8{=}\mathrm{rank}\,E_8{=}\dim A_2{=}\det R$ – the anchor-pairing is the $A_2$ ($\tau{=}\omega$) block dimension that reads $R$ in $E_6{\times}A_2$ (v405). [E] the pairings $\{2, 8, 121\}{=}\{|Z_2|, \mathrm{rank}\,E_8{=}\dim A_2{=}\det R, \|Pl(K)\|_1^2\}$ are the $\tau{=}\omega$ family-slice atoms (the $\mu_2$ sheet, the $A_2$ block, the Plucker frame). [E] the square selects the point ($\sigma$-pairing $11(11{+}t)$, a perfect square only at $t{=}0$). [C] deriving $(d,n)$ (R4′) folds into `SEAM.EQUIV.02` – $d{=}\frac{3}{2}a{-}\mathbf{21}$ anchor-derived (v139), the pairings the slice atoms. [O] the lone flagged residue "why $11{=}\|Pl(K)\|_1$" [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v408_phi0_puncture_heatkernel.py` | `HYP.PHI0.PUNCTURE.01`: name the $\varphi_0$ puncture term $\delta_{\text{top}}{=}48c_3^4$ as the order-4 local boundary heat-kernel (Seeley-DeWitt) contact term of the $\mu_4$ puncture divisor – the analytic complement of the icosahedral tree term (v396, `HYP.PHI0.01`). [E] THE SPLIT IS EXACT: $\varphi_0{=}\varphi_0^{\text{tree}}{+}\delta_{\text{top}}{=}1/(6\pi){+}48c_3^4$ (v396/`ARCH.QUAD.01`). [E] PUNCTURE COUNT $=\Omega_{\text{adm}}{=}N_{\text{fam}}\dim S^+{=}48$, order $=|\mu_4|{=}4$: $\delta_{\text{top}}{=}\Omega_{\text{adm}}c_3^{|\mu_4|}$. [E] CLOSED FORM $\delta_{\text{top}}{=}48c_3^4{=}3/(256\pi^4)$; $c_3{=}1/(|Z_2|4\pi)$ Gauss-Bonnet (v216/v342) so $c_3^4$ is a fourth-order boundary heat-kernel power (cf. v391). [E] HONEST NEGATIVE: $\delta_{\text{top}}$ not a rational hypergraph fraction ($\delta_{\text{top}}/\varphi_0^{\text{tree}}{<}1\%$). [O] THE TARGET: prove $\Omega_{\text{adm}}c_3^4$ IS the order-4 boundary Seeley-DeWitt contact term of the $\mu_4$ puncture summed over the $\Omega_{\text{adm}}$ admissible states. [C] two engines, two terms; a FACE of the seam boundary analysis (the $c_3$ datum), not a new gate [E][C][O] |
| `v409_coxeter_prime2_lemma.py` | `RES.COXETER.SYMMETRY.01` closed as a STRUCTURAL LEMMA: prime-2 is NECESSARY but NOT autonomous – no prime-2-only attractor exists (resolves the v394/`COXETER.ATOMS.01` open question's falsifiable corollary). [E] SHEET INVOLUTION $\sigma{=}\,\text{diag}(1,-1,-1)$, $\sigma^2{=}I$, spectrum $\{+1,-1\}$ (parity). [E] SHEET PROJECTOR $P{=}(I{+}\sigma)/2{=}\,\text{diag}(1,0,0)$, $P^2{=}P$, spectrum $\{0,1\}$ (projection). [E] NO CONTRACTION RATE: none of $\{-1,0,1\}$ lies in $(0,1)$, whereas the genuine rates $2/3$ and $(1/3)^6$ do – prime-2 alone gives parity/projection, never a nonzero rate. [E] FACTOR NOT RATE: $2/3{=}|Z_2|/N_{\text{fam}}$ needs prime-3, $(\varphi{+}2)/4{=}(\varphi{+}2)/|\mu_4|$ with $|\mu_4|{=}2^2$ needs the golden $\varphi$ (prime-5). [E] $30{=}g_{\text{car}}(2N_{\text{fam}})$, carrier static, 2·3 dynamic. [C] prime-2 is the universal dynamical parity factor, necessary in every sheet-paired transport but not an autonomous attractor; dynamics begins after coupling to family-3 or carrier-5 ($\mu_4{=}2^2$ the Galois bridge) [E][C] |
| `v410_sheet_generator_binary.py` | The *sheet generator* $V{=}Q\,\text{diag}(0,1,1)$ (rank-2 sheet axis, v95/v218) is a binary internal compiler: $V^n{=}\left(\begin{smallmatrix}0 & 2^{n-1} & 0\\ 0 & 2^n & 0\\ 0 & 2^{n+1}{-}2 & 1\end{smallmatrix}\right)$ (induction), so $V^n\mathbf{1}{=}(2^{n-1},2^n,2^{n+1}{-}1){=}(1,2,3),(2,4,7),(4,8,15),(8,16,31)$; bilinears $\mathbf{1}^TV^n\mathbf{1}{=}7{\cdot}2^{n-1}{-}1$, $\mathbf{1}^TV^na{=}7{\cdot}2^{n-1}$, $a^TV^na{=}11{\cdot}2^{n-1}$, $a^TV^n\mathbf{1}{=}11{\cdot}2^{n-1}{-}2$ give $(6,7,9,11)$ then $13{=}\Delta_Q$, $27{=}\mathbf{1}^TRa$, 55, $56{=}\dim\mathbf{56}_{E_7}$; the spine is FORCED by $\text{Spec}(V){=}\{0,1,2\}$, the carrier (rank $E_8{=}8$, $\dim S^+{=}16$, $248/8{=}31$) and theta ($\sigma_3(2){=}9{=}a^TV\mathbf{1}$, $\sigma_3(3){=}28{=}\mathbf{1}^TV^3a{=}\det(I{+}R)$, $\sigma_3(5){=}126$) readings audit-typed [E][C] |
| `v411_ud_ratio_vpower.py` | The quark $u/d$ ratio is a pure $V$-power readout: the established $c_u/c_d{=}g_{\text{car}}\|\,\text{Pl}(K)\|_1/(N_{\text{fam}}^2\Delta_Q){=}55/117$ (v94) re-expresses EXACTLY via the sheet generator (v410) as $\boxed{c_u/c_d=\frac{\mathbf{1}^TV^4\mathbf{1}}{(a^TV\mathbf{1})(\mathbf{1}^TV^2\mathbf{1})}}=\frac{55}{9\cdot13}$, with $55{=}\mathbf{1}^TV^4\mathbf{1}$, $9{=}a^TV\mathbf{1}{=}N_{\text{fam}}^2$, $13{=}\mathbf{1}^TV^2\mathbf{1}{=}\Delta_Q$; the same number as $5{\cdot}11/(9{\cdot}13)$, so the numerator repackages $g_{\text{car}}{\cdot}11{=}55$. An exact *re-encoding*: the value and the $u/d$ identification stay [C] (U_wall readout rigidity) [E][C] |
| `v412_sheet_source_corner_J.py` | The *sheet source corner* $J$: the determinant surface has two $s$-independent iso-volume walls (v135) $t{=}-1$ ($\det{=}4{=}|\mu_4|$, carries $K$) and $t{=}-2$ ($\det{=}2{=}|Z_2|$); the operator on the $Z_2$ wall, $J{=}M(1,-2){=}C{-}2V{=}\left(\begin{smallmatrix}4&1&0\\4&1&2\\5&1&1\end{smallmatrix}\right)$, completes the sheet axis $J,K,C,F{=}M(1,-2..1)$.<br>$\chi_J{=}\lambda^3{-}6\lambda^2{+}3\lambda{-}2\Rightarrow(\text{tr},e_2,\det){=}(6,3,2){=}(|R^+(A_3)|,N_{\text{fam}},|Z_2|)$;<br>$a^TJa{=}30{=}h(E_8)$, $\det(I{+}J){=}12{=}\dim\mathfrak{g}_{\text{SM}}{=}\chi_{E_8}(i)$, $\det(2I{+}J){=}40{=}|R(D_5)|$ (Lie readings audit-typed like the $G_2/F_4$ shadows of v95/v218) [E][C] |
| `v413_sheet_characteristic_calculus.py` | The sheet axis is a discrete *characteristic-difference calculator*: on $M_t{=}C{+}tV$ (v95/v218) the elementary-symmetric coefficients encode atoms as DIFFERENCE ORDERS – $e_1{=}3t{+}12$, $e_2{=}2t^2{+}15t{+}25$, $e_3{=}\det M_t{=}4t^2{+}14t{+}14$ give $\Delta e_1{=}3{=}N_{\text{fam}}$, $\Delta^2e_2{=}4{=}|\mu_4|$, $\Delta^2e_3{=}8{=}\text{rank}\,E_8$ (the $e_3$ curvature cross-cited to v218/v224; the full $(e_1,e_2,e_3)$ ladder is new); the anchor energy $a^TM_ta{=}52{+}11t$ (step $11{=}a^TVa$, v410) reads $30{\to}41{\to}52{\to}63{=}(h(E_8),10b_1,\dim F_4,7N_{\text{fam}}^2)$ [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v414_center_resolvent_portal.py` | The center $C$ is a *resolvent portal* $G_2 \to F_4 \to E_8$: $C=M(1,0)$ (v95) is the $t=0$ value of $V$; $\mathrm{tr}\, C=12=\dim \mathfrak{g}_{\mathrm{SM}}=\chi_{E_8}(i)$, $\det C=14=\dim G_2$, $\sum C=31=2^{g_{\mathrm{car}}}-1=248/8$; the shifted determinants climb $\det(C)=14=\dim G_2$, $\det(I+C)=52=\dim F_4$, $\det(2I+C)=120=\lvert R^+(E_8)\rvert$ – the shell ladder $G_2 \to F_4 \to E_8^+$ as $\det(C+kI)$, $k=0,1,2$ (the resolvent ladder is new; reading audit-typed) [E][C] |
| `v415_gaussian_operator.py` | The square CM-norm dictionary is *operator-realised* and the $\mu_4$ deck is an intrinsic diamond object: the Gaussian arithmetic of v222/v230 lifted from numbers to OPERATORS via $J=[U,V]/3$, the complex structure born from the commutator of the binary $V$ and ternary $U$ compilers (v410/v95). [E] $J=\left(\begin{smallmatrix}-1&1&0\\-2&1&0\\-3&1&0\end{smallmatrix}\right)$ is integer, $\mathrm{Spec}\, J=\{i,-i,0\}$, $J^2=-I$ on the rank-2 image, $J^3+J=0$ (a $\mu_4$ quarter-turn); [E] $3I+2J$ has spectrum $\{3+2i, 3-2i, 3\}$ and $5I+4J$ has $\{5+4i, 5-4i, 5\}$, so $3+2i$ ($\lvert\cdot\rvert^2=13=\Delta_Q$) and $5+4i$ ($\lvert\cdot\rvert^2=41=10b_1$) are literal operator eigenvalues; [E] $(3I+2J)(3I-2J)$ has $\mathrm{Spec}\{13,13,9\}$ and $(5I+4J)(5I-4J)$ has $\{41,41,25\}$ (kernel slot = real part$^2$: $9=N_{\mathrm{fam}}^2$, $25=g_{\mathrm{car}}^2$); [E] the intrinsic deck $D=-I+J-J^2$ is order 4 ($\mathrm{Spec}\{i,-1,-i\}$, $\chi=(x+1)(x^2+1)$) with $\chi_2$ line $\ker[U,V]=(0,0,1)=a-\mathbf{1}$. [C] the geometric-deck identification sharpens (does not close) the v140 $\mathbb{Z}_3$ residue [E][C] |
| `v416_atom_trichotomy.py` | The atom trichotomy: $\{2,3,5\}$ are ramified/inert/split in the two exceptional CM rings, and each atom is the unique ramified prime of one quadratic facet of $K=\mathbb{Q}(i,\sqrt{-3},\sqrt{5})$ (v403). [E] $\mathbb{Z}[i]$ (square/seam, $\tau=i$): $2=\lvert Z_2\rvert$ ramifies ($2=-i(1+i)^2$), $3=N_{\mathrm{fam}}$ inert (norm $9=N_{\mathrm{fam}}^2$), $5=g_{\mathrm{car}}$ splits ($(2+i)(2-i)$) – the carrier rank factors on the seam; [E] $\mathbb{Z}[\omega]$ (hex/flavor, $\tau=\omega$): $3=N_{\mathrm{fam}}$ ramifies ($N(1-\omega)=3$), 2 inert (norm 4), 5 inert (norm $25=g_{\mathrm{car}}^2$); [E] the ramified prime of each ring IS its atom ($\mathbb{Z}[i]$ at 2= sheet/seam, $\mathbb{Z}[\omega]$ at 3= family/flavor); [E] the three facets $\mathbb{Q}(i)/\mathbb{Q}(\sqrt{-3})/\mathbb{Q}(\sqrt{5})$ (disc $-4,-3,5$) ramify over one atom each, product $2\cdot3\cdot5=30=h(E_8)$; 2 imaginary (CM/dynamic) + 1 real (RM/static). [C] the seam/flavor/golden reading [E][C] |
| `v417_eisenstein_cp_operator.py` | The Eisenstein/CP operator (the hex dual of v415): the family rotation $P=(1\,2\,3)$ and the CP clock $W=-P^2$ put the $\tau=\omega$ flavor side on operators, unified with the seam into ONE order-30 clock. [E] $P^3=I$, $P^2+P+I=$ONES (Eisenstein $\omega^2+\omega+1=0$ on the plane $\perp(1,1,1)$); [E] $(3I+2P)(3I+2P^2)=7I+6$ ONES, $\mathrm{Spec}\{7,7,25\}$ ($7=N_{\mathbb{Z}[\omega]}(3+2\omega)=$ scalaron on the $\omega$-plane, $25=g_{\mathrm{car}}^2$) — the dual of v415's $\{13,13,9\}$; [E] the CP clock $W=-P^2$ ($=\rho$, v233) has $W^2=P$, $W^3=-I$, $\mathrm{Spec}\{-1,\zeta_6,\zeta_6^{-1}\}$, so $\delta_{\mathrm{CKM,lead}}=\arg\zeta_6=\pi/3$ and $\delta_{\mathrm{PMNS}}=\arg\zeta_6^4=4\pi/3$ ($W^4=P^2$); [E] all flavor clocks are powers of $\zeta_{30}$ ($\zeta_{30}^{15}=-1$, $\zeta_{30}^{10}=\omega$, $\zeta_{30}^{6}=\zeta_5$, $\zeta_{30}^{5}=\zeta_6$) while the seam $\mu_4=i$ is the Galois side $(\mathbb{Z}/30)^\times$ of order $\varphi(30)=8=\mathrm{rank}\, E_8$. [C] the carrier $\mu_5$ has no clean $3\times3$ operator (golden $\varphi$ an output-side trace, v313/v349); honest negatives included [E][C] |
| `v418_cyclotomic_norm_triple.py` | The cyclotomic norm triple: the carrier split $(3,2)$ read in the three atom-rings gives $(7,13,55)=$(scalaron, $\Delta_Q$, quark numerator), and the missing carrier-5 clock of v417 is the $4\times4$ $\Phi_5$-companion $C_5$. [E] $C_5$ has $\chi=x^4+x^3+x^2+x+1$, $C_5^5=I$, $\mathrm{tr}=-1$; the golden ratio is its real-part data ($\zeta_5+\zeta_5^4=1/\varphi$, $\zeta_5^2+\zeta_5^3=-\varphi$; output-side v313/v349; dim 4, not 3 — refines v417's gap); [E] $N_{\mathbb{Z}[\zeta_5]}(3+2\zeta_5)=\det(3I+2C_5)=55=5\cdot11=g_{\mathrm{car}}\lVert \mathrm{Pl}(K)\rVert_1=$ the quark numerator (v410/v411); [E] the triple $\det(3I+2\,\mathrm{Comp}(\Phi_n))=(7,13,55)$ for $n=3,4,5$ over $\mathbb{Z}[\omega],\mathbb{Z}[i],\mathbb{Z}[\zeta_5]$ (atoms 3, 2, 5, v416); [E] NEG control: the anchor $(5,4)\to(21,41,461)$, with $21=N_\omega(5,4)$ and $41=10b_1$ named (v222) but the carrier ring 461 prime/un-named, so the carrier ring separates $(3,2)\to55$ from $(5,4)\to461$. [C] HONEST: $(3,2)$ is FORCED (v14) not unique — the clean splits are $\{(2,1)\to(3,5,11),(3,2)\to(7,13,55)\}$, a small ladder; the sharper reading: only $(3,2)$ spans three physics sectors (gravity/flavor/masses), the unique cross-sector seed [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v419_seam_galois_carrier.py` | The seam $\mu_4$ *is* the carrier's Galois group: a positive resolution of the v409/`RES.COXETER.SYMMETRY.01` cyclic/Galois asymmetry. [E] $h(E_8)$=30=2·3·5 is squarefree, so $\mathbb{Z}/30$ has *no* order-4 element — the seam $\mu_4$ cannot be a cyclic hand and is forced onto the Galois side; [E] $(\mathbb{Z}/30)^\times=(\mathbb{Z}/2)^\times\times(\mathbb{Z}/3)^\times\times(\mathbb{Z}/5)^\times=\mu_4\times\mathbb{Z}_2$ (order $\varphi(30)$=8= rank $E_8$), where the $\mu_4$ ($\mathbb{Z}/4$) factor IS $(\mathbb{Z}/5)^\times$ (carrier 5) and the $\mathbb{Z}_2$ IS $(\mathbb{Z}/3)^\times$ (family/CP); [E] so the seam $\mu_4=\mathrm{Gal}(\mathbb{Q}(\zeta_5))=(\mathbb{Z}/5)^\times$, the Frobenius $\zeta_5\mapsto\zeta_5^2$, realised on the carrier clock $C_5$ (v418) by the explicit order-4 operator $G$ with $GC_5G^{-1}=C_5^2$, $G^4=I$, $G^2C_5G^{-2}=C_5^{-1}$. [C] resolves v409 (the $2^2$=4 is the order of $(\mathbb{Z}/5)^\times$; "no contraction rate" because the seam is an automorphism, not a hand); $\mathbb{Q}(\zeta_{30})$ carries both $E_8$ invariants (cyclic $h$=30, Galois rank =8). Figure `coxeter_galois` [E][C] |
| `v420_fpole_qed_running.py` | F_pole external: the Koide source→pole transfer is a *one-loop electromagnetic* effect, not an algebraic step — the honest external-physics follow-up to v93 and the clock No-Go. [E] Koide $Q$ is rescaling-invariant ($Q(c\,m)=Q(m)$), so flavor-*universal* running leaves it fixed and the source→pole shift is purely flavor-*split*; [C] the tree→pole gap $\frac{2}{3}-Q_{\mathrm{src}}$=0.00220 is one-loop-EM-sized ($=\varphi_0^{\mathrm{ret}}/24$ to 0.6%, $=\alpha/\pi$ to ∼5%), the PDG pole sits AT $\frac{2}{3}$ (the IR attractor); [C] a 1-loop QED per-lepton-log pole↔$\overline{\mathrm{MS}}$ estimate shifts Koide by EM-loop order toward $\frac{2}{3}$ (the v82 IR-attractor direction); [O] the EXACT flow time $t$∼2.84 (v93) is scheme/scale-dependent external QED, only sized. [C] F_pole stays [C], firewalled (v187): the lens forces the RATE $(2/3)^6$, this external EM supplies the flow. Numerical (mpmath), Python-only [E][C][O] |
| `v422_galois_net_bridge.py` | The Galois↔Net bridge: the seam $\mu_4=\mathrm{Gal}(\mathbb{Q}(\zeta_5))$ is the SAME cyclic $\mathbb{Z}/4$ as the $(E_8)_1$ simple-current glue — not a mere order-4 coincidence — linking the Galois gearbox of v419 to $G_{\mathrm{net}}$/`SEAM.EQUIV.01`. [E] $\mathrm{disc}(A_3)=\mathrm{disc}(D_5)=\mathbb{Z}/4$ (one Smith invariant factor 4 = cyclic, det =4); $D_n$ disc is $\mathbb{Z}/4$ for $n$ ODD, $\mathbb{Z}_2\times\mathbb{Z}_2$ for $n$ EVEN, so the carrier $D_5$ (rank 5=$g_{\mathrm{car}}$, odd) is cyclic while the control $D_8$ (rank 8, even) is Klein; [E] $\mathrm{Gal}(\mathbb{Q}(\zeta_5))=(\mathbb{Z}/5)^\times=\langle 2\rangle$ is cyclic order 4 (only the carrier prime 5 gives order 4); [E] in $\mathrm{disc}(D_5\oplus A_3)=\mathbb{Z}_4\times\mathbb{Z}_4$ (16= $\dim S^+$) the simple current $(1,1)$ is order 4 (cyclic) with Lagrangian quotient $16/4^2=1\Rightarrow(E_8)_1$ (holomorphic, v89/v154). [E] NEG control: the Klein $\mathbb{Z}_2\times\mathbb{Z}_2$ / order-2 halfway $\langle(2,2)\rangle$ gives $16/2^2=4=|\mathrm{disc}(D_8)|=SO(16)$ (det =4), NOT $E_8$ (det =1) — cyclic $\mathbb{Z}/4\to E_8$, Klein$\to D_8$. [C] the bridge: one cyclic $\mathbb{Z}/4$ threads the seam deck $\mu_4$ (v419), $\mathrm{Gal}(\mathbb{Q}(\zeta_5))$ and the $(E_8)_1$ glue (v1/v89/v154); cyclicity forced by 5, Klein excluded. [C] this is the four-fold $\mu_4$ identity (deck/divisor = Galois = discriminant = simple-current); per v428 a *compression* (one object read four ways), not four independent witnesses [E][C] |
| `v423_no_ambient_dependency.py` | The *No-Ambient-Dependency* certificate: no frozen TFPT readout is *computed* from the ambient QG measure `QG.AMB.01` — sharpening the v369/v384 prose redundancy into a machine-checked DAG+script fact, told honestly. [E] gap-decoupling margin $\Delta_{\mathrm{eff}}=6\ln(3/2)-31/(4\pi^2)\approx 1.648>0$ ($31=2^{g_{\mathrm{car}}}-1$), susceptibility $\chi=729/665$ finite (neg control gap$\to 0\Rightarrow\chi\to\infty$). [E] `QG.AMB.01` is a certification *sink*: every claim listing it as a dependency is a gravity-gate/QG row, none a frozen prediction. [E] it is not a value source — it has no producing script while each frozen readout is computed by a compiler script ($\alpha^{-1}$←v3, $\theta_{12}$←v9, $\theta_{13}$←v268, cosmology←v7). [E] the one transitive link is architectural: only $\theta_{13}$ references `QG.AMB.01`, via the 4D-QFT contract spine (`PS.DIRAC`→`CONTRACT.QFT4D`→`QG.G2`→`GATE.METRIC`), a coherence edge not a numerical input; $\alpha^{-1}/\theta_{12}$/cosmology have no route. [C] so `QG.AMB.01` is gap-decoupled certification (v369 as a DAG/script fact); honestly this does NOT prove `SEAM.EQUIV.01` — the readouts depend on the seam being $(E_8)_1$; `QG.AMB.01` specifically is dead [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v424_seam_lto_rp_reduction.py` | The *(LTO-RP) reduction* of the keystone: an honest MMST-style reduction of `SEAM.EQUIV.01`'s one open lemma — the raw collar's intrinsic Bisognano–Wichmann property (the v308/v309 residual) — to the cited commuting-projector theorem of Naaijkens–Penneys–Wallick (`arXiv:2605.10693`), with the central mapping *corrected*. [E] the $\mu_4$ clock is a modular *symmetry* ($[\rho,K]{=}0$, v309/v199), not the flow — $\sigma^{\psi}{=}\rho$ is type-wrong (a one-parameter group versus a finite order-4 element). [E] NPW26's axiom (LTO-RP) means $u_\Theta{=}J$ (reflection = modular conjugation) = the intrinsic BW condition $\theta K\theta{=}-K$, *distinct* from $\omega{\circ}\rho{=}\omega$: a positive character-block-diagonal $K$ satisfies $[\rho,K]{=}0$ yet never $\theta K\theta{=}-K$, so clock-invariance does *not* imply (LTO-RP). [O] make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* models (Toric Code $\lvert\det K\rvert{=}4$ via a trace; Levin–Wen anyonic), while the seam is invertible ($\lvert\det \mathrm{Cartan}(E_8)\rvert{=}1$, no anyons) and a $\beta{=}1$ KMS state (not a trace) — in neither proved bucket (neg control $D_8$ $\lvert\det\rvert{=}4$). [C] so `SEAM.EQUIV.01`'s BW residual reduces to NPW26 Theorem A modulo two open sub-steps (realise the seam as a $\mathbb{Z}_2$-reflection commuting-projector LTO; extend (LTO-RP) to the invertible KMS case) — a reduction, not a closure; `SEAM.EQUIV.01` stays [O] [E][C] |
| `v425_dyn_transfer_universal.py` | `DYN.TRANSFER.UNIVERSAL.01` — the *single-flow reduction* of $F_{\text{transfer}}$: the four frontier transfers ($F_{\text{pole}}$, $F_{\text{Boltzmann}}$, $F_{\text{relic}}$, $F_{\text{QCD}}$) do not each need a separate external solver; their *dynamics* is the ONE native flow of the theory — the seam modular/recovery semigroup (v238/v240, gap $6\ln(3/2)$) restricted to each sector. [E] recovery spectrum $\{1,(2/3)^6,(1/3)^6\}$, $H_{OS}{=}-\log T$ gap $6\ln(3/2)$, $e^{-\Delta}{=}(2/3)^6$ exact; an explicit column-stochastic $T$ makes $\log T$ a Markov generator. [E] $F_{\text{pole}}$ is this flow: the v99 generator $\dot q{=}(\Delta/N_{\text{fam}})(q{-}2)(q{-}5)$ linearised at $q{=}2$ has rate $-\Delta$, time-1 contraction $(2/3)^6$. [E] $F_{\text{Boltzmann}}$ washout = that contraction, CP phase $4\pi/3$ native (v320). [E] $F_{\text{QCD}}$ rate native ($b_3{=}{-}7$). [C] only the anchors ($v_{\text{geo}}$, $C_p$, $M_R$) external; $F_{\text{relic}}$'s cosmological IC $\theta_i$ the one wall. Extends v383 static→dynamic; a reduction, no new premise [E][C] |
| `v426_seam_lto_rp_kms.py` | `SEAM.EQUIV.LTORP.KMS.01` — the *invertible* $\beta{=}1$-*KMS* variant of (LTO-RP), the corner NPW26 leave open (they prove it for topologically-ordered models; the seam is invertible + $\beta{=}1$ KMS, not a trace). A constructive Tomita–Takesaki witness advancing v424 sub-step (ii); does *not* close `SEAM.EQUIV.01`. [E] build $K$ reflection-odd/$\mu_4$-even/gapped (gap $2k{=}6\ln(3/2)$); $C{=}1/(1{+}e^K)$ the $\beta{=}1$ KMS covariance, $C/(1{-}C){=}e^{-K}$ (genuine KMS, not trace). [E] $\Theta K\Theta^{-1}{=}{-}K$ ($u_\Theta{=}J$) AND $[\rho,K]{=}0$ both hold ⇒ (LTO-RP) satisfied by direct Tomita–Takesaki. [E] $\lvert\det\mathrm{Cartan}(E_8)\rvert{=}1$ (invertible, no anyons) vs anyonic $\lvert\det\rvert{=}4$; state not a trace ($\Delta{=}e^{-K}{\neq}I$) — neither NPW26 bucket. [E] neg controls: side-even reflection → $+K$ (BW fails), off-$\mu_4$-block $K$ breaks $[\rho,K]$. [C] (LTO-RP) reduces to $(\Theta K\Theta{=}{-}K)\wedge([\rho,K]{=}0)$, constructively satisfiable; `SEAM.EQUIV.01` stays [O] [E][C] |
| `v427_overdet_witness_map.py` | `OVERDET.WITNESS.MAP.01` — the multiply-vs-compress *framework* for over-determination: witnesses from *disjoint* grammars multiply, projections of the *same* generator compress. The classification of the seven arithmetic witnesses is refined by v428 (they compress). [E] seven grammar readouts, each from its own syntax landing on the carrier skeleton: Gauss $N(3{+}2i){=}13$, Eisenstein $N(3{+}2\omega){=}7$, cyclotomy $N(3{+}2\zeta_5){=}55$, Galois $\lvert(\mathbb{Z}/5)^\times\rvert{=}4$, $\lvert\det\mathrm{Cartan}(E_8)\rvert{=}1$, Pascal $\binom{4}{0}{+}\binom{4}{1}{+}\binom{4}{2}{=}11$ ($2^4{=}16{=}\dim S^+$), Coxeter $h(E_8){=}30{=}2{\cdot}3{\cdot}5$ with $\varphi(30){=}8{=}\mathrm{rank}\,E_8$. [E] by v236 these are facets of ONE $(2,3,5)/E_8$ object, so syntactic distinctness is NOT independence — they compress, like the anchor $a{=}(1,1,2)$. [C] the genuine multiplicative strength is the multiply-forced inputs + foreign witnesses (v428); the map does not by itself establish physics [E][C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v428_overdet_reclass.py` | `OVERDET.WITNESS.RECLASS.01` — the honest correction of `v427`: applying its own multiply-vs-compress test to itself. [E] the $(2,3,5)$ Brieskorn singularity (v236) is the ONE generator — Milnor $\mu{=}(2{-}1)(3{-}1)(5{-}1){=}8{=}\operatorname{rank} E_8$, $30{=}2{\cdot}3{\cdot}5{=}h(E_8)$, $\varphi(30){=}8$. [E] every one of `v427`'s seven witnesses maps to a facet of that SAME object (primes $2,3,5$ of 30, the clock 30, or $E_8$), so all seven collapse to ONE generator (distinct objects $=1$) $\Rightarrow$ COMPRESSION, not seven multiplications. [E] what honestly multiplies is the input forced four independent ways — the "8" in $c_3{=}1/(8\pi)$ from $\operatorname{rank} E_8$, $h(D_5){=}2(5{-}1)$, $\varphi(30)$ and the Milnor number — plus the foreign witness $\alpha^{-1}{\approx}137$ (prime, not in the $(2,3,5)$-skeleton). [C] refines and partly walks back `v427`; consistent with v236/v305/v313 [E][C] |
| `v429_axion_pentagon_phi.py` | `DM.AXION.PENTAGON.01` — the 'unmapped' icosahedral/golden $E_8$ structure surfaces in the ONE external cosmological input $\theta_i$, answering "why does idle $E_8$ structure exist?" for the one case where it has a job. [E] because $N_{\rm fam}{=}g_{\rm car}{-}2$, the spine quotient $\theta_i{=}\pi N_{\rm fam}/g_{\rm car}{=}(g_{\rm car}{-}2)\pi/g_{\rm car}$ is EXACTLY the interior angle of the regular $g_{\rm car}$-gon; for $g_{\rm car}{=}5$ the PENTAGON, $\theta_i{=}3\pi/5{=}108°$ (the carrier polygon, not a bare rational multiple of $\pi$). [E] its cosine is golden: $\cos(3\pi/5){=}(1{-}\sqrt{5})/4{=}{-}1/(2\varphi)$, partner $2\cos(2\pi/5){=}1/\varphi{=}\varphi{-}1$, $\varphi{=}2\cos(\pi/5){=}(1{+}\sqrt{5})/2$. [E] the golden character is UNIQUE to $g_{\rm car}{=}5$: among small $n$-gons only $n{=}5$ has an irrational interior-angle cosine ($n{=}3,4,6$ give $1/2, 0, {-}1/2$), so $\theta_i$ is golden BECAUSE the carrier is 5 (P2). [C] a re-reading bridging v354/v313 (the unmapped $g_{\rm car}{=}5$ golden signature) to v211 ($\theta_i{=}3\pi/5$, the one external input): it REFINES v354 from "$\varphi$ in no readout" to "$\varphi$ touches only the [C] spine misalignment angle", a geometric motive for the spine over the hilltop ${\sim}170°$; it does NOT upgrade `DM.AXION.SPINE.01` (stays [C]) and closes no gate. [E][C] |
| `v430_other_side_reverse_audit.py` | `E8.OTHERSIDE.AUDIT.01`: the 'other side of the seam' (the double-cover deck / conjugate sheet $S^-$) audited against $E_8$'s UNMAPPED Casimir degrees — the sheet/deck complement of v354/v355, answering "is the leftover where the second sheet lives?" with a disciplined NEGATIVE. [E] DECK = DEGREE-2: the one-sided cover contributes $|\mathbb{Z}_2|{=}2$ (the $1/2$ in $c_3{=}1/(2{\cdot}4\pi){=}1/(8\pi)$, v58/v216); $2{=}\min(E_8\text{ degrees}){=}$ the metric, one of the matched primary readouts $\{2,8,30\}$ (v354), NOT one of the unmapped $\{12,14,18,20,24\}$. [E] TWO SHEETS = 128-SPINOR: $\dim S^+{=}\dim S^-{=}16$, the spinor block $128{=}\operatorname{rank}\cdot\dim S^+{=}8{\cdot}16{=}2^{\operatorname{rank}-1}{=}\sum\deg$, the spinor half of $248{=}120{+}128$; $S^-$ the conjugate $(\overline{16},\overline{4})$, $128{=}2{\cdot}64$, not a spare singlet. [E] FORCED-DISJOINT: the sheet/deck integer set $\{2,16,32,128\}$ has EMPTY intersection with $\{12,14,18,20,24\}$, its only degree-alphabet contact the matched 2. [O] DISCIPLINED DECLINE: each unmapped degree needs an UNFORCED coefficient from the sheet atoms and admits $\geq 2$ readings (the v355 discriminator) — no forced sheet$\to$unmapped-degree identity. [E] '$S^-{=}$DM' DOWNGRADED + WIMP NO-GO: v227 replaced the over-strong '$S^-$ is dark matter'; $128{=}(16,4){+}(\overline{16},\overline{4})$ is the SAME matter spinor (no spare singlet), DM is the determinant-line axion (a phase). [O] the only live 'other side' question (the 128 phase/glue dark sector, v227) concerns MATCHED structure, not the unmapped degrees [E][O] |

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| Script | What it machine-checks *(continued)* |
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| `v431_e8_degree_ladder.py` | `E8.DEGREE.LADDER.01`: the reverse-audit 'unmapped' $E_8$ Casimir degrees are NOT diffuse overhead but the forced two-family decomposition $\deg(E_8)$=6·spine$\{2,3,4,5\} \sqcup (\{2\} \cup$ det-ladder$\{8,14,20\})$ — the structural complement of v354/v355/v430. [E] TWO-FAMILY SPLIT FORCED BY 30=2·3·5: the $E_8$ exponents are the $\varphi(30)$=8 totatives of $h$=30=$2N_{\mathrm{fam}}g_{\mathrm{car}}$ (coprime to 6 $\Rightarrow \pm 1$ mod 6), so the degrees occupy ONLY the residue classes $\{0,2\}$ mod 6. [E] $6k$ family $\{12,18,24,30\}/6=\{2,3,4,5\}=$ the v91 spine $\{e_3,e_1,e_2,p_0\}(a=(1,1,2))$. [E] $6k$+2 family $\{2,8,14,20\}$: $\{8,14,20\}=(\det R, \det C, \det L)=$ the winding line $\det M(s,0)$=$6s$+8 of the flavor diamond (v135), $C$=$R$+$Q$ diag$(1,0,0)$. [E] '18=$6N_{\mathrm{fam}}$' is NOT a holdout (spine family, not on the det-ladder). [E] SPECIAL TO $E_8$ among the simply-laced exceptionals ($E_6/E_7/D_5$ fail the clean $\{0,2\}$ mod 6 split). [C] HONEST v355 LINE: the arithmetic decomposition is exact, but the FUNCTORIAL claim '$6k$+2 Casimir stratum = flavor sheet operators' needs a representation-theoretic map ([C]); 12, 24 admit two canonical readings each, so this is a STRUCTURE theorem, not a forced functorial flavor map — it reclassifies v354 overhead to zero orphan degrees, closes no gate [E][C] |
| `v432_overdet_floor.py` | `OVERDET.FLOOR.01`: the UNCONDITIONAL improbability floor + the L1↔L2 bridge — the defensible number that survives even if a skeptic REJECTS the declared formula grammar, complementing the conditional null model (v100) and the multiply-vs-compress framework (v427/v428). [E] DISJOINT PIECES (v428): compression removed (v427's seven readouts are one $(2,3,5)/E_8$ object, NOT multiplied); what multiplies is the input '8' forced FOUR ways $\{\mathrm{rank}\,E_8,\ h(D_5)$=$2(g_{\mathrm{car}}-1),\ \varphi(30),\ \mathrm{Milnor}(2,3,5)\}$ plus the FOREIGN witness $\alpha^{-1}\sim$137 (prime, outside $(2,3,5)$). [C] CONSERVATIVE FLOOR: multiplying ONLY across disjoint pieces (input over-determination × the $\alpha$ grammar census 1/94500 (v100) × the $\varphi_0^{\mathrm{ret}}$-seed cluster counted ONCE × one downstream witness) the floor stays $\leq 10^{-6}$ across a grid of generous assignments, nominal $\sim 10^{-10}$ — 'extraordinary' WITHOUT the declared grammar. [E] THE BRIDGE: the floor ($\sim 10^{-10}$) sits ~20 orders ABOVE the conditional $10^{-30.7}$ (v100); that gap = the evidential value of proving the FORMS forced, so L1 (`ALPHA.QUILLEN.EXACT.01`, v382) and L2 (this floor) are the SAME lever. [C] HONEST SCOPE: bounds ONLY "is it numerology?", NOT the physics bridge (firewall v187); a conservative FLOOR, not a posterior [E][C] |
| `v433_alpha_quillen_heatkernel.py` | ALPHA.QUILLEN.PROGRESS.02: a *second honest step* on the external target ALPHA.QUILLEN.EXACT.01 (v382) – it GROUNDS and CONNECTS, it does NOT close it; ALPHA.QUILLEN.EXACT.01 stays [O] (external math, type A, v384), $\alpha^{-1}$ stays [E]. [E] a solvable 4D model (flat $\Delta = -\partial^2 + m^2$ on $T^4$) reaches the $a_4$ ($t^0$) conformal-anomaly order: $a_4 = Vm^4/(2(4\pi)^2)$ NON-ZERO – the order v391's 1D circle toy ($a_0, a_1$ only) could not reach. [E] it is the order v342 grounds (Gilkey $a_4$ gauge-curvature $30/360 = 1/12$), unifying v391 (a $\zeta$-det variation IS heat-kernel data) with v342 (the term IS the Gilkey $a_4$). [E] $(4\pi)^{-2} = 4c_3^2$ is $c_3$-built. [E] the v382 split re-verified at $\alpha^{-1} = 137.0359992$. [O] NOT CLOSED: the actual seam $a_4$ ($= 8b_1c_3^6$) and the cubic $\alpha^3$ Chern moment stay open (v384); the exact seam $F$-normalisation stays [C] (v342). Grounds/connects, does not close [E][O] |

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| Script | What it machine-checks *(continued)* |
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| `v434_alpha_quillen_betafunction.py` | ALPHA.QUILLEN.PROGRESS.03: a *third honest step* on ALPHA.QUILLEN.EXACT.01 (v382) – it settles the three residuals named after v433 and shows they are NOT three independent problems; the target stays [O], $\alpha^{-1}$ stays [E]. [E] $b_1$ IS THE U(1) $a_4$ HEAT-KERNEL COEFFICIENT (the $\beta$=$a_4$ theorem): from the carrier hypercharge content with spin weights ($\frac{2}{3}$ Weyl, $\frac{1}{3}$ complex scalar), $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$ (SM), $(3/5)\cdot$that $= 41/10 = b_1$ (GUT) – the $b_1$ in $8b_1c_3^6$ is the $a_4$ content coefficient, not a fitted multiplicity (same number as the $\beta$, v159/v246; same KIND as the $\alpha^2$ Calderón $a_4$, v342). [E] $8b_1c_3^6 = b_1 \cdot (\text{rank}\, E_8) \cdot c_3^6$ (rank $8 = \chi$=2 boundary, v216; $c_3^6$ six insertions, v342). [C] residual (1) "exact $a_4 = 8b_1c_3^6$" ≡ residual (3) "seam $F$-normalisation" – ONE obligation. [E] residual (2) located: $a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)$, so $a^3$ = the first moment of the Chern density (a level, order $c_3^0$); origin stays [O]. [E] net reduction: the EM-Ward residual is 1[C]+1[O], not three. [O] NOT CLOSED; $\alpha^{-1}$ stays [E]. Python-only [E][C][O] |
| `v435_alpha_quillen_chernlevel.py` | ALPHA.QUILLEN.PROGRESS.04: a *fourth honest step* on ALPHA.QUILLEN.EXACT.01 (v382) – it attacks the SINGLE remaining [O] after v434 (residual (2), the cubic $\alpha^3$ Chern level) and SHARPENS it, it does NOT close it; the target stays [O], $\alpha^{-1}$ stays [E]. [E] $\pi$-POWER / METRIC-INDEPENDENCE TEST: the three closure coefficients carry $\pi$-powers exactly $\{0, 3, 6\} = c_3$-orders ($k_0 = 1 \to \pi^0$; $-2c_3^3 = -1/(256\pi^3) \to \pi^3$; $-8b_1c_3^6 = -41/(327680\pi^6) \to \pi^6$), so the Maxwell $a^3$ is the UNIQUE $\pi^0/c_3^0$ (metric-INDEPENDENT) term – the signature of a topological level (a heat-kernel boundary coeff always carries the $(4\pi)^{-d/2}$ measure; a level carries none). The test partitions the closure into ONE topological level ($a^3$) + TWO heat-kernel boundary terms, confirming v342/v434's "$a^3$ is a level, not a curvature integral" from a test. [C] conditional on the CS/$\eta$ reading (EM1, v48), large-gauge invariance forces the level in $\mathbb{Z}$ and $k_0 = 1$ is the unit (minimal) level – an integer, not a free real. [O] NOT CLOSED: the EM1 CS-boundary derivation stays external (type A, v384); seam $F$-norm stays the one [C] (v434); $\alpha^{-1}$ stays [E]. [E][C][O] |
| `v436_overdet_floor2.py` | `OVERDET.FLOOR.02`: HARDENS `OVERDET.FLOOR.01` (v432) — the "is it numerology?" floor does NOT rest on v432's subjective chance assignments. [E] ASSUMPTION-MINIMAL FLOOR: the $\alpha$ cubic-class census (v100 layer D) is PURE COUNTING — of $N$=94500 complexity-matched $F_{U(1)}$ variants only ONE hits the $4 \times 10^{-8}$ CODATA window (the TFPT instance, $\alpha^{-1}$=137.0359992), so $p_\alpha \le 1/94500 = 1.06 \times 10^{-5} \sim 4.40\sigma$ with NO subjective probability, prior or multi-observable grammar. [E] MONOTONE CONCESSION LADDER: $-\log_{10} p$ full scorecard+$\alpha \sim 30.7 \to$ scorecard-only $\sim 25.8 \to \alpha$-census-only $\sim 4.98$ ($= 4.40\sigma$); the "not chance" verdict holds at the MOST adversarial rung. [E] INDEPENDENCE: strip ALL of v432's subjective factors ($0.1^3 \times 0.012 \times 0.15 = 1.8 \times 10^{-6}$) and the floor is still $4.40\sigma$ — they only sharpen it. [C] HONEST $5\sigma$ GAP: $5\sigma$ ($p = 5.7 \times 10^{-7}$) is crossed by the census times exactly ONE more independent factor $\le 0.054$ (the input-'8' four-fold over-determination or one foreign witness), stated openly. [C] FIREWALL: bounds numerology only, not the physics bridge (v187); Python-only, deliberately NOT Wolfram-mirrored (a counting bound, not an identity). [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v437_e8_degree_joint.py` | `E8.DEGREE.JOINT.01`: a CONSOLIDATION of v6/v66/v355/v431 deriving more from the $E_8$ Casimir degrees WITHOUT crossing the v354/v355 anti-numerology line — no new per-degree coincidence, only a joint lock. [E] $E_8$ IS FIXED BY ITS DEGREE MULTISET: from $\deg(E_8)=\{2,8,12,14,18,20,24,30\}$ alone — rank $=\#\deg=8$, $h=\max=30$, $\#$pos roots$=\sum\exp=120$, $\#$roots$=240$, $\dim=248$, $\lvert W\rvert=\prod\deg=696{,}729{,}600$, $\sum\deg=128=2^7=\dim S^+$. [E] THE JOINT QUADRATIC: the two structural integers $(g_{\rm car}, N_{\rm fam})$ are the UNIQUE root pair of $x^2-(\operatorname{rank}E_8)x+(h/2)=x^2-8x+15=(x-3)(x-5)$ — sum$=\operatorname{rank}E_8=8$ (rank-fill, v6), product$=h/2=15=g_{\rm car}N_{\rm fam}$ ($2g_{\rm car}N_{\rm fam}=h$, v431); $\{3,5\}$ the unique factor pair of 15 summing to 8, so BOTH are forced together by two degree-invariants. [E] the three mapped degrees $\{2,8,30\}=\min/\operatorname{rank}/\max$ (the canonical invariants); $g_{\rm car}=5=$ max prime of 30. [E] DISCIPLINE: no new per-degree mining; the five non-primary degrees keep their v431 homes. Exact integer/lattice, mirrored in wolfram. [E][C] |
| `v438_seam_hsmi_borchers.py` | `SEAM.EQUIV.BW.HSMI.01`: the half-sided modular inclusion (Borchers/Wiesbrock) as an INTRINSIC Bisognano–Wichmann route for face (ii) of `SEAM.EQUIV.01` — geometric modular flow WITHOUT presupposing covariance. [E] STANDARD-PAIR ALGEBRA (ladder, exact): boost $K=\operatorname{diag}(n)$ and translation $P$ (raising shift) satisfy $[K,P]=P$ and the Borchers scaling $\Delta^{it}P\Delta^{-it}=e^{-t}P$ (the ax+b/Moebius relation). [E] BW REFLECTION: $J(n\to -n)$ gives $J^2=I$, $JKJ=-K$ ($u_\Theta=J$) and $JPJ=P_-$ (the opposite-chirality translation), the standard-pair reflection $JU(s)J=U(-s)$. [E] POSITIVE ENERGY: in the unitary lowest-weight $sl(2,\mathbb{R})$ rep the line translation $P_{\rm line}=L_0-(L_++L_-)/2$ is POSITIVE (min eig $>0$), the positive-energy generator the $c_3$ one-sided origin demands; the boost direction is the indefinite negative control. [C] SYNTHESIS: $P\geq 0$ + $[K,P]=P$ + $J$ is a Longo STANDARD PAIR / +half-sided modular inclusion (Wiesbrock), so by Borchers the seam modular flow is geometric INTRINSICALLY (complementary to Adamo–Giorgetti–Tanimoto, `arXiv:2508.07109`); face (ii) becomes a STRUCTURE consequence, not a state condition. [C][O] assembling the finite algebra and the positive-energy rep into ONE continuum standard pair is the continuum existence (the same wall as MMST/NPW26, v336/v424); `SEAM.EQUIV.01` stays [O]. Python-only (numpy/scipy). [E][C][O] |
| `v439_seam_lieb_robinson.py` | `SEAM.EQUIV.CONTINUUM.LR.01`: the Lieb–Robinson/Nachtergaele–Sims continuum-existence MOTOR for the gapped seam collar — grounds the BULK half of the cited MMST existence (v336) in the explicit v367 $p+ip$ model. [E] UNIFORM GAP: $\Delta(N)=$const$>0$ across $N=20..160$ (the lattice image of the OS transfer gap $6\log\frac{3}{2}>0$, v302/v329). [E] EXPONENTIAL CLUSTERING: $\lvert\langle c_i^\dagger c_j\rangle\rvert\sim e^{-r/\xi}$ with $\xi$ finite and STABLE across $N$ (gapped $\Rightarrow$ Hastings–Koma decay), the $N$-independent locality. [E] LIEB–ROBINSON LIGHT CONE: the single-particle propagator $\lvert(e^{-iHt})_{ij}\rvert$ is confined to $\lvert i-j\rvert\lesssim v_{\rm LR}t$ with finite $v_{\rm LR}$ ($\sim 2t$), exponentially small beyond the cone. [C] NACHTERGAELE–SIMS LIFT: uniform gap + finite LR velocity $\Rightarrow$ exponential clustering with $N$-independent $\xi$ AND existence+uniqueness of the thermodynamic-limit ground state with a finite light cone — derived for OUR collar, not imported. [C][O] the bulk thermodynamic limit EXISTS by Lieb–Robinson; `SEAM.EQUIV.01` stays [O], the open analytic step narrowed to the CHIRAL massless EDGE scaling limit (the $c=8$ $(E_8)_1$ CFT), distinct from the bulk limit. Python-only (numpy/scipy). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v440_seam_lto_rp_beta.py` | `SEAM.EQUIV.LTORP.BETA.01`: the $\beta$-interpolation connecting the NPW26-PROVEN trace case ($\beta$=0) to the seam's $\beta$=1 KMS case along an obstruction-free homotopy — upgrading the single-point v426 witness for v424 sub-step (ii) to a PATH argument. [E] $\beta$-INDEPENDENT (LTO-RP) DATA: $\Theta K\Theta{=}{-}K$ ($u_\Theta{=}J$) and $[\rho,K]{=}0$ are conditions on $K$ alone, $\beta$-independent. [E] THE KMS PATH: $C_\beta{=}1/(1{+}e^{\beta K})$ has $\beta{=}0\to C{=}\frac{1}{2}I$ (the NPW26 TRACE state), $\beta{=}1\to$ the seam KMS state, $\beta\to\infty\to$ the ground-state projection; at every $\beta$ $\Theta C_\beta\Theta{=}1{-}C$ (reflection), $[\rho,C_\beta]{=}0$ (mu4), $0\le C_\beta\le 1$ (valid covariance/RP). [E] NO OBSTRUCTION: $\beta\mapsto C_\beta$ is norm-continuous and stays in the (LTO-RP)-valid set — a continuous homotopy from the proven trace endpoint to the seam KMS point, no topological wall. [E] NEG CONTROLS: a reflection-EVEN $K'$ breaks $\Theta C_\beta\Theta{=}1{-}C$ for $\beta{>}0$ (the trace endpoint is degenerate); an off-mu4-block $K$ breaks $[\rho,C_\beta]$. [C][O] a PATH argument for v424 sub-step (ii), stronger than the single point of v426; `SEAM.EQUIV.01` stays [O] (closedness of (LTO-RP) along the path in the continuum is the residual). Python-only (numpy/scipy). [E][C][O] |
| `v441_seam_applicability_ledger.py` | `SEAM.EQUIV.APPLICABILITY.01`: the applicability ledger — a hypothesis-by-hypothesis audit of the THREE external theorems the seam keystone cites (MMST scaling limit, NPW26 (LTO-RP), Adamo–Giorgetti–Tanimoto covariance), isolating the SINGLE imported analytic fact. [E] MMST HYPOTHESES: $D$=16 Majoranas ($=2^{g_{\rm car}-1}$), $c$=8 ($=g_{\rm car}{+}N_{\rm fam}$), rank $E_8$=8, range rank $\le c\le D$ reads $8\le 8\le 16$, gapped ($6\log\frac{3}{2}{>}0$), quasi-free CAR — all internal computed facts. [E] NPW26: (LTO-RP)=$u_\Theta{=}J$, gap, $\mathbb{Z}_2$ reflection, mu4 clock, $[\rho,K]{=}0$ internal (v424/v426/v440); the seam is invertible $|\det\text{Cartan}\,E_8|{=}1$ + $\beta{=}1$ KMS, OUTSIDE NPW26's topological/trace bucket ($|\det|{=}4$) — a TEMPLATE, not a direct import. [E] ADAMO–GIORGETTI–TANIMOTO (`arXiv:2508.07109`): DHR reps of nets extending loop-group/Virasoro are AUTOMATICALLY positive-energy + diff-covariant; the seam net extends $(D_5)_1{\times}(A_3)_1$ (v154), so covariance is a CONSEQUENCE — the v215 circularity is defused, covariance downgraded ASSUMPTION→[E]. [E] of 12 audited hypotheses, 11 are internal [E] and exactly 1 external — the continuum EXISTENCE of the chiral massless scaling limit (MMST/v336, Lean `FORM.SEAM.MMST.01`). [C][O] external dependence pinned to ONE analytic fact; `SEAM.EQUIV.01` stays [O]. Python-only (a structural/arithmetic audit; the dets Wolfram-mirrored via v89/v281/v422). [E][C][O] |
| `v442_seam_rigidity_uniform.py` | `SEAM.RIGIDITY.UNIFORM.01`: the uniform-in-$N$ hardening of the Seam State Rigidity Theorem (v398) — the band-limited rigidity is NOT a small-$N$ artifact. [E] EXACT UNIFORM 4-SECTOR COMMUTATION: $\rho{=}\,\text{diag}(i^n)$ has EXACTLY 4 distinct eigenvalues $\{1,i,-1,-i\}$ for all $N$, and a genuinely character-block-diagonal $K$ (random Hermitian within each $n$ mod 4 class, not merely diagonal) has $[\rho,K]{=}0$ EXACTLY for $N{=}4..256$ (extends v398's $N{\le}64$). [E] UNIFORM GAP: $\Delta(N){=}\text{const}{>}0$ across $N{=}20..160$ (the Nachtergaele–Sims uniform gap, shared with v439). [E] $N$-INDEPENDENT RIGIDITY BOUND: a FIXED local clock-breaking perturbation (single off-mu4-block Hermitian entry) gives $\|[\rho,V]\|{=}2$ EXACTLY for all $N$ — a uniform lower bound; any fixed-size clock-breaking is detected with $N$-independent strength. [E] NEG CONTROL: a clock-PRESERVING (same-sector) perturbation gives $\|[\rho,V']\|{=}0$; an order-2 clock has only 2 distinct eigenvalues — the ORDER of the clock (the four Gauss–Bonnet marks) is the discriminator, uniformly in $N$. [C][O] the band-limited rigidity of v398 is uniform in $N$; `SEAM.RIGIDITY.01`/`SEAM.EQUIV.01` stays [O]: that RP+gap+holomorphy FORCE $\Lambda_\Sigma$ block-diagonal on the full $L^2$ (vs merely permit) is the cited rigidity/MMST step. Python-only (numpy). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v443_seam_e8_discriminator.py` | `SEAM.E8.DISCRIMINATOR.01`: the empirical/structural DISCRIMINATOR bundle (mode D of the v347 closure-mode classification) — concrete FALSIFIABLE kill tests separating the claimed seam = $(E_8)_1$ from the nearest $c$=8 rival $SO(16)_1$=$(D_8)_1$. [E][X] PRIMARY/GSD: $\lvert\det \mathrm{Cartan}(E_8)\rvert$=1 (ONE primary, holomorphic) vs $\lvert\det \mathrm{Cartan}(D_8)\rvert$=4 (FOUR primaries $1, v, s, c$); torus GSD = #primaries = 1 vs 4 — a seam with GSD> 1 falsifies $(E_8)_1$. [E][X] CURRENT/THETA: level-1 current count $\dim E_8$=248 vs $\dim SO(16)$=120; the $E_8$ lattice theta has 240 norm-2 roots (= 112 integer +128 spinor, computed) vs $D_8$'s 112 — current spectrum and lattice theta DIFFER. [E] $c$=8 IS NOT A DISCRIMINATOR: both have $c$=8 (the v277 ambiguity); only the order-4 mu4 condensation + $\det K$=1 (the genus-1 statement, v90) selects $E_8$. [E] NON-VACUITY: the discriminators all fire on the rival ($\det 4 \neq 1$, currents $120 \neq 248$, roots $112 \neq 240$). [X][C] mode D realised as concrete kill tests; `SEAM.EQUIV.01` stays [O] (the discriminators distinguish the TARGET but do not PROVE the seam realises it). Python-only (numpy; the dets Wolfram-mirrored via v89/v281/v422). [E][C][O][X] |
| `v444_seam_edge_scaling.py` | `SEAM.EQUIV.CONTINUUM.EDGE.01`: the CHIRAL EDGE scaling-limit MOTOR — the edge half of the one `SEAM.EQUIV.01` continuum residual, attacking exactly the step v439 left open. On the SAME explicit v367/v368 $p+ip$ collar it shows the lattice edge carries the kinematic AND conformal data of a chiral $c=\frac{1}{2}$ Majorana CFT, converging under refinement. [E] CHIRAL LINEAR EDGE DISPERSION: the in-gap branch is linear near its zero-crossing (top $v$=0.983, $R^2$=0.9999) and the two boundaries carry OPPOSITE velocities — one chiral Majorana per edge (relativistic, chiral kinematics). [E] ALGEBRAIC EDGE TWO-POINT: the equal-time edge correlator decays as $r^{-\eta}$ with $\eta$=1.00 (the chiral free-fermion CFT exponent) — a CRITICAL edge. [E] EXPONENTIAL BULK (neg control): the bulk-row correlator decays exponentially ($\xi \sim 1.1$) — only the edge is critical, so the power law is the edge CFT, not a geometric artefact. [E] TRIVIAL-PHASE NEG CONTROL: $M$=3 (no edge, v368) has an exponential edge-row — the algebraic decay is the topological edge, not the boundary geometry. [E] SCALING-COLLAPSE / CONVERGENCE: $\eta$ is $N$-stable ($1.001 \pm 0.007$) and the edge correlator profile is Cauchy-convergent under refinement (successive spread shrinks 0.028→0.013, $N_x$=36..60) — an existing scaling limit, the edge analogue of v439's bulk thermodynamic limit. [C] EDGE CENTRAL CHARGE: $\eta$=1 ⇔ one chiral Majorana ($c=\frac{1}{2}$) per edge, so the $N_{\mathrm{Maj}}$=16-copy carrier has a chiral $c_-$=8 edge CFT (= $(E_8)_1$, $c_- = g_{\mathrm{car}} + N_{\mathrm{fam}}$). [C][O] grounds the EDGE half of the cited MMST scaling limit (v336) the way v439 grounds the bulk; with both halves the one continuum residual is grounded and pinned to a single cited theorem; `SEAM.EQUIV.01` stays [O] (the rigorous uniform-convergence theorem is not supplied). Python-only (numpy). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v445_seam_rigidity_forcing.py` | `SEAM.RIGIDITY.FORCING.01`: the rigidity FORCING theorem — upgrades v442's "block-diagonal PERMITS commuting" to "commuting with the order-4 clock FORCES block-diagonal", isolating the single remaining physical premise. [E] FORCING (commute ⇔ block-diagonal, EXACT): for $\rho=\mathrm{diag}(i^n)$, $[\rho,K]=0$ iff $K_{ij}=0$ whenever $i\not\equiv j \bmod 4$ — verified ENTRYWISE, swept $N$=4..128 (the converse of v442). [E] EXACT COMMUTANT DIMENSION: $\dim\{K:[\rho,K]=0\}=\sum_s n_s^2$ (the four sector sizes), verified as the $\mathrm{ad}_\rho$ nullspace; a PROPER subspace of the $N^2$ algebra ($N$=16: 64<256) — a genuine quantitative restriction. [E] OFF-BLOCK ⇔ NON-COMMUTING: every off-block matrix unit $E_{ij}$ has $\|[\rho,E_{ij}]\|=|i^i-i^j|\geq\sqrt{2}>0$ — zero slack. [E] THE ORDER IS THE DISCRIMINATOR: the order-4 clock gives 4 sectors ($\dim\sum n_s^2(4)$), the order-2 clock $\mathrm{diag}((-1)^n)$ a STRICTLY LARGER commutant ($N$=16: 64 vs 128) — the four Gauss–Bonnet marks (order $4=|\mu_4|=h(A_3)$) force strictly more structure than two; only the index-4 extension is $(E_8)_1$ (det 1; v281/v154/v351). [E] NONDEGENERACY: exactly 4 distinct eigenvalues $\{1,i,-1,-i\}$ for all $N\geq4$. [C] SYNTHESIS (force, not permit): commuting with the order-4 $\mu_4$ clock is EQUIVALENT to $\mu_4$-block-diagonality (iff, exact, uniform in $N$), so RP-definability (v54/v398) + clock-invariance FORCE the block-diagonal $\Lambda_\Sigma$ — v442's "permit" upgraded to "force". [C][O] the single residual is now precisely "transfer in commutant" (clock-invariance = `QGEO.SYM.01`, v323/v335); `SEAM.RIGIDITY.01`/`SEAM.EQUIV.01` stays [O]. Python-only (numpy; the det facts Wolfram-mirrored via v89/v281/v422). [E][C][O] |
| `v446_seam_clock_invariance.py` | `SEAM.RIGIDITY.CLOCKINV.01`: DERIVING clock-invariance ("transfer in the commutant") from RP-symmetry, closing v445's single residual one notch further — the residual drops from an OPERATOR commutation to the STRUCTURAL $\mu_4$-symmetry of the seam RP data. [E] OS RECONSTRUCTION INHERITS THE SYMMETRY: for a $\mu_4$-invariant RP covariance ($[\rho,C(t)]=0$), the OS canonical transfer $\Lambda=C(1)C(0)^{-1}=e^{-h}$ (the transfer FIXED by RP+reflection, v54) has $[\rho,\Lambda]=0$ — swept $N$=4..32 over random block-diagonal $h$. [E] SHARPNESS / NEG CONTROL: a $\mu_4$-BREAKING covariance gives $[\rho,\Lambda]\neq0$ — iff-sharp, so the inheritance is non-vacuous. [E] CONTRACTION / OS-VALIDITY: $\Lambda$ has spectrum in $(0,1]$ (a positive contraction, the physical canonical transfer). [E] MODULAR (Tomita–Takesaki): the free modular Hamiltonian $K=\log((1-C)C^{-1})$ (the v258 formula) of the $\mu_4$-invariant state has $[\rho,K]=0$ (the clock commutes with modular flow), the breaking control $[\rho,K]\neq0$. [C] SYNTHESIS: $[\rho,\Lambda_\Sigma]=0$ is a CONSEQUENCE of $\mu_4$-invariance of the seam RP data (the four Gauss–Bonnet marks, v216/v323) via the OS reconstruction (v54) + the modular fact (v258), NOT an extra assumption; with v445 (commute ⇔ block-diagonal), RP-definability PLUS the marks FORCE the block-diagonal $\Lambda_\Sigma$. [C][O] the residual reduces from an operator commutation to "the seam RP data is $\mu_4$-symmetric" (`QGEO.SYM.01`); the continuum Tomita–Takesaki implementation on the full $L^2$ is the cited backbone. `SEAM.RIGIDITY.01`/`SEAM.EQUIV.01` stays [O]. Python-only (numpy/scipy). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v447_seam_edge_chern.py` | `SEAM.EQUIV.EDGE.CHERN.01`: an INDEPENDENT (topological) reading of the edge $c_-$=8, corroborating v444's correlator route via the bulk–edge correspondence on the same v367/v368 $p+ip$ collar. [E] INTEGER CHERN NUMBER: the $M$=1 collar lower band has Chern $C$=+1 (the Fukui–Hatsugai–Suzuki lattice link-variable method, an EXACT integer, no fit) — one chiral channel per copy. [E] TRIVIAL-PHASE NEG CONTROL: $M$=3 (v368) has $C$=0 (no chiral edge), and $M$=−1 gives $C$=−1 (opposite chirality, matching v444's opposite per-edge velocities). [E] BULK–EDGE CORRESPONDENCE: the open strip carries EXACTLY $\|C\|$=1 in-gap edge branch crossing zero per boundary in the $M$=1 phase, 0 in the trivial phase — the chiral edge channel count = $\|C\|$. [E] LI–HALDANE: the half-cut bulk ground state's single-particle entanglement spectrum has in-gap modes at 1/2 in the topological phase and NONE in the trivial — the entanglement edge tracks the physical edge. [C] EDGE CENTRAL CHARGE: $c_-=N_{\mathrm{Maj}}\cdot\frac{1}{2}\cdot\|C\|=16\cdot\frac{1}{2}\cdot 1=8=g_{\mathrm{car}}+N_{\mathrm{fam}}$, from the Chern integer + bulk–edge correspondence — the SAME $c_-$=8 as v444's correlator route, from a logically independent (topological) observable. [C][O] corroborates v444's edge $(E_8)_1$ reading; `SEAM.EQUIV.01` stays [O] — the topology fixes $c_-$ and the channel count but not the cited continuum existence theorem. Python-only (numpy). [E][C][O] |
| `v448_seam_edge_fourpoint.py` | `SEAM.EQUIV.EDGE.FOURPOINT.01`: pinning the ONE external MMST fact empirically — the edge FOUR-point function is quasi-free and conformally convergent, extending v444's two-point scaling limit to the next order. [E] QUASI-FREE / WICK (CAR hypothesis): the ground-state correlation matrix is an exact projector $G^2$=$G$ ($\sim 10^{-13}$), so the state is quasi-free and all $n$-point functions are Wick/Pfaffian sums of the two-point — the MMST quasi-free CAR hypothesis (ledger entry 5) literally holds on the collar (structural: the BdG state is free). [E] FOUR-POINT CAUCHY CONVERGENCE: the edge four-point cross-ratio $Q$=$[G_{01}G_{23}]/[G_{02}G_{13}]$ at COMMENSURATE positions $(1,2,4,7)\cdot(N_x/12)$ is Cauchy-convergent under refinement (successive spread shrinks, $N_x$=36..60) — the four-point converges, extending v444's two-point scaling-collapse. [E] CONFORMAL FOUR-POINT FORM: $Q$ converges to the free-fermion CONFORMAL cross-ratio $\sin(5\pi/12)/\sin(\pi/12)$=$2+\sqrt{3}\approx 3.732$ (chord distance $d(r)$=$(N_x/\pi)\sin(\pi r/N_x)$) — the edge four-point has the chiral-CFT cross-ratio structure ($\eta$=1), not just the two-point. [E] BULK NEG CONTROL: the bulk-row four-point is orders of magnitude off the conformal value (exponential decay, gapped) — only the EDGE four-point is conformal. [C] the one external MMST fact (continuum existence of the chiral scaling limit) is now empirically supported at BOTH two-point (v444) and four-point order on our explicit collar; every measured $n$-point converges to its conformal limit. `SEAM.EQUIV.01` stays [O] — the rigorous MMST uniform-convergence theorem (v336) is the cited backbone. Python-only (numpy). [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v449_seam_mmst_uniform.py` | `SEAM.EQUIV.MMST.UNIFORM.01`: uniform-in-$N$ control of the edge scaling limit — the empirical content of the cited MMST uniform-convergence theorem, strengthened, on the same v367/v368 $p{+}ip$ collar. [E] CROSS-RATIO A CONVERGES: the edge four-point $Q(1,2,4,7)/12 \to \sin(5\pi/12)/\sin(\pi/12) = 2{+}\sqrt{3}$, the deviation shrinking monotonically under refinement ($N_x$=36, 48, 60). [E] CROSS-RATIO B (independent positions): $Q(1,3,5,9)/12 \to$ its OWN conformal value 2 — a SECOND observable with a DIFFERENT limit, so the convergence is not a single tuned point. [E] UNIFORM $1/N$ RATE: $\lvert Q(N){-}Q_\infty\rvert \cdot N \le 4$ for BOTH cross-ratios and every $N_x \in \{36,48,60\}$ — one $N$-independent constant bounds the deviation across two independent observables, the empirical hallmark of uniform-in-$N$ convergence. [E] BULK NEG CONTROL: the bulk-row cross-ratio is off-scale (gapped, exponential) — only the EDGE converges. [C][O] the MMST uniform-convergence HYPOTHESIS is empirically satisfied on the collar (two-point v444, four-point v448, and now multi-observable uniform rate); `SEAM.EQUIV.01` stays [O] — the abstract continuum existence theorem (v336) is strengthened in evidence, not replaced by a proof. Python-only (numpy). [E][C] |
| `v450_seam_edge_entanglement.py` | `SEAM.EQUIV.EDGE.ENTANGLEMENT.01`: a THIRD independent reading of the edge $c_-$=8 — the Calabrese–Cardy entanglement scaling $c$=1/2 per Majorana (after v444 correlator, v447 topology). [E] CALABRESE–CARDY LAW: the block entanglement entropy of the critical XX (Dirac) chain fits $S{=}(c/3)\ln[(L/\pi)\sin(\pi\ell/L)]{+}$const with $c_{\mathrm{Dirac}}$=1.000 (flat log slope) — the universal scaling-limit signature. [E] $c$ PER MAJORANA = 1/2: $c_{\mathrm{Dirac}}{=}1{=}2c_{\mathrm{Maj}}$ (Dirac = two Majoranas), so $c_{\mathrm{Maj}}$=1/2 — the chiral CFT of one $p{+}ip$ edge mode (v444). [E] GAPPED NEG CONTROL: a dimerised (gapped) chain has saturating, $\ell$-independent entanglement ($c$=0) — the log growth is the CRITICAL edge CFT. [C] EDGE CENTRAL CHARGE: $c_- {=} N_{\mathrm{Maj}} \cdot c_{\mathrm{Maj}} = 16 \cdot \frac{1}{2} = 8 = g_{\mathrm{car}}{+}N_{\mathrm{fam}}$, from entanglement — the SAME $c_-$=8 as v444 (correlator) and v447 (topology), from a third independent observable. [C][O] three observables agree; `SEAM.EQUIV.01` stays [O]. Python-only (numpy). [E][C] |
| `v451_seam_edge_cardy_tower.py` | `SEAM.EQUIV.EDGE.CARDY.01`: the Cardy finite-size conformal tower — the edge spectrum reproduces the EXACT Ising primary dimensions $\{1/16, 1/2\}$, on the critical Majorana ring (the v367/v368 edge universality class). [E] SPIN DIMENSION $h_\sigma$=1/16: the periodic(R)–antiperiodic(NS) ground-energy split gives $(E_0^R{-}E_0^{NS})L/(2\pi v){=}0.06250$ ($v$=2), stable across $L$=200, 400, 800 — the Ising spin field. [E] ENERGY DIMENSION $h_\varepsilon$=1/2: the lowest NS pair excitation gives $2\varepsilon_{\min}L/(2\pi v){=}0.50000$ — the Ising energy field. [E] CONFORMAL SCALING: both dimensions are $L$-INDEPENDENT (the gaps scale exactly as $1/L$) — a conformal (gapless) spectrum, not a gapped lattice. [C] ISING IDENTIFICATION: $\{c, h_\sigma, h_\varepsilon\}{=}\{1/2, 1/16, 1/2\}$ is the chiral Ising (free Majorana) CFT (the $c$=1/2 building block, v444/v450); with $N_{\mathrm{Maj}}$=16 copies and bulk–edge (v447), $c_-{=}16 \cdot \frac{1}{2}{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$. [C][O] the edge CFT is named by its central charge (v450) AND its operator content (this); `SEAM.EQUIV.01` stays [O]. Python-only (numpy). [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v452_seam_e8_modular.py` | `SEAM.EQUIV.E8.MODULAR.01`: the $(E_8)_1$ torus modular data — one primary, $S$-invariant, $T$-phase $e^{-2\pi i/3}$, the holomorphic $c$=8 signature (extends v156's $q$-expansion to the modular transformation data). [E] $S$-INVARIANCE: $\chi(-1/\tau)=\chi(\tau)$ for $\chi=E_4/\eta^8$ (weight 0, since $E_4$ wt 4 / $\eta^8$ wt 4), to $\sim 10^{-30}$ — ONE primary, the $S$-matrix is [1]. [E] $T$-PHASE: $\chi(\tau+1)/\chi(\tau)=e^{-2\pi i/3}=e^{-2\pi ic/24}$ ($c$=8), to $\sim 10^{-30}$ — the central charge appears in the modular $T$ eigenvalue. [E] LEADING POWER $\Rightarrow$ $c$=8: $\chi\, q^{1/3}\to 1$ so the leading power is $q^{-c/24}=q^{-1/3}\Rightarrow c=8=g_{\rm car}+N_{\rm fam}$. [E] HOLOMORPHIC CONSTRAINT: $c$=8=0 mod 8 — the minimal holomorphic (single-primary) chiral central charge, the even unimodular rank-8 lattice $E_8$. [C][O] the $(E_8)_1$ modular data (one primary, $S$-invariant, $T$-phase $e^{-2\pi i/3}$, $c$=8) is the torus signature of the seam net; with the edge readings (v444/v447/v450/v451) the $(E_8)_1$ identity is pinned on the torus AND the edge; `SEAM.EQUIV.01` stays [O] — the cited continuum existence theorem (v336) is the residual. Wolfram-mirrored (`tfpt_readouts_extension.wl`). [E][C] |
| `v453_seam_mu4_from_marks.py` | `SEAM.RIGIDITY.MU4FROMMARKS.01`: deriving `QGEO.SYM.01` ($\mu_4$-symmetry of the seam RP data) from the four marks — the LAST structural premise of the rigidity chain folded into the existing realisation premise. [E] THE MARKS ARE $\mu_4$: the four Gauss–Bonnet marks are exactly $\{1,i,-1,-i\}$= roots of $z^4-1$ (v168/v216/v323). [E] THE CLOCK PERMUTES THE MARKS: $z\mapsto iz$ is a 4-cycle of order 4 ($1\to i\to -1\to -i\to 1$), fixing the configuration setwise (the order-4 clock of v445). [E] MOEBIUS STABILISER: $z\mapsto iz$ preserves the marks' cross-ratio $\lambda$=2 (in the $D_4$ stabiliser, v168) — the symmetry is geometric. [E] THE FORM BASIS IS $\mu_4$-GRADED: $\omega_k=z^{k-1}dz/(z^4-1)\to i^k\omega_k$ under $z\mapsto iz$ ($k$=1,2,3 $\to\{i,-1,-i\}$, the three nontrivial $\mu_4$ characters; weights $(1,2,3)=A_3$ exponents = Spec$(Q_+)$) — the natural basis is $\mu_4$-graded, so any geometric datum is $\mu_4$-covariant. [C][O] `QGEO.SYM.01` ("the seam RP data is $\mu_4$-symmetric") is a CONSEQUENCE of marks=$\mu_4$ (this) + `QGEO.REALIZE.01` (seam = the marked geometry, the EXISTING premise), NO new premise; the rigidity chain (this → v446 → v445) closes modulo realisation, `SEAM.EQUIV.01` stays [O]. Wolfram-mirrored (`tfpt_readouts_extension.wl`). [E][C] |
| `v454_seam_edge_virasoro.py` | `SEAM.EQUIV.EDGE.VIRASORO.01`: the lattice realises the level-1 current algebra whose Sugawara central charge is EXACTLY 8 (G2 of the post-F next steps) — testing the cited MMST "Koo–Saleur lattice generators → continuum Virasoro with the correct $c$" fact on our own free-fermion edge. [E] CASIMIR $c$: the critical complex free-fermion ring $E_0(L)=e_\infty L-\pi vc/(6L)$ (Affleck–Cardy) fits $c_{\rm Dirac}$=1.000 $\Rightarrow$ 1/2 per Majorana (parameter-free, $L$=64..1024). [E] KAC–MOODY LEVEL: the lattice $U(1)$ current $J_n$ has $\langle 0|J_nJ_{-n}|0\rangle=n$ EXACTLY ($n$=1..8) — the affine level-1 central extension closes; a gapped (dimerised) control gives a vanishing non-linear term. [E] SUGAWARA EXACT: $c=\dim G/(1+h^\vee)$=8 for BOTH $SO(16)_1$ (120/15) and $(E_8)_1$ (248/31) — Wolfram-mirrored. [C] ASSEMBLY: 16 Majoranas $\times\frac{1}{2}$=8= the level-1 Sugawara $c$; the edge IS a $c_-$=8 level-1 current-algebra CFT. [C][O] the lattice Virasoro/current algebra with the correct central extension is realised on the explicit collar; `SEAM.EQUIV.01` stays [O] (the continuum existence theorem v336 is the cited backbone). Mixed numerical (Casimir, Kac–Moody) + exact (Sugawara, Wolfram-mirrored). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v455_seam_bw_uniform.py` | `SEAM.EQUIV.BW.UNIFORM.01`: a UNIFORM-IN-$N$ Tomita–Takesaki tower for the intrinsic Bisognano–Wichmann condition $u_\Theta{=}J$ (G1 of the post-F next steps) — the inductive-limit companion to the single finite witness v426, advancing v424 sub-step (i). One-particle: $\Delta{=}e^{-K}$, conjugation $J$, reflection $\Theta$; BW $\Leftrightarrow u_\Theta{=}J \Leftrightarrow \Theta K\Theta{=}-K$. [E] UNIFORM BW: the reflection-odd, $\mu_4$-even, gapped collar of size $4m$ ($m$=2..32, $N$=8..128) has $\|\Theta K\Theta{+}K\|{=}0$ to machine precision at EVERY $N$ — $u_\Theta{=}J$ uniform. [E] UNIFORM GAP: the gap of $K$ is $N$-independent ($6\log\frac{3}{2}{=}2N_{\mathrm{fam}}\log\frac{3}{2}$) so $\Delta{=}e^{-K}$ stays uniformly bounded away from 1 (a genuine KMS, not trace, state). [E] UNIFORM CLOCK SYMMETRY: $[\rho,K]{=}0$ exactly at every $N$. [E] $N$-INDEPENDENT NEG CONTROL: a side-even reflection gives $\|\Theta K\Theta{+}K\|{=}2\|K\|$ (a fixed per-mode margin, growing with $N$) and an off-$\mu_4$ $K$ breaks $[\rho,K]$ at every $N$ — the success is non-vacuous. [C][O] the inductive (continuum) limit inherits $u_\Theta{=}J$; `SEAM.EQUIV.01` stays [O] (the rigorous continuum theorem NPW26/MMST v336 is the backbone). Python-only (numpy). [E][C][O] |
| `v456_seam_chirality_from_c3.py` | `SEAM.S3.FROM-P1.01`: the edge chirality ($c_-{\neq}0$, the S3 topological phase) is FORCED by the one-sidedness that defines $c_3{=}1/(8\pi)$ (P1) (G6 of the post-F next steps) — so S3 is a CONSEQUENCE of axiom P1, not an independent input. [E] $c_3$'s "8" IS THE ONE-SIDED COUNT: $8\pi{=}|\mathbb{Z}_2|\int K{=}2(2\pi\chi(S^2)){=}2\cdot 4\pi$ ($\chi$=2), so $8{=}|\mathbb{Z}_2|\cdot(\int K/\pi){=}2\cdot 4$ (v73, Wolfram-mirrored). [E] REFLECTION REVERSES $C$: the explicit 2-band $p{+}ip$ Chern model has $C{=}{+}1$ and the orientation-reversed ($k_y{\to}{-}k_y$) model $C{=}{-}1$. [E] TWO-SIDED $\Rightarrow$ NON-CHIRAL: any GAPPED boundary carrying an orientation-reversing (reflection) symmetry must have $C{=}{-}C{=}0$ ($c_-{=}0$, gappable). [C] ONE-SIDED $\Rightarrow$ CHIRAL: the one-sided seam is precisely the $|\mathbb{Z}_2|$ quotient with NO reflection (the same $|\mathbb{Z}_2|$ that gives $c_3$ its $8\pi$), so $C{\neq}0$ is forced — S3 follows from P1. [C][O] magnitude $c_-{=}8{=}\operatorname{rank}E_8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$ is the SAME integer as $c_3$'s one-sided count; `SEAM.EQUIV.01`'s continuum EXISTENCE (v336) stays [O]. Mixed exact (the $8{=}|\mathbb{Z}_2|\int K/\pi$ arithmetic, Wolfram) + numerical (the reflection $C{\to}{-}C$ flip). [E][C][O] |
| `v457_seam_character_killtest.py` | `SEAM.EQUIV.KILLTEST.01`: an explicit $(E_8)_1$-vs-$SO(16)_1$ character/sector kill-test for the seam edge (G7 of the post-F next steps) — a FALSIFIABLE consistency test of the $(E_8)_1$ identification of `SEAM.EQUIV.01`. [E] $(E_8)_1$ CHARACTER TOWER: $\chi{=}E_4/\eta^8{=}q^{-1/3}(1{+}248q{+}4124q^2{+}34752q^3{+}\cdots)$ — the exact conformal-tower degeneracies ($E_4{\cdot}\prod(1{-}q^n)^{-8}$, Wolfram-mirrored). [E] CURRENT-COUNT DISCRIMINATOR: the weight-1 multiplicity is $248{=}\dim E_8{=}120(SO16){+}128(\text{spinor})$, NOT the 120 of the free $SO(16)_1$. [E] SECTOR-COUNT DISCRIMINATOR: $(E_8)_1$ is INVERTIBLE $|\det\text{Cartan}\,E_8|{=}1$ (one primary) vs $SO(16)_1{=}D_8$ $|\det\text{Cartan}\,D_8|{=}4$ (four sectors). [X] KILL CRITERION: a measured weight-1 count of 120, OR a sector count of 4, OR a tower coefficient $\neq\{1,248,4124,\dots\}$ would FALSIFY $(E_8)_1$; current data give $248/1/\{1,248,4124\}$, so PASSED. [C][O] the $(E_8)_1$ identification survives the kill-test; `SEAM.EQUIV.01`'s continuum EXISTENCE (v336) stays [O]. Exact (Wolfram-mirrored) + a numerical determinant cross-check. [E][C][O][X] |

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| Script | What it machine-checks *(continued)* |
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| `v458_seam_mmst_citation_audit.py` | `SEAM.EQUIV.MMST.AUDIT.01`: the EXACT hypothesis-by-hypothesis citation audit of the seam scaling-limit theorem (Osborne–Stottmeister, CMP **398** (2023) 219; `arXiv:2107.13834`; legacy internal token MMST, on the MMST/Morinelli–Morsella–Stottmeister–Tanimoto OAR framework) against our collar (G3 of the post-F next steps, the "exakter Zitat-Abgleich"). [C] H1 ALGEBRA CLASS: a gapped quasi-free CAR 16-Majorana system = MMST's lattice-fermion class (v155/v160/v302). [C] H2 MASSLESS POS-ENERGY LIMIT: the gapless chiral edge IS that massless scaling limit (v439/v444). [E] H3 PER-COPY ($c$=1/2): each of 16 chiral Majoranas is EXACTLY the proven Theorem A $c$=1/2 case. [C] H4 DECOUPLED ADDITIVITY: MMST's limit decouples into Majoranas ⇒ $L_k = \sum_i L_k^{(i)}$ on the common core ⇒ the free $c$=8 $SO(16)_1$ Virasoro follows ($16\cdot\frac{1}{2}=8$). [C] H5 WZW RANGE rank $\le c \le D$: $8 \le 8 \le 16$ IN range, the 120 $so(16)$ currents are bilinears — covered. [O] H6 RESIDUAL = THE 128-SPINOR EXTENSION: the index-4 $SO(16)_1 \to (E_8)_1$ adds 128 SPINOR currents (248=120+128), NOT bilinears — OUTSIDE MMST's method (the $\det K$ 4→1 holomorphy discriminator), handed to v459. [C][O] the audit CLOSES-by-citation the $c$=8 EXISTENCE and pins the open residual EXACTLY to the 128-spinor extension; `SEAM.EQUIV.01` stays [O], its open part now this one holomorphic-extension statement. Python (sympy; a structural/arithmetic audit). [C][O] |
| `v459_seam_lattice_voa_route.py` | `SEAM.EQUIV.LATTICEVOA.01`: the SECOND, independent route to $(E_8)_1$ — the lattice-VOA / conformal-net $A_Q(E_8)$ construction supplies EXACTLY the 128-spinor extension MMST (v458) leaves open (G5 of the post-F next steps). Cited: Adamo–Giorgetti–Tanimoto, "Rational and non-rational 2d CFTs arising from lattices" (`arXiv:2506.01008`); OS for unitary lattice VOAs (Adamo–Moriwaki–Tanimoto, `arXiv:2407.18222`). [E] $E_8$ EVEN UNIMODULAR: $\det \mathrm{Cartan}(E_8)=1$ and even ⇒ $A_{E_8}$ HOLOMORPHIC (one sector, the $\det K$=1 the residual demanded). [E] LATTICE CURRENTS 248=8+240: 8 Cartan + 240 roots = the full $(E_8)_1$ weight-1 space, from the lattice (not bilinears). [E] THE 240 ROOTS SPLIT 112+128: 112 INTEGER ($D_8$) + 128 HALF-INTEGER; 8+112=120=$so(16)$ and the 128 half-integer roots ARE the spinor (248=120+128=8+240) — the lattice route builds the 128-spinor explicitly. [C] AGT THEOREM: $A_Q$ of an even lattice IS a 2d conformal net; the holomorphic $SO(16)_1 \to (E_8)_1$ extension is CONSTRUCTED, not assumed. [C][O] TWO complementary routes now cover $(E_8)_1$ (MMST fermions v458 + the lattice net $A_{E_8}$); the residual reduces to the single realization input "the collar's scaling limit = $A_{E_8}$" (tied to P1 by v456). `SEAM.EQUIV.01` stays [O]. Exact ($E_8$ even unimodular; 240=112+128; 248=8+240=120+128, Wolfram-mirrored) + structural literature mapping. [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v460_seam_s3_lto.py` | `SEAM.S3.LTO.01`: the EXPLICIT flat-band commuting-projector (local-topological-order) realisation of the S3 input — advancing v424 sub-step (i) ("realise the raw seam as a $\mathbb{Z}_2$-reflection commuting-projector LTO") from abstract existence to an explicit gap-protected quasi-local projector net carrying the $\mathbb{Z}_2$ reflection and the $\mu_4$ clock. [E] FLAT-BAND PARENT: the flattened Bloch $Q{=}h/\lvert h\rvert$ of v367's $p{+}ip$ collar has $Q^2{=}I$ (flat $\pm1$ bands) and $P{=}(I{-}Q)/2$ is an exact projector EQUAL to the occupied ground-state projector — a frustration-free commuting-projector parent with the same gapped ground state. [E] GAP-PROTECTED QUASI-LOCALITY: the gapped projector kernel $\lvert P(n,0)\rvert$ decays rapidly ($\xi{\approx}1.14$ at $M{=}1$, exp $R^2{=}0.995$, $\lvert P(20,0)\rvert{<}10^{-6}$) and $\xi$ GROWS as the gap closes (2.7 at gap 0.02); at the GAPLESS point the kernel decays as a clean power law $\lvert R\rvert^{-2.1}$ ($R^2{=}0.999$, tail $\sim2{\times}10^{-4}$) — locality needs the gap. [E] $\mathbb{Z}_2$ REFLECTION + $\mu_4$ INHERITED: the flattened modular Hamiltonian $K_{\mathrm{flat}}{=}\mathrm{sign}(K)$ inherits $\Theta K_{\mathrm{flat}}\Theta{=}{-}K_{\mathrm{flat}}$ (BW, $u_\Theta{=}J$) and $[\rho,K_{\mathrm{flat}}]{=}0$ ($\mu_4$) EXACTLY (sign is odd; $\rho$ commutes with any $f(K)$). [E] NEG CONTROLS: a side-even reflection gives $2\lVert K_{\mathrm{flat}}\rVert{\neq}0$ (BW fails), an off-$\mu_4$ $K$ mixing opposite-sign marks breaks $[\rho,\mathrm{sign}(K)]{\neq}0$. [C][O] the chiral $c_-{=}8{\neq}0$ (FHS Chern $\lvert C\rvert{=}1$) obstructs STRICT finite-range commuting-projector locality (Kapustin–Fidkowski, `arXiv:1810.07756`; $C{\neq}0$ forbids exponentially-localised Wannier, Thouless), so the realisation is the quasi-local LTO net of NPW26 — the form sub-step (i) asks for. `SEAM.EQUIV.01` stays [O] (the continuum theorem MMST v336 is the backbone). Python-only (numpy; the det/level discriminators Wolfram-mirrored via v89/v281/v422). [E][C][O] |
| `v461_seam_strict_locality.py` | `SEAM.S3.LOCALITY.01`: the Kapustin–Fidkowski/Thouless strict-locality OBSTRUCTION made explicit — sharpening v460's verdict from a cited statement to an EXHIBITED topological obstruction on the same v367 $p{+}ip$ collar. [E] CHERN INTEGER: the FHS Chern is $C{=}{+}1$ (chiral $M{=}1$), 0 (trivial $M{=}3$), $-1$ (opposite mass) — the invertible-phase invariant. [E] WILSON-LOOP / WANNIER-CENTRE WINDING = CHERN: the hybrid Wannier centre $\theta(k_y)$ winds by $\lvert C\rvert{=}1$ (chiral) and 0 (trivial) over the BZ; exponentially-localised Wannier functions exist IFF the winding is 0 (Brouder–Panati et al., PRL 98 046402; Thouless), so the chiral collar has NONE. [E] OBSTRUCTION INTEGERS TIED: winding $=\lvert C\rvert{=}1{\neq}0$ and $c_-{=}16/2{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}{=}\mathrm{rank}\,E_8{\neq}0$ (the same 8 as $c_3$'s one-sided count, v456). [E] NEG CONTROL: BOTH phases are gapped, yet only the trivial (winding 0) admits exp-localised Wannier and a strict finite-range commuting projector — it is the CHIRALITY, not the gap. [C][O] Kapustin–Fidkowski (`arXiv:1810.07756`): a strictly finite-range commuting projector would force winding 0, $C{=}0$; since $C{=}1{\neq}0$ NONE exists, so v424 sub-step (i)'s realisation is PROVABLY the quasi-local NPW26 LTO net (strict locality topologically forbidden, not a missing premise) — upgrading v460's cited verdict. `SEAM.EQUIV.01` stays [O] (continuum theorem MMST v336 the backbone). Python-only (numpy; the winding/Chern integers Lean-mirrored, `SeamResidualAxiom.lean`). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v462_seam_spinor_continuum.py` | `SEAM.EQUIV.SPINOR.01`: the 128-spinor extension $SO(16)_1 \to (E_8)_1$ exhibited at CHARACTER level and as a finite-$L \to$ continuum convergence — the constructive companion of the v458 (MMST audit) / v459 (lattice-VOA) residual. [E] $(E_8)_1$ TOWER FROM $SO(16)_1$ SECTORS: the exact Jacobi/$E_8$ identity $\theta_2^8+\theta_3^8+\theta_4^8=2E_4$ gives $\chi_{(E_8)_1}=E_4/\eta^8=\frac{1}{2}[(\theta_3/\eta)^8+(\theta_4/\eta)^8]+\frac{1}{2}(\theta_2/\eta)^8=\chi_o^{SO16}+\chi_s^{SO16}$ — the holomorphic $(E_8)_1$ character IS the $SO(16)_1$ vacuum + spinor sector sum. [E] THE 128 SPINOR WEIGHT-1 STATES FILL 120→248: $\chi_o$ gives weight-1 multiplicity 120, $\chi_s$ gives 128, summing to the exact $(E_8)_1$ tower $\{1, 248, 4124, 34752\}$ — the spinor currents ARE the $(E_8)_1-SO(16)_1$ deficit (248=120+128). [E] FINITE 16-MAJORANA FOCK: $C(16,2)$=120 NS bilinear ($so(16)$) currents and $2^{16/2}$=256 Ramond ground states = 128 spinor + 128 cospinor — the spinor present already at finite $N$. [E] FINITE-$L \to$ CONTINUUM CONVERGENCE: the critical free-fermion ring Casimir $E_0(L)=e_\infty L-\pi vc/(6L)$ fits $c_{\text{Dirac}} \to 1$ ($c_-$=8) and the lowest scaled NS gap $x_1 \to 1$ with $\lvert x_1-1\rvert \sim 1/L^2$ ($L$=64..1024) — the lattice flows to the conformal tower. [C][O] the rigorous holomorphic extension ($\chi_o+\chi_s$ simple-current projection) is supplied by AGT/AMT (v459); `SEAM.EQUIV.01`'s continuum existence (v336) stays [O]. Mixed exact ($\theta$ identity + tower + 120=$C(16,2)$, 128=$2^7$, 248=120+128, Wolfram-mirrored) + numerical (the finite-$L$ convergence, Python-only). [E][C][O] |
| `v463_seam_c8_holomorphic_uniqueness.py` | `SEAM.EQUIV.UNIQ.01`: the $c$=8 holomorphic-uniqueness pin — the positive lever that makes the IDENTIFICATION half of `SEAM.EQUIV.01` classification-forced rather than assumed (B of the A+B round). [E] $c$=8 NOT UNIQUE AT LEVEL 1: $c=\dim \mathfrak{g}/(1+h^\vee)$=8 has THREE simple solutions — $A_8$ ($\dim 80$), $D_8=\mathfrak{so}(16)$ ($\dim 120$), $E_8$ ($\dim 248$); $c$=8 is necessary but NOT sufficient (the v457 $\det K$=4 $SO(16)$ rival is a real gleich-$c$ competitor). [E] HOLOMORPHY FORCES $\dim V_1$=248: the only weight-0 form with leading $q^{-1/3}$ and integer coefficients is $j^{1/3}=E_4/\eta^8$, whose $q^1$ coefficient (the 1-dim candidate space, vacuum $a_0$=1) is 248 — forced, not chosen. [E] $\dim V_1$=248+ level-1 $c$=8 ⇒ $E_8$ UNIQUELY (248/(1+30)=8); the gleich-$c$ rivals $A_8$ (80) and $D_8$ (120) are excluded. [E] the $(E_8)_1$ tower $\{1, 248, 4124, 34752, 213126\}$ are the $E_4/\eta^8$ coefficients, and $E_4/\eta^8=j^{1/3}$ (numeric $< 10^{-25}$). [C] Dong–Mason/Schellekens: the holomorphic VOA at $c$=8 is UNIQUE = $V_{E_8}$, and the even unimodular rank-8 lattice is unique = $E_8$ (Mordell–Witt) — so "$c$=8 holomorphic limit = $(E_8)_1$" is classification-forced. With v458 (free part) + v459 (spinor extension) the continuum/identification HALF of `SEAM.EQUIV.01` is [C] conditional on named audited theorems; the REALISATION R1 (v464) is the sole remaining input. `SEAM.EQUIV.01` stays [O]. Exact (sympy/integer $q$-series + Lie-algebra scan; mpmath cross-check), Wolfram-mirrored. [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v464_seam_oneparticle_rigidity.py` | `SEAM.EQUIV.ONEPARTICLE.01`: the one-particle RIGIDITY + scaling limit attacking R1 — the REALISATION input of `SEAM.EQUIV.01` (A of the A+B round). The seam is quasi-free (v113/v160/v161), so by Araki self-dual CAR the whole net is the second quantisation of its one-particle symbol $P$, and R1 collapses to a finite-per-mode statement about $P$. [C] ARAKI SELF-DUAL CAR: a gauge-invariant quasi-free state ↔ its symbol $P$ ($0\le P\le 1$) bijectively (Araki 1971; Shale–Stinespring 1965), so a realisation is fully determined by $P$. [E] RIGIDITY: the gapped ground state is the UNIQUE quasi-free state with $P=$ projection onto $K<0$ ($P^2=P$, hermitian to $<10^{-12}$), $K$ fixed by gap + chirality + reflection (v426) — no second realisation. [E] ONE-PARTICLE SCALING LIMIT EXISTS: the real-space kernel is Cauchy on a fixed window ($\max|P_{2N}-P_N|\sim 1/N\to 0$), so $P_\infty$ exists (Gaussianity is why a computation reaches it). [E] THE LIMIT IS $c=1$ DIRAC ($c_-=8$): entanglement $S(L)=(c/3)\ln L$ gives $c\to 1$ (per Majorana 1/2; Peschel/Calabrese–Cardy, ties v450), 16 copies → $c_-=8$. [C] Shale–Stinespring: gauge-invariant ⇒ no pairing block ⇒ the Bogoliubov map is implementable, GNS reps quasi-equivalent. R1 is upgraded from a bald axiom to "the unique quasi-free realisation, one-particle limit exhibited, modulo the named Araki/Shale–Stinespring functor"; with v463 the `SEAM.EQUIV.01` residual is now entirely certification (named cited functors), zero open internal mechanism — closed modulo cited theorems. `SEAM.EQUIV.01` stays [O]. Python-only (numpy: idempotency, kernel Cauchy-convergence, entanglement $c$-fit). [E][C][O] |
| `v465_seed_crosssector_joint.py` | `SEED.CROSSSECTOR.01`: the cross-sector one-parameter seed test — the $\theta_{13}$-independent, externally-runnable slice of v306. TFPT reads a CMB EB/TB birefringence angle $\beta$ and a quark Cabibbo angle $\lambda_C=\lvert V_{us}\rvert$ (unrelated in the SM) as two readouts of the ONE axiom seed $\varphi_0=1/(6\pi)+48c_3^4$. [E] plumbing: the axiom $\varphi_0$ reproduces the frozen $\lambda_C$, $\sin^2\theta_{12}$, $\beta$ to $10^{-9}$. [E] CROSS-SECTOR (headline): back-solving $\varphi_0$ *independently* from $\beta$ (CMB) and $\lambda_C$ (quark) gives the *same* seed within $0.01\sigma_{\rm comb}$ — a correlation only a shared-origin theory predicts. [E] one-parameter fit: the error-weighted mean of the three back-solved $\varphi_0$ reproduces the axiom to 0.06%; the joint $\chi^2$ at the axiom (zero free parameters, 3 dof) is 0.007, $\chi^2/\mathrm{dof}<1$, combined pull $0.08\sigma$. [E] $\theta_{13}$ EXCLUSION declared, not cherry-picked: adding $\sin^2\theta_{13}$ raises $\chi^2$ by $\sim 4$ (the documented $\sim 2\sigma$ pressure point, v328/v338), so this slice is *deliberately* $\theta_{13}$-independent. [E] negative control: a data↔formula shuffle explodes the cluster $\chi^2$ by $>5\times 10^4$. A consistency / out-of-sample statistic (a subset of v306); $\varphi_0$ is FIXED by the axioms, so it upgrades *no* prediction. Python-only. [E] |

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| Script | What it machine-checks *(continued)* |
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| `v466_seed_leptonmass_channel.py` | `SEED.LEPTONMASS.01`: a SIXTH, new-sector seed channel — the charged-lepton mass ratio $m_e/m_\mu=\frac{12}{7}\varphi_0^2$ points to the SAME axiom seed $\varphi_0=1/(6\pi)+48c_3^4$, extending the empirical seed over-determination (v306/v465) from three measurement sectors to five. **[E]** plumbing: the axiom $\varphi_0$ reproduces the frozen $m_e/m_\mu=\frac{12}{7}\varphi_0^2=0.00484673$ to $10^{-9}$. **[E]** NEW SECTOR (headline): back-solving $\varphi_0$ from the measured POLE ratio $m_e/m_\mu=0.004836$ (PDG) gives $\varphi_0=0.053113$, only $-0.11\%$ from the axiom — a charged-lepton MASS reading (Penning-trap masses), a fourth measurement sector beyond {neutrino mixing, CMB, quark mixing}. **[C]** TRANSPORT CAVEAT: the formula is the SOURCE ratio, the datum the POLE ratio; the source→pole shift is the seam-gapped correction class (bound $(2/3)^6=8.78\%$, v393) but largely CANCELS for $m_e/m_\mu$, so the 0.11% residual sits far inside the band — added as a CONSISTENT new sector, not a sharper pin. **[E]** EXTENDS THE OVER-DETERMINATION: the six-channel cluster (v306's five + this) has $\chi^2/\mathrm{dof}<1$ and the error-weighted mean stays at the axiom to <0.1%; with the corroborating heavy-quark $m_t/m_b$ ($-0.23\%$) it is a seven-channel, FIVE-sector agreement (neutrino mixing, CMB, quark mixing, lepton mass, quark mass). **[E]** negative control / power: a data↔formula shuffle explodes the cluster $\chi^2$ by >100×; and not every mass ratio qualifies — $m_c/m_s=(34/47)/\varphi_0$ is scheme-ambiguous (FLAG 11–14), a WEAK channel deliberately not counted. **[C]** HONEST: theoretically still the ONE seed (v305 multiplicity 1) — an EMPIRICAL strengthening (a new measurement sector agrees), NOT theoretical independence; $\varphi_0$ is FIXED by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only. **[E]**[C] |
| `v467_thirdgen_mixing_pattern.py` | FLAV.THIRDGEN.PATTERN.01: the three $\sim 2\sigma$ mixing tensions ($\theta_{13}$, $\theta_{23}$, $V_{cb}$) are ONE common $-\varphi_0^{\mathrm{ret}}$ suppression ($-5.36\% \pm 1.67\%$, $3.2\sigma$ non-zero, $-0.03\sigma$ from $-\varphi_0^{\mathrm{ret}}$); the ladder-base forms $\lambda_C^2 e^{-5/6}$ / $(1-\varphi_0^{\mathrm{ret}})/2$ / $\lambda_C^2/(1+\lambda_C)$ drop the joint $\chi^2$ $10.4 \to 0.11$ with zero parameters; post-hoc candidate [O], frozen record unchanged, JUNO-era reactors + Belle II decide |
| `v468_dm2_ratio_jarlskog.py` | FLAV.DM2RATIO.01: $\Delta m^2_{21}/\Delta m^2_{31} = \lvert J_{\mathrm{PMNS}}\rvert$ of the frozen matrix $= 0.029653$ vs measured 0.029805(792) $\to -0.19\sigma$ – a parameter-free candidate relation for the previously unpredicted splitting ratio; the informal $m_2/m_3 \sim \pi\varphi_0^{\mathrm{ret}}$ heuristic is now $-2.44\sigma$ and retyped; NO floor $\Sigma m_\nu = 0.0588$ eV consistent; post-hoc [O], JUNO decides |

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| Script | What it machine-checks *(continued)* |
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| `v469_seam_crossedproduct_route.py` | `SEAM.EQUIV.CROSSEDPRODUCT.01`: the net-level crossed-product certification of the 128-spinor extension + the R1→R1′ invariant-level reduction. [E] LR LOCALITY INTEGER (index-2): the $SO(16)_1$ spinor has $h_s{=}N/16{=}16/16{=}1{\in}\mathbb{Z}$ ⇒ statistics phase +1 — the Longo–Rehren criterion HOLDS, so the $\mathbb{Z}_2$ simple-current crossed product $SO(16)_1{\rtimes}\{1,s\}$ is a LOCAL index-2 extension (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1 $so(N)$ net with DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM $\mu(B){=}4/2^2{=}1$ ⇒ HOLOMORPHIC ⇒ $(E_8)_1$ by the v463 pin ($E_4/\eta^8$ $q^1{=}248$ re-verified). The 2024/2025-preprint leg (AGT/AMT, v459) is DEMOTED to an independent second witness — the extension certification now rests on 1995–2001 PEER-REVIEWED journals. [E] INDEX-4 PARALLEL: the $\mu_4$ glue $J{=}(s,\Lambda_1)$ has $h(J^k){=}\{1,1,1\}$ ALL integer ($\frac{5}{8}{+}\frac{3}{8}{=}1$; $\frac{1}{2}{+}\frac{1}{2}{=}1$) — the v125 isotropy $q(k,k){=}k^2$ at net level; $\mu{=}16/4^2{=}1$, the same endpoint. [E] R1′ INVARIANTS COMPUTED: per-copy FHS Chern $C(M{=}1){=}1$ vs $C(M{=}3){=}0$, $\nu{=}16{\times}|C|{=}16$, $c_-{=}\nu/2{=}8{=}g_{\rm car}{+}N_{\rm fam}$; $\nu$ mod 16=0 is EXACTLY Kitaev's class-D phase whose edge admits a purely bosonic description = the $(E_8)_1$ level-1 state (Ann. Phys. 321 (2006), sixteen-fold way). [E] THE CHAIN COMPOSES: T1 (phase classification: Kitaev + $c_-$ invariance, Kapustin–Spodyneiko PRB 101 (2020) / Kim et al. PRL 129 (2022)) → T2 (MMST CMP 397 (2023) + Böckenhauer 1996) → T3 (LR/BE + KLM CMP 219 (2001)) → T4 (Dong–Mason + v463) with matching in/out slots (v297 typing). [C] R1→R1′: `collar_realizes` (model-level fiat) replaceable by `collar_invariants` — {quasi-free [C] (v155/v160), gap [E] (v302), class D, $c_-{=}8$ [E] (v456, from P1)}; quasi-freeness stays the ONE [C] hypothesis; Lean mirror `FORM.SEAM.RESIDUAL.01`. [O] NOT CLOSED: the cited theorems stay cited (certification, not re-proof); `SEAM.EQUIV.01` stays [O]. Mixed exact (weights/$\mu$/tower, Wolfram-mirrored) + numerical (FHS Chern, Python-only). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v470_alpha_inflow_level.py` | `ALPHA.QUILLEN.INFLOW.01`: the $\alpha^3$ level = the COMPUTED bulk Chern invariant + the seam $F$-normalisation = the affine embedding index $k_Y$=5/3. [E] TARGET RE-VERIFIED: $\alpha^{-1}$=137.0359992168 unique root; the Quillen split $a^3-2c_3^3a^2=8b_1c_3^6\ln(1/\varphi_{\rm seam})$ at the root to $<10^{-35}$ (v341/v382). [E] THE INVARIANT COMPUTED: FHS Chern $C(M{=}1){=}1$, $C(M{=}3){=}0$ — the same integers as v367/v461, here read as the quantised $U(1)$ response of the collar phase. [C] $k_0{=}|C|{=}1$ UNDER THE EM1 READING: the $a^3$ coefficient EQUALS the computed bulk invariant of the SAME model that realises S3 — "why exactly 1" is answered by computation + the cited inflow identification (TKNN PRL 49 (1982) / Avron–Seiler–Simon PRL 51 (1983) quantisation; Callan–Harvey Nucl. Phys. B250 (1985) inflow; APS 1975 / Alvarez-Gaumé–Della Pietra–Moore 1985 / Witten Rev. Mod. Phys. 88 (2016) $\eta$=CS part of $\delta\log\det$; Quillen 1985 / Bismut–Freed CMP 106 (1986) determinant-line integrality) — REPLACING v435's minimality assumption. [E] $k_Y$=5/3 EXACT: $\mathrm{tr}_{\bar5}(Y^2)/\mathrm{tr}(T_3^2){=}(5/6)/(1/2){=}5/3$ (Ginsparg, Phys. Lett. B197 (1987)); $(3/5)\times\frac{41}{6}{=}\frac{41}{10}{=}b_1$ — the "GUT 3/5" IS the inverse embedding index, not an imported convention; carrier decomposition $\frac{41}{6}{=}\frac{20}{3}$ (fermions, 3×16-content) $+\frac16$ (Higgs doublet). [C] $F$-NORMALISATION RETYPED: once the seam net is $(E_8)_1$ level 1 (`SEAM.EQUIV.01`), the $U(1)_Y$ normalisation $\langle J(z)J(w)\rangle{=}k/(z{-}w)^2$ is FIXED by $k_Y$ — level-1 current-algebra rigidity, no continuous dial; the v434 [C] has ZERO independent content (a face of `SEAM.EQUIV.01`, per the v382 typing). [E] UNIFICATION: ONE invertible phase, TWO quantised responses — $c_-{=}16\times\frac12\times|C|{=}8{=}g_{\rm car}{+}N_{\rm fam}$ (gravitational, feeds `SEAM.EQUIV.01`/S3, v456) and $C{=}1$ ($U(1)$, feeds EM1) — "the one theorem and the one number" are two faces of the SAME phase; closing R1′ (v469) feeds both. [O] NOT CLOSED: the bridge lemma "$\delta\log\det_\zeta$(seam) = the inflow response" stays the named cited step; the EM1 reading stays [C]; `ALPHA.QUILLEN.EXACT.01` stays [O]; $\alpha^{-1}$ stays [E] regardless. Mixed exact ($k_Y/b_1$ identities, Wolfram-mirrored) + numerical (root/split mpmath, FHS Chern, Python-only). [E][C][O] |
| `v471_seam_horizon_replica.py` | `SEAM.EHMODEL.04`: the Seam–Horizon replica chain exercised NUMERICALLY on a discretized seam collar — the kernel-identification premise of `SEAM.THEOREM.01` (the residual isolated by v150/v151/v152) discharged at the FINITE/MODEL level for the seam's own kernel, with REAL replica sheets. [E] REPLICA⇒EH FORM on a real operator: $dG/d\ln m = 2C(\gamma)$ on deficit cones $\gamma/2\pi = \frac14, \frac12, \frac34$ (0.01–0.05%) AND replica sheets $n{=}2$ (0.8%), $n{=}3$ (1.5%). [E] COEFFICIENT CUTOFF-INDEPENDENT (0.5% drift) while the $\zeta$-scale $\mu_{\rm lat}$ drifts with the discretization = the v152 anchor exhibited on data; NO canonical $\ln(m/\mu){=}\frac34$ emerges (recorded honestly). [E] BFK SPLIT MEASURED ON THE KERNEL: Schur identity exact ($\sim10^{-16}$); the Calderón/Schur factor's deficit slope is a PURE CUT-EDGE law $E(1{-}q)$ with tip term 0 (power check: the same law fails on the full determinant); the halves carry the whole tip = $2C_{\rm cone}$ via Kac doubling (exact). [E] THE SEAM'S OWN SPECTRUM: the v302 transfer masses $m_2{=}6\ln\frac32$, $m_3{=}6\ln3$ sit ON the calibrated EH line ($\le0.05\%$); the Perron/attractor mode ($m{=}0$) is IR-DIVERGENT (runs exactly as $-2C\ln$(IR)) — the recovery gap $\Delta = 6\ln\frac32$ makes the induced-gravity coefficient finite: the same gap that makes the attractor unique makes $1/G$ finite. [O] HONEST SCOPE: finite/discretized level; `SEAM.THEOREM.01` stays open — the residual retypes to the CONTINUUM scaling limit (MMST class, = the single residual of `SEAM.EQUIV.01`, v336) + the v152 anchor. Numerical (sparse lattice determinants), Python-only by nature. [E][O] |

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| Script | What it machine-checks *(continued)* |
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| `v472_quillen_detline_moduli.py` | `ALPHA.QUILLEN.DETLINE.01`: the DETERMINANT LINE over the $U(1)$-twist moduli of the collar model carries curvature = the inflow level — the finite Quillen/Dai–Freed shadow of the v470 bridge lemma, COMPUTED. [E] BLOCH LEVEL RE-VERIFIED: BZ Chern $C(M{=}1){=}1$, $C(M{=}3){=}0$, $C(M{=}{-}1){=}{-}1$ (the v470/v367 integers). [E] THE QUILLEN OBJECT COMPUTED: the same collar $h(k)$ in real space on an $L\times L$ torus with TWISTED boundary conditions — the twist torus $(\theta_x,\theta_y)$ IS the moduli space of flat $U(1)$ connections — and the Fukui–Hatsugai–Suzuki curvature of the determinant line of the occupied one-particle frame (the free-fermion ground state) over that torus = 1 exactly ($10^{-9}$) at $M{=}1$, for $L{=}4$ AND $L{=}6$ (size-independent, as an integer must be). [E] CONTROLS: trivial collar $M{=}3\to 0$; orientation flip $M{=}{-}1\to{-}1$. [E] TWO TORI, ONE LEVEL: the twist-moduli integer equals the Bloch-BZ integer for all three $M$ — the Niu–Thouless–Wu identification (PRB 31 (1985)), exhibited on the collar model that realises S3 (v460/v461). [E] SECTION GLOBALLY DEFINED: the Fermi gap (= 2.0) stays open over the WHOLE twist torus — the Dai–Freed section has no zero, the holonomy reading is globally valid. [C] THE READING: det-line holonomy over the $U(1)$ moduli = inflow level $k_0{=}|C|{=}1$ — the v470 bridge lemma "$\delta\log\det_\zeta(\text{seam})$ = the inflow response" holds VERBATIM at the finite/model level, where log det of the occupied frame replaces $\log\det_\zeta$ (Quillen 1985; Bismut–Freed CMP 106 (1986); Dai–Freed JMP 35 (1994); FHS JPSJ 74 (2005)). [O] NOT CLOSED: the abstract-seam $\zeta$-determinant identification (the continuum leg, = the `SEAM.EQUIV.01` face); `ALPHA.QUILLEN.EXACT.01` stays [O]; $\alpha^{-1}$ stays [E] regardless. Structurally the same manoeuvre as v471 for `SEAM.THEOREM.01`'s replica side. Numerical (dense eigh + FHS twist grid), Python-only by nature. [E][C][O] |
| `v473_entropic_action_bridge.py` | `GRAV.ENTROPIC.ACTION.01`: the entropic-action bridge — Bianconi's *Gravity from entropy* (Phys. Rev. D 111, 066001 (2025); `arXiv:2408.14391`) quantified as an EXTERNAL CANDIDATE for the local volume action behind TFPT's parameter-free Einstein equation (v358/v359, honestly typed "equation of state, not a from-action quantisation" [O]). Her $S_{\rm B}=-\operatorname{Tr}_F\ln(\widetilde{G}\widetilde{g}^{-1})$ is the quantum relative entropy between the spacetime metric and the matter-induced metric (Araki-related, $\Delta^{1/2}=\widetilde{G}\widetilde{g}^{-1}$), a local covariant action whose variation gives Einstein + emergent nonnegative $\Lambda$ at low coupling — the missing variational layer IF the bridge holds. [E] CARRIER HODGE COUNT: her $\Omega^{0,1,2}$ fiber on the FIVE-SLOT carrier $\mathbb{C}^5$ (not 4d spacetime, where it is 1+4+6=11) counts $1{+}5{+}10{=}16{=}2^{g_{\rm car}-1}{=}\dim S^+_{D_5}$, Hodge-folded onto $\Lambda^{\rm even}\mathbb{C}^5$ (vector-space identity ONLY). [E] COEFFICIENT MATCH (the one quantitative pin): her weak-coupling $\mathcal{L}=3\beta R-\alpha\mathcal{L}_{\rm m}$ matched to $1/(16\pi G)=c_3/(2G)$ fixes her free constant EXACTLY, $\beta'_{\rm B}=c_3/6=1/(48\pi)$; the $6=3$ blocks $\times 2=p_2=|R^+(A_3)|$ (structural echo, not a proof). [E] ENTROPY POTENTIAL: $\Lambda_{\mathcal{G}}=\frac{1}{2\beta}\operatorname{Tr}(\mathcal{G}-I-\ln\mathcal{G})\ge 0$, quadratic at the fixed point $\Rightarrow$ with $\varepsilon_\star=e^{-\alpha^{-1}}Q$ the closed v60 branch $(3/4\pi^2)e^{-2\alpha^{-1}}$ is reproduced with the EXACT target $\operatorname{Tr}Q^2=32c_3^4=1/(128\pi^4)=\frac{2}{3}\,\delta_{\rm top}$ ([C] until $Q$ is constructed). [E] FRULLANI: $-\ln A=\int_0^\infty\frac{dt}{t}(e^{-tA}-e^{-t})$ — her log action is an all-scale heat-kernel action, the right genre against $S_{{\rm rel},\chi}$ (KMS cutoff). [E] $R^2$ KILL TEST: raw $(c_3/6)^2$ vs Starobinsky $1/(12c_3^7)$ ($M^2_{\rm scal}=c_3^7\bar{M}^2_{\rm Pl}$, v36) — EXACT gap $3(8\pi)^9\approx 1.2\times10^{13}$ (13 orders): a direct identification of the raw entropic action with the TFPT $R^2$ sector is FALSE without a mechanism (light $\mathcal{G}$-trace mode generating $c_3^{-9}$, or KMS-spectral renormalisation) — pre-registered. [C] COMPRESSION CONJECTURE: $P_\Sigma(\widetilde{G}\widetilde{g}^{-1})P_\Sigma=\Delta_\Sigma^{1/2}$ + the operator-level Hodge items (Clifford, Dirac, leaf involution, Calderón polarisation) stay open; Lorentzian positivity NOT automatic — TFPT's OS route is the sturdier construction. [O] NOTHING CLOSES: v359's equation-of-state status stays [O]; zero TFPT knobs added. Mixed exact (sympy, Wolfram-mirrored) + one quadrature (mpmath). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v474_entropic_hodge_carrier.py` | `GRAV.ENTROPIC.HODGE.01`: the OPERATOR level of the entropic-action bridge (work packages 1 + 4-algebra of v473). [E] CAR⇒CLIFFORD: the ten Γ's from $(a_k^\dagger, a_k)$ on the carrier Fock space $\Lambda^\bullet\mathbb{C}^5$ (dim $2^{g_{\rm car}}$=32) satisfy $\{\Gamma_a,\Gamma_b\}=2\delta_{ab}$ exactly (all 100 pairs, integer matrices) — the $\mathfrak{so}(10)=D_5$ Clifford algebra acts on the carrier exterior algebra. [E] $S^+=\Lambda^{\rm even}$ AS $D_5$-MODULE: the 45 bilinears $[\Gamma_a,\Gamma_b]/4$ commute with $\gamma=(-1)^N$, span exactly 45= dim $\mathfrak{so}(10)$ and leak NOTHING from the 16-dim even subspace — the half-spinor IS the even exterior algebra WITH its spinor structure (operator level of v2/v44/v197). [E] HODGE–DIRAC SYMBOL: $c(\xi)=a^\dagger(\xi)-a(\xi)$ satisfies $c(\xi)^2=-|\xi|^2\,\mathrm{Id}$ for SYMBOLIC $\xi$ — Bianconi's $D=d+\delta$ is Clifford multiplication at symbol level. [E] THE FOLD IS $5\to\bar5$: the traceless $\mathfrak{u}(5)$ weights of $\Lambda^4$ are the NEGATIVES of those of $\Lambda^1$, so the Hodge fold $\star:\Lambda^1\cong\Lambda^4$ conjugates the vector block — her fiber $1+5+10$ becomes the GUT half-spinor $16=1+\bar5+10$ (the SM carrier decomposition), not a mere dimension match. [E] HONEST OBSTRUCTION: $\Omega^{\le2}$ is NOT Clifford-invariant ($c(\xi)$ maps $\Lambda^2$ into $\Lambda^3$); only $\Lambda^{\rm even}$ is (even algebra) — the bridge must go through the fold, never through degree truncation (constraint on AP2). [E] $Q$-TARGET ENUMERATION (the v473 conjecture DECIDED): $\mathrm{Tr}\,Q^2=d\,k^2c_3^4=32c_3^4$ with INTEGER $k$ has exactly the supports $(d,k)=(2,4),(8,2),(32,1)=\{|\mathbb{Z}_2|,\ g_{\rm car}+N_{\rm fam}=\mathrm{rank}\,E_8,\ 2^{g_{\rm car}}=\dim\Lambda^\bullet\}$ — minimal uniform solution $q=c_3^2=1/(64\pi^2)$ per Fock mode; the two-form block $d=10$ (the "pair sector" conjecture) and the fiber $d=16$ need irrational multipliers $\sqrt{16/5}$, $\sqrt{2}\Rightarrow$ KILLED in uniform form; bonus identity $\mathrm{Tr}\,Q^2/\delta_{\rm top}=32/48=2/3=\dim\Lambda^\bullet/\Omega_{\rm adm}$. [C] the physical choice among the three admissible supports; [O] AP2 compression + $R^2$ mechanism stay open, no marker moves. Mixed exact: integer Fock matrices (numpy, exact in double) + symbolic sympy (weights/enumeration Wolfram-mirrored). [E][C][O] |
| `v475_entropic_scalaron_gate.py` | `GRAV.ENTROPIC.SCALARON.01`: the $R^2$ kill test of the entropic-action bridge EXECUTED (work package 5 of v473) + the explicit Lorentzian-positivity witness. [E] MAX-SYMMETRIC EIGENVALUES (her Appendix-B flattening verified): $\widetilde{\mathcal{R}}\widetilde{g}^{-1}$ has eigenvalues EXACTLY $\{R;\ R/4\times4;\ R/6\times6\}$ on a general diagonal 4d metric, incl. the 6×6 two-form flattening. [E] EXACT VACUUM $f(R)$: $\mathcal{L}_{\rm B}(R)=-[\ln(1-\beta R)+4\ln(1-\beta R/4)+6\ln(1-\beta R/6)]=3\beta R+\frac{17}{24}\beta^2R^2+O(R^3)$ — the 3 re-derives her Eq. (45) (= the v473 pin), $\frac{17}{24}=\frac12(1+\frac14+\frac16)$ is the EXACT tensorial $R^2$ factor. [E] RAW SCALARON TRANS-PLANCKIAN: with $\beta'=c_3/6$ both routes (Starobinsky matching AND $f'/(3f'')$) give $m_{\rm raw}^2=\frac{4608\pi^2}{17}M_{\rm Pl}^2$, $m_{\rm raw}\approx51.7\,M_{\rm Pl}\approx1.3\times10^{20}$ GeV — HEAVY, not light (TFPT: $c_3^{7/2}M_{\rm Pl}=3.06\times10^{13}$ GeV); required enhancement EXACTLY $m_{\rm raw}^2/m_{\rm TFPT}^2=\frac{72}{17}(8\pi)^9\approx1.69\times10^{13}$ in mass² — v473's interpretation 2 (the TFPT scalaron as the light $\mathcal{G}$-trace mode) is KILLED in naive form. [E] DOMAIN: first log pole at $R=1/\beta=384\pi^2M_{\rm Pl}^2$ (trans-Planckian) — the raw action is well-defined at all sub-Planckian curvatures; the obstruction is the MASS, not the domain. [E] LORENTZIAN WITNESS: a timelike gradient gives $Gg^{-1}$ eigenvalue $1-\alpha v^2\le0$ for $v^2\ge1/\alpha$ (ln undefined); spacelike/Euclidean positive for all $v$ — the v473 [C] caveat exhibited, the OS/reflection-positive route sturdier. [C] VERDICT: interpretation 1 survives (the entropic action covers the Einstein + Λ level), 2 is dead naive, 3 (KMS-spectral renormalisation) is the ONLY surviving $R^2$ route. [O] unchanged: AP2 compression + the $R^2$ gate; no marker moves. Exact (sympy, Wolfram-mirrored). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v476_entropic_compression_toy.py` | `GRAV.ENTROPIC.COMPRESS.01`: the AP2 compression conjecture of the entropic-action bridge (v473) made WELL-POSED on an exactly solvable model. Numerical facts on a staggered-mass chain: the ground-state covariance is a PROJECTION (pure bulk, thermal boundary — the seam situation); the compression $C_A{=}PCP$ has spectrum strictly inside $(0,1)$ with finite Peschel $K{=}\ln((1{-}C_A)/C_A)$ and $S{>}0$ (v155 in the bridge context). ORDER OBSTRUCTION (the reading DECIDED): on the pure bulk $f(x){=}x/(1{-}x)$ is SINGULAR on $\mathrm{spec}(C){=}\{0,1\}$ — the literal operator-side reading $Pf(C)P$ of the v473 conjecture is ILL-POSED while $f(PCP)$ is finite: the compression identity can only be read STATE-SIDE (build $\Delta_\Sigma$ from the COMPRESSED relative metric), matching Bianconi's own local-metric construction — AP2 retyped, not weakened. **[E]** BLOCK ALGEBRA (symbolic): $[C^2]_{AA}{-}(C_{AA})^2{=}BB^t$ and $[C^3]_{AA}{-}(C_{AA})^3{=}ABB^t{+}BB^tA{+}BDB^t$ — first order in the cross block vanishes identically, so for ANY analytic $f$ the two readings agree to SECOND order: the modular (entanglement) content lives exactly in that quadratic correction. GAP CONTROL at the bounded (covariance) level: the bulk identity $f(C_T){=}e^{-h/T}$ holds to $10^{-11}$ relative; the mismatch obeys the quadratic law $d \le x^2$ and falls with the gap in the gapped regime, the decoupled cut agrees EXACTLY — the two readings converge in the gap-dominated regime, where TFPT operates ($\Delta_{\mathrm{eff}}{>}0$, v76/v337). **[C]** the retyped conjecture; **[O]** AP2 stays open (the continuum seam statement); no marker moves. Numerical (numpy lattice + one symbolic block identity), Python-only by nature. **[C][O]** |
| `v477_entropic_scaleflow.py` | `GRAV.ENTROPIC.SCALEFLOW.01`: the scale-flow representation of the entropic action — v475's surviving interpretation 3 typed PRECISELY: the $(72/17)(8\pi)^9$ gap collapses to ONE moment condition on the scale weight, satisfied by the SAME KMS moment TFPT already fixes (v36: $f_0{=}6(4\pi)^2/c_3^7$); zero new dials, nothing closes. **[E]** SCALE-FLOW IDENTITY: $-\ln a = \int_0^\infty \frac{dt}{t}(e^{-ta}{-}e^{-t})$ (symbolic) $= 2\int_0^\infty \frac{d\chi}{\chi}(e^{-a/\chi^2}{-}e^{-1/\chi^2})$ ($t{=}1/\chi^2$, $10^{-20}$ quadrature) — Bianconi's log action IS the FLAT scale-integral of relative heat-kernel actions; TFPT's $S_{\mathrm{rel},\chi}$ evaluates ONE KMS scale with a weight $f$: the bridge is a choice of SCALE MEASURE. **[E]** MOMENT DICTIONARY: a weight $w(t)\,dt/t$ gives $3\beta R\,\mu_1 + \frac{17}{24}\beta^2R^2\mu_2$ ($\mu_1{=}\int we^{-t}dt$, $\mu_2{=}\int wte^{-t}dt$); the flat weight has $\mu_1{=}\mu_2{=}1$ = the v475 raw coefficients — the raw action is the unit-moment point of a moment family. **[E]** THE ONE MOMENT CONDITION: with the weight-dressed EH pin $\beta'\mu_1{=}c_3/6$ the scalaron is $m^2{=}\frac{4608\pi^2}{17}\frac{\mu_1^2}{\mu_2}M_{\mathrm{Pl}}^2$; demanding $m^2{=}c_3^7M_{\mathrm{Pl}}^2$ forces EXACTLY $\mu_2/\mu_1^2{=}\frac{72}{17}(8\pi)^9$ — the v475 kill-test number IS a moment ratio; and the closure identity $\frac{4608\pi^2/17}{(72/17)(8\pi)^9}{=}c_3^7$ holds IDENTICALLY: the 13 orders are a scale-measure datum, not a bridge mismatch. **[E]** v36-CONSISTENCY: $f_0{=}6(4\pi)^2/c_3^7 \Leftrightarrow M^2/M_{\mathrm{Pl}}^2{=}c_3^7$ — the same mass from both parametrisations, ZERO new dials. **[C]** a CONSISTENCY statement, not a derivation (the KMS weight is TFPT input; the entropic action supplies the Einstein + $\Lambda$ level and the scale-flow FORM, TFPT supplies the measure); negative side-record: the exact lattice-BW commutator attempt for AP2 (Peschel–Eisler simple forms) did NOT hold on the half-filled open chain (kept out of the suite, logged). **[O]** unchanged: the $R^2$ gate and v359's typing. Exact (sympy + one quadrature, Wolfram-mirrored). **[E][C][O]** |

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| Script | What it machine-checks *(continued)* |
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| `v478_entropic_continuum_legs.py` | `GRAV.ENTROPIC.LEGS.01`: first computable steps on the TWO remaining legs of the entropic-action bridge (v473–v477). LEG A (AP2 continuum): the compressed critical interval has $S(L)=(c/3)\ln L+k$ with $c_{\rm est}$=1.00089/1.00038/1.00021 (monotone → the Dirac $c=1=2\times\frac{1}{2}$, the v450 reading; gapped control $\sim 6\times10^{-8}$, area law); the Peschel modular Hamiltonian $K=\ln((1-C_A)/C_A)$ organises on the CHM weight $\beta(x)=x(L-x)/L$ (corr 0.87→0.99, plateau drift shrinking; constants deliberately NOT asserted — the lattice EH is only asymptotically local); even-range bands vanish identically (particle-hole, $\sim 10^{-36}$), odd tail 1.8% parabola-enveloped — the state-side $\Delta_\Sigma$ (v476's forced reading) FLOWS TO the CHM/BW geometric form that TFPT's Einstein derivation consumes (v323/v358): the bridge's modular side and TFPT's gravity side MEET in the continuum limit. LEG B (the measure): [E] single-scale corollary — $d\mu=\delta(t-t_0)dt$ gives $\mu_2/\mu_1^2=e^{t_0}$ exactly, so the v477 condition forces $t_0=\ln\frac{72}{17}+9\ln(8\pi)=30.461$ — ONE dimensionless KMS time, exact closed form; DECLINED: $t_0\neq h(E_8)=30$ (difference 0.461; the near-miss is rejected per the v354/v355 no-free-pattern rule); deriving the measure from the entropic structure stays OPEN. [C] the meeting point; [O] both legs, no marker moves. High-precision numerical (mpmath, 50 digits) + one exact corollary, Python-only by nature. [C][O] |
| `v479_kronheimer_quiver_bridge.py` | `SEAM.KRONHEIMER.01`: the R3 "combinatorial-to-geometric bridge" of the $\det K=1$ keystone (v344 — "the rewrite attractor graph IS the du Val singularity") is an ESTABLISHED THEOREM: Kronheimer's ALE hyper-Kähler quotient construction + Torelli classification (J. Diff. Geom. 29 (1989) 665 and 685), with ALL finite input data already compiler outputs. [E] MARKS = NULL = PERRON: $C_{\rm aff}\delta=0$ and $A\delta=2\delta$ on the affine-$E_8$ McKay graph; multiset $\{1,2,2,3,3,4,4,5,6\}$ = the $2I$ irrep dimensions = the v312 rewrite-attractor Perron vector. [E] DIMENSION COUNT: $\dim_{\mathbb{R}} M-4\dim_{\mathbb{R}} G=4(\sum_{\rm edges} d_i d_j-\sum d_i^2+1)=4$ — the hyper-Kähler quotient of the marks quiver is 4-real-dimensional (the $\mathbb{C}^2/\Gamma$ geometry); $\sum_{\rm edges} d_i d_j=\sum d_i^2=120$ forced by the null-vector identity. [E] GAUGE DATUM: $\sum\delta_i^2=120=|2I|=|\mu_4|\,h(E_8)$, $h(E_8)=30=|\mathbb{Z}_2|N_{\rm fam}g_{\rm car}$. [E] RESOLUTION: $8=\operatorname{rank}E_8$ exceptional spheres with intersection form $-\mathrm{Cartan}(E_8)$, $\det=1$ and SNF = identity (the Poincaré-sphere link = the $\det K=1$ face, v232/v344); deformation space $\mathfrak{h}\otimes\mathbb{R}^3$, $\dim 3\cdot 8=24$. [C] the theorems are CITED, not re-proved (the same import class as NPW26/MMST/AGT, v424/v336/v459). [O] the residual premise TRANSFORMS, it does not vanish: "the raw seam supplies ALE hyper-Kähler data asymptotic to the $(2,3,5)$ cone" = the SAME single arrow as `SEAM.EQUIV.01` (v346 L2); no gate closes. Exact (sympy; Wolfram mirror added, counted with the verified 2026-07-21 engine run). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v480_multilocal_four_interval.py` | `QGEO.MULTILOCAL.01`: the four $\mu_4$ marks make the seam circle a FOUR-INTERVAL free-fermion geometry — the class whose multi-interval modular theory is EXACTLY KNOWN (Casini–Huerta mixing term; Longo–Xu multi-interval subfactors; Rehren–Tedesco multilocal fermions; KLM: holomorphic $\Leftrightarrow$ trivial multi-interval inclusion, $\mu$=1 — the SAME one bit as $\det K$=1). On the exactly solvable antiperiodic critical ring ($N$=256, $Q$=64, $\ell$=48): [E] BINARY CLOCK: the quarter-turn clock is a signed permutation with $\rho^4 = -1$ EXACTLY — the fermionic $\mu_4$ clock is the order-8 double-cover lift (the same $\mathbb{Z}_2$-lift pattern as $2I$ over $I$). [E] EXACT SECTOR DECOUPLING: the primitive-8th-root character projectors resolve the identity and commute with the reduced covariance at $10^{-15}$ (the Rehren–Tedesco diagonal 4-fold-cover isomorphism at covariance level). [E] COVER/TWIST: each sector kernel is the SINGLE-interval quarter-ring kernel with boundary twist $\lambda_k$; the union of sector spectra = the full 4-interval spectrum at $10^{-14}$. [E] CLOCK INVARIANCE MANIFEST: $\rho C \rho^{-1}$=$C$ at $10^{-14}$ $\Rightarrow$ $[\rho, K]$=0 by functional calculus — $\omega \circ \rho = \omega$ HOLDS at the 4-interval state level; NEGATIVE CONTROL: displacing ONE interval by one site re-couples the sectors at $10^{-2}$. CONJUGATE-POINT MIXING (numerical): every cross-interval row of $K$ peaks on $\lvert s'-s\rvert \le 2$ at both sizes — the lattice shadow of the Casini–Huerta mixing term. [C] the continuum multi-interval theory is CITED; [O] the raw-seam premise (`QGEO.SYM.01`/`SEAM.EQUIV.01`) stays OPEN — mechanism exhibited, not closed. Numerical (numpy), Python-only by nature. [E][C][O] |
| `v481_seesaw_carrier_ladder.py` | `FLAV.NUSCALE.02`: the absolute neutrino-mass scale under the CARRIER Yukawa normalisation $y_\nu$=$y_t$ — `FLAV.NUSCALE.01`'s two-parameter $(y_\nu, M_R)$ trade-off collapses to ONE parameter and the required $M_R$ lands INSIDE the compiler's own PS window; the nearby integer $c_3$-rung is explicitly DECLINED at 1-loop (v354/v355). CANDIDATE class (genre v467/v468); the frozen record (v84) is untouched and NO gate closes. CARRIER NORMALISATION (conjectured premise; named premise): one $SO(10)$ **16** per family + minimal Yukawa sector ($\mathbf{10}_H$ / PS $(1,2,2)$) $\Rightarrow$ $y_\nu = y_t$ at the matching scale — the trade-off collapses to $M_R$ alone. INVERSION INSIDE THE WINDOW (numerical): the observed $m_3$=0.0503 eV demands $M_R$=$9.3 \times 10^{13}$ GeV, INSIDE $[M_{\rm PS}, M_{\rm GUT}] = [4.2 \times 10^{13}, 2.4 \times 10^{15}]$ (v249), at $\log_{c_3}(\bar{M}_{\rm Pl}/M_R) = 3.15$ ($y_\nu$=1 gives the structureless 2.58 — the old `FLAV.NUSCALE.01` number). [X] INTEGER-RUNG HONESTY GATE: $M_R = c_3^3 \bar{M}_{\rm Pl} = 1.53 \times 10^{14}$ GeV predicts $m_3 = 0.030$ eV (40% low at 1-loop; ADKLR $\kappa$ run-down factor 0.794) — NEAR the rung (3.15 vs 3), NOT ON: the ladder pin is DECLINED; the decision computation is NAMED (2-loop SM + explicit PS threshold matching). FALSIFIABLE BAND (numerical): over the PS window $m_3$ spans $[0.002, 0.115]$ eV bracketing the observation; the $M_{\rm PS}$ end ($\Sigma \sim 0.124$ eV) is already in tension with DESI $\sim$0.072 — cosmology cuts the window from below; the NO floor $\Sigma m_\nu = 0.059$ eV stays the falsifiable edge (v272). [C] / [O] candidate class; the absolute scale stays formally open. Numerical (explicit 1-loop RG: gauge/$y_t$/$\lambda$ up, ADKLR Weinberg-operator down), Python-only by nature. [C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v482_seesaw_rung_decision.py` | `FLAV.NUSCALE.03`: the decision computation NAMED by v481, EXECUTED in bracketed (assumption-robust) form — the integer rung $M_R = c_3^3 \bar{M}_{\rm Pl}$ is EXCLUDED as an unstructured exact pin, conditional on the carrier premise $y_\nu$=$y_t$. THE GAP: central $m_3$(rung)=0.0301 eV vs observed 0.0503 — rescue factor ×1.670 (+67%, 0.154 $c_3$-rungs). BRACKETED EXCLUSION: with $m_t(\overline{\rm MS}) \pm 3$ GeV ($> 3\sigma$), $\alpha_3^{-1} \mp 0.10$ ($> 3\sigma$), a ±10% band on the ADKLR $\kappa$ run-down and the PS-leg Yukawa beta scaled $\times[0.5, 1.5]$, the largest single shift is ×1.10 and the ALL-favourable combination reaches only ×1.165 — far below ×1.670: the v481 1-loop miss is NOT an RG artifact. [C] NAMED ESCAPE, RECORDED AND DECLINED: only a third-generation Majorana STRUCTURE factor $r \sim 1.67$ revives the rung; post-hoc $r_{\rm needed}$=1.6696 sits 0.18% from $g_{\rm car}/N_{\rm fam}$=5/3 — no mechanism forces a 3/5 Clebsch on $M_{R,3}$ today, so this is a named coincidence with a named missing mechanism (PS Majorana-sector Clebsch), DECLINED per v354/v355 (the $t_0$∼30 pattern of v478). [O] the v481 one-parameter window candidate (NO floor $\Sigma m_\nu$=0.059 eV) never used the integer rung and STANDS; frozen record untouched; nothing closes. Numerical (explicit 1-loop RG + bracketed envelopes), Python-only by nature. [C][O][X] |
| `v483_pillowcase_exact_traces.py` | `HYP.PHI0.GEOM.01`: the GEOMETRIC side of the $\varphi_0^{\rm ret}$ puncture target (`HYP.PHI0.PUNCTURE.01`/v408) is EXACT — every twisted/orbifold heat trace of the flat $\tau$=$i$ pillowcase is a $t$-INDEPENDENT RATIONAL, so the $\pi^{-4}$ in $d_{\rm top} = 48 c_3^4$ cannot come from flat orbifold geometry at ANY order. [E] TWISTED TRACES EXACTLY 1: $\mathrm{Tr}(\gamma e^{t\Delta}) = 1$ for $\gamma = \sigma\,(z{\mapsto}{-}z)$, $\rho\,(z{\mapsto}iz)$, $\rho^3$ at every tested $t \in [0.05, 3]$ (position-space Gaussian image sums / erf closed forms + 2d quadrature, residual $< 10^{-9}$; mode-space proof: only $n$=0 is $\gamma$-diagonal) — $t$-independent, no asymptotic tail. [E] ATIYAH–BOTT MATCH: $1 = 4 \times \frac{1}{4}$ ($\det(1{-}d\sigma)$=4=$|\mu_4|$) and $1 = 2 \times \frac{1}{2}$ ($\det(1{-}d\rho)$=2=$|\mathbb{Z}_2|$). [E] PILLOWCASE DATA EXACT: orbifold heat trace = $\frac{1}{2}\,\mathrm{Tr}_{T^2} + \frac{1}{2}$ (contact term EXACTLY $\frac{1}{2} = 4 \times \frac{1}{8}$ per mark, all $t$); clock-equivariant trace exactly 1 (all $t$). [E]/[O] THE REDUCTION: the analytic content of $d_{\rm top} = 48 c_3^4 = \frac{3}{256\pi^4}$ must sit in the per-insertion COUPLING normalisation — 4=$|\mu_4|$ insertions of weight $c_3$ give $c_3^4$, multiplicity $\Omega_{\rm adm}$=48; the bare $1/(2\pi)$-propagator 4-cycle differs by the exact RATIONAL $\frac{3}{16}$; the ONE remaining [O] is the rule "per-mark weight = $c_3$" (seam zeta/Green normalisation, the same residual class as `EM.WARD.02`). The target narrows from "derive the term" to "derive the weight"; `HYP.PHI0.PUNCTURE.01` stays open; no gate closes. Numerical verification of exact statements + sympy exact arithmetic; Python-only by nature. [E][O] |

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| Script | What it machine-checks *(continued)* |
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| `v484_seam_contact_unit.py` | `SEAM.CONTACT.UNIT.01`: the shared "$c_3$ per boundary insertion" rule of the two open analytic targets (`ALPHA.QUILLEN.EXACT.01` ladder $\{0,3,6\}$; `HYP.PHI0.PUNCTURE.01` per-mark weight $c_3$) is ONE identity and NOT a new unknown: it is the KMS seam unit $2\pi = 1/(4c_3)$ with $\frac{1}{4} = 1/|\mu_4|$ (v239) — one bare boundary propagator ORBIT-AVERAGED over the four marks. [E] UNIT CHAIN: $\frac{1}{2\pi} = 4c_3 = |\mu_4|c_3 = |\mathbb{Z}_2|\chi(S^2)c_3$. [E] MECHANISM DERIVED ON THE SEAM CIRCLE: $G(d) = -\frac{1}{\pi}\ln|2\sin(d/2)|$ (polylog identity, 40 digits) takes INTEGER multiples of $c_3 \ln 2$ at the $\mu_4$ separations — $G(\pi/2) = -4c_3 \ln 2$, $G(\pi) = -8c_3 \ln 2$, so $G_{\text{marks}} = -4c_3 \ln 2 \cdot \text{circ}(0,1,2,1)$ with spec $= \{4,-2,0,-2\}$; the $\mu_4$-invariant (orbit-averaged) insertion carries $(GV)|_{\text{marks}} = -(c_3 \ln 2)\, C$ EXACTLY, so the log-det contact expansion is exactly graded in $c_3$ per insertion with integer cycle combinatorics $\text{Tr}(C^k) = 4^k + 2(-2)^k$. [E] SUITE-WIDE GRAMMAR: the $F_{U(1)}$ ladder $\{0,3,6\}$ (v3); $\varphi_0^{\text{ret}}$ ladder $\{1,4\}$ (v37); the $\Lambda$ prefactor $\frac{3}{4\pi^2} = (8\pi)^2 d_{\text{top}} = 48c_3^2$ (v60) — the SAME multiplicity $\Omega_{\text{adm}}$=48 as the puncture, at TWO insertions; the $g-2$ seam loop divides by the same $2\pi = 1/(4c_3)$. [C]/[O] THE MERGE: the two tracked [O] targets collapse into ONE named analytic step — the diagonal (self-energy) zeta-renormalisation at the marks + the state-multiplicity matching ($48 = \Omega_{\text{adm}}$, $41 = 10b_1$); the finite cycle sector is proven. Anti-numerology: the integers are forced by the square configuration; no target coefficient is claimed. Exact (sympy + 40-digit mpmath polylog); Python-only by nature. [E][C][O] |
| `v485_contact_diagonal_closed.py` | `SEAM.CONTACT.UNIT.02`: the DIAGONAL channel of the merged analytic target (v484) settled at the computable level — the merged residual COLLAPSES ONTO THE KEYSTONE. [E] DIAGONAL ZERO AT THE KMS SCALE: the renormalised coincident-point Green function on circumference $\ell$ is $G_{\text{reg}}(0;\ell) = \frac{1}{\pi}\ln(\ell/2\pi)$ exactly — it VANISHES iff $\ell = 2\pi = 1/(4c_3)$, the v239 KMS seam circumference: the seam scale is the unique zero-self-energy scheme point. [E] THE LOG IS THE SCALE CHANNEL: rescaling gives $\frac{1}{\pi}\ln\kappa$ and nothing else — matching the functional structure ($F_{U(1)}$ carries exactly ONE log term, the transport $8b_1c_3^6 \ln(1/\varphi_{\text{seam}})$; the $\varphi_0^{\text{ret}}$ split is log-free); [C] reading: log = diagonal/scale channel, log-free = contact data at the KMS point. [E] CLOSED-FORM RESUMMATION (BFK route, v151 class): the finite-rank mark determinant resums exactly, $\det(I - uC) = (1-4u)(1+2u)^2$ with $u = \varepsilon c_3 \ln 2$ — all orders; the linear term is ABSENT precisely because the diagonal vanishes ($\text{Tr}\, C = 0$); the series matches v484's $\text{Tr}(C^k) = 4^k + 2(-2)^k$. [E] WEIGHT DICHOTOMY: the SAME $48 = 3 \times 16$ admissible states carry BOTH multiplicities — flat count $48 = \Omega_{\text{adm}}$ (the $\varphi_0^{\text{ret}}$ puncture) vs $Y^2$/loop/GUT-weighted $41 = 10b_1 = 40 + 1$ ($\text{Tr}_{16}\, Y^2 = \frac{10}{3}$ exact, $\nu^c = 0$; Ginsparg $(3/5)(41/6) = 41/10$, v470) — one state set, two response weights. [C]/[O] CONSEQUENCE: with v484 EVERY finite/computable piece of the merged `ALPHA.QUILLEN` + `PHI0.PUNCTURE` target is proven; the single remaining [O] is the identification of the abstract seam $\zeta$-det with this glued mark determinant — a face of `SEAM.EQUIV.01`, exactly the v382 typing, now substantiated computationally. No gate closes. Exact (sympy + 30-digit mpmath + exact charge sums); Python-only by nature. [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v486_transfer_full_rule.py` | `HYP.REWRITE.02`: the FULL physical transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (v54/v221) from ONE forced local rule — the lazy $\mathbb{Z}_2$-pair walk; the $(1/3)^6$ mode, previously generator-less, gets its local-rule origin (sharpens v327, whose uniform rule produces $\{1, \frac{2}{3}, 0\}$ and can never generate the third mode). [E] UNIQUENESS: for $M(s,h) = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & s & h \\ 0 & h & s \end{smallmatrix}\right)$ (one absorbing family channel + SYMMETRIC $\mathbb{Z}_2$ pair) the survival spectrum is $\{s{+}h,\ s{-}h\}$; demanding the physical pair $\{\frac{2}{3}, \frac{1}{3}\}$ forces UNIQUELY $(\text{stay}, \text{hop}, \text{leak}) = (\frac{1}{2}, \frac{1}{6}, \frac{1}{3}) = (1/\lvert\mathbb{Z}_2\rvert,\ 1/(\lvert\mathbb{Z}_2\rvert N_{\text{fam}}),\ 1/N_{\text{fam}})$ — atom form recorded (honest direction: spectrum → rates). [E] SIX-HAND GENERATION: over the order-6 $= 2N_{\text{fam}} = \lvert R^+(A_3)\rvert$ hand, $\text{eig}(B^6) = \{\frac{64}{729}, \frac{1}{729}\}$ exactly; decay gaps $6\ln\frac{3}{2}$ (the recovery gap, v76) and $6\ln 3$ (the subdominant gap, now generator-full). [E] SUPERSESSION: the uniform rule's zero mode persists under every power — no iterate reaches $(1/3)^6$; its $\lambda_2$ arity story ($\frac{2}{3} = \lvert\mathbb{Z}_2\rvert/N_{\text{fam}}$) is unchanged. [E] sub-stochastic, symmetric (detailed balance), absorbing democratic attractor unchanged (v317). [O] the arity $\{2,3\}$ stays the anchor input; the lazy-vs-uniform split is SELECTED by the physical spectrum, not yet first-principles-derived (named candidate: stay $= 1/\lvert\mathbb{Z}_2\rvert$ as the sheet-symmetric choice of the $\mathbb{Z}_2$ involution). Exact (sympy); mirrorable, deferred to the next verified engine run. [E][O] |
| `v487_transfer_clock_rungs.py` | `HYP.REWRITE.03`: the lazy split $(\frac{1}{2}, \frac{1}{6})$ is FORCED by the resummed clock — the v486 [O] 'split selection' closes onto v124 structure; sharpens `HYP.REWRITE.02`, no gate closes. [E] RUNG READING: the lazy rule's one-step spectrum is EXACTLY the complete clock ladder below the wall, $\text{eig}(M) = \{1 - n/3 : n = 0,1,2\} = \{1, \frac{2}{3}, \frac{1}{3}\}$; v327's uniform rule puts its second pair mode ON the wall ($0 = 1 - 3/3$) — the degeneracy re-read as 'odd channel collapsed to the wall'. [E] DECK PARITY = RUNG INDEX: $[B, \sigma] = 0$; the $\sigma$-even pair mode carries rung $n = 1$ (survival $\frac{2}{3}$), the $\sigma$-odd mode rung $n = 2$ (survival $\frac{1}{3}$) — each clock step flips the deck parity; the structural home of the v486 parity assignment ($\omega_1$ even-channel, $\omega_2$ odd-channel). [E] FORCING: faithful ladder (no rung missing, no physical mode on the wall — definitional for a full-spectrum generator) + $\mathbb{Z}_2$ equivariance + rates $\geq 0$ force UNIQUELY both the split $(\frac{1}{2}, \frac{1}{6})$ AND the assignment even → rung 1, odd → rung 2 (the swap needs $h = -\frac{1}{6} < 0$); the 'sheet-symmetric choice' candidate is superseded by this proof. [E] BEND COROLLARY: $\omega_1/\omega_2 = \text{rate}(2)/\text{rate}(1) = \log_{3/2} 3$ — the parity-comb tone ratio IS the v107/v124 bend of the pulsar-clock search. [O] the channel count $3 = N_{\text{fam}}$ is forced by the clock wall; the ladder stays an exact identity on frozen spectra (semiclassical derivation R1 open); the arity $\{2,3\}$ stays the anchor input. Exact (sympy); mirrorable, deferred to the next verified engine run. [E][O] |

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| Script | What it machine-checks *(continued)* |
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| `v488_majorana_clebsch_door.py` | `FLAV.NUSCALE.04`: the 3/5 CLEBSCH DOOR of the seesaw rung, ENUMERATED AND CLOSED — v482's named escape (a third-generation Majorana factor $r \approx 1.6696$, 0.18% from $g_{\rm car}/N_{\rm fam} = 5/3$) is decided NEGATIVE at the group-theory level; the v481 window candidate stands as the honest endpoint. [E] NO RENORMALISABLE CHANNEL: $\mathrm{sym}^2(\mathbf{16}) = 136 = \mathbf{10} + \mathbf{126}$; the $\overline{\mathbf{126}}$ is NOT in the $E_8$ hull $\{1, 10, 16, 45\}$ (v247) — $M_R$ only via the $d$=5 operator $(\mathbf{16}_F\overline{\mathbf{16}}_H)^2/\Lambda$ with the $E_8$-allowed $\mathbf{16}_H$. [E] CHANNEL LIST: $\mathbf{16} \times \overline{\mathbf{16}} = \mathbf{1} + \mathbf{45} + \mathbf{210}$; the hull admits singlet + **45** insertions (**210** absent). [E] CHARGE TABLE: $Y = T_{3R} + (B{-}L)/2$ on all 16 states, $\mathrm{Tr}_{16}\, Y^2 = \frac{10}{3}$ (v485); $(B{-}L, T_{3R}, Y)(\nu^c) = (+1, -\frac{1}{2}, 0)$. [E] THE $\{2,3\}$-SMOOTH THEOREM: every nonzero $\nu^c$ channel weight $\{1, \frac{1}{4}, 1, -\frac{1}{2}, \frac{3}{8}\}$ is $\{2,3\}$-smooth and the multiplicative closure of $\{2,3\}$-smooth rationals never contains $\frac{5}{3}$; the UNIQUE natural 5 of the embedding, $k_Y = \mathrm{Tr}\, Y^2 / \mathrm{Tr}\, T_{3L}^2 = \frac{5}{3}$ (Ginsparg, v470), is a full-multiplet trace whose direction has $Y(\nu^c) = 0$ EXACTLY — structurally decoupled from the Majorana operator. [E] GENERATION-BLINDNESS: the Clebsch is a family-space scalar; $\mathrm{diag}(1, 1, \frac{3}{5})$ cannot arise from group theory. [O] compiler-flavor inventory typed and declined: exact $\frac{3}{5}$'s only in wrong slots ($K$ lepton gen2/gen1; $R + L$ lepton gen3/gen2), no degenerate-base shape, no $M_R$ link (v354/v355). Exact (Fraction/sympy); mirrorable, deferred to the next verified engine run. [E][O] |
| `v489_seam_modular_commutator.py` | `SEAM.CCC.01`: the chiral central charge $c_-$=8 measured DIRECTLY from the bulk ground state of the v367 seam lattice model, via the Kim–Shehab–Kim modular commutator $J(A,B,C) = i\,\mathrm{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) = (\pi/3)\, c_-$ on a 120° pie tripartition (PRL 128 (2022)) — the FIRST ground-state witness for premise (i) of `SEAM.EQUIV.01` ($c_-$=8) INDEPENDENT of both the Chern route (v367 FHS curvature) and the counting route (v378 KLM/det-Cartan): $J$ is read off the entanglement data of the exact BdG ground state alone (Peschel covariance, $W = -2\,\mathrm{artanh}\,\Gamma$). [E] GROUND-STATE PURITY: $\mathrm{eig}(i\Gamma) = \pm 1$ to $6{\times}10^{-15}$. [E] PER-LAYER $c_- = \frac{1}{2}$: $c_-$=0.499998 at $L$=32, $R$=12 (0.499793 at $L$=16 → 0.500000 at $L$=48, dev $-1.8{\times}10^{-7}$). [E] CONTROLS: trivial $M$=3 → 0.000000; orientation flip $M$=−1 → −0.499986. [E] LAYER ADDITIVITY EXACT ($1.4{\times}10^{-14}$): $J$ additive over decoupled layers, so the carrier value is 16× the per-layer value. [E] 16-LAYER CARRIER $c_-$=8=$g_{\rm car}$+$N_{\rm fam}$: direct 16-layer run 7.9967 ($L$=16, tol $10^{-2}$; $L$=24 gives 7.999781), additivity route 8.0000 (tol $10^{-3}$). [E] CLIP INVARIANCE: $\varepsilon$=$10^{-8}$ vs $10^{-12}$ moves $c_-$ by $5{\times}10^{-8}$. [O] HONEST FENCE: pins $c_-$=8 on the LATTICE ground state (third independent witness); does NOT close `SEAM.EQUIV.01` — the continuum scaling limit (MMST v336) stays [O], the $\mu_4$ condensation $SO(16)_1 \to (E_8)_1$ stays the cited KLM step (v378). Numerical (numpy), Python-only by nature. [E][O] |

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| Script | What it machine-checks *(continued)* |
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| `v490_seam_parity_census.py` | `SEAM.MU.01`: the $T^2$ spin-structure fermion-parity census for the v367 seam lattice model — the lattice ground-state witness of "$\mu_{\rm pre}$=4 before condensation" (step 1 of the v378 KLM chain) + the 16-fold-way bridge $\theta_v$=1 with the MEASURED $\nu$=2$c_-$=16 (v489). Ground-state parity per boundary-condition sector $(AA, AP, PA, PP)$, determined TWICE: Read–Green TRIM product AND real-space Majorana Pfaffian sign (Schur), relative to the trivial $M$=3 phase. [E] REAL-SPACE = BLOCH SANITY: twisted spectra match to $9\times10^{-15}$. [E] PER-LAYER CENSUS $(+,+,+,-)$: TRIM ≡ Pfaffian in every sector — 3 even sectors, the $\nu$=1 Ising signature. [E] 16 LAYERS ⇒ $\mu_{\rm pre}$=4: parity$^{16}$=+ in ALL four sectors ⇒ gauged torus GSD=4=$\mu(SO(16)_1)$=$\lvert\det \mathrm{Cartan}(D_8)\rvert$. [E] 16-FOLD-WAY BRIDGE: $\theta_v$=$e^{2\pi i\nu/16}$=1 EXACTLY at $\nu$=2$c_-$=16 (vortex bosonic, condensable); rivals $\nu$=1, 2, 8 excluded (not holomorphically condensable). [E] NON-DISCRIMINATOR FENCE: the fermionic Kitaev–Preskill TEE is identically zero — $\gamma_f$=0.000001 ($M$=1) and 0.000000 ($M$=3) — an invertible fermionic phase has no TEE, so $\gamma_f$ does NOT discriminate $(E_8)_1$ vs $SO(16)_1$ (both live only after gauging). [C] the KLM step $\mu_{\rm post}$=4/2$^2$=1 stays cited (the condensed model is interacting); the numerics pin only $\{c_-$=8, $\mu_{\rm pre}$=4, $\theta_v$=1$\}$ — the premise package of the condensation. [O] the continuum limit (MMST v336); `SEAM.EQUIV.01` stays [O]. Python (numpy + scipy Schur + sympy exact $\theta_v$). [E][C][O] |
| `v491_p2_partition_corollary.py` | `P2.PARTITION.01`: $g_{\rm car}$=5 as a COROLLARY of the four marks — the partition sharpening of v53 (sympy exact, no floats). THE LEMMA: a multiset of exactly 3 POSITIVE INTEGERS summing to 4 is unique — $\{1,1,2\}$ — so $e$=$(e_1,e_2,e_3)$=$(4,5,2)$ and $g_{\rm car}$=$e_2$=5, $\lvert\mathbb{Z}_2\rvert$=$e_3$=2, $N_{\rm fam}$=$e_2$−$e_3$=3 are COROLLARIES, not inputs; consistency closes as a fixed point ($(2^4-1)/5$=3= rank $H^1$=#marks−1). NEGATIVE CONTROLS (every assumption load-bearing): drop positivity → 4 solutions, $e_2$ ambiguous $\{0,3,4,5\}$; sum 3 → $e_2$=3; sum 5 → not unique; 2/4 parts → $e_2\in\{3,4\}/\{6\}$; Mehta–Seshadri RATIONAL weights → $e_2$=$\frac{16}{3}$ resp. $\frac{44}{9}\neq5$ (integrality comes from the Birkhoff–Grothendieck SPLITTING-TYPE typing, not from MS weights). [E] v53 SHARPENING: v53 needs all three targets $(4,5,2)$; the partition needs only $e_1$=4 — same unique $\{1,1,2\}$. [E] SUM RULE: $4\times\frac{N-1}{2}$=4 iff $N$=3; the v115 route $\lvert a\rvert_1$=$\lvert\mu_4\rvert$ tr$(A_0)$=4. INDEPENDENCE CHAIN: #marks = 4 from v216 (P1-side, no $g_{\rm car}$ input); rank $H^1$=3 pure topology (v137/v228); $\lvert a\rvert_1$=4 $g_{\rm car}$-free (par-deg-0/(U), v115). [C] HONEST RESIDUAL: P2 is NOT made axiom-free — the weight-TYPING postulate remains (integrality = splitting exponents; positivity = $h^0$=0, correlate v137 check 1; sum rule "given (U)"); `AX.P2.01` stays a declared axiom; the $N_{\rm fam}$ direction reversal vs `tfpt_constants` is typed as an over-determination check, never double-counted. Exact (sympy); Wolfram-mirrored. [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v492_celestial_z4_orbifold.py` | `CELEST.WP1.01`: WP1 of the celestial-holographic research contract `CELEST.SEAM.01` — the $E_8$ $\mu_4$-glue grading (v128) IS the flat $\mathbb{Z}_4$ monodromy of the glue-equivariant celestial chiral algebra on the $A_3$ ALE space $\mathbb{C}^2/\mathbb{Z}_4$ (Bittleston–Homans–Sharma, `arXiv:2305.09451`, the $\mathbb{Z}_2$/Eguchi–Hanson case; Kronheimer–Nakajima flat connection at infinity). Sympy exact. [E] INNER GRADING: $h$=$(2,2,2,2,2;0^4)$ reads the glue class mod 4 on all 240 roots ⇒ glue = $\mathrm{Ad}(\exp(2\pi i h/4))$, an order-4 torus element = flat $\mathbb{Z}_4$ monodromy; the $A_3$-side detector $h'$=$(0^5;1,1,1,-3)$ reads the SAME class = the diagonal $(1,1)$ of the discriminant group $\mathbb{Z}_4\times\mathbb{Z}_4$ (the v92/v125 Lagrangian glue); $\mathbb{Z}_2$ halfway 120= $\dim SO(16)$, 128= spinor. [E] $\mathbb{C}^2/\mathbb{Z}_4$ IS the $A_3$ singularity $XY$=$Z^4$ (Molien table exact; Hilbert series $(1+t^4)/((1-t^2)(1-t^4))$; conjugation symmetry $N[d][1]$=$N[d][3]$). [E] EQUIVARIANT SECTOR CLOSES: $D[d]$=$60(d+1)$ even / $64(d+1)$ odd — uniform per parity ONLY because $\dim \mathfrak{g}_j$=$(60,64,60,64)$; zero modes ($d$=0) = $\mathfrak{g}_0$ = carrier $D_5\oplus A_3$+Cartan = 60; density 62/248=1/4=1/$\lvert\mathbb{Z}_4\rvert$; EH/$\mathbb{Z}_2$ halfway towers 120/128. [E] CHARACTER: $\Theta_{E_8}=\sum_j \Theta_{C_j}$; shell $1=(52,64,60,64)$ from the LATTICE; the 4 sector characters $\Theta_{C_j}/\eta^8$ sum exactly to $\chi_{(E_8)_1}$=$1+248q+4124q^2+34752q^3$ (v377); sector weights $h_{D_5}$=$(0,\frac{5}{8},\frac{1}{2},\frac{5}{8})$, $h_{A_3}$=$(0,\frac{3}{8},\frac{1}{2},\frac{3}{8})$ = the v92/v125 discriminant form $(5x^2+3y^2)/8$ mod 1; glue diagonal $h$=$(0,1,1,1)$ INTEGER = v125 isotropy = local extension to $(E_8)_1$. [E] CRITICAL CORRECTION: the $A_3$ deck $\mathrm{diag}(i,i^{-1})$ acts on the celestial sphere as $z\mapsto -z$ (projective order 2, the sheet flip), NOT as the order-4 clock $z\mapsto iz$; the clock is the $U(2)$ phase $\mathrm{diag}(i,1)$ (order 4, fixed $\{0,\infty\}$, free $\mu_4$ orbit — all v180 properties; normalises the deck, det =$i$ rotates the holomorphic symplectic form by the $\mu_4$ generator); exact spin bridge (spin clock)$^2$ = deck, spin group $\mathbb{Z}_8$, 8=2$\lvert\mu_4\rvert$ (the $c_3$=$1/(8\pi)$ winding integer). [E] clock-invariant $A_3$ deformation = the 1-parameter family $XY$=$Z^4$+$a_0$; its 4 branch points are ONE $\mu_4$ orbit with cross-ratio 2 (v179 clock orbit; v168/v214 pillowcase $j$=1728). [E] NEGATIVE CONTROLS: false spatial action $\mathrm{diag}(i,i)$ killed three ways (det =$-1\notin SL(2,\mathbb{C})$; Poisson closure fails 14/14; grades unsymmetric); false glue breaks additivity on 3112/6720 root pairs; rigidity: only $\mathrm{Aut}(\mathbb{Z}_4)$ relabelings additive. [C] VERDICT B: match under the ONE named identification clock$^2$=deck on the sphere (sheet-$\mathbb{Z}_2$ spin bookkeeping). [O] structural arithmetic ONLY — no celestial duality constructed; `CELEST.SEAM.01` (WP2–WP5) stays a research contract; `SEAM.EQUIV.01` stays [O], untouched. Exact (sympy). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v493_celestial_wp2_clock_deformation.py` | `CELEST.WP2.01`: WP2 of the celestial-holographic research contract `CELEST.SEAM.01` — the clock-invariant deformation $XY{=}Z^4{+}a_0$ of the $A_3$ seam orbifold: geometry, periods/monodromy, deformed algebra (Bittleston–Homans–Sharma $\mathbb{Z}_2 \to \mathbb{Z}_4$ transfer, `arXiv:2305.09451`). Sympy exact. [E] GEOMETRY: $P(iZ){=}P(Z)$ forces $a_3{=}a_2{=}a_1{=}0$ SHARPLY (two-sided: $\prod_k(Z - i^k w){=}Z^4{-}w^4$); smooth iff $a_0{\neq}0$ (disc $=256\,a_0^3$); branch points $Z^4{=}{-}a_0$ = ONE free $\mu_4$ orbit in every fibre; cross-ratio 2 / $j{=}1728$ for ALL $a_0$ (binary-quartic invariants $I{=}12a_0$, $J{=}0$ identically) $\Rightarrow$ $a_0$ is a PURE SCALE, the $\tau{=}i$ pillowcase (v168/v214) is FROZEN — no shape modulus. [E] PERIODS: $\Pi{=}t(i{-}1)(1, i, i^2)$, all 3 vanishing spheres carry the SAME volume $\sqrt{2}\,t$ (lockstep); the clock on $H_2$ is a COXETER element of $W(A_3)$ (char $x^3{+}x^2{+}x{+}1$, eigenvalues $\{i, -1, -i\}$ = the $A_3$ exponents $1, 2, 3$; $h(A_3){=}4{=}|\mu_4|{=}N_{\text{fam}}{+}1$); NO invariant cycle ($\det(A{-}1){=}{-}4$); the surviving direction is the weight-1 $\chi_1$-Fourier diagonal $(1, i, i^2)$, NOT the naive invariant diagonal; the clock IS the Picard–Lefschetz monodromy of the family ($a_0 \to e^{2\pi i}a_0$ gives the 4-cycle); versal weights $(a_3, a_2, a_1, a_0){=}(1, 2, 3, 0)$ = the REGULAR $\mu_4$ representation, invariant line = $a_0$ alone. [E] ALGEBRA ($\mathbb{Z}_2 \to \mathbb{Z}_4$): mode dictionary $\{w[m, n] : m{\equiv}n \bmod 4\} \leftrightarrow \{X^pY^qZ^r : r{\leq}3\}$; bracket $\{f, g\}{=}{-}4\,\text{Nambu}(XY{-}Z^4{-}a_0)$ anchored at the $a_0{=}0$ orbifold w-sector ($\{X, Y\}{=}16Z^3$, $\{Y, Z\}{=}{-}4Y$, $\{Z, X\}{=}{-}4X$), Jacobi + Casimir exact, uniform clock-degree $-1$; S-algebra corrections exactly LINEAR in $a_0$ at the $\mathbb{Z}_4$ wrap $r{+}r'{\geq}4$, grade-conserving (NO sector leak $\Rightarrow$ the v492 equivariant SDYM($E_8$) sector deforms consistently); $a_0 \leftrightarrow$ the BHS $c^2$ slot (weight 0); matrix factorisations (McKay modules) deform along every 1+3 and 2+2 orbit splitting; the clock fixes exactly the antipodal 2+2 sector (the EH/$\mathbb{Z}_2$ middle node, $4{=}2{\times}2$). [E] NEGATIVE CONTROLS: $a_1 Z \Rightarrow j{=}0$ (hexagonal CM point); pure $a_2Z^2$ stays SINGULAR ($A_1$ node); $a_2{\neq}0 \Rightarrow j{=}1556068/81{\neq}1728$ (exact instance, two routes agree); grading leak explicit ($Z^2{\cdot}Z^2$ picks up grades $\{0, 1, 2\}$). [C] three NAMED identifications (verdict B, not A): $-4$ Nambu = THE $k{=}4$ CCA bracket (BHS prove $k{=}2$); period = root difference (A-series residue normalisation); the seam-scale reading of $a_0$ presupposes the WP1 bridge clock$^2$=deck ($\mathbb{Z}_8$, $8{=}2|\mu_4|$). [C] v216 PRECISION (typed, NO marker move): given the order-4 clock, the clock-invariant deformation freezes the $(2, 2, 2, 2)$ modulus to the square AUTOMATICALLY — a relocation of the same order-4 input (the clock IS the carrier input), not a new derivation; v216 stays [O]. [O] full deformed twisted-sector OPEs, the $W(\mu)$-type scaling limit for $k{=}4$, the spacetime splitting-function check, WP3–WP5; `SEAM.EQUIV.01` stays [O], untouched. Exact (sympy). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v494_cosmo_running_mu_record.py` | `COSMO.RUNNING.01/COSMO.MUDIST.01`: the two 2026-07-20 prediction-of-record rows promoted to the suite — the SAME Starobinsky branch as $n_s, r, A_s$ (v7/v28/v86) read one derivative deeper and one observable further. **[E]** LO IDENTITIES (sympy exact): $\alpha_s = \mathrm{d}n_s/\mathrm{d}\ln k = -2/N_\star^2 = -r/6$, $\beta_s = -4/N_\star^3$ (NOT $-8/N^3$); LO band $N_\star \in [50, 60]$: $[-8.0, -5.6]\times10^{-4}$. **[E]** EXACT SLOW ROLL (third-order Lyth–Riotto + finite-difference cross-check $< 1\%$): $\alpha_s(51.4) = -7.09\times10^{-4}$, $\beta_s(51.4) = -2.70\times10^{-5}$; exact band edges $-7.48/-5.23\times10^{-4}$. **[C]** DATA: Planck 2018 ($-0.0045\pm0.0067$) $\to +0.57\sigma$; record leg P-ACT-LB ($+0.0062\pm0.0052$) $\to -1.33\sigma$ (watch channel: positive central value); $\beta_s$ run+runrun $\to -0.77\sigma$. KILL: robust $\alpha_s < -5\times10^{-3}$ or $> +3\times10^{-3}$ at $\geq 5\sigma$; any robust POSITIVE running kills the plateau branch. **[C]** $\mu$-DISTORTION (Chluba window, $+21\%$ calibrated on the $\Lambda$CDM anchor): sharp branch $N_\star = 51.4$ ($A_s = 1.76\times10^{-9}$) $\mu = 1.6\times10^{-8}$; profiled record $N_\star = 56.1$ $\mu = 1.94\times10^{-8}$; band $1.5$–$2.3\times10^{-8}$; branch split $\Delta\mu = 3.0\times10^{-9}$ (ratio 1.23) $\Rightarrow \geq 3\sigma$ $A_s/N_\star$ branch decision at $\sigma(\mu) = 10^{-9}$ (Voyage-2050 class); the PIXIE baseline $9\times10^{-8}$ does NOT reach ($0.18\times$ limit). KILL: robust $\mu < 0.9\times10^{-8}$ or $> 4\times10^{-8}$. **[E]** COMB BLINDNESS (honest S15): the frozen log-comb ($\omega = 2.5827$, $\varepsilon = 0.0173$) is structurally invisible in the $\mu$ window — $\lvert F(\omega)\rvert = 0.033$, $\delta\mu = 7.4\times10^{-12} = 0.007\sigma$ even at Voyage-2050. **[O]** FENCES: $N_\star$ stays an external reheating input (`GATE.NSTAR.01`); the frozen band $[50, 60]$ stays the surface of record. Python (sympy exact LO + numpy SR/quadrature), Python-only by nature. **[E][C][O]** |

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| Script | What it machine-checks *(continued)* |
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| `v495_celestial_wp3_gs_alignment.py` | `CELEST.WP3.01`: WP3 of the celestial-holographic research contract `CELEST.SEAM.01` — the Green–Schwarz axion coefficient $\lambda_{\mathfrak{g}}$ of celestial/twistorial SDYM (Costello, `arXiv:2111.08879`) against the TFPT seam constant $c_3{=}1/(8\pi)$; the pre-registered "alignment test only". Exact (Fraction/sympy, no floats). [E] OKUBO IDENTITY DERIVED, NOT QUOTED: $\sum_\alpha\langle\alpha,x\rangle^4=\frac{5}{2(\dim\mathfrak{g}+2)}\,(\sum_\alpha\langle\alpha,x\rangle^2)^2$ as an exact POLYNOMIAL identity for ALL 8 algebras on Costello's list ($\mathfrak{sl}_2,\mathfrak{sl}_3,\mathfrak{g}_2,\mathfrak{so}_8,\mathfrak{f}_4,\mathfrak{e}_6,\mathfrak{e}_7,\mathfrak{e}_8$; $\mathfrak{e}_6/\mathfrak{e}_7$ carved out of the v492 $E_8$ root set) — "no independent quartic Casimir"; NEGATIVE CONTROL: $\mathfrak{sl}_4$ (which HAS one) fails the proportionality (5/32 vs 3/32) — the list is sharp. [E] Killing $\sum_\alpha\langle\alpha,x\rangle^2=2h^\vee\langle x,x\rangle$ exact; Deligne dimension formula $\dim\mathfrak{g}=2(5h^\vee{-}6)(h^\vee{+}1)/(h^\vee{+}6)$; closed form $\tilde\lambda^2=10{h^\vee}^2/(\dim{+}2)=h^\vee{+}6$ EXACTLY $(8,9,10,12,15,18,24,36)$ — fixes Costello–Paquette eq. (1.5); Dynkin indices $i_R=(1,1,2,2,6,6,12,60)$ from explicit weight systems; $E_8$: $\tilde\lambda{=}6$, $\lambda_{\rm fund}{=}1/10$ EXACT; cross-checks $\mathfrak{sl}_2{=}8$, $\mathfrak{sl}_3{=}9$; SO8 SLIP FLAGGED: Costello's own weight content gives $\lambda^2(\mathfrak{so}_8){=}3$ (Okubo-consistent 12/4), the printed 3/2 is a factor-2 slip. [E] ALIGNMENT: $\kappa=\lambda_{\mathfrak{g}}/(8\pi\sqrt{3})\Rightarrow$ $(\kappa/c_3)^2=\lambda^2/3=12=|\mu_4|N_{\rm fam}$ (unit-trace) resp. 1/300 (adjoint-trace), both EXACT anchor rationals; $\kappa/c_3$ ITSELF is IRRATIONAL ($2\sqrt{3}$ resp. $1/(10\sqrt{3})$) — the pre-registered "$\lambda/\sqrt{3}$ rational?" reading FAILS, only the SQUARED alignment holds; anchor decomposition $h^\vee{=}30{=}|\mathbb{Z}_2|N_{\rm fam}g_{\rm car}$ (all three atoms once), $\varphi(30){=}8{=}\,$rank, $\dim{+}2{=}250{=}2\cdot5^3$, $\tilde\lambda{=}6{=}|\mathbb{Z}_2|N_{\rm fam}$, $\lambda_{\rm fund}{=}1/(|\mathbb{Z}_2|g_{\rm car})$. [E] LOOK-ELSEWHERE (the HARD FENCE — alignment $\neq$ selection): the alignment test AS SUCH passes 8/8 (ZERO selective power), $\{2,3,5\}$-smoothness 8/8, $\tilde\lambda$-integrality 2/8 (shared with $\mathfrak{sl}_3$), $\varphi(h^\vee){=}$rank 4/8; the ONLY $\mathfrak{e}_8$-selective single test is $g_{\rm car}{=}5\mid h^\vee$ (1/8); the isolating conjunction was assembled POST HOC — it quantifies where $\mathfrak{e}_8$ sits, it is NOT a preregistered discriminator. [E] KILL-TEST K3 does NOT fire (exact alignment EXISTS) but its scope is DEMOTED: the $c_3$-connection survives as convention-level compatibility + genuine $\lambda$-arithmetic, NOT as $\mathfrak{e}_8$-selective evidence. [C] TYPED: the $8\pi$ is the $2\pi$ convention of the anomaly polynomial (Costello's A.0.10 constant descended to spacetime), the $\sqrt{3}$ the box diagram's $\sqrt{48}=4\sqrt{3}$ — WP3 is an alignment yes/no ONLY, it can never derive $c_3$ from celestial data (contract wording, honoured). [O] WP5 (constructive holography) and `SEAM.EQUIV.01` stay [O], untouched. Verdict B. Exact (Fraction + sympy). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v496_celestial_wp4_character_shadow.py` | `CELEST.WP4.01`: WP4 of the celestial-holographic research contract `CELEST.SEAM.01` — the $(E_8)_1$ character $E_4/\eta^8$ against the glue-equivariant $E_8[\mathbb{C}^2]$ S-algebra: the contract's "type mismatch, head-on". Exact integer/Fraction/sympy, no floats. [E] BASE: v492 replication ($240 = 52{+}64{+}60{+}64$, dims with Cartan $(60, 64, 60, 64)$); equivariant tower $D[d] = 60(d{+}1)$ even / $64(d{+}1)$ odd; cumulative generator count QUADRATIC — $\sum_{d\le s} D[d] = 31s^2{+}92s{+}60$ (even) with leading $31 = k{+}h^\vee$ (the Sugawara denominator, an arithmetic note); [C] CP Table 1 (`arXiv:2201.02595`): $J[m, n]$ carries 2d spin $1-(m{+}n)/2$, UNBOUNDED BELOW along the tower ⇒ no rational VOA ($L_0$ bounded below) contains the towers as graded pieces. [E] HILBERT SERIES: $\chi_{\rm sp}(t) = (60{+}128t{+}60t^2)/(1{-}t^2)^2$ (palindromic numerator, value 248 at $t{=}1$); ALL $\mu_4$ collapses $w \in \{1, -1, i, -i\}$ and the $SU(2)_+$ primary slice $(60{+}64t)/(1{-}t^2)$ fail to start $(1, 248, \dots)$ — the "row sum" route fails at the first coefficient. [E] CHARACTER SIDE: $\chi_{(E_8)_1} = E_4/\eta^8 = (1, 248, 4124, 34752, 213126, 1057504, 4530744)$ exact through $q^6$ (machinery cross-check $qj = E_4^3/\eta^{24} = (1, 744, 196884, 21493760)$); theta split (shell $1 = (52, 64, 60, 64)$), sector currents $(60, 64, 60, 64)$, glue-diagonal $h = (0, 1, 1, 1)$ INTEGER (v125 isotropy = locality); Sugawara $c = 248/31 = 8 = 2|\mu_4| = g_{\rm car}{+}N_{\rm fam}$. [E] BRIDGES: zero-mode slice $D[0]{=}60{=}$ the $C_0$-sector current count EXACTLY; the single-line jet slice $z_1^d$ cycles $(60, 64, 60, 64)$ sector by sector; BOUNDARY/PERIOD LIMIT: one full $\mu_4$ period of loop energies carries $\dim \mathfrak{g}_{n \bmod 4} = (64, 60, 64, 60)$, sum = 248 EXACTLY — the single-particle level of the $248q$ stage; 248 is NOT a jet-tower dimension ($d \le 100$) — only the period collapse produces it; BUT the loop Fock at integer level 1 counts $897266 \gg 248$: the rational truncation is NOT in the jet/loop algebra. [E] MISMATCH LOCALISED THREE WAYS: (a) GRADING (spin unbounded below; 248 no tower dimension); (b) GROWTH: jet Fock $f = (1, 128, 8436, \dots)$ vs $\chi = (1, 248, 4124, \dots)$ TWO-SIDED ($f_1{<}\chi_1$ undercount, $f_2{>}\chi_2$ overcount), $f_n/\chi_n$ strictly increasing $n = 1..12$; Meinardus poles $s{=}2$ vs $s{=}1$ ⇒ growth exponents $n^{2/3}$ vs $n^{1/2}$, asymptotically DISJOINT; (c) TRUNCATION: the level-2 null ideal deletes EXACTLY $27000 = 30^3 = (h^\vee)^3$ ($\mathrm{Sym}^2(248) = 1{+}3875{+}27000$ [[C] standard]; the character keeps $4124 = 1{+}248{+}3875$) — NO analogue in the jet algebra (level 1 agrees: $248 = 248$). [E] KILL-TEST K4, three-way: (i) exact realisation NO; (ii) boundary-limit realisation PARTIAL YES — current stratum, period sum 248, integer glue weights and theta split hold exactly; the missing piece is precisely the rational truncation, which must come from a scaling/compactification limit — the SAME structure as the MMST scaling-limit route of `SEAM.EQUIV.01` (v336/v449: the rational net exists in the LIMIT of the lattice collar, not in the pre-limit algebra); (iii) the hard mismatch is exactly located; K4 does NOT degrade the contract to "$E_8$ admissible" (the sector arithmetic holds exactly). [C] VERDICT B(ii): the $(E_8)_1$ character is a BOUNDARY/LIMIT SHADOW of the equivariant S-algebra, not a conformal block of it in the bulk grading — WP4 sharpens into a statement of `SEAM.EQUIV.01` type. [O] the constructive boundary limit itself (which limit, which topology, convergence) stays OPEN and belongs to WP5/`SEAM.EQUIV.01`; `SEAM.EQUIV.01` stays [O], untouched. Exact (integer/Fraction/sympy). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v497_celestial_wp5a_null_ideal.py` | `CELEST.WP5A.01`: WP5a of the celestial-holographic research contract `CELEST.SEAM.01` — "the null ideal from the limit", the first constructive milestone of WP5 (the v496 boundary-limit shadow made precise). Exact integer/Fraction arithmetic, no floats. [E] BASE: v492/v496 replication (glue dims $(60, 64, 60, 64)$, equivariant tower $D[d]$, $\chi_{(E_8)_1} = E_4/\eta^8$ exact through $q^6$, $\mu_4$ period sum 248). [E] STABILISATION THEOREM: a one-parameter family $\chi_w$ of graded Fock characters on the CHIRAL jet generators (loop energy $m$, radial number $r$; $E_w = m + wr$ in quarter units $u = q^{1/4}$); $w$=2 IS the chiral jet grading (generator counts $64, 120, 128, 180, 192, 240, 256, 300$ per degree); the $u^n$ coefficient of $\chi_w$ equals the quarter-moded loop Fock for ALL $w \geq n{+}1$ and is STRICTLY larger for every $w \leq n$ ($n \leq 8$, $w \leq 10$): the boundary reading is the precise coefficientwise limit $\varepsilon = 1/w \to 0$ with explicit threshold $w = n{+}1$ — a limit of a series, not a slice; convention clarified: 897266 is the quarter-moded loop Fock at $u^4$ = INTEGER level 1 (v496). [E] NULL IDEAL DERIVED, NOT CITED: Freudenthal multiplicities (unit-tested on the adjoint) + Weyl dimension formula + generic character peeling in the standard $E_8$ frame DERIVE $\mathrm{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$ (peeling residual exactly zero; independent Weyl-orbit cross-check); $27000 = 30^3 = (h^\vee)^3$ exactly; $C_2 = (124, 96, 0)$, Dynkin indices 6750/750; level-1 integrability $\langle 2\theta, \theta^\vee \rangle = 4$, $\langle \omega, \theta^\vee \rangle = 2 \Rightarrow$ the only level-1 primary is the vacuum ([C] Kac, used through its counting consequence); level-2 quotient $31124 - 27000 = 4124 = 1 + 248 + 3875$ EXACTLY. [E] TWO ROUTES, ONE NUMBER: the $\mu_4$ theta-split sector characters summed at $q^2 = (1036, 1024, 1040, 1024)$ give the SAME 4124 — the independent lattice route hits the Fock quotient exactly; at $q^3$: $34752 = \chi_3$. [E] NEGATIVE CONTROLS: $SO(16)_1$ through the SAME pipeline — $\mathrm{Sym}^2(120) = 5304 + 1820 + 135 + 1$ derived (FOUR components vs $E_8$'s three), $5304 \neq 14^3 = 2744$ (the $h^{\vee\,3}$ identity is NOT generic), quotient $7380 - 5304 = 2076 = \Theta_{D_8}/\eta^8$ at $q^2$ (the pipeline validated on a second algebra), block weights $(0, \frac{1}{2}, 1, 1)$ do NOT fuse into one local character (four-block theory) vs $E_8$'s integer $(0, 1, 1, 1)$; false periodisation: only $P = 4 = |\mu_4|$ gives 248 ($P$=1: 64, $P$=2: 124, $P$=8: 496); swapped glue breaks the zero-mode anchor ($64 \neq 60$). [C] named identifications: "radial decoupling = MMST collar thickness $\to 0$" (the v336/v449 lattice-collar route has the same two-step shape), not a computed isomorphism; the Kac singular vector only via its counting consequence. [O] HONEST LIMIT: the limit does NOT generate the truncation — the singular vector as an OPERATOR in the limit net stays open (WP5b); the GNS limit state (kernel $\supseteq$ ideal) and local normality / Haag duality of the limit net stay open (WP5c/WP5d); the celestial route now has the SAME two-step shape as MMST (limit + maximal ideal), with the ideal quantitatively fixed; `SEAM.EQUIV.01` stays [O], untouched. Exact (integer + Fraction). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v498_celestial_wp5b_singular_vector.py` | `CELEST.WP5B.01`: WP5b of the celestial-holographic research contract `CELEST.SEAM.01` — "the singular vector as an operator", the second constructive milestone of WP5 (v497 fixed the ideal's size and location; WP5b constructs the deleting object). Exact integer/Fraction arithmetic, no floats, deterministic. [E] CONSTRUCTION: $\|s\rangle = (E^{\theta}_{-1})^2\|0\rangle$, $\theta$ the highest root, weight $2\theta$ (the top of the 27000), level $2 = 8$ quarter units $= q^2$ (an INTEGER level); the Chevalley/Frenkel–Kac basis is MACHINE-built (bimultiplicative 2-cocycle, asymmetry on all 57600 pairs; the $[e_\alpha, e_{-\alpha}]$ sign FORCED by Jacobi, SGN $= -1$; $\kappa$ DERIVED from invariance, $\kappa(\theta^\vee, \theta^\vee) = 2$; Jacobi exact on all 57600 $\mathfrak{sl}_2$-slice triples + 6000 seeded + all 496 $\theta$-adjacent; all $N_{\alpha\beta} = \pm 1$). [E] SINGULARITY: $J^a_1\|s\rangle = 0$ for ALL 248 basis elements (0 violations); case tally 190 (first bracket) / 57 (second) / 1 ($a = F^\theta$ via the central-term cancellation — the ONLY case that sees the level $k$); plus $J^a_2\|s\rangle = 0$ (248), $E^a_0\|s\rangle = 0$ (120 positive roots), Shapovalov norm 0 at $k$=1 (4 at $k$=2); the affine PBW engine unit-tested on all 61504 basis pairs. [E] LEVEL DIAL: the $F^\theta_1$ coefficient is $2(1-k)$ — 2 at $k$=0, 0 EXACTLY at $k$=1, $-2$ at $k$=2: without the central extension the deletion object does NOT exist. [E] $\mu_4$ DATA: $\theta$-grading dims $(1, 56, 134, 56, 1)$, $\mathfrak{g}_0 = \mathfrak{e}_7 \oplus \mathfrak{u}(1)$; the $h$-adapted chamber machine-built (Cartan det 1, $E_8$ Dynkin diagram, branch lengths $[1, 2, 4]$); $\langle\theta, h\rangle = 5 \Rightarrow j(\theta) = 1$, height $29 = h^\vee - 1$; $\|s\rangle$ in class 2 (sheet-EVEN, consistent with WP5a); clock phase $i^2 = -1$; cross-table $\theta$-grading $\times$ glue class consistent (rows $(1, 56, 126, 56, 1)$, columns $(52, 64, 60, 64)$); the deleting $\theta$-$\mathfrak{sl}_2$ crosses the sheet-ODD classes $(1, 3)$, its square is sheet-even. [E] LEVEL-2 MATCH: mult$(2\theta) = 1$ in the level-2 Fock $\Rightarrow U(\mathfrak{g})\|s\rangle$ is THE 27000; the direct $\mathfrak{g}_0$-orbit BFS reproduces the Freudenthal multiplicities of $V(2\theta)$ EXACTLY through depth 4; quotient $31124 - 27000 = 4124$; lattice reading $(2\theta, 2\theta)/2 = 4 > 2 \Rightarrow$ no momentum-$2\theta$ state at $q^2$ in $E_4/\eta^8$ — $\|s\rangle$ IS the kernel generator of Fock $\to E_4/\eta^8$. [E] NEGATIVE CONTROLS: $E^\theta_{-1}\|0\rangle$ NOT singular (one witness $F^\theta$, $\to -k\|0\rangle$) — the 248-current state that must REMAIN ($\chi_1 = 248$); 4 generic level-2 states non-singular (14–33 witnesses); $k$-dial: at $k$=2 the CUBE is singular (the $(E^\theta)^{k+1}$ mechanism = generic Kac level mechanics); $D_8/SO(16)_1$: the singular vector exists there TOO (level-1 generic) but the $h = \frac{1}{2}$ blocks do not fuse (3 comark-1 weights, $\dim V(2\theta_{D_8}) = 5304 \neq 14^3$) vs $E_8$ (comarks $\geq 2$, vacuum the ONLY level-1 primary, glue weights $(0, 1, 1, 1)$ integer) — the ONE-BLOCK closure is the $E_8/\mu_4$-specific part, honestly separated from the generic singular-vector mechanics. [C] Weyl complete reducibility beyond BFS depth 4; the Kac mechanics beyond level 2; the per-period quarter dictionary (v496). [O] HONEST LIMIT: in the twisted quarter-SLOT moding of the $\chi_w$ limit Fock two sector-$C_1$ modes never sum to 8 quarters (minimum $6 = q^{3/2}$) — the per-PERIOD dictionary (v496), not the per-slot identification, carries $\|s\rangle$ to $q^2$; exactly the precise WP5c question (GNS limit state, kernel $\supseteq$ ideal); WP5c–e stay [O]; `SEAM.EQUIV.01` stays [O], untouched. Exact (integer + Fraction). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v499_p2_weights_deligne_bg.py` | `P2.TYPING.01`: the P2 weight-typing postulate HARDENED (the v491 check-9 residual) — the anchor $a$=(1, 1, 2) typed as the Birkhoff–Grothendieck splitting exponents of the DELIGNE CANONICAL EXTENSION $E$ of the flavor connection; sympy exact, no floats. Given the [E] anchors (4 marks v216, rank 3 v137, cusp class + irreducibility v117, (U)): $\deg E = -4$ AS A THEOREM (cusp class ⇒ canonical residue exponents $\{0, \frac{1}{3}, \frac{2}{3}\}$, the unique lift in $[0, 1)$; trace 1 per mark × 4 marks; par-deg 0; $\infty$ APPARENT, monodromy product = **1** exactly); POSITIVITY AS A THEOREM (irreducibility EXACT: $\lvert\langle U, M_0\rangle\rvert$=24=$\lvert S_4\rvert$, $\sum_g \lvert\chi\rvert^2$=24; with (U) ⇒ Mehta–Seshadri stability ⇒ $h^0(E)$=0, all 324 section-saturation cases; BG dictionary $h^0$=0 ⇔ $a_i \geq 1$); INTEGRALITY = Birkhoff–Grothendieck, automatic once $E$ is typed as a bundle. UNIQUENESS + SELECTOR: {1, 1, 2} unique (v491); the Schur–Horn permutohedron of $4 \cdot \mathrm{spec}(A_0)$ has integer points exactly $\{(2, 2, 0), (2, 1, 1)\}$ (the two options of the "Global computation" theorem) and $h^0$=0 kills {0, 2, 2} (an $\mathcal{O}(0)$ summand has $h^0$=1) — the "balanced" selection is now TYPED as stability/positivity. THE $h^1$ DISCREPANCY, REAL + RESOLVED: $h^1(\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2) = 1 \neq 3$ — the naive "generations = $H^1(E)$" typing is FALSE (forcing it demands deg −6 and loses uniqueness: {1, 1, 4}, {1, 2, 3}, {2, 2, 2}, $e_2$ never 5); the correct TWO-OBJECT typing: $L$=$\mathcal{O}(-D)$, $D$=$\mu_4$, carries the generations ($h^1(L)$=3=$N_{\mathrm{fam}} \cong H^0(\Omega^1(\log D))$, the v137 basis), while $E$ carries them as its FIBER (rank 3) with the anchor as splitting type. BISWAS EXPLICIT (independent convergence witness): the weight denominator is 3=$N_{\mathrm{fam}}$, so the Biswas cover is the $\mathbb{Z}_3$ cusp cover $y^3$=$(z-1)(z-i)(z+1)^2(z+i)^2$, genus 2 (NOT the pillowcase double $w^2$=$z^4-1$, genus 1, $j$=1728 — the deck side); $p_*\mathcal{O}_Y = \mathcal{O} \oplus \mathcal{O}(-2)^2$ ($\chi$ cross-check −1= − 1); the regular-representation model bundle lands on {0, 2, 2} = the UNSTABLE Schur–Horn companion ($h^0$=1): integrality AND the sum rule COME OUT of the correspondence, and stability (irreducibility, v117) is the selector. NEGATIVE CONTROLS: drop positivity → 4 resp. 31 solutions; deg −3/ − 5 breaks; $n$-mark scan: $\{1, \ldots, 1, 2\}$ unique for EVERY $n \geq 3$ but $e_2 = (n-2)(n+1)/2 = 5$ ONLY at $n$=4 — the VALUE, not the uniqueness, is the $n$=4 content; MS-rational exponents $e_2 = \frac{16}{3}, \frac{44}{9}$. [C] RESIDUALS: R1 the module identification "generation space = this $H^1$ with this connection" [C] (`QGEO.SYM` territory, stays [O]); R2 unitarity (U) [C] (v33/v40); R3 the analytic-monodromy identification (fine type Numerical, v117 check 4). P2 stays an axiom (`AX.P2.01` declared, no marker moves); NON-CIRCULAR: neither v23's splitting-type input nor v115's diag($A_0$)=$a$/4 input is used. Sharpens v491. Exact (sympy); Wolfram-mirrored. [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v500_celestial_wp5c_gns_limit_state.py` | `CELEST.WP5C.01`: WP5c of the celestial-holographic research contract `CELEST.SEAM.01` — "the GNS limit state", the third constructive milestone of WP5 (v497 made the boundary limit precise, v498 built the deleting operator; WP5c constructs the STATE whose GNS kernel contains the ideal). Exact integer/Fraction arithmetic, no floats, deterministic. **[E]** CONSTRUCTION: the quasi-free family $\omega_w$ — loop sector ($r$=0) = the affine $k$=1 vacuum $n$-point functions via the MACHINE-determined compact anti-involution $\theta(e_\alpha)$=$-e_{-\alpha}$, $\theta(h)$=$h$ (the UNIQUE sign that is an anti-automorphism on ALL 61504 basis pairs; contravariance $\langle J^x_{-1}u, v\rangle = \langle u, J^{\theta x}_1 v\rangle$ exact); radial sector ($r\geq 1$) = oscillator pairings $x^{wr}$ (the exact Gibbs regulator $e^{-\beta E_w}$, kept $x$-adic). **[E]** CONVERGENCE, SHARP: $\omega_w = \omega_\infty \bmod x^{N+1}$ for ALL $w \geq N{+}1$ ($N$=8) and NOT yet at $w$=$N$ (the $r$=1 modes sit at order $N$) — the WP5a threshold $w = n{+}1$ holds IDENTICALLY at the state level; positivity of every $\omega_w$ checked. **[E]** THE KERNEL (core, fully machine): the FULL exact level-2 Gram — 31124 monomials in 9361 weight blocks (dim profile 164:1, 37:240, 7:2160, 1:6960), every block exact: level-1 rank 248 POSITIVE DEFINITE (Cartan block = the $E_8$ Cartan matrix, det 1); level-2 rank **4124**, kernel **27000**, every block PSD; rank per Weyl orbit (norm 8/6/4/2/0) = $(0, 0, 1, 8, 44)$; kernel = $V(2\theta)$ WEIGHT BY WEIGHT (Freudenthal cross-check on all 9361 weights); highest-weight norms $(3844, 49, 2, 0)$ at $(0, \omega, \theta, 2\theta)$; direct ideal checks $F_i|s\rangle$, $F_jF_i|s\rangle$ norm 0. **[E]** $\mu_4$-EQUIVARIANCE: $\omega_w \circ \text{clock} = \omega_w$ for ALL $w$; the clock unitary DESCENDS to GNS($\omega_\infty$); level-2 GNS rank split by clock class $= (1036, 1024, 1040, 1024) = \Theta_{C_j}/\eta^8$ at $q^2$ (the two-routes identity at the STATE level); level-1 split $(60, 64, 60, 64)$. RESOLUTION OF WP5b S4.10: $|s\rangle$ IS the ZERO VECTOR of GNS($\omega_\infty$) — the twisted two-$C_1$-mode floor (6 quarters $= q^{3/2}$) concerns only pre-limit objects (GNS has no state there: 6 mod 4 $\neq$ 0). **[E]** NEGATIVE CONTROLS: $k$=2 ($\langle s|s\rangle$=+4, dominant blocks FULL rank — NO ideal in the kernel); $k$=0 (level-1 Gram $\equiv 0$ — no 248 layer without the central extension); $D_8/SO(16)_1$ (level-2 rank 2076 = 7380 − 5304, kernel $= V(2\theta_{D_8})$ PSD weight by weight, but $h = (0, \frac{1}{2}, 1, 1)$: the vacuum GNS is 1 of 4 blocks — ONE-BLOCK completeness stays $E_8/\mu_4$-specific); wrong family $E'_w = mw{+}r$ (level-1 limit rank 0: the 248 layer ERASED, $u^4$ content 710955 $\neq$ 248); no damping (897266 $\neq$ 248); CCR OBSTRUCTION: at a FIXED radial pairing $c > 0$ positivity forces $\omega(bb^*) \geq c$ — NO $w$-uniform damping state exists: the family formulation is NECESSARY and the family EXISTS ⇒ the preregistered KILL is probed and NOT triggered. **[C]** the quasi-free/Gaussian extension to the full $*$-algebra; Kac positivity beyond level 2 (levels $\leq 2$ machine-checked); the module identification in the component route. **[O]** local normality of $\omega_\infty$, the net structure / Haag duality (WP5d), Costello–Li (WP5e), the operator-algebraic closure beyond the graded/$x$-adic topology; `SEAM.EQUIV.01` stays **[O]**, untouched. VERDICT: SUCCESS on the preregistered criterion (dim GNS level-2 = 4124 exactly). Exact (integer + Fraction). **[E][C][O]** |

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| Script | What it machine-checks *(continued)* |
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| `v501_celestial_wp5d_two_interval_index.py` | `CELEST.WP5D.01`: WP5d-$\alpha$ of the celestial-holographic research contract `CELEST.SEAM.01` — "the two-interval index from the lattice" (the local-net question, alpha stage). The roadmap discriminator: $(E_8)_1 \Rightarrow \mu{=}1$ (Haag-dual two-interval net, ONE sector); $SO(16)_1 \Rightarrow \mu{=}4$; preregistered KILL = "the two-interval $\mu$-offset does NOT vanish after the condensation". numpy/scipy Gaussian machinery + exact Fractions. [E] SETUP (machine identities): critical Majorana ring (v450 style, NS closure, $c{=}\frac{1}{2}$ per layer), 16 layers = the seam carrier $c_-{=}8{=}g_{\rm car}{+}N_{\rm fam}$ (v489); Peschel-exact; three prescriptions — fermionic $\rho$, orbifold sector sum $\{1, P_A, P_B, P_{AB}\}$, dual algebra $\{1, P_{AB}\}$; superselection $[\rho, P_{AB}]{=}0$ machine-exact; EVERY Gaussian formula validated against exact diagonalisation ($\leq 1.5{\times}10^{-15}$). [E] FERMIONIC MI EXTENSIVE (the $\mu{=}1$ reference; lattice witness): $I(x) = -(c/3)\ln(1{-}x)$ with $c$ fitted 0.5000; $\max|F(x)| = 5.97{\times}10^{-5}$ at $N{=}512$, $|F|$ falling monotonically to $1.3{\times}10^{-6}$ at $N{=}1024$ — Casini–Huerta extensivity on the lattice. [E] THE $\mu$-OFFSET (core): Rényi-2 exactly $\Delta_2 = -\ln[(1{+}R)/2]$; the 16-layer seam carrier saturates $\ln 2$ to MACHINE PRECISION ($|\Delta_2 - \ln 2| = 1.1{\times}10^{-15}$ at $N{=}512$; $R \sim N^{-0.500}$); index reading $[F : F_{\rm even}] = e^{\Delta_\infty} = 2.000000 \Rightarrow \mu_{\rm gauged} = e^{2\Delta_\infty} = 4.000000$ = the v490 parity census (TWO independent lattice witnesses); gauging every layer separately costs $16\ln 2$ — the offset measures ln[index of the subnet], not a universal blob. [E] HAAG-DUALITY INDICATOR: fermionic $S(E){=}S(E')$ to $8.2{\times}10^{-12}$ (graded-duality proxy); the orbifold BREAKS complementarity $(D(E) - D(E') = 0.427$ at $a{=}64$ — the direct duality-failure witness); the complementary-pair sum is exchange-symmetric and $\leq \ln 4 = \ln\mu(SO(16)_1)$ with max 1.2637 at the self-dual $x{=}\frac{1}{2}$ and $\to \ln 2$ at the squeezed ends: the $\ln 4$ KLM budget is a DOUBLE-LIMIT statement, NOT a fixed-$x$ identity (the first naive fixed-$x$ check was WRONG and was replaced — honestly typed). [E] CONDENSATION ARITHMETIC, LATTICE-ANCHORED: $\det\mathrm{Cartan}(D_5)\cdot\det\mathrm{Cartan}(A_3) = 4\cdot 4 = 16 = \mu$(carrier); KLM/Longo–Rehren quotients $\mu/|L|^2$: $16/4^2 = 1$ (one-step $\mu_4$ glue) and $16/2^2 = 4 \to 4/2^2 = 1$ (via $SO(16)_1$), both routes EXACT; $\Sigma d_i^2$ cross-checks (Ising 4, $SO(16)_1$ 4, $(E_8)_1$ 1); $\theta_v = e^{2\pi i\nu/16} = 1$ EXACTLY at $\nu = 2c_- = 16$, rivals $\nu = 1, 2, 8$ give $\theta_v \neq 1$. The anchored chain: $\mu = 1$ (graded, MEASURED) $\to 4$ (gauged, MEASURED + v490 census) $\to 1$ (arithmetic, [C]: the condensed net is interacting — the v490 fence). [E] NEGATIVE CONTROLS: $\nu{=}1$ single Majorana (offset non-trivial, $\Delta_2 = 0.6364$ at $N{=}2048$ rising, and $\theta_v = e^{i\pi/8} \neq 1$: NOT removable — the kill discriminator has teeth); trivial dimerised phase (everything $\leq 10^{-7}$); wrong sector sum $\{1, P_{AB}\}$ (offset EXACTLY 0 instead of $\ln 2$). [C] the continuum dictionary "offset = ln index" (KLM two-interval index; Longo–Xu; Casini–Huerta–Magan–Pontello entropic order parameters) + the $\theta_v{=}1$ condensation step itself. [O] continuum Haag duality, local normality, split/strong additivity of the quotient net (WP5d-$\beta$, running as exploration), Costello–Li (WP5e); `SEAM.EQUIV.01` stays [O], untouched. VERDICT: the $\mu{=}1$ indicator chain PASSES; the preregistered KILL is NOT triggered. Python (numpy + scipy + Fractions). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v502_celestial_wp5e_prefactor_level.py` | `CELEST.WP5E.ALPHA.01`: WP5e-$\alpha$ of the celestial-holographic research contract `CELEST.SEAM.01` — "prefactor and level bookkeeping", the CFT-side anchor stage of WP5e (the BCOV/Costello–Li twistor uplift: boundary partition function $= E_4/\eta^8$ *including* the $q^{-1/3}$ prefactor; preregistered kill: the anomaly inflow demands a level $\neq 1$). Exact sympy/Fraction. [E] PART A — the $q^{-1/3}$ prefactor is exact $\mu_4$ vacuum-energy bookkeeping: Hurwitz-$\zeta$ derivation ($\zeta(-1,\theta) = -B_2(\theta)/2$; $E_b(\theta) = -\frac{1}{24} + \theta(1-\theta)/4$ exactly); the clock is INNER ($h = (2,2,2,2,2;0^4)$ and the $A_3$ detector $h' = (0^5;1,1,1,-3)$ both read the class mod 4 on all 240 roots; $(h,h) = 20$, $(h',h') = 12$, sum 32) $\Rightarrow$ twist distribution $\theta = 0^8$ in ALL four sectors — a SHIFT orbifold (lattice cosets, integer-moded oscillators), *not* a rotation orbifold; $E_{\rm osc} = -\frac{1}{3} = -c/24$ sector-independent ($c = 8 =$ rank $= 248/31$); sector ground exponents $-\frac{1}{3} + (0,1,1,1)$. [E] discriminant form = Casimir energy, TWO routes: spectral flow $j^2(h,h)/32$ ($5j^2/8$, $3j^2/8$, sum $j^2 \equiv 0$ mod 1 = the isotropy/Lagrangian condition) and *exactly* via free fermions ($D_5 = 10$ Majorana, $A_3 = D_3 = 6$, glue diagonal $= 16 = 2^{g_{\rm car}-1}$, the WP5d seam carrier; R–NS shift $n/16$: $\frac{5}{8}, \frac{3}{8}, 1$); the ROTATION reading fails (the geometric $A_3$ deck gives $\frac{3}{16} \neq \frac{3}{8}$: the v492 "clock $\neq$ deck" lesson at the Casimir level). [E] character: $\sum_j \Theta_{C_j} = E_4$ and the four sector characters (leading coefficients $(1,64,60,64)$, currents $(60,64,60,64)$) sum exactly to $E_4/\eta^8 = q^{-1/3}(1,248,4124,34752)$. [E] PART B — $k = 1$ forced THREE independent ways: (i) the CURRENT condition $h(J) = k = 1$ (the $\mathbb{Z}_4$ extension closes onto the $E_8$ adjoint 60+64+60+64 = 248 only at spin 1); HONEST SHARPENING: glue-$h$ integrality $h(J^j;k) = k\,(0,1,1,1)$ holds for ALL $k = 1..8$ — the originally conjectured integrality route fixes *nothing*, the three other dials carry; (ii) conformal embedding $47k^2{+}219k{-}266 = 47(k{-}1)(k{+}266/47) = 0$ resp. $D_8$-in-$E_8$: $128k(1{-}k) = 0$; (iii) central charge $248k/(k{+}30) = 8 \Leftrightarrow 240(k{-}1) = 0$ (= the prefactor itself); plus the v498 singular-vector dial ($\|(E^\theta_{-1})^m 0\|^2 = m!\,k(k{-}1)\cdots(k{-}m{+}1)$, singular vector exactly at grade $k{+}1$; $k = 1$ deletes $V(2\theta)$: $31124 - 27000 = 4124$). [E] negative controls: $D_8/SO(16)_1$ (same rank, same $c = 8$, SAME $q^{-1/3}$ prefactor, but $h = (0,\frac{1}{2},1,1)$: the discriminator is glue-$h$ integrality, not the prefactor; $\Theta_{D_8} + \Theta_s = E_4$); $\mathbb{Z}_2$ rotation ($E = +\frac{1}{6}$, $h = \frac{1}{2}$: no $\mu_4$-shift sector there); $\mu_4$ rotation (sector-dependent $(-\frac{1}{3}, \frac{1}{24}, \frac{1}{6}, \frac{1}{24})$: the common prefactor breaks); wrong $k \in \{2,3,4\}$ fails all five dials coherently (residuals 360/814/1362; $\langle s|s\rangle = 4/12/24$; level-2 dim stays $31124 \neq 4124$). [C] Kac generation of the maximal submodule by the grade-$(k{+}1)$ singular vector for $k \geq 3$ ($k = 1,2$ machine-anchored by WP5b/WP5c). [O] the BCOV/Costello–Li twistor inflow itself (Costello's one-loop axion $\lambda_g/(8\pi\sqrt{3})$ on $\mathbb{PT}$, the Costello–Paquette–Sharma level-from-flux mechanism) stays open — v502 anchors the CFT side of WP5e only ($\alpha$ stage); the kill branch ("inflow demands a level $\neq 1$") is not triggered on the CFT side but its twistor side is untested; WP5e stays [O]; `SEAM.EQUIV.01` untouched, no marker moves. Exact (sympy + Fraction). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v503_qgeo_emergence_light.py` | `QGEO.EMERGE.LIGHT.01`: the v215 *deferred emergence test* (lever 7) executed in its first honest LIGHT version, plus the residual-convergence result QGEO-R1 ≡ P2-R1 (analysis/kill-test class; NO marker moves — `QGEO.REALIZE.01` stays [C]/[O]). NO DYNAMICAL SELECTION ($\zeta$-log-det via the Dedekind-$\eta$ formula, mpmath 40 digits): $\log F = \log(\tau_2\|\eta\|^4)$ is exactly $S$-invariant, so $t = 1$ is automatically critical on the rectangular family $\tau = it$; there a strict local maximum ($g''(1) = -1.298$) but a *minimum* in the $\tau_1$-direction ($h''(0) = +0.298$): $\tau = i$ is a SADDLE of the full moduli space; critical points = the CM points ($E_2^*(i) = E_2^*(\rho) = 0$ to 40 digits); the *global* det′ winner at fixed area is the HEXAGONAL torus ($\log F(\rho) = -1.03352 > \log F(i) = -1.05469$; Osgood–Phillips–Sarnak 1988) — free-field dynamics alone does *not* select $\tau = i$; criticality at the square is Palais symmetric criticality (the independent heat trace is equally critical). SHARP RIGIDITY (v280 lattice DtN): the mod-4 off-character indicator (the v215-K3 lever as a *modulus function*) has an isolated zero ($8\times10^{-15}$) exactly at $\tau = i$, strictly monotone in both directions (min $8.4\times10^{-2}$ away); $R^2 = \mathbb{Z}_2$ exactly as permutation matrices (the v492 $\mathbb{Z}_8$ bridge on the lattice); $j(it) \geq 1728$ with equality only at $t = 1$, $j(\rho) = 0$. [E] the $\mathbb{Z}_2$ chain is MODULUS-BLIND: 4 fixed points and $[\mathbb{Z}_2, \Delta_t] = 0$ on *every* torus (v216 confirmed without mark input); pillowcase det′ $= \frac{1}{2}$ torus det′ exactly (±-pair structure), so all $\tau$ conclusions transfer verbatim; marks/mark-locality/deck$^2$ exist on *every* pillowcase (v264/v280), only the order-4 clock distinguishes $\tau = i$; $\mathrm{Aut}(E_\rho) = \mathbb{Z}_6$ (element orders $1, 2, 3, 6$) contains *no* order-4 element — dynamics and clock are logically independent selectors; GIVEN the clock, $P(iZ) = P(Z)$ forces $a_3 = a_2 = a_1 = 0$ and $j = 1728$ identically (v493): the modulus is not a second input. [C]/[O] RESIDUAL CONVERGENCE (the core finding): the order-4 element on $H^1$ is a Coxeter element of $W(A_3)$ (char poly $\lambda^3+\lambda^2+\lambda+1$, order $h(A_3) = 4 = \|\mu_4\| = N_{\mathrm{fam}}+1$) — exactly the P2 cusp-class/carrier datum of the P2 hardening (v499 R1) ⇒ QGEO-R1 ≡ P2-R1: ONE common residual, "why does the raw seam carry the order-4 clock", not two scattered ones; chain: marks derived (v216 [E]), square derived *given* the clock (v493 [E]), clock = carrier input [C], rest = the dynamical realisation of the clock [O]; route classification (v347 analogue): B-rigidity in the sharpened form "RP seam data ⇒ order-4 clock" is the live route — free dynamics alone does not deliver the clock (the hexagonal finding). Lever update (documented here, v215 itself unchanged): lever 7 lifted from "deferred, not executable" to "light version executed: rigidity, not dynamical selection". Python (mpmath + numpy/scipy + sympy). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v504_celestial_wp5d_klm_completeness.py` | `CELEST.WP5DB.01`: WP5d-$\beta$ of the celestial-holographic research contract `CELEST.SEAM.01` — "split + strong additivity from the lattice", the two remaining Kawahigashi–Longo–Müger legs of complete rationality (the $\mu$-index is v501), witnessed for the orbifold prescription on the critical Majorana seam edge; the $U(1)$/Dirac subnet is the preregistered negative control (the current net, where strong additivity fails in the continuum). [E] STRONG ADDITIVITY, EXACT: shared boundary Majorana $\Rightarrow \mathrm{Even}(A) \vee \mathrm{Even}(B) = \mathrm{Even}(A \cup B)$ EXACTLY (GF2 spans full 64/64 up to 1024/1024; matrix rank 32/32); DISJOINT: exactly HALF (rank 16/32, index 2 — the missing odd $\otimes$ odd sector is the v501 $\ln 2$ bit, localised at the split point); the neutral $U(1)$ algebras do NOT generate the union even with the shared site (gaps 2/10/52, growing). [E] ENTROPIC WITNESS [N]: the fermionic touching MI carries the pure UV point-gap law ($(c/3)\ln 2$ increments, dev $1.2 \times 10^{-6}$ — no discriminator); the honest discriminator is BOUNDED vs DIVERGENT: the $\mathbb{Z}_2$/orbifold deficit stays $< \ln 2$ with $(\ln 2 - \Delta_2) \sim N^{-p}$, $p = 0.2444$ vs $2\Delta_\mu = \frac{1}{4}$ (Ising disorder dimension; "defect $\to 0$" would be FALSE at the sharp split — bounded $\Leftrightarrow$ finite index, Longo–Xu), while the $U(1)$ control BURSTS $\ln 2$ from $L = 128$ and grows as $\frac{1}{2}\ln \mathrm{Var}\, Q_A$ (Klich–Levitov 0.10134 vs $1/\pi^2$): infinite index. [E] SPLIT/NUCLEARITY [N]: coupling $\sigma_k$ fall at the elliptic-nome rate $\pi K(1-x)/K(x)$ to 1.3–2.0%; $\sigma_1 \sim x^{0.5044}$; trace norm summable; ORBIFOLD INHERITANCE EXACT: $P_A$ flips $C \to -C$ ($\sigma_k$ identical, $8.3 \times 10^{-17}$), the even-bilinear Gram is the compound $\Lambda^2 C$ ($\{\sigma_a\sigma_b\}$, $2.8 \times 10^{-14}$) — Longo heredity [C]. [E] PIMSNER–POPA: $E(a) - a/2 = PaP/2$ IDENTICALLY $\Rightarrow$ $\lambda = \frac{1}{2} = 1/[F : F_{\mathrm{even}}]$; integer attainment (16384 + 2048 monomial sweeps, 0 violations; pencil $\min \lambda = 0.5000000000$); $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ exactly; INDEX CONSISTENCY, two routes: $e^{\Delta_\infty} = 2.000000000 = 1/\lambda_{\mathrm{PP}}$ and $e^{2\Delta_\infty} = 4 = \mu$; $U(1)$: $\lambda = 1/(m{+}1) \to 0$. [E] CONTROLS + HONESTY: dimerised phase trivial ($3.6 \times 10^{-15}$); wrong prescription $\{1, P_{AB}\}$ deficit EXACTLY 0; finite-size honesty (14.9% below $\ln 2$ at $N = 4096$ — witness, not proof). [C] the KLM dictionary (Longo–Xu; Longo heredity; CHMP). [O] the continuum theorem "finite-group orbifolds of completely rational nets are completely rational" is XU'S THEOREM — CITED, NOT CLAIMED; the seam quotient net and the condensed $(E_8)_1$ net stay WP5e/Costello–Li; `SEAM.EQUIV.01` stays [O]. VERDICT: with v501, all THREE KLM ingredients witnessed on the lattice. Python (numpy + scipy); exact parts Wolfram-mirrored. [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v505_celestial_wp5e_beta_equivariant_ledger.py` | `CELEST.WP5E.BETA.01`: WP5e-$\beta$ of the celestial-holographic research contract `CELEST.SEAM.01` — "the equivariant anomaly ledger on twistor space", the twistor-side milestone of WP5e (the BCOV/Costello–Li uplift; preregistered kill: the anomaly inflow demands a level $\neq 1$); WP5e-$\alpha$ (v502) anchored the CFT side. The Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop box anomaly (Costello; Bittleston–Sharma–Skinner) evaluated exactly on $\mathbb{C}^2/\mathbb{Z}_4$ with the $E_8$ $\mu_4$-glue monodromy. Exact sympy/Fraction. [E] AB ledger: denominators $\det(1-g_j^{-1}) = (2,4,2)$; $\sum_j 1/\det_j = 5/4 = (\lvert\mathbb{Z}_4\rvert^2-1)/12$ (Dedekind); equivariant characters $\mathrm{tr}_{g_j}(\mathrm{Ad}\,E_8) = (248,0,-8,0)$ by two routes; invariant average $60$ = the $D_5{+}A_3$ carrier; Frobenius 61568; $\mathrm{tr}_{g_2} = -8 = 120-128$. [E] Okubo per sector (honest sharpening): the invariant sector is exactly Okubo-quadratic ($Q^{(0)} = 36\langle x,x\rangle^2$, $36 = \tilde{\lambda}^2_{e_8}$ — v495 re-derived through the glue grading); the twisted sectors do NOT reduce ($T_5$: $(-8,32,-8)$, $T_3$: $(48,-64,48)$); the AB-weighted sum cancels the $D_5$ quartic exactly and leaves the RIGID residual $32\,T_3$; the only quartic-free weighting is the uniform $\mathbb{Z}_4$ projector; the graded GS exchange is exactly rank-obstructed in sectors $1,2,3$; heterotic identities exact ($\mathrm{Tr}_{120}$, $\mathrm{Tr}_{128}$, $\mathrm{Tr}_{E_8} = 9(\mathrm{tr}^2)^2$). [E] index bridge (centrepiece): $f(m) = \frac{1}{4}\sum_j (i^{jm}-1)/\det_j = (0,-\frac{3}{8},-\frac{1}{2},-\frac{3}{8}) = \mathrm{ch}_2(T_m) = -(C^{-1})_{mm}/2$ EXACTLY — fixed-point ledger = McKay/Kronheimer intersection ledger; glue $\mathrm{ch}_2$ defect $-78$ by both routes (classes $(-24,-30,-24)$; $-30 = -h^\vee$, a [C] coincidence); $\sum_m h_{A_3}(m) = 5/4 = \sum_j 1/\det_j$; Killing sums $(320,320,240,320)$, total $1200 = 2h^\vee(h,h)$. [E] $k=1$ geometric: lattice current count $= 240$ at $k=1$, exactly 0 at $k=2,3,4$; embedding residual $47k^2{+}219k{-}266 = (0,360,814,1362)$; $h(J;k) = k$ integer for all $k$ — integrality alone fixes nothing (honest, as v502); one scale $\Rightarrow$ one level (v493 periods). [E] E3 REFUTED (prominent): $a_0 \in O(8)$ fills the GRAVITON slot $O(2)$, not the BSS axion slot $O(-2)$ ($X = -2 \neq +2 = Y$; mismatch $4 = \lvert\mu_4\rvert$, a [C] coincidence) — "the theory brings its GS axion as $a_0$" is refuted; instead the three $H^2$(ALE) classes carry exactly the Coxeter eigencharacters $\{i,-1,-i\}$ = the three twisted slots (bijection); slot count $1+3 = 4 = \lvert\mu_4\rvert$ soft. [E] negative controls: $\mathrm{diag}(i,i)$ breaks four ways ($\sum 1/\det = \frac{1}{4}$, $\omega$-weight $-1$, 5 generators); $SO(16)$ glue: characters $(120,0,+120,0)$, defect $-30 \neq -78$, bulk Okubo fails (independent quartic Casimir); $k=2$: count 0, prefactor $-31/48 \neq -\frac{1}{3}$. [C] dictionaries ($\mathrm{CS}(\rho)$ mod 1 = discriminant weight; $32 = (h,h){+}(h',h')$; $-30 = -h^\vee$; $\Omega$-contraction). [O] the global BCOV/Kodaira–Spencer quantisation on $\mathbb{PT}/\mathbb{Z}_4$, the $O(-2)$ bulk-axion construction, CPS level-from-flux, the pairing of the $32\,T_3$ residual with the three sphere axions. Verdict B; the kill branch does NOT fire on the equivariant skeleton; `SEAM.EQUIV.01` untouched, no marker moves. Exact (sympy + Fraction). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v506_seam_clock_rigidity.py` | `SEAM.CLOCK.RIGIDITY.01`: clock rigidity, Route B of the v503 classification — two exact theorems reducing the order-4 clock input to ONE alignment bit (Part A Moebius + Part B Fock in one module; `QGEO.REALIZE.01` stays [C]/[O], no marker moves). Exact sympy + exact Cl(16) Fractions. [E] THEOREM A (Moebius): the collar deck $z \mapsto -z$ has in $\mathrm{PGL}_2(\mathbb{C})$ exactly TWO square roots ($z \mapsto \pm iz$), both automatically of order 4 (complete case analysis: $\mathrm{tr} \neq 0 \Rightarrow$ diagonal $\Rightarrow a/d = \pm i$; $\mathrm{tr} = 0$ impossible) — "order 4" is no longer an input; a marks-preserving root exists $\iff$ the deck is the CENTRAL involution of the mark-$D_4$ (center $= \{\mathrm{id}, \mathrm{deck}\}$, exactly 2 order-4 elements = the clocks) $\Rightarrow$ then marks = one $\mu_4$ orbit, cross-ratio class $\{2, \frac{1}{2}, -1\}$, $j = 1728$, $\tau = i$, free 4-cycle, deck = clock$^2$, only the scale free; COUNTEREXAMPLE: $\{\pm 1, \pm(3{+}2\sqrt{2})\}$ is deck-free AND harmonic with full $D_4$, but the deck is non-central (edge involution) $\Rightarrow$ 0 roots — harmonicity alone is NOT sufficient; the RP filter selects nothing ($\mathrm{Aut}(\mathbb{Z}_4)$ ambiguity irreducible, consistent with v492). [E] B3/B4: free orbit $\Rightarrow n \mid 4$; mark transitivity $\Rightarrow n = 4$; faithfulness kills $n \geq 5$; $n = 3$ fixes a mark; $n = 6$ dies at cusp degree 3 (v117); the deck is modulus-blind (v503 B5 exact); controls: $n = 8$ impossible; hexagonal $A_4$ (0 order-4 elements, squares only $\{1, 3\}$); generic $V_4$ (exponent 2); stabiliser trichotomy $V_4/D_4/A_4$ with order-4 counts $(0, 2, 0)$. [E] THEOREM B (Fock, the core result): the unique $S^2$ spin structure gives NS on every circle (established P1 datum); one-particle $D^2 = -1$ (charpoly $(\lambda^2{+}1)^8$), quarter-shift $C^2 = D$, $C^8 = 1$ (charpoly $(\lambda^4{+}1)^4$); Fock: $\tilde{U} = \prod_j (1 - \gamma_j \gamma_{j+8})$, $\tilde{U}^2 = 256\,\gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F$ EXACTLY, and $(cU)^2 = c^2(-1)^F$ is never 1 (center of Cl(16) = scalars) — the $\mathbb{Z}_4$ lift is FORCED (nonsplit extension); $\mathbb{Z}_8$ tower: $V^2 \propto \tilde{U}$, $V^4 \propto (-1)^F$, projective order $8 = 2|\mu_4|$ — fermionically explained; R control $U_R^2 = +1$ (split — the forcing rests on NS = established geometry); 16-fold way $\theta_v = 1$ at $\nu = 16$; the forcing already holds at 8 sites. [C] THE ONE REST: "the collar deck is the central involution of the mark-$D_4$" ($\iff$ square-root existence on the mark data) — ONE alignment bit; the fermionic $\mathbb{Z}_4$ forces the ORDER, not the spatial alignment. No overclaim. Exact (sympy + Cl(16) Fractions). [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v507_seam_bit_origin.py` | `SEAM.BIT.ORIGIN.01`: the tautology attack on the v506 alignment bit, executed and refuted — where do the marks AND the deck come from, and does their common origin force the central alignment? (Part A torus origin + Part B fermion dichotomy in one module; NO marker moves — the bit stays [C], `SEAM.CLOCK.RIGIDITY.01` and `QGEO.REALIZE.01` text-precised only.) Exact sympy + exact Cl(16) Fractions. [E] ORIGIN THEOREM: the 4 marks are the fixed points of the hyperelliptic involution $w \mapsto -w$ on $\mathbb{C}/(\mathbb{Z}+\tau\mathbb{Z})$ = the four half-periods $\Lambda/2\Lambda$; a translation $t_c$ descends to the base sphere iff $2c \in \Lambda$ — exactly the THREE half-period translations (control $c = 1/3$ fails); the descended maps are the EXPLICIT rational involutions $\sigma_i(x) = e_i + (e_i - e_j)(e_i - e_k)/(x - e_i)$ with the exact curve-lift identity $\prod(\sigma_i - e_l) = \psi_i^2 \prod(x - e_l)$, $\psi_i$ RATIONAL; $\{\mathrm{id}, \sigma_1, \sigma_2, \sigma_3\}$ = Klein $V_4$; CONVERSELY, in ALL three stabiliser types the freely-acting involutions are EXACTLY the three $\sigma_c$ (generic $V_4$ $(4,3,3)$, harmonic $D_4$ $(8,5,3)$, hexagonal $A_4$ $(12,3,3)$) $\Rightarrow$ EVERY mark-free collar deck (v216) IS a half-period translation — the CLASS of the deck is forced, its POSITION is not. [E] CORE SOLVE (symbolic $\lambda$): $\sigma_c$ admits a marks-preserving $\mathrm{PGL}_2$ square root $\iff$ $\lambda \in \{-1\}$ $(\sigma_1)$, $\{2\}$ $(\sigma_2)$, $\{\frac{1}{2}\}$ $(\sigma_3)$ — per half-period exactly ONE solution; intrinsically: the two deck pairs separate each other HARMONICALLY (pair cross-ratio $= -1$); union over $c$ = the harmonic orbit $\{-1, 2, \frac{1}{2}\}$ $\iff$ $j = 1728$ $\iff$ $\tau = i$; generic modulus: $V_4$ abelian, ALL $\sigma_c$ central and yet clockless ("central alone is empty" — the bit is the CONJUNCTION $D_4$ AND central); at $\tau = i$ the stabiliser jumps to $D_4$ with $\sigma_1$ central (2 roots = the clock pair) and $\sigma_2/\sigma_3$ edge-type (0 roots); the v506 seam deck $z \mapsto -z$ lands on the central $\sigma_1 = -1/x$ under ALL 8 mark bijections; CM DICTIONARY: mult-by-$i$ fixes exactly ONE nonzero half-period $c^* = (1+\tau)/2$ (mult-by-$\rho$ 3-cycles all three); the clock lift needs the CM unit $i$ ($\psi = 2i/(x+1)^2$, $t^2+1$ irreducible over $\mathbb{Q}$) while half-period decks lift RATIONALLY: the clock root exists only WITH CM by $\mathbb{Z}[i]$ $\iff$ $\tau = i$. [E] EQUIVALENCE TETRAD + CIRCULARITY AUDIT: clock exists $\iff$ ($\tau = i$ AND deck = the CM-fixed half-period) $\iff$ (Stab = $D_4$ AND deck central) $\iff$ deck pairs harmonic $\iff$ the NS Fock lift is NONSPLIT; the v493 converse independently re-verified — two converses of ONE equivalence, no circularity. [E] FERMION DICHOTOMY (the new physical face): central arrangement (shift by 8): $\tilde{U}^2 = 256\,\gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F$, nonsplit $\mathbb{Z}_4$, 2 roots in the 32-element dihedral lift (the $\mathbb{Z}_8$ tower); edge arrangement (reflection $j \mapsto 4 - j$, fixed sites = non-marks = the silver $\{0, \infty\}$): the implementer EXISTS but $\tilde{U}_e^2 = +2^7 \cdot \mathbf{1} \Rightarrow U^2 = +1$, the extension SPLITS, 0 roots, no $\mathbb{Z}_8$; one-particle $(\lambda^2{+}1)^8$ vs $(\lambda^2{-}1)^8$; census: all 28 shift-invariant 4-sets cross, all 21 reflection-invariant do not; robust at 8 sites — the bit IS a physical extension class (Fidkowski–Kitaev-type), distinguished but not derived. [E] SILVER PLACEMENT: $\{\pm 1, \pm(3{+}2\sqrt{2})\}$ = "$\tau = i$ with deck = a non-CM-fixed half-period translation" (all 8 bijections $\to \sigma_2/\sigma_3$, never $\sigma_1$) — in-family witness against the tautology. [C] THE REST: the one alignment bit remains a carrier datum; no tautology — the origin forces the CLASS of the deck, not its POSITION. Exact (sympy + Cl(16) Fractions). [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v508_celestial_wp5e_gamma_pairing.py` | `CELEST.WP5E.GAMMA.01`: WP5e-$\gamma$ of the celestial-holographic research contract `CELEST.SEAM.01` — "the sphere-axion pairing check", an honest rigid NEGATIVE result: can the three twisted sphere axions $\eta_m$ (Coxeter eigencharacters $\{i,-1,-i\}$ on $H^2$(ALE), v505 S5.8) cancel the rigid $32\,T_3$ residual of the Atiyah–Bott ledger ($A_{\rm fix} = 9P_1 - 30P_2 - 15P_3 + 32\,T_3$, v505) by Green–Schwarz-type EXCHANGE with sector-compatible quadratic vertices? NO, with certificate. Exact sympy/Fraction. [E] VERTEX-SPACE THEOREM: the $W(D_5)\times W(A_3)$-invariant quadratics on the 8-dim glue Cartan are EXACTLY span$\{s_5, s_3\}$ (dim 2, bidegree dims $(1,0,1)$), the quartics EXACTLY span$\{P_1,P_2,P_3,T_5,T_3\}$ (dim 5, bidegree dims $(2,0,1,0,2)$) — Weyl nullspace arithmetic over all bidegrees, basis COMPLETE; PRODUCT THEOREM (the master kill): $(as_5{+}bs_3)(cs_5{+}ds_3) = ac\,P_1 + (ad{+}bc)P_2 + bd\,P_3$ identically — zero $T_5/T_3$ content, so EVERY exchange image (any axion count, any selection rule, any couplings) is annihilated by $\Phi_{T_3}$, while $\Phi_{T_3}(A_{\rm fix}) = 32 \neq 0$. [E] SELECTION ENUMERATION: the strict two-index reading COLLAPSES ($C_{j,j'} \neq 0$ only for $j' = -j$, so the rule forces $m = 0$ — the sphere axions have NO invariant bilinear Cartan vertex at all); twist insertion: $\eta_1 \leftrightarrow K^{(3)}$, $\eta_2 \leftrightarrow K^{(2)}$, $\eta_3 \leftrightarrow K^{(1)}$, $\eta_0 \leftrightarrow K^{(0)}$; channels $E_{13} = (16,-96,144,0,0)$, $E_{22} = (16,32,16,0,0)$, $E_{00} = 225\,E_{22}$; SIDE DISCOVERY: $K^{(0)} = -15\,K^{(2)}$ (the even-sector quadratics are PARALLEL). [E] UNSOLVABILITY WITH CERTIFICATE (the core result): exchange matrix $5\times 2$, rank 2, kernel 0; rank$([M \mid A_{\rm fix}]) = 3$; annihilator certificate $(\Phi_{T_5},\Phi_{T_3},\Phi_P)(A_{\rm fix}) = (0,32,72)$ with $A_{\rm fix} = (9,-30,-15,0,32)$ — two independent obstructions ($32\,T_3$ structural, rule-independent + $\Phi_P = 72$ rule-driven via the $K^{(0)} \parallel K^{(2)}$ collapse); $\ker M = 0$ forces the unique non-disturbing coupling $c = 0$: the sphere axions are RIGIDLY SILENT in the exchange ansatz; $\Phi_P(Q^{(0)}) = 0$ (the v505 sector-0 mechanism undisturbed); relaxed control: dropping charge conservation reaches the $P$-block (det $= -64$) but $T_5 = T_3 = 0$ persists. [E] NATURALNESS RESOLUTION: ch$_2$-natural $(9/64, 1/4)$ and AB-weight $(1/2, 1/4)$ couplings both carry certificate $(0,0)$ vs required $(32,72)$ — the failure is scale- and coupling-INDEPENDENT (no "unnatural factor": the ansatz space misses the target identically). [E] NEGATIVE CONTROLS: invariant axion only = v505 replication + sharpening; flipped rule $m{+}j{-}j'$: the $\Phi_{T_3}$ kill is flip-invariant, $\Phi_P$ is rule-dependent; $SO(16)$: odd classes EMPTY and the AB sum $(15,-18,-15,-4,40)$ does not even cancel $T_5$ ($E_8$ doubly special); $D_8$: $16\,T_8 + 12\,s_8^2$, no $T_3$ structure — the question cannot even be posed. [C] number fences: $32 = (h,h) + (h',h') = 20 + 12$; $72 = 2\tilde{\lambda}^2_{e_8}$. [O] SHARPENED FENCE: the exchange sub-branch is DEAD; the $32\,T_3$ residual MUST be carried by twisted BCOV contact terms — binary criterion WP5e-$\delta_1$: the one-loop coefficient on $\mathbb{C}^2/\mathbb{Z}_4$ must come out exactly $(9,-30,-15,0,32)$, COMPUTED, not fitted; named escape WP5e-$\delta_2$: the full-tensor ledger (the collapse is Cartan-restricted); slot bijection (v505 S5.8) untouched; the preregistered level-kill still does NOT fire; `SEAM.EQUIV.01` stays [O], no marker moves. Exact (sympy + Fraction). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v509_celestial_wp5e_eps2_level_from_flux.py` | `CELEST.WP5E.EPS2.01`: WP5e-$\varepsilon_2$ of the celestial-holographic research contract `CELEST.SEAM.01` — "the CPS level-from-flux dial", executed (verdict B): derive the boundary level $k = 1$ from flux quantisation on the lockstep spheres, following the Costello–Paquette–Sharma (CPS) Burns-holography mechanism (PRL **130**, 061602; `arXiv:2306.00940`). Exact sympy/Fraction, no floats. [E] CPS SKELETON REPLICATED EXACTLY: the $S^3$ period $\int \Omega_N = (2\pi i)^2 N$ (pullback density $-\sin 2\theta$, their eq. 3.33); the exceptional-sphere Kähler flux $\int_{\mathbb{P}^1} \omega = 2\pi N$; the boundary level magnitude $2N$ via the symplectic-boson double contraction ($N = 1, 2, 3$), Dynkin normalisation $T(\text{vector}) = 1$ — "level = flux quantum × Dynkin index". [E] PAIRING MATRIX PINNED, NOT POSTULATED: the v505 $\mathrm{ch}_2$ ledger + integrality + unimodularity + effectivity pin $P_{mi} = c_1(T_m).[\Sigma_i] = \delta_{mi}$ by COMPLETE enumeration (row candidates 8/6/8; 48 unimodular solutions of $PC^{-1}P^T = C^{-1}$; with $P \geq 0$ exactly 2 = {identity, $A_3$ diagram flip}); glue FLUXES $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$, total $188 = 248 - 60$, flip-invariant; flux per charged root current uniformly ONE quantum. [E] LEVEL FORMULA, honest: "level = total open-string flux" is KILLED by the lockstep test itself ($k_i = (64, 60, 64)$ unequal = the FLAVOUR multiplicity); what works: $k$ = flux per adjoint current = $F_i / \dim \mathfrak{g}_i = (1, 1, 1)$, anchored by the lattice current count $(240, 0, 0, 0)$ for $k = 1..4$ and the embedding index $j(D_5 \subset E_8) = j(A_3 \subset E_8) = 1$ ($60 = 2h^\vee$). ORD-4-VS-LEVEL-1 RESOLVED: the clock order counts FRACTIONAL flux sectors — $\mu = \det(D_5)\det(A_3) = 16 = \mathrm{ord}^2$, the $\mu_4$ glue diagonal isotropic + Lagrangian, KLM $16 \to 16/4^2 = 1$ (zero residual fractional flux); NEW DIAL: $\#\text{primaries}((E_8)_k) = (1, 3, 5, 10, 15, 27)$ for $k = 1..6$ from the affine marks — exactly ONE sector iff $k = 1$. [E] LOCKSTEP THEOREM (the lifting of v505 "one scale ⇒ one level" to a THEOREM): clock invariance forces $a_3 = a_2 = a_1 = 0$, one $\mu_4$ orbit of branch points, $\lvert\Pi_j\rvert = \sqrt{2}\,t$ EQUAL on all three spheres, monodromy = the pure phase $i$, $\det(A - 1) = -4$: no invariant flux vector — $k_1 = k_2 = k_3$ IS A THEOREM; NEGATIVE PROBE (new falsifier): the clock-forbidden family $Z^4 - Z$ (disc $-27$, branch points $\{0, 1, \omega, \omega^2\}$ not a $\mu_4$ orbit): 0/24 orderings lockstep; HONEST LIMIT: uniform flux doubling keeps the lockstep — EQUALITY, not the value (1 from the current/sector counters and the minimal quantum $N = 1$). [E] NEGATIVE CONTROLS: $k = 2$ (count 0, residual 360, prefactor $-31/48$, #prim 3 — the lockstep dial honestly does NOT fire); $SO(16)$ (fluxes $(0, 60, 0)$: spheres flux-dark, condensation stops at $4 = \det \mathrm{Cartan}(D_8)$); $A_2$ ($\mu = 12$ not a perfect square: NO Lagrangian glue, Coxeter $3 \neq 4$). [C] the CPS dictionary itself on $\mathbb{PT}/\mathbb{Z}_4$ (their geometry is the $\mathbb{C}^2$ blow-up with ONE $(-1)$-sphere and $\mathfrak{so}(8)$ at negative level — not the $A_3$ ALE). [O] the type-I B-model backreaction on $\mathbb{PT}/\mathbb{Z}_4$ (the $A_3$ $\Omega_N$) — the same fence as v502/v505; `SEAM.EQUIV.01` untouched; no marker moves. Exact (sympy + Fraction). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v510_seam_bit_freedom.py` | `SEAM.BIT.FREEDOM.01`: the freedom attack on the v506/v507 alignment bit — the POSITION half of the bit is topology (Part A dihedral/Fock + Part B Moebius/RP in one module; NO marker moves — the modulus datum stays [C]). Exact sympy + exact Cl(16) Fractions, no floats. [E] DIHEDRAL THEOREM: on the $N$-site seam circle the free involution is UNIQUE = the antipode ($N = 16$: 17 involutions in $D_{16}$, exactly 1 free = shift 8; all 16 reflections have exactly 2 circle fixed points; robustness $N = 8, 12$); the census is COMPLETE: $\mathrm{Aut}(C_{16}) = D_{16}$ exactly (brute force over all 8! permutations at $N = 8$, propagation at $N = 16$); $N = 9$ control: ZERO free involutions (even carrier count, consistent with $16 = 2^{g_{\mathrm{car}}-1}$); QUOTIENT TOPOLOGY (the P1 tie): circle/antipode = CIRCLE ($\chi = 0$, closed), circle/reflection = INTERVAL ($\chi = 1$, two corners) — the $6\pi = 2\pi+2\pi+2\pi$ Gauss–Bonnet budget counts three CLOSED circles, so "the deck is free on the seam circle" IS the established P1 datum "the seam is a closed circle", not a new premise. [E] NONSPLIT $\iff$ FREE (complete census, exact Cl(16)): ALL 17 involutions implemented — the free one's NS lift squares to $256\,\gamma_1 \cdots \gamma_{16} = (-1)^F$ (nonsplit, 2 roots = the $\mathbb{Z}_8$ clock tower); ALL 16 reflections have $U^2$ = SCALAR ($2^7$ site-, $2^8$ edge-axis; split, 0 roots); chain: free $\Rightarrow$ shift 8 $\Rightarrow$ crossing (all 28 invariant 4-sets) $\Rightarrow$ central class $\Rightarrow$ the edge/silver arrangement is EXCLUDED; Klein control (covering decks are free regardless of the gluing); branch control: the pillowcase involution has exactly 4 fixed points = the marks and restricts to a REFLECTION on invariant circles — the edge class IS the BRANCHED involution's restriction: silver mistakes the mark-creating branch $\mathbb{Z}_2$ for the seam deck. [E] REAL-STRUCTURE CLASSIFICATION: the deck-commuting antiholomorphic involutions of $\mathbb{P}^1$ are EXACTLY two families — $\mu\bar{z}$, $|\mu| = 1$ (fix LINE through the deck poles $\{0, \infty\}$: deck NOT free) and $\mu/\bar{z}$, $\mu > 0$ (fix circle $|z| = \sqrt{\mu}$ avoiding the poles: deck FREE; $\mu < 0$: empty); per configuration the mark-fixing real structure is UNIQUE; table: $\mu_4 \to$ equatorial $1/\bar{z}$, FREE (v506 A-S2.9); silver $\{\pm 1, \pm(3{+}2\sqrt{2})\} \to$ real axis THROUGH the poles, NOT free; generic-unit $\{\pm 1, \pm(3{+}4i)/5\} \to$ equatorial, free WITHOUT harmonicity; hexagonal $\to$ NO mark-fixing real structure (marks not concyclic: RP-incompatible, consistent v503); one-sidedness TYPES the $\mathbb{Z}_2$ as the free antipode (quotient $\mathbb{RP}^2$ = the $|\mathbb{Z}_2|$ of $c_3 = 1/(|\mathbb{Z}_2|\,2\pi\chi)$, v456). [E] HARMONIC + FREE $\Rightarrow \mu_4$ EXACTLY: the deck-pair cross-ratio $(1{-}b)^2/(1{+}b)^2 = -1$ solves exactly to $b = \pm i$ — the silver family (harmonic AND edge) is EMPTY on free circles, harmonicity becomes SUFFICIENT. [E] HONEST COUNTERWITNESSES: "nonsplit $\Rightarrow$ clock" is FALSE — discrete $\{0, 1, 8, 9\}$ and continuous $\{\pm 1, \pm(3{+}4i)/5\}$ (free, central, $j = 148176/25 \neq 1728$, no clock): the Fock dichotomy measures the CLASS, not the modulus. PREMISE TYPING: (P-i) deck = Moebius covering deck $\Rightarrow$ free [standard topology; origin_theory/v456]; (P-ii) marks on the $\Theta$-fix circle = existing REALIZE [C] content — no new input. [C] THE REDUCED REST (verbatim): the alignment bit reduces from "harmonic AND central" to the square-modulus datum $\tau = i$ alone; the position half is topology (edge class excluded), the modulus half stays the one carrier input. Exact (sympy + Cl(16) Fractions). [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v511_celestial_wp5e_delta2_tensor_ledger.py` | `CELEST.WP5E.DELTA2.01`: WP5e-$\delta_2$ of the celestial-holographic research contract `CELEST.SEAM.01` — "the full-tensor ledger", the named escape route of the v508 exchange no-go, executed (intermediate verdict, exact): does the FULL adjoint tensor structure of $\mathfrak{e}_8$ (non-Cartan external legs, higher-arity vertices) open a sphere-axion channel to the rigid $32\,T_3$ residual ($A_{\rm fix} = (9, -30, -15, 0, 32)$, v505)? The collapse is CONFIRMED full-tensorially, but one cubic door opens. Exact Kostant/Weyl weight arithmetic + sympy/Fraction. [E] STRUCTURE: $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$ is SEMISIMPLE with NO $\mathfrak{u}(1)$ factor (class-0 roots $40+12$, root span rank 8); the twisted sectors factorise multiplicity-free into minuscule Weyl orbits $\mathfrak{g}_1 = (16_s, 4)$, $\mathfrak{g}_2 = (10, 6)$, $\mathfrak{g}_3 = (\mathfrak{g}_1)^*$, each irreducible $(64, 60, 64)$; INNERNESS THEOREM: $h = (2, 2, 2, 2, 2; 0^4)$ lies in the Cartan OF $\mathfrak{g}_0$, so every $\mathfrak{g}_0$-invariant tensor carries total sector charge 0 mod 4 — the v508 collapse is a full-tensor statement AND holds across arities (all 15 non-neutral sector triples: Hom = 0). [E] HOM TABLES: bilinear $\dim \mathrm{Hom}_{\mathfrak{g}_0}(\mathfrak{g}_j \otimes \mathfrak{g}_{j'})$ nonzero ONLY at $j' = -j$ (dims 2/1/1/1, Killing blocks); trilinear neutral multisets $\{0,0,0\}:3$ (= 1 sym + 2 antisym), $\{0,1,3\}:2$, $\{0,2,2\}:2$, $\{1,1,2\}:1$, $\{2,3,3\}:1$ — ALL cross-sector trilinears are pure brackets (antisymmetric in the repeated legs); the ONLY totally symmetric survivor on all of $\mathfrak{e}_8$ is the $\mathfrak{su}(4)$ $d$-symbol ($\mathfrak{so}(10)$ has no cubic Casimir: $\mathrm{Inv}(S^3\,45) = 0$ protects $T_5$). [E] THE NEW CHANNEL: the quartic projection of the $d$-symbol exchange is $Q_{dd} = (0, 0, -\frac{1}{240}, 0, \frac{1}{60}) = \frac{1}{60}(T_3 - P_3/4)$, computed by TWO independent routes (Killing-propagator contraction with $2h^\vee = 60$ verified from the roots; hand route $|x^2 - (\mathrm{tr}\,x^2/4)\mathbf{1}|^2/60$); $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$ — the v508 master kill does NOT extend to cubic vertices; brackets, tadpoles and mixed $d \times f$ die exactly; no $T_5$ analogue. [E] NEW CERTIFICATE: $M = [E_{13}, E_{22}, E_{00}, Q_{dd}]$ has rank 3, augmented rank 4 — the charge reading remains OBSTRUCTED with the WEAKER certificate $\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}$, $\psi(A_{\rm fix}) = 72 - 8 = 64$; the RELAXED reading becomes EXACTLY solvable: $A_{\rm fix} = -u + 8v + 2w + 1920\,Q_{dd}$ (residual 0). [C] number fences: $64 = \dim \mathfrak{g}_1 = 2^6$; $1920 = |W(D_5)| = 2^7 \cdot 3 \cdot 5 = 8 \cdot 240$. [O] the arity $\geq 4$ ledger; the physical justification of the cubic GS term ($c_d = 1920$); $\mathrm{ch}_2$-naturalness for CUBIC vertices UNDECIDED; the burden on $\delta_1$ is SHARPENED to the $\psi = 64$ slice or the selection-rule relaxation. [E] NEGATIVE CONTROLS: Killing control = v508 replication; $SO(16)/D_8$: bilinears $(1, 0, 1)$, trilinears $(1, 0, 1, 0)$, $\mathrm{Inv}(S^3\,\mathrm{adj}_{D_8}) = 0$ — no symmetric cubic, no odd sectors, no $A_3$ block ($E_8/\mu_4$ doubly special); false $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_2 \oplus \mathfrak{u}(1)$: $(5, 2, 2) \neq (2, 1, 1)$ and $\mathrm{Inv}(S^3\mathfrak{g}_0) = 6 \neq 1$; false sector content kills every pairing. `SEAM.EQUIV.01` stays [O]; NO marker moves. Exact (sympy + Fraction). [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v512_seam_tau_flag.py` | `SEAM.TAU.FLAG.01`: the flag-transitivity equivalence web — the $\tau = i$ attack executed on the free RP seam circle: which ESTABLISHED seam datum excludes the free-but-non-harmonic configurations? (Test family $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$, $\delta \in (0, \pi/2]$; NO marker moves — the bit stays [C], `QGEO.REALIZE.01` and `P2.TYPING.01/AX.P2.01` text-precised only.) Exact sympy + mpmath 30 digits. [E] FALSIFICATION OF BARE TRANSITIVITY: on the free RP circle $V_4 = \{z, -z, b/z, -b/z\}$ acts SIMPLY TRANSITIVELY on the four marks for EVERY $\delta$ (pair-exchanging inversions; complete census: deck-commuting Möbius maps = diagonal/antidiagonal) — the expectation "transitivity ⇐ v216 local equality selects $\tau = i$" is REFUTED; hardening: ALL undirected per-mark data (arc-pair multiset $\{\delta, \pi - \delta\}$, distance multiset, cone germ) are identical for every $\delta$ — no local jet sees the bit. [E] THE SHARP SELECTOR: FLAG transitivity (marks AND their two sides indistinguishable) $\iff \delta = \pi/2 \iff \tau = i \iff$ the clock — stabiliser jump $1 \to 2$, flag/arc orbits $2 \to 1$, order-4 rotation ($b^2 = -1$ solveset $\{\pi/2\}$), mark mirror $z \mapsto 1/z \iff \cos\delta = 0$. [E] 12-FOLD EQUIVALENCE WEB (each link exact): clock rotation, $D_4$, mark mirror, flag transitivity, three spectral degeneracies (odd-doublet split $= (2/\pi)\ln\cot(\delta/2)$, zero only at $\pi/2$ — counterwitness exactly $(2/\pi)\ln 2$; arc-Laplacian degeneracy $4/\pi$; the v215-K3 indicator in CLOSED FORM $(2\sqrt{2}/\pi)|\ln\tan(\delta/2)|$ — the v503 lattice curve analytically), $\rho$-twist existence ($(m^2 - n^2)(t - 1/t) = 0 \iff t = 1$; Atiyah–Bott rationality is $\delta$-BLIND — $\mathrm{Tr}(\sigma e^{t\Delta}) = 1$ for ALL $\delta$, $j_{\mathrm{cw}} = 148176/25$ rational non-CM: rationality does NOT select), 4-cycle implementability (H4 rigidity: $V_4$ with element orders $\{1, 2, 2, 2\}$ cannot induce the cusp 4-cycle; centraliser scalar, $U^2$ nonscalar ⇒ no projective involution — projective order exactly 4, $\mathbb{Z}_8$-shadow consistent v510), harmonic, $j = 1728$. [E] NEGATIVE CONTROLS: hexagonal (not concyclic, fails the PRECONDITIONS); silver (harmonic but topologically dead — position/modulus halves independent, consistent v510); 3 marks (impossible on the free circle: 0 invariant 3-subsets vs 28 invariant 4-subsets; bare $n = 3$ transitivity already forces equal spacing — "4 counts": the deck-pairing makes bare transitivity free of charge). [C] THE REST (verbatim): the last input reduces from a continuous modulus to ONE discrete symmetry-lift bit ($V_4 \to D_4$: side-indistinguishability of the four equally-decorated marks); v216 local equality does NOT deliver it (no local jet can see the side bit); its negation is now concretely measurable (odd-split $(2/\pi)\ln\cot(\delta/2) \neq 0$). Scoreboard: the counterwitness passes all 7 established unsided tests, fails all 9 clock-equivalent ones. Exact (sympy + mpmath). [E][C] |

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| Script | What it machine-checks *(continued)* |
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| `v513_celestial_dterm_nonderivation.py` | `CELEST.DTERM.NONDERIV.01`: the $c_d$ negative certificate — can geometry derive $c_d = 1920$? An anti-numerology certificate in the v508 style (honest rigid NEGATIVE result): the v511 [C] fence "$1920 = \|W(D_5)\|$" stress-tested via four candidate routes with a PRE-DECLARED look-elsewhere ledger. Exact sympy/Fraction, no floats. [E] CONVENTION TABLE: $c_d \in \{1920, 120, 256, 32, 69120\}$ across five normalisations — only 1 of 5 hits $\|W(D_5)\|$; the convention-STABLE content is the factorisation $c_d = \Phi_{T_3}(A_{\text{fix}}) \times 2h^\vee(E_8) = 32 \times 60$, both factors [E] (Killing propagator 60 from $\sum_\alpha \langle \alpha, e\rangle^2 = 120$ over the actual roots; $T_3$-slot fixing via det(P-block) $= -64 \neq 0$). [E] K1 DEAD: the charge-legal flux pairings $(1,3)/(2,2)$ give 2048/1800 — they BRACKET 1920 and both MISS; the only flux hit $64 \cdot 60/2$ is the FORBIDDEN $(1,2)$ pairing (charge 3 mod 4, killed by the v511 innerness theorem). LOOK-ELSEWHERE LEDGER: 924 expressions from 21 [E] numbers, 11 hits on 1920; control targets $1800 \to 8$, $2048 \to 5$, $2016 \to 0$. [E] K2: quantisation delivers only lattices ($60\,\mathbb{Z}$: 1920 = the 32nd multiple, no minimality; strict $\text{ch}_3$-unit couplings $2160\,\mathbb{Z}$ EXCLUDE 1920); the cubic quantum is lattice-dependent (6 on coroots, 3/8 on weights); the residual integer 32 has 12 signed factorisations. [E] K3 PROVENANCE CLASH (the death blow): cubic indices $A_m = (0, 16, 0, -16)$, twisted sum $\sum_m i^m A_m = 32i$ NUMERICALLY $= \Phi_{T_3}(A_{\text{fix}})$ — BUT the quartic class decomposition $(10, -4, 30, -4)$ is dominated by classes 0/2 while the cubic 32 is built $16 + 16$ from classes 1/3: same number, disjoint mechanism = coincidence exposed; the Weyl side is WRONG (the $d$-channel is an $A_3$-side object, $\|W(A_3)\| = 24$; $D_5$ has no cubic invariant, $\text{Inv}(S^3\,45) = 0$; $\|W(E_8)\|/\|W(A_8)\| = 1920$ a second accidental 1920). [E] K4 NEGATIVE CONTROLS: $SO(16)/D_8$ fails every candidate formula consistently; $k = 2$ flux doubling scales the flux formula $\times 4$ (7680); false $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_2 \oplus \mathfrak{u}(1)$ has SIX symmetric cubics; $\mathfrak{so}(10)$ sanity ($\text{Tr}\, X^3 = 0$). VERDICT: NUMEROLOGY as a derivation claim — the v511 [C] fence stays a fence, now typed look-elsewhere-loaded; the honest [E] core is $c_d = 32 \times 60$; the physical generation of the 32 stays [O]. Exact (sympy/Fraction); no marker moves. [E] |

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| Script | What it machine-checks *(continued)* |
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| `v514_celestial_wp5e_eps1_axion_slot.py` | `CELEST.WP5E.EPS1.01`: WP5e-$\varepsilon_1$ of the celestial-holographic research contract `CELEST.SEAM.01` — "the $O(-2)$ bulk-axion slot", executed (verdict B): build the $s=0$ tower field explicitly, pin $\tilde{\lambda}=6$ as a residue with explicit factor bookkeeping on $\mathbb{PT}/\mathbb{Z}_4$, and take the first honest back-reaction step (Gibbons–Hawking $A_3$). Exact sympy/Fraction, no floats. [E] EQUIVARIANT $O(-2)$ SLOT: pushforward ledger $\pi_*O(-2)$ on $\mathbb{PT}'=\mathrm{Tot}(O(1)\oplus O(1)\to\mathbb{P}^1)$: per spacetime degree $d$ exactly $(d+1)^2$ = nullspaces of the wave operator $(1,4,9,16,25,36$ for $d=0..5)$; the v492 action on the fibre coordinates $(\mu_1,\mu_2)\mapsto(i\mu_1,i^{-1}\mu_2)$, incidence forces column weights $(+1,+1,-1,-1)$; the equivariant bookkeeping closes block by block (all $d\le 6$, all 4 characters); character series $P_0=1+3t^2+15t^4+21t^6+45t^8+\ldots$, $P_1=P_3=2t+8t^3+18t^5+\ldots$, $P_2=6t^2+10t^4+28t^6+\ldots$; the $d=0$ slot has multiplicity $(1,0,0,0)$ — the bulk axion survives the projection; Molien invariant ring $=(1-t^8)/((1-t^4)^2(1-t^2))$ — $Z\in O(2)$, $X,Y\in O(4)$, relation degree $8=$ the $a_0$ weight (v492/v493 bridge); E3 bijection sharpened: twisted characters $\{i,-1,-i\}$ with minimal content $(2t,6t^2,2t)$ and no degree-0 mode = the Coxeter eigenvalues on $H^2(\mathrm{ALE})$, $\det(A-1)=-4$; graviton control: the $O(+2)$ slot starts invariant only at fibre degree 4 with multiplicity $3=\{X,Z^2,Y\}$ (the cohomological form of v505 $X=-2\neq+2=Y$). [E] $\tilde{\lambda}=6$ TRIPLY PINNED: (1) Okubo: $\mathrm{Tr}_{\mathrm{adj}}X^4=36\langle x,x\rangle^2=(6\langle x,x\rangle)^2$ on the 240 glue roots, $36=h^\vee+6$, $\tilde{\lambda}=6=|\mathbb{Z}_2|\,N_{\mathrm{fam}}$; (2) measure cancellation: $\mu=1/4$ enters three times (vertex$^2$ $\mu^2$, propagator $1/\mu$, anomaly $\mu$) and cancels exactly — $\tilde{\lambda}^2_{\mathrm{eff}}=36$ for arbitrary $\mu$; wrong bookings quantified and excluded: anomaly-only $\Rightarrow\tilde{\lambda}=3$, missing propagator renormalisation $\Rightarrow\tilde{\lambda}=12$; (3) flux: minimal CPS quantum $N=1$, a single exchange channel exactly at $k=1$ (#primaries), $(\kappa/c_3)^2=36/3=12$ exact. [E] GH/$A_3$ BACK-REACTION STEP: invariant 4-centre configurations = exactly 2 branches (axis$^4$ or one free $\mu_4$ orbit); free orbit $\Rightarrow\prod_p(Z-i^pz_0)=Z^4-z_0^4\Rightarrow XY=Z^4+a_0$ with $a_0=-z_0^4$ (the v493 family re-derived from the geometry); monodromy $a_0\to e^{2\pi i}a_0=$ one clock step; the GH point of charge $k$ is exactly $\mathbb{R}^4/\mathbb{Z}_k$ ($k=4\Rightarrow$ seam boundary $S^3/\mathbb{Z}_4$); periods $\int\omega=4\pi(x_q-x_p)$, free orbit: $\Pi_j=4\pi t_0(i-1)(1,i,i^2)$, $|\Pi_j|$ equal (lockstep, counter-check of v509); the Coxeter clock from the geometry ($A^4=1$, $\det(A-1)=-4$, $\Pi\cdot A=i\Pi$); the CPS "$N\log$" analogue = source charge $4=|\mu_4|$ (flux $-4\pi$ per centre); honest sharpening: for the Ricci-flat ALE the asymptotic log coefficient is exactly 0 (EH: $uK'=\sqrt{u^2+a^4}$, $\det g=1$) — the log lives at the exceptional locus; Burns contrast ($K=u+N\log u\Rightarrow\det g\neq 1$, only scalar-flat); bonus: multipole selection rule $m\equiv 0 \bmod 4$, first symmetry-breaking multipole $(4,\pm4)$ carries $-a_0$ with amplitude $(35/16)\sin^4\theta$. [E] NEGATIVE CONTROLS: $\mathfrak{so}_8$ ($\tilde{\lambda}^2=12$ irrational; perfect squares in the Deligne series only $\{9,36\}=\{\mathfrak{sl}_3,\mathfrak{e}_8\}$); diag$(i,i)$ (Veronese, 5 generators, no hypersurface, the bijection collapses); $k=2$ (three channels; honest: $\tilde{\lambda}^2=36$ is $k$-independent, the kill is the closure dial). [C] conditional on Costello's flat-$\mathbb{PT}$ matching $\lambda_g^2E=A$. [O] the quantised BCOV one-loop coefficient on $\mathbb{PT}/\mathbb{Z}_4$; the twisted channels $(32\,T_3)$ are not covered by the bulk axion — M1–M3 preregistered verbatim: M1 the $A_3$ $\Omega_N$ on $\mathbb{PT}/\mathbb{Z}_4$ (success: $(2\pi i)^2$ periods with lockstep phases; kill: unequal fluxes $\Rightarrow$ the v509 dial dies); M2 the twisted KS measure (success: bulk 36 + sphere axions cancel $32\,T_3$; kill: a leftover $T_3$); M3 the $a_0$ uplift of the $(4,\pm4)$ multipole (success: log correction $\propto 4=|\mu_4|$; kill: decoupling from the centre count). No marker moves anywhere. |

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| Script | What it machine-checks *(continued)* |
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| `v515_celestial_wp5e_m1_omega_n.py` | `CELEST.WP5E.M1.01`: milestone M1 of the celestial-holographic research contract `CELEST.SEAM.01` (WP5e back-reaction programme) — "the $A_3$ $\Omega_N$", executed (SUCCESS on the preregistered v514 S8.1 criterion): the back-reacted holomorphic 3-form $\Omega_N$ on the twistor uplift of the deformed $A_3$ ALE, with exact periods. Exact sympy, no floats. [E] RESIDUE FORM DERIVED: on $F = XY - P(Z)$ all three ambient representatives satisfy $dF \wedge \omega_2 = -dX \wedge dY \wedge dZ$ and pull back to $\omega_2 = dX \wedge dZ/X$ (Poincaré residue, derived not assumed); cover pullback at $a_0 = 0$: $\omega_2 = 4\, dz_1 \wedge dz_2 = |\mu_4| \times$(flat form) — the residue normalisation carries the source charge 4; clock diag$(i, 1)$: $\omega_2 \to i\,\omega_2$ (v492 replicated); weight ledger $4{+}4{+}2{-}8 = 2$, $\Omega \in O(4) = -\deg K_{\mathbb{PT}}$; EH anchor $2 = |\mathbb{Z}_2|$; wrong-weight control $Z \in O(4)$ breaks three ways ($4 \neq 2$, $6 \neq 4$, $5 \neq 3$). [E] THE TWISTOR FAMILY CLOSED: from the four $O(2)$ centre sections $q_p(\lambda) = i^p t_0 - i^{-p} t_0 \lambda^2$ the family is exactly $XY = Z^4 + 4t_0^2\lambda^2 Z^2 - t_0^4(1-\lambda^4)^2$ ($e_1 = e_3 = 0$ identically; $\lambda = 0$ gives exactly the v493 clock family; $a_2 \in O(4)$ opens only off-seam, $a_2(0) = a_2(\infty) = 0$); clock lift $\gamma : (Z, \lambda) \mapsto (iZ, -i\lambda)$, $\gamma^4 = 1$, $\Omega \to +\Omega$ (CY-compatible). [E] $\Omega_N$ CLOSED + QUANTISED PERIODS: $\Omega_N = \Omega_0 + \sum_p N_p K_p$ with CPS/Bochner–Martinelli kernels on the centre twistor lines; $dK = 0$ away from the source, scale degree 0; period table: boundary $S^3/\mathbb{Z}_4$ at the orbifold point $(2\pi i)^2 \times 1$ (minimal invariant charge $4 = |\mu_4|$ — the lens fundamental domain forces $N \in 4\mathbb{Z}$), deformed phase $(2\pi i)^2 \times 4$, linking $S^3$ uniformly $(2\pi i)^2 \times N$, per-sphere flux vector $(2\pi i)^2 \times N(1,1,1)$ (forced uniform twice over: clock orbit + $K_4$ connectivity), seam-fibre phase vector $2\pi i\, t_0(i-1)(1, i, i^2)$; the undeformed $\Omega_0$ carries zero quantised 3-flux (all flux sourced); $(2\pi i)^2 =$ (fibre residue)$\times$(transverse residue) (density $-\sin 2\theta$). [E] CLOCK COVARIANCE: $\Pi_{j+1}(-i\lambda) = i\,\Pi_j(\lambda)$ for all $\lambda$; the 12 collision nodes sit exactly on the 8 eighth roots of unity ($8 = 2|\mu_4|$, the $\mathbb{Z}_8$ seam bridge), clock orbits 4+4+4, honest conifold points (Hessian det $512t_0^6$, quadric rank 4). [E] KILL PROBE (honest): the forbidden family $Z^4 - Z$: 0/24 orderings lockstep (vs 8/24), $\lambda$-support 12 twelfth roots $\neq \mathbb{Z}_8$, no antipodal pairing — but the $(2\pi i)^2$ quantisation holds there too: integrality alone is not the discriminator, the discriminator is the lockstep phase structure + the clock forcing of equal fluxes; the v509 dial survives. [E] ANCHORS + CONTROLS: EH/$\mathbb{Z}_2$ (2 nodes, charge $2 = |\mathbb{Z}_2|$ forced); CPS original (full $S^3$: every integer $N$ allowed); diag$(i, i)$: $\Omega_0 \to -\Omega_0$ (collapse at step zero) while $K$ stays invariant; fractional charges $N = 1, 2, 3$ excluded. [C] global kernel patching (Green function), Hitchin small resolution, CPS brane dictionary. [O] M2 (twisted KS measure), M3 ($a_0$ uplift), the quantised BCOV coefficient; `SEAM.EQUIV.TWISTOR.01` untouched (preparation, not construction). No marker moves. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v516_celestial_wp5e_m2_twisted_ks_measure.py` | `CELEST.WP5E.M2.01`: milestone M2 of the celestial-holographic research contract `CELEST.SEAM.01` (WP5e back-reaction programme) — "the twisted Kodaira–Spencer measure", executed (SUCCESS on the preregistered v514 S8.2 criterion, on the DECLARED completion measure — verdict B): the one-loop box coefficient per sector on $\mathbb{PT}/\mathbb{Z}_4$ under the completion contact term $\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$. Exact sympy/Fraction, no floats. [E] THE COMPLETION-WEIGHT IDENTITY (the centrepiece): per glue class the contact weights are $w_m = \sum_j (1 - i^{jm})/\det_j = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4| h_m = -4\,\text{ch}_2(T_m)$ EXACTLY ($h_m = m(4-m)/8$ the v502 sector Casimir energies, $\text{ch}_2(T_m)$ the McKay/Kronheimer charges, v505 S4.3 index bridge): the three sphere axions pair through their OWN $\text{ch}_2$ charges — an exact identity, NO free scale, no fit. [E] PARAMETER-FREE LOCKS: $T_5 = 0$ for ANY scale $c$ (the $\text{ch}_2$ pattern alone kills $T_5$); ratio $h_2 : h_1 = 4 : 3$ = the $\delta_1$-forced leading ratio (REPRODUCED, not fitted); the $T_3$ budget fixes $c = 4 = |\mu_4|$ uniquely. [E] THE CONTACT TERM: channel form $\sum_j (Q^{(0)} - Q^{(j)})/\det_j$ = class form $\sum_m 4h_m Q_m = (36, 120, 60, 0, -32)$ EXACTLY (one object, two bases); the contacts carry nonzero $T_5/T_3$ per channel (genuine contact, not exchange — the v508 product theorem replicated). [E] SUCCESS TABLE: $\text{skeleton}_j + \text{contact}_j = Q^{(0)}/\det_j = 36\langle x, x\rangle^2/\det_j$ — a PERFECT Okubo square ($T_5 = T_3 = 0$, $\tilde{\lambda}^2 = 36$) in EVERY twisted channel; total $A_{\text{fix}} + \text{contact} = 45\langle x, x\rangle^2 = \frac{5}{4} \times 36\langle x, x\rangle^2$ = Dedekind × Okubo = the UNIQUE quartic-free weighting (v505 S3.8 rigidity); both v508 certificates killed ($\Phi_{T_3} : 32 \to 0$, $\Phi_P : 72 \to 0$); the v511 slice $\psi = 64$ SUPPLIED EXACTLY — no cubic $d$-channel needed ($c_d$ free $= 0$); remainder inside the exchange span ($45\langle x, x\rangle^2 = \frac{45}{16} E_{22} = \frac{1}{80} E_{00}$), the bulk axion cancels every channel with the SAME $\tilde{\lambda}^2 = 36$; the v514 measure chain stays $\mu$-blind. [E] NEGATIVE CONTROLS: $c \in \{1, 2, 3, 5\}$ leaves $T_3 = (24, 16, 8, -8)$; the $h_1 \leftrightarrow h_2$ shuffle breaks $T_5 = -14$ AND $T_3 = 36$; $SO(16)$: $T_5 = 20$ / $T_3 = -40$ remain — the KILL fires there (Okubo failure inherited, $E_8$ doubly special); $\text{diag}(i, i)$ degenerates to the $\mathbb{Z}_2$ target and breaks the conjugation pairing + the index bridge; the $\mathbb{Z}_2$/Eguchi–Hanson anchor PASSES ($9\langle x, x\rangle^2 = \frac{1}{4} \times 36$, scale $2 = |\mathbb{Z}_2|$). [E] MULTIPLIER NOTE (cross-reference to $\delta_{1d}$): the contact weights are real rationals with no multiplier system of their own; the $\delta_{1c}$ residual phases = the SQUARES of the $A_3$ discriminant $T$-phases, orders $(1, 4, 1, 4)$. [C] THE COMPLETION READING (the fence): the loop of twisted states carries the UNPHASED sector trace with the same zero-mode normalisation — declared, supported by the $\delta_1$ modular-completion finding, NOT derived from the BCOV integral. [O] the BCOV-integral derivation ($\delta_{1b}/\delta_{1c}/\delta_{1d}$ territory); `SEAM.EQUIV.TWISTOR.01` untouched. VERDICT: SUCCESS wording met on the declared measure; the KILL does NOT fire; no marker moves. [E][C][O] |

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| Script | What it machine-checks *(continued)* |
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| `v517_celestial_wp5e_m3_a0_uplift.py` | `CELEST.WP5E.M3.01`: milestone M3 of the celestial-holographic research contract `CELEST.SEAM.01` (WP5e back-reaction programme) — "the $a_0$ uplift", executed (SUCCESS on the preregistered v514 S8.3 criterion): the $(4, \pm 4)$ GH multipole uplifted to the twistor-potential kernel $\chi = \log P_4$ (the log of the v515 family polynomial) with the induced Kähler-potential correction. Exact sympy, no floats. **[E]** KERNEL BRIDGE: the $O(2)$ section is a NULL coordinate — $\Delta_{3d}\, G(\eta_p) = 0$ identically for an ARBITRARY kernel $G$ (the uplift is well-typed); RESIDUE IDENTITY $\partial_x[\text{transform}(\log \eta_p)] = 1/r_p$ EXACTLY (residue $1/(2r)$): the GH potential $V = \sum_p 1/r_p$ IS the $x$-derivative of the contour transform of $\chi$; flux $-4\pi$ per centre, source charge $4 = \lvert\mu_4\rvert$ (the $V$-ledger matched). **[E]** FOUR COUPLED CENTRE-COUNT SCALES: (1) asymptotic kernel log $\chi = 4\log\eta + a_2/\eta^2 + (a_0 - a_2^2/2)/\eta^4 + \ldots$ — LOG COEFFICIENT $4 = \lvert\mu_4\rvert$ = centre count (the CPS "$N$ log" analogue at kernel level); at the clock-fixed seam fibre $\zeta = 0$ the $a_2$ slot closes and the first correction is EXACTLY $a_0/\eta^4$ = the $(4, \pm 4)$ multipole; exact $m$-grading ($m = \pm 4$ pieces PURE $a_0$, $m = 0$ piece $\varphi_0$-free); NO log ×power terms (the v514 honest zero preserved); (2) the GLT tower $\sum_p \eta_p \log \eta_p = 4\eta \log \eta + p_4/(12\eta^3) + \ldots$ with $p_n = 0$ unless $n \equiv 0 \bmod 4$ and $p_{4k} = 4(-a_0)^k$; (3) the exceptional-locus log $\chi(0) = \log a_0 = 4\log t_0 + i(4\varphi_0 + \pi)$, modulus response 1 at the locus vs power-law at infinity (v514 S6.3 uplifted); (4) periods: $d\log\Pi_j / d\log a_0 = 1/4 = 1/\lvert\mu_4\rvert$ uniform, integrated monodromy $e^{2\pi i/4} = i$ = ONE Coxeter clock step (v493 reproduced from perturbation theory); node $\lambda$-support $a_0$-RIGID (the $\mathbb{Z}_8$ seam bridge), $K_4$ discriminants $\sim t_0^2 \neq 0$ (the forced flux vector persists), BM kernel modulus-free with $\int K = (2\pi i)^2$. **[E]** NEGATIVE CONTROLS: $(4, 0)$ clock-invariant ($\varphi_0$-free in $V$ AND kernel: breaks nothing); $\mathbb{Z}_2$/EH analogue: EVERY dial reads $2 = \lvert\mathbb{Z}_2\rvert$ ($(2, \pm 2)$ amplitude $\frac{3}{2}\sin^2\theta$, kernel log 2, exceptional log $2\log t_0$, response $\frac{1}{2}$, monodromy $-1$); $k = 3, 5$ orbits: coefficient $k$, slot $O(6)/O(10) \neq O(8)$ — the observable MOVES with the centre count; forbidden family $Z^4 - Z$: $p_3 = 3 \neq 0$ (selection rule broken), $e_4 = 0$ (no $a_0$-log) — fails BOTH dials. **[C]** the GLT dictionary ($\eta_p \log \eta_p$ = the GH Kähler-potential kernel; Lindström–Roček / Ivanov–Roček). **[O]** the full nonlinear Kähler potential of the resolved $A_3$ ALE; `SEAM.EQUIV.TWISTOR.01` untouched. VERDICT: SUCCESS (log correction $\propto 4 = \lvert\mu_4\rvert$, coupled to the centre count on four scales); the KILL (decoupling) does NOT fire; no marker moves. **[E][C][O]** |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v518_celestial_wp5e_delta1_derived_kill.py` | `CELEST.WP5E.DELTA1.01`: the $\delta_1$ chain ($\delta_{1b}+\delta_{1c}+\delta_{1d}$) of the celestial-holographic research contract `CELEST.SEAM.01` DECIDED — "the derived chiral measure kills the $\delta_1$ route", an honest, decided NEGATIVE result (v508 style): does the Harvey–Moore/BCOV fundamental-domain integral on $\mathbb{PT}/\mathbb{Z}_4$, evaluated under a measure DERIVED from blockwise $SL(2,\mathbb{Z})$ covariance (solved for, not declared, not scanned), deliver $-A_{\rm fix}=-(9,-30,-15,0,32)$ or the $\psi=64$ slice? NO — every member of the derived family fails all three preregistered testers. **[E]** THE 16-COMPONENT WEIL COMPLETION ($\delta_{1c}$): discriminant module $\mathbb{Z}_4\times\mathbb{Z}_4$ with $q=(5x^2+3y^2)/8$; Gauss sums $2\zeta_8^5\times 2\zeta_8^3=4=\sqrt{16}$ (signature 0 mod 8 = rank $E_8$); $S$ symmetric + unitary, $S^2=C$, $(ST)^3=S^2$, $S^4=1$, $T^8=1$ EXACTLY in $\mathbb{Q}(\zeta_8)$; $S=S_{D_5}\otimes S_{A_3}$; the diagonal $H$ and anti-diagonal $H'$ are the two invariant Lagrangians (both counts = $E_4$: two $E_8$ gluings); $P_H S P_H=\frac{1}{4}J$ with 3/4 of every $S$-image weight on the 12 shifted classes; the 16-vector theta $S$-covariance CLOSES: the $\delta_{1b}$ E1.5 residual $2.91=O(1)$ drops to $\sim 10^{-39}$. **[E]** THE OBSTRUCTION IS A FINITE $\mu_4$ CHARACTER, IDENTIFIED ($\delta_{1d}$): gauge blocks $M=G[a,b]\,\eta^{-8}$ have constant transport $(t_p,s_p)$ (28+ digits) recognised as exact 1/960-grid phases; the $T$-fixed pairs $(0,b)$ carry $t=e(-1/2)=-1$ EXACTLY; loops land in $\Gamma_1(4)$ resp. $\Gamma_0(2)^{\pm}$, joint holonomy eigenspaces dim 8/16 resp. 6/16, every eigen-phase of order $\le 4$; KOBOUNDARY TEST: the relation defects $c_{S^4}$, $c_{(ST)^3S^{-2}}$, $c_{[S^2,T]}$ are $(1,1,1)$ EXACTLY on all 15 pairs — a CHARACTER of the stabilisers, NOT a genuine 2-cocycle on $SL(2,\mathbb{Z})$; on $\Gamma_1(4)$: $\lambda(\gamma)=i^{2B+C/4}$. **[E]** THE CANCELLATION IS THE TWISTED FIBRE BLOCK $f_1f_3$, NOT THE THREE SPHERE AXIONS: exact identity $G[a,b]=f_1f_3$ (30+ digits); the $T$-fix mechanism is one line of exact phase arithmetic: $(-1)\times e(-1/6)=e(1/3)=\chi_4(T)$ at every $T$-fixed node; CANCELLATION TABLE (exact, minimal residual order over $\eta^m\times\chi_k$): none $\to 4$, $f_1f_2f_3$ (the three sphere axions) $\to 4$, $f_2\to 6$, $f_1f_3\to 1$ (hol AND anti), wrong weights $\to 4$; strict solutions exactly $\chi_4$ (dims $(3,3)$) and $\chi_{10}$ (dims $(1,1)$) at $\eta^0$, certified on the dressed functions ($10^{-15}$, NO residual multiplier). **[E]** THE DECISION — ALL THREE TESTERS FAIL: under both derived solutions the integral fails $T_5(V)=0$ (fractions 0.5 / 0.83), fails $W_2:W_{13}=4:3$ ($W_{13}=0$), fails $V=-A_{\rm fix}N$ (spreads 1.05 / 1.47) and the $\psi=-64N$ slice; the honest $(N_1,N_2)$ rebalance: $\chi_4$ reaches $T_5=0$ only at $\psi>0$, $\chi_{10}$ only at $N_2/N_1=-4<0$ (outside the orbifold projector) — no branch fires in the positive cone. **[E]** NEGATIVE CONTROLS: $\mathbb{Z}_2$/Eguchi–Hanson (gauge-only residual order 2 — the $\mu_4$ defect is $\mathbb{Z}_4$-specific — and the SAME fibre dressing cancels it to 1); $SO(16)/D_8$ (only even sectors, all axion $b$-characters in $\mu_2$: the order-4 supply structurally absent); wrong form $(x^2+y^2)/4$ ($|G|^2=64\neq 16$, its $S$-rule fails at $O(1)$); wrong signature ($S^2=C$ breaks exactly). **[C]** the $f_1f_3$ = KS-weight identification (deck weights $(1,3)$ = the twistor fibre weights $(+1,-1 \bmod 4)$ forced by the v514 incidence ledger — exact match, NOT a complete BCOV derivation) + the kernel-family convention + the discrete $\chi_k$ choice (both evaluated, both fail). TENSION, stated honestly (the mandatory fence): the DECLARED completion reading (v516/M2) delivers the $\psi=64$ slice and cancels $32\,T_3$; the DERIVED chiral measure (this module) fails all three testers — both are exact; the sharp open question is which of the two is the true BCOV measure (the declared reading is supported by the $\delta_1$ modular-completion finding but not derived; the derived measure rests on the $f_1f_3$ identification). Neither result is hidden behind the other. **[O]** the BCOV-integral derivation on $\mathbb{PT}/\mathbb{Z}_4$ that decides between the two measures — the single open item of the $\delta_1$ strand; the twisted-contact route of v508 S6.2 is DECIDED NEGATIVE under the derived measure. NO marker moves anywhere. **[E][C][O]** |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v519_woit_theta_rp_free.py` | `WOIT.THETA.FREE.01`: the WOIT-$\alpha$ milestone of the OS twistor bridge `WOIT.OS.TWISTOR.01` — "the real structure exists, and free reflection positivity picks the SAME family" (Part W = the anti-linear $\Theta$: classification, exact relations, kill test 1 on the free/equivariant level; Part R = reflection positivity of the free 16-Majorana NS seam system under the pinned $\Theta$; NO marker moves — `WOIT.OS.TWISTOR.01` stays **[O]**: kill test 1 is discharged on the FREE level only and stays formally live on the interacting algebra). Exact sympy + exact Cl(16) Fractions; the RP definiteness certificates are 40-digit spectra of exact Hermitian matrices. **[E]** THETA CLASSIFICATION COMPLETE: exactly TWO families of anti-linear structures on $\mathbb{C}^2$ normalise the clock $\rho = \mathrm{diag}(i, 1)$ — family D ($z \mapsto \mu\bar{z}$, $\lvert\mu\rvert = 1$): $\Theta\rho\Theta = \rho^{-1}$ EXACTLY, $\Theta^2 = +1$ ($-1$ IMPOSSIBLE: $M\bar{M} = \mathrm{diag}(\lvert\mu\rvert^2, 1)$ — Kramers-free), inverts the deck, fixed set = the great circle through the deck poles; family A ($z \mapsto \mu/\bar{z}$): centralises the clock projectively (NEVER inverts), $\Theta^2 = \pm 1$ per $\mu$, fixed set $\lvert z\rvert = 1$ resp. empty. ROLE SEPARATION: family A DEFINES the euclidean section (Woit's $\rho_{\mathrm{tw}}$ side: $\rho_{\mathrm{tw}}^2 = -1$, no real points — replicated exactly on $\mathbb{C}^4$), family D REFLECTS it (the OS conjugation; $\sigma_{\mathrm{std}}$ = the standard conjugation with real points $\mathbb{RP}^3$). **[E]** MARK PINNING: $z \mapsto \mu\bar{z}$ preserves the $\mu_4$ marks iff $\mu \in \mu_4$ — a 4-element $\mu_4$-TORSOR (2 mark axes + 2 silver-midpoint axes); clock conjugation $\rho\,\Theta_\mu\,\rho^{-1} = \Theta_{-\mu} \Rightarrow$ 2 clock orbits. **[E]** SPIN/FOCK: the $\mathbb{Z}_8$ plane has no phase leaks ($\overline{\zeta_8^k} = \zeta_8^{-k}$); exact Cl(16): $\Theta_{\mathrm{Fock}} = U_r \circ K$ with $\Theta_{\mathrm{Fock}}^2 = 2^7 > 0$ (normalised $+1$), inverts the clock tower ($V \mapsto 4096\,V^{-1}$, exact scalar) and normalises the deck; KRAMERS DICHOTOMY: the deck-induced candidate $\Theta_t = \tilde{U} \circ K$ has $\Theta_t^2 = 256\,\gamma_1 \cdots \gamma_{16} = (-1)^F$ — the `v510` split/nonsplit dichotomy IS the $\Theta^2 = +1$ vs $(-1)^F$ dichotomy; the deck does NOT furnish $\Theta$, the seam-circle REFLECTION does. **[E]** KILL TEST 1 (free level): does NOT fire — $\Theta$ exists with $\Theta^2 = +1$ AND $\Theta\rho\Theta = \rho^{-1}$ and satisfies all three side conditions simultaneously (marks, deck normalisation, spin/Fock). **[E]** FREE RP: the chiral NS vacuum $C(d) = (2/N)/\sin(\pi d/N)$ has $C^2 = -1$ exact; preconditions exact ($RCR^T = -C$, $SCS^T = C$, $RSR^{-1} = S^{-1}$; the deck FAILS anti-invariance); BOND CUT: one-particle Gram $(8, 0, 0)$ PD (min eigenvalue $1.888 \times 10^{-3}$ at 40 digits), even deg $\leq 2$ sector $(29, 0, 0)$ PD, $N = 8$ complete half-sided algebra PD (even + odd each $(8, 0, 0)$) — free RP with NO degree truncation; the twist $\eta = +i$ is FORCED ($\eta = 1$ non-Hermitian); CONTROLS: the site cut fails exactly (det $= 0$, inertia $(3, 3, 1)$; continuum control: the Cauchy–Stieltjes factorisation is strictly PD — a lattice-PLACEMENT artifact: RP selects the site placement relative to the marks); the clock-CENTRALISING family-A structure fails RP structurally ($(4, 4, 0)$) — RP and $\Theta\rho\Theta = \rho^{-1}$ select the SAME family; bonus: the anti-chiral state flips the odd sector to $(0, 8, 0)$ ND (the free shadow of kill test 3). **[C]**/**[O]** NOT shown: the interacting algebra $\mathcal{A}_{\mathrm{hol}}$ (kill test 1 stays live there), gauge-fixed RP (kill test 2), the OS reconstruction, chirality without mirrors (kill tests 3/6 — only the free odd-sector shadow), $\mu_4$ incidence (kill test 7); WOIT-$\beta/\gamma$ roadmap: ($\beta_1$) $\Theta$ on the gauge-invariant subalgebra + gauge-fixed RP on the equivariant SDYM($E_8$) sector; ($\beta_2$) the OS quotient of the free system explicit; ($\beta_3$) the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality vs $\sigma_{\mathrm{std}}$ typed; ($\gamma$) the chirality theorem + the mark incidence. Two contract PRECISIONS (prose, no marker): (i) "$\Theta$ induced by the seam reflection" means the seam-circle REFLECTION, not the deck (the deck furnishes the $(-1)^F$ Kramers class — the `v510` bridge); (ii) the marks sit RP-side on the BOND MIDPOINTS of the 16-Majorana circle. **[E]**[**C**][**O**] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v520_celestial_wp5e_measure_decision.py` | `CELEST.WP5E.MEASURE.01`: the WP5e BCOV measure question DECIDED at probe level ($\delta_1$e + $\delta_1$f consolidated, v518 style) — "single-valuedness is derived, the completion reading wins": which of the two exact, contradictory measures on $\mathbb{PT}/\mathbb{Z}_4$ is the true BCOV measure, (A) the DECLARED completion reading (v516) or (B) the DERIVED chiral measure (v518)? ERFOLG-A on the decision layer. [E] INVARIANT SUBSPACE: $\dim_{\mathbb{Q}}\{v : \rho(T)v = v,\ \rho(S)v = v\} = 8 = 4 \times 2$ in the dressed 16-component Weil system, EVERY kernel vector in the $\mathbb{Q}(\zeta_8)$-span of $\{e_H, e_{H'}\}$ (the two Lagrangian gluings, both theta = $E_4$, level-1 quartic = $Q^{(0)}$ = the unphased trace); strict trivial-character solutions at $(m, k) = (0, 0)$: ZERO for every dressing (11 combinations). [E] SOURCE 1 (F-LEMMA): the only strict chiral families $\chi_4$ (dims $(3, 3)$) and $\chi_{10}$ (dims $(1, 1)$) carry nontrivial $T$-characters exactly; pointwise covariance of the canonical column member certified ($3.9 \times 10^{-16}$, $|G| \geq 59.6$); EXACT change of variables ⇒ fundamental-domain independence forces $\int = 0$ for EVERY chiral integrand — the v518 route was never a well-defined nonvanishing moduli integral (kill SHARPENED). [E] SOURCE 2 (QUILLEN PAIRING): $|t\bar{t} - 1| < 8.3 \times 10^{-40}$ on all 15 pairs/both generators; $|\chi_4|^2 = 1$ exact vs one-sided $\chi_4^2 = e(2/3) \neq 1$ ($T$-fixed: $640 \neq 0 \bmod 960$, 15/15 defective); the doubled 256-dim hol × antihol Weil system closes STRICTLY at trivial character and physical weight (nullspaces $(28, 17) \geq 11$), contains the unphased diagonal ($\sim 2 \times 10^{-15}$) AND the physical columns $w_b \otimes \bar{w}_b$ ($\sigma$ $4.9 \times 10^{-16}/8.3 \times 10^{-16}$); negative control hol ⊗ hol: EMPTY nullspace $(0, 0)$. [E] CONSISTENCY CLOSURE: the forced unphased numerator in the v518 scaffold reproduces $\sum_m V \propto (1, 2, 1, 0, 0) = \langle x, x\rangle^2$ exactly on every level $n \leq 8$; $J$-weighted spreads $\sim 5 \times 10^{-41}$, $T_5/T_3 \sim 7 \times 10^{-42}$, zero negative cells; $\psi$(skeleton) = $+0.2302 = -\psi$(contact), the $J$-weighted mirror image of $\psi(A_{\text{fix}}) = +64/\psi$(contact) = $-64$; linearity $4.6 \times 10^{-41}$ — the circle to v516 is CLOSED. [E] D3 $\mathbb{Z}_2$/EH ANCHOR: only the declared reading hits ($9\langle x, x\rangle^2 = (1/4) \times 36$, $\psi = -8$, scale tooth $c = 1/2 = |\mathbb{Z}_2|h^{A_1}$); all three derived instantiations fail (spreads 1.12/0.895/0.516, $\psi$ sign wrong). [E] D1 COLUMN CANONICALISATION: the physical AB column $\sum_m i^{bm} e_{(m,m)}$ is EXACTLY contained in the $\chi_4$ family ($\sigma$ $1.8 \times 10^{-16}/1.7 \times 10^{-16}$, one scalar per orbit — canonicalises the $(N_1, N_2)$ freedom); the canonical member still fails all testers. [E] NEGATIVE CONTROLS: $SO(16)/D_8$ ($T_5 = 20/T_3 = -40$ exact, $\mu_2$-only), wrong form ($|\text{Gauss}|^2 = 64$), wrong weight (dev 0.78), shuffle ($T_5 = -14/T_3 = 36$), planted family detected ($1.5 \times 10^{-16}$), hol ⊗ hol empty. [C] TYPED PREMISES (the mandatory fence): TP-1 (F-independent moduli integral), TP-2 (nonvanishing — must source $\psi = 64$), TP-3 (v518 kernel convention), TP-4 (Quillen structure of BCOV $F_1$). [O] the constructive BCOV derivation of the $w_m$ normalisation itself. VERDICT: ERFOLG-A per the preregistered criteria (both sources independent, the consistency closure carries); the [O] question "which is the true BCOV measure" is DECIDED at probe level in favour of (A) under TP-1..TP-4. `SEAM.EQUIV.TWISTOR.01` stays Open; NO marker moves. [E][C][O] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v521_seam_bit_rp_blind.py` | `SEAM.BIT.RPBLIND.01`: the RP/Θ attack on the alignment bit DECIDED — "free OS positivity does NOT see $\delta$", an honest, decided NEGATIVE result (v508 style): does free reflection positivity (v519) or anti-linear Θ existence break for $\delta \neq \pi/2$ on the deformed mark family $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$, which would derive the v512 alignment bit from OS positivity? NO — the free RP/Θ battery is the EIGHTH side-blind test on the v512 scoreboard ($7 \to 8$); the bit remains genuine discrete input. [E] CENSUS (exact solvesets, symbolic $\delta$): for every $\delta$ exactly the two mark-SWAPPING reflections $\Theta_{\pm e^{i\delta}}$ exist (axes $\delta/2$, $\delta/2 + \pi/2$; all 20 cut/mark incidence solvesets EMPTY); mark-FIXING reflections exist exactly at $\delta = \pi/2$ (solveset $\cos\delta = 0$, lost immediately at $\pi/2 - \varepsilon$); the v510/v512 counterwitness $\{\pm 1, \pm(3{+}4i)/5\}$ carries the swap pair exactly. [E] RP IS $\delta$-BLIND: bond-cut Gram inertias $(8, 0, 0)$ PD for all tested $\delta$, spectra IDENTICAL to 40 digits (deviation 0.0) — the ENTIRE free-RP spectral datum carries ZERO $\delta$ information; the $N = 16$ odd-$m$ failure (site axes, $(3, 3, 1)$, $\det = 0$) is a lattice-PLACEMENT artifact: at $N = 32$ ALL $m = 1..7$ ($\delta = m\pi/8$) have mark-compatible BOND axes with PD inertia $(16, 0, 0)$; continuum: the OS kernel in axis-adapted coordinates is EXACTLY $1/\sin((s{+}t)/2)$, the axis position drops out identically (Cauchy–Stieltjes PD); the counterwitness PASSES free RP (6-point Gram $(6, 0, 0)$). [E] Θ EXISTENCE $\delta$-BLIND: $\Theta^2 = +1$, deck normalisation and Fock implementability ($U^2 = +2^7/2^8 > 0$, exact Cl(16)) for every $\delta$; the ONLY $\delta$-sensitive v519 clause ($\Theta\rho\Theta = \rho^{-1}$) is for $\delta \neq \pi/2$ not violated but NOT FORMULABLE (a mark-compatible order-4 rotation exists iff $\delta = \pi/2$) — it PRESUPPOSES the bit; same for the clock-tower clause. [E] the battery has teeth at every $\delta$: family A/antipode fails RP structurally (zero diagonal, $(4, 4, 0)$ indefinite), chirality–$\eta$ pinning persists ($(0, 8, 0)$ ND) — RP separates the two real-structure FAMILIES, never the SIDES. [E] precondition controls: hexagonal (marks not concyclic, no seam circle), silver (deck not free), lifting audit (B = index relabelling, $\delta$-blind by construction; B' fails the $\pi/2$ negative control — discarded). [C] FACE #13 of the v512 equivalence web: the group generated by the admissible OS reflections is $V_4$ for every $\delta < \pi/2$ and closes to $D_4$ exactly at $\delta = \pi/2$ ($\Theta_i \circ \Theta_1 =$ the clock) — $12 \to 13$ faces, a typed REFORMULATION of the bit, not a derivation. [O] honest gap: marks decorating the STATE (defect/twist insertions, interacting $\mathcal{A}_{\mathrm{hol}}$) = exactly v519's [O] scope — there, and only there, could positivity still see the bit. VERDICT: KILL exactly as preregistered; the alignment bit is NOT derivable from OS positivity and remains genuine discrete input; no marker moves. [E][C][O] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v522_woit_beta1_gso_gauge.py` | `WOIT.BETA1.GSO.01`: the WOIT-$\beta_1$ milestone of the OS twistor bridge `WOIT.OS.TWISTOR.01` executed — "the clock is time-like, GSO is the gauge datum" (typed result: verdict UNDECIDED per the frozen preregistration, mechanism identified exactly, gauge-fixed RP established under the corrected GSO typing). [E] HERMITICITY WITNESS: the $\mu_4$ clock average violates Hermiticity EXACTLY (the one-step insertion $T_1 = \langle\theta(e_a), \alpha_S(e_b)\rangle$ has entry $-i/(8\sin(5\pi/16))$ where the conjugate transpose demands $+i/(8\sin(5\pi/16))$); the pairing is complex-SYMMETRIC (745 entries matching, 96 anti-matching, 0 violations) — "OS-symmetric" is strictly weaker than Hermitian: "positivity after gauge fixing" is not well-posed for the clock reading. [E] TIME-LIKE TYPING: ALL 16 dihedral reflection axes of the NS seam circle invert the clock lift ($RSR^{-1} = S^{-1}$), NONE commutes — the $\mu_4$ clock IS Woit's euclidean rotation (the time translation itself); the gaugeable ($\theta$-compatible) part of the $\mathbb{Z}_8$ tower is EXACTLY its 2-torsion $\{1, (-1)^F\}$ (Hermiticity census over all powers; $\alpha^4$ = the full wrap = the $2\pi$ rotation of the seam = fermion parity, lattice spin-statistics). [E] POSITIVE PART — GSO-FIXED RP HOLDS EXACTLY: $N = 16$ even deg $\leq 2$ parity-projected bond-cut Gram $(29, 0, 0)$ PD (min eigenvalue $1.78 \times 10^{-6}$ at 40 digits); $N = 8$ complete half algebra: the projected Gram kills all odd rows/columns exactly, even block $(8, 0, 0)$ PD ($3.35 \times 10^{-3}$), no degree truncation; the site-cut defect SURVIVES gauge fixing ($(7, 9, 6)$) — the RP bond placement is gauge-invariant information; family A keeps failing ($(17, 12, 0)$ indefinite vs family D's $(29, 0, 0)$); Kramers: $\Theta_t^2 = \mathrm{id}$ on $\mathcal{A}^G$, $\Theta_{\mathrm{Fock}}$ restricts with $\Theta^2 = +1$; Ramond control: the wrap $+1$ "true $\mathbb{Z}_4$" lift fails state invariance — the NS/$\mathbb{Z}_8$ tower is forced. [E] KILL-TEST BOOKKEEPING: kill test (1) stays discharged also gauge-invariantly (RP separates the families on the invariant sector); kill test (2)'s free shadow does NOT fire (no legitimate Hermitian gauge-fixed form has a negative direction); both stay formally live on the interacting algebra $\mathcal{A}_{\mathrm{hol}}$. [E] WRONG-PROJECTION CONTROLS: invariant-dimension chain parity 120 > deck 64 > clock 32 strict, but Hermiticity singles out PARITY alone (deck and clock averages non-Hermitian): the finer "gauge" projections are structurally ill-posed pre-quotient. TRANSPARENCY: the first frozen run scored 6/13 and the preregistered deg-2 invariant dimension 30 was a combinatorial error (correct census: $28 + 4 = 32$) — documented in full; verdict by the frozen [C-3] logic (Hermiticity failure ⇒ UNDECIDED). [C] CONTRACT PRECISION (iii), prose only: "gauge-invariant subalgebra" at the free level means the GSO/fermion-parity $\mathbb{Z}_2$ — the only $\theta$-compatible part of the $\mu_4$ tower (the free-fermion shadow of the $E_8$ glue: $(E_8)_1$ = even NS of $SO(16)_1$ + R spinor); the clock is the euclidean rotation itself (timelike), its equivariant statement moves to $\beta_2$ (OS quotient first, then the clock as reconstructed transfer operator). [O] $\beta_1$ proper stays open via $\beta_2$; `WOIT.OS.TWISTOR.01` stays [O]; no marker moves. [E][C][O] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v523_celestial_wp5e_wm_derived.py` | `CELEST.WP5E.WM.01`: the constructive derivation of the $w_m$ normalisation — "$1/\det_j$ is the Atiyah–Bott / zeta-determinant fixed-point factor, COMPUTED from three independent sources", closing the residual [O] of v516/v520 at the fibre-zero-mode level (verdict ERFOLG per the frozen preregistration). [E] ATIYAH–BOTT (route i): the equivariant mode ledger of the fibre $\mathbb{C}^2$ equals the closed form $1/((1-q\,i^{j})(1-q\,i^{-j}))$ exactly in $\mathbb{Q}(i)$ for all $n \le 120$; REGULAR at $q=1$ with Abel value $(1/2, 1/4, 1/2) = 1/\det_j$, $\det_j = \det(1-g^j) = (2,4,2)$; $(C,2)$ Cesaro convergence in exact arithmetic; the fixed-point factor splits off EXACTLY at every truncation level $d \in \{7, 12, 25\}$, for the phased (skeleton) AND the v520-forced unphased (completion) trace alike — the Abel-limit $\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$ is the v516 contact vector COMPUTED, not declared. [E] ZETA/QUILLEN (routes ii/iii): $\det_\zeta$ of the $g^j$-twisted circle Laplacian $= 4\sin^2(\pi j/4) = \det_j$ EXACTLY (Lerch + reflection formula symbolically; Hurwitz certificates $\sim 10^{-41}$); Quillen split $\det\Delta = \det_j^2$, $|\det\bar\partial| = \det_j$, with the real positive holomorphic section UNIQUE via the $SU(2)$ conjugation pairing; the $\delta_{1f}\ f_1 f_3$ block at the $(0,b)$ nodes carries the exact constant term $1/\det_b$ (deviations $\le 5.2\times10^{-28}$). [E] CONSISTENCY CHAIN: the derived weight reproduces the v516 chain number by number — $w_m = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4|h_m = -4\,\mathrm{ch}_2(T_m)$; $T_5 = 0$ at every scale; $h_2 : h_1 = 4 : 3$; the $T_3$ budget $32 - 8c$ forces $c = 4 = |\mu_4|$ uniquely; per-sector Okubo squares $(18, 9, 18)\langle x,x\rangle^2$; total $45\langle x,x\rangle^2 = \frac{5}{4}\times 36\langle x,x\rangle^2$; $\Phi_{T_3}: 32 \to 0$, $\Phi_P: 72 \to 0$; $\psi(A_{\text{fix}}) = +64$, $\psi(\text{contact}) = -64$. [E] NEGATIVE CONTROLS: $1/\det^2$ gives $w' = (0, 5/8, 1, 5/8)$ with leftovers $(2, 12)$ and a Quillen contradiction; $1/|1-i^j|$ gives $w_1'' = 1+\sqrt{2} \notin \mathbb{Q}$; the $\mathbb{Z}_2$/Eguchi–Hanson anchor PASSES with the same derivation ($\det = 4$, $w = 1/2 = |\mathbb{Z}_2|h^{A_1}$, $9\langle x,x\rangle^2 = \frac{1}{4}\times 36$, $c = 2 = |\mathbb{Z}_2|$); the $SO(16)/D_8$ KILL fires; $\mathrm{diag}(i,i)$ breaks the zeta = AB identity itself; $k = 3, 5$: $w_m = m(k-m)/2 = k\,h_m$ — the weights wander correctly with the centre count, only $k=4$ carries the $E_8$-glue chain. [C] TYPED PREMISES (the mandatory fence): TP-REG (Abel/Cesaro/zeta regularisation IS the definition of the equivariant character), TP-Q (Quillen structure of BCOV $F_1$ = v520 TP-4), TP-NUM (unphased numerator = the v520 ERFOLG-A decision layer), TP-CH (the insertion channel $(0,j)$ computes the box-channel weight). [O] the GLOBAL BCOV integral on $\mathbb{PT}/\mathbb{Z}_4$ beyond the fibre zero-mode factor; the resolved-ALE route (v514/v517) not built; `SEAM.EQUIV.TWISTOR.01` untouched. NO marker moves. [E][C][O] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v524_woit_beta2_os_quotient.py` | `WOIT.BETA2.OS.01`: the WOIT-$\beta_2$ milestone of the OS twistor bridge `WOIT.OS.TWISTOR.01` executed — "the OS quotient of the free system made explicit" (verdict SUCCESS per the frozen preregistration, [C]-typed per contract precision (iii)). [E] $\mathcal{H}_{\text{phys}}$ EXPLICIT: $N = 16$ (deg $\leq 2$): the 37×37 bond-cut OS Gram is exactly Hermitian, parity-block-diagonal, inertia $(37, 0, 0)$ PD (min eigenvalue $1.7801 \times 10^{-6}$ at 40 digits), null space $\{0\}$, $\dim 37 = 29 \oplus 8$; $N = 8$ (complete half algebra): $(16, 0, 0)$ PD (min $3.3471 \times 10^{-3}$), $\dim 16 = 8 \oplus 8 = 4^2$ — compact euclidean time reconstructs a THERMAL (KMS) representation; exact certificate $\sin^2(3\pi/8) - \sin(\pi/8)\sin(5\pi/8) = 1/2$. [E] TRANSFER SEMIGROUP (Klein–Landau): $\tau_k$ exactly Hermitian on all domains ($k = 1..7$, dims $29, 22, 16, 11, 7, 4, 2$), chain identities exact, vacuum fixed; the positivity pattern IS the site/bond dichotomy: even $k$ PSD via the exact square identity $T(2j) = A_j^* A_j$ ($(22, 0, 0)/(11, 0, 0)/(4, 0, 0)$), odd $k$ indefinite ($(10, 12, 7)/(5, 6, 5)/(2, 2, 3)$) — the one-step transfer is NOT positive (the free chirality datum, kill-test-3 shadow sharpened), its even powers ARE. [E] THE CLOCK $= T^{N/4}$: $N = 16$: $\tau_4$ PD $(11, 0, 0)$ (min $6.6821 \times 10^{-5}$), compressed spectrum $[0.0194517, 1.64144]$, vacuum eigenvalue exactly 1; $N = 8$: spectrum EXACTLY $\{1, \sqrt{2} - 1\} = \{1, 1/\delta_S\}$ in both parity sectors — the silver axes of the v519 $\mu_4$ torsor return as the clock-transfer eigenvalue; spectral projections $\sim 10^{-40}$; the reconstructed rotation group $U(s) = e^{isH}$ is unitary with the group law. [E] v522 RESOLUTION: the census $(745, 96, 0)$ reproduced; EVERY anti-matching entry is a wrap overlap, every overlap-free entry matches — the pre-quotient non-Hermiticity is EXACTLY the domain/wrap artifact; on the quotient the clock has spectral calculus. Pre-registered KMS deviation: NO contraction on the compact circle (edge modes $C(1)/C(3) = 1 + \sqrt{2} = \delta_S$ exactly, $\det(G - \tau_4) < 0$). [E] GSO/$\Theta$: the grading survives; the perpendicular torsor mirror descends anti-unitarily ($\eta_\perp \in \{\pm i\}$), $\Theta^2_{\text{phys}} = +1$ on every sector (Kramers-free), $\theta_{\text{cut}} \circ \theta_\perp = \alpha_{N/2}$ exactly. [E] CONTROLS: site cut indefinite ($(3, 3, 1)/(2, 2, 4)$) — the contract kill branch fires there; family A $(4, 4, 0)$; anti-chiral $(8, 8, 0)$ no quotient; wrong direction non-Hermitian (mismatches all wrap-overlap). KILL-TEST BOOKKEEPING: (1), (2) strengthened, (3) shadow sharpened, none fires; (4)–(7) untouched; all seven stay live on $\mathcal{A}_{\text{hol}}$. [C] FENCE: "the clock acting unitarily" is operationalised per contract precision (iii) as the reconstructed transfer/rotation structure. TRANSPARENCY: run 1 scored 18/20 (a sympy simplify failure on a TRUE $\pi/16$ product formula, fixed by a Laurent-polynomial certificate; no criterion changed). [O] $\beta_3$ is the next milestone; `WOIT.OS.TWISTOR.01` stays [O]; no marker moves. [E][C][O] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v525_seam_bit_twist_blind.py` | `SEAM.BIT.TWISTBLIND.01`: the twist-state attack on the alignment bit DECIDED — "mark-decorated states do not see $\delta$", an honest, decided NEGATIVE result (v508 style): the NINTH side-blind test on the v512 scoreboard ($8 \to 9$); the free-plus-twist class (everything Wick-computable) is EXHAUSTED. [E] ALL LIFTINGS WELL-DEFINED AND BLIND: (a) $\sigma$-gauge arc decoration ($\mathbb{Z}_2$): $\sigma \circ r = \sigma$ on the swap axes makes the twisted Gram a diagonal unitary congruence $DMD$ — spectra IDENTICAL to untwisted (dev 0.0); (b) Kadanoff–Ceva 4-twist insertion $\omega(D\,\cdot)/\omega(D)$: well-defined on $N = 32$ ($\omega(D)$ real nonzero; the $N = 16$ odd-$m$ degeneration is the $C(\text{even}) = 0$ checkerboard artifact); the RP SPECTRA depend on $\delta$ for the first time in the program (pairwise up to 1.05), but the INERTIA is $(16, 0, 0)$ PD with the SAME $\eta = +i$ for every $m$ — the decoration sees $\delta$, positivity does not; (c) the $\mu_4$ twist at $\beta = \pi/2$ is pure plane gauge; (c′) genuine Bogoliubov defect $\beta = \pi/4$: $\Theta$-compatibility selects exactly 2 of 16 sign patterns (NS-wrap forced), the winner genuinely non-monomial — still PD, spectra identical. STRUCTURE THEOREM (empirical, mechanism exact): EVERY $\Theta$-compatible quasi-free mark decoration is RP-spectrally INVISIBLE (the defect planes never straddle a cut bond; orthogonal congruence). [E] SUB-RESULTS: the defect energy $E_{\text{def}} = 1.2752872$ for EVERY $m$ (spread $4 \times 10^{-40}$) — $\delta$-blind; the 4-twist Casimir $\ln\lvert\omega(D)\rvert$ measures $\delta$ ($-0.890\,..\,-1.348$) but is exactly mirror-symmetric under $m \leftrightarrow 8 - m$ — a family gauge, never a side selector; Kramers $\langle\Gamma\rangle_{\text{tw}} = +1$ for every $m$. [E] TWO BIT-PRESUPPOSING $\pi/2$ STRUCTURES (the harvest): the $\eta$ flip $+i \to -i$ on the mark-fixing axes ($\sigma \circ r = -\sigma$, odd sector exactly negated); the twist frustration of the mark-fixing mirror ($(4, 4, 0)$ indefinite for both $\eta$ — the lattice shadow of the Ising twist field's mirror-oddness) — both formulable ONLY on the fixing axes existing iff $\delta = \pi/2$: they PRESUPPOSE the bit (v512 facet class), no selector. [E] NEGATIVE CONTROLS: the twist-free limit = v519/v521 exactly; the site cut keeps failing in every decorated state; the v512 counterwitness is arc-equivalent at $N = 16$ to the blind member $\pi/4$ — no state-level selector can exclude it; a single twist has $\omega(D) = 0$ identically (twists exist only in mark pairs); the deck-paired FOUR-twist decoration is the minimal well-defined one. [O] the door narrows to a genuinely INTERACTING $\mathcal{A}_{\text{hol}}$ whose OS data are NOT Pfaffian-reducible to the chiral vacuum — beyond every Wick-computable class. VERDICT: KILL exactly as preregistered; the alignment bit remains genuine discrete input; no marker moves. [E][C][O] |
| `v526_seam_thermal_kms_nariai_bridge.py` | `SEAM.THERMAL.KMS.01`: the thermal seam — KMS of the reconstructed free OS quotient CLOSED onto the Nariai/BH temperature chain (the THIRD LEG of $c_3$). Verdict SUCCESS. [E] $\beta$ MEASURED = $N$ at both levels ($N = 8$: $\omega = \operatorname{arcsinh}(2^{-3/4})$, $\kappa = q^{-8}$ exact; $N = 16$: pairing exponent $(N/2) + 1 = 9$); $T_{\text{seam}} = 1/N$ per clock step, $\beta_{\text{angle}} = 2\pi$ EXACT $= 1/(4c_3)$, $T_{\text{seam}} = 4c_3 =$ Bisognano–Wichmann/Hawking normalisation (v208); modular Hamiltonian closed ($\nu_1 = \cos(\pi/24)$, $\nu_2 = \cos(7\pi/24)$; Tomita KMS-$\beta = 1$ on all 256 pairs); $c_3$ third leg class-1 selected; entropy $S(8/16/32) = 0.52187/0.63510/0.75006$, $\Delta S/\ln 2 \to 1/6$, horizon fractions do NOT transfer; [C] silver $= C(3)/C(1)$ thermal kernel ratio; non-compact $\Rightarrow T = 0$. NO marker moves. [E][C] |
| `v527_seam_clock_silver_spectrum.py` | `SEAM.CLOCK.SILVER.01`: the silver clock eigenvalue EXPLAINED and DEMYSTIFIED (double verdict). STRUCTURE = SUCCESS: $\{1, \sqrt{2} - 1\}$ closed via diagonal Hankel pencil at $N = 8$, $\tan(\pi/8) = C(3)/C(1)$, doubling = exact second quantisation, carrier = silver-midpoint axes, gap $= \operatorname{arcsinh}(1) = \text{gd}^{-1}(\pi/4)$; $N$-scaling = pure $N = 8/16$ lattice datum. CONNECTION = KILL: preregistered 11466-candidate census over 16 frozen numbers: 0 hits above null; placebos comparable. Cross-link v526: $\sqrt{2} - 1 =$ thermal kernel ratio. NO marker moves. [E] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v528_seam_bit_twist_class_definition.py` | `SEAM.BIT.TWISTCLASS.01`: the alignment bit as an exact measurable TWIST-CLASS choice (the physical definition). Verdict SUCCESS. [E] Facets #14 ($\eta$-flip $\Leftrightarrow \delta = \pi/2$) and #15 (twist-frustration $\Leftrightarrow \delta = \pi/2$) both directions exact (cardinality lemma $\lvert A_+\rvert = \lvert A_-\rvert \Rightarrow m = 4$; $g$-class census 224 axes: flip class exactly at $\{15, 31\}$); order parameter $O = 1/2$ at $m = 4$ else 0; unification #14=#15=flag-transitivity = ONE $\mathbb{Z}_2$ invariant (v512 web $13 \to 15$); Kramers = POSITION half (derived), twist class = MODULUS half (the input). [C] measurement sketch (interferometric in principle); the bit remains formal input. NO marker moves. [E][C] |
| `v529_seam_interacting_toy_fk.py` | `SEAM.INT.FKTOY.01`: the minimal interacting Fidkowski–Kitaev seam toy — the FIRST firing Kill-Test-2 shadow (triple-verdicted). F1 = KILL: $\Theta$ exists so Kill-Test 1 does NOT fire; Kill-Test 2 FIRES at toy level — RP breaks for every $g > 0$ (inertia ladder $(37, 0, 0) \to (29, 8, 0)$); STRADDLE LAW (24/24). F2 = KILL: the TENTH side-blind test ($9 \to 10$) — $\Theta_{15}$ maps $H(m) \to H(8 - m)$ EXACTLY. F3 = ERFOLG: grading + $\mathbb{Z}_8$ tower + FK anchor + constructive OS quotient at $\pi/2$. [C] fence: one toy / one interaction class — narrows the Woit route, does NOT close it; `WOIT.OS.TWISTOR.01` stays Open. NO marker moves. [E][C][O] |
| `v530_center_quotient_compiler.py` | center quotient compiler: $Cv{=}v$, $\mathbb{Z}^3/\mathbb{Z}v$ quotient $= \left(\begin{smallmatrix} 8 & 2 \\ 5 & 3 \end{smallmatrix}\right)$ (atoms); row sums $(7, 11, 13)$ self-code; CP code $(1, 7, 11, 13)$; $\det(C{+}kI){=}(k{+}1)\det(A{+}kI)$ |
| `v531_coxeter_tensor_stage_a.py` | Stage A tensor clock: $T_{30}{=}\,\mathrm{Comp}(\Phi_5){\otimes}\,\mathrm{Comp}(\Phi_6)$, order 30, $\chi{=}\Phi_{30}$; traces $c_{30}(k){=}c_5(k)c_6(k)$; atom readout $(2, 4, 8)$ |
| `v532_e8_degree_modular_checksum.py` | dual-degree checksum: $(60, 192, 240, 252)$: sum 744=3·248, prod $\lvert W(E_8)\rvert$, gcd 12; $j{=}q^{-1}{+}744{+}\ldots$; $D_{16}$ control 1488=3·496; primes $< 30$ fingerprint [audit] |
| `v533_ftransfer_disc7_norm.py` | sheet transfer as a disc $=-7$ norm line: $D(t){=}N(\alpha_t)$, $\alpha_{t+1}{=}\alpha_t{+}2$, norms $(2, 4, 14, 32)$ on $J{\to}K{\to}C{\to}F$ |
| `v534_seam_straddle_cone.py` | `SEAM.STRADDLE.CONE.01`: the RP cone on the interacting FK quartets — the FIRST DYNAMICAL SELECTION of the alignment bit (double-verdicted). V1 = KILL: the literal leading-order cone is EMPTY everywhere (the linear pencil cannot carry the selection). V2 = ERFOLG — THE SELECTION LAW: reflection positivity holds for EXACTLY ONE member × sign of the equivariant FK family — $\delta = \pi/2$ with positive coupling, PD $(37, 0, 0)$ on all four cuts incl. the straddled clock axes 7/15 for $g \in \{1/32..8\}$, minimum eigenvalue LIFTED off the RP boundary; asymmetric members die. Exact geometry: $\pi/2$ is the unique symmetric-straddle member (quartet partition, reflection-closed straddles, sign +1). [C] fence: one toy / one interaction class — NO marker moves |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v561_cp_channel_identities.py` | CP channel/invariant identities (T177): the hexagonal channel identity — on the CM unit $\rho = e^{i\pi/3}$ the $E_8$ split $248 = 120 + 128$ has an exact fiber realisation, $\sum_k \rho^{d_k} = 4\rho$ (arg $= \pi/3$ exactly, sheet-EVEN because all degrees are even) and $\sum_k \rho^{m_k} = 4$ (arg $= 0$ exactly, sheet-ODD because all exponents are odd) — the phase-channel angle is sheet-blind; $\arg(\prod c) = 0$ exactly on both TFPT magnitude sectors and the $\mu_3$ branch group is the kernel because its order divides $N_{\text{fam}} = 3$ (the `REDTEAM.D.01` residual is a *fiber* datum, not a magnitude datum); Tier-1 invariance is EMPTY by the blindness theorem (both kernels commute with conjugation, so every $\pi/3$-carrier is $\kappa$-odd); the frame removal — $\operatorname{Im}\det(\mathbf{1}, d, \rho n) = 21\sin(\pi/3)$ is homogeneous of degree $+1$ in the frame normalisation (frame data), while $\operatorname{Im}(\rho\det)/\det = \operatorname{Im}\rho = \sin(\pi/3)$ exactly (the invariant core is the fiber datum, the sign a declared orientation); and the two inequivalent Tier-2 characters $(-1,+1)$ vs $(-1,-1)$ carrying the same $\pi/3$ (the DEGENERATE verdict as algebra, neither frame-coupled), anchored by Jarlskog 1985 ($J = 3.327 \times 10^{-5}$ = `v88`, passing rephasing/CP-oddness, failing only the $\pi/3$-carrier test) — four ledger readings narrowed, no marker moves, no CP derivation claimed; R1/R2 typed OPEN **[E]** |
| `v564_cp_fiber_pin.py` | The frame door of the CP mirror, decided (T178): the Euler factorisation — the fiber acts by a scalar on the $n$-row, so $P(\rho n) = \rho^m P(n)$ for fully generic homogeneous blocks and every scale-free functional is $\sum_k A_k \operatorname{Im}(\rho^k)$ with real degree-0 frame ratios $A_k$; the natural (metric-free) class is FIBER-ONLY by theorem (the bracket enumeration on $\{\mathbf{1}, d, n\}$ has one generator: six permutations $= \pm\Delta$, the other 21 vanish identically in a generic $n$; Weyl 1946 FFT the cited class boundary) — T177's R1 obstruction is now a theorem; the metric door is NON-CANONICAL ($\Delta/\langle d, n\rangle$, $\Delta/\langle \mathbf{1}, n\rangle$ at $(-1,-1,+1)$ vs $\Delta^2/\langle n, n\rangle$ at $(-1,-1,-1)$; the continuum $q_t = t\,n_1^2 + n_2^2 + n_3^2$ gives values $441\sin(\pi/3)/(25t + 117)$ — no canonical frame factor, anti-numerology wired as a check); the SINGLE-SINE LEMMA — $\{|\operatorname{Im}\rho_n^k|\}$ has one nonzero element iff $n \in \{1, 2, 3, 4, 6\}$ (crystallographic, $\varphi(n) \le 2$), by $\sin(4\pi/n) - \sin(2\pi/n) = 2\sin(\pi/n)\cos(3\pi/n)$: the two compiler clocks $\mu_4$ ($j = 1728$) and $\mu_6$ ($j = 0$) are the two maximal single-sine fibers, so T177's DEGENERATE verdict was FORCED; and the ANCHOR SELECTION — Jarlskog carries the full tuple $(-1,-1,-1)$ (certified at 40 dps), matching the sheet-odd orientation class and not the sheet-even channel class: the unique anchor-matching canonical carrier is the fiber-pure $\operatorname{Im}\rho = \sin(\pi/3)$ — R1 decided, R2 selected conditional on the anchor, no marker moves **[E]** |
| `v565_woit_beta3_pt_duality.py` | The WOIT-$\beta_3$ milestone executed: the $\mathbb{PT} \leftrightarrow \mathbb{PT}^*$ duality typed against $\sigma_{\text{std}}$ — clock compatibility forces the two-block Hermitian class ($16 \to 8$ parameters); the $\rho_{\text{tw}}$ pairing splits it into an EUCLIDEAN branch ($\sigma_{\text{std}}$'s home; seam norm $a(|z|^2 + 1)$, no OS cut ever) and a MINKOWSKI branch whose null locus meets the seam line EXACTLY in the seam circle (the OS cut forced) with total signature $(2, 2)$ by the mirror theorem ($\Phi_1 = -\overline{\Phi_i}$, charpoly mirrored) — Woit's Minkowski signature a theorem, not an input; the kill branch is EMPTY BY ALGEBRA (every compatible duality inverts clock and deck — family D; family A would need $\rho_4^2 =$ scalar); every seam-preserving Minkowski member induces the SAME conjugation $z \to -\bar{z}$ ($\mu = -1$, fixed points $\pm i$), perpendicular to $\sigma_{\text{std}}$'s $z \to +\bar{z}$, and under the declared RP-side placement it IS $\theta_\perp$ — the v524 $\Theta_{\text{phys}}$ carrier (regressions re-verified: Gram PD, anti-unitary, $\Theta^2 = +1$ per sector, time reversal, torsor closure onto the deck); verdict SUCCESS, the contract stays **[O]**, $\gamma$ is next **[E][C][O]** |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v566_parabolic_anchor_selfcode.py` | The parabolic anchor self-code: the sheet-diamond directions $U = Q\,\mathrm{diag}(1,0,0)$, $V = Q\,\mathrm{diag}(0,1,1)$ generate the FULL 2\|1 parabolic algebra $\mathcal{A}$ (dim 7, word determinant $-81$, multiplicatively closed); the anchor $a = (1,1,2)$ is the spectrum of $H = I + V(V-I)/2$ and $\chi_H = x^3 - 4x^2 + 5x - 2$ carries the DOUBLE code $(4,5,2) = (\|\mu_4\|, g_{\mathrm{car}}, \|\mathbb{Z}_2\|) = (\mathrm{flag}, \mathrm{Levi}, \mathrm{radical})$; the uniqueness theorem $(x-m)(x^2 - mx + 1)$ forces $m = 2$ — $a = (1,1,2)$ is the only anchor this family can code; the saturated integral order has index $[P_{\mathbb{Z}} : \mathbb{Z}\langle U, V\rangle] = 3^4 = 81$ = the v218 determinant-staircase sum (cokernel $(\mathbb{Z}/3)^4$; mark-local mechanism [O]); the basic algebra is the $A_2$ path algebra with $\dim \mathrm{Ext}^1 = 1$ (two module classes: the alignment bit gains a canonical carrier, selection stays with RP); $p_n(a) = 1 + \mathrm{tr}(V^n)$ (AnchorLadder re-encoded, 240 and rank 8 as trace readouts); the CR $= 4/3$ preregistered test for `CONTRACT.F.01`; the Lorentz/pencil form of the Paper-II rank-3 classification — NO P2 reduction (the 'No Carrier Reconstruction' audit is the named [O] door) [E][C][O] |
| `v567_carrier_free_q_reconstruction.py` | The No-Carrier-Reconstruction audit (Stage A): with the v11 carrier budgets FORBIDDEN (rows $(4,5,6)$, columns $(9,5,1)$, $\chi_Q$ with its 5), the carrier-free inputs — sheet involution, winding normalisation $Q(I+\Sigma) = 2N_{\mathrm{fam}}W$, $A_3$ exponent spectrum $\{1,2,3\}$, monodromy budget $Q_-^2 = N_{\mathrm{fam}}P$, integrality, the seam-line demand — admit exactly $52 \to 28$ (parabolic) reconstructions, and ALL 28 code the SAME anchor ($\chi_{H'} = x^3 - 4x^2 + 5x - 2$, spectrum $(1,1,2) = a$): **the anchor self-code is carrier-free**; the single family invariant (saturation index $N_{\mathrm{fam}}^4 = 81$) then pins $Q$ to a TWO-element residual family ($q_{32} \in \{2 = \|\mathbb{Z}_2\|, 0\}$, difference $2E_{32}$) on which every classical carrier-free certificate coincides — the one separating certificate is the forbidden column budget $(9,5,1)$: one binary sheet-coupling choice remains (typed resonance with the split/nonsplit alignment bit). Winding ablation $28 \to 1376$; bound-stable; verdict RECONSTRUCTS-TO-FAMILY; the audit relocates to `GATE.QGEO.01` and the binary selector — no P2 reduction, markers untouched [E][C][O] |
| `v568_bit_selector_ladder.py` | The reconstruction bit located and given its canonical witness: the two 81-twins generate DIFFERENT integral orders (equal index, different lattices — the bit is integrally visible); the dominant mode of the sheet direction is $v(c) = (1, 2, 2c)$ identically, and the LADDER SELF-CONSISTENCY — the dominant mode reproduces its own eigenvalue as the geometric ladder $(2^0, 2^1, 2^2) = (1,2,4)$ — selects $c = 2$ (the compiler $Q$) UNIQUELY on all 28 members of the carrier-free class, with the witness anchored in prior structure (AnchorLadder steps $2^n$, $2 = \|\mathbb{Z}_2\|$, the trace code); the split geometry sits at $q_{32} = 1$ and is EXCLUDED (algebra dim 5; the $A_2$ Ext arrow is nonzero on the whole class — the bit toggles WHICH nonsplit order, not split/nonsplit); Priority-2 candidates: the $\mu_4$-grading fails, the $\mathbb{F}_3[V]$-action is computed — charpoly $x(x-1)^2(x+1)$ mod 3, NOT the regular $\mu_4$ pattern; the 81 stays a fingerprint — the physical selection stays open (the RP-selection slot), no P2 reduction, markers untouched [E][C][O] |
| `v571_cr_discrete_corners.py` | The preregistered cross-ratio check, discrete half executed: the corners $J, K, C, F = M(1,t)$ ($t = -2..1$, dets $(2,4,14,32)$) read $(26,35,44,53)$ under the declared Koide functional $\ell(M) = a^T M \mathbf{1}$ — CR $= 4/3$ exactly, and $y_F = 53$ is Möbius-forced by the three upstream corners (fit-free); $\ell$ is affine on the path (step $9 = N_{\mathrm{fam}}^2$; every linear functional passes — stated honestly); negative controls: determinants 28/25, dominant eigenvalue 1.3222; the continuous solver states stay open per `CONTRACT.F.01` [E][C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v572_rp_bit_form.py` | The RP form on the 81-twins: spectrally blind (identical inertia for every declared state; never PSD — no kinematic positivity selection, honest negative) but integrally ANCHOR-VISIBLE — the Smith towers of the anchor-state Gram separate the twins in one slot (18 vs 2, a factor $9 = N_{\text{fam}}^2$; ladder state identical, anonymous states blind); trace discriminant $-81^2$ on both; blindness robust under the compiler $V$-flow, but the anchor-Hamiltonian GIBBS PENCIL separates: the compiler twin is Gibbs-critical at $s^* = (8 - 4\sqrt{19})/15$ (det constant $81^2$), the alt never — a rank-drop marker whose critical mode is the seam-row, sheet-flow-fixed direction (the $A_2$-Ext carrier); the positivity selection stays open [E] |
| `v574_stageb_winding.py` | Stage B, first narrowing: the involution TYPE is forced (rank theorem — a 2-dim-positive involution cannot carry the binary trace code), $(E)$ bounds $N \leq 3$, the ladder self-consistency selects one member per $N$, and sat $= N^4$ holds ONLY at $N = 3$ ($32 \neq 1$, $512 \neq 16$, $81 = 3^4$): the double self-consistency pins $(N, Q) = (3, \text{compiler } Q)$ — $N_{\text{fam}} = 3$ relocates from declared to consistency-forced; spectrum ablation: the surviving family is exactly $\{1, 2, N\}$ with the compiler shape — the self-consistency forces the sheet block, $(E)$ contributes $N = 3$; redundancy: $(M)$ and $(L)$ fall relative to the self-code demands (class 1274, compiler $Q$ still unique) — the maximal ablation drops $(E)$ too — winding column + integrality + self-code pin the compiler alone; declared: involution existence, the winding UNIFORMITY (without it a four-column family survives, exactly one uniform — the compiler), $N_{\text{fam}} = 3$ (E8-glue); `GATE.QGEO.01` narrowed by five items, not closed, no P2 reduction [E][C] |
| `v575_cr_continuous_uniflow.py` | The continuous half of the cross-ratio check, executed: the native exports $\{(2/3)^6, (2/3)^6, 1, -7\}$ are forced degenerate by the single-flow theorem (v425), and a repeated quadruple gives CR $\in \{0, 1, \infty\}$ in every assignment — never $4/3$: the continuous half closes onto the single-flow reduction; the anchors stay [O] walls, the discrete half (v571) carries the content [E][C] |
| `v578_native_jets.py` | The jet contract executed natively: the seam flow has $\{q, t\} = -\Delta^2/2$ exactly (Moebius in its own clock $u = e^{\Delta t}$), the 1-loop QCD flow has Schwarzian 0 in its own RG time — no shared wall-time jet invariant; each instance projective in its own clock, the obstruction is the clock; external anchor-level jets remain the future test [E][C] |
| `v584_ext_source_response.py` | The interacting follow-up of v572: rank-one Ext-source coupling on the Gibbs pencil separates the 81-twins directionally inside the unit coupling box – exact rational flip thresholds whose numerators are the v572 pencil factors; positive coupling $J \leq 15/16 < 1$ heals exactly the compiler twin, the alt twin is rigid; the response pole sits on the golden polynomial $s^2 - s - 1$; a response-level selection, not a positivity theorem [E][C] |
| `v590_involution_existence.py` | The sheet-involution existence of GATE.QGEO.01: the integral $V$-commuting involutions are exactly the spectral sign group $(\mathbb{Z}/2)^3$ on both twins (rigidity); the v574 type theorem, seam reversal and zero-mode fixing single out $S_0$ uniquely with exact $\Sigma$-signature – existence is unconditional; the two demands' continuum derivation stays the named open [E][C] |
| `v597_mu4_cover_model.py` | Strategy I/II first step: the $\mu_3$-cover $y^3 = x^4 - 1$ carries four compiler fingerprints – rank-3 Eisenstein module, projective $D_4$ via the deck ($r^4 = \omega$), cusp-class group $(\mathbb{Z}/3)^4$ of order 81, and the unique invariant hermitian form of Lorentz signature $(1, 2)$ matching Chevalley–Weil and Paper II; the $Q/V$ dictionary stays the named open [E][C] |
| `v598_cover_dictionary.py` | The cover dictionary, round 2: three compiler objects located exactly – $U \sim G_2$, $Q_- \sim G_1 - G_3$ (the sheet coupling as the diamond-diagonal Gram difference), the binary ladder $\sim 1 + G_1 + G_3$; opposite-twist products nilpotent; the grading half absent from 7296 scanned canonical combinations (kill not triggered; the real structure is the named suspect) [E][C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v599_real_structure.py` | The real structure on the $\mu_3$-cover: the anti-holomorphic $D_4$ reflection constructed exactly ($R$ unique up to scale, $R^2 = +1$, four twist intertwinings); the grading functor closes the three v598 dictionary gaps at char level ($V$, $H$ with the anchor spectrum $(1, 1, 2)$, $Q_+$); the $\Gamma$-decomposition carries 5, 9, 27; the joint embedding stays open (quantified negative); the invariant form has $\det J = 8$ and Lorentz signature $(1, 2)$ [E][C] |
| `v600_joint_embedding.py` | The joint embedding: the compiler pair $(U, V)$ realized on the $\mu_3$-cover as a $\Gamma$-even degree-2 pair — $\mathbb{Z}[\omega]$-integral, seven-word rank 7, simultaneous conjugation to the compiler pair solved exactly, integer restrictions on the saturated $\mathrm{Fix}(R)$ lattice, and saturation index $81 = 3^4$ with the exact v566 divisor pattern $(1, 1, 1, 3, 3, 3, 3)$; canonicity typed open [E][C] |
| `v601_equivariant_dual.py` | The canonical equivariant realization: the J-adapted cusp projectors rotate exactly under $D_4$; the pair family $U_k = 3\,GJ_k$, $V_k = 1 + GJ_{k-1} - GJ_kGJ_{k+1}$ is $\mathbb{Z}[\omega]$-integral, compiler-exact and equivariant by construction (mark choice = winding choice); it realizes the DUAL parabolic exactly (conjugate to the transposed pair; the direct intertwiner is singular), and the v590 Klein four-group lives integrally on the curve; the Weil-twisted RP comparison stays open [E][C] |
| `v602_duality_forms.py` | The duality-and-forms round: the J-adjoint exchanges the two parabolic types inside one equivariant structure; the dictionary completes in the J-projector language (the corpus anchor formula $H = 1 + V(V-1)/2$ holds verbatim); Weil positivity holds exactly ($JC$ positive definite, the Riemann bilinear relations on the $\omega$-sheet); the canonical bilinear form $B = M^\dagger J$ is symmetric with $\det B = 8$; the corpus-transpose and mark-local questions typed open; Lean certificate `FORM.QGEO.04` [E][C] |
| `v603_seam_marks.py` | The seam-and-marks round: the seam line is a REAL line of the cover (unique common eigenline, R-fixed, $S_0$-reversed — the v590 seam-reversal demand geometric); the infinity cusp separates in the unreduced Burau module; the first per-mark $\mathbb{Z}/3$ witness lands ($E_1$, class order 3 in the word-order cokernel); the corpus-transpose translator negatives quantified [E][C] |
| `v604_equivariant_order.py` | The equivariant order and the infinity bridge: $R$ IS an anti-isometry ($M^\dagger JM = J^T$ exact; correction of a diary side remark), $J$ restricts to the integer form $J_{\mathrm{fix}}$ with $\det = 72$; the four mark frames generate a FULL order of index $27 = 3^3$ in $M_3(\mathbb{Z}[\omega])$; the 3-torsion localizes at infinity ($\mathrm{coker}(1 - T_\infty) = \mathbb{Z}^2 \oplus (\mathbb{Z}/3)^3$, finite cusps torsion-free): 81 at the marks, 27 at infinity; translator negatives extended (the $\bar{\omega}$-sheet named) [E][C] |
| `v605_translator.py` | The translator round: the corpus RP anti-automorphism IS the J-adjoint twisted by an explicit inner element $c$ ($\varphi(A) = cAc^{-1}$ exact, intertwiner space 1-dim, $14c$ integral over $\mathbb{Z}[\omega]$); the seam is a NONZERO pure 3-torsion class at infinity (torsion coords $(2, 2, 0)$); the 81 survives frame-joining as the saturation defect (chain indices $3^5$, $3^4$); the $J_{\mathrm{fix}}$ duality certified with a Lean addendum [E][C] |
| `v606_mode_separation.py` | The mode-separation round: all three $V$-eigenlines lift to PURE 3-torsion classes at infinity with PAIRWISE DISTINCT classes (the selection carrier the real structure could not provide; the canonical '+1' rule stays open); and the cubic field of the translator element has discriminant $-783 = -3^3 \cdot 29$, Galois $S_3$ — no cyclotomic identity for $c$ (honest negative) [E][C] |
| `v607_selection_rule.py` | The selection rule: the infinity torsion is a genuine $\mathbb{F}_3$ space, the deck rotation acts with char $(x + 1)(x^2 + 1)$ mod 3 ($x^2 + 1$ irreducible) — EXACTLY ONE projective eigenline, and it IS the top mode; the three modes carry mutually exclusive canonical labels (seam = $R$-real, top = deck-eigen, zero = unlabeled) and $S_0$ flips exactly the labeled pair: both v590 demands expressed through cover structure [E][C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v608_gamma_free_slice.py` | The free gamma slice: mirror modes are STRICTLY OS-negative (the anti-chiral odd sector is negative definite – no mirror survives reconstruction, kill test 3 cannot fire at the free level); the chiral generation is multiplicity-free (clock spectrum $\{1, \sqrt{2}-1\}$); and the mark transport exists ($\tau_2$ is the mark-A→B transport, PD) – kill test 7 does not fire at the free level; the gamma milestone stays open on $\mathcal{A}_{\text{hol}}$ [E][C] |
| `v609_translator_factorization.py` | The translator factorization: the mirror intertwiner $d$ ($dA = \Gamma(A)d$) is UNIQUE, unimodular, $\mathbb{Z}[\omega]$-integral and $\Gamma$-involutive ($\Gamma(d)d = 1$); the v605 element factorizes exactly, $c = \lambda K_1 d$ with $K_1 = J^{-1}\overline{F_2}^{T} M^{-1}$ — $\theta_{\text{corpus}} = J\text{adj} \circ \text{inner}(K_1) \circ \text{inner}(d)$: the geometric identity of $c$ is closed [E][C] |
| `v610_curve_landscape.py` | The curve landscape: the Jacobian splits (differential characters of orders $(12, 12, 6)$) into a $\mathbb{Q}(\zeta_{12})$-CM abelian surface — the field of deck AND marks — times the $\mathbb{Q}(\omega)$-CM elliptic curve $E : y^3 = u^2 - 1$; point counts confirm the splitting ($a_p(C) = a_p(E)$ for all tested $p \not\equiv 1 \bmod 12$); the integer-spectrum census honestly relativizes single-char hits (the anchor class is the most frequent nontrivial family; the compiler's distinction is the joint structure) [E][C] |
| `v611_periods.py` | The periods round: the five basic period integrals are EXACT Beta values with all Gamma denominators on the $\zeta_{12}$ grid; the $\mu_4$ rotation acts on periods with the exact characters $i^{k+1}$; $\Omega_E = B(1/3, 1/6)/2$ exactly (the classical $\omega$-CM period); the cross-factor ratio admits no low-degree integer relation (PSLQ) — the two CM factors are period-independent, as the v610 splitting demands [E][C] |
| `v612_polarization_frame.py` | The polarization frame: $r$ has three simple eigenvalues $\zeta_{12} \cdot \{1, -1, -i\}$; in the rotation eigenbasis $J$ becomes exactly diagonal with signature $(1, 2)$, and THE UNIQUE POSITIVE (Hodge) DIRECTION IS THE PRIMITIVE $\zeta_{12}$ EIGENSPACE — the analytic Hodge structure and the finite rotation agree on where positivity lives; the canonical period normalization (Beta weights) is the named next step [E][C] |
| `v613_canonical_periods.py` | The canonical period normalization – the analytic bridge closes: the quarter rotation acts DECK-FREE on the four branch segments (12 wrap factors exactly 1); THE EXACT DICTIONARY $r = \text{deck} \circ \text{rotation}$: $\text{char}(r) = \text{char}(\omega M)$ exactly, explaining $r^4 = \omega$ structurally; the period rows (hol $dx/y^2$, hol $x\,dx/y^2$, antihol $\overline{dx/y}$) realize the rotation eigenframe; the Hodge weights are explicit Gamma monomials (complex Beta formula, $h_1 = (3\sqrt{3}/16\pi)\Gamma(1/3)^2\Gamma(1/6)^2$); and $J = -h$ PER CHARACTER: the compiler polarization is minus the analytic Hodge form — Riemann–Hodge positivity realized [E][C] |
| `v614_transport_deck.py` | The Gauss-Manin transport round – THE DECK STEP IS THE BOUNDARY MONODROMY: transporting the segment cycles along the rigid quarter-rotation loop, the discriminant $\lambda(\tau) = e^{2\pi i\tau}$ winds once and the periods factorize exactly; at $\tau = 1$ the deck factor is $\omega = t^4$ on every sheet (the exponent $4 = \lvert\mu_4\rvert$ forced by the puncture count); the canonical transport matrix is EXACTLY $\omega M$ — the v613 dictionary $r \sim \omega M$ is CANONICAL, and $(\omega M)^4 = \omega$ re-derives $r^4 = \omega$ [E][C] |
| `v615_gamma_toy_interacting.py` | The interacting gamma slice at toy level: the gamma battery executed ON the RP-surviving interaction (the alignment bit, v534) – THE INTERACTING MIRROR FLIP: at the forced twist the mirror state's odd sector is STRICTLY NEGATIVE DEFINITE on every cut and every coupling (kill test (3) cannot fire on the surviving class – v608 upgraded from g = 0 to the full grid); the mark transport exists interacting (kill test (7) does not fire); the odd transfer spectrum stays multiplicity-free; the dead member has no RP home: bit selection and chirality data CO-LOCATE [E][C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v616_su2_internal_kinematic.py` | The internal $SU(2)$, kinematic separation at twistor level: Woit's Euclidean structure IS the internal $j$-direction ($\rho_{\text{tw}}$ = right-$j$ verbatim); the internal $\mathfrak{su}(2)$ preserves every twistor fiber, commutes exactly with the spacetime quaternion action and intersects it in $\{0\}$ — kill test (6)'s KINEMATIC branch does not fire; THE CLOCK FACTORIZES as left-$u$∘right-$u$ exactly ($u = e^{i\pi/4}$: spacetime half-quarter × internal half-quarter), the ALE deck is purely spacetime-side, and the complex choice breaks the internal $SU(2)$ to $U(1)$ with the v519 mark torsor as its $\mu_4$ [E][C] |
| `v617_seam_cover_forcing.py` | The seam forcing round — THE CONFORMAL SEAM AXIOMS FORCE THE $\mu_3$-COVER: $\mathbb{Z}_4$-Möbius rigidity (every $\mathbb{Z}_4$-cyclic configuration is Möbius-equivalent to $\mu_4$; harmonic cross-ratio orbit $\{2, 1/2, -1\}$), the weight census forces UNIFORM weights $j \in \{1, 2\}$ (the conjugate sheet pair), $j = 1$ gives $y^3 = x^4 - 1$ exactly with genus 3, and the seam reflections match the cover's real structure (the v599 $(k, 5-k)$ pattern, the two v519 cut points); the bedrock residue narrows to the cyclic-3 choice and the physical↔conformal seam identification [E][C] |
| `v620_cyclic_n_census.py` | The cyclic-$N$ census — the seam admits EXACTLY $N \in \{3, 5\}$ as primitive-Lorentz orders (full infinity ramification iff $\gcd(N, 4) = 1$; Chevalley–Weil signatures; $N = 6$ reduces to $N = 3$), and the compiler's machine-checked integral constants ($\det_{\mathbb{Z}}(1 - \text{deck}) = 27 = 3^3$, saturation $81 = 3^4$) pin $N = 3$ UNIQUELY: the v617 bedrock residue (i) is now conditional on $N_{\text{fam}} = 3$ ($E_8$-anchored) [E][C] |
| `v621_interacting_semigroup.py` | The interacting Klein–Landau semigroup — at $g = 0$ the v524 pattern reproduces exactly at $N = 16$; at $g > 0$ exact Hermiticity survives PRECISELY on the $\mu_4$-commensurate steps $\{4, 6, 7\}$ and the clock step stays a positive transfer $((11, 0, 0))$ over the whole grid: the interaction breaks translation invariance to the quartet stabilizer exactly, and the reconstructed rotation group of the interacting toy is the CLOCK tower, not the full circle [E][C] |
| `v622_seam_identification.py` | The seam identification round — THE PHYSICAL SEAM IS THE CONFORMAL SEAM: the v519 kernel is EXACTLY the discrete NS chiral mode sum (including the even-distance zeros — the flat band IS the NS spectrum; Ramond must-fail), the mark-bond midpoints sit exactly at $\mu_4$, the group dictionary is exact (clock = $z \mapsto iz$, cuts = diameters), and the straddle law is literal geometry (straddled = axis through marks); the remaining residue is the covering-level identification [E][C] |
| `v623_covered_seam.py` | The covered seam round — the lattice skeleton of the covering-level identification: the lifted seam is ONE 48-site NS circle for uniform weights (the alternating pair DISCONNECTS it — a new independent kill), the lifted clock has EXACT ORDER 12 with $L^4 =$ deck ($r^4 = \omega$ as lattice combinatorics), the v622 kernel identity generalizes to $N = 48$, and the deck-character mode grid is the 12-grid with the untwisted sector equal to the base seam verbatim and the twisted pair carrying $\omega$ on bilinears [E][C] |
| `v624_external_lattice_audit.py` | The external-review lattice audit: THE FOUR-MARK CLOSURE (three independent selectors force $d = 4$ under the $D_{d+1} \oplus A_{d-1}$ glue architecture — disc-group census, the norm equation $(d{+}1)/4 + (d{-}1)/d = 2$, the Pythagorean $(3, 4, 5)$ selector; the two-input presentation reduces to ONE boundary degree plus $\pi$, conditional on the architecture); THE ANCHOR BYTECODE ($\chi = (t{-}1)^2(t{-}2)$, two-state recursion, $240 + 8 = 248$); and THE LORENTZ CONGRUENCE ($P^{\mathsf{T}} J_{\det} P = J_{\text{fix}}$ exactly, index 6) — the prime-front determinant form and the cover polarization lattice are the SAME rational quadratic form; the review's $B$-test is retyped as ill-posed ($B$ complex symmetric) [E][C] |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v626_e8_code.py` | The E8 code round – E8 IS an error-correcting code: Construction A on the self-dual extended Hamming code $[8, 4, 4]$ yields the even unimodular rank-8 lattice (= E8, by uniqueness) with shells 240/2160 matching the v625 theta identity; every single-bit error is exhaustively correctable; the compiler ties ($8 =$ rank, $4 = d = \lvert\mu_4\rvert$, $16 =$ half-spinor, $14 = 2 \times 7$) are typed observations. |
| `v628_orbifold_ casimir.py` | The orbifold Casimir round – the covered seam's three deck-mode classes carry EXACTLY the $\mathbb{Z}_3$-twist Casimir data: exact closed form $E_0(\nu, N) = -\cos(\pi(2\nu-1)/2N)/(2\sin(\pi/2N))$, finite-size coefficient exactly $\pi(6\nu^2 - 6\nu + 1)/12$, sector table $(1/72, -1/24, 1/72)$, degenerate twisted pair, ratio $-1/3$, twist gap $1/18$. |
| `v629_root_ incidence.py` | The root incidence census – the naive $48 \times 5$ incidence to the E8 roots is KILLED (orbit census $19 \times 12 + 3 \times 4$ for the compiler-canonical order-12 element, not 20 free $\zeta_{12}$ orbits), and the positive residue is sharp: the $\mu_4$ clock alone acts freely with EXACTLY 60 orbits ($60 = D_{\text{start}}$), and the family 3-cycle fixes exactly 12 roots (the lifted marks). |
| `v632_ftransfer_ pgl2.py` | The $F_{\text{transfer}}$ $\mathrm{PGL}_2$ round – the v533 preregistration executed in its literal group form: the four external transfer solvers share ONE Moebius action on the disc $-7$ norm line where they act at all (the Koide reading is the parabolic step $+9 = N_{\text{fam}}^2$, exactly intertwined with the base translation $T$; $g_{\text{QCD}}$ explicitly conjugate to $T$ in the $1/\alpha$ coordinate; $g_{\text{Boltzmann}} = g_{\text{pole}}$ by the single-flow theorem; relic degenerate); the kill criterion 'four incompatible arithmetic actions' is NOT met; five must-fail controls have teeth. |
| `v633_orbit60_ quotient.py` | The $\mu_4$-quotient round – the quotient of the 240 E8 roots by the free $\mu_4$ clock IS the hermitian-unimodular $\mathbb{Z}[i]$-E8 with the 60-hyperplane system of ST31 (hermitian-unimodular basis among the reps, $\sigma$ census $19 \times 3 + 3$ on the lines, line group of order $11520 = 46080/4$; ST32 killed by its 40-hyperplane count); the $60 \to 6$ cascade route on the quotient is KILLED (Gram rank 4, regular graphs, no canonical pair rule) – a typed negative. |
| `v634_st31_ structure.py` | The ST31 structure round – $G_{31}$ is the FULL unitary stabilizer of the $\mu_4$-quotient structure (exact count 46080), the degrees $(8, 12, 20, 24)$ are unique via Molien, the compiler data embeds canonically ($\sigma = c^4$, $J = c^9$, $\langle c\rangle = Z_{12}$), and the computed structure is $G_{31} = (Z_4 \circ 2^{1+4}).\mathrm{Sp}_4(2)$ – while the numerology $\lvert G_{31}\rvert = \lvert W(D_5)\rvert \times \lvert W(A_3)\rvert$ carries NO structure (not isomorphic, not a (semi)direct product, and $W(D_5)$ does not even embed: minimal faithful dimension $5 > 4$); HONEST: the v629 compiler clock is NOT the Springer-regular 12-element (census $19 \times 12 + 3 \times 4$, $\lvert C(c)\rvert = 72$). |
| `v638_code_ semantics.py` | The code-semantics rounds (kill + stage 2) – the unique compiler-equivariant Hamming placement $C^*$ IS Reed–Muller $\mathrm{RM}(1, 3)$ on $\mathrm{AG}(3, 2)$ and carries an exact dictionary: information bits one per $\mu_4$ pair (3 families cycled + 1 anchor fixed; the unique minimal $Z_6$-orbit of information sets), syndrome = in-pair bit + pair pointer with the $\sigma$ 3-cycle (anchor = sum of families), decode = projection verbatim on the single-error class (3840/3840, exact arithmetic) – and all of it dies for every non-equivariant placement (only 2/30 carry the structure). |
| `v639_twist_ope.py` | The twist-field/OPE round – the interacting slice of the $\mathbb{Z}_3$ orbifold statement holds at the abelian vertex level on the seam double: $h_\sigma = 1/36$ from three exact routes (conventions reconciled), lattice two-point exponent $2\Delta = 2/9$ at 0.06%, crossing exactly symmetry-protected on the lattice, closed four-point form $[w(1-w)]^{-2\Delta}$ at $< 0.1\%$, OPE $c_1 = 2/9$ model-bound at 0.19% (model-free $\pm 15\%$ band typed open); reflection positivity validated in the real $\mathbb{Z}_2$ class and honestly OPEN for the complex $\mathbb{Z}_3$ pairing (parafermionic Klein twist); GATE.QGEO does not move. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v645_klein_rp.py` | Parafermionic Klein-twist reflection positivity – the $\mathbb{Z}_3$ twist sector IS reflection positive on the seam-double lattice under the Klein pairing (mirror COMPOSED WITH charge conjugation, $\omega^{kq}$ zero-mode phases, $\eta = (1,1)$ UNIQUE among 36 scanned pairs), resolving the named open item of QGEO.TWISTOPE.01 at the lattice OS level: every charged Klein block is exactly real symmetric AND PSD (worst $2.9 \times 10^{-12}$), $N$-stable 48..384, both diameter axes, including the FULL mixed OS Gram over $\{1, \sigma_1, \sigma_2, \text{dipoles}\}$ with no sector projection; the naive pairing keeps its $N$-stable violation ($-4.5 \times 10^{-3}$, must-fail); continuum OS reconstruction and modularity stay typed open; GATE.QGEO does not move. |
| `v646_rm13_ reverse.py` | The RM(1,3) reverse reading – are compiler numbers CODE operations? An honestly TWO-PART answer: the $\sigma$-fixed syndromes carry the anchor decomposition EXACTLY (coset-leader weights $(0,1,1,2)$, norm-square sum 6, anchor = family sum $P_3 = P_0{+}P_1{+}P_2$ – the SELECTION is equivariance-specific, while the $(1,1,2)$/norm-6 weights are generic for any $[8,4,4]$ placement), but the bytecode sequence $p_n = 2 + 2^n$ has NO code reading (0/11 natural families; 18/27/81 without any hit – the bingo is buried); the kill control fires selectively (the naive placement loses $q$-universality and the $\sigma$ action but keeps the generic sector weights, all pure-code invariants, and even the $\sigma$-fixed-root count 12 – honestly retyped). |
| `v647_st31_ degree24.py` | The ST31 degree-24/-20 round – the $\mu_4$ center is a POWER of both regular clocks ($w^6 = \pm J$ for the regular 24-element with $\chi_w = x^4 + ix^2 - 1$ exact, all four eigenvalues primitive $\zeta_{24}$; $u^5 = \pm J$ central for the regular 20-element), and the clock question $c = w^2$ is KILLED by a PARITY THEOREM: the 12-cycles of any square come in pairs from 24-cycles, and census$(c) = 19{\times}12 + 3{\times}4$ has ODD counts – impossible even in $S_{240}$ (proof-grade); $u^4$ acts freely with census {5:48}, $48 = 240/5$ exactly, lines {5:12}; the equality $48 =$ seam sites stays a typed OBSERVATION. |
| `v649_discipline_ audit.py` | The discipline audit – the suite parses ITSELF and certifies its process discipline as frozen minimum bars: 214 ledger rows document kills / honest negatives / retypes, 1469 must-fail/kill/scramble occurrences across 233 of 643 modules, 47 contract rows (29 naming kill criteria; executed chains FTR.PGL2 v533 → v632 and the W1 chain → v643 verified), deterministic replay 12/12 on the declared sample v55..v648 (two fresh runs, identical up to timing lines), five classical anchors recomputed against the literature (Jacobi $r_4$, Construction A → 240 minimal vectors, the Shephard–Todd G31 degrees, the Suzuki archimedean constants to 25+ digits, the completed-zeta functional equation), and 54 committed Lean proof modules; the honest limit: process discipline, not physics truth – the open gates stay open (278 class-O rows). |
| `v650_orbifold_ modular.py` | The modular data of the $\mathbb{Z}_3$ orbifold of the seam on the lattice: all nine twisted-sector partition functions $Z[g,h]$ match $\lvert\theta[g/3, h/3]/\eta\rvert^2$ at $< 1\%$, the modular S-covariance is MEASURED with a clean common $N^{-4}$ rate (the $N^{-2}$ lattice artifact is itself S-covariant and cancels in the S-relation), T is the exact discrete Dehn twist $(g,h) \to (g, g{+}h) \times (-1)^F$ at machine precision, the twisted characters carry $h_\sigma = 1/36$ with the parafermionic 1/6 gap, and deck$^3 = (-1)^F$ closes exactly as $\mathbb{Z}_6$ trace arithmetic including the cover factorization; chiral phases, continuum T-phases and continuum OS stay typed open; GATE.QGEO does not move. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v651_orbifold_assembly.py` | The full $\mathbb{Z}_3$-orbifold assembly of the seam: $H^2(\mathbb{Z}_3, U(1)) = 0$ by machine enumeration (27 cocycles, all symmetric – NO discrete torsion), and of six assembly candidates the survivors of {integer spectrum, unique vacuum, S measured clean, T exact} are exactly $\{B, C3\}$ – the naive $\mathbb{Z}_3$ dies by an exact T violation $3.5 \times 10^{-3}$ (= its charge-3 content), C1 by S stalling, C2 has no vacuum, A' additionally by the wrong first twisted exponent; the seam data then discriminate B from C3 (B keeps every measured $c = 0$ sector ground and the Klein phase $\eta = (1, 1)$): THE SEAM ASSEMBLY IS B, the $\mathbb{Z}_6$ deck quotient with uniform charge-0 projection (data-backed, v652 hardens it structurally); the chiral phase $e^{-2\pi i \tilde{a} b}$ is measured; GATE.QGEO does not move. |
| `v652_orbifold_arf.py` | Arf/GSO hardening of the B-vs-C3 discriminator: the $6^6$ classification of all sector-character wirings leaves exactly $\{B, C3\}$, the single-Fock-space pin forces B, and the $\mu_6$ defect ladder measures the C3-excised states DIRECTLY – ladder exponents $2\Delta = (1/18, 2/9, 1/2)$ at $< 8 \times 10^{-4}$, exact charge-resolved minima $(0, 1, 4, 9, 4, 1)/36$; C3's odd-sector predictions 85/18 and 81/18 exceed the measurements by factors 85.0 and 9.0 (the defect string is exactly charge-neutral: C3 excises states that exist on the lattice seam with positive Klein norm); the earlier 2/9 read alone is B-vs-C3-blind (both coincide exactly on the even sectors) – the new $\tau$ datum: OPE exponent 1/18; the bond-defect premise passes to v653; GATE.QGEO does not move. |
| `v653_bond_defect.py` | The bond-defect premise becomes a LATTICE THEOREM: twist sector = deck boundary condition = bond defect, canonically and exactly (unitary equivalence $A_r = \sqrt{3}\, RP_r$ on all modes, exact in $\mathbb{Q}(\zeta_6)$, at 16/48 and 48/144), with $D_{\text{can}} = L^4$ and $D_{\text{can}}^3 = -1$ as operator identities (the quarter clock fixes the canonical section lift); the defect position is pure gauge (symbolic telescoping theorem in all 16 free bond phases) with the gauge class pinned by the mark geometry (sheet reduction = one uniform twist unit per mark; cocycle support = the mark set); the exclusion census is complete – the base chain has NO other $\mathbb{Z}_3$, traces/holonomy/v628-Casimir kill everything outside, including the HONEST isospectral NN chain (it exists at $7.8 \times 10^{-16}$: purely spectral data cannot exclude it, but it is no $\mathbb{Z}_3$-twist realization and the measured string data discriminate it); continuum CFT and the seam identification stay typed; GATE.QGEO does not move. |
| `v654_st31_degree8.py` | The ST31 degree-8 round plus the unified $d/4$ theorem: every Springer-regular $d$-clock of $G_{31}$ satisfies $x^{d/4} = \pm i\,\mathrm{Id}$, a GENERATOR of $\mu_4$, never $\pm 1$ (one formula from the exponent systematics: all exponents of $G_{31}$ are 3 mod 4); the CONVERSE is exact at $d \in \{8, 20, 24\}$, and the unique exception at $d = 12$ is the COMPILER-CLOCK class (1280 extra elements = the class of $c$ and its Galois twin; $\chi_c = (x - i)^2(x^2 + ix - 1)$ is half the regular-12 Springer block, all eigenvalues satisfy $\lambda^3 = -i$ – exactly why $c^3 = J^3$ is central: the compiler-clock classes are the ONLY non-regular clocks in all of $G_{31}$ whose $d/4$-th power generates the center); the order-8 landscape is complete (10560 elements, 480 regular = 2 Galois classes, $\chi = (x^2 + i)^2$ exact) with the honest kill: "free ⇒ regular" is FALSE (4320 free order-8 elements noncentral; already at order 4); degrees = free orders divisible by 4 above 4; no marker moves. |

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| Script | What it machine-checks *(continued)* |
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| `v664_look_elsewhere.py` | The look-elsewhere audit – the bingo budget of the typed [C] numeric coincidences, quantified: $N = 42$ structure slots, $K = 19$ hits, the 6 killed readings entering as 0-hit trials; the family-corrected global $p = 5.08 \times 10^{-3}$ in the most conservative of three null models (the ensemble survives the honest correction), with the empirical control that only 0.2%/0.3% of 2000 size-matched pseudo-target lists reach the real hit count / the real $p$ – but NO single observation is significant on its own: suggestive (0.01..0.1) only $60 = D_{\text{start}}$, 12 = marks, the G31 degrees, $48 = \Omega_{\text{adm}}$; unremarkable ($p > 0.1$): $6 = p_2(a)$ ($p = 0.169$) and the CODESEM number battery ($p = 0.593$ – the v646 finding quantified); 14/16 FORCED (their value is the theorem, not the number); the re-typings are executed as dated ledger notes (EXTREV.LATTICE.01, E8.CODESEM.01, E8.CODE.01); seed 20260802, 100k MC, exact Poisson-binomial tails. |
| `v679_orbifold_continuum_os.py` | The first continuum-OS slice of the seam orbifold B: correlator convergence with UNDERSTOOD rates – all six $\sigma$-channel observables share the uniform raw rate 0.659 = the predicted Fisher–Hartwig 2/3 branch, the $\tau$-channel contrast measures 1.298 vs 4/3 (the channel law $\rho = 2(1-\beta)^2 - 2\beta^2$: rates understood, not fitted); after removing the ONE known FH term the extrapolated limits hit the CFT values at $< 0.05\%$; RP SURVIVES THE LIMIT: $\lambda_{\min}$ extrapolations bounded away from zero ($\sigma$ $1.36 \times 10^{-3}$, margin 7×; $\tau$ $2.80 \times 10^{-4}$, margin 358×; mixed OS Gram $1.34 \times 10^{-3}$, margin 31×; must-fail $\eta = -1$ flips); the cluster property holds in space AND Euclidean time ($2\Delta = 0.222220$ vs 2/9 at $8 \times 10^{-6}$, cylinder/cosh forms pointwise, time gap = the $\varepsilon$ level at 0.99933); the continuum characters 1/36, 1/9, 5/18, 4/9 extrapolate at $10^{-12}$ with the measured $N^{-2}$ rate; typed honestly: OS axioms E0/E2/E3/E4 are MEASURED at the convergence level, E1 only discretely exact; the formal-limit remainder is named (uniform bounds, the GNS limit Hilbert space, operator convergence, R sectors, the B assembly) – GATE.QGEO does NOT move. |
| `v689_gaussian_code_bridge.py` | The Gaussian code bridge (review contract E8.GAUSSIAN.CODE.01) – $E_8(\mathbb{Z}[i])/(1+i) \cong \mathbb{F}_2^4$ EXACTLY (Smith normal form; standard model: $\lvert\det\rvert = 256 = 2^8$, SNF $[1^4, 2^4]$ – presentation-independent): the $240 = 15 \times 16$ census (the zero class PROVABLY empty, each class = 4 of the 60 G31 lines), $\sigma = c^4$ acts on the quotient as the RM(1,3) family permutation (3-cycle $F_1 \to F_2 \to F_3$ + fixed anchor, exactly the v638 info-bit action), residual gauge = the order-18 centralizer in $GL(4, \mathbb{F}_2)$, coordinate block = the $F_1$+$F_2$+$F_3$ label. All must-fail controls fire: the naive placement keeps the census but loses the $\sigma$ semantics (off-lattice witness); a non-$\pi J$ placement has no $\mathbb{Z}[i]$-module; $\mathbb{Z}[i]^4$ trivializes (16 vectors, 11 classes empty); exact integer/Fraction arithmetic; no marker move. |
| `v690_quartic_half.py` | The quartic invariant restriction (review contract E8.QUARTIC.HALF.01) – is G31 the holomorphic $\mu_4$ shadow of the E8 compiler? ALIVE, with the honest split: the vanishing half $\{2, 14, 18, 30\}$ is TRUE but TRIVIAL ($\mu_4$ orbit factor, proven symbolically: $W(J\alpha) = -i\,W(\alpha) \Rightarrow$ every $J$-orbit contributes $t^d(1 + (-i)^d + (-1)^d + i^d) = 0$ for $d \not\equiv 0 \bmod 4$ – holds for $6, 10, 22, 26$ too). THE SUBSTANCE: $F_8\rvert, F_{12}\rvert, F_{20}\rvert, F_{24}\rvert$ are G31-invariant (60/60 mirrors exact), ALGEBRAICALLY INDEPENDENT (exact Jacobian $\neq 0$), degrees Molien-unique with product $46080 = \lvert G31\rvert \Rightarrow$ by Chevalley [cited] BASIC INVARIANTS: the holomorphic restriction of the $W(E_8)$ ring generates the FULL G31 ring. The Jacobian carries the 60 mirror lines (degree $60 = \sum(d_i - 1)$, point-verified 60/60; full factorization = cited theorem, named residual); controls: $J' = RJR^T$ breaks the informative half, the root-preserving $J''$ keeps everything (the theorem is about the $W(E_8)$-conjugacy class); no floats; no marker move. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v700_orbit60_census.py` | The ORBIT60 census – the 60 $\mu_4$-lines of E8 as a combinatorial object: quotient graph, $\sigma$/clock/Coxeter dynamics on the ORBIT SPACE, and a STRICT park test for the cascade $60 \to 6$. All five routes adjudicated: (i) degree rules DEAD (all quotient graphs regular); (ii) $\sigma$ structure DEAD (orbit sizes $\{3,1\}$, no 2-blocks); (iii) THE INVOLUTION WITNESS DOES NOT EXIST: the full involution census of the line group LG (order 11520; 375 involutions, 21 classes) shows NO element of cycle type $2^{27}1^6$ (fixed-line census $\{0{:}120, 4{:}180, 12{:}15, 16{:}60\}$); (iv) W(E8) Coxeter does not descend; (v) the MUST-FAIL numerology control fires: rule-free lexicographic pair removal reproduces $60 - 2n$ in exactly 27 steps, $> 10^{70}$ equally 'successful' sequences – the counting carries no structure. A5 one-liner: order-5 elements exist (2304) but no visible natural A5 action ($\langle \sigma, \text{order-5} \rangle$ has order 960); PARKED per the review protocol; deterministic. |
| `v705_chain_deck_sector.py` | Original v705 D1 physical NS circle: 48-site antiperiodic shift, order-12 clock, projective deck cube -1 and three 16-mode deck sectors. |
| `v715_qgeo_car_continuum.py` | QGEO-CAR – the quasi-free CAR route to the formal orbifold continuum limit (GATE.QGEO), first slice. QGEO-CAR-RATES-SUMMABLE: all 6+1 deck/twist channels are Schatten/shadow-norm Cauchy on $\varepsilon$-separated configurations ($N$ up to 3072/6144) with geometrically summable rates; the FH renormalization is FIT-FREE LATTICE-DERIVED (Barnes G, $8.8 \times 10^{-7}$); the RP cone does not degenerate along the ladder; all four kill criteria are clear. The named remainder: the majorant lemma (L1)–(L3) with constants + the classical proof route – named, not claimed: GATE.QGEO does NOT move. |
| `v722_phys_ramified_ns_r.py` | PHYS moonshot slice (T2 / $G_{net}$) – is the Ramond projection $(1+i)$-adic? RAMIFIED-SEES-NSR: $(1+i)L$ lies in the even/NS sublattice ($\mathbb{Z}$-basis proof), the 240 roots fall into $15 \times 16$ classes with PURE parity (7 NS + 8 R), parity is a homomorphism $\mathbb{F}_2^4 \to \mathbb{F}_2$ – the NS/R grading IS the parity character of $E8(\mathbb{Z}[i])/(1+i) = \mathbb{F}_2^4$; the first exact witness for the groupoid reading of $G_{net}$ (the ONE ramified edge of the tower is the Gaussian prime $(1+i)$); controls fire: the $\mathbb{Z}[i]^4$ census is NOT $15 \times 16$, scrambled labels break purity. |
| `v723_phys_modular_clock.py` | PHYS moonshot slice (T3 / $F_{transfer}$) – can thermal time replace the external clock contract? The conjugacy-class KILL test. STT-KILLED + INTERNAL-PAIR-CONFIRMED: the $F_{QCD}$ 1-loop clock map is PARABOLIC and not diagonalizable (single Jordan block), and exp of a real-diagonalizable generator is diagonalizable (symbolic lemma) – no constant-rescaled modular flow produces the QCD clock; the pole multiplier is EXACTLY $(2/3)^6 = 64/729$ (hyperbolic, dimensionless $\Rightarrow$ internal); typed consequence: the external clock contract is CONFIRMED and sharpened to the anchor census {one unit ($v_{geo}$-class), one angle $\theta_i$, one lattice $O(1)$} – T3's clock wall == T1's unit wall, counted once. |
| `v724_phys_t3b_modular_flows.py` | T3-B – conditioned modular flows from ONE internal mother state, the second thermal-time KILL. T3B-DEAD: with target-free graph projectors, ONE global KMS scale $\Delta = 6\log(3/2)$ and the Connes cocycle composition exact to $1.7 \times 10^{-15}$ along J–K–C–F, all four conditioned flows come out LOXODROMIC/COMPLEX – the frozen v578/v632 solver classes (hyperbolic / parabolic / degenerate) do NOT arise from conditioned modular flows of one internal mother state; the probe honestly refuses the NEEDS-LINDBLAD escape (the v238 GKSL variant is also all-loxodromic); with v723 BOTH thermal-time routes are closed: the external clock contract stands. |

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| Script | What it machine-checks *(continued)* |
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| `v725_phys_vgeo_torsor_audit.py` | T1 CLOSURE – the calibration/scale-torsor audit of the $v_{geo}$ interface. TORSOR CLOSURE: the $v_{geo}$ interface is structurally CLOSED as an $\mathbb{R}_+$ scale torsor – (1) complete export table $O_i = r_i\, v_{geo}^{d_i}$, (2) dimension matrix rank 1 (conditional [A] on the flavor block, named), (3) all consistency conditions $\lambda$-homogeneous (the machine form of the No-Unit theorem v153), (4) two-anchor leave-one-out compatible ($G$ vs $\rho_\Lambda$: 0.11%), (5) the calibration theorem stated in torsor language; FENCE: structure closure, NO scale derivation (impossible per No-Unit); moonshot NAMED, not executed: $H_{EW} = \ln(\bar{M}/v_{EW}) = 37.1776$. |
| `v726_phys_car_pp_index.py` | PHYS T2 main slice – Pimsner-Popa / Watatani index 4 on the CAR ladder, with the $\mu_4$ action DERIVED and the Ramond zero-mode test. T2-SLICE-GO: the optimal PP constant is EXACTLY 1/4 (explicit rank-one minimizer + pinching theorem) and the Watatani index EXACTLY 4 via an explicit quasi-basis ($\sum vv^* == 4 \cdot 1$, weights FORCED – uniform weights break scalarness, the $\mathbb{Z}_2$ average gives index 2); $\mu_4$ is NOT inserted but derived from the clock $L = S^{N/12}$ (order 4 forced by the NS spin structure); the Ramond sector degenerates undressed (index 2) and the v679 Klein dressing restores index 4 STATE-PRESERVINGLY ($[\rho, U'] = 0$); the $(1+i)$ dictionary to v722 closes; the GNET.RAMIFIED.01 contract candidate is printed. |
| `v736_orbit_packet.py` | The Gaussian ORBIT PACKET lemma. ORBIT-PACKET-EXACT: Burnside $7 = (15+3+3)/3$; the canonical $5 \times 48$ partition (fixed block + 4 moved blocks of 12 Gaussian lines each, label-free canonical – packets $= 1+4 = 5 = g_{\mathrm{car}}$, $240 = g_{\mathrm{car}} \times 48$); anchor $(1,1,2)$ from the fixed-syndrome leader weights $(0,1,1,2)$, leader pair $e_6/e_7$ = the v638 anchor pair; $q$ normal form $240 = 16 \cdot 15$, $248 = 240 + 8$; circularity fence: P1/P2 come back as EXACT-INTERNAL-RECONSTRUCTION (reconstruction, NOT derivation); $\sigma$-class specificity: the 160 free order-3 elements of $G_{31}$ give Burnside 5, not 7; companion Lean module OrbitPacket.lean (34 theorems, kernel-checked). |
| `v737_seam48_intertwiner.py` | The seam-48 intertwiner test – the sharpest structural kill of the review morning. SEAM48-DEAD: the $48 \times 5$ incidence matrix is NOT identically 1 (entry distribution 0:96 / 1:64 / 2:64 / 3:16), representative-independent (0 of 2304 order-5 elements of $G_{31}$) and the must-fail control fires (0 of 25 random operators); valuable residues: the 4 moved blocks carry the v623 character table EXACTLY, the fixed block deviates exactly by the 12 lifted marks; the 5-period lemma $u^5 = \pm J$ holds exactly; $240 = 48 \cdot 5$ remains a factorization WITHOUT coordinatization. |
| `v745_qgeo_car_l1_sector_lemma.py` | The (L1) sector lemma of the CAR continuum majorant. L1-SECTOR-PROVEN-MODULO-ELEMENTARY: $\|K_{2N}^{(q)} - K_N^{(q)}\|_{S_1(\Lambda_\varepsilon)} \le B_q(\varepsilon)/N$ for all 6 undressed sectors on the fixed grid, proof-near – the occupied-arc mode sums are EXACT geometric sums (exact summation replaces Euler–Maclaurin), the entire $1/N$ structure is the sublattice embedding offset, the leading term is the exact matrix $(iu/2MN)\, f_q'(s)$ (ratio $\to 1$ at $6 \times 10^{-4}$), $\gamma = 2$ is DERIVED, $39744 + 108$ inequalities at 0 violations, $r_1 = 1$ sharp; status typed PROVEN-MODULO-ELEMENTARY with explicit constants; named residues: (a) grid-sup [elementary], (b) L2 uniform FH/RH [hard-technical], (c) L3 uniformity [medium]; GATE.QGEO does not move. |
| `v746_phys_gnet_local_functor.py` | The $G_{\mathrm{net}}$ local net functor. GNET-LOCAL-FUNCTOR-ALIVE: the net axioms are EXACT (isotony, graded locality, clock covariance on the quotient circle); the Watatani index of the local fixed-point inclusion is LOCALLY EXACTLY 4 for every interval $\ell \ge 2$ – the $[D, A]$ index statement is interval-local, not merely global; Ramond = half-line/bond defect (solitonic; local footprint, leakage $\sqrt{2/N}$); 4 must-fail controls fire; missing continuum theorems NAMED, not claimed (GNS limit net, index continuity Pimsner–Popa/Longo, Longo Q-system identification to GATE.METRIC.11, twisted DHR/Longo–Roberts sectors, Haag duality/split). |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v752_projective_hamming_incidence.py` | The projective Hamming incidence lemma on $V = L/(1+i)L$ – the symplectic $\mathbb{F}_2^4$ of the unimodular hermitian E8 lattice, and the v737 resolution. PROJ-HAMMING-EXACT: $h = H/4$ is $\mathbb{Z}[i]$-valued with hermitian Gram determinant 1 ($\mathbb{Z}[i]$-UNIMODULAR); $\bar{h}$ alternating (doubly-even), nondegenerate $\Rightarrow (V, \bar{h})$ symplectic with explicit Darboux basis; $\lvert H_y\rvert = 7$, pair intersections 3; $BB^T = 4I + 3J$ EXACT, characteristic polynomial $(x-7)(x-2)^9(x+2)^5$ – the multiplicities $(1, 9, 5)$ are the SYMPLECTIC signature; $K^2 = (4/49)I + (45/49)\Pi_0$ exactly ($K$ itself honestly not psd); NS/R $\chi = \bar{h}(\cdot, y_0)$ for the UNIQUE $\sigma$-fixed point $y_0$ = FSUM (the anchor bit), $112 + 128 = 240$, +8 Cartan: $248 = 120 + 128$; THE v737 RESOLUTION: the distribution 0:96/1:64/2:64/3:16 derived exactly (16× the $3 \times 5$ label table) AND the $5 \times 48$ coordinatization exists via exactly ONE $\bar{u}$-invariant spread (kill explained, not revised); two honest corrections to the review derivation documented; companion Lean module ProjectiveHamming.lean (32 theorems). |
| `v757_fano_disc7_bridge.py` | The preregistered Fano → discriminant($-7$) bridge – the three-cycle half is real, the parity half dies arithmetically. FANO-DISC7-DECORATIVE (break exactly at condition 2): the NS/R parity bit has NO arithmetic carrier in $\mathcal{O}_{-7}$ – 2 SPLITS in $\mathbb{Q}(\sqrt{-7})$ ($-7 \equiv 1 \bmod 8$, $\mathcal{O} \otimes \mathbb{Z}_2 = \mathbb{Z}_2 \times \mathbb{Z}_2$), every 2-adic ideal quotient has 2-rank $\leq 2 < 3$ (complete census of all norm-8/16 ideals): the code $\mathbb{F}_2$-structures are arithmetically UNREALIZABLE; the REAL half (COND1, exact): equivariant lifts exist for both canonical difference-set Fano models, the canonical order-3 action is $x \to 2x$ (the difference-set multiplier, 2 = the transfer step), every lift sends the unique $\bar{\sigma}$ fixed point to $0 = \mathrm{red}(\alpha_0)$ = the C corner (norm $14 = \dim G_2$); the seven-correlation permanently typed DECORATIVE; controls fire; the one [FAIL] = the preregistered COND2 break. |
| `v774_arf_spinor_compiler.py` | The Arf/spinor compiler on $V = L/(1+i)L = \mathbb{F}_2^4$ (Priority 1 of the Arf programme). ARF-SPINOR-EXACT: the parity lift $\iota$: $V \cong C_{\mathrm{even}}(5)$ carries $\beta = \bar{h}$ EXACTLY in all 256 cells (crosslink 1); brute force over all $2^{16}$ functions: exactly 16 quadratic refinements; Arf census $6 + 10$ with $\lvert Sp(4,2)\rvert = 720$ and a faithful+surjective $S_6$ action; the frozen selector picks a UNIQUE $q^*$ (census $4 \to 2 \to 1$) and the weight form IS $q^*$ (offset 0); $\mathrm{Stab}(q^*) \cong S_5$ gives $16 = 1 + \bar{5} + 10$ on refinements AND words; the $K_6$ duad model is an equivariant bijection onto $E(K_6)$; $B = I + A_{KG(6,2)}$ in all 225 cells with the exact ovoid eigenvector; all 6 isotropic spreads have signature $(1, 2)$; the hypercharge table $X = (-2, -2, -2, 3, 3)$ reproduces the v310/v14 dictionary; moments $(16, 0, 120, 0)$ with the chain 40/41/240/248; Pati–Salam $8 + 8$ with $B - L = 1 - 2n_c/3$; $\chi_{\mathrm{NSR}}$ = the anchor bit. $A_3^{\mathrm{Fam}} \neq A_3^{\mathrm{PS}}$ naming discipline; the code→matter reading FENCED (killed by v775). Lean companion ArfSpinorCompiler.lean (23 kernel theorems). |
| `v775_gaussian_class_d5_purity.py` | THE FIRED PHYSICAL KILL of the Arf compiler programme. ROOTCLASS-MIXED: no $D_5 \oplus A_3$ / $SU(5)$ convention makes even one full set of 15 Gaussian classes pure per the frozen code→matter table – the embedding-INDEPENDENT counting bound (every embedding has exactly 128 spinor-side roots, $240 > 128$) forces $\geq 7$ adjoint-side classes ALWAYS; the exhaustive scan (56 splits × 240 assignments × 2 chiralities) finds ZERO passing conventions; all 8 spinor classes mix 8 distinct 16-weight pairs. BINDING FENCE: the code→matter reading must NOT enter any main claim (incl. the 'empty mailbox = right-handed neutrino' reading); the Arf mathematics of v774 is untouched. All three controls fire. |

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| Script | What it machine-checks *(continued)* |
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| `v776_boundary_hamming_uniqueness.py` | Boundary code uniqueness. BOUNDARY-CODE-UNIQUE: the non-circular chain boundary → code → $E_8$ → $\mathbb{F}_2^4$ → 5 slots closes with $g_{\mathrm{car}} = 5$ as OUTPUT – census $135 \to 30 \to 14 \to 2 \to 1$ rebuilt without Hamming/$E_8$ input; the flag datum is ONE deck-invariant $\mathbb{Z}_2$ (the product of the four side orientations; all three predeclared formalizations agree); the boundary gauge group (order 12) makes the flagless twofold pure gauge; Construction A → $E_8$ exact (240/2160 theta, Cartan det = 1, arms 1-2-4); $q^*$ unique; packets $1 + 4$. Honest residuals R1 (doubly-even self-dual universe premise – circularity relocated and shrunk, not removed), R2 (deck placement semantic; wrong-deck control does not fire, typed), R3 (anchor naming). NO P2-removal claim; the open contract form registered as ARF.BOUNDARY.CODE.01 [O]. |
| `v777_ftransfer_clock_jets.py` | FTRANSFER.CLOCKS.01 EXECUTED + the fibered refinement of CONTRACT.F.01, ONE module from two probes. NO-COMMON-CONNECTION + FTRANSFER-FIBERED-CARRIES: the preregistered K1 kill fires EXACTLY as the frozen prior expected (prereg YAML byte-verified; both data tables SHA-frozen BEFORE the first run); all four dictionaries construction-valid (Koide scheme check 571×, washout drift $1.15 \times 10^{-4}$, QCD overlay 0.83%, relic slope 0.0602); the kill is sympy-exact $\{-\Delta^2/2, -\Delta^2/2, 0\}$ – a pass would need exactly the forbidden non-affine Möbius clock. The discrete layer coincides EXACTLY (cross-ratio 4/3 symbolic, deck unit $\lvert\Delta\sigma\rvert = 1$, unit parabolic class – the v632 lattice-explicit half); the obstruction is the CONSTANT seam cocycle with values exactly $\{0, \pm\Delta^2/2\}$; classification = the predeclared thermal-vs-RG marker; relic degenerate unclassified; controls incl. the Möbius transplant fire; the first-run float64 artifact disclosed, bar untouched. The jet route of CONTRACT.F.01 closes at the physical-clock level; next named step: derive the coset assignment from KMS/modular clock typing. |
| `v778_qgeo_gridsup_logsum.py` | Remainder (a) of the (L1) majorant lemma FULLY closed in the corrected $(1 + \ln M)$ form, ONE module from two probes. GRIDSUP-SPLIT-CLOSED-LOG-LAW + FOURIER-LOGSUM-CLOSED: the v745 residue (a) splits four ways – (a1) the continuum entrywise supremum CLOSED (∼1.26e6 census inequalities, 0 violations), (a2) the configuration-$S_1$ closure with explicit $B_{\mathrm{conf}}(\varepsilon)$ (packing $+\zeta(2)/\zeta(3)$), (a3) the refinement remainder with RHO($\varepsilon$) via exact $\csc^3$ integrals, (a4) the leading part honestly LOG-divergent – the literal v715 clause 'one $B(\varepsilon)$ for ALL grid refinements' is FALSE at $r_1 = 1$ and repaired to $(1 + \ln M)$; the surviving inequality CERTIFIED at theorem level via the exact twisted-circulant diagonalisation + two exact summation-by-parts: $\lVert F\rVert_1 \le c_0 + (J/4)\ln(M{+}3)$, a fortiori $(J/\pi)\ln M$; 162-case census to $M = 6144$, zero violations; repaired (L1) clause verified on all 252 transfer cases; Araki summability explicit. Honest: the measured slope is $J/\pi^2$ – valid, not sharp. Residues (b)/(c) open; GATE.QGEO does not move. |
| `v779_gnet_gns_arf.py` | The $G_{\mathrm{net}}$ continuum strand, ONE module from two probes. GNS-PRECURSORS-COHERENT + GNET-ARF-NO-CORRESPONDENCE: the Haar scaling isometry intertwines exactly ($\le 2.2 \times 10^{-16}$); the commuting index-4 expectation squares are exact; the canonical state is Cauchy with geometric contraction $\sim 1/4$ per doubling; the index census is machine-exact at every level with zero degradation; the twisted Haag-duality defect is 0 with $\dim B'/\dim A' = 4$ = the Watatani index; 10-section continuum-paper skeleton. The exact dim law $d_c(\ell) = 4^{\ell-1} + 2^{\ell-1}\cos(\pi c/2)$ makes the $\mathbb{C}[\mathbb{Z}_4]$ quasi-basis obstruction exactly $2^{-\ell}$ (explains the $\ell = 1$ anomaly); explicit 185-element quasi-basis at $\ell = 2$; uniform sector-resolved split bounds. THE HONEST NEGATIVE: the 16 = 1+5+10 counting echo is exact but non-geometric (rank 3 vs 4, $GL(4,2)$ census 0/20160); $K_{\mathrm{matter}}$ does not govern tower transport; the two fours stay in their registers. All five controls fire. |

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| Script | What it machine-checks *(continued)* |
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| `v781_one_object_clock_census.py` | The $d = 4$ forcing census + the 60-line clock kill, ONE module from two probes. ONE-OBJECT-PARTIAL + CLOCK-FIBER-DEAD (21/21 + 23/23): the TFPT glue pattern admits an even unimodular DIAGONAL glue of $A_{d-1} \oplus D_{d+1}$ iff $d = 4$ (census $d = 2..12$, exact fractions; at $d = 4$ exactly the two v92 Lagrangians $k \mapsto k$, $k \mapsto 3k$; explicit $E_8$ certificate with 240 norm-2 vectors; $d = 9$ dies at isotropy; both relaxation controls fire) — corollaries $g_{\rm car} = d{+}1 = 5$, $N_{\rm fam} = d{-}1 = 3$, rank $2d = 8$, index $4 = d = \lvert\mu_4\rvert$; honesty gate: $c_3 = 1/(2\pi d)\lvert_{d=4} = 1/(8\pi)$ is a corpus-legitimate REWRITING, NOT a new derivation of P1. The clock kill: the fiber-product action $\mathbb{Z}_{12} \times_{\mathbb{Z}_6} \mathbb{Z}_{30}$ on the 60 Gaussian lines is dead at three levels (full line-compatible group order 23040, max element order 12 — orders 15/30/60 absent for every realization; the fiber product is cyclic $\mathbb{Z}_{60}$, exact SNF); all controls fire. |
| `v782_e8_transition_bus.py` | The transition-bus structure theorem. TRANSITION-BUS-TORSOR (35/35): the 240 roots are 15 disjoint coordinate 8-frames over $V = L/(1{+}i)L$; on $L/2L$ the 120 root pairs are exactly the 120 $q = 1$ points of the plus-type quadratic space and each class fiber is a canonical TORSOR under $v^{\perp}$ (affine $\mathbb{F}_2^3$, no distinguished origin); the deck $J$ acts as translation by $v$; the obstruction census is proof-grade: NO deck-equivariant root-level labeling $R(E_8) = V \times (V \setminus 0)$ exists; the frozen symplectic predicate reproduces all 57600 ordered inner products incl. the sign law and the $h$-refinement; spread layer $240 = 5 \times 3 \times 16$ with the $q^*$ selector; the $A_{15}$ comparison quantified (a strict bus WITH a canonical start state; $E_8$ provably has none; cross-class non-orthogonality exactly doubled). Controls fire. |
| `v783_two_qubit_clifford.py` | The two-qubit stabilizer identification of the compiler carrier. CLIFFORD-PARTIAL(H4) (32/32): under the single predeclared chart $U$ the 60 Gaussian $E_8$ lines ARE the 60 pure two-qubit stabilizer rays, entry-exactly over $\mathbb{Z}[i]$ (60/60; support census $4 + 24 + 32$); $UG_{31}U^{-1}$ is an index-2 subgroup of the two-qubit Clifford matrix group ($\lvert C_2\rvert = 92160 = 2 \times 46080$, coset $\zeta_8 G_{31}$); $46080 = 4 \times 16 \times 720 = \mu_4 \times \mathrm{Pauli}_{16} \times Sp(4,2)$ with the 15 classes = the 15 Pauli contexts ($W(3,2) = GQ(2,2)$) and the 6 spreads = the MUB pentads; H4 fails as predeclared: the index bit is the Galois $\sqrt{2}$/Hadamard bit, NOT $J \leftrightarrow -J$; all controls fire. Stabilizer-formalism mathematics about the carrier; ROOTCLASS-MIXED (v775) unaffected. |
| `v784_curve_code_2torsion.py` | The curve/code 2-torsion decider. CURVE-CODE-PARTIAL (33/33): the frozen CM-model route through $y^3 = x^4 - 1$ (every generator period certified as an integral $\mathcal{O} = \mathbb{Z}[\zeta_{12}]$ recognition, dps 120, HNF index 1; $A[2] = \mathcal{O}/2\mathcal{O}$ exact; $\tau = \zeta^4$ DERIVED as the only order-3 class; minimal ODD $\tau$-invariant polarization type $(1,3)$, no principal form in the $\tau$-invariant cone; $q_\Theta$ the unique $\tau$-fixed refinement): the correspondence DIES at two exact layers — (1) $\tau$ acts freely (cycle type $[3,3]$, syntheme side) vs $\bar\sigma$ fixing a 2-plane ($[3,1,1,1]$, duad side), exchanged only by the OUTER $S_6$ automorphism, census 0; (2) $q_\Theta$ EVEN (Arf 0, 1+6+9) vs $q^*$ ODD (Arf 1, 1+5+10), census 0; joint census 0 with positive control 6. All controls fire. |
| `v792_ftransfer_kms_coset.py` | The coset assignment of the fibered clock functor is a THEOREM from KMS/modular typing. KMS-COSET-THEOREM (18/18): $S = -(\ln\lambda)^2/2$ with $\lambda$ the time-1 $\mathrm{PGL}_2$ holonomy multiplier per clock e-fold; the classifier uses NO Schwarzian and NO derivatives (non-circularity gated on its own source): functional-identity holonomy witness + $\mathrm{PGL}_2$ class + detailed-balance multiplier cross-ratio (the discrete KMS signature); Lemmas A/B/C sympy-exact; BLIND protocol: $F_{\rm pole}/F_{\rm Boltzmann}$ modular with $\lvert\ln\lambda\rvert = \Delta = 6\log(3/2)$, $F_{\rm QCD}$ autonomous (parabolic), $F_{\rm relic}$ degenerate; unblinding EXACT (predicted cosets = transported Schwarzians, sympy zero); controls: fake KMS caught, affine invariance, the v578 Moebius clock as the positive case — the fiber classification of the CONTRACT.F.01 refinement is derived, not case-inspected. |

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| Script | What it machine-checks *(continued)* |
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| `v796_duad_syntheme_bridge.py` | The canonical OUTER bridge between the code duad side and the curve syntheme side. BRIDGE-CANONICAL (24/24): the Sylvester duality $\beta$: code odd forms $\to$ curve spreads with $\beta\sigma = \tau\beta$ and $\beta(q^*) = S^*$ (the UNIQUE $\langle\tau, \rho\rangle$-fixed spread) exists and is unique up to the order-6 code joint stabilizer (census $18 + 18 \to 6 + 6$, ONE orbit incl. the $\tau/\tau^2$ swap by an inner code symmetry); the outer witness exact ($[3, 1, 1, 1] \leftrightarrow [3, 3]$ uniformly); $q_\Theta$ pulls back CONSTANTLY to $q^*_{\text{even}}$ with $q^* = q^*_{\text{even}} + \bar{h}(\cdot, A)$ — the anchor shift resolves the v784 Arf mismatch canonically; $\bar{5}/10$ and 3+2 transport exactly onto the $S^*$-line geometry; $t_A \mapsto \rho = J$ on every bridge; $\chi_{\text{NSR}}$ an honest exact TWO-variant torsor (pinned by the v752 NS/R parity). Controls: inner censuses 0/0, odd-Arf pullback 0, scrambled $\sigma$ 0. The v784 INNER identification stays dead; no matter semantics. |
| `v798_seam_clifford_modular_s.py` | The missing Clifford bit as the metaplectic lift of modular $S$ at $\tau = i$. MODULAR-S-PARTIAL (27/27): COSET POSITIVE — the per-binary-factor Hadamard $K_1 = H \otimes I$ IS the Weil lift of $S$ at $\tau = i$ and lands exactly in the missing coset $\zeta_8 G_{31}$ (exact relations $S^2 = I$, $T^2 =$ Pauli parity, $(TK)^3 = \zeta_8 I$); the TOTAL Fourier $H \otimes H$ lies INSIDE $G_{31}$ with metaplectic anomaly $(T_3 K_3)^3 = iI = J$ — the deck IS the anomaly of the total modular lift (R2 placed); $\zeta_8 = (1+J)/\sqrt{2}$ the real half-deck swapping the seam pair $\{L, \zeta_8 L\}$; $F_4$ typed out; OS reflection gate exact ($\tau = i$ fixed). THE STRICT UNIQUENESS CENSUS IS EMPTY (0/46080): no $\Theta$-real metaplectic involution in the coset fixes $q^*$ — the Arf defect is unavoidable; retyped census: 32 survivors = ONE $G_{31}$-conjugacy class; R3 NAMED, not fixed: the $\sigma$-orbit defects are exactly $\{A + F_\Sigma + F_i\}$ with XOR = the anchor $A$ itself. Controls fire. |
| `v799_seam_code_typeii.py` | Type II FORCED from three physical seam axioms — the direct attack on residual R1 of ARF.BOUNDARY.CODE.01. TYPEII-FORCED (19/19): exhaustively over ALL 308,993 subspaces of $\mathbb{F}_2^8$ (dim 0..4, Gaussian binomials; dim $> 4$ by rank-nullity): mutual locality of the Construction-A vertex words $\iff$ self-orthogonal (135 at dim 4; the lift-shift braiding identity exhaustive); integer conformal spin $\iff$ doubly even (30); holomorphy/index one ($\lvert\text{disc}\rvert = 1$, exact Fraction Gram determinants) $\iff$ self-dual — the survivor set is EXACTLY the 30 $\hat{e}_8$ copies; no boundary-legal non-Type-II code exists; every non-self-dual doubly-even code has a fractional-weight discriminant sector ($[8, 3]$ witness with exact coset thetas, $\theta_{L^*} \neq 2\theta_L$). R1 $\to$ R1′ = A1 mutual locality + A2 integer spin + A3 holomorphy/index one; imported lattice-VOA/theta ingredients cited and machine-verified in every instance used. Controls fire. NO P2 marker move. |
| `v800_e8_torsor_fourier.py` | The projective Fourier modes of the 15 Gaussian torsors — carrier purity at CHARACTER level with the mutually-unbiased cap. TORSOR-FOURIER-PARTIAL (30/30): 120 canonical projective rays (phase law exact on 960 origin shifts), orthogonal basis (Gram $= 16I$); deck $J$ and all 16 Paulis diagonal by characters, $G_{31}$ monomial through $Sp(4, 2)$ (kernel $64 = \mu_4 \times \text{Pauli}_{16}$; orbits $15 + 45 + 60$). THE CARRIER BLOCK: the 8 R-fibers' 64 rays = (1+5+10) $\boxtimes$ (1+3) under $W(D_5) \times W(A_3)$ (Clebsch charpoly $(x-5)(x-1)^{10}(x+3)^5$ exact; $S_4$-natural tetrad with $\sigma =$ family 3-cycle + anchor) — the carrier Pascal at CHARACTER level, impossible at point level (v775); P3 fails as predeclared: the weight and character bases are MUTUALLY UNBIASED ($\lvert\langle\text{pair}\rvert\text{mode}\rangle\rvert^2 = 1/8$ exactly) — ROOTCLASS-MIXED intact; the 45 deck-even modes ARE the 45 (context, Pauli) incidences under the certified chart. Controls fire. |

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| Script | What it machine-checks *(continued)* |
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| `v802_gnet_martingale.py` | The martingale route to the $G_{\text{net}}$ GNS limit — finite hypotheses MEASURED, all met. MARTINGALE-HYPOTHESES-MET (15/15): the filtration is EXACT (Haar isometry with support coherence, leak 0; Watatani quasi-basis 185/3041 at $l = 2/3$ with $\sum vv^* = 4 \cdot 1$; ONE index-4 expectation for the whole tower); the state defects are SUMMABLE at measured rate ∼0.25 per doubling — one power better than the $2^{-l}$ dim-law envelope (reported, not forced); the unique limit follows by telescoping and is verified on every rung; the split/normality precursors are uniform (determinant witness $D \geq 0.985$, stabilizing at 0.9928). Steps 5–7 (GNS, local normality, split + $\mu_4$ simple-current extension) typed NOT executed. QGEO note: the open majorant residues (b)/(c) are bypassed for limit EXISTENCE and relocated into the typed normality step — a reallocation, not a proof. Controls fire (decimation converges to the WRONG degenerate limit — the coherent bond is load-bearing). |
| `v803_e8_jetcode_affine_nsr.py` | The jet-code normal form of the Gaussian E8 compiler + the NS/R bit as an affine 8-point geometry. JETCODE-EXACT + AFFINE-NSR-EXACT (18/18 + 15/15): $\mathbb{Z}[i]/(2) \cong \mathbb{F}_2[\epsilon]/(\epsilon^2)$ with $\epsilon = 1 + i$ (2 RAMIFIES; unique iso, the split ring $\mathbb{F}_2 \times \mathbb{F}_2$ fails all 24 bijections); $W = L/2L$ is FREE of rank 4 over the dual numbers (explicit $\mathbb{Z}[i]$-basis by Hermite reduction, covolume-certified); $0 \to \epsilon W \to W \to V \to 0$ is an exact SELF-extension with $\ker \epsilon = \text{im}\, \epsilon$; the deck IS the jet unit $J = 1 + \epsilon$, re-deriving the v782 torsor on all 240 roots; ZERO $\mu_4$-equivariant splittings among all 65536 linear sections — the no-origin obstruction IS non-splitness; the 120 root pairs = the $q = 1$ unit shell ($240 = 15 \times 8 \times 2$). PART 2: $V \setminus \{0\} = 7$ NS + an affine 8-point torsor carrying 112 $D_8$ + 128 spinor roots ($248 = (112{+}8){+}128$); $\text{Stab}(\chi_{\text{NSR}}) = AGL(3, 2)$ of order $1344 = \text{Aut}(H_8)$ with the SAME natural 8-point action; affine functions reconstruct $RM(1, 3) = H_8$ with exactly 1344 equivalences. Typed internal reconstruction — no P2 removal; no matter semantics (ROOTCLASS-MIXED). |
| `v805_e8_syndrome_algebra.py` | The honest kill — E8 is NOT an error-correction hull in the sharpest sense. SYNDROME-DEAD (13/13, the kill fires as preregistered): the $\mu_4$-linear-lift realization family reaches max rank 222/256 over ALL FOUR frozen phase conventions (sympy-exact over $\mathbb{Q}(i)$); the $\pm r$ proportionality obstruction is STRUCTURAL for the whole diagonal-phase family ($c(-r) = -c(r)$; collision witness printed), and both post-hoc escape directions score WORSE (nonlinear 215, $\zeta_8$-linear 160) — the death is robust. SURVIVES: the syndrome-table counting $16 + 240 = 256$, the neutral sector (the 16 v112 kernels ARE the diagonal MASA; same-class products $A_0$-valued with $\mu_4$ coefficients EXACTLY), the fine groupoid law (associative 0/3375) and the $W(3, 2)$ commutation skeleton; NO scalar V-cocycle (0/210). Successor criterion named: root-individual operator data. Controls fire. |
| `v807_lorentz_nullselector.py` | Original v807 S1 RR/Serre and null-spinor algebra: d(4-d), d=4 rank one, conditional on stated functor assignment; E8.ONEOBJECT.01 addendum. |
| `v808_petersen_sixthroot.py` | The K5 edge machine + the bit-exact sixth-root decider. PETERSEN-RADIAL-EXACT + SIXTHROOT-SPECTRAL-ONLY (19/19 + 16/16): $X = (-2, -2, -2, 3, 3)$ unique primitive traceless block-constant; the $E(K_5)$ charge census $\{-4{:}\,3,\ +1{:}\,6,\ +6{:}\,1\}$ = the $SU(5)$ decuple against the certified v774 table; Petersen $SRG(10, 3, 0, 1)$ certified ($\lvert\text{Aut}\rvert = 120$ census = $S_5$); distance shells $(1, 3, 6)$ CHARGE-PURE; the flavor hexagon = the distance-2 shell [E neu]; equitable quotient with hypercharge eigenvector $(6, -4, 1) \to -2$; $\text{spec}(P^6) = \{1, 64/729, 1/729\}$ = the deployed transport spectrum EXACTLY (v54/v56/v124); the 10-dim multiplicity prediction ($\times 5 = g_{\text{car}}$ fast, $\times 4 = \lvert\mu_4\rvert$ slow) typed [H neu]. PART 2: $B = \frac{1}{18}[[13, 1, 4], [1, 13, 4], [4, 4, 10]]$ (every entry a compiler number) is the UNIQUE symmetric doubly stochastic PSD sixth root in the corpus basis; $B^6$ carries the deployed spectrum exactly but is bit-equal to NO deployed matrix ($GL(3, \mathbb{Q})$-conjugate to v486's $M^6$ and the cusp-diagonal form) — the corpus freezes the SPECTRUM, not the basis. |

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| Script | What it machine-checks *(continued)* |
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| `v809_curve_code_doily.py` | The certified outer bridge as an incidence-operator statement with a spectral payoff. DOILY-EXACT (21/21): $NN^T = B + 2I$ ENTRYWISE ($N$ = the $K_6$ doily incidence, 15 duads × 15 synthemes; $B$ = the certified v752/v774 symplectic incidence); spec = $\{9, 4^9, 0^5\}$ exact; the singular values of $N/3$ are $\{1, 2/3, 0\}$: the RECOVERY BASE RATE 2/3 (v221) as the unique nontrivial singular value of the canonical duad-to-syntheme incidence — the $(2/3)^6$ identification typed [H neu], pending the six-step composition; Petersen edges = synthemes 15/15 at the $q^*$ point; EVERY census bridge factors EXACTLY as $N_{\text{code}} = P_\Delta^T N_{\text{curve}}^T P_\Gamma$ — a DUALITY, not an isomorphism (the inner route stays dead, 0/720); the Arf switch set $T_6$ = the six within-block duads; $N/3$ doubly stochastic, typed classical, not CPTP. |
| `v810_cutcode_moment.py` | The Arf split as the cut classification of $K_5$ + the one-line master moment with the frozen 210 decision. CUTCODE-EXACT + MOMENT-INCIDENCE-PARTIAL (17/17 + 16/16): the cut map $\delta$ is injective-isometric on all 256 cells (kernel = the repetition code); the Arf split $16 = 1 + \bar{5} + 10$ IS the cut classification $(0\|5) + (1\|4) + (2\|3)$ [E neu]; the Clebsch 40 edges grade $5 + 20 + 15$ over the Arf shells with forbidden blocks EMPTY and within-10 = Petersen (isomorphism census 120); the 240 roots / 60 $\mu_4$-lines grade $80 + 160$ / $20 + 40$ — Arf/cut grading, NOT matter. PART 2: the symbolic identity $\sum_{i<j}(x_i+x_j)^2 - 3\sum x^2 = (\sum x)^2$ yields $\mathrm{Tr}_{S^+} X^2 = 4\|X\|^2 = 4\,h(E_8) = 120$ in one line; charpoly $(t+2)^3(t-3)^2$ exact, 72 audit-only; the $210 = \binom{10}{4}$ map DECORATIVE — equivariant bijections killed by the fixed-point obstruction (exhaustive); the partial $80 \to 5 \times 16$ miss-slot map and the Newton tie survive as audit. |
| `v812_flavor_graph_filtration.py` | The flavor-matrix filtration $Q \to K \to L$ as an incidence-compiler fingerprint. FILTRATION-FINGERPRINT-ONLY (16/16): the row budgets are EXACT, order-exact counts of the $K_{3,2}$/prism/$K_5/H_8$ incidence chain — $Q$ $(4, 5, 6)$, $K$ $(6, 9, 10)$ (line graph ≅ triangular prism ≅ $C_6(2, 3)$), $L$ $(10, 14, 16)$ — with the level-wise additivity $(4, 5, 6) + (6, 9, 10) = (10, 14, 16)$ mirroring the deployed $L = K + Q$; but NO entry-level reconstruction: none of the frozen structural candidates (25/20/47 per level, entry-exact, 6 column permutations) reproduces any deployed matrix; the exhaustive false-positive rates 0.748%/0.035%/0.005% make the misses INFORMATIVE. Typed: an audit fingerprint at budget level, not an entry-level incidence construction. |
| `v813_p1_index_kms.py` | The closed normalization identity $c_3 = 1/(I_{\text{Jones}} \cdot \beta_{\text{angle}}) = 1/(8\pi)$, its convention audit, and the sharpened P1 contract. P1-IDENTITY-CLOSED (13/13): $I\beta c_3 = 4 \cdot 2\pi \cdot 1/(8\pi) = 1$ EXACT (formal Fraction × $\pi^k$ arithmetic); dependency audit: the index $I = 4 = \|\mu_4\|$ is an INDEPENDENT $c_3$-free theorem (zero occurrences in all four sources, gated by file scan); $\beta = 2\pi$ independently MEASURED (v526, wrong $\beta$ excluded; the v519 radian pin is the one shared convention and cancels); $c_3$ STAYS the P1 axiom the identity EXPLAINS; the finite evenness precursor measured: $w_1 = w_3 = 1/4$ exactly at every level, symmetric zero-sum offset $\pm 2^{-(l+1)}$ (the no-anomaly shape); the sea state does NOT equidistribute (dev 0.188) — the bridge must normalize the MODULAR RESPONSE; contract registered [O] with the one open bridge and the kill criterion; controls: index $2 \to 1/(4\pi)$, booked elsewhere by the corpus. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v814_k5_sixstep_transport.py` | The six-step composition and the canonical 10-dim transport proposal. SIXSTEP-CLOCK-ONLY + TRANSPORT10-PROPOSED (23/23): the rate $(2/3)^6$ arises canonically on three spatial routes (doily alternation, hexagon compression, hexagon circulation), each blocked by a typed obstruction (no unit mode / norm obstruction, bipartite sign $-2/3$, wrong space) — the deployed six is the CLOCK exponent (v124 $\lambda_n = (1 - n/3)^6$, v486 hand, fixed eigenbasis); sixth-power blindness [E neu]: only per-step data distinguish walk from clock; dated correction on the v808 row: $T_{v221} = B^6$ BIT-EXACT, so SIXTHROOT upgrades to BITEXACT-vs-v221; the canonical proposal $T_{10} = P_{10}^6$ in the $\Lambda^2$ basis passes all gates (spectrum $\{1, (1/3)^6 \times 5, (2/3)^6 \times 4\}$, equitable v774 quotient $= (Q_{\mathrm{Pet}}/3)^6$ bit-exact) — a promotion-ready proposal, no deploy claim. |
| `v817_positive_descent_master.py` | Abstract positive-descent theorem and G_net finite hypothesis checks; the prime packet instance is maintained by Prime Front. |
| `v832_anchor_flavor_checksum.py` | The anchor affine normal form and the flavor bidirectional checksum: $p_n(a) = 2 + 2^n$ obeys $p_{n+1} = 2p_n - 2$ identically; $T(x) = 2x - 2$ has the unique fixed point $2 = \lvert\mathbb{Z}_2\rvert$ and the $T$-orbit of 4 is the compiler quintet $(4, 6, 10, 18, 34)$ – the whole budget (240, 248, 30, 40, 48, 41) from ONE recursion. Part B: $Ra = (4, 10, 13)$, $R^{\mathsf{T}}a = (6, 18, 8)$ – the same anchor decodes both sides; of 12 sibling candidates exactly ONE passes, and the anchor contraction $(Ra)_2 = 10 = p_3$ kills both siblings. ANCHOR-AFFINE-EXACT + FLAVOR-BIDIR-CHECKSUM (selective), $18/18 + 15/15$ exact. |
| `v833_gaussian_ramification_ladder.py` | The $(1 + i)$ ramification ladder: the four documented roles of the ramified Gaussian prime $\pi_2 = 1 + i$ are ONE machine-checked object – (i) norm doubling $(1 + J)^{\mathsf{T}}(1 + J) = 2I_8$ with the empty zero class (0/240, $240 = 15 \times 16$); (ii) the 4-bit address $L/(1 + i)L \cong \mathbb{F}_2^4$ with the deck acting trivially; (iii) the non-split jet $\mathbb{Z}[i]/(2) \cong \mathbb{F}_2[\varepsilon]/(\varepsilon^2)$, deck $= 1 + \varepsilon$, exactly 0 of 65536 sections deck-equivariant; (iv) the metaplectic lift $\zeta_8 = (1 + i)/\sqrt{2}$ with $(SH)^3 = \zeta_8 I$, the exact Clifford census $\lvert C_2/\mu_8\rvert = 11520$ and the RM-CSS phase bit $\chi(H \otimes I) = 1$. RAMIFICATION-LADDER-EXACT, 33/33 exact. |
| `v834_pg32_flag_completion.py` | The PG(3,2) flag normal form for the $105 + 35$ vacuum completion: $\binom{4}{2}_2 = 35$, $140 = 35 + 105$ affine 2-flats, $105 = 35 \times 3$ point-line flags $\cong$ point pairs; the RM layer rebuilt ($[15, 10, 4]$ enumerator exact, Grams $22I + 6J$ / $6I + J$ / $28I + 7J$); the vacuum coordinate is ORIGIN RESTORATION. The incidence kill typed as a split: $105 = 45 + 60$ is the $\mathrm{Sp}(4, 2)$-orbit decomposition on both sides; on the 60 non-isotropic rungs the symplectic polarity gives a canonical incidence-preserving dictionary; on the 45 doily rungs the obstruction is typed ($(t + U) \cap U$ empty for 45/45, flag Gram $18I + 3J \neq 22I + 6J$) – '105 checks = 105 flags' is exact as a deck-equivariant bijection, NOT an incidence isomorphism. PG32-FLAGS-EXACT + INCIDENCE-SPLIT, 26/26 exact. |
| `v844_message_doily_rank.py` | The E8 addressing formula and the Doily–Pascal rank theorem, two exact corpus compressions: $M_n = 15 \cdot 2^n = (15, 30, 60, 120, 240)$ certified against the rebuilt objects with one NAMED structure per rung ($h(E_8) = 30$ derived from the rebuilt root system itself: highest root height 29, marks $\{2, 2, 3, 3, 4, 4, 5, 6\}$; 60 Gaussian lines; $120 = \lvert R^+\rvert$; $240 = 15 \times 16$ census; coda $248 = 15 \cdot 16 + 8$; wrong anchor/multiplier controls fire); and $NN^T = B + 2I$ entrywise for the Cremona–Richmond duad–syntheme incidence under the K6 duad bijection, so charpoly$(NN^T) = (x - 9)(x - 4)^9 x^5$ exactly: $\operatorname{rank} N = 10 = A_\Lambda$, $\dim\ker N = 5 = g_{\mathrm{car}}$ (the six ovoid indicators SPAN the kernel), multiplicity $9 = N_{\mathrm{fam}}^2$, top value $3 = N_{\mathrm{fam}}$, recovery $2/3 = (N_{\mathrm{fam}} - 1)/N_{\mathrm{fam}}$, six-step $(2/3)^6 = 64/729$ exact – the P2 integers are the singular-value data of the doily incidence; NO $(1/3)^6$ doily mode (v814 typed, no upgrade). MESSAGE-LADDER-EXACT + DOILY-PASCAL-RANK-EXACT, $23/23 + 16/16$. |

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| Script | What it machine-checks *(continued)* |
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| `v845_cfin_normal_form.py` | The finite compiler normal form $C_{\mathrm{fin}} = (V, \hbar, q^*, \sigma, \iota)$ assembled as ONE object read off the Gaussian quotient: the unique selector-picked refinement (16 refinements, Arf $6 + 10$), the order-3 family cycle, the parity lift onto $C_{\mathrm{even}}(5)$ with $\beta \equiv \hbar$ and $q_{\mathrm{wt}} \equiv q^*$; derived counting $16 = 1 + 5 + 10$ with $\lvert\mathrm{Sp}(4,2)\rvert = 720$ and $\mathrm{Stab}(q^*) = 120$ faithful $S_5$, the Pascal reading $(1, 10, 5) = (\binom{5}{0}, \binom{5}{2}, \binom{5}{4})$, the ovoid projective reading, and the budget $240 = 16 \cdot 15$, $30 = 2 \cdot 15 = h(E_8)$, $248 = 16 \cdot 15 + 8$ against the actual lattice census. Built-in negative: the exact 128-spinor counting bound ($\geq 7$ of 15 classes always adjoint-side; the concrete census saturates $8 + 7$ with zero mixed classes) and the weight-level kill re-verify v775 ROOTCLASS-MIXED, not overturned. P2 ($g_{\mathrm{car}} = 5 =$ the $\iota$ slot count) narrowed to the v799 residual R1′ ([C]), NOT eliminated. NORMAL-FORM-ASSEMBLED, 28/28 (SPEC v2 declared). |
| `v849_cfin_unique_cofinal_lean.py` | CFIN.UNIQUE.01: finite compiler uniqueness, one orbit and C6 automorphisms; cofinal Weil mirror moved to Prime Front. |
| `v852_arf_vacuum_ovoid.py` | The Arf-vacuum syndrome theorem and the integral five-mode decoder: the Arf bit $a(t + U) = \hbar(u, v)$ on ALL 140 affine flats (position-independent); the two-bit syndrome table $(s_{\mathrm{vac}}, s_{\mathrm{Arf}})$ splits them EXACTLY into 15/20/45/60; the LOCAL Hecke theorem at all 15 points ($12 + 16 = 28 = \sigma_3(3)$, $12 - 16 = -4 = a_3$, isotropic directions $\leftrightarrow$ points of $x^{\perp}/\langle x\rangle$); the code pencil $[15, 11, 3] \supset \{[15, 10, 4],\ 2 \times [15, 10, 3]\} \supset [15, 9, 4]$ with the canonicity lemma (all 16 refinements induce ONE Arf functional on $H$) and the vacuum bit as the $\mathrm{RM}(2, 4)$ parity bit; the doily kernel gains the integral inverse $N^+ = N^T/4 - J/36$ (four Penrose axioms exact in $\mathbb{Q}$; $\mathrm{SNF}(N) = \mathrm{diag}(1^{10}, 0^5)$ torsion-free with a $\det \pm 1$ minor exhibited), the projector $P_5 = I/2 - B/4 + J/12$ ($\mathrm{tr} = 5 = g_{\mathrm{car}}$), the exact closed-form decoder $\hat{m} = (I - P_5)m$, the $S_6$ standard representation (11/11 classes), and the three-ring correspondence ($\mathbb{R}$ ovoid span / $\mathbb{Z}$ torsion-free / $\mathbb{F}_2$ code $A_q = [15, 5, 6]$ via $v_a \bmod 2 = q_a$). ARF-VACUUM-SYNDROME-EXACT + OVOID-DECODER-EXACT, $24/24 + 23/23$ exact; NO RH claim. |
| `v853_bent_css_mangoldt.py` | ARF.BENT.CSS.01: exact bent phase and CSS code identities; Mangoldt incidence experiment moved to Prime Front. |
| `v857_simplex_fourier_winding.py` | The E8 audits I: the Gaussian census IS a Fourier law ($\hat{r} = (240, -16^{15})$; the uniform-nonzero channel $P = (J - I)/15$ with spectrum $\{1, (-1/15)^{15}\}$ and two-step contraction $1/225$, exact in Fractions), the packet numerator EQUALS the code Walsh transform at every level $\leq 16$ (Eisenstein normal form $15\,\Theta_{D8} = 7\,\Theta_L + 8\,\Theta_0$), and $mh_2(n) = -1/15$ for EVERY odd $n \leq 16500$ (integer identity; all 1911 odd primes covered) – v817's per-prime measurement is the prime restriction of a CHARACTER THEOREM of the code, typed honestly SPECTRAL-only (the gradings do not refine each other); the tfpt_2 winding line collapses to ONE adjugate quadratic $q_{\mathrm{wind}} = e_1^{\mathsf{T}}\,\mathrm{adj}(tI - R)\,\mathbf{1} = t^2 - g_{\mathrm{car}}\,t + \lvert Z_2\rvert$ with the triple lock $s = 6$ (trace/det/Coxeter each alone; $\{6\} = \{\lvert R^+(A_3)\rvert\}$), $\Delta(6) = 39200 = 2 \cdot 140^2$ (NOT a collision point; the sole $2\times$square hit) and the reality threshold $s^* \approx 2.825$ (the unique $\Delta$-root in $(2, 3)$; deployed $\approx 2.83$). |
| `v858_g31_clock_alphabet.py` | The compiler clock alphabet as a NORMAL FORM of $G_{31}$: $\mathrm{Deg}(G_{31})/4 = \{2, 3, 5, 6\} = \{\lvert Z_2\rvert, N_{\mathrm{fam}}, g_{\mathrm{car}}, \lvert R^+(A_3)\rvert\}$ with $\gcd = 4 = \lvert\mu_4\rvert$, $\mathrm{lcm} = 120 = \lvert R^+(E_8)\rvert$, $\mathrm{lcm}/\gcd = 30 = h(E_8)$ and the audit identity $16 \times 15 = 240 = \lvert R(E_8)\rvert$ REALIZED on the roots (the 60 reflections = the 60 Gaussian lines, 4 per class); $(8, 12, 20, 24)$ the UNIQUE integer quadruple with product 46080 and sum 64; the 607-group rank-4 kill scan: $G_{31}$ the SOLE full-battery passer (weak-triple impostors and raw-sum-64 coincidences typed by name; the honesty line kept: degree-only audits cannot see the reducible torus); the $W(D_5) \times W(A_3)$ fence re-killed by COMPUTED centers 4 vs 1 (v634 cited for the structural refutation). |

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| Script | What it machine-checks *(continued)* |
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| `v874_feedback_decoder_lean.py` | TWO EXACT COMPILER THEOREMS: the integral basis $P = [2 \cdot \mathbf{1} \mid e_3 \mid Re_3]$ (det $-4$, SNF $(1,2,2)$) conjugates the whole winding line to the gain-plus-companion form $[[s{+}4,0,3],[0,0,-2],[4,1,5]]$ exactly (symbolic in $s$), so $\chi_{R_s} = (t - s - \lvert\mu_4\rvert)\, q_{\text{wind}} - 12t$ with $12 = N_{\text{fam}}\lvert\mu_4\rvert = \dim g_{\text{SM}}$ and the honest typing $\mathbb{Z}^3/P\mathbb{Z}^3 = \mathbb{Z}_2^2$ (cardinality 4, NOT $\mu_4$); and the winding quadratic decodes the clock alphabet, $q(\{2,3,5,6\}) = (-4,-4,2,8)$, $D(y) = (y{+}4)^2(y{-}2)(y{-}8)$ with budget coefficients $(2,48,32,256)$, and $\lvert\mu_4\rvert\, q(\lvert Z_2\rvert)/\lvert R(E_8)\rvert = -1/15$ EXACTLY the Walsh message eigenvalue – an audit theorem, not a functor. Lean mirror: FlavorFeedback.lean, 11 theorems, standard axioms. |
| `v875_shell_global_lean.py` | THE GLOBAL SHELL THEOREM: the $-1/15$ census law upgraded from the finite census ($n \le 16500$) to a proof for ALL shells – $\mathrm{Sp}(4,2)$ transitivity (60 transvections, identical to the shell-module action) + $(1{+}i)$ doubling + $\Theta_L = 240\sigma_3$ give $\hat{\Theta}_u(n)/\Theta_L(n) = -1/15$ at every ODD $n$ exactly; a non-transitive control breaks the law (transitivity load-bearing), the even-shell formula differs correctly (7/135 at $n = 2$). Lean skeleton GaussianShells.lean: the consequence chain kernel-checked from the named ShellSystem hypotheses (7 theorems, exact integer identities, no division); the census range retained as the exact ward. |
| `v878_ledger_integrity.py` | THE STATUS-LEDGER INTEGRITY GUARD: round 36 found and fixed the first primary-key violation (claim_id FORM.QGEO.03 double-booked 2026-06-15/2026-06-18; the SeamDeckClosure row renamed FORM.QGEO.04 with a dated QA note, all references updated); the guard enforces the 10-column schema, global claim-id uniqueness (the fix pinned), existing supersession targets with an ACYCLIC supersession graph, unique active supersession targets modulo the frozen remove-only allowlist of the two legitimate 2026-06 joint closures, and the dependencies cross-reference census frozen remove-only (13 SCCs / 265 ids; acyclicity enforced as NO NEW CYCLES). |
| `v880_finite_anchor_theorems.py` | the four finite anchor theorems: the census-free normal form $q^*(x) = \sum_{i<j} x_i x_j + \sum_i x_i = \binom{\lvert x\rvert+1}{2} \bmod 2$ (weight-only; the selector forced on the linear part); the EXCLUSIVE anchor classification of $PG(3,2)$ lines (all 15 doily lines carry $\{2,1,1\} = a$, per-line Vieta $(4,5,2)$ with $e_2 = g_{\text{car}}$, $e_3 = \lvert\mathbb{Z}_2\rvert$; 10 secants carry the WRONG anchor $(1,2,2)$, 10 external lines $(1,1,1)$); the doily as the multiplication table $\Lambda^2 E \times \Lambda^2 E \to \Lambda^4 E$ (wedge nonzero EXACTLY on doily lines); one-step quadratic exhaustion FORCES $g = 5$ ([E] theorem; the Calderón premise stays [O]). |
| `v888_finite_pfaffian_signature.py` | the complete finite Pfaffian/$K_6$ signature: Specht channel scalars $(1,0,2/3)$ exact; the 15 doily lines ARE the 15 Plücker monomials of $Gr(2,5)$ with perfect sign match; the $(16,6,2,10)$ Hadamard anchor; the $S_6 \to S_5$ vacuum selection; THE FLAVOR HEXAGON IS THE $\mathrm{Aut}(C_{\text{fin}})$ ORBIT ($g^2 = \sigma$ exact, $(2/3)^6 =$ the recovery value); the $K_6$ motif theorem ($35 = 15$ matchings $+20$ triangles); the fermionic Pfaffian signs GAUGE-MATCHED via the canonical character $\chi(i) = (-1)^{i+1}$; the self-hosting counting theorem $\binom{g+1}{2} = g!!$ forces $g = 5$ uniquely (nontrivially) with $N_{\text{fam}} = 3$; the mod-30 clock is CLOCK-DISTINGUISHED. |
| `v896_wick_block_functor.py` | the Wick functor arc of the finite compiler: the six roles exist in the deployed seam (sector law, $q^*$ grading, $C_6$ intertwining exact) but the vacuum kernel is channel-diagonal; the PROVED scalar obstruction (the $C_6$ 2-cycle $\{4,5\}$ forces a duad zero); the block resolution (BLOCKFUNCTOR-CONSTRUCTED: 15/15 blocks and Wick monomials with the canonical sign law) and the decisive fact SEAM-DIAGONAL – the value level awaits a physical mixing mechanism; the [O] premise is unmoved. |

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| Script | What it machine-checks *(continued)* |
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| `v898_kms_schur_mixing.py` | the channel-mixing state the block Wick functor awaits: the $C_6$-covariant mixing KMS state EXISTS ($u = 1$, $t = 1/8$, $\beta = 1$; all 15 blocks at the floor, grading $5/5 + 10/10$, canonical Pf4 signs; $\beta = 2$ robust); EXACT (sympy): Schur elimination of the boundary GENERATES all 10 carrier duads from the bare diagonal state – on both fronts the real correlation lives in the ELIMINATED boundary; the frozen 15-signature table is the registered prediction; RP-THETA-OPEN named; the [O] premise is unmoved. |
| `v903_seam_rp_exclusion.py` | the seam RP/modular exclusion and the two dead readings: strict RP forces $t = 0$ under BOTH spatial reflections (odd-sector Gram eigenvalues exactly $\pm\lvert a_J\rvert$: OS positivity $\iff$ the v898 mixing floor falls – mutually exclusive on this family); $u \geq t$ forced; the $2\pi$-KMS locus is the point ($t$=0, $\beta$=$2\pi$); the complete twisted-OS census is exclusionary (0/6); the dilation route splits (marginal witness ON the cone boundary, Schur compression carrying the full Pfaffian mixing with $t^2 \cdot 3m/(1-m^2) = 1/200$ exactly; the strict collar route obstructed by an exact linear $2t$ law); $t_{\mathrm{gap}} = 0.230949$ theorem-shaped ($\mathrm{Pf} = -(t-1)(3t^2-1)(9t^3+21t^2-t-1)$ exactly, Sturm uniqueness on $(0, 1/4]$, kernel dimension exactly 2; no root is 1/8); the two dead readings booked as first-class negatives: '$t = 1/8$ is a compiler value' is a DECORATION (PIN-ABSENT), '$N_{\mathrm{fam}} = 3$ as minimal mediation rank' is REFUTED (rank 1 suffices; deployed rank exactly 2; purity = dimension bookkeeping); the v898 [O] premise is unmoved. |
| `v908_seam_equilibrium_wiring.py` | the seam equilibrium wiring (the physics closure of the round-60 questions): strict 2-cycle RP is IMPOSSIBLE on the whole C6-covariant class (the seat law reduces exactly to $M_1 = \mathrm{diag}(a_J, -a_J)$, interior EMPTY – the marginal witness is neither a limit of covariant strict equilibria nor isolated); the only escape is the covariance-breaking $X$ coordinate at exact linear price $2\varepsilon$, with the floor exchange $\mathrm{Pf}_4(\varepsilon) = (\varepsilon - \frac{1}{200})(\varepsilon + \frac{1}{200})$ exactly – the canonical sign gate IS the negative effective RP margin; the strict-collar obstruction is a two-seat linear law with kernel $\{J, Z\}$, the deployed wiring is PURE-I (maximally obstructed), and equilibrium witnesses $V_J/V_Z$ carry the full 1/200 mixing at ZERO entropy production – NO NESS NEEDED (stationary $\Rightarrow \sigma \equiv 0$ exactly; drive buys no RP); the registered next contract – is PURE-I compiler-forced? (SEAM.STATE.WIRING.SELECTOR.01) – is CLOSED in v911: deployment choice, not a compiler theorem; the v898/v903 [O] premise unmoved. |
| `v911_wiring_freedom.py` | the wiring freedom theorem (closes the v908-registered contract SEAM.STATE.WIRING.SELECTOR.01): the exact Groebner census determines EXACTLY ONE admissible 3-dim wiring component mod rule gauge (the rotation cell $C_{\mathrm{rot}}$); Z/X wirings are excluded gauge-invariantly by orientation propagation; pure-I is an INTERIOR point, connected to the pure-J ray by the integer gauge element $(\oplus_5 J_2) \oplus I_6$ – PURE-I IS A DEPLOYMENT REPRESENTATIVE, not a compiler theorem; THETA-CONVENTIONAL: no compiler-side demand distinguishes the $\theta_S$ sheet-swap frame in the 9-dim rule-gauge algebra (five demand classes silent-on-angle), and the deployed parent PASSES strict collar RP exactly in the integer-conjugated frame – the round-58/59 obstruction is FRAME-RELATIVE; RP-FRAME-COVARIANT: the strict-collar RP Grams transform exactly covariantly under the rule gauge, the pure-I exclusion is a frame statement (option 2), NOT a wiring no-go, while the Z/X exclusion IS orbit-invariant; the canonical sentence is frozen and 'pure-I is compiler-forced' is FORBIDDEN; the v898/v903 [O] premise stays [O]; zero RH content. |

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| Script | What it machine-checks *(continued)* |
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| `v957_ccc_crossover_kinematics.py` | CCC.SEAM.KINEMATICS.01 (rounds DL–DLV, 2026-08-24): THE CCC CROSSOVER KINEMATICS DICTIONARY ($28+29+18+21+14+7$ gates, six probes embedded byte-exact). (1) Penrose's reciprocal gauge $\Omega\,\tau(\Omega)=-1$ is the element $\tau=\rho^2\sigma$ of the PROVEN $\mu_4$ stabiliser $D_4$ ($\tau$ reverses the clock = the $\mathbb{Z}_2$ sheet parity; the crossover circle carries all four marks). (2) The clock alone leaves exactly ONE conformal modulus on the 2D seam and the reciprocal $\mathbb{Z}_2$ kills it (Liouville-exact; $so(3,1)$ commutants 2/0). (3) The Stinespring dilation of the v221 transport is explicit: $V^\dagger V=1$, $\|T^n-P_\star\|=(2/3)^{6n}$ EXACT, global overlap invariant – entropy reset = reduced channel, no information destroyed. (4) The kinematic 2D→3D→4D uplift: stereographic(Hopf) $=z_1/z_2$ EXACT, crossover $X=S^3/\mathbb{Z}_4$ = the lens boundary of the corpus $A_3$ ALE (deck free; marks = four linked fibre circles in the two sheet orbits, Gauss linking 1; $so(4,1)$ deck-commutant = 4 Killing / 0 boosts – Yamabe rigidity with NO extra condition, global via the Obata family); the 4D bridge is the cone with the inversion as conformal aeon swap; the Lorentz shadow $(1/\cos^2 T)(-dT^2+g_{S^3})$ is Einstein $R_{ab}=3g_{ab}$ with a spacelike $S^3$ crossover and $\hat{\Omega}\hat{\Omega}=-1$. (5) KMS closure: detailed balance exact, spec$(-\ln T)=\{0,6\ln(3/2),6\ln 3\}$, $T=e^{-H_{\rm mod}}$, $\beta_{\rm angle}=2\pi=1/(4c_3)$ = Gibbons–Hawking on the fibre. (6) Holonomy: pole strings carry the full $\mu_4$, every seam string carries EXACTLY the 2-torsion $-1$ = the v522 GSO/fermion-parity wrap (typed [C]). (7) Selector necessity: $M(\delta)$ admissible reflections close to $D_4$ iff $\delta=\pi/2$; Majorana operator level: clock axes $\Theta$-real ONLY at $k=4$ – the UNIQUENESS half of the v534 selector DERIVED (survival + sign cited). (8) The frozen Gate-6 kernel: causal top-hat disc $\theta_{\max}\approx 1.16^\circ$, rates $\ln(729/64)/\ln 729$, $\kappa$ derived from v526 (declared band, contrast $\le 3.3\times10^{-4}$) – feeds the preregistered search `ccc-crossover-disc` (no data touched). The cyclic/CCC reading stays [C]; WOIT.OS.TWISTOR.01 stays [O]. |
| `v971_markov_embedding_generator.py` | DYN.MARKOV.EMBED.01 (finite kernel half [E]; external-review round 2026-08-27, 'three meanings of dynamics'): $Q=\log T$ of the v221 seam transfer kernel is an EXACT classical Markov generator – sympy-exact spectral log with $e^Q=T$, row sums symbolically zero, off-diagonals exactly $\{\ln(9/8),2\ln 3,2\ln 3\}=\{0.117783,2.197225,2.197225\}>0$, diagonals $-\ln(81/8),-\ln(81/8),-4\ln 3$; uniformization certificate $Q+(4\ln 3)I\ge 0$ entrywise $\Rightarrow e^{tQ}$ positivity-preserving doubly stochastic for ALL $t\ge 0$ ($T$ embeddable, every root $T^{1/n}$ stochastic);<br>spec$(Q)=-$spec$(H_{\rm mod})=\{0,-6\ln(3/2),-6\ln 3\}$ (gap $=\Delta$), detailed balance w.r.t. the uniform state. NEGATIVE control: second eigenvalue $-1/10$ gives det $=-1/200<0$ – provably NOT embeddable (embeddability is not generic; the RP/OS spectral positivity is the enabling fact). FIREWALL: closes relaxation→semigroup (a)→(b) at the finite kernel level ONLY; the local, size-consistent unitary dilation (b)→(c) is the separate open contract DYN.UNITARY.DILATION.01; the shared F_transfer relaxation shape is a universal CONTRACTION CLASS, not one physical clock (v723/v724/v777). |

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| Script | What it machine-checks *(continued)* |
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| `v972_seam_interaction_front.py` | SEAM.INT.ODDSPLIT.01 [E] + SEAM.INT.OSAVOID.01 [E] + SEAM.STATE.RPMIXING.01 [E] (+ updates to SEAM.INT.FKTOY.01 / WOIT.OS.TWISTOR.01 / SEAM.STATE.DERIVATION.01; three exploration rounds of 2026-08-27 consolidated, the probes embedded byte-exact with pinned source SHA-256 and executed verbatim, 20/20 + 18/18 + 25/25, verdicts CLASS_EMPTY_ON_BASIS + OS_VIABLE_ON_AVOIDING + RP_ADMITS_MIXING): THE SEAM INTERACTION FRONT – (1) the Jaffe–Pedrocchi sufficient condition for Majorana reflection positivity (arXiv:1406.1384; Wei–Li–Xiang arXiv:1601.01994) made DECIDABLE on the v529 FK toy: the crossing matrix $B$ is UNIQUE (left-monomial pairs biject onto crossing monomials), so '$B$ Hermitian AND $-B$ PSD' is a finite check; the free FK Hamiltonian certified on all 8 bond cuts ($\eta = -i$ forced); the confusion matrix vs measured RP PERFECT over 35 cells (23/12/0/0); the $\mu_4$-covariant quartic space (dim 464) EMPTY on the basis, the support-level exclusion pinning the full 1820-dim space – THE DISCOVERY: the v529 straddle law REFINES to the ODD-SPLIT LAW (JP predicted RP survival on CENTER-straddled cuts, $m = 4$ model, axes 7/15, 2+2 mirror split; measurement confirmed $(37, 0, 0)$ at $g = 0.5/1/2$; the v529 anchor ladder reproduced exactly). (2) the quartet-AVOIDING cut family of the clock-invariant $m = 4$ member generates the FULL even algebra (rank 15, dim $32768 = 2^{15}$; single-cut members $m = 2/6$ typed deficit exactly 1) and carries a POSITIVE OS/GNS structure at $g = 1$ (min $+4.43 \times 10^{-5}$, GNS 37/37, distances monotone) while the straddled control genuinely fails ($-6.94 \times 10^{-2}$); the $\mu_4$ clock maps the avoiding pair $\{3, 11\}$ onto itself exactly – the kill-test-2 counterweight NARROWED, not removed. (3) the v903 exclusivity is RAY- and REFLECTION-relative, not slice-wide: on the full 33-dim $C_6$-covariant mixing slice $\theta_{abT}$ sees exactly ONE coordinate ($a_J$, kernel value $-1/(1 + c^2) = -0.8240271368$ at $c = \tanh(1/2)$), 32/33 invisible ⇒ a 32-dim MARGINAL RP hyperplane $\{a_J = 0\}$; the $\theta_S$ defect map has rank 16 ⇒ a 17-dim first-order-neutral subspace with STRICT finite-$s$ witnesses ($\lambda_{\min} = 0.1558$ at $s = 1/8$); the v903 anchor reproduced ($a_J = +0.04068013$). Module-own exact S0 (sympy/GF(2), no probe imports): the JP decidability lemma on a 4-Majorana toy (9 pairwise-distinct crossing masks ⇒ unique $B$; the wrong-sign toy fails), the odd-split arithmetic (centered ⇔ 2+2 exact on $\mathbb{Z}_{16}$; the odd 1+3 block forces $b = 0$ – the obstruction; the flipped-sign-law mutant caught), the GF(2) generation bookkeeping (pair rank 15, $2^{15} = 32768$; within-halves rank 14 – deficit exactly 1), the window arithmetic ($33 = 24 + 9$, $32 + 1$, $33 - 16 = 17$; kernel value $\equiv -\cosh^2(1/2)/\cosh(1)$ symbolically) plus 4 verdict bars with tipping mutants. HONEST: no marker moves; SEAM.EQUIV.01 stays [O]; the v529 fence (one toy, one interaction class, [C] flat-band parent) inherited by all three results; JP sufficient-only; the S1 census sector-truncated. NO RH claim. |

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| Script | What it machine-checks *(continued)* |
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| `v973_seam_route_narrowing.py` | SEAM.EQUIV.MMST.01 [O update] + SEAM.EQUIV.TWISTOR.01 [O update] + SEAM.EQUIV.SKELETON.01 [E, NEW] + SEAM.EQUIV.01 [O update] (seam adjudication wave, 2026-08-27): THE SEAM ROUTE NARROWING ($19/19 + 20/20 + 19/19$, three probes embedded byte-exact with pinned SPEC SHAs, plus the module-own exact section S0). (1) MMST COVERAGE CARVED: the cited Osborne–Stottmeister theorem (CMP 398 (2023) 219, arXiv:2107.13834) COVERS the bilinear half of the seam target – $\mathfrak{so}(16)_1$ fermion-bilinear WZW currents converge STRONGLY on the finite-particle core (Thm 5.1 + Cauchy–Schwarz Eq. 229 incl. Majorana algebras), Virasoro/Koo–Saleur (Thms A/4.11/4.16), correlators (Thms B/C), polynomial errors $\text{Error}^2 \le C\,2^{-2\delta N}$, $\delta \in [0,2]$ – and does NOT cover four NAMED items: N1 the $\mu_4$/GSO extension $SO(16)_1 \to (E_8)_1$ as a NET statement (the 128 spinor currents are not bilinears), N2 the twist/spin-field sector, N3 the one-sided chiral edge net of a 2+1D bulk, N4 the holomorphy selector $\det K = 1$ vs 4; NEW MEASUREMENT: the collar edge-correlator rate $p = 3.48$ (Richardson 3.1–4.2, preregistered band $[0.5, 4.0]$), error $6.3 \times 10^{-2} \to 2.7 \times 10^{-6}$ over $N_x = 16..256$, the $M = 3$ control does NOT converge (min rel dev 0.836); exact-integer coverage gates ($248 = 120 + 128$, $240 = 112 + 128$, det Cartan $4 \to 1$, index $2 \times 2 = 4 = |\mu_4|$). (2) TWISTOR MEASURE RIGIDITY: on the computable v508/v515 sector the 18-parameter deformation of the declared measure data has executed-constraint ranks 9/8/2/2, joint 13, 4 gauge directions ⇒ the moduli kernel is EXACTLY 1 = the uniform sector-weight shift $dv = (1,1,1,1)$, mapping to the untwisted Okubo square $Q^{(0)} = 36(1,2,1,0,0)$, annihilated by every executed certificate functional BY CONSTRUCTION; ablation ledger $(1,3,1,2,3,10)$; the $W_2 : W_{13}$ tester honestly NOT-TRANSCRIBABLE. (3) OS SKELETON SHARED: all FIVE WOIT kill-test main invariants ($\alpha$: forced $\eta = +i$, $\Theta^2 = +1$, bond RP; $\beta_1$: GSO 2-torsion; $\beta_2$: OS quotient + KMS $\lambda_{\max} > 1$; $\beta_3$: perp descent; $\gamma$: mirror flip, multiplicity-free, mark incidence) hold VERBATIM on the p+ip collar edge; ALL divergences control-typed to ONE mechanism (the twistor lattice's exact chiral checkerboard $C(\text{even}) = 0$ vs the collar's $K(\text{even})$ up to 0.1736 – confirming v519-R4 from the second route); the top edge realizes the anti-chiral state EXACTLY ($7.5 \times 10^{-16}$); the $M = 3$ collar loses the skeleton first at GAMMA.MARK.INCIDENCE. Module-own S0 exact: root census + Cartan determinants + index arithmetic; the uniqueness linear algebra in exact rationals (ranks $(9,8,2,2) \to 13 \to$ kernel 1, ablation ledger exact, the dropped-C4 mutant opens the kernel to 3 – CAUGHT); the skeleton mechanisms exact (bipartite $SHS = -H$ ⇒ odd functions vanish on even distances; $PHP = -H$ on a strip ⇒ $K_{\text{top}} = -K_{\text{bot}}$); verdict bars with tipping mutants (a fake COVERAGE_FULL is REFUSED by the nonempty N1–N4 gate; hash mutation CAUGHT). HONEST: the coverage gates are executable transcriptions of a literature adjudication; the rate SUPERCONVERGES ($p > \delta_{\max} = 2$) vs the cited upper bound (band calibration-informed); the sector uniqueness is NOT a BCOV derivation; skeleton fidelity typed per test. NO marker moves: MMST and TWISTOR stay [O], the parent stays [O]; the only new row is SEAM.EQUIV.SKELETON.01 [E]. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v974_alpha_faces_computed.py` | ALPHA.QUILLEN.EXACT.01 [O update] + SEAM.CONTACT.UNIT.01/.02 + EM.WARD.01 / ALPHA.QUILLEN.INFLOW.01 + ALPHA.QUILLEN.FACES.01 [E, NEW] (seam/alpha promotion wave 2026-08-27): both named faces of the alpha target now carry computed witnesses — three probes embedded byte-exact, sha256-pinned and executed verbatim (21/21 MATCH_MODULO_LOCAL_FACTOR + 20/20 PUMP_QUANTIZED_MATCHES_CHERN + 18/18 CHAIN_SHADOW_OK), plus the module-own exact section S0 (sympy/Fraction/mpmath, no probe imports). (1) The BFK face EXECUTED: the Burghelea–Friedlander–Kappeler gluing constant is exactly $2^{-N}$ (derived symbolically $N = 1..4$, general interval lengths); massless reassembly $\det'_\zeta(\Delta_{S^1}) = l^2 = 2^{-N}\prod(2a_j)\,(l/N)\,\det'(R_0)$; at the KMS scale $l = 2\pi = 1/(4c_3)$: $R_0 = 16c_3\,\mathrm{circ}(2,-1,0,-1)$, $\det' R_0 = 2^{16}c_3^3$, glued product $(2\pi)^2$; $R(m) = (G_m|_{\mathrm{marks}})^{-1}$ exactly; vs the v484 mark matrix NO global proportionality (kernels in swapped $\mu_4$ channels: gauge zero mode vs sign rep) but channel-wise EXACT (same $\mu_4$ Fourier eigenbasis, \|spec\| multisets $\{0,2,2,4\}$, doublet ratio $4/\ln 2$ with $c_3$ cancelling, $\mathrm{Tr}\,B^k = 4^k + 2\cdot 2^k$ mirroring $\mathrm{Tr}\,C^k = 4^k + 2(-2)^k$); the v485 scale channel reproduced exactly via the KMS unit. (2) EM1 COMPUTED: pumped charge == edge spectral flow == Streda == Chern == 1 per flux quantum on the v367 collar (conjugation mutant flips all four readings simultaneously; wrong-band projector breaks quantization). (3) The finite Quillen identity TWO-SIDED: Berry integral == Poincaré–Lelong divisor integral == +1 over the twist torus, difference an exact form ($3.3\times10^{-16}$), transition winding == Chern; typed conditional on the quasi-free rigidity premise (measured, not proven — SEAM.STATE.RPMIXING.01). S0 re-derives everything independently: the BFK closed forms from $\zeta_R$ values, the alpha anchor at 50 digits ($8b_1 = 164/5$ exact, unique root, $\alpha^{-1} = 137.03599921684071$, split residual $6.4\times10^{-58}$), the inflow bookkeeping ($c_- = 8 = g_{\mathrm{car}} + N_{\mathrm{fam}}$, $k_Y = 5/3$, $(1/k_Y)\,41/6 = 41/10 = b_1$), a probe-free coarse FHS Chern certificate. Mutants: hash mutation caught by the sha256 pin; the wrong KMS unit ($2\pi \to 4\pi$) breaks the scale identity by exactly $-2\ln 2$; the $b_1$ mutant caught. NO marker moves — the target stays [O]: remaining content = {rigidity premise} + {channel-swap question} + {continuum Bismut–Freed identification}; $\alpha^{-1}$ stays [E] regardless. |
| `v975_dimension_selector_4d.py` | DIMENSION.SELECTOR.4D.01 [C, NEW] (master-route wave 2026-08-27): the CONDITIONAL 4D dimension selector, 16/16 sympy-exact. Under the named TFPT-compatible axiom set A1 ($d > 2$: propagating local gauge fields), A2 (dimensionless Yang–Mills coupling), A3 (real self-dual/anti-self-dual 2-form sectors: the Euclidean Hodge star an involutive endomorphism of 2-forms), A4 (chiral Weyl representations: even $d$), the dimension $d = 4$ is UNIQUE and MINIMAL on the exact scan window $d = 2..12$ — and the selector is OVERDETERMINED: A2 alone = $\{4\}$ ($[g_{\mathrm{YM}}] = (4-d)/2$ forced symbolically, unique zero 4) and A3 alone = $\{4\}$ (the explicit $6\times6$ star on $\Lambda^2(\mathbb{R}^4)$, $*^2 = I$, eigenvalues $\pm1$ at multiplicities 3/3), while A1 & A4 alone give $\{4,6,8,10,12\}$ — chirality does NOT select 4 by itself (honest ablation ledger). Weyl chirality certified by exact Clifford algebras $d = 2..7$ (even $d$: $\gamma_*$ anticommutes, $\gamma_*^2 = I$, traceless; odd $d$: the product is CENTRAL and scalar). Signature typed honestly: Lorentzian $*^2 = -1$ on 2-forms — the real split is a EUCLIDEAN/OS-side statement. Mutants caught ($[g] = (6-d)/2$ selects $\{6\}$; the 'A4 alone' overclaim refused). FIREWALL: $\mu_4$ and $g_{\mathrm{car}}$ enter NOWHERE — '$\mu_4 \Rightarrow$ 4D' stays forbidden (DIMENSION.UPLIFT.FIREWALL.01); axiom provenance from $\{c_3, g_{\mathrm{car}}\}$ is the OPEN half (the kill criterion in the ledger row); no 4D construction contract is advanced. |

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| Script | What it machine-checks *(continued)* |
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| `v976_seam_lift_birkhoff.py` | TRANSFER.HIDDEN.CIRCULATION.01 [E, executed half] + the EB no-go for DYN.UNITARY.DILATION.01 (external-review round 2026-08-27): the Birkhoff fibre of the seam single step $B$ is EXACTLY a line, its hidden parameter is the sign character, and the deployed Kraus dilation is entanglement breaking — $B = \frac{1}{2}I + \frac{1}{18}P_{12} + \frac{2}{9}P_{13} + \frac{2}{9}P_{23}$ bit-exact ($B^6$ = the v221 kernel), the decomposition map has a 1-dim kernel ($I + C_{123} + C_{132} = P_{12} + P_{13} + P_{23} = J$), complete family $w_t$ with $t \in [0, 1/18]$; $\widehat{w_t}(\mathrm{sgn}) = 6t$, populations exactly $t$-free, no fibre member chiral ($w(C_{123}) = w(C_{132})$ forced). DISCRIMINATOR: one-step coherence dynamics = the weighted left regular $S_3$ representation, spectrum $\{1, 6t, 2/3 \times 2, 1/3 \times 2\}$ exactly; the triangle loop current is an exact eigen-observable $\Psi_t(A) = 6t\,A$ (population-blind: one step + one current readout measures $t$, no ancilla; tomography rank 6). CONTRAST QUBIT: $\mathrm{spec}(\mathrm{Choi}^{T_2}) = \{-3t/2,\ 1/6 + 3t/2,\ 1/3 + 3t/2,\ 1/2 - 3t/2\}$ symbolically; CRITICALITY $\lambda_{\min} = (1 - \lambda_2 - \lambda_3 - 6t)/4$ with $2/3 + 1/3 = 1$ EXACTLY ($t = 0$ precisely on the EB boundary; mutated spectra miss by $\mp 1/12$ — not generic); the squared channel is PPT on the whole range (endpoint minima 1/9, 1/12) ⇒ qubit EB index 1 ($t = 0$) / 2 ($t > 0$) [PPT ⇔ EB in $2 \times 2$ + Lami–Giovannetti, literature-typed]. FULL LEVEL: $\lambda_{\min}(\mathrm{Choi}(\Psi_t^n)^{T_2}) = -(1/3)(2/3)^n$, $t$-INDEPENDENT (exact for $n \in \{1, 2, 6\}$, float $n = 1..8$) — never EB at 3 levels. THE NO-GO: the deployed Kraus dilation $K_{ij} = \sqrt{T_{ij}}\,|i\rangle\langle j|$ is measure-and-prepare ⇒ entanglement breaking [HSR] ⇒ quantum capacity 0; $C_0(T) = 0$ exactly; it is NOT in the Birkhoff fibre. BINARY MESSAGE: $p^* = (1/2, 1/2, 0)$ is KKT-exactly optimal for BOTH $T$ and $B$ ($C(T) = 0.0083580191$ bit, $C(B) = 0.4890415237$ bit, ratio 1.71%). [C] observation: the weights $(9, 1, 4, 4)/18$ = squares/18 = the v652 defect-ladder values $g^2/18$ (multiplicity matches only at $g = 2$; the forcing functor is the OPEN contract TRANSFER.DEFECT.BIRKHOFF.01). FIREWALL: finite matrix algebra + literature-typed channel theory; the fibre-point selection stays [O]. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v977_transfer_unistochastic_wilson.py` | TRANSFER.COHERENT.WILSON.01 [E, executed half] (external-review round 2026-08-27, companion of v976): $B$ is UNISTOCHASTIC — an exact $SU(3)$ lift carrying the 5-12-13 Wilson code, the center bridge and a classically invisible orientation bit — angles forced by $B$ ($\sin^2\theta_{13} = 2/9$, $\sin^2\theta_{12} = 1/14$, $\sin^2\theta_{23} = 2/7$), $\cos\delta = -4/\sqrt{65}$ unique, $e^{i\delta} = (-4 \pm 7i)/\sqrt{65}$, lift set $= \{U, U^*\}$ up to rephasing; $U^\dagger U = I$, $\det U = 1$, $\lvert U_{ij}\rvert^2 = B_{ij}$ entrywise exact [coherification iff unistochastic, literature]. $J = \mathrm{Im}(\text{plaquette}) = 1/27$ EXACT; Re of every plaquette fixed by $B$ (unitarity) $\Rightarrow$ the full rephasing-invariant Wilson data is classical up to the $\mathbb{Z}_2$ ORIENTATION BIT $\operatorname{sgn} J$ ($U^*$ lifts the same $B$ with $J = -1/27$; observable = the plaquette interference readout). THE CODE: $Q_{12;12} = (-5 + 12i)/324$, normalized $(-5 + 12i)/13$ with $5^2 + 12^2 = 13^2$ (atoms $5 = g_{\mathrm{car}}$, $12 = \dim \mathfrak{g}_{\mathrm{SM}}$, $13 = \Delta_Q = N(3 + 2i)$); all nine plaquettes close on four Gaussian phase types with norms exclusively $\{10, 13, 130, 169\} = \{2 \cdot 5, 13, 2 \cdot 5 \cdot 13, 13^2\}$; every Im $= \pm 1/27$; the four fundamental plaquettes generate all nine exactly. CENTER BRIDGE: $\operatorname{disc}\chi_A = 65 = \det(A - I)^2 + (\det A/2)^2 = 4^2 + 7^2$ for the v530 atom matrix $A = [[8, 2], [5, 3]]$; $e^{i\delta} = (-\det(A - I) \pm i \det A/2)/\sqrt{\operatorname{disc}}$ EXACTLY equals the transport phase; $-4 + 7i = (1 + 2i)(2 + 3i)$ (the v738 tower primes, norms 5 and 13); the v530 siblings $A_K/A_Q$ FAIL the square-sum identity (the bridge selects $A$). POWER LAW: $J_n^2 = (1 - b)^2(1 + 2b - 3a^2)/108$, every $B^n$ unistochastic ($U_6$ built, $\lvert U_6\rvert^2 = T$ exact); $\lvert J_6\rvert = 0.0951088904 = 0.9884006\, J_{\max}$; the uniform limit lifts to the Fourier matrix $F_3$ at $J_{\max}$ exactly; $J_n$ RATIONAL ONLY at $n = 1$ ($3(1 + 2b - 3a^2) = 1$ exactly $\Rightarrow J = 1/27$) — the Gaussian–Pythagorean code selects the single step. NO FIXED MICROSTEP: $B^2 \neq \lvert U^2\rvert^2$ exact; Dirichlet recurrence witness $n^* = 967909$ ($\min \mathrm{diag}\, \lvert U^{n^*}\rvert^2 = 0.999999$ while $B^n$ is uniform to $5.6 \times 10^{-17}$) — $U_6 \neq U_1^6$ (clock-vs-walk sharpened coherently). CENSUS (hard kill passed): 238 valid family points, 4 with rational $J$, EXACTLY ONE full-code point = the deployed $(2/3, 1/3)$ ($0.42\% \ll 10\%$). PMNS NEAR-MISS stays NEGATIVE (branch phase $240.2551^\circ$ vs deployed $240^\circ$; $-1/27 \neq -0.02965$; all angles differ). FIREWALL: the $\operatorname{sgn} J$ branch selection, CP/time readings and the $A \to$ angles intertwiner stay [O]. |
| `v983_simple_current_generator.py` | `SEAM.SIMPLECURRENT.GENERATOR.01` [O update: the generator IDENTIFIED, its lattice-arithmetic shadow complete] (external master-object review wave 2, 2026-08-28): THE ONE GLUE VECTOR $\lambda = (\omega_s, \omega_f)$ GENERATES THE FULL 128 SPINOR SECTOR — exact at the lattice level — $\lVert\omega_s\rVert^2 = 5/4$, $\lVert\omega_f\rVert^2 = 3/4$, $\lVert\lambda\rVert^2 = 2$, conformal weight $h_\lambda = 1$ (integer $\Rightarrow$ trivial self-monodromy); $[\lambda]$ has ORDER FOUR in the discriminant group $\mathbb{Z}_4 \times \mathbb{Z}_4$ of $D_5 \oplus A_3$ ($4\lambda \in L_0$, $\lambda, 2\lambda \notin$); $\lambda \cdot L_0 \subset \mathbb{Z} \Rightarrow$ isotropic glue, the glued lattice is EVEN; det $= 16/4^2 = 1 \Rightarrow$ UNIMODULAR rank $8 \Rightarrow E_8$ (classification [C]; explicit certificate v1). COSET ROOT CENSUS $[52, 64, 60, 64]$ (total 240; $+8$ Cartan $= 248 = (45 + 15) + (16, 4) + (10, 6) + (\overline{16}, \bar{4})$); the odd fusion powers carry $64 + 64 = 128 =$ the FULL missing spinor sector — ONE field generates all four cosets. KILL TESTS FIRE: the vector-class glue (norm 7/4) and the mismatched glue (norm 9/4) both break evenness. HOLOMORPHY SHADOW: $\det 1 \Rightarrow$ one sector vs $\det(D_8) = 4 \Rightarrow SO(16)_1$ has four (the lattice shadow of N4 only). FIREWALL: the analytic half (implementer convergence S1/S2, crossed-product/scaling exchange S3, operator-algebraic holomorphy S4) stays [O]; 1+1D in and out; no marker move. |

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| Script | What it machine-checks *(continued)* |
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| `v984_markov_qca_dilation.py` | `DYN.UNITARY.DILATION.01` [O update: the DISCRETE leg realized constructively] (external master-object review wave 2, 2026-08-28): THE EXACT COLLISION MODEL — $B^n$ as the reduced dynamics of an exactly unitary, strictly local QCA band — the collision step $V = C\,(U_B \otimes I)$ with the mod-3 record $C\|j,k\rangle = \|j,j+k\rangle$ is unitary on system + ancilla; the STINESPRING IDENTITY $\Phi(\rho) = \mathrm{Tr}_A[V(\rho \otimes \|0\rangle\langle 0\|)V^\dagger] = \sum_j P_j U_B \rho U_B^\dagger P_j$ holds as a symbolic operator identity (Kraus completeness $\sum_j K_j^\dagger K_j = I$); $\mathrm{diag}\,\Phi(\mathrm{diag}\,p) = Bp$ exactly with vanishing off-diagonals; SIX fresh ancillas give $\mathrm{diag}\,\Phi^6 = B^6 p = Tp$ exactly — the v977 macro-gate as reduced dynamics of a radius-1 collision band. MUST-FAIL FIRES: the closed unitary has $\|U_B^2\|^2 \neq B^2$ (31/54 vs 0.2450) — the dilation is necessary. TYPED REMARK (probe finding AGAINST the review text): a REUSED mod-3 adder ancilla still gives $\mathrm{diag} = B^2 p$ exactly (the record $j_1 + i$ stays distinguishable) — fresh ancillas are needed for the coherences, not the populations. The conjugate branch $V^*$ dilates the SAME $B$ ($\mathrm{sgn}\,J = \pm 1/27$ stays classically invisible; selection stays [O]). FIREWALL: the discrete finite leg only; quasi-local Hamiltonian dilation, OS continuation to $e^{-itH}$ and any 3+1D reading stay [O]; no marker move. |
| `v985_quillen_channel_swap.py` | `ALPHA.QUILLEN.EXACT.01` [O update: the CHANNEL-SWAP face resolved at the finite level by $\mu_4$-character grading] (external master-object review wave 2, 2026-08-28; v974 anchors read-only): THE DECK GRADING RESOLVES THE v974 KERNEL SWAP EXACTLY — the jump pattern $\mathrm{circ}(2,-1,0,-1)$ has $\mu_4$ eigenvalues $[0,2,4,2]$ (kernel $k=0$), the mark pattern $\mathrm{circ}(0,1,2,1)$ has $[4,-2,0,-2]$ (kernel $k=2$) — the v974 swapped-kernel fact symbolically; the DECK SWAP $r \to r+2$ ($P_0 \leftrightarrow P_2$, $P_1 \leftrightarrow P_3$) maps the jump \|spectrum\| onto the mark pattern EXACTLY — after the swap both kernels sit in the same channel and the graded comparison is well defined; the ODD swap $r \to r+1$ does NOT match (must-fail fires); the graded $\det'$ difference is $c_3$-FREE: $\mathrm{gr}(\text{jump}) - \mathrm{gr}(\text{mark}) = \log(\ln 2/4)$ exactly; the channelwise unit ratio is UNIFORM $= 4/\ln 2$ in every matched nonzero channel (the v974 doublet ratio extended to full uniformity). FIREWALL: finite $4 \times 4$ $\mu_4$ circulants only; continuum Bismut–Freed, KMS rigidity, the $2^{-4}$/Quillen-metric compensation and the exactness of $F_{U(1)}$ stay [O]; $\alpha^{-1}$ untouched; no marker move. |
| `v986_nu_scalaron_rung.py` | `FLAV.NUSCALE.05` [C]/[N] candidate (external master-object review wave 2, 2026-08-28): THE SCALARON-TRACE RUNG $M_R = N_{\mathrm{fam}} M_{\mathrm{scal}} = 3\,c_3^{7/2}\,\bar{M}_{\mathrm{Pl}}$ — a structured point INSIDE the v481 window, 1.8 % from the 1-loop requirement — $3M_{\mathrm{scal}} = 9.180 \times 10^{13}$ GeV vs the v481-required $9.346 \times 10^{13}$ GeV (same frozen 1-loop inversion, $y_\nu = y_t$ [P]) — deviation $-1.8$ % (the review said 'about 1 %'; the honest value is 1.8), $m_3$ at the candidate = 0.0512 eV (+1.9 %, inside the $> 50$ % v482 RG envelope); EXACT ladder position $3\,c_3^{7/2} = (3\sqrt{c_3})\,c_3^3$ with $3\sqrt{c_3} = 3/\sqrt{8\pi} = 0.598413$, 0.26 % from the 3/5 of the v482/v488 story — ADJACENCY TYPED HONESTLY: v488 killed 3/5 as an $SO(10)$ CLEBSCH; the scalaron-trace reading is a DIFFERENT mechanism class (wave-3 retype: $\mathrm{Tr}\,I = 3$ is loose, $\mathrm{Spec}(I) = \{1,1,1\}$; operator reading is $Q_+$ $\mathrm{Spec}\{1,2,3\}$; texture `DATA_CONSTRAINS_TEXTURE`) and itself stays [O]; $\sim 20\times$ sharper than the declined integer rung $c_3^3 \bar{M}_{\mathrm{Pl}}$ (0.019 vs 0.401 in $m_3$); the same $M_{\mathrm{scal}}$ already carries $f_a = M_{\mathrm{scal}}/128$ (v185) — no new scale atom. KILL TEST: $\Sigma m_\nu$ (NO) = 0.059 eV; a robust bound below, or a $\gtrsim 5$ % required-$M_R$ shift, kills the candidate. FIREWALL: $y_\nu = y_t$ stays [P]; NO seesaw closure; the v481 window candidate and the v488 negative untouched; no PMNS claim. |
| `v987_os_dilation_package.py` | DYN.UNITARY.DILATION.01 [O update: OS/dilation ladder executed] (promote round 2026-08-28): kernel OS continuation exact ($B = P_1 + \frac{2}{3}P_2 + \frac{1}{3}P_3 > 0$, $H = -\log B$ PSD, gap $\log\frac{3}{2}$, Wick $U(-in) = B^n$); size-uniform free band; congruence interacting family SPD; quantum SWAP $d(\theta) \sim \theta^2$, CPTP, LR radius 1. Continuous field-level OS stays [O]. Python-only / Wolfram mirror deferred. |

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| Script | What it machine-checks *(continued)* |
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| `v988_psi_lambda_reduction.py` | SEAM.SIMPLECURRENT.GENERATOR.01 [O update: S3 exact + S1/S2 skeleton measured + lemma REDUCED, not closed] (promote round 2026-08-28): S3 crossed-product/tower exchange exact (naturality); S1 window Cauchy + sector collapse; S2 Shale plateau + topological $a + b \log N$ vs trivial saturation; energy envelope $f_n < C^{n+1}(n!)^2$. Reduction to cited quasi-free theorems + MMST identification – not closure. Python-only / Wolfram mirror deferred. |
| `v989_lattice_gate_battery.py` | TFPT4D.LATTICE.ACTION.01 + CHIRAL4D.NOMIRROR.01 [O update: T1–T7 harness + Euclidean T2 kill + Hamiltonian route clears T2] (promote round 2026-08-28): T4 exact SM anomalies; T5/T6 2D overlap index $q$; T7 toy $\kappa_4 \neq 0$. Euclidean T2 kill ($N_t$-exact Z2: violation absent at $N_t = 2$, appears at $N_t = 4$ det-only $-0.249$; Wilson PSD). Hamiltonian route clears T2 by construction. Chiral Gauss census: single-exponent iff $q_L = q_R$. Contracts stay [O]. Python-only / Wolfram mirror deferred. |
| `v990_wj_transduction_mechanism.py` | FTRANSFER.GENERATING.01 + OBS.TRANSDUCTION.01 [O update: content shadow + finite $W[J]$ + transduction] (promote round 2026-08-28): Z6-centre $6Y$ table; $dW/dJ = \langle O \rangle$ ($6{\times}10^{-11}$), Hessian = Kubo ($10^{-7}$); mixed $\partial^2 W/\partial J \partial\theta$ seam-coupled $-9.3{\times}10^{-3}$, $e^c$ control exactly 0. Contracts stay [O]. Python-only / Wolfram mirror deferred. |
| `v991_detline_bulk_edge.py` | SEAM.DETLINE.UNIFICATION.01 [O update: finite bulk-edge shadow [E]; continuum Bismut–Freed stays Open] (promote round 2026-08-28): occupied-frame FHS Chern +1 equals seam holonomy winding +1 (diff $6{\times}10^{-16}$) on the v367 QWZ collar; $M = 3$ control $0 = 0$; conjugation mutant flips both signs. Continuum identification stays [O]. Python-only / Wolfram mirror deferred. |
| `v992_joint_likelihood_v1.py` | PRED.JOINTLIKELIHOOD.01 [O update: v1 executed [N]; promotion conditions typed; contract stays Open] (promote round 2026-08-28): nine-observable joint likelihood $\chi^2 = 11.80/\text{dof}\,9$ ($p = 0.225$), $\nu_{\text{eff}} = 1$, TFPT at 0th percentile of 200 scrambled decoders, LOO stable. Holdouts declared not consumed. Python-only / Wolfram mirror deferred. |
| `v993_minimal_defect_selector.py` | AX.P2.01 / AX.P1.01 / SEAM.SIMPLECURRENT.GENERATOR.01 [O update: census lifts v624 architecture conditionality at lattice level; axiom typing UNCHANGED] (external review wave 3, 2026-08-28): full rank-8 ADE census (SNF) selects $(D_5, A_3)$ uniquely – the $D{+}A$ split is an OUTPUT of (rank 8 + cyclic $\mathbb{Z}_4$ both sides). $[\lambda]^2 = [v]$ exact (parity shadow of $U^2 = (-1)^F$). Unique finite CE (module property needed; kernel 36/720 without it). Axioms unchanged. Python-only / Wolfram mirror deferred. |
| `v994_mmst_criteria_battery.py` | SEAM.EQUIV.MMST.01 / SEAM.SIMPLECURRENT.GENERATOR.01 [O update: 5/7 MMST identification criteria executed [E-measured]; C6 written-out [C]; C7 cited/open] (completeness wave, 2026-08-28): measured $c = 8.005$ (16 Majorana copies); conformal edge spectrum $[0.506, 1.516, 2.523, 3.524]$ with degeneracies $[1, 0, 1, 1, 2]$; current normalisation $k = 1.000000000000$; sector collapse $\det D_8 = 4 \to \det E_8 = 1$; FLAGSHIP: $\mu_4$/GSO-projected lattice character sums reproduce the $(E_8)_1$ vacuum character coefficients $[1, 0, 248, 0, 4124, 0, 34752]$ three orders deep. C6 = the quasi-free convergence theorem is now WRITTEN OUT (`articles/2026-08-28/psi_lambda_convergence_theorem_en.tex`; $2/\pi^2 = 0.20264$ vs measured 0.203, 0.18 %). C7 modular invariance stays cited. Python-only / Wolfram mirror deferred. |
| `v995_kms_subnet_rigidity.py` | SEAM.STATE.RPMIXING.01 [E update: finite KMS rigidity] + ALPHA.QUILLEN.EXACT.01 [O note: rigidity premise finite] (completeness wave, 2026-08-28): ALL 33 v972 mixing directions killed by KMS ALONE on the finite CAR algebra ($\text{CAR}_8 = M_{256}$); covariance provably redundant under full KMS (`COVARIANCE_DROP_REDUNDANT_UNDER_FULL_KMS`); unique $E$-invariant crossed-product state extension ($196608 \to 0$). Type-III/continuum stays cited/open. Python-only / Wolfram mirror deferred. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v996_gauge_grammar_kill.py` | GAUGE.DETLINE.FIXPOINT.01 [O update: naive grammar extension excluded [E]; genuinely nonabelian structure required – typed] + ALPHA.QUILLEN.EXACT.01 [O note: ln-2-rate shadow] (completeness wave, 2026-08-28): H0 STRUCTURAL KILL – negative $b_2/b_3$ flip the stationary-root signs ($SU(2)$ inverse $-151.2$, $SU(3)$ $-115.9$); 64 conventions 0 hits; $U(1)$ control 137.0359992168 survives all mutants. Detline curvature $\to 2\pi$ exponentially, rate $0.6936 \sim \ln 2$ (numerical Bismut–Freed shadow, not the continuum theorem). $\alpha_s(M_Z)$ remains an external input. Python-only / Wolfram mirror deferred. |
| `v997_spin2_smg_witnesses.py` | GRAV.SPIN2.EMERGENCE.01 [O update: quadratic witnesses [E]; content pinned on the untruncated form factor] + CHIRAL4D.NOMIRROR.01 [O update: SMG mechanism toy-proved [E]; 4D Spin(10) theorem stays [O]] (completeness wave, 2026-08-28): exact Barnes–Rivers decomposition – $R{+}R^2$ clean (one massless TT pole, two helicities, scalaron at $M_{\rm scal}$ exact); local $a_4$ Weyl$^2$ truncation NECESSARILY carries a spin-2 ghost (opposite residue) – typed. FK SMG toy: $N = 8$ unique Spin(7)-singlet ground state, Schmidt rank 1, volume-independent gap 14 at $L = 1, 2, 3$; $N = 4$ contrast $2^L$-degenerate; physical chiral edge keeps its $\pi/L$ spectrum exactly. Python-only / Wolfram mirror deferred. |
| `v998_seam_modular_closure.py` | SEAM.EQUIV.MMST.01 / SEAM.SIMPLECURRENT.GENERATOR.01 [O update: C7 arithmetic shadow now exact; net-level C7 stays cited] (review wave 4, 2026-08-29): two-edge HS remainder $\leq 2.11$ on the plateau $N_x \in (16, 24, 32, 48, 64, 96)$; lattice character $\theta_{E_8}/\eta^8$ EXACT four orders $[1, 0, 248, 0, 4124, 0, 34752]$; $\Theta_{E_8}(\tau) = E_4(\tau)$ $q$-expansion $\geq 8$ orders exact (Jacobi vs Eisenstein); $j^{1/3}$ numeric transform at 60 digits; $S$-weight bookkeeping. Net-level C7 stays cited/open. Python-only / Wolfram mirror deferred. |
| `v999_weak_collision_hp.py` | DYN.UNITARY.DILATION.01 [O update: finite ladder now COMPLETE – discrete/kernel/field-free/interacting-classical/quantum-coupled/CONTINUOUS-TIME; open = thermodynamic field limit] (review wave 4, 2026-08-29): weak-collision HP – Richardson $1.098 \times 10^{-10}$, $\operatorname{diag} L = Q$ symbolic, $e^Q = B$ exact, GNS splitting defect 0, locality $1.355 \times 10^{-20}$, strong-collision rank 9/81. Contract stays [O]. Python-only / Wolfram mirror deferred. |
| `v1000_spin2_sk_witnesses.py` | GRAV.SPIN2.EMERGENCE.01 [O update: quadratic Hamiltonian half executed] + FTRANSFER.SK.RHO0.01 [O update: SK mechanism executed] (review wave 4, 2026-08-29): lattice graviton – positivity exact, 2 helicities at all 5852 momenta ($N = 6..12$), $\omega = k - k^3/24$ exact. SK – KMS/FDT $\sim 10^{-17}$, $\rho_0$ contrast $3.550 \times 10^{-3}$ vs static $10^{-16}$, saddle 5-fold degenerate, $3\pi/5$ unique in the $\mu_4$ lift [C]. Both contracts stay [O]. Python-only / Wolfram mirror deferred. |
| `v1001_nu_pentagon_chain.py` | FLAV.NUSCALE.06 Candidate [C]/[N] (review wave 4, 2026-08-29; NO closure; mechanism = $Q_+$-to-flavor operator [O]): $U_e = I$ inventory + analytic 13/23 misalignment; pentagon double hit $\phi = 288° = 4(2\pi/5)$, $\theta = 2\pi/35$ unique LEE survivor 2.7 %, max pull 0.557; v3 chain SHA-16 `a4c28732fa687620`: $\Sigma = 0.0599$ eV, $m_\beta = 9.0$ meV, $m_{\beta\beta} \in [1.5, 3.8]$ meV, $\delta_{\rm CP} = 287.66°$; v270–$\theta_{23}$ tension $1.85\sigma$ TYPED; census 0/1607. Kills: DESI floor, DUNE $\delta_{\rm CP}$ 287.7 vs 240, JUNO. FLAV.NUSCALE.05 unmoved. Python-only / Wolfram mirror deferred. |
| `v1002_spin10_mirror_projector.py` | CHIRAL4D.MIRROR.DET16.01 Candidate [C] (review wave 5, 2026-08-29; exact finite-cluster gapping; open: dynamical gauge fields, domain-wall geometry, 4D volume theorem). [E-finite] DET16 rank-one projector on $2^{16}$: $\Omega^\dagger\Omega = P_{16}$, $\Omega\Omega^\dagger = P_0$, $P_\phi$ hermitian idempotent $2.8{\times}10^{-18}$, $h_{\rm mir}$ spectrum $\{0^1, 1^{65535}\}$ gap 1, 45 $\mathfrak{so}(10)$ Fock commutators 0 ($\det \rho_{16} = 1$). [N] hopping gaps 0.90/0.86 at $t = 0.2$. [X] number-preserving Casimir route excluded (census $N$=8:0, $k$=2:27, $k$=3:1463, $\lvert W(D_5)\rvert$=1920). Python-only / Wolfram deferred. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v1003_nu_structural_nulls.py` | FLAV.NU.TEXTURE.MECHANISM.01 [O] (review wave 5, 2026-08-29; binding constraints, no texture). [E] charged-sector $h = (1, 1, 1)$, $d = (64, 60, 64)$. [X] Schur-texture NULL: 8/8 $(K, B)$ fail; D4 reflection forces $\lambda_1 = \lambda_3$; required $K \sim [3536.998, 1768.499, 1/3]$ matches none of 9 comparators. [E] orientation-doubling: sgn $J$ odd sector generically lifts degeneracy. [X] scale NULL: 12/12 fail; $1 - \lvert c/a \rvert = 1.885{\times}10^{-4}$ not supplied. Mechanism must be D4-odd AND supply $\sim 2{\times}10^{-4}$ cancellation; pure seam data excluded. Python-only / Wolfram deferred. |
| `v1004_milnor_frobenius_gray.py` | SEAM.MILNOR.LOCALRING.01 [O] (review wave 6, 2026-08-30; finite algebra; corpus identification stays open). CLASSICAL (cite, no claim): Milnor algebra of $x^2 + y^3 + z^5 \cong \mathbb{Q}[y, z]/(y^2, z^4)$ dim 8; spectral numbers = $E_8$ exponents $\{1, 7, 11, 13, 17, 19, 23, 29\}$. NEW-IN-CORPUS EXACT [E-finite]: rank clock $(2 - a)(4 - b)$, table sum $30 = h(E_8)$; $S = pqr \cdot \mu/8$; self-clock iff $\mu = 8$; $(2, 3, 5)$ unique among ordered spherical triples; Galois-Gray $11/7/-1$; CP = socle/Frobenius; integral $D^4$ bridge basis-dependent. [C] $PG(3, 2) = AG(3, 2) \sqcup PG(2, 2)$ interpretive only. Contract stays [O]. Python-only / Wolfram deferred. |
| `v1005_relative_pencil_bridges.py` | Relative pencil bridges (review wave 6, 2026-08-30; circularity-flagged). EXACT [E-finite]: Hankel $p_n p_{n+2} - p_{n+1}^2 = 2^{n+1}$ + Jacobi spec $\{1, 2\}$ diag $(4/3, 5/3)$ (v106/v119); winding spec $\{5/2, 1, 1\}$ + fixed plane (v4). EXACT BUT [C] (circularity-flagged): unique $t = 2$, $\lambda = 7/3$, CM $J_0^2 = -I$, index $25 = g_{\text{car}}^2$ SNF $(1, 1, 25)$ – BUT carrier 5 already sits in the column sums of $Q_2$ and $C$: NOT a P2 selector. [C] Klein-four SNF $(2, 2)$ only after $D = 2d$. AX.P2.01 unmoved. Python-only / Wolfram deferred. |
| `v1006_mmst_lemma_battery.py` | SEAM.MMST.TYPEIII.CHARGED.01 [O]: L3 crossed-product; UGF $K_G = \pi^2/4$; TEL-a $N^{-2}$ isometries; integer D5+A3 pairings; Z4 outerness with finite-inner correction. Round-4 guard requires v1026's $\lVert R_N \rVert_{\text{HS}} < 2.995906 < 3$ at fixed $M = 1$, $N_y = 8$ for every even $N \geq 16$, and explicitly open FE-GEN/ALG-EXH/MMST. Historical finite-grid 2.11 is not the proof. 68/68; no TOE closure. |
| `v1007_det16_stability_battery.py` | CHIRAL4D.MIRROR.DET16.01 Candidate [C] (review wave 7, 2026-08-30). T1/T2 proved (v1002); T3 Michalakis–Zwolak cited-verified (LTQO, $t_* > 0$, gap $\geq 1/2$). [N] counting line $n = 2, 3$; $n = 2$ closing scan through $t = 4$; dynamical Z2/Z4 uniquely gapped (full-probe min 0.728854416); SW $1 - \alpha t^2 + O(t^4)$; $c_{\text{geo}}/L$ bounded. [X] ordinary hopping not block-diagonal so 1/192 boxed; BDL supplies no volume-uniform gap remainder. Stays [C]. Python-only / Wolfram deferred. |
| `v1008_master_assembly_scaffold.py` | QFT4D.LATTICE.FUNDAMENTAL.01 Decision / GAUGE.DETLINE.FIXPOINT.01 [O] (review wave 7, 2026-08-30; 2+1D scaffold-coherent). Prototype $K = 0$ plaquette-exponent theorem; matter frozen-link $q^2 \sim 4$, $K(q = 0) = 0$, unique-contractive $\lvert R' \rvert < 1$; wall/mirror QWZ slope $\sim -0.9989$, DET gap $\geq 1.5$ through $t = 0.6$; assembly Z6 $(1, 2, 3, 0)$, clock $[0, 1, 1, 2]$, $\chi_{\text{phys}}$ vs $\chi_{\text{mirror}}$. No marker moves. Python-only / Wolfram deferred. |
| `v1009_rho0_minimizer.py` | FTRANSFER.SK.RHO0.01 [O] (review wave 7, 2026-08-30). [E-finite] unique KMS compression (Klein), Hessian strictly positive, centre dim 1, identity $\sim 8{\times}10^{-17}$. [X typed] V1000_CONSEQUENCE_MISMATCH (compression kills the SK response vs frozen contrast $3.550{\times}10^{-3}$) + CENTER_FLATNESS_PREMISE_FALSE. Canonical formula needs a weaker admissible set. Contract stays [O]. Python-only / Wolfram deferred. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v1010_simplicity_bridge_census.py` | Simplicity census + W/Q canonicality + Milnor strong-bridge double obstruction (afternoon harvest, 2026-08-30). Derivation matrix; unique remainder min S4=2; anchor $(1,1,2)$ not among 3 declared Milnor maps; $\varphi_0$ four novel NULL. Bridge Q CANONICAL: ladder forces sums $(9,5,1)$; C's 5 inherited from $R$ – v1005 circularity flag corrected as forced provenance. Bridge W CHOICE-but-conditionally-unique ($1078 \to 1$). Milnor all-structure bridge NONEXISTENT at $\mathbb{F}_2$ and all upstairs homes; coarse $D$+pairing exists; $\lvert\mathrm{Aut}(X)\rvert = 36864$, $\lvert\mathrm{Aut}(Y)\rvert = 8$. `SEAM.MILNOR.LOCALRING.01` stays [O]; `AX.P2.01` stays an axiom. Python-only / Wolfram deferred. |
| `v1011_3p1d_ladder_battery.py` | 3+1D ladder B6–B9 (afternoon harvest, 2026-08-30). Minimal 3+1D viable (Gauss 32768, confinement, exact isotropy, $g_\star \sim 1.915$); exact tree-fixing $524288 \to 4096$, tree-independence, DET-under-coupling, back-reaction; periodicity restores $K$; charged channels $\sim 2.014\times$ at weak $g$; $K$ collapses in the confined phase. BINDING: matching scale of the coupling functional must sit in the deconfined/weak regime. `QFT4D.LATTICE.FUNDAMENTAL.01` stays Decision; `GAUGE.DETLINE.FIXPOINT.01` stays [O]. Python-only / Wolfram deferred. |
| `v1012_toe_gate_witnesses.py` | Evening TOE-gate wave (2026-08-30). T5 IR witnesses + typed SPLIT (physical $c \to 1$, $\xi\cdot$gap $\sim 1.089$ vs power 2.034, $v_{\mathrm{LR}} = 5/3$, curvature $\to 1/6$; holonomy+clock not propagating modes). T8 $\rho_0$ 8-candidate `NO_SET_WORKS` (hard sectors response-dead; KMS-affine sets selection-vacuous). T6 $\kappa$ six-fold NULL ($K_4$ 18.8× too large; 4D functional required). T7 interacting TT (conservation/positivity/sum-rules/$Z$-persistence; gapped). FIREWALL: `TFPT.TOE.COMPLETE.01` / `FTRANSFER.SK.RHO0.01` / `FLAV.NU.TEXTURE.MECHANISM.01` / `GRAV.SPIN2.EMERGENCE.01` stay [O]. Python-only / Wolfram deferred. |
| `v1013_thermodynamic_dynamics.py` | Thermodynamic dynamics (late-evening harvest, 2026-08-30). Mandatory-dynamics leg CLOSED at Hamiltonian-class level (theorem memorandum). Uniform LR bound $J = 12/5$, $R = 2$, $z = 4$, $\kappa = 16043/450$, $v_{\mathrm{LR}} = 32086e/225 \sim 387.64$; norm-convergent $\tau_t$; gauge-invariant quasilocal $\mathcal{A}^G$; existence of ground/$\beta$-KMS states. Numeric twin: measured $v = 2.0 \ll$ proved bound; convergence bound dominates buffers 8/16/32. Remaining: phase uniqueness, limit gap, IR universality. `QFT4D.LATTICE.FUNDAMENTAL.01` stays Decision. Python-only / Wolfram deferred. |
| `v1014_bridge_refinements.py` | Bridge refinements (late-evening harvest, 2026-08-30). Finite detline restriction isomorphism VERIFIED ($c_{\mathrm{phase}}$ $3.7\times10^{-16}$; orientation $2 \to 1$; residual constant $U(1)$ fixed by the $A_0$ reference normalization). W-bridge: P-anch DERIVED ($e_1 = R^{-1}a$ via winding lock); P-dem PARTIAL ($\mathbb{Z}_4$ average annihilates nontrivial characters; missing = character-blind determinant-response map, MMST/Quillen externalized leg). `ALPHA.QUILLEN.EXACT.01` / `AX.P2.01` unmoved. Python-only / Wolfram deferred. |
| `v1015_axiom_core_closure.py` | Axiom-core closure (Monday-morning harvest, 2026-08-31). Character-blind P-dem response DERIVED ($r = (1,1,1)$ exactly at collar sizes 12 and 16, democracy residual 0; mutants $(1,0,0)$ and $(5/4,1,5/4)$). Together with P-anch (v1014) BOTH W-bridge premises are derived at finite level; axiom-core remainder ZERO modulo the externalized MMST identification. `AX.P2.01` stays an axiom (finite shadows; continuum identification is the same external leg). T1 note: structure-postulate route fully premise-supported at finite level. Python-only / Wolfram deferred. |

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| Script | What it machine-checks *(continued)* |
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| `v1016_state_gap_batteries.py` | State/gap batteries (Monday-morning harvest, 2026-08-31). $R_4$ seam-modular cap unique finite-level non-entropic selector ($D_{\mathrm{tr}} = 0.1434$, SK RMS $3.038\times10^{-3}$, saddle preserved; finite-proxy caveat). TEL-B certified-tail BLOCKED ($N^2 D_N$ 530 → 19032); scalar dyadic $A_R < 43$ with factor-350 headroom; $N$-uniform CAR nuclearity proved ($C_\beta = 67.97/29.73/12.15$); ALG-EXH unchanged. DFP volume-uniform gap cited-verified for $\mathbb{Z}_2/\mathbb{Z}_4$ open chains ($a = 0.0233$, $a/16 = 0.00146$); remaining rotor/wall/3+1D. `FTRANSFER.SK.RHO0.01` stays [O] / `SEAM.MMST.TYPEIII.CHARGED.01` stays [O] / `CHIRAL4D.MIRROR.DET16.01` stays [C]. Python-only / Wolfram deferred. |
| `v1018_e8_directed_readout.py` | E8.DIRECTED.READOUT.01 [E] (2026-09-02, r609): **seven exact $E_8$ readout cells** (C7 open) – re-derived (no probe imports). Seifert $S + S^{\mathsf{T}} = A_{E_8}$, charpoly$(-S^{-1}S^{\mathsf{T}}) = \Phi_{30}$, rank$(S - S^{\mathsf{T}}) = 8$; Hamming $1 + 14y^4 + y^8$, $240 = 16 + 14 \cdot 16$, $S(3,4,8)$; srg$(120, 56, 28, 24)$ eig $56/8/-4$ mult $1/35/84$; $N(n) = 240\sigma_3(n)$ for $n \le 10$; $A_P(n)/A_P(1) = \tau(n)$ for $n \le 8$, $E_4^3 - E_6^2 = 1728\Delta$; $(\mathbb{Z}/30\mathbb{Z})^\times \cong C_2 \times C_4$, $D_\chi = L(s,\chi)L(s-3,\chi)$ for $n \le 200$; $E_6 \oplus A_2$ index 3, Smith $(1^6, 3, 3)$, glue 78/81/81. C7 Gauss-code transform stays OPEN (`rem:c7-audit`). NO RH CLAIM. Python-only / Wolfram deferred. |
| `v1019_coxeter_euler_completion.py` | E8.COXETER.EULER.COMPLETION.01 [E]+Numerical (2026-09-02, r617): **Coxeter–Euler completion** – re-derived (no probe imports). Exact: $\det(I - xC) = \Phi_{30}$, $\mathrm{Tr}\, C = -1$; Moebius $\Phi_{30} = (1-x^2)(1-x^3)(1-x^5)(1-x^{30})/[(1-x)(1-x^6)(1-x^{10})(1-x^{15})]$ (classical); $U = 1 \oplus C$, $\det(I - xU) = (1-x)\Phi_{30} = 1 - x^2 - x^3 + x^6 + x^7 - x^9$, $\mathrm{Tr}\, U = 0$. Global $Z_C = \zeta(2s)\zeta(3s)\zeta(5s)\zeta(30s)/[\zeta(s)\zeta(6s)\zeta(10s)\zeta(15s)]$, $D_{E_8} = \zeta(6s)\zeta(10s)\zeta(15s)/[\zeta(2s)\zeta(3s)\zeta(5s)\zeta(30s)]$, abs. conv. $\mathrm{Re}\, s > 1/2$. Numerical ($X = 10^5$): $D_{E_8}$ residual $4.76\times10^{-4}$ at $s = 0.75$, $4.16\times10^{-12}$ at $s = 1.5$. Class: vanishing linear term generic from $\mathrm{Tr}\, C = -1$; $E_8$ selects only the divisor set of 30. Beurling-generic. FENCE: The trace-free completion is zero-free and pole-free in $\mathrm{Re}\, s > 1/2$. The splitting into the scalar zeta channel and the Coxeter channel is open and RH-equivalent. No RH claim. NO-GO `E8.COXETER.REGULARIZED_SPLIT.NO_GO.01`: $\det_2$ cannot isolate $1/\zeta(s)$. Python-only / Wolfram deferred. |
| `v1020_spectator_sector_obstruction.py` | Spectator-sector obstruction (r644 audit harvest, 2026-09-04): finite free-fermion witness, re-derived (no probe imports). Stacking a decoupled trivial gapped sector $C$ ($M$=3, Chern 0, gap $\geq$1.095) onto the seam model $B$ ($M$=1, $N_Y$=8, 16 modes) leaves every seam datum unchanged ($A1$–$A5$: Chern $C_B$=$C_{B\oplus C}$=−1, $C_C$=0; zero modes 2/0/2 with $B$-support 1.0; edge dispersion coincident to $1.4\times10^{-15}$; seam-restricted covariance identical to 0.0 both grids; edge-profile overlap 1.0) and changes the bulk ($A6$: dim 16 vs 32, $E_0(C)$=−25.31, mid-strip rank 4 vs 8, $\|\Gamma_0 - \Gamma_\alpha\|_{\mathrm{HS}}$ 0.752536 vs 0.760202). Repaired criterion ($B$): Krylov generatedness spectator_dim($B\oplus C$) − spectator_dim($B$)=16 at all 128 momenta; commutant dim 1 ($B$) vs 17 ($B\oplus C$) generically, $\Pi_C$=0⊕1 in commutant (residual 0.0); selection unique ($B$ selected, $B\oplus C$ rejected). Honest: spectator_dim($B$)=10 at $p$=0 only – criterion generated a.e. in $p$. `SEAM.BULK4D.RECON.01` stays **[O]**; `BULK.PRIME_COMPLETION.01` registered **[O]**; NO marker move. `DIMENSION.UPLIFT.FIREWALL.01` F2: 2+1D free-fermion hypothesis, re-scoped 4D conclusion. The 2D modular-invariant uniqueness of the holomorphic $(E_8)_1$ net is untouched and is NOT a 4D no-spectator theorem. Python-only / Wolfram mirror deferred. |

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| Script | What it machine-checks *(continued)* |
|---|---|
| `v1022_telb_round3_certificates.py` | TEL-B round-3 certificates (2026-09-05): native 256-bit Arb/Acb over 5812 rational cells gives $\mathrm{TV}_p(F'') \le 15.817695962988924\ldots < 15.928674$ and $K_3^{\mathrm{TV}} < 2.517464500835$. The analytic C2b scalar reduction plus executable Arb arithmetic gives $c_\epsilon = 0.288547359238\ldots$, $c_\zeta = 0.110677064901\ldots$, total $0.399224424139\ldots < 0.400$; full C2b remains open exactly at CF $\le 0.139/\sqrt{N}$ and DG $\le 0.060/\sqrt{N}$. An exact one-mode countermodel refutes the old ALG2 Vandermonde implication; FE-GEN or a precise gauge/code projection is still required. T2/TOE stay [O]. Native `python-flint==0.9.0`, not Pyodide. NO RH CLAIM. |
| `v1023_derivative_filtered_wall.py` | Derivative-filtered wall block (2026-09-05): $H_{\mathrm{df}}$ preserves $\ker D_{\mathrm{GW}}$ exactly; block determinant $\lambda g^2 \Delta s^2$; the adiabatically mirror-continuated high band is $\ge \Delta + \lambda$ and the low branch is quadratic. The bare mirror operator is not invariant and has low-band weight for $s > 0$–no bare-mirror response gap and no Weyl cone. Gauge defect $2.41\times10^{-15}$, spectral defect $1.78\times10^{-15}$. Finite quasi-free block lemma only; canonical DET16/DET32 realization, many-body parent, admissibility/quasilocality, measure/anomaly/index control, universality and the 4D limit remain open. No marker move. NO RH CLAIM. |
| `v1024_parent_internal_boundaries.py` | Parent-internal T6–T8 boundaries (2026-09-05): finite $A_3$–$D_4$ seam/neutrino-pair intertwiner with no new factor; positive, even, gauge compatible, non-additive, local gap 0.25 conditional on the inherited full-parent bound, mixed resolvent 0.104712041885. Internal 12D $\mathbb{Z}_5$ cap gap 0.197423715893, but arbitrary, $D_4$-breaking and not a global $\rho_0$. Exact boundaries: projective full-Fock $D_4$ lift; relative commutant under unital enlargement; a full $\mathbb{Z}_5$ Weyl pair needs $5 \mid D$ only unprojected; a uniform full gap excludes a massless spin-2 pole; exact TT projection is not finite-range. 49/49; T6–T8/TOE stay [O]. NO RH CLAIM. |
| `v1025_telb_c1_c2a_certificates.py` | TEL-B C1/C2a certificates (2026-09-05): native `python-flint==0.9.0`. C1: 100-bit Arb/Acb over 3888 analytic rectangles plus a closed axis-strip majorant and exact imaginary contribution 1/12 gives $C_\star \le 0.1520453048\ldots < 0.1525$. C2a: exact cotangent reindexing, Peano monotonicity and the shifted-Euler remainder reduce the theorem for every even $N \ge 16$ to interval comparisons; $N \sup_k \lvert\varphi_N(k)\rvert \le 0.5082287792\ldots < 0.509$. C1/C2a close; C2b CF/DG, ALG-EXH, MMST scaling and TOE remain open. Native-only, not Pyodide. NO RH CLAIM. |
| `v1026_telb_round4_closure.py` | uniform TEL-B norm $\lVert R_N\rVert_{\mathrm{HS}} < 2.995906 < 3$ at fixed $M = 1$, $N_y = 8$ for all even $N \ge 16$; native interval CF/DG/CROSS and Fourier aliases; ALG-EXH open |
| `v1027_signed_det_car_wall.py` | canonical DET-CAR rotation and signed $z = 1$ wall; fixed classical backgrounds only; high band $\ge 4$ in the stated example; no full chiral continuum |
| `v1028_gauss_d4_cap.py` | Gauss site/cube residues 3/0 (mod 5) with full 192-mode sites and nine-irrep cube links; selected-pair-space $D_4$ cap gap one; Majorana obstruction; T6/T8 open |
| `v1029_local_tensor_constraints.py` | local first-class tensor target: two positive $z = 1$ modes at nonzero momentum; unstable zero mode requires global removal; T7 open |
| `v1030_alg2_sector_frame.py` | sector-connectivity criterion and uniform section $2MD^{3/2}/\sigma$; actual microscopic words and joint-adjoint tails required; ALG2 open |
| `v1031_quantum_tensor_curvature.py` | free zero-mean Weyl/Fock target with $C = r^4 P_{TT}$ and distribution limit; TFPT parent open |
| `v1032_covariant_curvature.py` | positive local covariant free curvature field with helicities $\pm2$; nonlinear gravity and shared parent open |
| `v1033_charged_disorder.py` | auxiliary $\mathbb{Z}_4$ flux corners and $h = 1/4 \ne h_\lambda = 1$; microscopic MMST open |
| `v1034_parent_mirror.py` | factorized gap $1 + \sqrt{3}/2$ and bare-hopping relative-bound obstruction; interacting mirror gap open |

*continued on next page*

| Script | What it machine-checks *(continued)* |
|---|---|
| `v1035_matter_coupling.py` | physical Fourier Ward propagation and prescribed-source response; dynamical and universal quantum coupling open |